



# Model Test Results Report

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 20-March-2026

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# 1 Executive Summary

This report documents the results of automated model tests run on **20 March 2026 at 12:15:19**.

Table 1: Overall Test Results

|                |         |
|----------------|---------|
| Total Tests    | 5844    |
| Passed         | 5826    |
| Failed         | 3       |
| Skipped        | 15      |
| Pass Rate      | 99.7%   |
| Total Duration | 381.05s |

## 1.1 Test Environment

|            |                                     |
|------------|-------------------------------------|
| Platform   | macOS-26.3.1-arm64-arm-64bit-Mach-O |
| Python     | 3.13.1                              |
| pytest     | 8.3.4                               |
| Coverage   | 96.3%                               |
| Git Commit | d904bf3dc530                        |

## 1.2 Results by Model

Table 2: Test Results by Model

| Model ID   | Model Name                   | Total | Pass | Fail | Skip | Time (s) |
|------------|------------------------------|-------|------|------|------|----------|
| MKM-DD-001 | Flood Risk (Depth-Damage)    | 128   | 128  | 0    | 0    | 0.01     |
| MKM-GH-001 | GEV Hazard Model             | 16    | 16   | 0    | 0    | 0.30     |
| MKM-MP-001 | Mortgage Pricer              | 67    | 67   | 0    | 0    | 0.30     |
| MKM-PR-001 | PRS Pricing Model            | 61    | 61   | 0    | 0    | 7.35     |
| MKM-PV-001 | Property Valuation           | 55    | 54   | 1    | 0    | 21.64    |
| MKM-SG-001 | Storm-Gauge Response Model   | 81    | 81   | 0    | 0    | 0.04     |
| MKM-SI-001 | Storm Intensity Distribution | 105   | 105  | 0    | 0    | 0.24     |
| MKM-SS-001 | Storm Sequence Generator     | 52    | 52   | 0    | 0    | 0.24     |
| MKM-TD-001 | Trading Desk                 | 278   | 278  | 0    | 0    | 2.31     |
| CDM-ALL    | CDM Schema Validation        | 76    | 76   | 0    | 0    | 0.01     |
| PLATFORM   | Platform Infrastructure      | 4925  | 4908 | 2    | 15   | 261.29   |

## 2 Test Traceability Matrix

The following table maps every executed test to its assigned model component, providing evidence of coverage required under SR 11-7 §4.2 and SS1/23.

| Model      | Module                                      | Test Name                                  | Status |
|------------|---------------------------------------------|--------------------------------------------|--------|
| MKM-DD-001 | risk_assessor.py::TestCombinedRiskScore     | test_score_50_to_100                       | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_over_100                        | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_basic_returns_float             | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_construction_concrete_lower     | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_construction_steel_lowest       | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_construction_timber_higher      | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_construction_wood_equals_timber | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_high_ltv_multiplier             | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_ltv_60_to_80_multiplier         | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_ltv_80_to_95_multiplier         | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_ltv_no_multiplier               | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_unknown_construction_type       | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_unknown_flood_level             | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_very_high_flood_capped          | Pass   |
|            | risk_assessor.py::TestCombinedRiskScore     | test_score_very_low_flood_lowest_score     | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_at_zero_is_max                        | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_close_band                            | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_far_band                              | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_medium_band                           | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_monotone_decrease                     | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_near_band                             | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_negative_returns_default              | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_none_returns_default                  | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_very_close                            | Pass   |
|            | risk_assessor.py::TestDistanceRiskEstimator | test_very_far                              | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_at_high_threshold                     | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_at_low_threshold                      | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_at_medium_threshold                   | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_below_low_threshold                   | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_high_range                            | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_medium_range                          | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_monotone_with_depth                   | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_negative_returns_unknown              | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_none_returns_unknown                  | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_very_high                             | Pass   |
|            | risk_assessor.py::TestFloodRiskLevel        | test_zero_is_very_low                      | Pass   |
|            | risk_assessor.py::TestGaugeReliability      | test_daily_minus_15                        | Pass   |
|            | risk_assessor.py::TestGaugeReliability      | test_decommissioned                        | Pass   |
|            | risk_assessor.py::TestGaugeReliability      | test_fully_operational                     | Pass   |

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| Model | Module                                           | Test Name                               | Status |
|-------|--------------------------------------------------|-----------------------------------------|--------|
|       | risk_assessor.py::TestGaugeReliability           | test_hourly_minus_5                     | Pass   |
|       | risk_assessor.py::TestGaugeReliability           | test_maintenance_required               | Pass   |
|       | risk_assessor.py::TestGaugeReliability           | test_monthly_minus_50                   | Pass   |
|       | risk_assessor.py::TestGaugeReliability           | test_realtime_no_adjustment             | Pass   |
|       | risk_assessor.py::TestGaugeReliability           | test_result_keys                        | Pass   |
|       | risk_assessor.py::TestGaugeReliability           | test_score_clamped_at_zero              | Pass   |
|       | risk_assessor.py::TestGaugeReliability           | test_temporarily_offline                | Pass   |
|       | risk_assessor.py::TestGaugeReliability           | test_unknown_status                     | Pass   |
|       | risk_assessor.py::TestGaugeReliability           | test_weekly_minus_30                    | Pass   |
|       | risk_assessor.py::TestGenerateRecommendations    | test_all_recommendations_are_strings    | Pass   |
|       | risk_assessor.py::TestGenerateRecommendations    | test_deep_flood_extra_items             | Pass   |
|       | risk_assessor.py::TestGenerateRecommendations    | test_high_risk_four_or_more             | Pass   |
|       | risk_assessor.py::TestGenerateRecommendations    | test_high_risk_mentions_flood_insurance | Pass   |
|       | risk_assessor.py::TestGenerateRecommendations    | test_low_risk                           | Pass   |
|       | risk_assessor.py::TestGenerateRecommendations    | test_mediation_risk                     | Pass   |
|       | risk_assessor.py::TestGenerateRecommendations    | test_very_low_risk                      | Pass   |
|       | risk_assessor.py::TestInsurancePremiumCalculator | test_base_rates_by_level                | Pass   |
|       | risk_assessor.py::TestInsurancePremiumCalculator | test_deep_flood_10_pct                  | Pass   |
|       | risk_assessor.py::TestInsurancePremiumCalculator | test_deep_flood_increases_premium       | Pass   |
|       | risk_assessor.py::TestInsurancePremiumCalculator | test_deep_flood_multiplier_formula      | Pass   |
|       | risk_assessor.py::TestInsurancePremiumCalculator | test_unknown_level_uses_default         | Pass   |
|       | risk_assessor.py::TestInsurancePremiumCalculator | test_zero_depth_no_effect               | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_at_high_boundary                   | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_at_low_boundary                    | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_at_moderate_boundary               | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_critical_ltv                       | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_high_ltv                           | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_low_ltv                            | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_moderate_ltv                       | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_none_returns_unknown               | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_normalises_95_pct                  | Pass   |
|       | risk_assessor.py::TestLTVRiskLevel               | test_normalises_percentage_input        | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_elevated_ltv_medium_flood          | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_high_flood_overrides               | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_high_ltv_medium_flood              | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_low_flood_risk                     | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_medium_flood_default               | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_minimal_risk                       | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_negative_value_critical            | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_negative_value_high                | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_negative_value_moderate            | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_none_ltv_no_crash                  | Pass   |
|       | risk_assessor.py::TestMortgageRisk               | test_very_high_flood_overrides          | Pass   |
|       | risk_assessor.py::TestPropertyVulnerability      | test_billage_to_water_increases_score   | Pass   |

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| Model | Module                                      | Test Name                                        | Status |
|-------|---------------------------------------------|--------------------------------------------------|--------|
|       | risk_assessor.py::TestPropertyVulnerability | test_flooded_depth_computed                      | Pass   |
|       | risk_assessor.py::TestPropertyVulnerability | test_highly_risk_recommendations_present         | Pass   |
|       | risk_assessor.py::TestPropertyVulnerability | test_no_flood                                    | Pass   |
|       | risk_assessor.py::TestPropertyVulnerability | test_no_elevation_returns_unknown                | Pass   |
|       | risk_assessor.py::TestPropertyVulnerability | test_no_flood_level_returns_unknown              | Pass   |
|       | risk_assessor.py::TestPropertyVulnerability | test_no_result_dict_keys                         | Pass   |
|       | risk_assessor.py::TestPropertyVulnerability | test_nothing_capped_at_100                       | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_base_pct_wins_when_higher_than_depth_factor | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_depth_1_to_2m                               | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_depth_half_to_1m                            | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_depth_over_2m                               | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_depth_under_half_m                          | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_negative_property_value                     | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_risk_level_percentages                      | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_zero_depth_uses_base_pct                    | Pass   |
|       | risk_assessor.py::TestValueAtRisk           | test_zero_property_value                         | Pass   |
|       | velocity.py::TestAttenuation                | test_at_length_scale                             | Pass   |
|       | velocity.py::TestAttenuation                | test_far_distance_near_zero                      | Pass   |
|       | velocity.py::TestAttenuation                | test_negative_distance                           | Pass   |
|       | velocity.py::TestAttenuation                | test_zero_distance_full_strength                 | Pass   |
|       | velocity.py::TestAttenuation                | test_zero_length                                 | Pass   |
|       | velocity.py::TestBuildPropertyHydrograph    | test_no_hours_present                            | Pass   |
|       | velocity.py::TestBuildPropertyHydrograph    | test_depth_never_negative                        | Pass   |
|       | velocity.py::TestBuildPropertyHydrograph    | test_each_entry_has_required_keys                | Pass   |
|       | velocity.py::TestBuildPropertyHydrograph    | test_empty_readings                              | Pass   |
|       | velocity.py::TestBuildPropertyHydrograph    | test_high_floor_prevents_flooding                | Pass   |
|       | velocity.py::TestBuildPropertyHydrograph    | test_no_flooding_before_arrival                  | Pass   |
|       | velocity.py::TestBuildPropertyHydrograph    | test_output_length_matches_input                 | Pass   |
|       | velocity.py::TestBuildPropertyHydrograph    | test_peak_produces_flooding                      | Pass   |
|       | velocity.py::TestComputeSlope               | test_basic_slope                                 | Pass   |
|       | velocity.py::TestComputeSlope               | test_min_slope_clamp                             | Pass   |
|       | velocity.py::TestComputeSlope               | test_slope_is_absolute                           | Pass   |
|       | velocity.py::TestComputeSlope               | test_zero_distance                               | Pass   |
|       | velocity.py::TestManningVelocity            | test_deeper_water_faster                         | Pass   |
|       | velocity.py::TestManningVelocity            | test_negative_depth_returns_zero                 | Pass   |
|       | velocity.py::TestManningVelocity            | test_negative_slope_uses_abs                     | Pass   |
|       | velocity.py::TestManningVelocity            | test_positive_depth_and_slope                    | Pass   |
|       | velocity.py::TestManningVelocity            | test_slope_clamped_to_min                        | Pass   |
|       | velocity.py::TestManningVelocity            | test_steeper_slope_faster                        | Pass   |
|       | velocity.py::TestManningVelocity            | test_zero_depth_returns_zero                     | Pass   |
|       | velocity.py::TestManningVelocity            | test_zero_roughness_returns_zero                 | Pass   |
|       | velocity.py::TestTravelTime                 | test_basic_travel_time                           | Pass   |
|       | velocity.py::TestTravelTime                 | test_farther_takes_longer                        | Pass   |
|       | velocity.py::TestTravelTime                 | test_zero_depth_infinite                         | Pass   |

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| Model      | Module                                     | Test Name                                   | Status |
|------------|--------------------------------------------|---------------------------------------------|--------|
|            | velocity.py::TestTravelTime                | test_zero_distance                          | Pass   |
| MKM-GH-001 | hazard_builder.py::TestGEVFit              | test_exceedance_probability_in_range        | Pass   |
|            | hazard_builder.py::TestGEVFit              | test_fit_returns_three_floats               | Pass   |
|            | hazard_builder.py::TestGEVFit              | test_force_gumbel_sets_shape_zero           | Pass   |
|            | hazard_builder.py::TestGEVFit              | test_higher_threshold_lower_probability     | Pass   |
|            | hazard_builder.py::TestGEVFit              | test_return_level_increases_with_period     | Pass   |
|            | hazard_builder.py::TestGaugeResponseModule | test_all_responses_returns_dict             | Pass   |
|            | hazard_builder.py::TestGaugeResponseModule | test_response_returns_gauge_response        | Pass   |
|            | hazard_builder.py::TestGaugeResponseModule | test_response_threshold_flags               | Pass   |
|            | hazard_builder.py::TestHazardCurveBuilder  | test_all_flood_probabilities_in_range       | Pass   |
|            | hazard_builder.py::TestHazardCurveBuilder  | test_build_returns_dict_of_curves           | Pass   |
|            | hazard_builder.py::TestHazardCurveBuilder  | test_build_curve_is_gauge_hazard_curve      | Pass   |
|            | hazard_builder.py::TestTermStructure       | test_cumulative_prob_increasing             | Pass   |
|            | hazard_builder.py::TestTermStructure       | test_probabilities_sum_to_one               | Pass   |
|            | hazard_builder.py::TestTermStructure       | test_returns_correct_length                 | Pass   |
|            | hazard_builder.py::TestTermStructure       | test_returns_term_structure_points          | Pass   |
|            | hazard_builder.py::TestTermStructure       | test_survival_prob_decreasing               | Pass   |
| MKM-MP-001 | credit.py::TestCreditSpread                | test_custom_tax_rate                        | Pass   |
|            | credit.py::TestCreditSpread                | test_floor_at_10bps                         | Pass   |
|            | credit.py::TestCreditSpread                | test_higher_affordability_higher_spread     | Pass   |
|            | credit.py::TestCreditSpread                | test_negative_income_fallback               | Pass   |
|            | credit.py::TestCreditSpread                | test_spread_bounded_by_schedule             | Pass   |
|            | credit.py::TestCreditSpread                | test_zero_income_fallback                   | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_batch_passes_flood_risk                | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_case_insensitive                       | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_flood_factor_in_result                 | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_flood_risk_increases_spread            | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_flood_risk_multipliers[High-1.4]       | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_flood_risk_multipliers[Low-1.05]       | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_flood_risk_multipliers[Medium-1.2]     | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_flood_risk_multipliers[Very High-1.75] | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_flood_risk_multipliers[Very Low-1.0]   | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_no_flood_risk_defaults_to_unity        | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_none_returns_unity                     | Pass   |
|            | credit.py::TestFloodRiskFactor             | test_unknown_category_returns_unity         | Pass   |
|            | credit.py::TestLTVRiskFactor               | test_ltv_thresholds[0.5-1.0]                | Pass   |
|            | credit.py::TestLTVRiskFactor               | test_ltv_thresholds[0.8-1.0]                | Pass   |
|            | credit.py::TestLTVRiskFactor               | test_ltv_thresholds[0.85-1.1]               | Pass   |
|            | credit.py::TestLTVRiskFactor               | test_ltv_thresholds[0.9-1.1]                | Pass   |
|            | credit.py::TestLTVRiskFactor               | test_ltv_thresholds[0.91-1.3]               | Pass   |

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| Model | Module                                       | Test Name                                        | Status |
|-------|----------------------------------------------|--------------------------------------------------|--------|
|       | credit.py::TestLTVRiskFactor                 | test_ltv_thresholds[0.95-1.3]                    | Pass   |
|       | credit.py::TestLTVRiskFactor                 | test_ltv_thresholds[0.96-1.5]                    | Pass   |
|       | credit.py::TestLTVRiskFactor                 | test_ltv_thresholds[1.0-1.5]                     | Pass   |
|       | credit.py::TestLTVRiskFactor                 | test_zero_property_value                         | Pass   |
|       | payment.py::TestMonthlyPayment               | test_higher_rate_higher_payment                  | Pass   |
|       | payment.py::TestMonthlyPayment               | test_shorter_term_higher_payment                 | Pass   |
|       | payment.py::TestMonthlyPayment               | test_standard_mortgage                           | Pass   |
|       | payment.py::TestMonthlyPayment               | test_zero_rate                                   | Pass   |
|       | payment.py::TestTotalCost                    | test_positive_rate_exceeds_principal             | Pass   |
|       | payment.py::TestTotalCost                    | test_zero_rate_equals_principal                  | Pass   |
|       | pricing.py::TestBatchPricing                 | test_default_ids_assigned                        | Pass   |
|       | pricing.py::TestBatchPricing                 | test_error_isolation                             | Pass   |
|       | pricing.py::TestBatchPricing                 | test_prices_multiple_mortgages                   | Pass   |
|       | pricing.py::TestConstructor                  | test_custom_tax_rate                             | Pass   |
|       | pricing.py::TestConstructor                  | test_default_tax_rate                            | Pass   |
|       | pricing.py::TestDebugAndEdgeCases            | test_debug_also_in_credit_spread                 | Pass   |
|       | pricing.py::TestDebugAndEdgeCases            | test_debug_true_does_not_crash                   | Pass   |
|       | pricing.py::TestDebugAndEdgeCases            | test_debug_with_zero_income                      | Pass   |
|       | pricing.py::TestDebugAndEdgeCases            | test_flood_risk_none_with_debug                  | Pass   |
|       | pricing.py::TestDebugAndEdgeCases            | test_zero_interest_rate                          | Pass   |
|       | pricing.py::TestPortfolioMetrics             | test_all_errors_returns_error                    | Pass   |
|       | pricing.py::TestPortfolioMetrics             | test_high_risk_count                             | Pass   |
|       | pricing.py::TestPortfolioMetrics             | test_mixed_results_exclude_errors                | Pass   |
|       | pricing.py::TestPortfolioMetrics             | test_valid_results                               | Pass   |
|       | pricing.py::TestPriceMortgage                | test_balance_converges_to_zero                   | Pass   |
|       | pricing.py::TestPriceMortgage                | test_expected_losses_non_negative                | Pass   |
|       | pricing.py::TestPriceMortgage                | test_fair_value_less_than_par                    | Pass   |
|       | pricing.py::TestPriceMortgage                | test_hazard_rates_positive                       | Pass   |
|       | pricing.py::TestPriceMortgage                | test_high_ltv_increases_spread                   | Pass   |
|       | pricing.py::TestPriceMortgage                | test_higher_haircut_more_loss                    | Pass   |
|       | pricing.py::TestPriceMortgage                | test_input_validation_clamps                     | Pass   |
|       | pricing.py::TestPriceMortgage                | test_ltv_factor_in_result                        | Pass   |
|       | pricing.py::TestPriceMortgage                | test_raises_on_invalid_types                     | Pass   |
|       | pricing.py::TestPriceMortgage                | test_recovery_haircut_zero_low_ltv               | Pass   |
|       | pricing.py::TestPriceMortgage                | test_survival_first_period_less_than_one         | Pass   |
|       | pricing.py::TestPriceMortgage                | test_survival_monotonically_decreasing           | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_id                    | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_quality               | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_rate                  | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_term                  | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_ltv                   | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_haircut               | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_survival              | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_flood_risk            | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_interest_rate         | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_has_all_properties        | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_output_matches_properties | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_output_format             | Pass   |
|       | mortgage_generator.py::TestMortgageGenerator | test_generate_mortgage_output_unique             | Pass   |

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| Model      | Module                             | Test Name                                 | Status |
|------------|------------------------------------|-------------------------------------------|--------|
| MKM-PR-001 | maturity.py::TestComputeMaturity   | test_date_from_feb_2026                   | Pass   |
|            | maturity.py::TestComputeMaturity   | test_date_from_feb_2026                   | Pass   |
|            | maturity.py::TestComputeMaturity   | test_date_from_jan_2027                   | Pass   |
|            | maturity.py::TestComputeMaturity   | test_date_from_sep_2026                   | Pass   |
|            | maturity.py::TestComputeMaturity   | test_date_from_feb_2026                   | Pass   |
|            | maturity.py::TestComputeMaturity   | test_date_from_sep_2026                   | Pass   |
|            | maturity.py::TestComputeMaturity   | test_maturity_is_friday                   | Pass   |
|            | maturity.py::TestComputeMaturity   | test_roll_jump_at_august                  | Pass   |
|            | maturity.py::TestCurrentRollDate   | test_aug_period                           | Pass   |
|            | maturity.py::TestCurrentRollDate   | test_dec_uses_aug                         | Pass   |
|            | maturity.py::TestCurrentRollDate   | test_feb_period                           | Pass   |
|            | maturity.py::TestCurrentRollDate   | test_jan_uses_prev_aug                    | Pass   |
|            | maturity.py::TestCurrentRollDate   | test_jul_still_feb_roll                   | Pass   |
|            | maturity.py::TestCurrentRollDate   | test_mar_still_feb_roll                   | Pass   |
|            | maturity.py::TestDefaultDates      | test_compute_maturity_default             | Pass   |
|            | maturity.py::TestDefaultDates      | test_current_roll_date_default            | Pass   |
|            | maturity.py::TestDefaultDates      | test_maturity_table_default               | Pass   |
|            | maturity.py::TestLastFridayOfMonth | test_aug_2026                             | Pass   |
|            | maturity.py::TestLastFridayOfMonth | test_feb_2026                             | Pass   |
|            | maturity.py::TestLastFridayOfMonth | test_may_2029                             | Pass   |
|            | maturity.py::TestLastFridayOfMonth | test_nov_2029                             | Pass   |
|            | maturity.py::TestLastFridayOfMonth | test_result_is_friday                     | Pass   |
|            | maturity.py::TestMaturityTable     | test_actual_years_increasing              | Pass   |
|            | maturity.py::TestMaturityTable     | test_all_may_for_feb_roll                 | Pass   |
|            | maturity.py::TestMaturityTable     | test_all_nov_for_aug_roll                 | Pass   |
|            | maturity.py::TestMaturityTable     | test_maturity_str_format                  | Pass   |
|            | maturity.py::TestMaturityTable     | test_table_has_5_entries                  | Pass   |
|            | maturity.py::TestMaturityTable     | test_tenors_sequential                    | Pass   |
|            | maturity.py::TestTenorFromMaturity | test_all_tenors_round_trip                | Pass   |
|            | maturity.py::TestTenorFromMaturity | test_closest_match_fallback               | Pass   |
|            | maturity.py::TestTenorFromMaturity | test_exact_match                          | Pass   |
|            | basis.py::TestBasisWaterfall       | test_basis_waterfall_in_generated_output  | Pass   |
|            | basis.py::TestBasisWaterfall       | test_composition_flat_penalty             | Pass   |
|            | basis.py::TestBasisWaterfall       | test_composition_pre2000_penalty          | Pass   |
|            | basis.py::TestBasisWaterfall       | test_distance_basis_capped                | Pass   |
|            | basis.py::TestBasisWaterfall       | test_distance_basis_scales_linearly       | Pass   |
|            | basis.py::TestBasisWaterfall       | test_elevation_above_gauge_reduces_spread | Pass   |
|            | basis.py::TestBasisWaterfall       | test_elevation_below_gauge_no_benefit     | Pass   |
|            | basis.py::TestBasisWaterfall       | test_elevation_capped_at_3bp              | Pass   |
|            | basis.py::TestBasisWaterfall       | test_model_uncertainty_is_fixed           | Pass   |
|            | basis.py::TestBasisWaterfall       | test_terrain_zone1_low_risk               | Pass   |
|            | basis.py::TestBasisWaterfall       | test_terrain_zone3_high_risk              | Pass   |
|            | basis.py::TestBasisWaterfall       | test_total_basis_sums_components          | Pass   |
|            | basis.py::TestEdgeCases            | test_empty_propertyts_dir                 | Pass   |

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| Model      | Module                               | Test Name                                  | Status |
|------------|--------------------------------------|--------------------------------------------|--------|
|            | basis.py::TestEdgeCases              | test_no_gauge_hazard_curves                | Pass   |
|            | basis.py::TestEdgeCases              | test_no_propertyts_dir                     | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_base_rate_calculation              | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_base_rate_thresholds               | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_generator_produces_output          | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_generator_structure                | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_generator_mission_rate             | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_generator_structure                | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_properties_with_gev                | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_survival_decreases_with_tenor      | Pass   |
|            | generator.py::TestPropertyHazard     | test_no_survival_structure                 | Pass   |
|            | pricing.py::TestPRSSpreadComp        | test_no_positive_spread                    | Pass   |
|            | pricing.py::TestPRSSpreadComp        | test_no_spread_increases_with_hazard       | Pass   |
|            | pricing.py::TestPRSSpreadComp        | test_no_spread_values_at_different_tenors  | Pass   |
|            | pricing.py::TestPRSSpreadComp        | test_no_zero_hazard_rate                   | Pass   |
|            | pricing.py::TestRecoveryRates        | test_recovery_rates_by_trigger             | Pass   |
|            | pricing.py::TestRecoveryRates        | test_recovery_reduces_spread               | Pass   |
| MKM-PV-001 | property_generator.py::TestBuild     | test_no_dict_schema_returns_empty          | Pass   |
|            | property_generator.py::TestBuild     | test_no_hips_type_field                    | Pass   |
|            | property_generator.py::TestDate      | test_no_date_to_isoformat                  | Pass   |
|            | property_generator.py::TestDate      | test_no_float                              | Pass   |
|            | property_generator.py::TestDate      | test_no_int                                | Pass   |
|            | property_generator.py::TestDate      | test_no_list                               | Pass   |
|            | property_generator.py::TestDate      | test_no_unknown_type_raises                | Pass   |
|            | property_generator.py::TestGenerator | test_no_base_rate_has_required_key         | Pass   |
|            | property_generator.py::TestGenerator | test_no_base_rate_returns_correct_count    | Pass   |
|            | property_generator.py::TestGenerator | test_no_base_rate_returns_elevation        | Pass   |
|            | property_generator.py::TestGenerator | test_no_properties_basic                   | Pass   |
|            | property_generator.py::TestGenerator | test_no_properties_output_file_created     | Pass   |
|            | property_generator.py::TestProperty  | test_no_general_data_quality               | Pass   |
|            | property_generator.py::TestProperty  | test_no_general_data_quality               | Pass   |
|            | property_generator.py::TestProperty  | test_no_general_data_exists                | Pass   |
|            | property_generator.py::TestProperty  | test_no_general_data_is_valid_json         | Pass   |
|            | property_generator.py::TestProperty  | test_no_general_data_count                 | Pass   |
|            | property_generator.py::TestProperty  | test_no_general_data_returns_expected_keys | Pass   |
|            | property_generator.py::TestProperty  | test_no_general_data_output_format         | Pass   |
|            | property_generator.py::TestProperty  | test_no_general_data_output_unique         | Pass   |
|            | property_generator.py::TestProperty  | test_no_generate_grid_reference_format     | Pass   |
|            | property_generator.py::TestProperty  | test_no_generate_owner_name_two_words      | Pass   |
|            | property_generator.py::TestProperty  | test_no_generate_past_date_format          | Pass   |
|            | property_generator.py::TestProperty  | test_no_generate_past_date_range           | Pass   |
|            | rand_property_mortgage.py::Test      | test_no_date_rate_insurance_premium_flood  | Pass   |
|            | rand_property_mortgage.py::Test      | test_no_date_rate_insurance_premium_no_are | Pass   |
|            | rand_property_mortgage.py::Test      | test_no_date_rate_insurance_premium_no_val | Pass   |

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| Model      | Module                                                                    | Test Name                                                          | Status    |
|------------|---------------------------------------------------------------------------|--------------------------------------------------------------------|-----------|
|            | rand_property_mortgage.py::TestCalculateMortgageMonthlyPayment            | Test calculate mortgage monthly payment all types                  | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageMonthlyPayment            | Test calculate mortgage monthly payment no area                    | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageMonthlyPayment            | Test calculate mortgage monthly payment no value                   | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageMonthlyPayment            | Test calculate mortgage monthly payment positive                   | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageMonthlyPayment            | Test calculate mortgage monthly payment area all types             | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageAdaptivePrice             | Test calculate mortgage adaptive price invalid date                | Fail back |
|            | rand_property_mortgage.py::TestCalculateMortgageAdaptivePrice             | Test calculate mortgage adaptive price no current value            | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageAdaptivePrice             | Test calculate mortgage adaptive price positive                    | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageAdaptivePrice             | Test calculate mortgage adaptive price with sale date 1 to 2 years | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageAdaptivePrice             | Test calculate mortgage adaptive price with sale date over 2 years | Pass      |
|            | rand_property_mortgage.py::TestCalculateMortgageAdaptivePrice             | Test calculate mortgage adaptive price with sale date recent       | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageFinancialsReturnDict       | Test generate mortgage financials return dict                      | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageFinancialDataKeys          | Test generate mortgage financial data keys                         | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageRateRange                  | Test generate mortgage rate range                                  | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageRateLessThanValue          | Test generate mortgage rate less than value                        | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageRateFormat                 | Test generate mortgage rate format                                 | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageRateConsistencyCheck       | Test generate mortgage rate consistency check                      | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageInsurancePremiumPositive   | Test generate mortgage insurance premium positive                  | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgagePropertyAreaPositive       | Test generate mortgage property area positive                      | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgagePropertyValueInRange       | Test generate mortgage property value in range                     | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageBedroomsDetached           | Test generate mortgage bedrooms detached                           | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageBedroomsFlat               | Test generate mortgage bedrooms flat                               | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgagePostcodeFormat             | Test generate mortgage postcode format                             | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageVfieldValueDecimal         | Test generate mortgage vfield value decimal                        | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageVfieldValueText            | Test generate mortgage vfield value text                           | FAIL      |
|            | rand_property_mortgage.py::TestGenerateMortgageConstructionYearInRange    | Test generate mortgage construction year in range                  | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgagePropertyIdFormat           | Test generate mortgage property id format                          | Pass      |
|            | rand_property_mortgage.py::TestGenerateMortgageReturnDictWithRequiredKeys | Test generate mortgage return dict with required keys              | Pass      |
| MKM-SG-001 | forward_model.py::TestComputeABRResponseIdsMatch                          | Test compute ABR response ids match                                | Pass      |
|            | forward_model.py::TestComputeABRResponseGaugeHigherResponse               | Test compute ABR response gauge higher response                    | Pass      |
|            | forward_model.py::TestComputeABRResponseOneResponsePerGauge               | Test compute ABR response one response per gauge                   | Pass      |
|            | forward_model.py::TestComputeDestAllKernelsDecreaseWithDistance           | Test compute Dest all kernels decrease with distance               | Pass      |
|            | forward_model.py::TestComputeDestAllKernelsIn0.1Range                     | Test compute Dest all kernels in 0.1 range                         | Pass      |
|            | forward_model.py::TestComputeDestExponentialAtHalf                        | Test compute Dest exponential at half                              | Pass      |
|            | forward_model.py::TestComputeDestExponentialAtOneLambda                   | Test compute Dest exponential at one lambda                        | Pass      |
|            | forward_model.py::TestComputeDestGaussianAtOneSigma                       | Test compute Dest gaussian at one sigma                            | Pass      |
|            | forward_model.py::TestComputeDestGaussianAtTwoSigma                       | Test compute Dest gaussian at two sigma                            | Pass      |
|            | forward_model.py::TestComputeDestLinearBeyondCutoffIsZero                 | Test compute Dest linear beyond cutoff is zero                     | Pass      |
|            | forward_model.py::TestComputeDestLinearMidpoint                           | Test compute Dest linear midpoint                                  | Pass      |
|            | forward_model.py::TestComputeDestZeroDistanceReturnsOneAllKernels         | Test compute Dest zero distance returns one all kernels            | Pass      |
|            | forward_model.py::TestComputeDestSiftArcGaugesLastPoint                   | Test compute Dest sift Arc gauges last point                       | Pass      |
|            | forward_model.py::TestComputeDestSiftArcGaugesNonzero                     | Test compute Dest sift Arc gauges nonzero                          | Pass      |
|            | forward_model.py::TestComputeDestSiftArcGaugesFirstPoint                  | Test compute Dest sift Arc gauges first point                      | Pass      |
|            | forward_model.py::TestComputeDestSiftArcGaugesReturnsZero                 | Test compute Dest sift Arc gauges returns zero                     | Pass      |

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| Model | Module                                   | Test Name                                              | Status |
|-------|------------------------------------------|--------------------------------------------------------|--------|
|       | forward_model.py::TestComputeIntensity   | test_intensity_track_is_lower                          | Pass   |
|       | forward_model.py::TestComputeResponse    | test_accumulation_nonnegative                          | Pass   |
|       | forward_model.py::TestComputeResponse    | test_duration_nonnegative                              | Pass   |
|       | forward_model.py::TestComputeResponse    | test_encoded_flag_correct                              | Pass   |
|       | forward_model.py::TestComputeResponse    | test_high_intensity_storm_floods                       | Pass   |
|       | forward_model.py::TestComputeResponse    | test_peak_level_at_least_base                          | Pass   |
|       | forward_model.py::TestComputeResponse    | test_peak_level_is_max_of_series                       | Pass   |
|       | forward_model.py::TestComputeResponse    | test_response_ids_match                                | Pass   |
|       | forward_model.py::TestComputeResponse    | test_timeseries_length                                 | Pass   |
|       | forward_model.py::TestComputeResponse    | test_dict_roundtrip                                    | Pass   |
|       | forward_model.py::TestComputeResponse    | test_dict_without_timeseries                           | Pass   |
|       | forward_model.py::TestFindNearTrackPoint | test_track_point_is_positive                           | Pass   |
|       | forward_model.py::TestFindNearTrackPoint | test_track_point_nearest_of_three                      | Pass   |
|       | forward_model.py::TestFindNearTrackPoint | test_track_point_outtrack_point                        | Pass   |
|       | forward_model.py::TestHaversine          | test_equatorial_degree_approx_111km                    | Pass   |
|       | forward_model.py::TestHaversine          | test_london_to_paris_approx_340km                      | Pass   |
|       | forward_model.py::TestHaversine          | test_same_point_is_zero                                | Pass   |
|       | forward_model.py::TestHaversine          | test_symmetric                                         | Pass   |
|       | forward_model.py::TestIntensityLevel     | test_notone_increasing                                 | Pass   |
|       | forward_model.py::TestIntensityLevel     | test_negative_intensity_returns_zero                   | Pass   |
|       | forward_model.py::TestIntensityLevel     | test_positive_intensity_positive_level                 | Pass   |
|       | forward_model.py::TestIntensityLevel     | test_sensitivity_scales_output                         | Pass   |
|       | forward_model.py::TestIntensityLevel     | test_uses_historical_high_when_set                     | Pass   |
|       | forward_model.py::TestIntensityLevel     | test_zero_intensity_returns_zero                       | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder  | test_encoder_codes_datetime                            | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder  | test_encoder_codes_numpy_float32                       | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder  | test_encoder_codes_numpy_floating                      | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder  | test_encoder_codes_numpy_integer                       | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder  | test_encoder_codes_numpy_ndarray                       | Pass   |
|       | gauge_generator.py::TestDateTimeEncoder  | test_encoder_decode_raises_for_unknown                 | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_ordering                                | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_quality                                 | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_caption_propagates                      | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_end_time_hits_end_time_branch           | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_build_section_non_dict_schema_returns_empty       | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_build_section_skips_metadata_keys                 | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_exceeds_available_points_capped         | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_exceeds_raises_value_error              | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_gauges_convenience_function             | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_single_gauge_failure_count              | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_logs_works                              | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_missing_values_creates_missing_sections | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_false_sets_warning_level                | Pass   |
|       | gauge_generator.py::TestGaugeGenerator   | test_attr_data_file_exists                             | Pass   |

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| Model      | Module                                 | Test Name                                   | Status |
|------------|----------------------------------------|---------------------------------------------|--------|
|            | gauge_generator.py::TestGaugeGenerator | test_output_file_is_valid_json              | Pass   |
|            | gauge_generator.py::TestGaugeGenerator | test_output_json_has_correct_count          | Pass   |
|            | gauge_generator.py::TestGaugeGenerator | test_main_block                             | Pass   |
|            | gauge_generator.py::TestGaugeGenerator | test_tag_output_format                      | Pass   |
|            | gauge_generator.py::TestGaugeGenerator | test_tag_ids_unique                         | Pass   |
|            | gauge_generator.py::TestGaugeGenerator | test_tag_output_correct_count               | Pass   |
|            | gauge_generator.py::TestGaugeGenerator | test_tag_output_returns_expected_keys       | Pass   |
|            | gauge_generator.py::TestGaugeGenerator | test_locations_have_coordinates             | Pass   |
|            | rand_gauge.py::TestFloodRiskAssessment | test_choose_to_thames_high_risk             | Pass   |
|            | rand_gauge.py::TestFloodRiskAssessment | test_choose_from_thames_lower_risk          | Pass   |
|            | rand_gauge.py::TestGenerateFieldValues | test_date_field_returns_date_string         | Pass   |
|            | rand_gauge.py::TestGenerateFieldValues | test_decimal_field_returns_float            | Pass   |
|            | rand_gauge.py::TestGenerateFieldValues | test_gauge_name_includes_area               | Pass   |
|            | rand_gauge.py::TestGenerateFieldValues | test_menu_field_returns_valid_option        | Pass   |
|            | rand_gauge.py::TestGenerateFieldValues | test_text_field_returns_string              | Pass   |
|            | rand_gauge.py::TestGenerateGauges      | test_location_name_stored                   | Pass   |
|            | rand_gauge.py::TestGenerateGauges      | test_flood_stage_ordering                   | Pass   |
|            | rand_gauge.py::TestGenerateGauges      | test_gauge_id_format                        | Pass   |
|            | rand_gauge.py::TestGenerateGauges      | test_historical_high_level_range            | Pass   |
|            | rand_gauge.py::TestGenerateGauges      | test_install_date_format                    | Pass   |
|            | rand_gauge.py::TestGenerateGauges      | test_returns_dict_with_required_keys        | Pass   |
| MKM-SI-001 | distribution.py::TestConstruction      | test_custom_parameters                      | Pass   |
|            | distribution.py::TestConstruction      | test_default_parameters                     | Pass   |
|            | distribution.py::TestConstruction      | test_from_params_object                     | Pass   |
|            | distribution.py::TestConstruction      | test_from_scenario_baseline                 | Pass   |
|            | distribution.py::TestConstruction      | test_from_scenario_unknown_raises           | Pass   |
|            | distribution.py::TestConstruction      | test_seed_gives_deterministic_output        | Pass   |
|            | distribution.py::TestConstruction      | test_tail_prob_between_0_and_1              | Pass   |
|            | distribution.py::TestExceedance        | test_above_max_is_zero                      | Pass   |
|            | distribution.py::TestExceedance        | test_above_threshold_uses_pareto            | Pass   |
|            | distribution.py::TestExceedance        | test_below_max_intensity_is_zero            | Pass   |
|            | distribution.py::TestExceedance        | test_below_min_intensity_is_one             | Pass   |
|            | distribution.py::TestExceedance        | test_below_threshold_continuous             | Pass   |
|            | distribution.py::TestExceedance        | test_below_min_is_one                       | Pass   |
|            | distribution.py::TestExceedance        | test_below_threshold_uses_normal            | Pass   |
|            | distribution.py::TestExceedance        | test_below_monotonically_decreasing         | Pass   |
|            | distribution.py::TestInverseExceedance | test_one_prob_returns_min                   | Pass   |
|            | distribution.py::TestInverseExceedance | test_result_in_bounds                       | Pass   |
|            | distribution.py::TestInverseExceedance | test_roundtrip_above_threshold              | Pass   |
|            | distribution.py::TestInverseExceedance | test_roundtrip_below_threshold              | Pass   |
|            | distribution.py::TestInverseExceedance | test_zero_prob_returns_max                  | Pass   |
|            | distribution.py::TestParetoInternal    | test_fat_tail_higher_exceedance             | Pass   |
|            | distribution.py::TestParetoInternal    | test_pareto_above_threshold_between_0_and_1 | Pass   |
|            | distribution.py::TestParetoInternal    | test_pareto_at_threshold_is_one             | Pass   |

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| Model | Module                                    | Test Name                              | Status |
|-------|-------------------------------------------|----------------------------------------|--------|
|       | distribution.py::TestParetoInterpretation | test_pareto_below_threshold_is_one     | Pass   |
|       | distribution.py::TestReturnPeriod         | test_100yr_greater_than_10yr           | Pass   |
|       | distribution.py::TestReturnPeriod         | test_ordering_10_50_100                | Pass   |
|       | distribution.py::TestReturnPeriod         | test_result_in_bounds                  | Pass   |
|       | distribution.py::TestReturnPeriod         | test_storms_per_year_affects_result    | Pass   |
|       | distribution.py::TestSampling             | test_different_seeds_different_output  | Pass   |
|       | distribution.py::TestSampling             | test_sample_n_returns_correct_length   | Pass   |
|       | distribution.py::TestSampling             | test_sample_single                     | Pass   |
|       | distribution.py::TestSampling             | test_samples_within_bounds             | Pass   |
|       | distribution.py::TestSampling             | test_seeded_reproducibility            | Pass   |
|       | distribution.py::TestSerialisation        | test_describe_returns_string           | Pass   |
|       | distribution.py::TestSerialisation        | test_params_roundtrip                  | Pass   |
|       | distribution.py::TestSerialisation        | test_tail_probability_matches_internal | Pass   |
|       | distribution.py::TestSerialisation        | test_to_dict_keys                      | Pass   |
|       | distribution.py::TestSummaryStatistics    | test_exceedance_probs_decreasing       | Pass   |
|       | distribution.py::TestSummaryStatistics    | test_thin_tail_higher_p99_than_thin    | Pass   |
|       | distribution.py::TestSummaryStatistics    | test_mean_in_reasonable_range          | Pass   |
|       | distribution.py::TestSummaryStatistics    | test_percentiles_ordered               | Pass   |
|       | distribution.py::TestSummaryStatistics    | test_required_keys_present             | Pass   |
|       | storm_duration.py::TestDurationDefinition | test_all_categories_defined            | Pass   |
|       | storm_duration.py::TestDurationDefinition | test_catastrophic_extended             | Pass   |
|       | storm_duration.py::TestDurationDefinition | test_each_tier_doubles_approx          | Pass   |
|       | storm_duration.py::TestDurationDefinition | test_extreme_extended                  | Pass   |
|       | storm_duration.py::TestDurationDefinition | test_moderate_extended                 | Pass   |
|       | storm_duration.py::TestDurationDefinition | test_severe_extended                   | Pass   |
|       | storm_duration.py::TestHelpers            | test_get_duration_range                | Pass   |
|       | storm_duration.py::TestHelpers            | test_get_max_duration                  | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_invalid_category                  | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_mode_is_most_common               | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_rng_uses_default                  | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_reproducibility                   | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_within_bounds[baseline]           | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_within_bounds[catastrophic]       | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_within_bounds[extreme]            | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_within_bounds[minimal]            | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_within_bounds[moderate]           | Pass   |
|       | storm_duration.py::TestSampleDefinition   | test_within_bounds[severe]             | Pass   |
|       | storm_gap.py::TestFitsInWindow            | test_constants                         | Pass   |
|       | storm_gap.py::TestFitsInWindow            | test_exact_boundary                    | Pass   |
|       | storm_gap.py::TestFitsInWindow            | test_exceeds_window                    | Pass   |
|       | storm_gap.py::TestFitsInWindow            | test_simple_doublet_fits               | Pass   |
|       | storm_gap.py::TestFitsInWindow            | test_single_storm                      | Pass   |
|       | storm_gap.py::TestGapParams               | test_long_gaps                         | Pass   |
|       | storm_gap.py::TestGapParams               | test_medium_gaps                       | Pass   |

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| Model      | Module                                         | Test Name                                   | Status |
|------------|------------------------------------------------|---------------------------------------------|--------|
|            | storm_gap.py::TestGapParams                    | test_short_gaps                             | Pass   |
|            | storm_gap.py::TestGetGapRange                  | test_doublet_medium                         | Pass   |
|            | storm_gap.py::TestGetGapRange                  | test_isolated_zero                          | Pass   |
|            | storm_gap.py::TestSampleGap                    | test_doublet_uses_medium                    | Pass   |
|            | storm_gap.py::TestSampleGap                    | test_isolated_returns_zero                  | Pass   |
|            | storm_gap.py::TestSampleGap                    | test_no_rng_uses_default                    | Pass   |
|            | storm_gap.py::TestSampleGap                    | test_persistent_uses_short                  | Pass   |
|            | storm_gap.py::TestSampleGap                    | test_reproducibility                        | Pass   |
|            | storm_gap.py::TestSampleGap                    | test_within_bounds[SequenceType.CLUSTER]    | Pass   |
|            | storm_gap.py::TestSampleGap                    | test_within_bounds[SequenceType.DOUBLET]    | Pass   |
|            | storm_gap.py::TestSampleGap                    | test_within_bounds[SequenceType.PERSISTENT] | Pass   |
|            | storm_gap.py::TestValidation                   | test_duration_mismatch                      | Pass   |
|            | storm_gap.py::TestValidation                   | test_exceeds_precip_window                  | Pass   |
|            | storm_gap.py::TestValidation                   | test_invalid_intensity_category             | Pass   |
|            | storm_gap.py::TestValidation                   | test_invalid_sequence_type                  | Pass   |
|            | storm_gap.py::TestValidation                   | test_overlapping_storms                     | Pass   |
|            | storm_gap.py::TestValidation                   | test_storm_count_mismatch                   | Pass   |
|            | storm_gap.py::TestValidation                   | test_too_many_storms                        | Pass   |
|            | storm_gap.py::TestValidation                   | test_valid_doublet                          | Pass   |
|            | storm_sequence.py::TestEnums                   | test_gap_types                              | Pass   |
|            | storm_sequence.py::TestEnums                   | test_sequence_gap_mapping                   | Pass   |
|            | storm_sequence.py::TestEnums                   | test_sequence_types                         | Pass   |
|            | storm_sequence.py::TestEnums                   | test_storm_count_ranges                     | Pass   |
|            | storm_sequence.py::TestIDGenerator             | test_ids_unique                             | Pass   |
|            | storm_sequence.py::TestIDGenerator             | test_id_sequence_id_format                  | Pass   |
|            | storm_sequence.py::TestIDGenerator             | test_id_sequence_ids_unique                 | Pass   |
|            | storm_sequence.py::TestIDGenerator             | test_id_storm_id_format                     | Pass   |
|            | storm_sequence.py::TestSequenceStorm           | test_create_basic                           | Pass   |
|            | storm_sequence.py::TestSequenceStorm           | test_from_dict_round_trip                   | Pass   |
|            | storm_sequence.py::TestSequenceStorm           | test_precipitation_rounding                 | Pass   |
|            | storm_sequence.py::TestSequenceStorm           | test_no_dict                                | Pass   |
|            | storm_sequence.py::TestStormSequence           | test_no_antecedent_defaults                 | Pass   |
|            | storm_sequence.py::TestStormSequence           | test_create_doublet                         | Pass   |
|            | storm_sequence.py::TestStormSequence           | test_from_dict_round_trip                   | Pass   |
|            | storm_sequence.py::TestStormSequence           | test_isolated_no_gaps                       | Pass   |
|            | storm_sequence.py::TestStormSequence           | test_spatial_correlation_defaults           | Pass   |
|            | storm_sequence.py::TestStormSequence           | test_spatial_correlation_enabled            | Pass   |
|            | storm_sequence.py::TestStormSequence           | test_to_dict                                | Pass   |
| MKM-SS-001 | batch_generator.py::TestGenerateStormSequences | test_generate_storm_sequences               | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_doublet_propagated                     | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_doublet_count                          | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_doublet_seeds_differ                   | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_doublet_sequence_type_doublet          | Pass   |
|            | batch_generator.py::TestGenerateStormSequences | test_doublet_sequence_type_isolated         | Pass   |

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| Model | Module                                        | Test Name                                           | Status |
|-------|-----------------------------------------------|-----------------------------------------------------|--------|
|       | batch_generator.py::TestGenerateEventSeqBasic | test_event_seq_basic_with_same_seed                 | Pass   |
|       | batch_generator.py::TestGenerateEventSeqBasic | test_event_seq_basic_dist                           | Pass   |
|       | batch_generator.py::TestGenerateEventSeqBasic | test_event_seq_basic_highlights_change_distribution | Pass   |
|       | batch_generator.py::TestGenerateEventSeqBasic | test_event_seq_basic_categories_present             | Pass   |
|       | batch_generator.py::TestGenerateEventSeqBasic | test_event_seq_basic_hypertension_distribution      | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_parent_dirs                         | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_missing_file_raises                 | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_returns_correct_count               | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_wrong_schema_raises                 | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_sequences_still_valid               | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_trip_num_storms                     | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_trip_precipitation                  | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_trip_sequence_type                  | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_trip_storm_fields                   | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_trip_total_duration                 | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_creates_file                        | Pass   |
|       | batch_generator.py::TestSaveLoadBatch         | test_batch_load_schema_version_in_file              | Pass   |
|       | batch_generator.py::TestSaveSummary           | test_summary_creates_file                           | Pass   |
|       | batch_generator.py::TestSaveSummary           | test_summary_duration_stats_present                 | Pass   |
|       | batch_generator.py::TestSaveSummary           | test_summary_num_sequences_matches                  | Pass   |
|       | batch_generator.py::TestSaveSummary           | test_summary_precipitation_stats_present            | Pass   |
|       | batch_generator.py::TestSaveSummary           | test_summary_schema_version                         | Pass   |
|       | batch_generator.py::TestSaveSummary           | test_summary_type_counts_sum_to_total               | Pass   |
|       | batch_generator.py::TestSaveSummary           | test_summary_type_fractions_sum_to_one              | Pass   |
|       | batch_generator.py::TestValidation            | test_all_sequences_valid                            | Pass   |
|       | batch_generator.py::TestValidation            | test_medium_batch_all_valid                         | Pass   |
|       | batch_generator.py::TestValidation            | test_no_negative_durations                          | Pass   |
|       | batch_generator.py::TestValidation            | test_no_negative_precipitation                      | Pass   |
|       | batch_generator.py::TestValidation            | test_timing_constraint                              | Pass   |
|       | sequence_gen.py::TestBuildIsolated            | test_isolated_duration_clamped                      | Pass   |
|       | sequence_gen.py::TestBuildIsolated            | test_isolated_has_one_storm                         | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_stats                             | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_id_in_storms                      | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_sequence                          | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_force                             | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_force                             | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_category_ofen_isolated            | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_calls                             | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_sequence                          | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_storm_sequence                    | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_fits_in_window                    | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_correct_fields                    | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_catchment                         | Pass   |
|       | sequence_gen.py::TestSequenceGenerator        | test_agg_generate_default_catchment                 | Pass   |

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| Model      | Module                                   | Test Name                                       | Status |
|------------|------------------------------------------|-------------------------------------------------|--------|
|            | sequence_gen.py::TestSequenceGenerator   | test_base_seed_reproducibility                  | Pass   |
|            | sequence_gen.py::TestSequenceGenerator   | test_unknown_catchment_defaults_to_35           | Pass   |
|            |                                          |                                                 |        |
| MKM-TD-001 | leg_pvs.py::TestComputeLegPV             | test_higher_spread_means_higher_premium         | Pass   |
|            | leg_pvs.py::TestComputeLegPV             | test_positive_hazard_rate                       | Pass   |
|            | leg_pvs.py::TestComputeLegPV             | test_zero_hazard_rate                           | Pass   |
|            | market_making.py::TestGenerateBook       | test_balanced_book                              | Pass   |
|            | market_making.py::TestGenerateBook       | test_balanced_notional                          | Pass   |
|            | market_making.py::TestGenerateBook       | test_counterparties_distributed                 | Pass   |
|            | market_making.py::TestGenerateBook       | test_files_saved                                | Pass   |
|            | market_making.py::TestGenerateBook       | test_payer_trades_below_fair                    | Pass   |
|            | market_making.py::TestGenerateBook       | test_receiver_trades_above_fair                 | Pass   |
|            | market_making.py::TestGenerateBook       | test_reproducible_with_seed                     | Pass   |
|            | market_making.py::TestGenerateBook       | test_small_mtm                                  | Pass   |
|            | market_making.py::TestGenerateBook       | test_valid_cdm_structure                        | Pass   |
|            | market_making.py::TestPrintBook          | test_summary_runs                               | Pass   |
|            | market_making.py::TestTradeSpreadBook    | test_book_making_spreads_are_integers           | Pass   |
|            | market_making.py::TestTradeSpreadBook    | test_book_making_central_spreads_are_integers   | Pass   |
|            | thames_central.py::TestThamesCentralBook | test_book_saved                                 | Pass   |
|            | thames_central.py::TestThamesCentralBook | test_book_calendar_spreads                      | Pass   |
|            | thames_central.py::TestThamesCentralBook | test_book_reproducible_with_seed                | Pass   |
|            | thames_central.py::TestThamesCentralBook | test_book_count                                 | Pass   |
|            | thames_central.py::TestThamesCentralBook | test_book_central_gauges                        | Pass   |
|            | thames_central.py::TestThamesCentralBook | test_book_valid_cdm_structure                   | Pass   |
|            | thames_central.py::TestThamesCentralBook | test_book_minster_is_net_short                  | Pass   |
|            | engine.py::TestDeltaEngine               | test_build_risk_grid                            | Pass   |
|            | engine.py::TestDeltaEngine               | test_enrich_receiver_trade_negative_dv01        | Pass   |
|            | engine.py::TestDeltaEngine               | test_enrich_trade                               | Pass   |
|            | engine.py::TestDeltaEngine               | test_payer_receiver_dv01_opposite               | Pass   |
|            | engine.py::TestHazardCurveRevaluation    | test_curve_change_moves_fair_spread             | Pass   |
|            | engine.py::TestHazardCurveRevaluation    | test_curve_change_moves_mtm                     | Pass   |
|            | engine.py::TestHazardCurveRevaluation    | test_curve_change_pnl_consistent_with_fair      | Pass   |
|            | engine.py::TestHazardCurveRevaluation    | test_curve_change_produces_daily_pnl            | Pass   |
|            | formulas.py::TestBasisDelta              | test_basis_delta_positive                       | Pass   |
|            | formulas.py::TestBasisDelta              | test_basis_delta_similar_to_gauge_delta         | Pass   |
|            | formulas.py::TestGaugeDelta              | test_delta_spread_approximately_one             | Pass   |
|            | formulas.py::TestGaugeDelta              | test_higher_notional_means_higher_dv01          | Pass   |
|            | formulas.py::TestGaugeDelta              | test_positive_dv01_for_long_position            | Pass   |
|            | formulas.py::TestGaugeDelta              | test_zero_hazard_rate_gives_zero_delta          | Pass   |
|            | formulas.py::TestMarkToMarket            | test_mtm_calculation_correct                    | Pass   |
|            | formulas.py::TestMarkToMarket            | test_mtm_negative_when_spread_widens_for_seller | Pass   |
|            | formulas.py::TestMarkToMarket            | test_mtm_positive_when_spread_widens_for_buyer  | Pass   |
|            | formulas.py::TestMarkToMarket            | test_mtm_zero_when_spreads_equal                | Pass   |
|            | formulas.py::TestRiskyAnnuity            | test_longer_tenor_means_higher_annuity          | Pass   |

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| Model | Module                               | Test Name                                   | Status  |
|-------|--------------------------------------|---------------------------------------------|---------|
|       | formulas.py::TestRiskyAnnuity        | test_positive_hazard_rate_reduces_annuity   | Pass    |
|       | formulas.py::TestRiskyAnnuity        | test_reasonable_range                       | Pass    |
|       | formulas.py::TestRiskyAnnuity        | test_zero_hazard_rate                       | Pass    |
|       | market_state.py::TestEffectiveRates  | test_effective_rates_include_all_gauges     | Pass    |
|       | market_state.py::TestEffectiveRates  | test_effective_rates_mark_adjusted          | Pass    |
|       | market_state.py::TestEffectiveRates  | test_effective_rates_merge_correctly        | Pass    |
|       | market_state.py::TestMarketState     | test_base_rates_match_gaugehc               | Pass    |
|       | market_state.py::TestMarketState     | test_initial_load_creates_state             | Pass    |
|       | market_state.py::TestMarketState     | test_load_persists                          | Pass    |
|       | market_state.py::TestMarketState     | test_reset_all                              | Pass    |
|       | market_state.py::TestMarketState     | test_reset_single_gauge                     | Pass    |
|       | market_state.py::TestMarketState     | test_update_effective_rate_with_adjustment  | Pass    |
|       | market_state.py::TestMarketState     | test_updated_gauge_returns_base             | Pass    |
|       | market_state.py::TestMarketState     | test_update_gauge_rate                      | Pass    |
|       | market_state.py::TestMissingCoverage | test_commit_yield_curve                     | Pass    |
|       | market_state.py::TestMissingCoverage | test_get_yield_rate_above_max_tenor         | Pass    |
|       | market_state.py::TestMissingCoverage | test_get_yield_rate_below_min_tenor         | Pass    |
|       | market_state.py::TestMissingCoverage | test_get_yield_rate_exact_tenor             | Pass    |
|       | market_state.py::TestMissingCoverage | test_get_yield_rate_interpolated            | Pass    |
|       | market_state.py::TestMissingCoverage | test_load_base_curves_gauge_with_no_id      | Skipped |
|       | market_state.py::TestMissingCoverage | test_load_base_curves_gauges_key            | Pass    |
|       | market_state.py::TestMissingCoverage | test_load_base_curves_no_file               | Pass    |
|       | market_state.py::TestMissingCoverage | test_load_base_curves_unknown_format        | Pass    |
|       | market_state.py::TestMissingCoverage | test_reset_yield_curve                      | Pass    |
|       | market_state.py::TestMissingCoverage | test_update_gauge_rate_with_notes           | Pass    |
|       | daily_pnl.py::TestDailyPnL           | test_closed_trades_excluded                 | Pass    |
|       | daily_pnl.py::TestDailyPnL           | test_daily_pnl_without_previous_eod         | Pass    |
|       | daily_pnl.py::TestDailyPnL           | test_pnl_decomposition_sums                 | Pass    |
|       | daily_pnl.py::TestDailyPnL           | test_running_pnl_calculation                | Pass    |
|       | daily_pnl.py::TestPhantomPnLFile     | test_market_pnl_nonzero_with_eod_mark       | Pass    |
|       | daily_pnl.py::TestPhantomPnLFile     | test_market_pnl_zero_for_missing_eod_trades | Pass    |
|       | daily_pnl.py::TestPhantomPnLFile     | test_new_trade_has_inception_pnl            | Pass    |
|       | daily_pnl.py::TestPhantomPnLFile     | test_pnl_decomposition_with_prev_eod        | Pass    |
|       | eod.py::TestEodSnapshot              | test_eod_history                            | Pass    |
|       | eod.py::TestEodSnapshot              | test_generate_snapshot                      | Pass    |
|       | eod.py::TestEodSnapshot              | test_idempotent_snapshot                    | Pass    |
|       | eod.py::TestEodSnapshot              | test_pnl_series                             | Pass    |
|       | eod.py::TestEodSnapshot              | test_snapshot_persists                      | Pass    |
|       | marks.py::TestTradeMarks             | test_close_trade                            | Pass    |
|       | marks.py::TestTradeMarks             | test_load_empty_marks                       | Pass    |
|       | marks.py::TestTradeMarks             | test_update_and_load_mark                   | Pass    |
|       | stress_hydrograph.py::TestStressKest | test_koi_price_stays_at_notional_after_pos  | Pass    |
|       | stress_hydrograph.py::TestStressKest | test_koi_trigger_triggers_koi_independently | Pass    |
|       | stress_hydrograph.py::TestStressKest | test_koi_trigger_below_trigger              | Pass    |

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| Model | Module                                      | Test Name                                        | Status |
|-------|---------------------------------------------|--------------------------------------------------|--------|
|       | stress.hydrograph.py::TestStress            | test_close_out_logico_cash_is_negative           | Pass   |
|       | stress.hydrograph.py::TestStress            | test_close_out_logico_clocks_out_at_trigger      | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_base                             | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_1m_hours                         | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_rise                             | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_peak                             | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_after_peak                       | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_168_hours                        | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_uniform                          | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_base_level                       | Pass   |
|       | stress.hydrograph.py::TestSynthesis         | test_hydrograph_condition                        | Pass   |
|       | blotter_routes.py::TestActiveGauges         | test_all_gauges_with_trades_are_returned         | Pass   |
|       | blotter_routes.py::TestActiveGauges         | test_empty_blotter_returns_empty_list            | Pass   |
|       | blotter_routes.py::TestActiveGauges         | test_gauge_without_trade_is_not_returned         | Pass   |
|       | blotter_routes.py::TestActiveGauges         | test_lambeth_appears_when_it_has_an_open_trade   | Pass   |
|       | blotter_routes.py::TestActiveGauges         | test_lambeth_excluded_after_its_trade_is_closed  | Pass   |
|       | blotter_routes.py::TestActiveGauges         | test_other_gauges_unaffected_by_lambeth_closeout | Pass   |
|       | blotter_routes.py::TestActiveGauges         | test_response_is_sorted                          | Pass   |
|       | blotter_routes.py::TestBlotterClosedTrades  | test_filters_closed_trades_by_default            | Pass   |
|       | blotter_routes.py::TestBlotterClosedTrades  | test_include_closed                              | Pass   |
|       | blotter_routes.py::TestBlotterErrorHandling | test_handle_gauges_corrupt_file_skipped          | Pass   |
|       | blotter_routes.py::TestBlotterErrorHandling | test_handle_engine_error_returns_500             | Pass   |
|       | blotter_routes.py::TestBlotterErrorHandling | test_handle_trade_engine_error_returns_500       | Pass   |
|       | blotter_routes.py::TestBlotterErrorHandling | test_handle_trade_pdf_regen_exception_catch      | Pass   |
|       | blotter_routes.py::TestBlotterGauges        | test_gauge_name_injected_when_missing            | Pass   |
|       | blotter_routes.py::TestCloseOut             | test_close_trade_has_final_pnl                   | Pass   |
|       | blotter_routes.py::TestCloseOut             | test_close_trade_marks_closed                    | Pass   |
|       | blotter_routes.py::TestCloseOut             | test_close_trade_not_found                       | Pass   |
|       | blotter_routes.py::TestCloseOut             | test_close_trade_requires_closeout_spread        | Pass   |
|       | blotter_routes.py::TestCloseOut             | test_close_trade_settlement_uses_closeout_spread | Pass   |
|       | blotter_routes.py::TestCloseOut             | test_close_trade_success                         | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_blotter_empty_when_no_trades                | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_blotter_fs01_sign_payer_positive            | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_blotter_fs01_sign_receiver_negative         | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_blotter_gauge_name_injected                 | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_blotter_has_pnl_fields                      | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_blotter_has_prev_fair_spread                | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_blotter_has_required_fields                 | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_blotter_summary_fields                      | Pass   |
|       | blotter_routes.py::TestGetBlotter           | test_get_blotter_returns_trades                  | Pass   |
|       | curves_routes.py::TestCurveHistory          | test_curve_history_empty_when_no_eods            | Pass   |
|       | curves_routes.py::TestCurveHistory          | test_curve_history_requires_gauge_id             | Pass   |
|       | curves_routes.py::TestCurveHistory          | test_curve_history_with_eod                      | Pass   |
|       | curves_routes.py::TestCurvesErrorHandling   | test_handle_hazard_ts_error_returns_500          | Pass   |

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| Model | Module                                    | Test Name                                        | Status |
|-------|-------------------------------------------|--------------------------------------------------|--------|
|       | curves_routes.py::TestCurvesErrorHandler  | test_handle_history_error_returns_500            | Pass   |
|       | curves_routes.py::TestCurvesErrorHandler  | test_handle_hazard_ts_error_returns_500          | Pass   |
|       | curves_routes.py::TestCurvesErrorHandler  | test_handle_yield_curve_error_returns_500        | Pass   |
|       | curves_routes.py::TestCurvesErrorHandler  | test_handle_series_error_returns_500             | Pass   |
|       | curves_routes.py::TestCurvesErrorHandler  | test_handle_hazard_ts_error_returns_500          | Pass   |
|       | curves_routes.py::TestCurvesErrorHandler  | test_handle_yield_curve_error_returns_500        | Pass   |
|       | curves_routes.py::TestCurvesErrorHandler  | test_update_yield_curve_error_returns_500        | Pass   |
|       | curves_routes.py::TestCurvesNullBody      | test_commit_hazard_ts_null.body                  | Pass   |
|       | curves_routes.py::TestCurvesNullBody      | test_commit_yield_curve_null.body                | Pass   |
|       | curves_routes.py::TestCurvesNullBody      | test_update_hazard_ts_null.body                  | Pass   |
|       | curves_routes.py::TestCurvesNullBody      | test_update_yield_curve_null.body                | Pass   |
|       | curves_routes.py::TestHazardTSControl     | test_commit_hazard_ts                            | Pass   |
|       | curves_routes.py::TestHazardTSControl     | test_commit_hazard_ts_missing.gauge              | Pass   |
|       | curves_routes.py::TestHazardTSControl     | test_commit_hazard_ts_missing.rates              | Pass   |
|       | curves_routes.py::TestHazardTSControl     | test_commit_hazard_ts_only_affects.target.gauge  | Pass   |
|       | curves_routes.py::TestHazardTSReset       | test_reset_hazard_ts.all                         | Pass   |
|       | curves_routes.py::TestHazardTSReset       | test_reset_hazard_ts.single.gauge                | Pass   |
|       | curves_routes.py::TestHazardTermStructure | test_get_hazard_term.structure                   | Pass   |
|       | curves_routes.py::TestHazardTermStructure | test_update_hazard_ts.invalid.tenor              | Pass   |
|       | curves_routes.py::TestHazardTermStructure | test_update_hazard_ts_missing.gauge              | Pass   |
|       | curves_routes.py::TestHazardTermStructure | test_update_hazard_ts.single.point               | Pass   |
|       | curves_routes.py::TestPnLSeries           | test_pnl_series.after.eod                        | Pass   |
|       | curves_routes.py::TestPnLSeries           | test_pnl_series.empty                            | Pass   |
|       | curves_routes.py::TestYieldCurve          | test_get_yield_curve                             | Pass   |
|       | curves_routes.py::TestYieldCurve          | test_get_yield_curve.default.shape               | Pass   |
|       | curves_routes.py::TestYieldCurve          | test_update_yield_curve.invalid.tenor            | Pass   |
|       | curves_routes.py::TestYieldCurve          | test_update_yield_curve.no.body                  | Pass   |
|       | curves_routes.py::TestYieldCurve          | test_update_yield_curve.single.tenor             | Pass   |
|       | curves_routes.py::TestYieldCurveControl   | test_commit_yield_curve                          | Pass   |
|       | curves_routes.py::TestYieldCurveControl   | test_commit_yield_curve.affected.count           | Pass   |
|       | curves_routes.py::TestYieldCurveControl   | test_commit_yield_curve.empty.rates              | Pass   |
|       | curves_routes.py::TestYieldCurveControl   | test_commit_yield_curve.no.body                  | Pass   |
|       | curves_routes.py::TestYieldCurveReset     | test_reset_yield_curve                           | Pass   |
|       | eod_routes.py::TestEODErrorHandler        | test_handle_eod_history_engine_error_returns_500 | Pass   |
|       | eod_routes.py::TestEODErrorHandler        | test_get_eod_pdf_error_returns_500               | Pass   |
|       | eod_routes.py::TestEODErrorHandler        | test_submit_eod_engine_error_returns_500         | Pass   |
|       | eod_routes.py::TestEODHistory             | test_eod_history_after_snap                      | Pass   |
|       | eod_routes.py::TestEODHistory             | test_eod_history_empty                           | Pass   |
|       | eod_routes.py::TestEODHistory             | test_multiple_eod.snaps                          | Pass   |
|       | eod_routes.py::TestEODNoDateBody          | test_submit_eod_no_json_uses.today               | Pass   |
|       | eod_routes.py::TestEODPdf                 | test_eod_pdf_found.serves.file                   | Pass   |
|       | eod_routes.py::TestEODPdf                 | test_eod_pdf_not_found                           | Pass   |
|       | eod_routes.py::TestEODPdfGenerator        | test_submit_eod_import_error.still.succeeds      | Pass   |
|       | eod_routes.py::TestEODPdfGenerator        | test_submit_eod_pdf_exception.still.succeeds     | Pass   |

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| Model | Module                                              | Test Name                                                 | Status |
|-------|-----------------------------------------------------|-----------------------------------------------------------|--------|
|       | eod_routes.py::TestPnLResetAfterEOD                 | test_pnl Uses eod as baseline                             | Pass   |
|       | eod_routes.py::TestSubmitEOD                        | test_eod idempotent same day                              | Pass   |
|       | eod_routes.py::TestSubmitEOD                        | test_submit_eod                                           | Pass   |
|       | eod_routes.py::TestSubmitEOD                        | test_submit_eod returns summary                           | Pass   |
|       | eod_routes.py::TestSubmitEOD                        | test_submit_eod saves snapshot file                       | Pass   |
|       | eod_routes.py::TestSubmitEOD                        | test_submit_eod snapshot has market state                 | Pass   |
|       | eod_routes.py::TestSubmitEOD                        | test_submit_eod snapshot has positions                    | Pass   |
|       | market_state_routes.py::TestGetMarketState          | test_get_market_state                                     | Pass   |
|       | market_state_routes.py::TestGetMarketState          | test_get_market_state has effective rates                 | Pass   |
|       | market_state_routes.py::TestGetMarketState          | test_get_market_state initial load creates defaults       | Pass   |
|       | market_state_routes.py::TestMarketStateErrorHandler | test_market_state_engine_error returns 500                | Pass   |
|       | market_state_routes.py::TestMarketStateErrorHandler | test_market_state_engine_error returns 500                | Pass   |
|       | market_state_routes.py::TestMarketStateErrorHandler | test_market_state null body returns 400                   | Pass   |
|       | market_state_routes.py::TestResetMarketState        | test_reset_market_state all                               | Pass   |
|       | market_state_routes.py::TestResetMarketState        | test_reset_market_state single gauge                      | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_update_market_state persists                         | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_update_market_state                                  | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_update_market_state missing gauge                    | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_update_market_state no body                          | Pass   |
|       | market_state_routes.py::TestUpdateMarketState       | test_update_market_state returns pnl impact               | Pass   |
|       | risk_routes.py::TestRiskErrorHandler                | test_risk_grid engine error returns 500                   | Pass   |
|       | risk_routes.py::TestRiskErrorHandler                | test_trade_map engine error returns 500                   | Pass   |
|       | risk_routes.py::TestRiskGrid                        | test_risk_grid has gauges                                 | Pass   |
|       | risk_routes.py::TestRiskGrid                        | test_risk_grid returns grid                               | Pass   |
|       | risk_routes.py::TestRiskGrid                        | test_risk_grid with no trades                             | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map fs01 by tenor                              | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map gauge has coordinates                      | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map includes fs01                              | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map includes notional                          | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map includes pnl                               | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map num trades                                 | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map returns positions                          | Pass   |
|       | risk_routes.py::TestTradeMap                        | test_trade_map with no trades                             | Pass   |
|       | risk_routes.py::TestTradeMapWithProperties          | test_trade_map gauge not in gaugehc loads from gauge json | Pass   |
|       | risk_routes.py::TestTradeMapWithProperties          | test_trade_map loads property locations                   | Pass   |
|       | risk_routes.py::TestTradeMapWithProperties          | test_trade_map skips closed trades                        | Pass   |
|       | blotter_detail.py::TestBlotterEndpoint              | test_blotter fetches correct endpoint                     | Pass   |
|       | blotter_detail.py::TestBlotterEndpoint              | test_blotter close endpoint                               | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_clear_filters                                        | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_external_filter_api                                  | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_filter_function_exists                               | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_get_filtered_trades                                  | Pass   |
|       | blotter_detail.py::TestFilterBar                    | test_remove_single_filter                                 | Pass   |
|       | blotter_detail.py::TestHazardRateArrow              | test_arrow                                                | Pass   |

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| Model | Module                                   | Test Name                                       | Status |
|-------|------------------------------------------|-------------------------------------------------|--------|
|       | blotter_detail.py::TestHazardRateArrow   | test_prev_fair_spread_comparison                | Pass   |
|       | blotter_detail.py::TestHazardRateArrow   | test_group_arrow                                | Pass   |
|       | blotter_detail.py::TestMarketPnLColumn   | test_market_pnl_column                          | Pass   |
|       | blotter_detail.py::TestMarketPnLColumn   | test_market_pnl_label                           | Pass   |
|       | blotter_detail.py::TestPnLBar            | test_pnl_bar_daily                              | Pass   |
|       | blotter_detail.py::TestPnLBar            | test_pnl_bar_from_market                        | Pass   |
|       | blotter_detail.py::TestPnLBar            | test_pnl_bar_net_notional                       | Pass   |
|       | blotter_detail.py::TestPnLBar            | test_pnl_bar_running                            | Pass   |
|       | blotter_detail.py::TestSignedNotional    | test_is_payer_referenced                        | Pass   |
|       | blotter_detail.py::TestSignedNotional    | test_signed_notional_direction                  | Pass   |
|       | blotter_detail.py::TestSortAndNavigation | test_sort_toggle                                | Pass   |
|       | blotter_detail.py::TestSortAndNavigation | test_make_view_click                            | Pass   |
|       | blotter_detail.py::TestTableStructure    | test_gauge_fs01_column                          | Pass   |
|       | blotter_detail.py::TestTableStructure    | test_maturity_formatting                        | Pass   |
|       | eod_detail.py::TestEODEndpoint           | test_eod_blotter_fetch                          | Pass   |
|       | eod_detail.py::TestEODEndpoint           | test_eod_history_fetch                          | Pass   |
|       | eod_detail.py::TestEODEndpoint           | test_eod_pnl_series_fetch                       | Pass   |
|       | eod_detail.py::TestEODEndpoint           | test_eod_submit_endpoint                        | Pass   |
|       | eod_detail.py::TestEODFunction           | test_attribution_section                        | Pass   |
|       | eod_detail.py::TestEODFunction           | test_chart_rendering                            | Pass   |
|       | eod_detail.py::TestEODFunction           | test_cleanup_function                           | Pass   |
|       | eod_detail.py::TestEODFunction           | test_download_pdf_function                      | Pass   |
|       | eod_detail.py::TestEODFunction           | test_submit_function                            | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_attribution_wrap                           | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_cards_container                            | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_date_input                                 | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_history_wrap                               | Pass   |
|       | eod_detail.py::TestEODUIStructure        | test_submit_button                              | Pass   |
|       | market_detail.py::TestActionButtonFont   | test_button_font_matches_tabs                   | Pass   |
|       | market_detail.py::TestActionButtonFont   | test_button_padding_matches_tabs                | Pass   |
|       | market_detail.py::TestActionButtonFont   | test_buttons_in_shared_container                | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_format_button_blue                         | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_format_history_button_grey_not_blue        | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_format_individual_border_radius_on_buttons | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_format_hist_button_grey_not_blue           | Pass   |
|       | market_detail.py::TestActionButtonFormat | test_format_reset_button_grey                   | Pass   |
|       | market_detail.py::TestCommitButton       | test_commit_button_calls_function               | Pass   |
|       | market_detail.py::TestCommitButton       | test_commit_button_id_present                   | Pass   |
|       | market_detail.py::TestCommitButton       | test_commit_button_not_hidden                   | Pass   |
|       | market_detail.py::TestCommitButton       | test_commit_function_defined                    | Pass   |
|       | market_detail.py::TestCommitEndpoint     | test_commit_posts_hazard_ts                     | Pass   |
|       | market_detail.py::TestCommitEndpoint     | test_commit_posts_yield_curve                   | Pass   |
|       | market_detail.py::TestCommitEndpoint     | test_commit_shows_pnl_impact                    | Pass   |
|       | market_detail.py::TestCommitEndpoint     | test_commit_switches_to_blotter                 | Pass   |

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| Model   | Module                                | Test Name                                           | Status |
|---------|---------------------------------------|-----------------------------------------------------|--------|
|         | market_detail.py::TestDirtyState      | test_hazard_dirty_keys                              | Pass   |
|         | market_detail.py::TestDirtyState      | test_hazard_committed_label                         | Pass   |
|         | market_detail.py::TestDirtyState      | test_hazard_dirty_flag                              | Pass   |
|         | market_detail.py::TestMarketUIState   | test_cleanup_function                               | Pass   |
|         | market_detail.py::TestMarketUIState   | test_gauge_list_function                            | Pass   |
|         | market_detail.py::TestMarketUIState   | test_market_canvas                                  | Pass   |
|         | market_detail.py::TestMarketUIState   | test_mode_selector                                  | Pass   |
|         | market_detail.py::TestNewTradeButton  | test_new_trade_button_grey                          | Pass   |
|         | market_detail.py::TestNewTradeButton  | test_new_trade_button_label                         | Pass   |
|         | market_detail.py::TestNewTradeButton  | test_new_trade_button_present                       | Pass   |
|         | market_detail.py::TestNewTradeButton  | test_new_trade_calls_function                       | Pass   |
|         | market_detail.py::TestNewTradeButton  | test_new_trade_function_opens_prs                   | Pass   |
|         | market_detail.py::TestNewTradeButton  | test_new_trade_left_of_history                      | Pass   |
|         | market_detail.py::TestNewTradeButton  | test_no_border_radius_on_new_trade_btn              | Pass   |
|         | market_detail.py::TestResetEndpoint   | test_reset_function_exists                          | Pass   |
|         | market_detail.py::TestResetEndpoint   | test_reset_hazard_ts_endpoint                       | Pass   |
|         | market_detail.py::TestResetEndpoint   | test_reset_yield_curve_endpoint                     | Pass   |
| CDM-ALL | gauge_mapping.py                      | test_all_cdm_fields_present                         | Pass   |
|         | gauge_mapping.py                      | test_all_types_valid                                | Pass   |
|         | gauge_mapping.py                      | test_all_values_valid                               | Pass   |
|         | mortgage_mapping.py                   | test_all_cdm_fields_present                         | Pass   |
|         | mortgage_mapping.py                   | test_all_types_valid                                | Pass   |
|         | mortgage_mapping.py                   | test_all_values_valid                               | Pass   |
|         | property_mapping.py                   | test_all_cdm_fields_present                         | Pass   |
|         | property_mapping.py                   | test_all_types_valid                                | Pass   |
|         | property_mapping.py                   | test_all_values_valid                               | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_required_fields_returns_list[gauge_cdm]    | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_required_fields_returns_list[mortgage_cdm] | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_required_fields_returns_list[property_cdm] | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_required_fields_returns_list[prs_cdm]      | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_required_fields_returns_list[storm_cdm]    | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_required_fields_returns_list[stormts_cdm]  | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_all_fields_returns_paths[gauge_cdm]        | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_all_fields_returns_paths[mortgage_cdm]     | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_all_fields_returns_paths[property_cdm]     | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_all_fields_returns_paths[prs_cdm]          | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_all_fields_returns_paths[storm_cdm]        | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_get_all_fields_returns_paths[stormts_cdm]      | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_repr_returns_string[gauge_cdm]                 | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_repr_returns_string[mortgage_cdm]              | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_repr_returns_string[property_cdm]              | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_repr_returns_string[prs_cdm]                   | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_repr_returns_string[storm_cdm]                 | Pass   |
|         | schema_basics.py::TestCDMSchemaBasics | test_repr_returns_string[stormts_cdm]               | Pass   |

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| Model | Module                              | Test Name                                                         | Status |
|-------|-------------------------------------|-------------------------------------------------------------------|--------|
|       | schema_basics.py::TestCDMSchema     | test_create_mapping_returns_non_empty_dict[gauged_cdm]            | Pass   |
|       | schema_basics.py::TestCDMSchema     | test_create_mapping_returns_non_empty_dict[mortgage_cdm]          | Pass   |
|       | schema_basics.py::TestCDMSchema     | test_create_mapping_returns_non_empty_dict[property_cdm]          | Pass   |
|       | schema_basics.py::TestCDMSchema     | test_create_mapping_returns_non_empty_dict[storm_cdm]             | Pass   |
|       | schema_basics.py::TestCDMSchema     | test_create_mapping_returns_non_empty_dict[storm_event_cdm]       | Pass   |
|       | schema_basics.py::TestCDMSchema     | test_create_mapping_returns_non_empty_dict[storm_time_series_cdm] | Pass   |
|       | schemas.py::TestFloodGaugeCDM       | test_create_mapping_returns_flat_dict                             | Pass   |
|       | schemas.py::TestFloodGaugeCDM       | test_create_mapping_with_empty_input                              | Pass   |
|       | schemas.py::TestFloodGaugeCDM       | test_get_field_info_for_gauge_id                                  | Pass   |
|       | schemas.py::TestFloodGaugeCDM       | test_required_fields_include_gauge_id                             | Pass   |
|       | schemas.py::TestFloodGaugeCDM       | test_schema_has_flood_gauge_root                                  | Pass   |
|       | schemas.py::TestFloodGaugeCDM       | test_schema_has_header_section                                    | Pass   |
|       | schemas.py::TestFloodGaugeCDM       | test_validate_empty_data_returns_errors                           | Pass   |
|       | schemas.py::TestFloodGaugeCDM       | test_validate_valid_gauge_data                                    | Pass   |
|       | schemas.py::TestMortgageCDM         | test_create_mapping_returns_flat_dict                             | Pass   |
|       | schemas.py::TestMortgageCDM         | test_required_fields_include_mortgage_id                          | Pass   |
|       | schemas.py::TestMortgageCDM         | test_schema_has_mortgage_root                                     | Pass   |
|       | schemas.py::TestMortgageCDM         | test_validate_valid_mortgage                                      | Pass   |
|       | schemas.py::TestPhysicalRiskSwapCDM | test_create_mapping_returns_flat_dict                             | Pass   |
|       | schemas.py::TestPhysicalRiskSwapCDM | test_all_fields_has_gauge_set                                     | Pass   |
|       | schemas.py::TestPhysicalRiskSwapCDM | test_required_fields_include_swap_id                              | Pass   |
|       | schemas.py::TestPhysicalRiskSwapCDM | test_schema_has_root                                              | Pass   |
|       | schemas.py::TestPropertyCDM         | test_create_mapping_returns_flat_dict                             | Pass   |
|       | schemas.py::TestPropertyCDM         | test_required_fields_not_empty                                    | Pass   |
|       | schemas.py::TestPropertyCDM         | test_schema_has_property_root                                     | Pass   |
|       | schemas.py::TestPropertyCDM         | test_validate_valid_property                                      | Pass   |
|       | schemas.py::TestStormEventCDM       | test_create_mapping_exception_raises                              | Pass   |
|       | schemas.py::TestStormEventCDM       | test_create_mapping_full_data                                     | Pass   |
|       | schemas.py::TestStormEventCDM       | test_create_mapping_returns_flat_dict                             | Pass   |
|       | schemas.py::TestStormEventCDM       | test_get_required_fields_returns_list                             | Pass   |
|       | schemas.py::TestStormEventCDM       | test_required_fields_include_storm_id                             | Pass   |
|       | schemas.py::TestStormEventCDM       | test_schema_has_storm_root                                        | Pass   |
|       | schemas.py::TestStormEventCDM       | test_validate_empty_data                                          | Pass   |
|       | schemas.py::TestStormEventCDM       | test_validate_exception_returns_error_dict                        | Pass   |
|       | schemas.py::TestStormEventCDM       | test_validate_missing_catchment_id                                | Pass   |
|       | schemas.py::TestStormEventCDM       | test_validate_missing_storm_id                                    | Pass   |
|       | schemas.py::TestStormEventCDM       | test_validate_valid_storm                                         | Pass   |
|       | schemas.py::TestStormTimeSeriesCDM  | test_create_mapping_exception_raises_validation_error             | Pass   |
|       | schemas.py::TestStormTimeSeriesCDM  | test_create_mapping_full_data                                     | Pass   |
|       | schemas.py::TestStormTimeSeriesCDM  | test_create_mapping_returns_flat_dict                             | Pass   |
|       | schemas.py::TestStormTimeSeriesCDM  | test_create_reading_mapping                                       | Pass   |
|       | schemas.py::TestStormTimeSeriesCDM  | test_get_required_fields_returns_list                             | Pass   |
|       | schemas.py::TestStormTimeSeriesCDM  | test_required_fields_include_ts_id                                | Pass   |
|       | schemas.py::TestStormTimeSeriesCDM  | test_schema_has_root                                              | Pass   |

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| Model    | Module                                  | Test Name                                   | Status |
|----------|-----------------------------------------|---------------------------------------------|--------|
|          | schemas.py::TestStormTimeSeriesGDM      | test_validate_empty_data                    | Pass   |
|          | schemas.py::TestStormTimeSeriesGDM      | test_validate_exception_returns_error_dict  | Pass   |
|          | schemas.py::TestStormTimeSeriesGDM      | test_validate_missing_catchment_id          | Pass   |
|          | schemas.py::TestStormTimeSeriesGDM      | test_validate_missing_ts_id                 | Pass   |
|          | schemas.py::TestStormTimeSeriesGDM      | test_validate_valid_data                    | Pass   |
| PLATFORM | catchment_integration.py                | test_config                                 | Pass   |
|          | catchment_integration.py                | test_config_endpoints_include_health_check  | Pass   |
|          | catchment_integration.py                | test_html_file                              | Pass   |
|          | catchment_integration.py                | test_routes                                 | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_empty_report                           | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_fmt_stats_has_python                   | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_largest_line_numbers_are_integers      | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_largest_line_numbers_not_question_mark | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_largest_sorted_descending              | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_sum_clones_correct                     | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_returns_dict                           | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_total_present                          | Pass   |
|          | test_duplication_report.py::TestAnalyze | test_worst_populated                        | Pass   |
|          | test_duplication_report.py::TestMain    | test_main_handles_jscpd_error               | Pass   |
|          | test_duplication_report.py::TestMain    | test_main_prints_stats                      | Pass   |
|          | test_duplication_report.py::TestMain    | test_main_produces_pdf                      | Pass   |
|          | test_duplication_report.py::TestMakePdf | test_good_rating_for_low_pct                | Pass   |
|          | test_duplication_report.py::TestMakePdf | test_high_rating_report                     | Pass   |
|          | test_duplication_report.py::TestMakePdf | test_pdf_header                             | Pass   |
|          | test_duplication_report.py::TestMakePdf | test_pdf_nonempty                           | Pass   |
|          | test_duplication_report.py::TestMakePdf | test_pdf_returns_bytes                      | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_curve_points_present                   | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_each_gauge_has_required_fields         | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_file_exists                            | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_flood_trigger_ordering                 | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_gauge_count_at_least_40                | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_gauge_ids_are_well_formed              | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_has_hazard_curves_key                  | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_hazard_rate_ordering                   | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_hazard_rates_positive                  | Pass   |
|          | test_blotter_data.py::TestGaugeData     | test_term_structure_present                 | Pass   |
|          | test_blotter_data.py::TestMarketData    | test_base_rates_not_empty                   | Pass   |
|          | test_blotter_data.py::TestMarketData    | test_file_exists                            | Pass   |
|          | test_blotter_data.py::TestMarketData    | test_has_required_keys                      | Pass   |
|          | test_blotter_data.py::TestMarketData    | test_hazard_term_structure_has_triggers     | Pass   |
|          | test_blotter_data.py::TestMarketData    | test_hazard_term_structure_not_empty        | Pass   |
|          | test_blotter_data.py::TestMarketData    | test_data_updated_present                   | Pass   |
|          | test_blotter_data.py::TestMarketData    | test_yield_curve_has_all_tenors             | Pass   |
|          | test_blotter_data.py::TestMarketData    | test_yield_curve_rates_in_range             | Pass   |

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| Model | Module                                            | Test Name                                          | Status |
|-------|---------------------------------------------------|----------------------------------------------------|--------|
|       | test_blotter_data.py::TestPRSTradeData            | test_data_trade_has_physical_swap_root             | Pass   |
|       | test_blotter_data.py::TestPRSTradeData            | test_data_trade_has_swap_id                        | Pass   |
|       | test_blotter_data.py::TestPRSTradeData            | test_data_directory_exists                         | Pass   |
|       | test_blotter_data.py::TestPRSTradeData            | test_data_swap_id_matches_filename                 | Pass   |
|       | test_blotter_data.py::TestPRSTradeData            | test_data_trade_count_thames_central               | Pass   |
|       | test_blotter_data.py::TestPRSTradeData            | test_data_trade_statuses_are_valid                 | Pass   |
|       | test_blotter_data.py::TestPRSTradeData            | test_data_trades_are_present                       | Pass   |
|       | test_blotter_data.py::TestPRSTradeData            | test_data_trades_have_gauge_set                    | Pass   |
|       | test_id_consistency.py::TestClassifierConsistency | test_classifier_gauge_ids_match                    | Pass   |
|       | test_id_consistency.py::TestClassifierConsistency | test_classifier_filters_exist                      | Pass   |
|       | test_id_consistency.py::TestClassifierConsistency | test_classifier_summary_exists                     | Pass   |
|       | test_id_consistency.py::TestDataLensing           | test_dependency_graph_complete                     | Skip   |
|       | test_id_consistency.py::TestDataLensing           | test_manifest_exists                               | Pass   |
|       | test_id_consistency.py::TestDataLensing           | test_stale_inputs                                  | Pass   |
|       | test_id_consistency.py::TestDataLensing           | test_output_hashes_current                         | Pass   |
|       | test_id_consistency.py::TestDeterministicIDs      | test_ids_are_deterministic                         | Skip   |
|       | test_id_consistency.py::TestGaugeHC               | test_gauge_json_has_gauges                         | Pass   |
|       | test_id_consistency.py::TestGaugeHC               | test_gauges_match_gauges                           | Pass   |
|       | test_id_consistency.py::TestGaugeHC               | test_hazard_curves_match_gauges                    | Pass   |
|       | test_id_consistency.py::TestGaugeHC               | test_hazard_curves_subset_of_gauges                | Pass   |
|       | test_id_consistency.py::TestGaugeHC               | test_all_gauges_exist_in_gauge_json                | Pass   |
|       | test_id_consistency.py::TestGaugeHC               | test_all_gauges_have_hazard_curves                 | Pass   |
|       | test_id_consistency.py::TestPropertyTypes         | test_ids_in_header                                 | Pass   |
|       | test_id_consistency.py::TestPropertyTypes         | test_gauge_json_has_properties                     | Pass   |
|       | test_id_consistency.py::TestPropertyTypes         | test_files_match_properties                        | Pass   |
|       | test_id_consistency.py::TestStormHC               | test_gauges_storms_from_sequences                  | Pass   |
|       | test_id_consistency.py::TestStormHC               | test_properties_storms_from_sequences              | Pass   |
|       | test_storm_data.py::TestGaugetsData               | test_each_file_has_required_keys                   | Pass   |
|       | test_storm_data.py::TestGaugetsData               | test_flood_simulation_has_readings                 | Pass   |
|       | test_storm_data.py::TestGaugetsData               | test_gauge_id_field_matches_filename               | Pass   |
|       | test_storm_data.py::TestGaugetsData               | test_gaugets_dir_exists                            | Pass   |
|       | test_storm_data.py::TestGaugetsData               | test_gaugets_gauge_id_format                       | Pass   |
|       | test_storm_data.py::TestGaugetsData               | test_gaugets_not_empty                             | Pass   |
|       | test_storm_data.py::TestGaugetsData               | test_readings_count_matches_simulation             | Fail   |
|       | test_storm_data.py::TestPipelineComplete          | test_snapshots_exist                               | FAIL   |
|       | test_storm_data.py::TestPipelineComplete          | test_gauges_match_gauge_count                      | Pass   |
|       | test_storm_data.py::TestPipelineComplete          | test_hazard_curves_match_gauge_count               | Pass   |
|       | test_storm_data.py::TestPipelineComplete          | test_property_hazard_curves_match_properties_count | Pass   |
|       | test_storm_data.py::TestPipelineComplete          | test_trades_gauges_exist_in_gauge_json             | Pass   |
|       | test_storm_data.py::TestPipelineComplete          | test_trades_exist                                  | Pass   |
|       | test_storm_data.py::TestStormDataConsistency      | test_filters_cover_gaugehc_gauges                  | Skip   |
|       | test_storm_data.py::TestStormDataConsistency      | test_filters_exist_for_stress_gauges               | Skip   |
|       | test_storm_data.py::TestStormDataConsistency      | test_storm_gauges_in_gaugehc                       | Skip   |
|       | test_storm_data.py::TestStormDataConsistency      | test_storm_ids_overlap_properties                  | Skip   |

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| Model | Module                                       | Test Name                                         | Status |
|-------|----------------------------------------------|---------------------------------------------------|--------|
|       | test_storm_data.py::TestStormDataConsistency | storm_ids_subset_of_gauges                        | Skip   |
|       | test_storm_data.py::TestStressClassifier     | all_gauges_trained                                | Pass   |
|       | test_storm_data.py::TestStressClassifier     | avg_auc_above_threshold                           | Pass   |
|       | test_storm_data.py::TestStressClassifier     | classifier_count                                  | Pass   |
|       | test_storm_data.py::TestStressClassifier     | classifier_count_matches_summary                  | Pass   |
|       | test_storm_data.py::TestStressClassifier     | classifier_filenames_well_formed                  | Pass   |
|       | test_storm_data.py::TestStressClassifier     | classifiers_not_empty                             | Pass   |
|       | test_storm_data.py::TestStressClassifier     | stress_model_dir_exists                           | Pass   |
|       | test_storm_data.py::TestStressClassifier     | training_summary_exists                           | Pass   |
|       | test_storm_data.py::TestStressClassifier     | training_summary_has_required_keys                | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_unique_gauge_ids_in_responses            | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_gauge_ids_use_gauge_prefix               | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_storm_ids_use_valid_prefix               | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_at_least_one_storm_exceeds_alert         | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_each_storm_has_gauge_responses           | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_each_storm_has_required_fields           | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_file_exists                              | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_gauge_ids_well_formed                    | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_gauge_responses_have_required_fields     | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_has_storms                               | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_intensity_categories_present             | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_minimum_storm_count                      | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_peak_level_exceeds_base                  | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_peak_levels_positive                     | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_storm_ids_unique                         | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_storm_ids_well_formed                    | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_storms_cover_multiple_gauges             | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_storms_sorted_by_severity_descending     | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_total_storms_key_matches_list_length     | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_trigger_summary_max_trigger_valid        | Pass   |
|       | test_storm_data.py::TestStressStorms         | test_152_trigger_summary_populated_for_all_storms | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Basic                                             | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Bad gauges key also works                         | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Bad id all returns list                           | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Bad id missing file returns empty                 | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | checking works                                    | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Cache                                             | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Cache reload                                      | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Lists up false                                    | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Lists up true                                     | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Lookup by id missing returns none                 | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Lookup by id returns entity                       | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | List all returns summaries                        | Pass   |
|       | test_gauge_loader.py::TestGaugeLeader        | Status returns dict                               | Pass   |

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| Model                                            | Module                                           | Test Name                                 | Status |
|--------------------------------------------------|--------------------------------------------------|-------------------------------------------|--------|
| test_mortgage_loader.py::TestMortgageLoaderBasic | test_mortgage_loader.py::TestMortgageLoaderBasic | test_load_all_baseturns_summaries         | Pass   |
| test_mortgage_loader.py::TestMortgageLoaderBasic | test_mortgage_loader.py::TestMortgageLoaderBasic | test_load_all_baseturns_list              | Pass   |
| test_mortgage_loader.py::TestMortgageLoaderBasic | test_mortgage_loader.py::TestMortgageLoaderBasic | test_find_by_lookup                       | Pass   |
| test_mortgage_loader.py::TestMortgageLoaderBasic | test_mortgage_loader.py::TestMortgageLoaderBasic | test_find_by_property_nested_structure    | Pass   |
| test_mortgage_loader.py::TestMortgageLoaderBasic | test_mortgage_loader.py::TestMortgageLoaderBasic | test_load_all_lookup_returns_none         | Pass   |
| test_property_county.py::TestFindByCounty        | test_property_county.py::TestFindByCounty        | test_find_by_county_case_insensitive      | Pass   |
| test_property_county.py::TestFindByCounty        | test_property_county.py::TestFindByCounty        | test_find_by_county_empty_portfolio       | Pass   |
| test_property_county.py::TestFindByCounty        | test_property_county.py::TestFindByCounty        | test_find_by_county_exact_match           | Pass   |
| test_property_county.py::TestFindByCounty        | test_property_county.py::TestFindByCounty        | test_find_by_county_missing_county_field  | Pass   |
| test_property_county.py::TestFindByCounty        | test_property_county.py::TestFindByCounty        | test_find_by_county_no_match              | Pass   |
| test_property_county.py::TestFindByCounty        | test_property_county.py::TestFindByCounty        | test_find_by_county_partial_match         | Pass   |
| test_property_loader.py::TestPropertyLoaderBasic | test_property_loader.py::TestPropertyLoaderBasic | test_load_all_baseturns_list              | Pass   |
| test_property_loader.py::TestPropertyLoaderBasic | test_property_loader.py::TestPropertyLoaderBasic | test_load_all_baseturns_empty             | Pass   |
| test_property_loader.py::TestPropertyLoaderBasic | test_property_loader.py::TestPropertyLoaderBasic | test_find_by_lookup                       | Pass   |
| test_property_loader.py::TestPropertyLoaderBasic | test_property_loader.py::TestPropertyLoaderBasic | test_load_all_lookup_returns_property_id  | Pass   |
| test_registry.py::TestCreateRegistry             | test_registry.py::TestCreateRegistry             | test_creates_registry                     | Pass   |
| test_registry.py::TestCreateRegistry             | test_registry.py::TestCreateRegistry             | test_with_file_overrides                  | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_clear_all_caches                     | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_all_status                       | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_clear_all_caches                     | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_data_files_all_missing               | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_data_files_present                   | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_data_file_overrides                  | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_data_dir                         | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_available_loaders                | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_gauge_loader                     | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_loader_by_name                   | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_loader_unknown_name_returns_none | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_mortgage_loader                  | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_property_loader                  | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_storm_loader                     | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_get_timeseries_loader                | Pass   |
| test_registry.py::TestLoaderRegistry             | test_registry.py::TestLoaderRegistry             | test_loader_cached_on_repeated_access     | Pass   |
| storm_loader.py::TestFindByBasin                 | storm_loader.py::TestFindByBasin                 | test_atl_basin                            | Pass   |
| storm_loader.py::TestFindByBasin                 | storm_loader.py::TestFindByBasin                 | test_case_insensitive                     | Pass   |
| storm_loader.py::TestFindByBasin                 | storm_loader.py::TestFindByBasin                 | test_unknown_basin_returns_empty          | Pass   |
| storm_loader.py::TestFindByBasin                 | storm_loader.py::TestFindByBasin                 | test_wpac_basin                           | Pass   |
| storm_loader.py::TestFindByCategory              | storm_loader.py::TestFindByCategory              | test_all_category_4                       | Pass   |
| storm_loader.py::TestFindByCategory              | storm_loader.py::TestFindByCategory              | test_category_3                           | Pass   |
| storm_loader.py::TestFindByCategory              | storm_loader.py::TestFindByCategory              | test_category_5                           | Pass   |
| storm_loader.py::TestFindByCategory              | storm_loader.py::TestFindByCategory              | test_no_matches_returns_empty             | Pass   |
| storm_loader.py::TestFindByName                  | storm_loader.py::TestFindByName                  | test_case_insensitive                     | Pass   |
| storm_loader.py::TestFindByName                  | storm_loader.py::TestFindByName                  | test_exact_match                          | Pass   |

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| Model | Module                           | Test Name                                               | Status |
|-------|----------------------------------|---------------------------------------------------------|--------|
|       | test_lookups_gauge_property.py:: | TestClassifyPropertyRisk                                | Pass   |
|       | test_lookups_gauge_property.py:: | TestClassifyPropertyRisk                                | Pass   |
|       | test_lookups_gauge_property.py:: | TestClassifyPropertyRisk                                | Pass   |
|       | test_lookups_ids.py::            | TestAnalyzeRelationships_returns_zeros                  | Pass   |
|       | test_lookups_ids.py::            | TestAnalyzeRelationships_patch_data                     | Pass   |
|       | test_lookups_ids.py::            | TestAnalyzeRelationships_portgages_counted              | Pass   |
|       | test_lookups_ids.py::            | TestAnalyzeRelationships_properties_overlap             | Pass   |
|       | test_lookups_ids.py::            | TestAnalyzeRelationships_required_keys                  | Pass   |
|       | test_lookups_ids.py::            | TestExtractGauges_returns_ids                           | Pass   |
|       | test_lookups_ids.py::            | TestExtractGauges_missing_gauge_id_skipped              | Pass   |
|       | test_lookups_ids.py::            | TestExtractGauges_returns_empty_set                     | Pass   |
|       | test_lookups_ids.py::            | TestExtractMortgages_returns_ids                        | Pass   |
|       | test_lookups_ids.py::            | TestExtractMortgages_returns_empty_set                  | Pass   |
|       | test_lookups_ids.py::            | TestExtractMortgages_property_ids_from_mortgage         | Pass   |
|       | test_lookups_ids.py::            | TestExtractMortgages_returns_empty_set                  | Pass   |
|       | test_lookups_ids.py::            | TestExtractProperties_returns_items_returns_empty_set   | Pass   |
|       | test_lookups_ids.py::            | TestExtractProperties_returns_ids                       | Pass   |
|       | test_lookups_ids.py::            | TestExtractProperties_missing_property_id_skipped       | Pass   |
|       | test_lookups_ids.py::            | TestExtractProperties_returns_empty_set                 | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildAllLookups_returns_empty_dicts                 | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildAllLookups_returns_four_keys                   | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildAllLookups_populates_lookups                   | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildEmptyLookup_returns_empty                      | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_loan_details_fallback           | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_out_property_id_skipped         | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_mortgages                       | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_returns_empty                   | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_empty                           | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_mortgage                        | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_info                            | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_mortgage_or_property_id_skipped | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_info_uses_unknown               | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_info_returns_empty_struct       | Pass   |
|       | test_lookups_mortgage.py::       | TestBuildMortgageLookup_info_calculation                | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderBasics_number_of_gauge_files        | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderBasics_gauge_ids                    | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderBasics_gauge_ids_missing_dir        | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderBasics_returns_summaries            | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderBasics_directory                    | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderBasics_returns_empty                | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderCache                               | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderCache                               | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderCache                               | Pass   |
|       | test_timeseries_loader.py::      | TestTimeSeriesLoaderCache_skipped_all_gauges_passed     | Pass   |

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| Model                                           | Module                                          | Test Name                                     | Status |
|-------------------------------------------------|-------------------------------------------------|-----------------------------------------------|--------|
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_bad_timestamp_skipped_gauge_path         | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_readings_in_range_all_gauges         | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_readings_in_range_for_gauge          | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_latest_reading                       | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_latest_reading_missing_returns_none  | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_peak_reading                         | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_peak_reading_missing_returns_none    | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_readings_for_gauge                   | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_readings_for_gauge_by_name_fall_back | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_readings_missing_gauge_returns_empty | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_sensor_responses                     | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_sensor_responses_missing_gauge       | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_gauge_statistics_missing_gauge       | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_gauge_statistics_no_numeric_level    | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_gauge_statistics_with_data           | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_status_by_timestamp                  | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_status_not_found_returns_none        | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_file_status_alias                    | Pass   |
| test_timeseries_loader.py::TestTimeSeriesLoader | test_timeseries_loader.py::TestTimeSeriesLoader | test_get_status_returns_dict                  | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_appends_entry                            | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_bad_audit_path_does_not_raise            | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_corrupted_file_restarted                 | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_creates_audit_file                       | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_entry_has_required_keys                  | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_existing_file_appended                   | Pass   |
| audit.py::TestLogModelUsage                     | audit.py::TestLogModelUsage                     | test_multiple_calls_accumulate                | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_case_insensitive                         | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_depth_damage_maps_to_dd                  | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_distribution_maps_to_si                  | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_mkm_prefix_returned_as_is                | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_shorthand_gev_resolved                   | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_shorthand_intensity_resolved             | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_shorthand_mortgage_resolved              | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_shorthand_prs_resolved                   | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_spatial_maps_to_sp                       | Pass   |
| audit.py::TestResolveModelId                    | audit.py::TestResolveModelId                    | test_unknown_shorthand_returned_as_is         | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_bool_passed_through                      | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_empty_params                             | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_float_passed_through                     | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_int_passed_through                       | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_long_list_summarised                     | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_none_passed_through                      | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_object_converted_to_string               | Pass   |
| audit.py::TestSanitiseParams                    | audit.py::TestSanitiseParams                    | test_short_list_converted_to_str              | Pass   |

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| Model | Module                                    | Test Name                                              | Status |
|-------|-------------------------------------------|--------------------------------------------------------|--------|
|       | audit.py::TestSanitiseParams              | test_string_passed_through                             | Pass   |
|       | audit.py::TestSanitiseParams              | test_tuple_treated_like_list                           | Pass   |
|       | depth_damage.py::TestCalculateFloodDamage | test_depth_severe_level_some_positive                  | Pass   |
|       | depth_damage.py::TestCalculateFloodDamage | test_depth_severe_level_returns_zeros                  | Pass   |
|       | depth_damage.py::TestCalculateFloodDamage | test_depths_capped_at_5m                               | Pass   |
|       | depth_damage.py::TestCalculateFloodDamage | test_depth_gauge_data_returns_zeros                    | Pass   |
|       | depth_damage.py::TestCalculateFloodDamage | test_depth_gauge_data_returns_zeros                    | Pass   |
|       | depth_damage.py::TestCalculateFloodDamage | test_depths_correct_length                             | Pass   |
|       | depth_damage.py::TestDepthDamage          | test_flood_damage_higher_damage_than_residential       | Pass   |
|       | depth_damage.py::TestDepthDamage          | test_flood_level_reduces_damage                        | Pass   |
|       | depth_damage.py::TestDepthDamage          | test_flood_depth_all_near_one                          | Pass   |
|       | depth_damage.py::TestDepthDamage          | test_flood_clip_to_0_1                                 | Pass   |
|       | depth_damage.py::TestDepthDamage          | test_flood_returns_array_same_length                   | Pass   |
|       | depth_damage.py::TestDepthDamage          | test_flood_depths_all_zero_damage                      | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_max_depth_returns_one                      | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_between_control_points_interpolated        | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_between_0_and_1                            | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_meter_damage                               | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_max_depth_returns_one                      | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_monotonically_increasing                   | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_negative_depth_returns_zero                | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_meter_damage                               | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_returns_float                              | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_small_depth_returns_small_damage           | Pass   |
|       | depth_damage.py::TestScalarDepthDamage    | test_damage_small_depth_returns_zero                   | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_construction                     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_correlation_matrix_built         | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_correlation_params               | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_elevation_warns_and_defaults     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_floor_level_defaults_to_zero     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_geometry_raises                  | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_property_id_auto_assigned        | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_property_type_warns_and_defaults | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_value_raises                     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_properties_copied                | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_gauge_data                       | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_clusters_returns_df              | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_spatial_concentration_returns_df | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_advanced_metrics_returns_dict    | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_direct_impacts_returns_array     | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_flood_depths_returns_array       | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_distance_delegate                | Pass   |
|       | flood_risk_model.py::TestFloodRiskModel   | test_flood_risk_model_portfolio_impact_returns_dict    | Pass   |
|       | spatial.py::TestBuildCorrelationMatrix    | test_correlation_matrix_diagonal_is_one                | Pass   |

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| Model | Module                                                                           | Test Name                                                     | Status |
|-------|----------------------------------------------------------------------------------|---------------------------------------------------------------|--------|
|       | spatial.py::TestBuildCorrelationMatrix                                           | test_off_diagonal_between_zero_and_one                        | Pass   |
|       | spatial.py::TestBuildCorrelationMatrix                                           | test_shape_is_n_by_n                                          | Pass   |
|       | spatial.py::TestBuildCorrelationMatrix                                           | test_is_symmetric                                             | Pass   |
|       | spatial.py::TestBuildSpatialIndex                                                | test_empty_dataframe_returns_none                             | Pass   |
|       | spatial.py::TestBuildSpatialIndex                                                | test_none_gauge_data_returns_none                             | Pass   |
|       | spatial.py::TestBuildSpatialIndex                                                | test_with_gauges_returns_kdtree                               | Pass   |
|       | spatial.py::TestHaversineDistance                                                | test_equatorial_degree_approx_111km                           | Pass   |
|       | spatial.py::TestHaversineDistance                                                | test_london_paris_approx_340km                                | Pass   |
|       | spatial.py::TestHaversineDistance                                                | test_north_south_degree_approx_111km                          | Pass   |
|       | spatial.py::TestHaversineDistance                                                | test_returns_float                                            | Pass   |
|       | spatial.py::TestHaversineDistance                                                | test_same_point_is_zero                                       | Pass   |
|       | spatial.py::TestHaversineDistance                                                | test_symmetry                                                 | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_all_zero_weights_returns_zero                            | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_closer_gauge_has_higher_weight                           | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_empty_list_returns_zero                                  | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_equal_distance_gauges_averages_wse                       | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_near_gauge_returns_wse_directly                          | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_power_one_changes_result                                 | Pass   |
|       | spatial.py::TestIDWInterpolate                                                   | test_single_far_gauge                                         | Pass   |
|       | spatial.py::TestSpatialInterpolateWSE                                            | test_at_gauge_location_returns_wse_directly                   | Pass   |
|       | spatial.py::TestSpatialInterpolateWSE                                            | test_empty_gauge_data_returns_zero                            | Pass   |
|       | spatial.py::TestSpatialInterpolateWSE                                            | test_idw_returns_weighted_value                               | Pass   |
|       | spatial.py::TestSpatialInterpolateWSE                                            | test_single_gauge                                             | Pass   |
|       | test_depth_damage_coverage.py::TestCycloneGaleForceDapdesNegatesToSynthetic      | test_cyclone_gale_force_dapdes_negates_to_synthetic           | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageFunctionReturns0.9Cfactor          | test_depth_damage_function_returns_0.9C_factor                | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageAdjustment                         | test_depth_damage_adjustment                                  | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageFunctionReturnsCorrectLength       | test_depth_damage_function_returns_correct_length             | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageFunctionReturns10.9Cfactor         | test_depth_damage_function_returns_10.9C_factor               | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamagePredictionInterval                 | test_depth_damage_prediction_interval                         | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDamageDefaultCoverage0                   | test_depth_damage_default_coverage_0                          | Pass   |
|       | test_depth_damage_coverage.py::TestDepthDepthsFunctionCoverage                   | test_depth_depths_function_coverage                           | Pass   |
|       | test_depth_damage_coverage.py::TestScaledDepthDamageFallback                     | test_scaled_depth_damage_fallback                             | Pass   |
|       | test_depth_damage_coverage.py::TestScaledDepthDamageFallbackDepth                | test_scaled_depth_damage_fallback_depth                       | Pass   |
|       | test_depth_damage_coverage.py::TestSyntheticFloodReturnsZeros                    | test_synthetic_flood_returns_zeros                            | Pass   |
|       | test_depth_damage_coverage.py::TestSyntheticFloodAt15m                           | test_synthetic_flood_at_15m                                   | Pass   |
|       | test_depth_damage_coverage.py::TestSyntheticFloodDapdesApplied                   | test_synthetic_flood_dapdes_applied                           | Pass   |
|       | test_depth_damage_coverage.py::TestSyntheticFloodReturnsZeros                    | test_synthetic_flood_returns_zeros                            | Pass   |
|       | test_depth_damage_coverage.py::TestSyntheticFloodPropertyGetsZero                | test_synthetic_flood_property_gets_zero                       | Pass   |
|       | test_depth_damage_coverage.py::TestSyntheticFloodProducesPositiveDepth           | test_synthetic_flood_produces_positive_depth                  | Pass   |
|       | test_depth_damage_coverage.py::TestSyntheticFloodReturnsZeros                    | test_synthetic_flood_returns_zeros                            | Pass   |
|       | test_spatial_coverage.py::TestBuildRegionalMatrixCoverage                        | test_build_regional_matrix_coverage                           | Pass   |
|       | test_spatial_coverage.py::TestBuildRegionalMatrixCorrelationIncreasesOffDiagonal | test_build_regional_matrix_correlation_increases_off_diagonal | Pass   |
|       | test_spatial_coverage.py::TestBuildRegionalMatrixCoverageIncreasesCorrelation    | test_build_regional_matrix_coverage_increases_correlation     | Pass   |
|       | test_spatial_coverage.py::TestBuildRegionalMatrixPositive                        | test_build_regional_matrix_positive                           | Pass   |

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| Model | Module | Test Name                                                                  | Status |
|-------|--------|----------------------------------------------------------------------------|--------|
|       |        | test_spatial_coverage.py::TestBuildSpatialRelationshipMatrixCoverage       | Pass   |
|       |        | test_spatial_coverage.py::TestBuildSpatialRelationshipMatrixCoverage       | Pass   |
|       |        | test_spatial_coverage.py::TestBuildSpatialIndexCoverage_radians            | Pass   |
|       |        | test_spatial_coverage.py::TestBuildSpatialIndexEdgeBuilder_tree            | Pass   |
|       |        | test_spatial_coverage.py::TestBuildSpatialIndexDataFrame_returns_kdtree    | Pass   |
|       |        | test_spatial_coverage.py::TestIDWeightCustomPowerCases                     | Pass   |
|       |        | test_spatial_coverage.py::TestIDWeightEmptyEdgeCasesZero                   | Pass   |
|       |        | test_spatial_coverage.py::TestIDWeightSingleEdgeCases_exact_min_distance   | Pass   |
|       |        | test_spatial_coverage.py::TestIDWeightThreeEdges_weighted                  | Pass   |
|       |        | test_spatial_coverage.py::TestIDWeightPolyEdge_Gauge_returns_directly      | Pass   |
|       |        | test_spatial_coverage.py::TestSpatialInterpolateVoronoiLocation_returns_ws | Pass   |
|       |        | test_spatial_coverage.py::TestSpatialInterpolateVoronoiEdgeReturns_zero    | Pass   |
|       |        | test_spatial_coverage.py::TestSpatialTwoGauges_VoronoiEdges_coverage       | Pass   |
|       |        | test_builder_coverage.py::TestBuildFixedProportionCapped_at_001            | Pass   |
|       |        | test_builder_coverage.py::TestBuildLogVerbose_false_no_log                 | Pass   |
|       |        | test_builder_coverage.py::TestBuildLogVerbose_true_emits_log               | Pass   |
|       |        | test_builder_coverage.py::TestBuildProgressLog_every_10_gauges             | Pass   |
|       |        | test_distribution_coverage.py::TestTypeNormalEdges_one_returns_pos_inf     | Pass   |
|       |        | test_distribution_coverage.py::TestTypeNegativeEdges_returns_neg_inf       | Pass   |
|       |        | test_distribution_coverage.py::TestTypeNormalEdges_pos_inf                 | Pass   |
|       |        | test_distribution_coverage.py::TestTypeNormalEdges_neg_inf                 | Pass   |
|       |        | test_distribution_coverage.py::TestTypeHighNormalHighP                     | Pass   |
|       |        | test_distribution_coverage.py::TestTypeHighNormalHighP_98                  | Pass   |
|       |        | test_distribution_coverage.py::TestTypeHighNormalHighFive                  | Pass   |
|       |        | test_distribution_coverage.py::TestTypeInverseNormalHighTrip_high_p        | Pass   |
|       |        | test_distribution_coverage.py::TestNormalHalf_pdf_at_zero                  | Pass   |
|       |        | test_distribution_coverage.py::TestNormalHalf_pdf_decreases_from_center    | Pass   |
|       |        | test_distribution_coverage.py::TestNormalHalf_pdf_positive                 | Pass   |
|       |        | test_distribution_coverage.py::TestNormalHalf_pdf_symmetric                | Pass   |
|       |        | test_io_coverage.py::TestBuildHazardMissingNoGauges_raises                 | Pass   |
|       |        | test_io_coverage.py::TestBuildHazardMissingNoHighSequences_raises          | Pass   |
|       |        | test_io_coverage.py::TestBuildHazardVerbosFalse_no_log                     | Pass   |
|       |        | test_io_coverage.py::TestBuildHazardVerbosTrue_logs_messages               | Pass   |
|       |        | test_prs_analytical_coverage.py::TestBasicWagaugeEdge_elevation_zero_diff  | Pass   |
|       |        | test_prs_analytical_coverage.py::TestBasicWagaugeEdge_gauges               | Pass   |
|       |        | test_prs_analytical_coverage.py::TestCanQuitPyZeroChalEdgask               | Pass   |
|       |        | test_prs_analytical_coverage.py::TestCanQuitPySquardnReturns_zero          | Pass   |
|       |        | test_prs_analytical_coverage.py::TestCanQuitPyJdSmonvdeEdges               | Pass   |
|       |        | test_prs_analytical_coverage.py::TestCanQuitPyZerodlmgzzero                | Pass   |
|       |        | test_prs_analytical_coverage.py::TestInemptydictRetatEdfailback            | Pass   |
|       |        | test_prs_analytical_coverage.py::TestInemptydictRetRtmsEdfailback          | Pass   |
|       |        | test_prs_analytical_coverage.py::TestInfailchrtVghHtateEdgesop             | Pass   |
|       |        | test_prs_analytical_coverage.py::TestIntcapolamndByrEdrgns_last            | Pass   |
|       |        | test_prs_analytical_coverage.py::TestIntcapolamYkldPttmEdrgast             | Pass   |

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| Model | Module                                                  | Test Name                              | Status |
|-------|---------------------------------------------------------|----------------------------------------|--------|
|       | test_synthetic_coverage.py::TestHistoricalHighInjection | test_high_injected_post_loop           | Pass   |
|       | test_synthetic_coverage.py::TestHistoricalHighInjection | test_high_not_in_range_skipped         | Pass   |
|       | test_synthetic_coverage.py::TestHistoricalHighInjection | test_high_date                         | Pass   |
|       | cli.py::TestCompareScenarios                            | test_contains_header                   | Pass   |
|       | cli.py::TestCompareScenarios                            | test_contains_scenario_names           | Pass   |
|       | cli.py::TestCompareScenarios                            | test_none_uses_all_scenarios           | Pass   |
|       | cli.py::TestCompareScenarios                            | test_returns_string                    | Pass   |
|       | cli.py::TestCompareScenarios                            | test_single_scenario                   | Pass   |
|       | cli.py::TestCompareScenarios                            | test_two_scenarios                     | Pass   |
|       | cli.py::TestMain                                        | test_compare_mode                      | Pass   |
|       | cli.py::TestMain                                        | test_default_describe_mode             | Pass   |
|       | cli.py::TestMain                                        | test_exceedance_json_mode              | Pass   |
|       | cli.py::TestMain                                        | test_exceedance_mode                   | Pass   |
|       | cli.py::TestMain                                        | test_json_default_mode                 | Pass   |
|       | cli.py::TestMain                                        | test_return_period_json_mode           | Pass   |
|       | cli.py::TestMain                                        | test_return_period_mode                | Pass   |
|       | cli.py::TestMain                                        | test_samples_json_mode                 | Pass   |
|       | cli.py::TestMain                                        | test_samples_mode                      | Pass   |
|       | cli.py::TestMain                                        | test_scenario_mode                     | Pass   |
|       | cli.py::TestMain                                        | test_seed_reproducible                 | Pass   |
|       | prs_hazard_curve.py::TestCreateFlatHazardCurve          | test_flat_hazard_curve_counter         | Pass   |
|       | prs_hazard_curve.py::TestCreateFlatHazardCurve          | test_flat_hazard_curve_and_quote       | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityFlatHazard    | test_sensitivity_flat_hazard           | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityFlatHazard    | test_sensitivity_flat_hazard_counter   | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityFlatHazard    | test_sensitivity_flat_hazard_start     | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityFlatHazard    | test_sensitivity_flat_hazard           | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityFlatHazard    | test_sensitivity_flat_hazard_rate      | Pass   |
|       | prs_hazard_curve.py::TestCreateSensitivityFlatHazard    | test_sensitivity_flat_hazard_rate      | Pass   |
|       | prs_hazard_curve.py::TestLoadHazardCodes                | test_hazard_codes_data_has_gauge       | Pass   |
|       | prs_hazard_curve.py::TestLoadHazardCodes                | test_hazard_codes_from_file            | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_alert_trigger                     | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_custom_notional                   | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_flat_hazard_curve                 | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_result_has_fair_spread_bps        | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_result_has_npv                    | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_result_has_survival_probabilities | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_returns_dict                      | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_severe_trigger                    | Pass   |
|       | prs_hazard_curve.py::TestPricePricing                   | test_term_structure_true               | Pass   |
|       | prs_hazard_curve.py::TestPrintPricingResults            | test_print_results_gauge_name          | Pass   |
|       | prs_hazard_curve.py::TestPrintPricingResults            | test_print_results_without_error       | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHazardCurve      | test_flat_hazard_curve_accepted        | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHazardCurve      | test_flat_hazard_curve_additional      | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHazardCurve      | test_flat_hazard_curve_additional      | Pass   |

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| Model | Module                                                  | Test Name                                     | Status |
|-------|---------------------------------------------------------|-----------------------------------------------|--------|
|       | test_load_and_curves.py::TestCreateFlatHazardCurve      | testFlatHazardCurveProbabilityFormula         | Pass   |
|       | test_load_and_curves.py::TestCreateFlatHighHazardRate   | testFlatHighHazardRateNearZeroSurvival        | Pass   |
|       | test_load_and_curves.py::TestCreateSurvival360DayDoomer | testSurvival360DayDoomerProducesCurve         | Pass   |
|       | test_load_and_curves.py::TestCreateSimpleCurve          | testSimpleCurveAdditionalFormula              | Pass   |
|       | test_load_and_curves.py::TestCreateSurvivalCurve        | testSurvivalCurveAdditionalPositiveHazard     | Pass   |
|       | test_load_and_curves.py::TestCreateSinglePoint          | testSinglePointAdditionalRiskTs               | Pass   |
|       | test_load_and_curves.py::TestCreateSinglePoint          | testSinglePointAdditionalStructure            | Pass   |
|       | test_load_and_curves.py::TestCreateSurvivalCurve        | testSurvivalCurveAdditionalPositive           | Pass   |
|       | test_load_and_curves.py::TestLoadHazardCurve            | testLoadHazardCurveFromObjectReturnsEmptyDict | Pass   |
|       | test_load_and_curves.py::TestLoadHazardCurve            | testLoadHazardCurveRaisesJsonDecodeError      | Pass   |
|       | test_load_and_curves.py::TestLoadHazardCurve            | testLoadHazardCurveRaisesFileNotFound         | Pass   |
|       | test_load_and_curves.py::TestLoadHazardCurve            | testLoadHazardCurveRaisesFileStructure        | Pass   |
|       | test_price_prs.py::TestPricePrsGauge                    | testPassingHazardRateMatchesInput             | Pass   |
|       | test_price_prs.py::TestPricePrsGauge                    | testPassingInResult                           | Pass   |
|       | test_price_prs.py::TestPricePrsGauge                    | testPassingInResult                           | Pass   |
|       | test_price_prs.py::TestPricePrsGauge                    | testPassingAnnualHazardRateMatchesInput       | Pass   |
|       | test_price_prs.py::TestPricePrsLeg                      | testSignpvIsSumOfLegs                         | Pass   |
|       | test_price_prs.py::TestPricePrsLeg                      | testSignPremiumLegNegative                    | Pass   |
|       | test_price_prs.py::TestPricePrsLeg                      | testSignProtectionLegPositive                 | Pass   |
|       | test_price_prs.py::TestPricePrsRecoveryRate             | testRecoveryRateStored                        | Pass   |
|       | test_price_prs.py::TestPricePrsRecoveryRate             | testRecoveryRateAffectsFairSpread             | Pass   |
|       | test_price_prs.py::TestPricePrsRiskFreeRate             | testHigherRiskFreeRateChangesNpv              | Pass   |
|       | test_price_prs.py::TestPricePrsRiskFreeRate             | testRiskFreeRateNotInOutput                   | Pass   |
|       | test_price_prs.py::TestPricePrsType                     | testFairSpreadIsFloat                         | Pass   |
|       | test_price_prs.py::TestPricePrsType                     | testFairUpfrontIsFloat                        | Pass   |
|       | test_price_prs.py::TestPricePrsType                     | testNpvIsFloat                                | Pass   |
|       | test_price_prs.py::TestPricePrsType                     | testRunningSpreadBpsIdentity                  | Pass   |
|       | test_price_prs.py::TestPricePrsType                     | testSurvivalProbsAllInZeroOne                 | Pass   |
|       | test_price_prs.py::TestPricePrsType                     | testTenor1YrGivesOneSurvivalEntry             | Pass   |
|       | test_price_prs.py::TestPricePrsType                     | testUseTermStructureStoredFalse               | Pass   |
|       | test_price_prs.py::TestPricePrsType                     | testUseTermStructureStoredTrue                | Pass   |
|       | test_print_and_cli.py::TestMain                         | testCustomNotionalAndSpread                   | Pass   |
|       | test_print_and_cli.py::TestMain                         | testDefaultGaugeSelectionReturnsZero          | Pass   |
|       | test_print_and_cli.py::TestMain                         | testExplicitGaugeReturnsZero                  | Pass   |
|       | test_print_and_cli.py::TestMain                         | testFlatFlagDispatchesFlatMode                | Pass   |
|       | test_print_and_cli.py::TestMain                         | testListFlagLogsGaugeNames                    | Pass   |
|       | test_print_and_cli.py::TestMain                         | testListFlagReturnsZero                       | Pass   |
|       | test_print_and_cli.py::TestMain                         | testMissingGaugeLogsAvailableGauge            | Pass   |
|       | test_print_and_cli.py::TestMain                         | testMissingGaugeLogsError                     | Pass   |
|       | test_print_and_cli.py::TestMain                         | testMissingGaugeReturnsOne                    | Pass   |
|       | test_print_and_cli.py::TestMain                         | testTriggerAlertExplicit                      | Pass   |
|       | test_print_and_cli.py::TestMain                         | testTriggerSevereExplicit                     | Pass   |
|       | test_print_and_cli.py::TestMain                         | testTriggerWarningExplicit                    | Pass   |
|       | test_print_and_cli.py::TestPrintPrs                     | testPrintPrsGaugeInLog                        | Pass   |

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| Model | Module                                   | Test Name                                    | Status |
|-------|------------------------------------------|----------------------------------------------|--------|
|       | data_structures.py::TestGaugeRecord      | test_extends_timeseries_when_requested       | Pass   |
|       | data_structures.py::TestGaugeRecord      | test_extends_bool                            | Pass   |
|       | data_structures.py::TestGaugeRecord      | test_has_required_keys                       | Pass   |
|       | data_structures.py::TestGaugeRecord      | test_extends_timeseries_by_default           | Pass   |
|       | data_structures.py::TestGaugeRecord      | test_extends_dict                            | Pass   |
|       | data_structures.py::TestGaugeRecord      | test_value_is_rounded                        | Pass   |
|       | data_structures.py::TestIntensityProfile | test_beta_value                              | Pass   |
|       | data_structures.py::TestIntensityProfile | test_gaussian_value                          | Pass   |
|       | data_structures.py::TestIntensityProfile | test_triangular_value                        | Pass   |
|       | data_structures.py::TestStormTotal       | test_custom_kernel                           | Pass   |
|       | data_structures.py::TestStormTotal       | test_decay_kernel_serialised_as_string       | Pass   |
|       | data_structures.py::TestStormTotal       | test_has_storm_id                            | Pass   |
|       | data_structures.py::TestStormTotal       | test_has_track                               | Pass   |
|       | data_structures.py::TestStormTotal       | test_intensity_profile_serialised_as_string  | Pass   |
|       | data_structures.py::TestStormTotal       | test_returns_dict                            | Pass   |
|       | data_structures.py::TestStormTotal       | test_track_has_coords                        | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_auto_generated_id_starts_with_storm     | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_decay_kernel_stored                     | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_decay_parameter_stored                  | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_default_num_track_points                | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_default_start_time_is_recent            | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_duration_hours_stored                   | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_explicit_id_preserved                   | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_explicit_name                           | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_explicit_start_time                     | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_footprint_km_stored                     | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_num_track_points                        | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_peak_intensity_stored                   | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_returns_storm_object                    | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_speed_is_positive                       | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_track_end_time_equals_duration          | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_track_longitude_moves_from_start_to_end | Pend   |
|       | storm_factory.py::TestCreateStorm        | test_track_points_are_trackpoint_instances   | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_track_start_time_is_zero                | Pass   |
|       | storm_factory.py::TestCreateStorm        | test_track_time_increases                    | Pass   |
|       | storm_factory.py::TestIntensityProfile   | test_all_profiles_produce_valid_storm        | Pass   |
|       | storm_factory.py::TestIntensityProfile   | test_beta_profile_creates_storm              | Pass   |
|       | storm_factory.py::TestIntensityProfile   | test_gaussian_profile_peak_near_midpoint     | Pass   |
|       | storm_factory.py::TestIntensityProfile   | test_triangular_max_equals_peak_intensity    | Pass   |
|       | storm_factory.py::TestIntensityProfile   | test_triangular_profile_peak_near_midpoint   | Pass   |
|       | storm_factory.py::TestLoadGauge          | test_empty_portfolio_returns_empty_list      | Pass   |
|       | storm_factory.py::TestLoadGauge          | test_loads_gauge_from_portfolio              | Pass   |
|       | flood_classifier.py::TestFloodClassifier | test_all_avg_auc_in_range                    | Pass   |
|       | flood_classifier.py::TestFloodClassifier | test_all_creates_summary_file                | Pass   |

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| Model | Module                                                | Test Name                                             | Status |
|-------|-------------------------------------------------------|-------------------------------------------------------|--------|
|       | flood_classifier.py::TestFloodClassifier              | test_train_all_returns_summary                        | Pass   |
|       | flood_classifier.py::TestFloodClassifier              | test_train_all_with_mixed_gauges                      | Pass   |
|       | flood_classifier.py::TestFloodClassifier              | test_train_gauge_creates_joblib_file                  | Pass   |
|       | flood_classifier.py::TestFloodClassifier              | test_train_gauge_has_metrics                          | Pass   |
|       | flood_classifier.py::TestFloodClassifier              | test_train_gauge_returns_dict                         | Pass   |
|       | flood_classifier.py::TestFloodClassifier              | test_train_gauge_single_class_skipped                 | Pass   |
|       | flood_classifier.py::TestFloodClassifier              | test_train_gauge_status_trained                       | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_available_gauges                           | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_available_gauges_empty_dir                 | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_load_summary                               | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_load_summary_cached                        | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_caches_model                       | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_high_level_higher_probability      | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_in_range                           | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_missing_model_raises               | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_returns_float                      | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_series_all_in_range                | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_series_default_derivatives         | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_series_returns_list                | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_series_with_derivatives            | Pass   |
|       | flood_classifier.py::TestFloodPredictor               | test_train_predict_surface_returns_grid               | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_levels_positive                         | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_levels_length_approximately             | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_levels_are_sequential                   | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_levels_different_seeds_different_output | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_obs_has_date_and_level                  | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_gauge_data_uses_defaults                | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_historical_high_date_in_range           | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_historical_high_date_invalid_format     | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_historical_high_date_out_of_range       | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_missing_flood_stages_uses_defaults      | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_returns_list                            | Pass   |
|       | synthetic.py::TestGenerateSyntheticData               | test_generate_series_reproducible                     | Pass   |
|       | test_hazard_pricing.py::TestCreateHazardPricingEngine | test_create_engine_one                                | Pass   |
|       | test_hazard_pricing.py::TestCreateHazardPricingEngine | test_create_engine_with_engines                       | Pass   |
|       | test_hazard_pricing.py::TestCreateHazardPricingEngine | test_create_hazard_approval                           | Pass   |
|       | test_hazard_pricing.py::TestCreateHazardPricingEngine | test_create_hazard_approval_decreases                 | Pass   |
|       | test_hazard_pricing.py::TestCreateHazardPricingEngine | test_create_hazard_approval_survival                  | Pass   |
|       | test_hazard_pricing.py::TestHazardPricingEngine       | test_points_to_ready_count                            | Pass   |
|       | test_hazard_pricing.py::TestHazardPricingEngine       | test_points_to_hazard_gives_low_survival              | Pass   |
|       | test_hazard_pricing.py::TestHazardPricingEngine       | test_points_to_ready_curve                            | Pass   |
|       | test_hazard_pricing.py::TestHazardPricingEngine       | test_points_to_ready_decreases                        | Pass   |
|       | test_hazard_pricing.py::TestHazardPricingEngine       | test_points_to_hazard_gives_survival_near_one         | Pass   |
|       | test_engine_coverage.py::TestProperty                 | test_base_property_has_basis_info                     | Pass   |

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| Model | Module                                                                           | Test Name                                              | Status |
|-------|----------------------------------------------------------------------------------|--------------------------------------------------------|--------|
|       | test_engine.coverage.py::TestRenderEngineTypeEdgeCases                           | testRenderEngineTypeEdgeCases returns_5                | Pass   |
|       | test_engine.coverage.py::TestRenderEngineInvalidEdgeCaseFormat                   | testRenderEngineInvalidEdgeCaseFormat returns_5        | Pass   |
|       | test_engine.coverage.py::TestRenderEngineInvalidEdgeCaseDate                     | testRenderEngineInvalidEdgeCaseDate returns_5          | Pass   |
|       | test_engine.coverage.py::TestRenderEngineInvalidEdgeCaseSet                      | testRenderEngineInvalidEdgeCaseSet returns_5           | Pass   |
|       | test_engine.coverage.py::TestRenderEngineInvalidEdgeCaseYears                    | testRenderEngineInvalidEdgeCaseYears returns_years     | Pass   |
|       | test_engine.coverage.py::TestReverseDefaultMarketState                           | testReverseDefaultMarketState market_state             | Pass   |
|       | test_pnl.coverage.py::TestGenerateSnapshotsFromDefaultDate                       | testGenerateSnapshotsFromDefaultDate                   | Pass   |
|       | test_pnl.coverage.py::TestGetEodSnapshotExistingFromSnapshotReturnsData          | testGetEodSnapshotExistingFromSnapshot returns_data    | Pass   |
|       | test_pnl.coverage.py::TestGetEodSnapshotNonexistentFromSnapshotReturnsNone       | testGetEodSnapshotNonexistentFromSnapshot returns_none | Pass   |
|       | test_pnl.coverage.py::TestGetTradeVlaGetExistingTradeMark                        | testVlaGetExistingTradeMark                            | Pass   |
|       | test_pnl.coverage.py::TestGetTradeVlaGetNonexistentTradeMark                     | testVlaGetNonexistentTradeMark                         | Pass   |
|       | test_pnl.coverage.py::TestGetTradeVlaGetTradeMarkAfterMultipleUpdates            | testVlaGetTradeMarkAfterMultipleUpdates                | Pass   |
|       | test_market_state.coverage.py::TestCommitYieldCurveNoExistingYieldCurveIfMissing | testCommitYieldCurveNoExistingYieldCurveIfMissing      | Pass   |
|       | test_market_state.coverage.py::TestGetHazardRateForFlatRate                      | testGetHazardRateForFlatRate                           | Pass   |
|       | test_market_state.coverage.py::TestGetHazardRateForFlatRateToZero                | testGetHazardRateForFlatRateToZero                     | Pass   |
|       | test_market_state.coverage.py::TestGetYieldCurveEmptyCurveRiskFreeRate           | testGetYieldCurveEmptyCurveRiskFreeRate                | Pass   |
|       | test_market_state.coverage.py::TestGetYieldCurveEmptyCurveReturnsRiskFreeRate    | testGetYieldCurveEmptyCurveReturnsRiskFreeRate         | Pass   |
|       | test_market_state.coverage.py::TestGetInterestRateFromFlatRate                   | testGetInterestRateFromFlatRate                        | Pass   |
|       | test_market_state.coverage.py::TestUpdateYieldCurveNoExistingYieldCurveIfMissing | testUpdateYieldCurveNoExistingYieldCurveIfMissing      | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorFactorsApplied                          | testEstimatorFactorsApplied                            | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorBasicCalculation                        | testEstimatorBasicCalculation                          | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorClaimsFactorIncreasesPremium            | testEstimatorClaimsFactorIncreasesPremium              | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorCappedToMax                             | testEstimatorCappedToMax                               | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorCappedToMin                             | testEstimatorCappedToMin                               | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorContentsPremiumAdded                    | testEstimatorContentsPremiumAdded                      | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorHighFloodRiskIncreasesPremium           | testEstimatorHighFloodRiskIncreasesPremium             | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorReturnsFloat                            | testEstimatorReturnsFloat                              | Pass   |
|       | insurance.py::TestApplyInsuranceEstimatorSecurityFactorReducesPremium            | testEstimatorSecurityFactorReducesPremium              | Pass   |
|       | insurance.py::TestGetAgePremiumEstimator100YearsIsHeritage                       | testEstimator100YearsIsHeritage                        | Pass   |
|       | insurance.py::TestGetAgePremiumEstimator10YearsIsModern                          | testEstimator10YearsIsModern                           | Pass   |
|       | insurance.py::TestGetAgePremiumEstimator30YearsIsEstablished                     | testEstimator30YearsIsEstablished                      | Pass   |
|       | insurance.py::TestGetAgePremiumEstimatorDefaultCurrentYear                       | testEstimatorDefaultCurrentYear                        | Pass   |
|       | insurance.py::TestGetAgePremiumEstimatorEstablished                              | testEstimatorEstablished                               | Pass   |
|       | insurance.py::TestGetAgePremiumEstimatorExactly29YearsModern                     | testEstimatorExactly29YearsModern                      | Pass   |
|       | insurance.py::TestGetAgePremiumEstimatorExactly99YearsEstablished                | testEstimatorExactly99YearsEstablished                 | Pass   |
|       | insurance.py::TestGetAgePremiumEstimatorExactly9YearsNewBuild                    | testEstimatorExactly9YearsNewBuild                     | Pass   |
|       | insurance.py::TestGetAgePremiumEstimatorHeritageOld                              | testEstimatorHeritageOld                               | Pass   |
|       | insurance.py::TestGetAgePremiumEstimatorModern                                   | testEstimatorModern                                    | Pass   |
|       | insurance.py::TestGetAgePremiumEstimatorNewBuild                                 | testEstimatorNewBuild                                  | Pass   |
|       | insurance.py::TestGetAreaPremiumEstimatorRange100ThirdBand                       | testEstimatorRange100ThirdBand                         | Pass   |
|       | insurance.py::TestGetAreaPremiumEstimatorRange50SecondBand                       | testEstimatorRange50SecondBand                         | Pass   |
|       | insurance.py::TestGetAreaPremiumEstimatorJustBelow100SecondBand                  | testEstimatorJustBelow100SecondBand                    | Pass   |
|       | insurance.py::TestGetAreaPremiumEstimatorJustBelow50FirstBand                    | testEstimatorJustBelow50FirstBand                      | Pass   |
|       | insurance.py::TestGetAreaPremiumEstimatorRangeFourthBand                         | testEstimatorRangeFourthBand                           | Pass   |

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| Model | Module                                   | Test Name                                          | Status |
|-------|------------------------------------------|----------------------------------------------------|--------|
|       | insurance.py::TestGetAreaPremiumFactors  | test_base_area.second.band                         | Pass   |
|       | insurance.py::TestGetAreaPremiumFactors  | test_base_area.tuple                               | Pass   |
|       | insurance.py::TestGetAreaPremiumFactors  | test_base_area.first.band                          | Pass   |
|       | insurance.py::TestInsuranceFactors       | test_base_premium_factors.all_positive             | Pass   |
|       | insurance.py::TestInsuranceFactors       | test_base_area.bands_are_sorted                    | Pass   |
|       | insurance.py::TestInsuranceFactors       | test_base_rate_range                               | Pass   |
|       | insurance.py::TestInsuranceFactors       | test_base_construction_type_factors.present        | Pass   |
|       | insurance.py::TestInsuranceFactors       | test_base_flood_risk_factors.increase_with_risk    | Pass   |
|       | insurance.py::TestInsuranceFactors       | test_base_flood_risk_factors.present               | Pass   |
|       | insurance.py::TestInsuranceFactors       | test_base_min_max_premium.boundaries               | Pass   |
|       | insurance.py::TestInsuranceFactors       | test_base_property_type_factors.present            | Pass   |
|       | property_value.py::TestApplyValueFactors | test_base_factors_applied                          | Pass   |
|       | property_value.py::TestApplyValueFactors | test_base_factor_calculation                       | Pass   |
|       | property_value.py::TestApplyValueFactors | test_base_factor_to_max                            | Pass   |
|       | property_value.py::TestApplyValueFactors | test_base_factor_to_min                            | Pass   |
|       | property_value.py::TestApplyValueFactors | test_base_factor_risk_discount_reduces_value       | Pass   |
|       | property_value.py::TestApplyValueFactors | test_base_factor_applied                           | Pass   |
|       | property_value.py::TestApplyValueFactors | test_base_returns_float                            | Pass   |
|       | property_value.py::TestFactorTable       | test_base_age_band_factors.within_reasonable_bands | Pass   |
|       | property_value.py::TestFactorTable       | test_base_area_ranges.all_positive                 | Pass   |
|       | property_value.py::TestFactorTable       | test_base_price_per_sqm.all_positive               | Pass   |
|       | property_value.py::TestFactorTable       | test_base_condition_factors.ordered_by_severity    | Pass   |
|       | property_value.py::TestFactorTable       | test_base_epc_factors.a_highest                    | Pass   |
|       | property_value.py::TestFactorTable       | test_base_flood_risk_factors.decrease_with_risk    | Pass   |
|       | property_value.py::TestFactorTable       | test_base_min_max_value.boundaries                 | Pass   |
|       | property_value.py::TestFactorTable       | test_base_proximity_zones.have_range_key           | Pass   |
|       | property_value.py::TestFactorTable       | test_base_rent_per_sqm.all_present                 | Pass   |
|       | property_value.py::TestFactorTable       | test_base_rental_yield_rates.all_present           | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_100_years.is_heritage                    | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_10_years.is_modern                       | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_25_years.is_established                  | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_50_years.is_period                       | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_default.current_year                     | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_established                              | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_exactly_24_years.is_modern               | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_exactly_49_years.is_established          | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_exactly_99_years.is_period               | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_exactly_9_years.is_new_build             | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_heritage_old                             | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_modern                                   | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_new_build                                | Pass   |
|       | property_value.py::TestGetAgeBand        | test_base_period                                   | Pass   |
|       | property_value.py::TestGetProximityZones | test_base_high_risk_high                           | Pass   |
|       | property_value.py::TestGetProximityZones | test_base_high_risk_medium                         | Pass   |

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| Model | Module                                  | Test Name                                               | Status  |
|-------|-----------------------------------------|---------------------------------------------------------|---------|
|       | property_value.py::TestGetProximityZone | testZone_high_risk_very_high                            | Pass    |
|       | property_value.py::TestGetProximityZone | testZone_low_risk_low                                   | Pass    |
|       | property_value.py::TestGetProximityZone | testZone_low_risk_very_low                              | Pass    |
|       | property_value.py::TestGetProximityZone | testZone_exactly_200m_is_medium_zone                    | Pass    |
|       | property_value.py::TestGetProximityZone | testZone_exactly_500m_is_far                            | Pass    |
|       | property_value.py::TestGetProximityZone | testZone_far_zone                                       | Pass    |
|       | property_value.py::TestGetProximityZone | testZone_medium_zone                                    | Pass    |
|       | property_value.py::TestGetProximityZone | testZone_medium_zone_upper_bound                        | Pass    |
|       | display.py::TestBlotterTradeExistence   | testblotter_glob_pattern_matches_prs_id_protocol        | Pass    |
|       | display.py::TestBlotterTradeExistence   | testblotter_returns_trades_with_fixture                 | Pass    |
|       | display.py::TestStormDisplayFormat      | test_all_files_include_storm_id_in_label                | Pass    |
|       | display.py::TestStormDisplayFormat      | test_all_files_use_sentinel_or_pipe_separator           | Pass    |
|       | display.py::TestStormDisplayFormat      | test_canonical_format_defined_in_config                 | Format  |
|       | display.py::TestStormDisplayFormat      | test_no_file_uses_old_precipitation_label               | Pass    |
|       | display.py::TestStormDisplayFormat      | test_no_file_uses_old_severe_levels_format              | Pass    |
|       | display.py::TestStormDisplayFormat      | test_port_stress_uses_sentinel_with_warning             | Warning |
|       | display.py::TestStormDisplayFormat      | test_sp_table_uses_sentinel_pattern                     | Pass    |
|       | display.py::TestStormDisplayFormat      | test_storm_option_js_custom_var                         | Pass    |
|       | display.py::TestStormDisplayFormat      | test_storm_option_js_peak_variant                       | Pass    |
|       | display.py::TestStormDisplayFormat      | test_storm_option_js_warning_variant                    | Pass    |
|       | display.py::TestStormDisplayFormat      | test_trading_stress_uses_sentinel_with_peak             | Peak    |
|       | display.py::TestStormDisplayFormat      | test_visual_files_use_sentinel_not_hardcoded_format     | Code    |
|       | fixtures.py::TestProtocolUniformities   | test_all_entity_suffixes_have_same_length               | Pass    |
|       | fixtures.py::TestProtocolUniformities   | test_all_uuid4_suffixes_produce_lowercased_hex          | Pass    |
|       | fixtures.py::TestProtocolUniformities   | test_gauge_prop_mort_storm_share_same_suffix_derivation | Fix     |
|       | fixtures.py::TestTestFixtureGaugeIDs    | test_backward_compat_gauge_ids_are_documented           | Doc     |
|       | fixtures.py::TestTestFixtureGaugeIDs    | test_gauge_json_and_gaugehc_ids_are_consistent          | Pass    |
|       | fixtures.py::TestTestFixtureGaugeIDs    | test_gaugehc_new_entries_follow_protocol                | Pass    |
|       | fixtures.py::TestTestFixtureGaugeIDs    | test_new_conf_test_gauge_ids_match_protocol             | Pass    |
|       | fixtures.py::TestTestFixtureGaugeIDs    | test_stress_storm_gauge_ids_are_in_known_set            | Set     |
|       | fixtures.py::TestTestFixtureStormIDs    | test_port_stress_routes_storm_constants_match_protocol  | Match   |
|       | fixtures.py::TestTestFixtureStormIDs    | test_sample_port_stress_storms_ids_match_protocol       | Pass    |
|       | fixtures.py::TestTestFixtureStormIDs    | test_sample_stress_storms_storm_ids_match_protocol      | Pass    |
|       | fixtures.py::TestTestFixtureStormIDs    | test_stress_routes_storm_constants_match_protocol       | Pass    |
|       | generators.py::TestGaugeIDGenerator     | test_generator_does_not_use_sequential_format           | Format  |
|       | generators.py::TestGaugeIDGenerator     | test_generator_source_uses_gauge_prefix                 | Pass    |
|       | generators.py::TestGaugeIDGenerator     | test_generator_source_uses_uuid                         | Pass    |
|       | generators.py::TestGaugeIDGenerator     | test_uuid4_produces_protocol_matching_ids               | Pass    |
|       | generators.py::TestMortgageIDGenerator  | test_generator_does_not_use_sequential_format           | Format  |
|       | generators.py::TestMortgageIDGenerator  | test_generator_source_uses_mort_prefix                  | Pass    |
|       | generators.py::TestMortgageIDGenerator  | test_generator_source_uses_uuid                         | Pass    |
|       | generators.py::TestMortgageIDGenerator  | test_uuid4_produces_protocol_matching_ids               | Pass    |
|       | generators.py::TestPRSTradeIDGenerator  | test_thames_uses_prs_prefix                             | Pass    |
|       | generators.py::TestPRSTradeIDGenerator  | test_thames_uses_uuid                                   | Pass    |

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| Model | Module                                                     | Test Name                                                  | Status |
|-------|------------------------------------------------------------|------------------------------------------------------------|--------|
|       | generators.py::TestPRSTradeIDGenerator                     | test_turid4_hex_upper_produces_protocol_matching_ids       | Pass   |
|       | generators.py::TestPropertyIDGenerator                     | test_generator_does_not_use_sequential_format              | Pass   |
|       | generators.py::TestPropertyIDGenerator                     | test_generator_source_uses_prop_prefix                     | Pass   |
|       | generators.py::TestPropertyIDGenerator                     | test_generator_source_uses_uuid                            | Pass   |
|       | generators.py::TestPropertyIDGenerator                     | test_turid4_produces_protocol_matching_ids                 | Pass   |
|       | generators.py::TestStormIDGenerator                        | test_multi_storm_make_storm_id_follows_protocol            | Pass   |
|       | generators.py::TestStormIDGenerator                        | test_single_storm_generator_does_not_use_sequential_format | Pass   |
|       | generators.py::TestStormIDGenerator                        | test_single_storm_generator_uses_storm_prefix              | Pass   |
|       | generators.py::TestStormIDGenerator                        | test_single_storm_generator_uses_uuid                      | Pass   |
|       | generators.py::TestStormIDGenerator                        | test_uuid4_produces_protocol_matching_ids                  | Pass   |
|       | patterns.py::TestGaugeIDPattern                            | test_accepts_valid_uuid_derived_ids                        | Pass   |
|       | patterns.py::TestGaugeIDPattern                            | test_rejects_non_hex_chars                                 | Pass   |
|       | patterns.py::TestGaugeIDPattern                            | test_rejects_sequential_integers                           | Pass   |
|       | patterns.py::TestGaugeIDPattern                            | test_rejects_uppercase_hex                                 | Pass   |
|       | patterns.py::TestGaugeIDPattern                            | test_rejects_wrong_length                                  | Pass   |
|       | patterns.py::TestMortgageIDPattern                         | test_accepts_valid                                         | Pass   |
|       | patterns.py::TestMortgageIDPattern                         | test_rejects_sequential                                    | Pass   |
|       | patterns.py::TestPRSTradeIDPattern                         | test_accepts_lowercase                                     | Pass   |
|       | patterns.py::TestPRSTradeIDPattern                         | test_accepts_uppercase                                     | Pass   |
|       | patterns.py::TestPRSTradeIDPattern                         | test_rejects_non_hex_suffix                                | Pass   |
|       | patterns.py::TestPropertyIDPattern                         | test_accepts_valid                                         | Pass   |
|       | patterns.py::TestPropertyIDPattern                         | test_rejects_sequential                                    | Pass   |
|       | patterns.py::TestPropertyIDPattern                         | test_rejects_uppercase                                     | Pass   |
|       | patterns.py::TestStormIDPattern                            | test_accepts_valid                                         | Pass   |
|       | patterns.py::TestStormIDPattern                            | test_rejects_descriptive_labels                            | Pass   |
|       | patterns.py::TestStormIDPattern                            | test_rejects_sequential_integers                           | Pass   |
|       | patterns.py::TestStormIDPattern                            | test_rejects_uppercase                                     | Pass   |
|       | patterns.py::TestStormIDPattern                            | test_rejects_wrong_prefix_format                           | Pass   |
|       | test_book_common_coverage_b6.py::TestBorrowerIDPattern     | test_invalid_id_logged                                     | Pass   |
|       | test_book_common_coverage_b6.py::TestEmptyCounterpartyFile | test_bankes_fallback                                       | Pass   |
|       | test_book_common_coverage_b6.py::TestEmptyCounterpartyFile | test_bankes_fallback                                       | Pass   |
|       | test_book_common_coverage_b6.py::TestEmptyCounterpartyFile | test_bankes_fallback                                       | Pass   |
|       | test_book_common_coverage_b6.py::TestEmptyCounterpartyFile | test_bankes_fallback                                       | Pass   |
|       | test_book_coverage.py::TestEmptySturges                    | test_raises_value_error                                    | Pass   |
|       | test_book_coverage.py::TestFallbackEmptyCounterpartyFile   | test_fallback                                              | Pass   |
|       | test_book_coverage.py::TestFallbackMissingCounterpartyFile | test_fallback                                              | Pass   |
|       | test_book_coverage.py::TestGenerateProfileFailure          | test_returns_empty_list                                    | Pass   |
|       | test_book_coverage.py::TestGenerateProfileFailure          | test_returns_success                                       | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMapping          | test_fields_extracted                                      | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMapping          | test_mapping_sent                                          | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMapping          | test_data_returns_dict                                     | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMapping          | test_flat_dict                                             | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMapping          | test_raises_value_error                                    | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMapping          | test_value_included                                        | Pass   |
|       | ctpy_mapping.py::TestCounterpartyCBMCreateMapping          | test_required_fields_returns_list                          | Pass   |

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| Model | Module                                       | Test Name                                             | Status |
|-------|----------------------------------------------|-------------------------------------------------------|--------|
|       | ctpy_mapping.py::TestCounterparty            | test_CDMSchema.dict                                   | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.counterparties_adds_error           | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.data_returns_errors                 | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.counterparty_role_adds_error        | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.counterparty_role_adds_error        | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.party_id_adds_error                 | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.party_name_adds_error               | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.party_ref_adds_error                | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.same_role_adds_error                | Pass   |
|       | ctpy_mapping.py::TestCounterparty            | test_CDMValidator.no_errors                           | Pass   |
|       | class_attrs.py::TestClassAttribute           | test_metadata_categories_contains_expected_keys       | Fail   |
|       | class_attrs.py::TestClassAttribute           | test_metadata_categories_station_has_required_subkeys | Fail   |
|       | class_attrs.py::TestClassAttribute           | test_schema_version_is_string                         | Pass   |
|       | class_attrs.py::TestGaugeHistoricalData      | test_daily_inavailable_flag_is_bool                   | Pass   |
|       | directory_main.py::TestMain                  | test_main_skips_bad_file_without_raising              | Pass   |
|       | directory_main.py::TestMain                  | test_main_with_csv_files_runs_without_error           | Pass   |
|       | directory_main.py::TestMain                  | test_main_with_no_csv_files_runs_without_error        | Error  |
|       | filter.py::TestFilterByYears                 | test_empty_flows_returns_empty                        | Pass   |
|       | filter.py::TestFilterByYears                 | test_filter_1_year_keeps_only_recent                  | Pass   |
|       | filter.py::TestFilterByYears                 | test_filter_keeps_recent                              | Pass   |
|       | filter.py::TestFilterByYears                 | test_none_years_returns_all                           | Pass   |
|       | filter.py::TestFilterByYearsBound            | test_all_dates_within_range_kept                      | Pass   |
|       | filter.py::TestFilterByYearsBound            | test_zero_years_returns_only_most_recent_year         | Pass   |
|       | generate.py::TestGenerate                    | test_daily_flows_filtered_by_years                    | Pass   |
|       | generate.py::TestGenerate                    | test_explicit_gauge_id_overrides_metadata             | Pass   |
|       | generate.py::TestGenerate                    | test_output_file_written                              | Pass   |
|       | generate.py::TestGenerate                    | test_returns_dict_with_required_keys                  | Pass   |
|       | generate.py::TestGenerateUsesInstanceDefault | test_default_years_overrides_instance_default         | Pass   |
|       | generate.py::TestGenerateUsesInstanceDefault | test_instance_years_used_when_no_years_passed         | Pass   |
|       | generate.py::TestGenerateWithNoOutputPath    | test_no_output_path_none_uses_config_input_dir        | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_returns_expected_keys                            | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_for_floats_data_quality_notes                    | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_for_ordered_keys_present                         | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_for_missing_keys_always_none                     | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_for_custom_flow_keys                             | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_for_base_flow_keys_from_station_metadata         | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_for_base_flow_keys_m3_per_s                      | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_forphy_missing_quality_notes_when_no_description | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_for_missing_keys_source_returns_none             | Pass   |
|       | metadata.py::TestGetNrfaMetadata             | test_for_missing_keys_return_none                     | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_flow_values_are_floats                           | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_flows_have_date_and_flow_cumeecs                 | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_invalid_rows_skipped                             | Pass   |
|       | parse_csv.py::TestParseNrfaCsv               | test_metadata_contains_station_name                   | Pass   |

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| Model | Module                                     | Test Name                                         | Status |
|-------|--------------------------------------------|---------------------------------------------------|--------|
|       | parse_csv.py::TestParseNrfaCsv             | test_metadata_station_id                          | Pass   |
|       | parse_csv.py::TestParseNrfaCsv             | test_returns_metadata_and_flows                   | Pass   |
|       | parse_csv.py::TestParseNrfaCsv             | test_short_row_skipped                            | Pass   |
|       | parse_csv.py::TestParseNrfaCsv             | test_metadata_categories_parsed                   | Pass   |
|       | parse_csv.py::TestParseNrfaCsv             | test_metadata_with_non_numeric_value_skipped      | Pass   |
|       | parse_csv.py::TestParseNrfaCsv             | test_metadata_row_with_no_value_uses_empty_string | Pass   |
|       | parse_csv.py::TestParseNrfaCsv             | test_metadata_row_with_three_columns_stored_value | Pass   |
|       | process_directory.py::TestProcessDirectory | test_empty_directory_returns_empty_list           | Pass   |
|       | process_directory.py::TestProcessDirectory | test_nonmatching_files_skipped                    | Pass   |
|       | process_directory.py::TestProcessDirectory | test_output_dir_none_uses_input_dir               | Pass   |
|       | process_directory.py::TestProcessDirectory | test_output_files_created                         | Pass   |
|       | process_directory.py::TestProcessDirectory | test_process_directory_error_continues            | Pass   |
|       | process_directory.py::TestProcessDirectory | test_processes_matching_files                     | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_annual_maxima_present                        | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_basic_stats_keys                             | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_empty_returns_empty_dict                     | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_extreme_events_above_p99                     | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_mean_correct                                 | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_min_max_correct                              | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_monthly_means_present                        | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_percentiles_present                          | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_single_flow_no_std_dev_error                 | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_99_percentile_keys_present                   | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_99_percentile_maxima_events_capped_at_ten    | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_extreme_events_top_10_capped                 | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_flow_date_correctly_identified               | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_flow_equals_p95                              | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_flow_equals_p5                               | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_flow_start_and_end_match_data                | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_flow_year_annual_maxima_std_dev_zero         | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_flow_years_approximate                       | Pass   |
|       | statistics.py::TestCalculateStatistics     | test_flow_std_dev_non_zero                        | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_required_fields_default                  | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_info_leaf_returns_dict                   | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_info_not_found_returns_none              | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_info_partial_path_returns_none           | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_info_returns_none_for_non_dict_leaf      | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_fields_nested_metadata_keys              | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_fields_prs                               | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_fields_returns_nested_path               | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_fields_skips_metadata_keys               | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_get_coverage                                 | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_validation_returns_validation_error          | Pass   |
|       | test_block1.coverage.py::TestBaseCoverage  | test_validation_menu_value_produces_sensor_error  | Pass   |

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| Model | Module                                                                                | Test Name                                                | Status |
|-------|---------------------------------------------------------------------------------------|----------------------------------------------------------|--------|
|       | test_block1.coverage.py::TestGatewaysValidMultipleInvalidMenuValues                   | testValidMultipleInvalidMenuValues                       | Pass   |
|       | test_block1.coverage.py::TestGatewaysValidMenuValuesNoSensorErrors                    | testValidMenuValuesNoSensorErrors                        | Pass   |
|       | test_block1.coverage.py::TestMockedGDMMappingExceptionRaisesValueError                | testGDMMappingExceptionRaisesValueError                  | Pass   |
|       | test_block1.coverage.py::TestMockedGDMValidAdditionalInvalidReturnsEmpty              | testGDMValidAdditionalInvalidReturnsEmpty                | Pass   |
|       | test_block1.coverage.py::TestMockedGDMValidExceptionReturnsErrorDict                  | testGDMValidExceptionReturnsErrorDict                    | Pass   |
|       | test_block1.coverage.py::TestMockedGDMValidHeaderErrorsCollected                      | testGDMValidHeaderErrorsCollected                        | Pass   |
|       | test_block1.coverage.py::TestMockedGDMValidMissingCatchmentId                         | testGDMValidMissingCatchmentId                           | Pass   |
|       | test_block1.coverage.py::TestMockedGDMValidMissingMortgageId                          | testGDMValidMissingMortgageId                            | Pass   |
|       | test_block1.coverage.py::TestMockedGDMValidMissingOriginalLoan                        | testGDMValidMissingOriginalLoan                          | Pass   |
|       | test_block1.coverage.py::TestMockedGDMValidMissingPropertyId                          | testGDMValidMissingPropertyId                            | Pass   |
|       | test_block1.coverage.py::TestMockedGDMValidTermsErrorsCollected                       | testGDMValidTermsErrorsCollected                         | Pass   |
|       | test_block1.coverage.py::TestPRSDMCustomGaugeBasketSize                               | testPRSDMCustomGaugeBasketSize                           | Pass   |
|       | test_block1.coverage.py::TestPRSDMDefaultGaugeBasketSize                              | testPRSDMDefaultGaugeBasketSize                          | Pass   |
|       | test_block1.coverage.py::TestPRSDMMappingExceptionRaisesValueError                    | testPRSDMMappingExceptionRaisesValueError                | Pass   |
|       | test_block1.coverage.py::TestPRSDMMappingFullData                                     | testPRSDMMappingFullData                                 | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationEmptyData                                 | testPRSDMValidationEmptyData                             | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationEmptyGaugeBasket                          | testPRSDMValidationEmptyGaugeBasket                      | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationExceptionReturnsErrorDict                 | testPRSDMValidationExceptionReturnsErrorDict             | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationGaugeErrorsAppended                       | testPRSDMValidationGaugeErrorsAppended                   | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationHeaderErrorsAppended                      | testPRSDMValidationHeaderErrorsAppended                  | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationMissingCatchmentId                        | testPRSDMValidationMissingCatchmentId                    | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationMissingGaugeSetId                         | testPRSDMValidationMissingGaugeSetId                     | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationMissingSwapId                             | testPRSDMValidationMissingSwapId                         | Pass   |
|       | test_block1.coverage.py::TestPRSDMValidationValidReturnsEmpty                         | testPRSDMValidationValidReturnsEmpty                     | Pass   |
|       | test_block1.coverage.py::TestPropertyValidationReturnsValidationErrors                | testPropertyValidationReturnsValidationErrors            | Pass   |
|       | test_block1.coverage.py::TestPropertyValidationRequiredFields                         | testPropertyValidationRequiredFields                     | Pass   |
|       | test_block1.coverage.py::TestPropertyValidationErrorsCollected                        | testPropertyValidationErrorsCollected                    | Pass   |
|       | test_block1.coverage.py::TestPropertyValidationLatitude                               | testPropertyValidationLatitude                           | Pass   |
|       | test_block1.coverage.py::TestPropertyValidationLongitude                              | testPropertyValidationLongitude                          | Pass   |
|       | test_block1.coverage.py::TestPropertyValidationReturnsEmpty                           | testPropertyValidationReturnsEmpty                       | Pass   |
|       | test_block2.cdm.coverage.py::TestGaugeMappingCoverageMappingExceptionRaisesValueError | testGaugeMappingCoverageMappingExceptionRaisesValueError | Pass   |
|       | test_block2.cdm.coverage.py::TestGaugeMappingCoverageReturnsExpectedList              | testGaugeMappingCoverageReturnsExpectedList              | Pass   |
|       | test_block2.cdm.coverage.py::TestGaugeInDirectoryCoverage                             | testGaugeInDirectoryCoverage                             | Pass   |
|       | test_block2.cdm.coverage.py::TestGaugeInDirectoryCoverage                             | testGaugeInDirectoryCoverage                             | Pass   |
|       | test_block2.cdm.coverage.py::TestGaugeInDirectoryCoverageDefaultOutput                | testGaugeInDirectoryCoverageDefaultOutput                | Pass   |
|       | test_block2.cdm.coverage.py::TestGaugeInDirectoryCoverageFileErrorLogged              | testGaugeInDirectoryCoverageFileErrorLogged              | Pass   |
|       | test_block2.cdm.coverage.py::TestGaugeMappingCoverageMappingDefaultElement            | testGaugeMappingCoverageMappingDefaultElement            | Pass   |
|       | test_block2.cdm.coverage.py::TestGaugeMappingCoverageMappingExceptionRaisesValueError | testGaugeMappingCoverageMappingExceptionRaisesValueError | Pass   |
|       | test_ctpy_coverage_b6.py::TestValidAllocationExceptionMessageIsString                 | testValidAllocationExceptionMessageIsString              | Pass   |
|       | test_ctpy_coverage_b6.py::TestValidAllocationWithIntegerReturnsValidationErrors       | testValidAllocationWithIntegerReturnsValidationErrors    | Pass   |
|       | test_ctpy_coverage_b6.py::TestValidAllocationWithNonDictReturnsValidationErrors       | testValidAllocationWithNonDictReturnsValidationErrors    | Pass   |
|       | counterparty_generator.py::TestCounterpartyGeneratorOutputFile                        | testCounterpartyGeneratorOutputFile                      | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator                                  | testCounterpartyGenerator                                | Pass   |
|       | counterparty_generator.py::TestCounterpartyGeneratorUsesAllCounterparties             | testCounterpartyGeneratorUsesAllCounterparties           | Pass   |

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| Model | Module                                               | Test Name                                                   | Status |
|-------|------------------------------------------------------|-------------------------------------------------------------|--------|
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.pathlib_path                      | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.has_total_generated               | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.is_valid_json                     | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.dict_with_required_keys           | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.raises_error_when_returns_empty   | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.raises_error_when_raises_function | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.raises_error_when_raises_function | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.has_csa                           | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.parties_list_has_two_parties      | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.has_counterparty_set              | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.agreement_is_true                 | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.accounts                          | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.contact_info                      | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.uid                               | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.name                              | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.natural_person                    | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.has_credit_rating                 | Pass   |
|       | counterparty_generator.py::TestCounterpartyGenerator | TestCounterpartyGenerator.party_ids                         | Pass   |
|       | test_counterparty_coverage_b6.py::TestVerboseLogging | TestVerboseLogging.no_per_counterparty_logs                 | Pass   |
|       | test_counterparty_coverage_b6.py::TestVerboseLogging | TestVerboseLogging.logs_per_counterparty                    | Pass   |
|       | test_counterparty_coverage_b6.py::TestVerboseLogging | TestVerboseLogging.logs_summary                             | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.chronological_order                        | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.ends_before_start                          | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.large_count                                | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.last_day_is_day_before_start               | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.no_weekends                                | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.raises_error_when_returns_correct_count    | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.single_day                                 | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.start_on_monday_skips_weekend              | Pass   |
|       | business_days.py::TestBusinessDays                   | TestBusinessDays.start_on_wednesday                         | Pass   |
|       | business_days.py::TestSimulation                     | TestSimulation.business_days_is_three_months                | Pass   |
|       | business_days.py::TestSimulation                     | TestSimulation.daily_vol_is_small                           | Pass   |
|       | business_days.py::TestSimulation                     | TestSimulation.floats_is_positive                           | Pass   |
|       | business_days.py::TestSimulation                     | TestSimulation.move_at_least_ten_daily_vols                 | Pass   |
|       | business_days.py::TestSimulation                     | TestSimulation.move_bounds_vol                              | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.produces_no_file                   | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.have_date                          | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.have_tenors                        | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.fill_by_gauge                      | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.fill_by_trigger                    | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.generated_at                       | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.snapshots                          | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.gauges_in_metadata                 | Pass   |
|       | history_files.py::TestGenerateHistoryFiles           | TestGenerateHistoryFiles.snapshots_produce_multiple_entries | Pass   |

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| Model | Module                                   | Test Name                                      | Status |
|-------|------------------------------------------|------------------------------------------------|--------|
|       | history_files.py::TestGenerateHaze       | testCourtHistoryFile metadata and history      | Pass   |
|       | history_files.py::TestGenerateHaze       | testSingleHazeSnapshot creates file            | Pass   |
|       | history_files.py::TestGenerateTrade      | testHistoryFile produces no file               | Pass   |
|       | history_files.py::TestGenerateTrade      | testHistoryFile entry structure                | Pass   |
|       | history_files.py::TestGenerateTrade      | testHazeFile has generated at                  | Pass   |
|       | history_files.py::TestGenerateTrade      | testHazeFile num snapshots                     | Pass   |
|       | history_files.py::TestGenerateTrade      | testHistoryFile days builds history vector     | Pass   |
|       | history_files.py::TestGenerateTrade      | testHistoryFile trades in snapshot             | Pass   |
|       | history_files.py::TestGenerateTrade      | testHistoryFile metadata and trades            | Pass   |
|       | history_files.py::TestGenerateTrade      | testHistoryFile missing swap id is skipped     | Pass   |
|       | history_files.py::TestGenerateTrade      | testHingeFile snapshot creates file            | Pass   |
|       | history_files.py::TestGenerateTrade      | testHistoryFile keyed by swap id               | Pass   |
|       | history_files.py::TestGenerateTrade      | testHistoryFile metadata fields                | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testDefault simulation hours is 168            | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testGenerate default argument is 168           | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testGenerate gaugets function default          | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testPeak hour max within simulation window     | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testPeak hour min less than max                | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testGenerate produces more readings than 60    | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testConfigure known param                      | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testConfigure unknown param ignored            | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testExplicit 168 hours produces 168 readings   | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testExplicit 60 hours stores 60 in meta        | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testMissing gauge file raises                  | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testSimulation hours stored in output          | Pass   |
|       | gaugets.py::TestGaugeTimeSeries          | testStale gauge files removed                  | Pass   |
|       | gaugets.py::TestGaugetsDateTime          | testEncoder datetime                           | Pass   |
|       | gaugets.py::TestGaugetsDateTime          | testEncoder numpy float32                      | Pass   |
|       | gaugets.py::TestGaugetsDateTime          | testEncoder numpy integer                      | Pass   |
|       | gaugets.py::TestGaugetsDateTime          | testEncoder numpy ndarray                      | Pass   |
|       | gaugets.py::TestGaugetsDateTime          | testEncoder raises for unknown                 | Pass   |
|       | gaugets.py::TestGaugetsMain              | test_main_block                                | Pass   |
|       | gaugets.py::TestGaugetsRandom            | testRandom default peak hour max within window | Pass   |
|       | gaugets.py::TestGaugetsRandom            | testRandom default simulation hours is 168     | Pass   |
|       | gaugets.py::TestGenerateGaugets          | testGenerate gaugets calls generator           | Pass   |
|       | test_gauge_random_coverage.py::testField | testField returns empty string                 | Pass   |
|       | test_gauge_random_coverage.py::testField | testField returns empty string                 | Pass   |
|       | test_gauge_random_coverage.py::testField | testField returned as is                       | Pass   |
|       | test_gauge_random_coverage.py::testField | testField default is text                      | Pass   |
|       | test_gauge_random_coverage.py::testField | testField type returns empty                   | Pass   |
|       | test_gauge_random_coverage.py::testField | testField fails back to area                   | Pass   |
|       | test_gauge_random_coverage.py::testField | testField fails back to area                   | Pass   |
|       | test_gauge_random_coverage.py::testField | testField integer field returns int            | Pass   |
|       | test_gauge_random_coverage.py::testField | testField default returns empty                | Pass   |

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| Model | Module                                                                       | Test Name                                              | Status |
|-------|------------------------------------------------------------------------------|--------------------------------------------------------|--------|
|       | test_gaugehd_synthetic.py::TestGenerateFromConfigDefault                     | test_generate_from_config_default                      | Pass   |
|       | test_gaugehd_synthetic.py::TestGenerateFromGaugePortfolio                    | test_generate_from_gauge_portfolio                     | Pass   |
|       | test_gaugehd_synthetic.py::TestGenerateFromGaugePortfolio                    | test_generate_from_gauge_portfolio                     | Pass   |
|       | test_gaugets_random_coverage.py::TestGenerateNonDefaultParams                | test_generate_non_default_params                       | Pass   |
|       | test_gaugets_random_coverage.py::TestGenerateParamsLevelDefaultParams        | test_generate_params_level_default_params              | Pass   |
|       | test_gaugets_random_coverage.py::TestDefaultParamsDefaults_data              | test_default_params_defaults_data                      | Pass   |
|       | test_gaugets_random_coverage.py::TestFromInstallationStringParsedTime        | test_from_installation_string_parsed_time              | Pass   |
|       | test_gaugets_random_coverage.py::TestGaugerReadingReadingsDefaultFields      | test_gauger_reading_readings_default_fields            | Pass   |
|       | test_gaugets_random_coverage.py::TestGaugerReadingReadingsDefaultTimestamp   | test_gauger_reading_readings_default_timestamp         | Pass   |
|       | test_gaugets_random_coverage.py::TestClearReturnsSeriesAsInStructure         | test_clear_returns_series_as_in_structure              | Pass   |
|       | test_gaugets_random_coverage.py::TestRemovingZeroingZero_base_amplitude      | test_removing_zeroing_zero_base_amplitude              | Pass   |
|       | test_gaugets_random_coverage.py::TestEmptyGaugeReadingsSkipped               | test_empty_gauge_readings_skipped                      | Pass   |
|       | test_nrfa_parsing.py::TestFilterByYear5_year_filter                          | test_filter_by_year_5_year_filter                      | Pass   |
|       | test_nrfa_parsing.py::TestFilterByYearempty_list_returns_empty               | test_filter_by_year_empty_list_returns_empty           | Pass   |
|       | test_nrfa_parsing.py::TestFilterByYearlarge_years_returns_all                | test_filter_by_year_large_years_returns_all            | Pass   |
|       | test_nrfa_parsing.py::TestFilterByYearone_years_returns_all                  | test_filter_by_year_one_years_returns_all              | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromOutputPath                             | test_generate_from_output_path                         | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromNrfasJsonFile                          | test_generate_from_nrfas_json_file                     | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromNrfasIdOverride                        | test_generate_from_nrfas_id_override                   | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromReturnsDict                            | test_generate_from_returns_dict                        | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromSchemaVersionInOutput                  | test_generate_from_schema_version_in_output            | Pass   |
|       | test_nrfa_parsing.py::TestGenerateFromStationMetadataExtracted               | test_generate_from_station_metadata_extracted          | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfasEmptyFile                                | test_parse_nrfas_empty_file                            | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfasFlowsHaveDateAndFlow                     | test_parse_nrfas_flows_have_date_and_flow              | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfasMetadataContainsStationId                | test_parse_nrfas_metadata_contains_station_id          | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfasMetadataContainsStationName              | test_parse_nrfas_metadata_contains_station_name        | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfasReturnsTuple                             | test_parse_nrfas_returns_tuple                         | Pass   |
|       | test_nrfa_parsing.py::TestParseNrfasSkipsShortRows                           | test_parse_nrfas_skips_short_rows                      | Pass   |
|       | test_stress_storms_core.py::TestGenerateIdPresentInAllResponses              | test_generate_id_present_in_all_responses              | Pass   |
|       | test_stress_storms_core.py::TestGenerateMultipleGaugesHaveCorrectIds         | test_generate_multiple_gauges_have_correct_ids         | Pass   |
|       | test_stress_storms_core.py::TestGenerateStormsFoundWhenGaugetsEmpty          | test_generate_storms_found_when_gaugets_empty          | Pass   |
|       | test_stress_storms_core.py::TestGenerateIndexHasGaugeIdsAlert                | test_generate_index_has_gauge_ids_alert                | Pass   |
|       | test_stress_storms_core.py::TestGenerateIndexHasStormsRequiredTopLevelKeys   | test_generate_index_has_storms_required_top_level_keys | Pass   |
|       | test_stress_storms_core.py::TestGenerateStormsOfFileHasGaugeResponses        | test_generate_storms_of_file_has_gauge_responses       | Pass   |
|       | test_stress_storms_core.py::TestGenerateStormsPrefixUsesConfigConstant       | test_generate_storms_prefix_uses_config_constant       | Pass   |
|       | test_stress_storms_core.py::TestGenerateSubIndexStormsExcluded               | test_generate_sub_index_storms_excluded                | Pass   |
|       | test_stress_storms_core.py::TestGenerateSubIndexStormsMatchesListLength      | test_generate_sub_index_storms_matches_list_length     | Pass   |
|       | test_stress_storms_core.py::TestGenerateStressIndexSandStormFiles            | test_generate_stress_index_sand_storm_files            | Pass   |
|       | test_stress_storms_core.py::TestReturnReturnsDictWithTotalStorms             | test_return_returns_dict_with_total_storms             | Pass   |
|       | test_stress_storms_core.py::TestReturnReturnsElapsedS                        | test_return_returns_elapsed_s                          | Pass   |
|       | test_stress_storms_coverage.py::TestCorruptGaugeFileNoStorms                 | test_corrupt_gauge_file_no_storms                      | Pass   |
|       | test_stress_storms_coverage.py::TestCorruptGaugeFileFileSkipped              | test_corrupt_gauge_file_file_skipped                   | Pass   |
|       | test_stress_storms_coverage.py::TestCorruptStormsJsonDoesNotCrash            | test_corrupt_storms_json_does_not_crash                | Pass   |
|       | test_stress_storms_coverage.py::TestDenegatesByCategoryWarningReturnBaseline | test_denegates_by_category_warning_return_baseline     | Pass   |

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| Model | Module                                                 | Test Name                                           | Status |
|-------|--------------------------------------------------------|-----------------------------------------------------|--------|
|       | hazard.io.py::TestSaveHazardC                          | test_metadata_merged                                | Pass   |
|       | business_days.py::TestBusinessDays                     | test_all_days_before_start                          | Pass   |
|       | business_days.py::TestBusinessDays                     | test_returns_ascending_order                        | Pass   |
|       | business_days.py::TestBusinessDays                     | test_returns_business_days_only                     | Pass   |
|       | business_days.py::TestBusinessDays                     | test_returns_correct_count                          | Pass   |
|       | business_days.py::TestBusinessDays                     | test_single_day                                     | Pass   |
|       | business_days.py::TestBusinessDays                     | test_skips_weekends                                 | Pass   |
|       | business_days.py::TestBusinessDays                     | test_zero_days                                      | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_days_are_weekdays                    | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_existing_eod_files                   | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_eod_dir                              | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_trades_returns_zero                  | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_int_failure_is_skipped               | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_trades_adds_more_snapshot            | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_integer                              | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_reproducible                         | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_trade_generates_snapshots            | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_trade_writes_eod_files               | Pass   |
|       | eod_series.py::TestGenerateHistoricalEODSeries         | test_EODSeries_trade_writes_history_files           | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_generate_history_output_file                   | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_HazardHistory_history_has_date                 | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_HazardHistory_history_gauges                   | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_HazardHistory_multiple_gauges                  | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_HazardHistory_returns_early                    | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_HazardHistory_has_history                      | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_HazardHistory_has_metadata                     | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_HazardHistory_read_numerically                 | Pass   |
|       | hazard_history.py::TestGenerateHistoricalHazardHistory | test_HazardHistory_has_two_history_entries          | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_output_file                         | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_entry_has_required_keys             | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_multiple_eods_accumulate            | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_multiple_trades_in_same_eod         | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_returns_early                       | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_snapshots_in_metadata               | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_has_metadata                        | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_has_trades                          | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_position_without_swap_id_skipped    | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_has_history                         | Pass   |
|       | pnl_history.py::TestGenerateTradePnLHistory            | test_PnLHistory_metadata_preserved                  | Pass   |
|       | test_mortgage_builder.py::TestBuildEmptySchema         | test_empty_schema_returns_empty_dict                | Pass   |
|       | test_mortgage_builder.py::TestBuildEmptySchema         | test_empty_dict_without_type_recurses               | Pass   |
|       | test_mortgage_builder.py::TestBuildEmptySchema         | test_empty_dict_schema_returns_empty_dict           | Pass   |
|       | test_mortgage_builder.py::TestBuildEmptySchema         | test_empty_schema_returns_empty_dict                | Pass   |
|       | test_mortgage_builder.py::TestBuildEmptySchema         | test_empty_value_from_generate_field_value_excluded | Pass   |

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| Model | Module                                        | Test Name                                                 | Status |
|-------|-----------------------------------------------|-----------------------------------------------------------|--------|
|       | integration.py::TestEndToEndPipeline          | test_inproperty_count_propagates                          | Pass   |
|       | property_energy.py::TestCalculateAnnualEnergy | test_annual_energy_gives_larger_energy                    | Pass   |
|       | property_energy.py::TestCalculateAnnualEnergy | test_annual_energy_positive_value                         | Pass   |
|       | property_energy.py::TestCalculateAnnualEnergy | test_annual_energy_default_area_when_missing              | Pass   |
|       | property_energy.py::TestCalculateAnnualEnergy | test_annual_energy_floats                                 | Pass   |
|       | property_energy.py::TestCalculateAnnualEnergy | test_annual_energy_area_tends_to_give_larger_emissions    | Pass   |
|       | property_energy.py::TestCalculateAnnualEnergy | test_annual_energy_positive_value                         | Pass   |
|       | property_energy.py::TestCalculateAnnualEnergy | test_annual_energy_default_area_when_missing              | Pass   |
|       | property_energy.py::TestCalculateEnergyBill   | test_energy_bill_calculation                              | Pass   |
|       | property_energy.py::TestCalculateEnergyBill   | test_energy_bill                                          | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_positive_value                          | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_defaults_when_missing                   | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_age_returns_zero                        | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_less_than_remaining_energy              | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_negative_when_electricity_exceeds_total | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_returns_non_negative                    | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_defaults_when_missing                   | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_fraction_of_total                       | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_positive_value                          | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_default_energy_when_missing             | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_calculation_returns_zero                | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_calculation_returns_positive            | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_calculation_returns_zero                | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_calculation_returns_zero                | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_calculation_returns_positive            | Pass   |
|       | property_energy.py::TestCalculateEnergyBills  | test_energy_bills_calculation_default_area_when_missing   | Pass   |
|       | test_geometry.py::TestEnsureOffset            | test_close_point_pushed_away                              | Pass   |
|       | test_geometry.py::TestEnsureOffset            | test_far_point_unchanged                                  | Pass   |
|       | test_geometry.py::TestEnsureOffset            | test_returns_tuple_of_two_floats                          | Pass   |
|       | test_geometry.py::TestHaversine               | test_approximate_one_degree_lat                           | Pass   |
|       | test_geometry.py::TestHaversine               | test_approximate_one_degree_lon_at_equator                | Pass   |
|       | test_geometry.py::TestHaversine               | test_same_point_returns_zero                              | Pass   |
|       | test_geometry.py::TestHaversine               | test_symmetry                                             | Pass   |
|       | test_geometry.py::TestMinRiverDistance        | test_constant_point_returns_larger_distance               | Pass   |
|       | test_geometry.py::TestMinRiverDistance        | test_point_on_gauge_returns_small_distance                | Pass   |
|       | test_geometry.py::TestMinRiverDistance        | test_returns_float                                        | Pass   |
|       | test_geometry.py::TestNearestOnSegment        | test_generate_segment_returns_start                       | Pass   |
|       | test_geometry.py::TestNearestOnSegment        | test_distance_is_non_negative                             | Pass   |
|       | test_geometry.py::TestNearestOnSegment        | test_midpoint_projects_to_midpoint                        | Pass   |
|       | test_geometry.py::TestNearestOnSegment        | test_point_before_start_clamps_to_start                   | Pass   |
|       | test_geometry.py::TestNearestOnSegment        | test_point_past_end_clamps_to_end                         | Pass   |
|       | test_geometry.py::TestPerpendicularOffset     | test_generate_segment_falls_back_to_parallel_offset       | Pass   |
|       | test_geometry.py::TestPerpendicularOffset     | test_offset_moves_point                                   | Pass   |
|       | test_geometry.py::TestPerpendicularOffset     | test_offset_returns_two_floats                            | Pass   |

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| Model | Module                                                       | Test Name                                                  | Status |
|-------|--------------------------------------------------------------|------------------------------------------------------------|--------|
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_boundary_values                     | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_boundary_value                      | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_low_value                           | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_high_value                          | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_range_value                         | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_upper_value                         | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_lower_value                         | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_house_paths_1_or_2                  | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_house_paths_2_or_3                  | Pass   |
|       | test_location_coverage.b6.py::TestGenerateLocation           | test_generate_location_house_paths_2_or_3                  | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_each_location_has_required_keys                       | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_elevation_is_non_negative                             | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_fallback_used_when_fewer_than_3_gauge_points          | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_fallback_used_when_no_gauge_points                    | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_primary_idx_zero_uses_gauges_012                      | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_reference_gauge_indices_sorted                        | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_reference_gauge_indices_three_elements                | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_returns_correct_count                                 | Pass   |
|       | test_locations.py::TestGenerateLocation                      | test_fallback_no_floating_elevation_use_random             | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorm                | test_cluster_max_depth_is_from_highest Storm               | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorm                | test_cluster_produces_three_flood_event                    | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorm                | test_cluster_three_distinct_storm_ids_in_events            | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorm                | test_distinct_storms_create_flood_event                    | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorm                | test_distinct_storms_not_in_flood_event                    | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorm                | test_persistent_sequential_events_have_reading             | Pass   |
|       | test_multi_storm_cluster.py::TestClusterStorm                | test_persistent_sequential_events_recorded                 | Pass   |
|       | test_multi_storm_doublet.py::TestDoubletStorm                | test_doublet_each_event_has_independent_damage             | Pass   |
|       | test_multi_storm_doublet.py::TestDoubletStorm                | test_doublet_events_have_distinct_storm_ids                | Pass   |
|       | test_multi_storm_doublet.py::TestDoubletStorm                | test_doublet_produces_two_flood_events                     | Pass   |
|       | test_multi_storm_doublet.py::TestDoubletStorm                | test_doublet_queries_all_doublet_floods                    | Pass   |
|       | test_multi_storm_portfolio.py::TestGeneratePortfolio         | test_generate_portfolio_with_storms_generate_zero_floods   | Pass   |
|       | test_multi_storm_portfolio.py::TestGeneratePortfolio         | test_generate_portfolio_with_storms_generate_all_sequences | Pass   |
|       | test_multi_storm_portfolio.py::TestGeneratePortfolio         | test_generate_portfolio_with_storms_generate_for_cluster   | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_generate_random_coverage_exception_falls_back_type    | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_integer_type_returns_int                              | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_float_type_returns_float                              | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_string_with_options_picks_from_list                   | Pass   |
|       | test_property_random_coverage.py::TestGenerateRandomCoverage | test_string_with_options_returns_empty                     | Pass   |
|       | test_property_values.py::TestGauging                         | test_gauging_gauge_json_does_not_crash                     | Pass   |
|       | test_property_values.py::TestGauging                         | test_gauging_map_empty_when_no_gauge_json                  | Pass   |
|       | test_property_values.py::TestGauging                         | test_gauging_map_loaded_when_gauge_json_present            | Pass   |
|       | test_property_values.py::TestGauging                         | test_gauging_map_works_with_gauges_key                     | Pass   |
|       | test_property_values.py::TestLogging                         | test_debug_does_not_raise                                  | Pass   |
|       | test_property_values.py::TestLogging                         | test_error_does_not_raise                                  | Pass   |
|       | test_property_values.py::TestLogging                         | test_info_does_not_raise                                   | Pass   |
|       | test_property_values.py::TestLogging                         | test_unhandled_access_falls_back_gracefully                | Pass   |

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| Model | Module                                                        | Test Name                                     | Status |
|-------|---------------------------------------------------------------|-----------------------------------------------|--------|
|       | test_property_values.py::TestLogMethod                        | warning does not raise                        | Pass   |
|       | test_property_values.py::TestProtectingStatefulProperties     | is tracked                                    | Pass   |
|       | test_property_values.py::TestPropertyOutputFileStructure      | contains generation metadata                  | Pass   |
|       | test_property_values.py::TestPropertyOutputFileStructure      | contains properties key                       | Pass   |
|       | test_property_values.py::TestPropertyOutputFileStructure      | has property_id                               | Pass   |
|       | test_property_values.py::TestPropertyOutputFileStructure      | contains locations                            | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | in attrs                                      | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | sets ground_level                             | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | reference gauge indices no ref key            | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | location                                      | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | location does not set ground_level            | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | reference gauge indices no ref key            | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | in attrs                                      | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | fallback when not in map                      | Pass   |
|       | test_property_values.py::TestSetSpecificAttribute             | set from gauge_id map                         | Pass   |
|       | test_propertyhc_generate.py::TestGenerateAggregates           | is float                                      | Pass   |
|       | test_propertyhc_generate.py::TestGenerateAggregates           | rate between 0 and 1                          | Pass   |
|       | test_propertyhc_generate.py::TestGenerateAggregates           | stats                                         | Pass   |
|       | test_propertyhc_generate.py::TestGenerateAggregates           | events counted                                | Pass   |
|       | test_propertyhc_generate.py::TestLogMethod                    | does not raise                                | Pass   |
|       | test_propertyhc_generate.py::TestLogMethod                    | with empty string                             | Pass   |
|       | test_propertyhc_generate.py::TestTestSurvivalModel            | spread not less than seven                    | Pass   |
|       | test_propertyhc_generate.py::TestTestSurvivalModel            | increases with longer tenor or non-decreasing | Pass   |
|       | test_propertyhc_generate.py::TestTestSurvivalModel            | between zero and one                          | Pass   |
|       | test_propertyhc_helpers.py::TestGetMissingSpreads             | defaults to zero rate                         | Pass   |
|       | test_propertyhc_helpers.py::TestGetMissingSpreads             | returns all zeros                             | Pass   |
|       | test_propertyhc_helpers.py::TestGetMissingSpreads             | returns positive spreads                      | Pass   |
|       | test_propertyhc_helpers.py::TestGetMissingSpreads             | has one entry per tenor                       | Pass   |
|       | test_propertyhc_helpers.py::TestGetZeroRate                   | returns all zeros                             | Pass   |
|       | test_propertyhc_helpers.py::TestGetP                          | returns none when quantlib unavailable        | Pass   |
|       | test_propertyhc_helpers.py::TestTestDefault                   | time encoded as isoformat                     | Pass   |
|       | test_propertyhc_helpers.py::TestTestDefault                   | float32 encoded                               | Pass   |
|       | test_propertyhc_helpers.py::TestTestDefault                   | float encoded                                 | Pass   |
|       | test_propertyhc_helpers.py::TestTestDefault                   | int32 encoded                                 | Pass   |
|       | test_propertyhc_helpers.py::TestTestDefault                   | int encoded                                   | Pass   |
|       | test_propertyhc_helpers.py::TestTestDefault                   | ndarray encoded as list                       | Pass   |
|       | test_propertyhc_helpers.py::TestTestDefault                   | unknown type raises type_error                | Pass   |
|       | test_propertyhc_helpers.py::TestLoadMissingHazardCurves       | returns empty and default storms              | Pass   |
|       | test_propertyhc_helpers.py::TestLoadMissingHazardCurves       | returns curves                                | Pass   |
|       | test_propertyhc_helpers.py::TestLoadMissingHazardCurves       | returns num_storms                            | Pass   |
|       | test_propertyhc_helpers.py::TestTestTangshibbbForecastTrigger |                                               | Pass   |
|       | test_propertyhc_helpers.py::TestTestModerateToSevereTrigger   |                                               | Pass   |
|       | test_propertyhc_helpers.py::TestTestSevereToExtremeTrigger    |                                               | Pass   |
|       | test_propertyhc_helpers.py::TestTestUnknownKey                | returns warning                               | Pass   |

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| Model                            | Module                    | Test Name                                                  | Status |
|----------------------------------|---------------------------|------------------------------------------------------------|--------|
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseIncreasesPeak                     | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseDeclining                         | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseHigherLevel                       | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseSameFirstStorm                    | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseDuringStorm                       | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseReversion                         | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseWindow                            | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseShape                             | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseBaselineByHour168                 | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponsePeakAboveBase                     | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponsePeakMatchesStormIndices           | Pass   |
| sequence_response.py::TestCompu  | testSequenceGaugeResponse | testSequenceGaugeResponseOnePeakPerStorm                   | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationStep                                    | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationInitialStateAffectsOutput               | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationQuickflow                               | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationPositive                                | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationRunSeries                               | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationDecaysWithoutRain                       | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationNoPrecipNoQuickflow                     | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationWaterAccumulatesFromSoil                | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationPrecipTreatedAsZero                     | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationIncreasesSoilMoisture                   | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationProducesQuickflow                       | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationSoilMoistureDrainsWithoutRain           | Pass   |
| sequence_response.py::TestHydro  | testModelSimulation       | testModelSimulationRaisesQuickflow                         | Pass   |
| sequence_response.py::TestMake   | testPrecipitation         | testPrecipitationIsolatedSingleStormPeak                   | Pass   |
| sequence_response.py::TestMake   | testPrecipitation         | testPrecipitationOutsideStorms                             | Pass   |
| sequence_response.py::TestMake   | testPrecipitation         | testPrecipitationNegative                                  | Pass   |
| sequence_response.py::TestMake   | testPrecipitation         | testPrecipitationShape                                     | Pass   |
| sequence_response.py::TestMake   | testPrecipitation         | testPrecipitationPrecipConserved                           | Pass   |
| storm_intensity.py::TestGetStorm | testStormIntensity        | testStormIntensityClusterReturns3or4                       | Pass   |
| storm_intensity.py::TestGetStorm | testStormIntensity        | testStormIntensityDoubletReturns2                          | Pass   |
| storm_intensity.py::TestGetStorm | testStormIntensity        | testStormIntensityIsolatedReturns1                         | Pass   |
| storm_intensity.py::TestGetStorm | testStormIntensity        | testStormIntensityPersistentReturns4or5                    | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityCatastrophicTendsTowardClusterPersistent | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityCategories[catastrophic]                 | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityCategories[extreme]                      | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityCategories[moderate]                     | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityCategories[severe]                       | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensitySequenceType                             | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityUnknownCategoryUsesDefaultWeight         | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityIncreasingPathSubsequentStorm            | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityStormDownwardVariation                   | Pass   |
| storm_intensity.py::TestSample   | testStormIntensity        | testStormIntensityStormReturnsPositive                     | Pass   |

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| Model | Module                                                 | Test Name                                           | Status |
|-------|--------------------------------------------------------|-----------------------------------------------------|--------|
|       | storm_intensity.py::TestSampleStormIntensity           | test_intensity_within_20_percent_range              | Pass   |
|       | storm_intensity.py::TestSampleStormIntensity           | test_subsequent_storm_mostly_up                     | Pass   |
|       | storm_intensity.py::TestSampleStormIntensity           | test_subsequent_storm_positive                      | Pass   |
|       | storm_intensity.py::TestSampleStormIntensity           | test_intensity_low_base_clamped_to_01               | Pass   |
|       | storm_intensity.py::TestShouldGenerateSequences        | test_at_least_defined_categories                    | Pass   |
|       | storm_intensity.py::TestShouldGenerateSequences        | test_at_least_catastrophic_high_probability         | Pass   |
|       | storm_intensity.py::TestShouldGenerateSequences        | test_at_least_mild_low_probability                  | Pass   |
|       | storm_intensity.py::TestShouldGenerateSequences        | test_at_least_storms_cool                           | Pass   |
|       | storm_intensity.py::TestShouldGenerateSequences        | test_at_least_cooler_category_defaults_to_015       | Pass   |
|       | test_classifier.py::TestPortfolioClassifier            | test_portfolio_classifier_summary_unchanged         | Pass   |
|       | test_classifier.py::TestPortfolioClassifier            | test_portfolio_classifier_trains_all_gauges         | Pass   |
|       | test_classifier.py::TestPrintClassifierReady           | test_ready_trained_no_metrics_printed               | Pass   |
|       | test_classifier.py::TestPrintClassifierReady           | test_ready_trained_status                           | Pass   |
|       | test_classifier.py::TestPrintClassifierReady           | test_ready_trained_shows_feature_importance         | Pass   |
|       | test_classifier.py::TestPrintClassifierReady           | test_ready_trained_shows_label_info                 | Pass   |
|       | test_classifier.py::TestPrintClassifierReady           | test_ready_trained_status_prints_metrics            | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression          | test_gauge_id_in_output                             | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression          | test_gauge_id_in_output_sequence_types_reported     | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression          | test_gauge_id_in_output_category_skipped            | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression          | test_gauge_id_in_output_contains_gauge_id           | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression          | test_gauge_id_in_output_severe_marker_in_output     | Pass   |
|       | test_classifier.py::TestPrintGaugeProgression          | test_gauge_id_in_output_warning_marker_in_output    | Pass   |
|       | test_classifier.py::TestTrainGaugeSeveritiesClassifier | test_fail_classifier_to_alert_when_no_severe_events | Pass   |
|       | test_classifier.py::TestTrainGaugeSeveritiesClassifier | test_model_classifier_written                       | Pass   |
|       | test_classifier.py::TestTrainGaugeSeveritiesClassifier | test_gauges_classifier_gauge_id                     | Pass   |
|       | test_classifier.py::TestTrainGaugeSeveritiesClassifier | test_gauges_classifier_label_threshold              | Pass   |
|       | test_classifier.py::TestTrainGaugeSeveritiesClassifier | test_gauges_classifier_result_dict                  | Pass   |
|       | test_classifier.py::TestTrainGaugeSeveritiesClassifier | test_gauges_classifier_branch                       | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_gauge_ids                     | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_json_written                  | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_num_gauges                    | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_schema_version                | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_sequence_count                | Pass   |
|       | test_generate.py::TestGenerateSequences                | test_generate_sequences_file_loadable               | Pass   |
|       | test_generate.py::TestGenerateSequences                | test_generate_sequences_json_written                | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_json_written                  | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_count_non_negative            | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_positive                      | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_gauges                        | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_sequences                     | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_dict                          | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_severe_alert_warning          | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_severe_alerts_present         | Pass   |
|       | test_generate.py::TestGenerateSummary                  | test_generate_summary_severe_alert                  | Pass   |

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| Model | Module                                 | Test Name                                            | Status |
|-------|----------------------------------------|------------------------------------------------------|--------|
|       | test_generate.py::TestMissingGauges    | test_no_crash_when_gauge_json_absent                 | Pass   |
|       | test_generate.py::TestMissingGauges    | test_num_gauges_zero_without_gauge_json              | Pass   |
|       | test_generate.py::TestMissingGauges    | test_sequences_still_written_without_gauge_json      | Pass   |
|       | test_generate.py::TestNoValidGauges    | test_returns_gracefully_when_all_gauges_invalid      | Pass   |
|       | test_generate.py::TestNoValidGauges    | test_sequences_still_written_when_all_gauges_invalid | Pass   |
|       | test_generate.py::TestVerboseProgress  | test_verbose_false_no_mid_loop_output_at_small_count | Pass   |
|       | test_generate.py::TestVerboseProgress  | test_verbose_run_completes                           | Pass   |
|       | test_helpers.py::TestBuildSummary      | test_counts_default_to_zero                          | Pass   |
|       | test_helpers.py::TestBuildSummary      | test_num_gauges                                      | Pass   |
|       | test_helpers.py::TestBuildSummary      | test_num_sequences                                   | Pass   |
|       | test_helpers.py::TestBuildSummary      | test_returns_required_keys                           | Pass   |
|       | test_helpers.py::TestExtractGauges     | test_dict_format                                     | Pass   |
|       | test_helpers.py::TestExtractGauges     | test_empty_list                                      | Pass   |
|       | test_helpers.py::TestExtractGauges     | test_list_format                                     | Pass   |
|       | test_helpers.py::TestExtractGauges     | test_missing_key_returns_empty                       | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_base_level_derived_from_alert                   | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_flood_alert                                     | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_flood_warning                                   | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_gauge_id                                        | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_lat_lon                                         | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_missing_alert_returns_none                      | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_missing_gauge_id_returns_none                   | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_missing_lat_returns_none                        | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_severe_warning                                  | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_valid_record_returns_dict                       | Pass   |
|       | test_helpers.py::TestParseGaugetest    | test_warning_defaults_when_absent                    | Pass   |
|       | test_records.py::TestIntensityCategory | test_batch_generator_populates_category              | Pass   |
|       | test_records.py::TestIntensityCategory | test_from_dict_missing_category_defaults_to_empty    | Pass   |
|       | test_records.py::TestIntensityCategory | test_from_dict_pound_trip                            | Pass   |
|       | test_records.py::TestIntensityCategory | test_from_dict_periad_trip_sequences_have_category   | Pass   |
|       | test_records.py::TestIntensityCategory | test_from_dict_trip_includes_intensity_category      | Pass   |
|       | test_records.py::TestSequenceRecord    | test_alert_flags_length_matches_num_gauges           | Pass   |
|       | test_records.py::TestSequenceRecord    | test_all_have_intensity_category                     | Pass   |
|       | test_records.py::TestSequenceRecord    | test_all_have_sequence_id                            | Pass   |
|       | test_records.py::TestSequenceRecord    | test_all_have_sequence_type                          | Pass   |
|       | test_records.py::TestSequenceRecord    | test_all_have_total_precip                           | Pass   |
|       | test_records.py::TestSequenceRecord    | test_peak_hours_in_valid_range                       | Pass   |
|       | test_records.py::TestSequenceRecord    | test_peak_hours_length_matches_num_gauges            | Pass   |
|       | test_records.py::TestSequenceRecord    | test_peak_hours_length_matches_num_gauges            | Pass   |
|       | test_records.py::TestSequenceRecord    | test_peak_hours_positive                             | Pass   |
|       | test_records.py::TestSequenceRecord    | test_severe_flags_length_matches_num_gauges          | Pass   |
|       | test_records.py::TestSequenceRecord    | test_severe_implies_warning                          | Pass   |
|       | test_records.py::TestSequenceRecord    | test_starting_flags_length_matches_num_gauges        | Pass   |
|       | test_records.py::TestSequenceRecord    | test_starting_implies_alert                          | Pass   |

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| Model | Module                                                       | Test Name                                                | Status |
|-------|--------------------------------------------------------------|----------------------------------------------------------|--------|
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_message_lists_available_ids              | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_gauge_id_raises                  | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_file_gauge_ids                   | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_file_peaks_length_one            | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_file_schema_version              | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_file_sequence_count              | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_output_file_written              | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_primary_num_gauges_is_one        | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_classifier_result_dict           | Pass   |
|       | test_single_gauge.py::TestSingleGauge                        | TestSingleGauge_invalid_train_classifier                 | Pass   |
|       | test_spatial_correlation_matrix.py::TestCholesky             | TestCholesky_cache_key_rounding                          | Pass   |
|       | test_spatial_correlation_matrix.py::TestCholesky             | TestCholesky_cached                                      | Pass   |
|       | test_spatial_correlation_matrix.py::TestCholesky             | TestCholesky_lower_triangular                            | Pass   |
|       | test_spatial_correlation_matrix.py::TestCholesky             | TestCholesky_reconstruction                              | Pass   |
|       | test_spatial_correlation_matrix.py::TestCholesky             | TestCholesky_cholesky_reconstruction                     | Pass   |
|       | test_spatial_correlation_matrix.py::TestCovarianceMatrix     | TestCovarianceMatrix                                     | Pass   |
|       | test_spatial_correlation_matrix.py::TestCovarianceMatrix     | TestCovarianceMatrix_higher_correlation                  | Pass   |
|       | test_spatial_correlation_matrix.py::TestCovarianceMatrix     | TestCovarianceMatrix_off_diagonal                        | Pass   |
|       | test_spatial_correlation_matrix.py::TestCovarianceMatrix     | TestCovarianceMatrix_infinite_interval                   | Pass   |
|       | test_spatial_correlation_matrix.py::TestCovarianceMatrix     | TestCovarianceMatrix_infinite                            | Pass   |
|       | test_spatial_correlation_matrix.py::TestCovarianceMatrix     | TestCovarianceMatrix                                     | Pass   |
|       | test_spatial_correlation_matrix.py::TestCovarianceMatrix     | TestCovarianceMatrix_positive_definite                   | Pass   |
|       | test_spatial_correlation_matrix.py::TestEfficientRange       | TestEfficientRange_density_longer_range                  | Pass   |
|       | test_spatial_correlation_matrix.py::TestEfficientRange       | TestEfficientRange_density_no_reduction                  | Pass   |
|       | test_spatial_correlation_matrix.py::TestEfficientRange       | TestEfficientRange_density_equals_base                   | Pass   |
|       | test_spatial_correlation_matrix.py::TestEfficientRange       | TestEfficientRange_with_rho                              | Pass   |
|       | test_spatial_coverage_b6.py::TestCholesky                    | TestCholesky_callback_adds_jitter                        | Pass   |
|       | test_spatial_coverage_b6.py::TestFromFromRangePortfolioFile  | TestFromFromRangePortfolioFile                           | Pass   |
|       | test_spatial_coverage_b6.py::TestSpatialMultipliersWithNoRng | TestSpatialMultipliersWithNoRng                          | Pass   |
|       | test_spatial_multipliers.py::TestApplyToSequence             | TestApplyToSequence_negative                             | Pass   |
|       | test_spatial_multipliers.py::TestApplyToSequence             | TestApplyToSequence_mean_preserved_per_hour              | Pass   |
|       | test_spatial_multipliers.py::TestApplyToSequence             | TestApplyToSequence_shape                                | Pass   |
|       | test_spatial_multipliers.py::TestApplyToSequence             | TestApplyToSequence_variance_present                     | Pass   |
|       | test_spatial_multipliers.py::TestApplyToSequence             | TestApplyToSequence_precipitation_conserved              | Pass   |
|       | test_spatial_multipliers.py::TestApplyToSequence             | TestApplyToSequence_precip_hours_remain_zero             | Pass   |
|       | test_spatial_multipliers.py::TestSpatialMultipliers          | TestSpatialMultipliers                                   | Pass   |
|       | test_spatial_multipliers.py::TestSpatialMultipliers          | TestSpatialMultipliers_correlation_consistent_with_model | Pass   |
|       | test_spatial_multipliers.py::TestSpatialMultipliers          | TestSpatialMultipliers_density_lower_variance            | Pass   |
|       | test_spatial_multipliers.py::TestSpatialMultipliers          | TestSpatialMultipliers                                   | Pass   |
|       | test_spatial_multipliers.py::TestSpatialMultipliers          | TestSpatialMultipliers_lognormal_mean                    | Pass   |
|       | test_spatial_multipliers.py::TestSpatialMultipliers          | TestSpatialMultipliers_reproducible_with_seed            | Pass   |
|       | test_spatial_multipliers.py::TestSpatialMultipliers          | TestSpatialMultipliers                                   | Pass   |
|       | test_spatial_params_distance.py::TestConfigFile              | TestConfigFile_constants_valid                           | Pass   |
|       | test_spatial_params_distance.py::TestConfigFile              | TestConfigFile_structure                                 | Pass   |

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| Model | Module                                           | Test Name                                           | Status |
|-------|--------------------------------------------------|-----------------------------------------------------|--------|
|       | classification.py::TestFloodDuration             | test_duration_above_threshold                       | Pass   |
|       | classification.py::TestFloodDuration             | test_duration_hours_calculation                     | Pass   |
|       | classification.py::TestFloodDuration             | test_no_duration_normal_readings                    | Pass   |
|       | classification.py::TestGaugeLevelAnalysis        | test_current_level                                  | Pass   |
|       | classification.py::TestGaugeLevelAnalysis        | test_level_trend_falling                            | Pass   |
|       | classification.py::TestGaugeLevelAnalysis        | test_level_trend_rising                             | Pass   |
|       | classification.py::TestGaugeLevelAnalysis        | test_level_trend_stable                             | Pass   |
|       | classification.py::TestGaugeLevelAnalysis        | test_peak_level_detection                           | Pass   |
|       | classification.py::TestGaugeLevelAnalysis        | test_peak_timing                                    | Pass   |
|       | classification.py::TestGaugeStatusClassification | test_classification_status                          | Pass   |
|       | classification.py::TestGaugeStatusClassification | test_classification_thresholds                      | Pass   |
|       | classification.py::TestGaugeStatusClassification | test_classification_status                          | Pass   |
|       | classification.py::TestGaugeStatusClassification | test_classification_status                          | Pass   |
|       | classification.py::TestGaugeStatusClassification | test_classification_status                          | Pass   |
|       | cli.py::TestErrorHandling                        | test_bad_gauge_file_exits_with_1                    | Pass   |
|       | cli.py::TestErrorHandling                        | test_invalid_json_exits_with_1                      | Pass   |
|       | cli.py::TestListCategories                       | test_list_categories_does_not_open_gauge_file       | Pass   |
|       | cli.py::TestListCategories                       | test_list_categories_runs_without_error             | Pass   |
|       | cli.py::TestListPages                            | test_list_pages_does_not_open_gauge_file            | Pass   |
|       | cli.py::TestListPages                            | test_list_pages_runs_without_error                  | Pass   |
|       | cli.py::TestOptionalLoading                      | test_gauge_id_filters_data                          | Pass   |
|       | cli.py::TestOptionalLoading                      | test_timeseries_file_is_loaded                      | Pass   |
|       | cli.py::TestReportGeneration                     | test_monitoring_without_timeseries_hits             | Pass   |
|       | cli.py::TestReportGeneration                     | test_report_type_analysis                           | Pass   |
|       | cli.py::TestReportGeneration                     | test_report_type_basic                              | Pass   |
|       | cli.py::TestReportGeneration                     | test_report_type_full_hits_else_branch              | Pass   |
|       | cli.py::TestReportGeneration                     | test_report_type_monitoring_with_timeseries         | Pass   |
|       | cli.py::TestReportGeneration                     | test_success_logging_executes                       | Pass   |
|       | convenience.py::TestEdgeCases                    | test_completely_empty_gauge_data                    | Pass   |
|       | convenience.py::TestEdgeCases                    | test_gauge_data_with_hazard_curve_and_timeseries    | Pass   |
|       | convenience.py::TestEdgeCases                    | test_none_flood_gauge_header                        | Pass   |
|       | convenience.py::TestEdgeCases                    | test_output_is_non_empty_bytes                      | Pass   |
|       | convenience.py::TestEdgeCases                    | test_report_with_all_pages                          | Pass   |
|       | convenience.py::TestEdgeCases                    | test_single_page_report                             | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_without_known_key_raises              | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_gauges_key_first_gauge                | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_gauges_key_last_gauge                 | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_not_found_raises                      | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_integer_input_raises                  | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_input_used_directly                   | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_entry_missing_flood_gauge_key_skipped | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_entry_with_none_skipped               | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_flood_gauge_dict_returns_itself       | Pass   |
|       | convenience.py::TestFindGaugeByAddition          | test_addition_string_input_raises                   | Pass   |

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| Model | Module                                            | Test Name                                           | Status |
|-------|---------------------------------------------------|-----------------------------------------------------|--------|
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_generate_analy                                 | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_returns_path                                   | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_handles_exception                              | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_open_succeeds                                  | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_generate_basic                                 | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_monitoring_type                                | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_timeseries_callable_monitoring                 | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_timeseries_fall_back                           | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_path_function                                  | Pass   |
|       | convenience.py::TestGenerateGaugeReportModuleFunc | test_generate_report                                | Pass   |
|       | elements.py::TestGenerateElement                  | test_empty_page_list_returns_empty                  | Pass   |
|       | elements.py::TestGenerateElement                  | test_exception_in_page_continues                    | Pass   |
|       | elements.py::TestGenerateElement                  | test_first_page_no_page_break                       | Pass   |
|       | elements.py::TestGenerateElement                  | test_mixed_valid_and_unknown_pages                  | Pass   |
|       | elements.py::TestGenerateElement                  | test_multiple_pages_have_page_break                 | Pass   |
|       | elements.py::TestGenerateElement                  | test_none_timeseries_passed_to_page                 | Pass   |
|       | elements.py::TestGenerateElement                  | test_returns_list                                   | Pass   |
|       | elements.py::TestGenerateElement                  | test_timeseries_passed_to_page                      | Pass   |
|       | elements.py::TestGenerateElement                  | test_unknown_page_skipped                           | Pass   |
|       | elements.py::TestGenerateReport                   | test_all_unknown_pages_still_creates_pdf            | Pass   |
|       | elements.py::TestGenerateReport                   | test_auto_select_called_when_pages_none             | Pass   |
|       | elements.py::TestGenerateReport                   | test_auto_select_not_called_when_pages_provided     | Pass   |
|       | elements.py::TestGenerateReport                   | test_custom_output_filename                         | Pass   |
|       | elements.py::TestGenerateReport                   | test_custom_pages_to_include                        | Pass   |
|       | elements.py::TestGenerateReport                   | test_empty_gauge_data                               | Pass   |
|       | elements.py::TestGenerateReport                   | test_pdf_exists                                     | Pass   |
|       | elements.py::TestGenerateReport                   | test_pdf_is_bytes                                   | Pass   |
|       | elements.py::TestGenerateReport                   | test_pdf_non_empty                                  | Pass   |
|       | elements.py::TestGenerateReport                   | test_pdf_suffix                                     | Pass   |
|       | elements.py::TestGenerateReport                   | test_returns_path                                   | Pass   |
|       | elements.py::TestGenerateReport                   | test_with_timeseries_data                           | Pass   |
|       | init.py::TestAutoSelectPages                      | test_all_pages_exist                                | Pass   |
|       | init.py::TestAutoSelectPages                      | test_always_includes_gauge_info_pages               | Pass   |
|       | init.py::TestAutoSelectPages                      | test_always_includes_operational_pages              | Pass   |
|       | init.py::TestAutoSelectPages                      | test_always_includes_summary_pages                  | Pass   |
|       | init.py::TestAutoSelectPages                      | test_returns_list                                   | Pass   |
|       | init.py::TestAutoSelectPages                      | test_with_hazard_curve_includes_analysis_pages      | Pass   |
|       | init.py::TestAutoSelectPages                      | test_with_timeseries_includes_analysis_pages        | Pass   |
|       | init.py::TestAutoSelectPages                      | test_without_timeseries_or_hazard_no_analysis_pages | Pass   |
|       | init.py::TestGaugeReportGenerator                 | test_categories_dict_initialized                    | Pass   |
|       | init.py::TestGaugeReportGenerator                 | test_explicit_output_dir_path                       | Pass   |
|       | init.py::TestGaugeReportGenerator                 | test_explicit_output_dir_string                     | Pass   |
|       | init.py::TestGaugeReportGenerator                 | test_output_dir_uses_config                         | Pass   |
|       | init.py::TestGaugeReportGenerator                 | test_output_dir_created                             | Pass   |
|       | init.py::TestGaugeReportGenerator                 | test_pages_dict_initialized                         | Pass   |

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| Model | Module                                               | Test Name                                       | Status |
|-------|------------------------------------------------------|-------------------------------------------------|--------|
|       | init.py::TestGenerateFilename                        | test_contains_gauge_id                          | Pass   |
|       | init.py::TestGenerateFilename                        | test_ends_with_pdf                              | Pass   |
|       | init.py::TestGenerateFilename                        | test_missing_gauge_id_uses_unknown              | Pass   |
|       | init.py::TestGenerateFilename                        | test_partial_path_missing_gauge_id              | Pass   |
|       | init.py::TestGenerateFilename                        | test_partial_path_missing_header                | Pass   |
|       | init.py::TestGenerateFilename                        | test_returns_string                             | Pass   |
|       | init.py::TestGenerateFilename                        | test_timestamp_in_filename                      | Pass   |
|       | specialised.py::TestHeaderFooter                     | test_header_footer_callable                     | Pass   |
|       | specialised.py::TestHeaderFooter                     | test_header_footer_draws_confidential_footer    | Pass   |
|       | specialised.py::TestHeaderFooter                     | test_header_footer_draws_header_strings         | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_analysis_report_custom_filename   | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_analysis_report_exists            | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_analysis_report_with_timeseries   | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_basic_report_custom_filename      | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_basic_report_exists               | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_basic_report_non_empty            | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_basic_with_timeseries             | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_monitoring_report_custom_filename | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_monitoring_report_exists          | Pass   |
|       | specialised.py::TestSpecialisedReportGenerator       | test_generate_monitoring_report_no_timeseries   | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_get_page_categories_contains_expected_keys | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_get_page_categories_returns_copy           | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_get_page_categories_returns_dict           | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_list_available_pages_contains_expected     | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_list_available_pages_non_empty             | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_list_available_pages_returns_list          | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_validate_pages_all_invalid                 | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_validate_pages_all_valid                   | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_validate_pages_empty_list                  | Pass   |
|       | specialised.py::TestUtilityMethods                   | test_validate_pages_mixed                       | Pass   |
|       | metrics.py::TestGaugeMetrics                         | test_level_range                                | Pass   |
|       | metrics.py::TestGaugeMetrics                         | test_max_level                                  | Pass   |
|       | metrics.py::TestGaugeMetrics                         | test_mean_level                                 | Pass   |
|       | metrics.py::TestGaugeMetrics                         | test_min_level                                  | Pass   |
|       | metrics.py::TestGaugeMetrics                         | test_rate_of_rise                               | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_does_not_crash    | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_not_crash         | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_output            | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_section_header    | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_section_header    | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_missing           | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_present           | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_present           | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_present           | Pass   |
|       | page_01_title_overview.py::TestGaugePageOverviewPage | test_gauge_page_overview_page_calculated        | Pass   |

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| Model | Module                          | Test Name                                         | Status |
|-------|---------------------------------|---------------------------------------------------|--------|
|       | page_01_title_overview.py::Test | Guest Hold Overview Page does not crash           | Pass   |
|       | page_01_title_overview.py::Test | Guest Hold One List Page                          | Pass   |
|       | page_01_title_overview.py::Test | Guest Title Overview Page present                 | Pass   |
|       | page_01_title_overview.py::Test | Guest Link Overview Page fallback                 | Pass   |
|       | page_01_title_overview.py::Test | Guest With Overview Page does not crash           | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page does not crash          | Pass   |
|       | page_02_sensor_details.py::Test | Guest Gauge Details Subtitle                      | Pass   |
|       | page_02_sensor_details.py::Test | Guest Gauge Detail Page on section                | Pass   |
|       | page_02_sensor_details.py::Test | Guest Gauge Details Missing                       | Pass   |
|       | page_02_sensor_details.py::Test | Guest Gauge Details Present                       | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page                         | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page or statistics section   | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page information section     | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page configuration section   | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page does not crash          | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page                         | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page                         | Pass   |
|       | page_02_sensor_details.py::Test | Guest Sensor Details Page does not crash          | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France close to thames              | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France zero on thames               | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France data does not crash          | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France risk section                 | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France graphic coordinates section  | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France tables                       | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France central london               | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France north london                 | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France south london                 | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France info does not crash          | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France toning summary section       | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France pages list                   | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France category in location profile | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France core formatted as out of 10  | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France context section              | Pass   |
|       | page_03_location.py::Test       | Gauge Test in France present                      | Pass   |
|       | page_04_measurements.py::Test   | Guest Empty data Page does not crash              | Pass   |
|       | page_04_measurements.py::Test   | Guest Empty reading Page does not crash           | Pass   |
|       | page_04_measurements.py::Test   | Guest Measurement Missing                         | Pass   |
|       | page_04_measurements.py::Test   | Guest Measurement Present                         | Pass   |
|       | page_04_measurements.py::Test   | Guest Measurement Page                            | Pass   |
|       | page_04_measurements.py::Test   | Guest Measurement Statistics section              | Pass   |
|       | page_04_measurements.py::Test   | Guest Measurement Configuration section           | Pass   |
|       | page_04_measurements.py::Test   | Guest Measurement Shows message                   | Pass   |
|       | page_04_measurements.py::Test   | Guest Measurement Status section                  | Pass   |
|       | page_04_measurements.py::Test   | Guest Measurement Page                            | Pass   |

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| Model | Module                           | Test Name                                                | Status |
|-------|----------------------------------|----------------------------------------------------------|--------|
|       | page_06_risk_assessment.py::Test | test_should_alert_range                                  | Pass   |
|       | page_06_risk_assessment.py::Test | test_shard_analyze_range                                 | Pass   |
|       | page_06_risk_assessment.py::Test | test_hide_and_isrange                                    | Pass   |
|       | page_06_risk_assessment.py::Test | test_create_alert_status                                 | Pass   |
|       | page_06_risk_assessment.py::Test | test_normal_status                                       | Pass   |
|       | page_06_risk_assessment.py::Test | test_severe_alert_status                                 | Pass   |
|       | page_06_risk_assessment.py::Test | test_timeseries_status_no_level                          | Pass   |
|       | page_06_risk_assessment.py::Test | test_timeseries_status_no_readings                       | Pass   |
|       | page_06_risk_assessment.py::Test | test_waiting_status                                      | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_friedman_regression                        | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_statistics_not_crash                       | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_highsignificance                           | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_lowsignificance                            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_no_exceedances_conclusion                  | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_nonsignificance                            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_summary_section                            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_augendy_passing                            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_augendy_present                            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_augendy_passing                            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_augendy_present                            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_cables_page                                | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_highfriedman_regression                    | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_highmax_peds_severe_significance           | Fail   |
|       | page_07_data_summary.py::Test    | test_get_data_highwith_lagerange_significance            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_atomics_slider_section                     | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_sisknoncclusion                            | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_saturnis_pageconclusion                    | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_operational_status_conclusion              | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_operational_status_summary_section         | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_stationonlysigns_section                   | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_summary_page                               | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_seriesqual_pageconclusion                  | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_seriesnoqual_pageconclusion                | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_seriesusd_pageconclusion                   | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_serieswd_pageconclusion                    | Pass   |
|       | page_07_data_summary.py::Test    | test_get_data_ispresent_page                             | Pass   |
|       | page_08_flood_history.py::Test   | test_flow_in_historical_graph_no_dates_fallback          | Pass   |
|       | page_08_flood_history.py::Test   | test_flow_in_historical_graph_with_thresholds            | Pass   |
|       | page_08_flood_history.py::Test   | test_flow_history_responses_shows_message                | Pass   |
|       | page_08_flood_history.py::Test   | test_flow_history_page_data_shows_no_data_message        | Pass   |
|       | page_08_flood_history.py::Test   | test_flow_in_historical_pages_no_exceedances             | Pass   |
|       | page_08_flood_history.py::Test   | test_flow_in_historical_pages_with_exceedances           | Pass   |
|       | page_08_flood_history.py::Test   | test_flow_in_historical_page                             | Pass   |
|       | page_08_flood_history.py::Test   | test_flow_in_historical_page_daily_returns_more_elements | Fail   |

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| Model | Module                                         | Test Name                                               | Status |
|-------|------------------------------------------------|---------------------------------------------------------|--------|
|       | page_08_flood_history.py::TestGauge            | testFloodHistoryPageResponses_includes_table            | Pass   |
|       | page_08_flood_history.py::TestGauge            | testFloodHistoryPageSeverity_classification             | Pass   |
|       | page_08_flood_history.py::TestGauge            | testFloodHistoryPageTable_top_10                        | Pass   |
|       | page_09_hazard_curves.py::TestGauge            | testEmptyCurvesPagePeriod_levels_no_extra_table         | Pass   |
|       | page_09_hazard_curves.py::TestGauge            | testHazardCurvesPageShows_fallback                      | Pass   |
|       | page_09_hazard_curves.py::TestGauge            | testHazardCurvesPage                                    | Pass   |
|       | page_09_hazard_curves.py::TestGauge            | testHazardCurvesPageBasic                               | Pass   |
|       | page_09_hazard_curves.py::TestGauge            | testHazardCurvesPagePeriod_levels_renders_Table         | Table  |
|       | page_10_prs_pricing.py::TestGauge              | testPRSPricingPageCurve_shows_fallback                  | Pass   |
|       | page_10_prs_pricing.py::TestGauge              | testPRSPricingPage                                      | Pass   |
|       | page_10_prs_pricing.py::TestGauge              | testPRSPricingPageRates_renders_spreads                 | Pass   |
|       | page_10_prs_pricing.py::TestGauge              | testPRSPricingPageRates_renders_na                      | Pass   |
|       | page_11_current_risk.py::TestGauge             | testEmptyRiskPageHistorical_daily_shows_fallback        | Pass   |
|       | page_11_current_risk.py::TestGauge             | testCurrentRiskPageData_shows_fallback                  | Pass   |
|       | page_11_current_risk.py::TestGauge             | testCurrentRiskPage                                     | Pass   |
|       | page_11_current_risk.py::TestGauge             | testCurrentRiskPageWithRiskPredictions_includes_content | Skip   |
|       | page_11_current_risk.py::TestGauge             | testCurrentRiskPageWithRiskPredictions_returns_elements | Skip   |
|       | page_12_trading.py::TestBuildBlock             | testSection_trades_returns_message                      | Pass   |
|       | page_12_trading.py::TestBuildBlock             | testSection_trade_direction                             | Pass   |
|       | page_12_trading.py::TestBuildBlock             | testSection_capitalized                                 | Pass   |
|       | page_12_trading.py::TestBuildBlock             | testSection_trades_returns_table                        | Pass   |
|       | page_12_trading.py::TestBuildBlock             | testSection_trades_shows_count                          | Pass   |
|       | page_12_trading.py::TestBuildModule            | testGrowthSectionState_returns_message                  | Pass   |
|       | page_12_trading.py::TestBuildModule            | testGrowthSection_handles_none_rates                    | Pass   |
|       | page_12_trading.py::TestBuildModule            | testGrowthSection_returns_extra_table                   | Pass   |
|       | page_12_trading.py::TestBuildModule            | testGrowthSection_returns_table                         | Pass   |
|       | page_12_trading.py::TestDeriveTable            | test_both_missing_returns_na                            | Pass   |
|       | page_12_trading.py::TestDeriveTable            | test_five_year_tenor                                    | Pass   |
|       | page_12_trading.py::TestDeriveTable            | test_invalid_date_format_returns_na                     | Pass   |
|       | page_12_trading.py::TestDeriveTable            | test_missing_end_returns_na                             | Pass   |
|       | page_12_trading.py::TestDeriveTable            | test_missing_start_returns_na                           | Pass   |
|       | page_12_trading.py::TestDeriveTable            | test_valid_dates_returns_years                          | Pass   |
|       | page_12_trading.py::TestGenerateTable          | testEmpty_gauge_id_key_returns_message                  | Pass   |
|       | page_12_trading.py::TestGenerateTable          | testEmpty_returns_list                                  | Pass   |
|       | page_12_trading.py::TestGenerateTable          | testWith_gauge_id_returns_trading_header                | Pass   |
|       | page_12_trading.py::TestLoadGaugeTable         | testReports_dir_raises_fallback                         | Pass   |
|       | page_12_trading.py::TestLoadGaugeTable         | testReports_dir_not_exist_returns_empty                 | Pass   |
|       | page_12_trading.py::TestLoadGaugeTable         | testTrade_matching_gauge_id_appended                    | Pass   |
|       | page_12_trading.py::TestLoadGaugeTable         | testTrade_not_matching_gauge_id_skipped                 | Pass   |
|       | page_12_trading.py::TestLoadMarketState        | testMarket_state_exception_returns_empty                | Pass   |
|       | test_base_page_format.py::TestFormatCoordinate | test_coordinate_six_decimals                            | Pass   |
|       | test_base_page_format.py::TestFormatCoordinate | test_negative_coordinate                                | Pass   |
|       | test_base_page_format.py::TestFormatCoordinate | test_none_delegates_to_format_value                     | Pass   |
|       | test_base_page_format.py::TestFormatCoordinate | test_hidden_replacements                                | Pass   |

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| Model | Module                                                                        | Test Name                                                                     | Status |
|-------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|
|       | test_base_page_format.py::TestFormatFieldNames                                | test_base_page_format.py::TestFormatFieldNames_split                          | Pass   |
|       | test_base_page_format.py::TestFormatForNumericDelegates                       | test_base_page_format.py::TestFormatForNumericDelegates                       | Pass   |
|       | test_base_page_format.py::TestFormatPositiveReturnsTimes                      | test_base_page_format.py::TestFormatPositiveReturnsTimes                      | Pass   |
|       | test_base_page_format.py::TestFormatErrorReturnsNever                         | test_base_page_format.py::TestFormatErrorReturnsNever                         | Pass   |
|       | test_base_page_format.py::TestFormatIntegerValue                              | test_base_page_format.py::TestFormatIntegerValue                              | Pass   |
|       | test_base_page_format.py::TestFormatNonNumericIdDelegatesToFormatValue        | test_base_page_format.py::TestFormatNonNumericIdDelegatesToFormatValue        | Pass   |
|       | test_base_page_format.py::TestFormatNumericValueWithCustomUnit                | test_base_page_format.py::TestFormatNumericValueWithCustomUnit                | Pass   |
|       | test_base_page_format.py::TestFormatNumericValueWithDefaultUnit               | test_base_page_format.py::TestFormatNumericValueWithDefaultUnit               | Pass   |
|       | test_base_page_format.py::TestFormatEmptyStringReturnsNotSpecified            | test_base_page_format.py::TestFormatEmptyStringReturnsNotSpecified            | Pass   |
|       | test_base_page_format.py::TestFormatFalseReturnsNo                            | test_base_page_format.py::TestFormatFalseReturnsNo                            | Pass   |
|       | test_base_page_format.py::TestFormatFloatWholeNumberStripsDecimal             | test_base_page_format.py::TestFormatFloatWholeNumberStripsDecimal             | Pass   |
|       | test_base_page_format.py::TestFormatFloatWithDecimalsFormatted                | test_base_page_format.py::TestFormatFloatWithDecimalsFormatted                | Pass   |
|       | test_base_page_format.py::TestFormatIntegerReturnsStr                         | test_base_page_format.py::TestFormatIntegerReturnsStr                         | Pass   |
|       | test_base_page_format.py::TestFormatNoneReturnsNotSpecified                   | test_base_page_format.py::TestFormatNoneReturnsNotSpecified                   | Pass   |
|       | test_base_page_format.py::TestFormatNormalStringReturned                      | test_base_page_format.py::TestFormatNormalStringReturned                      | Pass   |
|       | test_base_page_format.py::TestFormatValueReturnsYes                           | test_base_page_format.py::TestFormatValueReturnsYes                           | Pass   |
|       | test_current_risk.py::TestCurrentRiskEmptyIdElementObservedCasesKey           | test_current_risk.py::TestCurrentRiskEmptyIdElementObservedCasesKey           | Pass   |
|       | test_current_risk.py::TestCurrentRiskEmptyIdElementEdgeCases                  | test_current_risk.py::TestCurrentRiskEmptyIdElementEdgeCases                  | Pass   |
|       | test_current_risk.py::TestCurrentRiskFullPathNetworkIsLargeElements           | test_current_risk.py::TestCurrentRiskFullPathNetworkIsLargeElements           | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneralDataElementsEdgeCasesCriticalLoop | test_current_risk.py::TestCurrentRiskGeneralDataElementsEdgeCasesCriticalLoop | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneralDataElementsEdgeCasesObs          | test_current_risk.py::TestCurrentRiskGeneralDataElementsEdgeCasesObs          | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneralDataElementsEdgeCasesAllLevels    | test_current_risk.py::TestCurrentRiskGeneralDataElementsEdgeCasesAllLevels    | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneralDataElementsEdgeCases             | test_current_risk.py::TestCurrentRiskGeneralDataElementsEdgeCases             | Pass   |
|       | test_current_risk.py::TestCurrentRiskGeneralDataElementsIsLoopSkipped         | test_current_risk.py::TestCurrentRiskGeneralDataElementsIsLoopSkipped         | Pass   |
|       | test_flood_history.py::TestBuildFloodsClassification                          | test_flood_history.py::TestBuildFloodsClassification                          | Pass   |
|       | test_flood_history.py::TestBuildFloodsInvalidDateInExceedanceSkipped          | test_flood_history.py::TestBuildFloodsInvalidDateInExceedanceSkipped          | Pass   |
|       | test_flood_history.py::TestBuildFloodsDayEventUpdatesPeak                     | test_flood_history.py::TestBuildFloodsDayEventUpdatesPeak                     | Pass   |
|       | test_flood_history.py::TestBuildFloodsDayRangeDateFormat                      | test_flood_history.py::TestBuildFloodsDayRangeDateFormat                      | Pass   |
|       | test_flood_history.py::TestBuildFloodsEventsReturnsMessage                    | test_flood_history.py::TestBuildFloodsEventsReturnsMessage                    | Pass   |
|       | test_flood_history.py::TestBuildFloodsClassification                          | test_flood_history.py::TestBuildFloodsClassification                          | Pass   |
|       | test_flood_history.py::TestBuildFloodsDayEventDateFormat                      | test_flood_history.py::TestBuildFloodsDayEventDateFormat                      | Pass   |
|       | test_flood_history.py::TestBuildFloodsSeparateEvents                          | test_flood_history.py::TestBuildFloodsSeparateEvents                          | Pass   |
|       | test_flood_history.py::TestBuildFloodsFloodClassification                     | test_flood_history.py::TestBuildFloodsFloodClassification                     | Pass   |
|       | test_gauge_generator_extras.py::TestAutoOpenCalled                            | test_gauge_generator_extras.py::TestAutoOpenCalled                            | Pass   |
|       | test_gauge_generator_extras.py::TestAutoOpenFalseDoesNotRaise                 | test_gauge_generator_extras.py::TestAutoOpenFalseDoesNotRaise                 | Pass   |
|       | test_gauge_generator_extras.py::TestAutoOpenSuccess                           | test_gauge_generator_extras.py::TestAutoOpenSuccess                           | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugeByLidFloodGaugesList             | test_gauge_generator_extras.py::TestFindGaugeByLidFloodGaugesList             | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugeByLidGaugesKey                   | test_gauge_generator_extras.py::TestFindGaugeByLidGaugesKey                   | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugeByLid                            | test_gauge_generator_extras.py::TestFindGaugeByLid                            | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugeByLidGaugeObject                 | test_gauge_generator_extras.py::TestFindGaugeByLidGaugeObject                 | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugeByLidRaisesValueError            | test_gauge_generator_extras.py::TestFindGaugeByLidRaisesValueError            | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugeByLidRaisesValueError            | test_gauge_generator_extras.py::TestFindGaugeByLidRaisesValueError            | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugeByLidListRaisesValueError        | test_gauge_generator_extras.py::TestFindGaugeByLidListRaisesValueError        | Pass   |
|       | test_gauge_generator_extras.py::TestFindGaugeByLidRaisesValueError            | test_gauge_generator_extras.py::TestFindGaugeByLidRaisesValueError            | Pass   |

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| Model | Module                                                                  | Test Name                                                                                                                     | Status |
|-------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------|
|       | test_generator_direct.py::TestFindGaugesById                            | test_generator_direct.py::TestFindGaugesById gauges.key_found                                                                 | Pass   |
|       | test_generator_direct.py::TestFindGaugesById                            | test_generator_direct.py::TestFindGaugesById dict_structure_raises                                                            | Pass   |
|       | test_generator_direct.py::TestFindGaugesById                            | test_generator_direct.py::TestFindGaugesById type_raises                                                                      | Pass   |
|       | test_generator_direct.py::TestFindGaugesByInput                         | test_generator_direct.py::TestFindGaugesByInput found                                                                         | Pass   |
|       | test_generator_direct.py::TestFindGaugesById                            | test_generator_direct.py::TestFindGaugesById found gauge_entry_skipped                                                        | Pass   |
|       | test_generator_direct.py::TestFindGaugesById                            | test_generator_direct.py::TestFindGaugesById found raises_value_error                                                         | Pass   |
|       | test_generator_direct.py::TestFindGaugesById                            | test_generator_direct.py::TestFindGaugesById gauge_dict_returns_self                                                          | Pass   |
|       | test_generator_direct.py::TestGenerateReportDirect                      | test_generator_direct.py::TestGenerateReportDirect report_direct                                                              | Pass   |
|       | test_generator_direct.py::TestGenerateReportFnAnalysis                  | test_generator_direct.py::TestGenerateReportFnAnalysis report_fn_analysis                                                     | Pass   |
|       | test_generator_direct.py::TestGenerateReportFnAutoOpen                  | test_generator_direct.py::TestGenerateReportFnAutoOpen report_fn_auto_open                                                    | Pass   |
|       | test_generator_direct.py::TestGenerateReportFnBasic                     | test_generator_direct.py::TestGenerateReportFnBasic report_fn_basic                                                           | Pass   |
|       | test_generator_direct.py::TestGenerateReportFnFallback                  | test_generator_direct.py::TestGenerateReportFnFallback report_fn_fallback                                                     | Pass   |
|       | test_generator_direct.py::TestGenerateReportFnMonitoring                | test_generator_direct.py::TestGenerateReportFnMonitoring report_fn_monitoring                                                 | Pass   |
|       | test_generator_direct.py::TestGenerateReportNoPagesUsesAutoSelect       | test_generator_direct.py::TestGenerateReportNoPagesUsesAutoSelect report_no_pages_uses_auto_select                            | Pass   |
|       | test_generator_direct.py::TestGenerateReportOpenFile                    | test_generator_direct.py::TestGenerateReportOpenFile returns_false                                                            | Pass   |
|       | test_generator_direct.py::TestGenerateReportNoAnalysisData              | test_generator_direct.py::TestGenerateReportNoAnalysisData no_analysis_data                                                   | Pass   |
|       | test_generator_direct.py::TestGenerateReportWithHazard                  | test_generator_direct.py::TestGenerateReportWithHazard with_hazard                                                            | Pass   |
|       | test_generator_direct.py::TestGenerateReportDailyReport                 | test_generator_direct.py::TestGenerateReportDailyReport daily_report                                                          | Pass   |
|       | test_generator_direct.py::TestGenerateReportExceptionContinue           | test_generator_direct.py::TestGenerateReportExceptionContinue exception_continue                                              | Pass   |
|       | test_generator_direct.py::TestGenerateReportSkipsUnknownPage            | test_generator_direct.py::TestGenerateReportSkipsUnknownPage skips_unknown_page                                               | Pass   |
|       | test_generator_direct.py::TestGenerateReportNoGaugeId                   | test_generator_direct.py::TestGenerateReportNoGaugeId no_gauge_id                                                             | Pass   |
|       | test_generator_direct.py::TestGenerateReportWithGaugeId                 | test_generator_direct.py::TestGenerateReportWithGaugeId with_gauge_id                                                         | Pass   |
|       | test_generator_direct.py::TestGenerateReportMonitoringReport            | test_generator_direct.py::TestGenerateReportMonitoringReport monitoring_report                                                | Pass   |
|       | test_generator_direct.py::TestGenerateReportPages                       | test_generator_direct.py::TestGenerateReportPages pages                                                                       | Pass   |
|       | test_generator_direct.py::TestGenerateReportAvailablePages              | test_generator_direct.py::TestGenerateReportAvailablePages available_pages                                                    | Pass   |
|       | test_generator_direct.py::TestGenerateReportOutputDirDoesNotExist       | test_generator_direct.py::TestGenerateReportOutputDirDoesNotExist output_dir_does_not_exist                                   | Pass   |
|       | test_generator_direct.py::TestGenerateReportPagesInvalid                | test_generator_direct.py::TestGenerateReportPagesInvalid pages_invalid                                                        | Pass   |
|       | test_generator_direct.py::TestGenerateReportPagesInvalidWithInvalid     | test_generator_direct.py::TestGenerateReportPagesInvalidWithInvalid pages_invalid_with_invalid                                | Pass   |
|       | test_integrator.py::TestGenerateReportForKeyStructureWorks              | test_integrator.py::TestGenerateReportForKeyStructureWorks report_for_key_structure_works                                     | Pass   |
|       | test_integrator.py::TestGenerateReportFindGaugeReturnsPath              | test_integrator.py::TestGenerateReportFindGaugeReturnsPath report_find_gauge_returns_path                                     | Pass   |
|       | test_integrator.py::TestGenerateReportFindGaugeNotfoundReturnsNone      | test_integrator.py::TestGenerateReportFindGaugeNotfoundReturnsNone report_find_gauge_not_found_returns_none                   | Pass   |
|       | test_integrator.py::TestGenerateReportFindGaugeFileReturnsNone          | test_integrator.py::TestGenerateReportFindGaugeFileReturnsNone report_find_gauge_file_returns_none                            | Pass   |
|       | test_integrator.py::TestGenerateReportFindGaugeContainsGaugeId          | test_integrator.py::TestGenerateReportFindGaugeContainsGaugeId report_find_gauge_contains_gauge_id                            | Pass   |
|       | test_integrator.py::TestGenerateReportFindGaugeDirCreated               | test_integrator.py::TestGenerateReportFindGaugeDirCreated report_find_gauge_dir_created                                       | Pass   |
|       | test_integrator.py::TestGenerateReportFindGaugeFileCreated              | test_integrator.py::TestGenerateReportFindGaugeFileCreated report_find_gauge_file_created                                     | Pass   |
|       | test_integrator.py::TestGetAvailableGaugesUsingFileReturnsEmptyList     | test_integrator.py::TestGetAvailableGaugesUsingFileReturnsEmptyList get_available_gauges_using_file_returns_empty_list        | Pass   |
|       | test_integrator.py::TestGetAvailableGaugesReturnsAllIds                 | test_integrator.py::TestGetAvailableGaugesReturnsAllIds get_available_gauges_returns_all_ids                                  | Pass   |
|       | test_integrator.py::TestGetAvailableGaugesReturnsCorrectId              | test_integrator.py::TestGetAvailableGaugesReturnsCorrectId get_available_gauges_returns_correct_id                            | Pass   |
|       | test_integrator.py::TestGetAvailableGaugesReturnsList                   | test_integrator.py::TestGetAvailableGaugesReturnsList get_available_gauges_returns_list                                       | Pass   |
|       | test_integrator.py::TestValidateGaugeExistsKeyStructure                 | test_integrator.py::TestValidateGaugeExistsKeyStructure validate_gauge_exists_key_structure                                   | Pass   |
|       | test_integrator.py::TestValidateGaugeExistsExistingGaugeReturnsTrue     | test_integrator.py::TestValidateGaugeExistsExistingGaugeReturnsTrue validate_gauge_exists_existing_gauge_returns_true         | Pass   |
|       | test_integrator.py::TestValidateGaugeExistsMissingFileReturnsFalse      | test_integrator.py::TestValidateGaugeExistsMissingFileReturnsFalse validate_gauge_exists_missing_file_returns_false           | Pass   |
|       | test_integrator.py::TestValidateGaugeExistsNonexistentGaugeReturnsFalse | test_integrator.py::TestValidateGaugeExistsNonexistentGaugeReturnsFalse validate_gauge_exists_nonexistent_gauge_returns_false | Pass   |
|       | test_integrator_exception.py::TestGenerateReportForGaugeException       | test_integrator_exception.py::TestGenerateReportForGaugeException report_for_gauge_exception                                  | Pass   |

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| Model | Module                                       | Test Name                                    | Status |
|-------|----------------------------------------------|----------------------------------------------|--------|
|       | gauge_pdf.py::TestGaugePdfGenerator          | test_gauge_pdf_filename_contains_gauge_pdf   | Pass   |
|       | gauge_pdf.py::TestGaugePdfGenerator          | test_gauge_pdf_has_reasonable_size           | Pass   |
|       | gauge_pdf.py::TestGaugePdfGenerator          | test_gauge_pdf_starts_with_pdf_header        | Pass   |
|       | gauge_pdf.py::TestGaugePdfGenerator          | test_gauge_report_with_minimal_data          | Pass   |
|       | gauge_pdf.py::TestGaugeReportGenerator       | test_report_generator_initialises            | Pass   |
|       | gauge_pdf.py::TestGaugeReportGenerator       | test_list_available_pages                    | Pass   |
|       | gauge_pdf.py::TestGaugeReportGenerator       | test_validate_pages                          | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_full_property_report_generates_pdf      | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_mortgage_focused_report_generates_pdf   | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_output_dir_created_if_missing           | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_pdf_filename_contains_property_id       | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_pdf_has_reasonable_size                 | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_pdf_starts_with_pdf_header              | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_property_only_report_generates_pdf      | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_report_with_minimal_property_data       | Pass   |
|       | property_pdf.py::TestPropertyPdfGenerator    | test_risk_focused_report_generates_pdf       | Pass   |
|       | property_pdf.py::TestPropertyReportGenerator | test_generator_select_pages_with_mortgage    | Pass   |
|       | property_pdf.py::TestPropertyReportGenerator | test_generator_select_pages_without_mortgage | Pass   |
|       | property_pdf.py::TestPropertyReportGenerator | test_report_generator_initialises            | Pass   |
|       | property_pdf.py::TestPropertyReportGenerator | test_list_available_pages                    | Pass   |
|       | property_pdf.py::TestPropertyReportGenerator | test_validate_pages_valid                    | Pass   |
|       | server_flow.py::TestServerReportGenerator    | test_gauge_report_via_route_function         | Pass   |
|       | server_flow.py::TestServerReportGenerator    | test_gauge_report_with_real_data             | Pass   |
|       | server_flow.py::TestServerReportGenerator    | test_property_report_via_route_function      | Pass   |
|       | server_flow.py::TestServerReportGenerator    | test_property_report_with_real_data          | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis      | test_find_nearest_gauge                      | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis      | test_full_analysis_high_risk_property        | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis      | test_full_analysis_low_risk_property         | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis      | test_load_gauge_data                         | Pass   |
|       | analysis.py::TestSinglePropertyAnalysis      | test_load_property_data                      | Pass   |
|       | calculations.py::TestDamageRatioCalculator   | test_commercial_higher_damage                | Pass   |
|       | calculations.py::TestDamageRatioCalculator   | test_damage_capped                           | Pass   |
|       | calculations.py::TestDamageRatioCalculator   | test_damage_increases_with_depth             | Pass   |
|       | calculations.py::TestDamageRatioCalculator   | test_no_damage_no_flood                      | Pass   |
|       | calculations.py::TestDistanceCalculator      | test_haversine_known_distance                | Pass   |
|       | calculations.py::TestDistanceCalculator      | test_haversine_same_point                    | Pass   |
|       | calculations.py::TestDistanceCalculator      | test_haversine_teddington_to_richmond        | Pass   |
|       | calculations.py::TestFloodDepthCalculator    | test_flood_above_trigger                     | Pass   |
|       | calculations.py::TestFloodDepthCalculator    | test_flood_attenuates_with_distance          | Pass   |
|       | calculations.py::TestFloodDepthCalculator    | test_high_elevation_no_flood                 | Pass   |
|       | calculations.py::TestFloodDepthCalculator    | test_no_flood_below_trigger                  | Pass   |
|       | calculations.py::TestRiskClassifier          | test_high_risk                               | Pass   |
|       | calculations.py::TestRiskClassifier          | test_low_risk                                | Pass   |
|       | calculations.py::TestRiskClassifier          | test_medium_risk                             | Pass   |

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| Model | Module                                  | Test Name                                    | Status |
|-------|-----------------------------------------|----------------------------------------------|--------|
|       | calculations.py::TestRiskClassification | test_minimal_risk                            | Pass   |
|       | formatters.py::TestConstants            | test_margin_positive                         | Pass   |
|       | formatters.py::TestConstants            | test_page_dimensions                         | Pass   |
|       | formatters.py::TestFmtGbp               | test_large_value                             | Pass   |
|       | formatters.py::TestFmtGbp               | test_negative                                | Pass   |
|       | formatters.py::TestFmtGbp               | test_non_numeric_fallback                    | Pass   |
|       | formatters.py::TestFmtGbp               | test_none_fallback                           | Pass   |
|       | formatters.py::TestFmtGbp               | test_positive_float                          | Pass   |
|       | formatters.py::TestFmtGbp               | test_zero                                    | Pass   |
|       | formatters.py::TestLayouts              | test_body_style_alignment                    | Pass   |
|       | formatters.py::TestLayouts              | test_body_style_bold                         | Pass   |
|       | formatters.py::TestLayouts              | test_body_style_default                      | Pass   |
|       | formatters.py::TestLayouts              | test_build_header_footer_callable            | Pass   |
|       | formatters.py::TestLayouts              | test_white_hdr_style_returns_paragraph_style | Pass   |
|       | formatters.py::TestSeqTypeColor         | test_case_insensitive                        | Pass   |
|       | formatters.py::TestSeqTypeColor         | test_cluster                                 | Pass   |
|       | formatters.py::TestSeqTypeColor         | test_doublet                                 | Pass   |
|       | formatters.py::TestSeqTypeColor         | test_isolated                                | Pass   |
|       | formatters.py::TestSeqTypeColor         | test_none_returns_white                      | Pass   |
|       | formatters.py::TestSeqTypeColor         | test_persistent                              | Pass   |
|       | formatters.py::TestSeqTypeColor         | test_unknown_returns_white                   | Pass   |
|       | formatters.py::TestSetupStyles          | test_required_keys                           | Pass   |
|       | formatters.py::TestSetupStyles          | test_returns_stylesheet                      | Pass   |
|       | formatters.py::TestSetupStyles          | test_styles_are_paragraph_styles             | Pass   |
|       | generator.py::TestClaimReportGenerator  | test_claim_ref_contains_prop_id_suffix       | Pass   |
|       | generator.py::TestClaimReportGenerator  | test_generate_empty_sequence_lookup          | Pass   |
|       | generator.py::TestClaimReportGenerator  | test_generate_returns_bytes                  | Pass   |
|       | generator.py::TestClaimReportGenerator  | test_generate_without_mortgage               | Pass   |
|       | generator.py::TestClaimReportGenerator  | test_minimal_prop_data                       | Pass   |
|       | generator.py::TestClaimReportGenerator  | test_output_is_valid_pdf                     | Pass   |
|       | generator.py::TestClaimReportGenerator  | test_pdf_size_reasonable                     | Pass   |
|       | generator.py::TestCompatShim            | test_shim_class_is_same                      | Pass   |
|       | generator.py::TestCompatShim            | test_shim_imports_generator                  | Pass   |
|       | generator.py::TestPublicAPI             | test_all_exports                             | Pass   |
|       | generator.py::TestPublicAPI             | test_import_from_package                     | Pass   |
|       | pages1.3.py::TestPage1Cover             | test_contains_claim_ref_paragraph            | Pass   |
|       | pages1.3.py::TestPage1Cover             | test_no_flood_events_still_renders           | Pass   |
|       | pages1.3.py::TestPage1Cover             | test_non_float_coordinates                   | Pass   |
|       | pages1.3.py::TestPage1Cover             | test_returns_list                            | Pass   |
|       | pages1.3.py::TestPage1Cover             | test_stats_reflect_events                    | Pass   |
|       | pages1.3.py::TestPage1Cover             | test_without_mortgage                        | Pass   |
|       | pages1.3.py::TestPage2Chronology        | test_events_with_no_sequence_id              | Pass   |
|       | pages1.3.py::TestPage2Chronology        | test_no_flood_events                         | Pass   |
|       | pages1.3.py::TestPage2Chronology        | test_returns_list                            | Pass   |

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| Model | Module                                           | Test Name                                  | Status |
|-------|--------------------------------------------------|--------------------------------------------|--------|
|       | pages1.3.py::TestPage2Chronology                 | test_stats_bar_present                     | Pass   |
|       | pages1.3.py::TestPage2Chronology                 | test_table_included                        | Pass   |
|       | pages1.3.py::TestPage2Chronology                 | test_unknown_sequence_id_handled           | Pass   |
|       | pages1.3.py::TestPage3Damage                     | test_contains_tables                       | Pass   |
|       | pages1.3.py::TestPage3Damage                     | test_high_damage_rows_highlighted          | Pass   |
|       | pages1.3.py::TestPage3Damage                     | test_multiple_sequences_grouped            | Pass   |
|       | pages1.3.py::TestPage3Damage                     | test_no_flood_events                       | Pass   |
|       | pages1.3.py::TestPage3Damage                     | test_returns_list                          | Pass   |
|       | pages1.3.py::TestPage3Damage                     | test_zero_property_value                   | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_contains_tables                       | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_negative_equity_highlights_row        | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_no_flood_events_with_mortgage         | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_with_mortgage                         | Pass   |
|       | pages4.5.py::TestPage4Mortgage                   | test_without_mortgage                      | Pass   |
|       | pages4.5.py::TestPage5Determination              | test_contains_reference_table              | Pass   |
|       | pages4.5.py::TestPage5Determination              | test_damage_calculation                    | Pass   |
|       | pages4.5.py::TestPage5Determination              | test_negative_equity_flag                  | Pass   |
|       | pages4.5.py::TestPage5Determination              | test_returns_list                          | Pass   |
|       | pages4.5.py::TestPage5Determination              | test_signature_block_present               | Pass   |
|       | pages4.5.py::TestPage5Determination              | test_without_mortgage                      | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_building_page_calculated              | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_empty_property_does_not_crash         | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_get_property_id_missing_nested_key    | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_get_property_id_present               | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_has_tables                            | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_has_title_paragraph                   | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_proper_pid_in_output                  | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_proper_page_overview_section_header   | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_proper_page_summary_section_header    | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_returns_list                          | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_unknown_property_id_fallback          | Pass   |
|       | page_01_title_overview.py::TestTitleOverviewPage | test_with_page_data_does_not_crash         | Pass   |
|       | page_02_location.py::TestLocationPage            | test_address_subsection_header             | Pass   |
|       | page_02_location.py::TestLocationPage            | test_administrative_subsection_header      | Pass   |
|       | page_02_location.py::TestLocationPage            | test_area_classification_subsection_header | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_high                          | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_low                           | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_low_medium                    | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_medium                        | Pass   |
|       | page_02_location.py::TestLocationPage            | test_density_very_high                     | Pass   |
|       | page_02_location.py::TestLocationPage            | test_empty_property_does_not_crash         | Pass   |
|       | page_02_location.py::TestLocationPage            | test_geographic_subsection_header          | Pass   |
|       | page_02_location.py::TestLocationPage            | test_has_section_header                    | Pass   |
|       | page_02_location.py::TestLocationPage            | test_has_tables                            | Pass   |

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| Model | Module                                            | Test Name                                    | Status |
|-------|---------------------------------------------------|----------------------------------------------|--------|
|       | page_02_location.py::TestLocationPage             | test_no_coordinates_skips_geo_rows           | Pass   |
|       | page_02_location.py::TestLocationPage             | test_returns_list                            | Pass   |
|       | page_02_location.py::TestLocationPage             | test_with_mortgage_data_does_not_crash       | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_contemporary                       | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_historic                           | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_mature                             | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_modern                             | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_basic_info_subsection              | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_building_age_subsection            | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_condition_fair_recommendation      | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_condition_good_recommendation      | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_condition_poor_recommendation      | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_condition_unknown_falls_back       | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_empty_property_does_not_crash      | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_external_features_subsection       | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_section_header                     | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_tables                             | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_construction_year_skips_age        | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_physical_dimensions_subsection     | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_returns_list                       | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_room_composition_subsection        | Pass   |
|       | page_03_attributes.py::TestAttributePage          | test_page_high_mortgage_data_does_not_crash  | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_empty_time_series_returns_short_list    | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_page_has_title                    | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_page_gauge_name_does_not_crash    | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_time_series_message_included      | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_time_series_returns_short_list    | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_time_series_list                  | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_time_series_mixed_statuses        | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_time_series_with_thresholds       | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_time_series_returns_more_elements | Pass   |
|       | page_03_gauge_time_series.py::TestGaugeTimeSeries | test_gauge_time_series_table_included        | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_basics_year_shown                  | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_empty_properties_does_not_crash    | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_basics_year_shown                  | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_has_page_for_header                | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_basics                             | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_construction_data_returns_message  | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_construction_year                  | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_basics_with_data                   | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_basics_shown                       | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_drift_type_analysis                | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_type_analysis                      | Pass   |
|       | page_04_construction.py::TestConstructionPage     | test_page_drift_type_analysis                | Pass   |

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| Model | Module                                                                       | Test Name                                              | Status |
|-------|------------------------------------------------------------------------------|--------------------------------------------------------|--------|
|       | page_05_risk_assessment.py::TestRiskScreeningScenarioLevel                   | TestRiskScreeningScenarioLevel                         | Pass   |
|       | page_05_risk_assessment.py::TestRiskScreeningCalculationAdds1                | TestRiskScreeningCalculationAdds1                      | Pass   |
|       | page_05_risk_assessment.py::TestRiskScreeningCalculationFactorsMessage       | TestRiskScreeningCalculationFactorsMessage             | Pass   |
|       | page_05_risk_assessment.py::TestRiskScreeningCalculationAdds2                | TestRiskScreeningCalculationAdds2                      | Pass   |
|       | page_05_risk_assessment.py::TestRiskScreeningCalculationKeys                 | TestRiskScreeningCalculationKeys                       | Pass   |
|       | page_05_risk_assessment.py::TestRiskScreeningCalculationOverAdds2            | TestRiskScreeningCalculationOverAdds2                  | Pass   |
|       | page_05_risk_assessment.py::TestRiskScreeningCalculationRiskAdds4            | TestRiskScreeningCalculationRiskAdds4                  | Pass   |
|       | page_05_risk_assessment.py::TestRiskScreeningCalculationLevel                | TestRiskScreeningCalculationLevel                      | Pass   |
|       | page_05_risk_assessment.py::TestRiskScreeningCalculationAdds3                | TestRiskScreeningCalculationAdds3                      | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity0to500m                         | TestRiskProximity0to500m                               | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity500to1000m                      | TestRiskProximity500to1000m                            | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity1000to1500m                     | TestRiskProximity1000to1500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity1500to2000m                     | TestRiskProximity1500to2000m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity2000to2500m                     | TestRiskProximity2000to2500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity2500to3000m                     | TestRiskProximity2500to3000m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity3000to3500m                     | TestRiskProximity3000to3500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity3500to4000m                     | TestRiskProximity3500to4000m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity4000to4500m                     | TestRiskProximity4000to4500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity4500to5000m                     | TestRiskProximity4500to5000m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity5000to5500m                     | TestRiskProximity5000to5500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity5500to6000m                     | TestRiskProximity5500to6000m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity6000to6500m                     | TestRiskProximity6000to6500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity6500to7000m                     | TestRiskProximity6500to7000m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity7000to7500m                     | TestRiskProximity7000to7500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity7500to8000m                     | TestRiskProximity7500to8000m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity8000to8500m                     | TestRiskProximity8000to8500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity8500to9000m                     | TestRiskProximity8500to9000m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity9000to9500m                     | TestRiskProximity9000to9500m                           | Pass   |
|       | page_05_risk_assessment.py::TestRiskProximity9500to10000m                    | TestRiskProximity9500to10000m                          | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicPropertyDoesNotCrash             | TestFinancialPageBasicPropertyDoesNotCrash             | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicFinancialHeader                  | TestFinancialPageBasicFinancialHeader                  | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicTable                            | TestFinancialPageBasicTable                            | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicAreaSkipsValuePerSqM             | TestFinancialPageBasicAreaSkipsValuePerSqM             | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicPropertyValue                    | TestFinancialPageBasicPropertyValue                    | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicValuationData                    | TestFinancialPageBasicValuationData                    | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicValuationFieldWithValueInName    | TestFinancialPageBasicValuationFieldWithValueInName    | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicValuationFieldWithoutValueInName | TestFinancialPageBasicValuationFieldWithoutValueInName | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicList                             | TestFinancialPageBasicList                             | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicNonEmptyList                     | TestFinancialPageBasicNonEmptyList                     | Pass   |
|       | page_06_financial.py::TestFinancialPageBasicPerSqMShownWhenAreaPresent       | TestFinancialPageBasicPerSqMShownWhenAreaPresent       | Pass   |
|       | page_06_financial.py::TestInvestmentRatingAverage2to4pct                     | TestInvestmentRatingAverage2to4pct                     | Pass   |
|       | page_06_financial.py::TestInvestmentRatingLentOver8pct                       | TestInvestmentRatingLentOver8pct                       | Pass   |
|       | page_06_financial.py::TestInvestmentRating4to6pct                            | TestInvestmentRating4to6pct                            | Pass   |
|       | page_06_financial.py::TestInvestmentRating6to8pct                            | TestInvestmentRating6to8pct                            | Pass   |
|       | page_06_financial.py::TestInvestmentRatingUnder2pct                          | TestInvestmentRatingUnder2pct                          | Pass   |
|       | page_06_financial.py::TestMarketSegmentAnalysis1Under250k                    | TestMarketSegmentAnalysis1Under250k                    | Pass   |
|       | page_06_financial.py::TestMarketSegmentAnalysis250kto500k                    | TestMarketSegmentAnalysis250kto500k                    | Pass   |
|       | page_06_financial.py::TestMarketSegmentAnalysisOver1m                        | TestMarketSegmentAnalysisOver1m                        | Pass   |
|       | page_06_financial.py::TestMarketSegmentAnalysisIdle500kto1m                  | TestMarketSegmentAnalysisIdle500kto1m                  | Pass   |
|       | page_06_financial.py::TestTransactionMonthlyRentAndAnnualRent                | TestTransactionMonthlyRentAndAnnualRent                | Pass   |
|       | page_06_financial.py::TestTransactionTransactionData                         | TestTransactionTransactionData                         | Pass   |
|       | page_06_financial.py::TestTransactionFieldComputed                           | TestTransactionFieldComputed                           | Pass   |
|       | page_06_financial.py::TestTransactionDateShown                               | TestTransactionDateShown                               | Pass   |
|       | page_06_financial.py::TestTransactionPriceAndAppreciation                    | TestTransactionPriceAndAppreciation                    | Pass   |

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| Model | Module                              | Test Name                                             | Status |
|-------|-------------------------------------|-------------------------------------------------------|--------|
|       | page_06_financial.py::TestTransact  | test_SalesSection_rendered                            | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageNaturalMeasureRecommendation                 | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageNaturalMeasureSection_rendered               | Pass   |
|       | page_07_protection.py::TestProtect  | test_EmptyNoProperty_returns_list                     | Pass   |
|       | page_07_protection.py::TestProtect  | test_EmptyNoData_section_measures_dict_shows_fallback | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageNoData_section_measures_shows_fallback       | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageProtectionFlow_recommendation                | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRateProtection_recommendation_returns_dict   | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRecommendationMeasures_in_recommendations    | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRecommendationNatural_false                  | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRecommendationNatural_re_eligible            | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRecommendationMissing_critical               | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRecommendationSection_always_rendered        | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageInstalledMeasure_rating                      | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageSingleMeasure_results                        | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRadianceAllBands_80pct                       | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRadiance40pct                                | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRadiance60pct                                | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRadianceMed_20pct                            | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRadiancePoor_0pct                            | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRadianceResults                              | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRadianceResults                              | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRadianceMeasure_still_shown                  | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRiskAssessment_shown                         | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRiskAssessment_shown                         | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRiskAssessment_shown                         | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRiskAssessment_include_flood_re              | Pass   |
|       | page_07_protection.py::TestProtect  | test_PageRiskAssessment_section_rendered              | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_analyze_epc_rating_all_bands                     | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_analyze_epc_rating_case_insensitive              | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_energy_intensity_calculation                     | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_full_data_no_crash                               | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_no_energy_data_fallback_message                  | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_returns_list                                     | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_with_building_fabric                             | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_with_energy_usage                                | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_with_ratings_data                                | Pass   |
|       | page_08_energy.py::TestEnergyPage   | test_with_ratings_includes_table                      | Pass   |
|       | page_09_history.py::TestHistoryPage | test_assess_historical_risks_active_sub               | Pass   |
|       | page_09_history.py::TestHistoryPage | test_assess_historical_risks_high_env                 | Pass   |
|       | page_09_history.py::TestHistoryPage | test_assess_historical_risks_low_env                  | Pass   |
|       | page_09_history.py::TestHistoryPage | test_assess_historical_risks_minor_sub                | Pass   |
|       | page_09_history.py::TestHistoryPage | test_assess_historical_risks_moderate_env             | Pass   |

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| Model | Module                                                 | Test Name                                          | Status |
|-------|--------------------------------------------------------|----------------------------------------------------|--------|
|       | page_09_history.py::TestHistoryPage                    | test_assess_historical_risks_moderate_flood_damage | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_assess_historical_risks_no_flood_damage       | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_assess_historical_risks_severe_flood_damage   | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_assess_historical_risks_stable_ground         | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_build_flood_history_bad_json                  | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_build_flood_history_no_flood_days             | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_build_flood_history_no_hd_file                | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_build_flood_history_no_reference_gauges       | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_build_flood_history_with_flood_data           | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_empty_flood_events                            | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_empty_property_does_not_crash                 | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_has_history_header                            | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_no_flood_history                              | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_returns_list                                  | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_with_environmental_issues                     | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_with_fire_incidents                           | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_with_ground_conditions                        | Pass   |
|       | page_09_history.py::TestHistoryPage                    | test_with_mortgage_data                            | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_average                             | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_below_average                       | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_excellent                           | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_good                                | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_avg_yield_poor                                | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_empty_property_does_not_crash                 | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_empty_transactions                            | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_has_transactions_header                       | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_appreciation                  | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_few_vacancies                 | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_many_vacancies                | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_marketing_above_30            | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_marketing_above_90            | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_never_rented                  | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_no_vacancy                    | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_market_analysis_yield                         | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_history                          | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_transactions_list                             | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_extended_marketing                            | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_quick_sale                                    | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_very_long_marketing                           | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_with_mortgage_data                            | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_with_rental_data                              | Pass   |
|       | page_10_transactions.py::TestTransactionsPage          | test_with_singles_data                             | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_application_is_subject                        | Pass   |
|       | page_11_mortgage_overview.py::TestMortgageOverviewPage | test_when_default                                  | Pass   |

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| Model | Module                         | Test Name                                       | Status |
|-------|--------------------------------|-------------------------------------------------|--------|
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage.when not default       | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage.dict does not crash    | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage.accepted               | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage                        | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage.subsection             | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage loan and purchase      | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage subsection             | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage subsection             | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage returns message        | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage returns message        | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage data                   | Pass   |
|       | page_11_mortgage_overview.py:: | TestMortgageOverviewPage present                | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage field                   | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage subsection              | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage risk                    | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage subsection              | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage subsection              | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage of purchase calculation | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage risk                    | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage                         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage risk                    | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage returns message         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage returns message         | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage with data               | Pass   |
|       | page_11a_mortgage_details.py:: | TestMortgageDetailsPage present                 | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage subsection                | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                           | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                           | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage burden                    | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                           | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage high burden               | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage analysis                  | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage dict does not crash       | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage                           | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage reduces net cost          | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage terms returns message     | Pass   |
|       | page_11b_mortgage_costs.py::   | TestMortgageCostsPage data returns message      | Pass   |

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| Model | Module                                                                | Test Name                                   | Status |
|-------|-----------------------------------------------------------------------|---------------------------------------------|--------|
|       | page_11b_mortgage_costs.py::TestMortgageCostsPageSetupSubsection      | testMortgageCostsPageSetupSubsection        | Pass   |
|       | page_11b_mortgage_costs.py::TestMortgageCostsPageSubsection           | testMortgageCostsPageSubsection             | Pass   |
|       | page_11b_mortgage_costs.py::TestMortgageCostsPageWithData             | testMortgageCostsPageWithData               | Pass   |
|       | page_11b_mortgage_costs.py::TestMortgageCostsPagePresent              | testMortgageCostsPagePresent                | Pass   |
|       | page_11b_mortgage_costs.py::TestMortgageCostsPagePercentageOfLoan     | testMortgageCostsPagePercentageOfLoan       | Pass   |
|       | page_11b_mortgage_costs.py::TestMortgageCostsPageOverviewSubsection   | testMortgageCostsPageOverviewSubsection     | Pass   |
|       | page_11c_regulatory.py::TestContentSectionsTogether                   | testContentSectionsTogether                 | Pass   |
|       | page_11c_regulatory.py::TestContentSectionHandledGracefully           | testContentSectionHandledGracefully         | Pass   |
|       | page_11c_regulatory.py::TestContentSectionMortgageStructure           | testContentSectionMortgageStructure         | Pass   |
|       | page_11c_regulatory.py::TestContentRegulatedFlag                      | testContentRegulatedFlag                    | Pass   |
|       | page_11c_regulatory.py::TestContentBusinessPurposeFalse               | testContentBusinessPurposeFalse             | Pass   |
|       | page_11c_regulatory.py::TestContentBusinessPurposeTrue                | testContentBusinessPurposeTrue              | Pass   |
|       | page_11c_regulatory.py::TestContentRegulatorySectionReturnsElements   | testContentRegulatorySectionReturnsElements | Pass   |
|       | page_11c_regulatory.py::TestContentRegulatoryReferenceNumber          | testContentRegulatoryReferenceNumber        | Pass   |
|       | page_11c_regulatory.py::TestContentRegulatoryRegulatoryFlags          | testContentRegulatoryRegulatoryFlags        | Pass   |
|       | page_11c_regulatory.py::TestHmdaSectionMinimalFields                  | testHmdaSectionMinimalFields                | Pass   |
|       | page_11c_regulatory.py::TestHmdaSectionRateSpreadFormatted            | testHmdaSectionRateSpreadFormatted          | Pass   |
|       | page_11c_regulatory.py::TestHmdaSectionCreatesTable                   | testHmdaSectionCreatesTable                 | Pass   |
|       | page_11c_regulatory.py::TestMcoSectionAdviceRejectionReason           | testMcoSectionAdviceRejectionReason         | Pass   |
|       | page_11c_regulatory.py::TestMcoSectionAllFieldsReturnsList            | testMcoSectionAllFieldsReturnsList          | Pass   |
|       | page_11c_regulatory.py::TestMcoSectionCreatesTables                   | testMcoSectionCreatesTables                 | Pass   |
|       | page_11c_regulatory.py::TestMcoSectionWithMinimalFields               | testMcoSectionWithMinimalFields             | Pass   |
|       | page_11c_regulatory.py::TestErrorMessageMortgageMessage               | testErrorMessageMortgageMessage             | Pass   |
|       | page_11c_regulatory.py::TestErrorMessageMortgageReturnsList           | testErrorMessageMortgageReturnsList         | Pass   |
|       | page_11c_regulatory.py::TestErrorMessageRegulatoryReturnsMessage      | testErrorMessageRegulatoryReturnsMessage    | Pass   |
|       | page_11c_regulatory.py::TestErrorMessageRegulatoryList                | testErrorMessageRegulatoryList              | Pass   |
|       | page_12_current_status.py::TestContentBasicsPageBasics                | testContentBasicsPageBasics                 | Pass   |
|       | page_12_current_status.py::TestContentCurrentPayBasicShown            | testContentCurrentPayBasicShown             | Pass   |
|       | page_12_current_status.py::TestContentStatusPageHeader                | testContentStatusPageHeader                 | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasics                | testContentStatusPageBasics                 | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsShown           | testContentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsDateShown       | testContentStatusPageBasicsDateShown        | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsShown           | testContentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsShown           | testContentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsReturnsMessage  | testContentStatusPageBasicsReturnsMessage   | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsReturnsMessage  | testContentStatusPageBasicsReturnsMessage   | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBalanceShown          | testContentStatusPageBalanceShown           | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsShown           | testContentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsShown           | testContentStatusPageBasicsShown            | Pass   |
|       | page_12_current_status.py::TestContentStatusPageBasicsReturnsList     | testContentStatusPageBasicsReturnsList      | Pass   |
|       | page_12_current_status.py::TestLeverageRatioClassPercentageNotDecimal | testLeverageRatioClassPercentageNotDecimal  | Pass   |
|       | page_12_current_status.py::TestLeverageRatioLower60LowMedium          | testLeverageRatioLower60LowMedium           | Pass   |
|       | page_12_current_status.py::TestLeverageRatioLower70Medium             | testLeverageRatioLower70Medium              | Pass   |
|       | page_12_current_status.py::TestLeverageRatioLower80MediumHigh         | testLeverageRatioLower80MediumHigh          | Pass   |

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| Model | Module                                            | Test Name                                     | Status |
|-------|---------------------------------------------------|-----------------------------------------------|--------|
|       | page.12_current_status.py::TestLiveBranch         | TestLiveBranchLow_90_high_risk                | Pass   |
|       | page.12_current_status.py::TestLiveBranch         | TestLiveBranchUnder_60_low_risk               | Pass   |
|       | page.12_current_status.py::TestPastallHoliday     | TestPastallHolidayFieldsShown                 | Pass   |
|       | page.12_current_status.py::TestPastnoHoliday      | TestPastnoHolidayStakenShown                  | Pass   |
|       | page.12_current_status.py::TestPastallHtHoliday   | TestPastallHtHolidayDateShown                 | Pass   |
|       | page.12_current_status.py::TestPastnoHoliday      | TestPastnoHolidayDataNoSection                | Pass   |
|       | page.12_current_status.py::TestPastnoHoliday      | TestPastnoHolidayKdaysShown                   | Pass   |
|       | page.12_current_status.py::TestPastnearPerforssed | TestPastnearPerforssedPaymentsShown           | Pass   |
|       | page.12_current_status.py::TestPastnearPerforssed | TestPastnearPerforssedPaymentsPoor            | Pass   |
|       | page.12_current_status.py::TestPastnearPerforssed | TestPastnearPerforssedPaymentsConcerning      | Pass   |
|       | page.12_current_status.py::TestPastnearPerforssed | TestPastnearPerforssedPaymentsGood            | Pass   |
|       | page.12_current_status.py::TestPastnearPerforssed | TestPastnearPerforssedPaymentsExcellent       | Pass   |
|       | page.12_current_status.py::TestPastnearPerforssed | TestPastnearPerforssedPaymentsConcerning      | Pass   |
|       | page.12_current_status.py::TestPastnearPerforssed | TestPastnearPerforssedPaymentsGood            | Pass   |
|       | page.12_current_status.py::TestPastnearPerforssed | TestPastnearPerforssedPaymentsExcellent       | Pass   |
|       | page.13_risk_analysis.py::TestCreditRiskBranch    | TestCreditRiskBranchCredit_800_plus           | Pass   |
|       | page.13_risk_analysis.py::TestCreditRiskBranch    | TestCreditRiskBranchCredit_580_to_669         | Pass   |
|       | page.13_risk_analysis.py::TestCreditRiskBranch    | TestCreditRiskBranchCredit_670_to_739         | Pass   |
|       | page.13_risk_analysis.py::TestCreditRiskBranch    | TestCreditRiskBranchCredit_under_580          | Pass   |
|       | page.13_risk_analysis.py::TestCreditRiskBranch    | TestCreditRiskBranchCredit_740_to_799         | Pass   |
|       | page.13_risk_analysis.py::TestDebtStagnant        | TestDebtStagnantDti_under_28                  | Pass   |
|       | page.13_risk_analysis.py::TestDebtStagnant        | TestDebtStagnantDti_36_to_43                  | Pass   |
|       | page.13_risk_analysis.py::TestDebtStagnant        | TestDebtStagnantDti_28_to_36                  | Pass   |
|       | page.13_risk_analysis.py::TestDebtStagnant        | TestDebtStagnantDti_unknown                   | Pass   |
|       | page.13_risk_analysis.py::TestDebtStagnant        | TestDebtStagnantDti_over_43                   | Pass   |
|       | page.13_risk_analysis.py::TestHealthMonitor       | TestHealthMonitorMonitoringHighScore          | Pass   |
|       | page.13_risk_analysis.py::TestHealthMonitor       | TestHealthMonitorMonitoringLowScore           | Pass   |
|       | page.13_risk_analysis.py::TestHealthMonitor       | TestHealthMonitorMonitoringMediumScore        | Pass   |
|       | page.13_risk_analysis.py::TestHealthMonitor       | TestHealthMonitorRecommendationsHighScore     | Pass   |
|       | page.13_risk_analysis.py::TestHealthMonitor       | TestHealthMonitorRecommendationsLowScore      | Pass   |
|       | page.13_risk_analysis.py::TestHealthMonitor       | TestHealthMonitorRecommendationsMediumScore   | Pass   |
|       | page.13_risk_analysis.py::TestHealthMonitor       | TestHealthMonitorIdentifyKeyFactorsHighRisk   | Pass   |
|       | page.13_risk_analysis.py::TestHealthMonitor       | TestHealthMonitorIdentifyKeyFactorsMediumRisk | Pass   |
|       | page.13_risk_analysis.py::TestInterestRateBatch   | TestInterestRateBatchRate_5_to_7              | Pass   |
|       | page.13_risk_analysis.py::TestInterestRateBatch   | TestInterestRateBatchRate_over_7              | Pass   |
|       | page.13_risk_analysis.py::TestInterestRateBatch   | TestInterestRateBatchRate_under_3             | Pass   |
|       | page.13_risk_analysis.py::TestInterestRateBatch   | TestInterestRateBatchRate_3_to_5              | Pass   |
|       | page.13_risk_analysis.py::TestInterestRateBatch   | TestInterestRateBatchRate_unknown             | Pass   |
|       | page.13_risk_analysis.py::TestLTVRiskBranch       | TestLTVRiskBranchLtv_90_to_95                 | Pass   |
|       | page.13_risk_analysis.py::TestLTVRiskBranch       | TestLTVRiskBranchLtv_under_70                 | Pass   |
|       | page.13_risk_analysis.py::TestLTVRiskBranch       | TestLTVRiskBranchLtv_70_to_80                 | Pass   |
|       | page.13_risk_analysis.py::TestLTVRiskBranch       | TestLTVRiskBranchLtv_80_to_90                 | Pass   |
|       | page.13_risk_analysis.py::TestLTVRiskBranch       | TestLTVRiskBranchLtv_over_95                  | Pass   |

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| Model | Module                                               | Test Name                                     | Status |
|-------|------------------------------------------------------|-----------------------------------------------|--------|
|       | page.13_risk_analysis.py::TestOverview               | testRiskLevel                                 | Pass   |
|       | page.13_risk_analysis.py::TestOverview               | testRiskRisk                                  | Pass   |
|       | page.13_risk_analysis.py::TestOverview               | testRiskProperty_only_high_flood_risk         | Pass   |
|       | page.13_risk_analysis.py::TestOverview               | testRiskProperty_only_very_low_flood_risk     | Pass   |
|       | page.13_risk_analysis.py::TestPayment                | testTitleForeclosuresHomeLies                 | Pass   |
|       | page.13_risk_analysis.py::TestPayment                | testTitleForeclosuresPayments_over_3          | Pass   |
|       | page.13_risk_analysis.py::TestPayment                | testTitleForeclosuresPayments_good            | Pass   |
|       | page.13_risk_analysis.py::TestPayment                | testTitleForeclosuresPayments_1_to_3          | Pass   |
|       | page.13_risk_analysis.py::TestRiskAnalysis           | testAnatomyPageInside_with_mortgage           | Pass   |
|       | page.13_risk_analysis.py::TestRiskAnalysis           | testAnatomyPageProperty_does_not_crash        | Pass   |
|       | page.13_risk_analysis.py::TestRiskAnalysis           | testAnalysisPageHeader                        | Pass   |
|       | page.13_risk_analysis.py::TestRiskAnalysis           | testAnalysisPageBasics                        | Pass   |
|       | page.13_risk_analysis.py::TestRiskAnalysis           | testAnatomyPagePropertyBasics_mode            | Pass   |
|       | page.13_risk_analysis.py::TestRiskAnalysis           | testAnalysisPageBasics                        | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranching_retirement_over_55       | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_under_25                  | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_25_to_55                  | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_approaching_retirement    | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_daily_career              | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_years                     | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_high_for_poor_score       | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_low_for_high_score        | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_low_medium_for_good_score | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_medium_for_fair_score     | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_risk_established_2_to_5   | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_risk_recent_under_2       | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_risk_stable_5_plus        | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_risk_low_for_high_income  | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_risk_low_medium_adequate  | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_risk_medium_limited       | Pass   |
|       | page.14_borrower_profile.py::TestAgeRisk             | testAgeRiskBranches_risk                      | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageShown                  | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageHeader                 | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageBasics                 | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePagePosition_shown         | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageStatusShown            | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageBasics                 | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageBasics_returns_early   | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageBusiness_message       | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageStatusShown            | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageStatusShown            | Pass   |
|       | page.14_borrower_profile.py::TestBorrowerProfilePage | testBorrowerProfilePageReturns_list           | Pass   |
|       | page.14_borrower_profile.py::TestCreditScore         | testCreditScoreBranch800plus                  | Pass   |
|       | page.14_borrower_profile.py::TestCreditScore         | testCreditScoreBranch580_to_669               | Pass   |

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| Model                                            | Module                                                                         | Test Name                                             | Status |
|--------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------|--------|
| page_14_borrower_profile.py::TestCreditScore     | page_14_borrower_profile.py::TestCreditScoreBelow700                           | TestCreditScoreBelow700                               | Pass   |
|                                                  | page_14_borrower_profile.py::TestCreditScoreBetween500                         | TestCreditScoreBetween500                             | Pass   |
|                                                  | page_14_borrower_profile.py::TestCreditScoreBetween700to799                    | TestCreditScoreBetween700to799                        | Pass   |
|                                                  | page_14_borrower_profile.py::TestEmploymentLengthUnder1year                    | TestEmploymentLengthUnder1year                        | Pass   |
|                                                  | page_14_borrower_profile.py::TestEmploymentLength1to5years                     | TestEmploymentLength1to5years                         | Pass   |
|                                                  | page_14_borrower_profile.py::TestEmploymentLength5to20years                    | TestEmploymentLength5to20years                        | Pass   |
|                                                  | page_14_borrower_profile.py::TestEmploymentLength20to30years                   | TestEmploymentLength20to30years                       | Pass   |
|                                                  | page_14_borrower_profile.py::TestLoanAnalysisUnder30kto50k                     | TestLoanAnalysisUnder30kto50k                         | Pass   |
|                                                  | page_14_borrower_profile.py::TestLoanAnalysisUnder50kto100k                    | TestLoanAnalysisUnder50kto100k                        | Pass   |
|                                                  | page_14_borrower_profile.py::TestLoanAnalysisOver100k                          | TestLoanAnalysisOver100k                              | Pass   |
| page_14_borrower_profile.py::TestLoanAnalysis    | page_14_borrower_profile.py::TestLoanAnalysisUnder30k                          | TestLoanAnalysisUnder30k                              | Pass   |
|                                                  | page_14_borrower_profile.py::TestLoanAnalysisOver30kwithsecondary              | TestLoanAnalysisOver30kwithsecondary                  | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisCompletenesspercentage_computed           | TestAnalysisCompletenesspercentage_computed           | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataCompletenessincluded                  | TestAnalysisDataCompletenessincluded                  | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataCompletenessexcluded                  | TestAnalysisDataCompletenessexcluded                  | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataCompletenesssection_present           | TestAnalysisDataCompletenesssection_present           | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataCompletenesssection_excluded          | TestAnalysisDataCompletenesssection_excluded          | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataCompletenesssection_high              | TestAnalysisDataCompletenesssection_high              | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataCompletenesssection_low               | TestAnalysisDataCompletenesssection_low               | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataCompletenesssection_medium            | TestAnalysisDataCompletenesssection_medium            | Pass   |
| page_15_data_summary.py::TestAnalysisDataQuality | page_15_data_summary.py::TestAnalysisDataQualityover_90                        | TestAnalysisDataQualityover_90                        | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataQuality75to90                         | TestAnalysisDataQuality75to90                         | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataQuality40to60                         | TestAnalysisDataQuality40to60                         | Pass   |
|                                                  | page_15_data_summary.py::TestAnalysisDataQuality40                             | TestAnalysisDataQuality40                             | Pass   |
|                                                  | page_15_data_summary.py::TestBothSummaryPageBasics                             | TestBothSummaryPageBasics                             | Pass   |
|                                                  | page_15_data_summary.py::TestSummaryPageBasicsdoes_not_crash                   | TestSummaryPageBasicsdoes_not_crash                   | Pass   |
|                                                  | page_15_data_summary.py::TestSummaryPageBasics                                 | TestSummaryPageBasics                                 | Pass   |
|                                                  | page_15_data_summary.py::TestSummaryPageBasics                                 | TestSummaryPageBasics                                 | Pass   |
|                                                  | page_15_data_summary.py::TestSummaryPageBasics                                 | TestSummaryPageBasics                                 | Pass   |
|                                                  | page_15_data_summary.py::TestSummaryPageBasics                                 | TestSummaryPageBasics                                 | Pass   |
| page_15_data_summary.py::TestGeneratedData       | page_15_data_summary.py::TestGeneratedDatafrom50to75gets_enhancement_rec       | TestGeneratedDatafrom50to75gets_enhancement_rec       | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedDatafrom75to100gets_recommendation       | TestGeneratedDatafrom75to100gets_recommendation       | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedDataRecommendations                      | TestGeneratedDataRecommendations                      | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedDataRecommendations50gets_recommendation | TestGeneratedDataRecommendations50gets_recommendation | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedDataRecommendations                      | TestGeneratedDataRecommendations                      | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedReportMetadatatimestamp                  | TestGeneratedReportMetadatatimestamp                  | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedReportMetadataextracted                  | TestGeneratedReportMetadataextracted                  | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedReportMetadataextracted                  | TestGeneratedReportMetadataextracted                  | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedReportMetadataextracted                  | TestGeneratedReportMetadataextracted                  | Pass   |
|                                                  | page_15_data_summary.py::TestGeneratedReportMetadataextracted                  | TestGeneratedReportMetadataextracted                  | Pass   |

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| Model | Module                                                                                | Test Name                                                                     | Status |
|-------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|
|       | page_15_data_summary.py::TestGenerateReportMetadata                                   | test_generate_report_metadata                                                 | Pass   |
|       | page_15_data_summary.py::TestGenerateReportMetadataWithMortgage                       | test_generate_report_metadata_with_mortgage                                   | Pass   |
|       | page_15_data_summary.py::TestGenerateReportMetadataOnly                               | test_generate_report_metadata_only                                            | Pass   |
|       | page_15_data_summary.py::TestGenerateReportPropertyIdWhenMissing                      | test_generate_report_property_id_when_missing                                 | Pass   |
|       | page_15_data_summary.py::TestGenerateReportMetadata                                   | test_generate_report_metadata                                                 | Pass   |
|       | cli.py::TestErrorHandling                                                             | test_bad_property_file_exits_with_1                                           | Pass   |
|       | cli.py::TestErrorHandling                                                             | test_invalid_json_exits_with_1                                                | Pass   |
|       | cli.py::TestListCategories                                                            | test_list_categories_does_not_open_property_file                              | Pass   |
|       | cli.py::TestListCategories                                                            | test_list_categories_runs_without_error                                       | Pass   |
|       | cli.py::TestListPages                                                                 | test_list_pages_does_not_open_property_file                                   | Pass   |
|       | cli.py::TestListPages                                                                 | test_list_pages_runs_without_error                                            | Pass   |
|       | cli.py::TestOptionalLoading                                                           | test_mortgage_file_is_loaded                                                  | Pass   |
|       | cli.py::TestOptionalLoading                                                           | test_property_id_filters_without_mortgage                                     | Pass   |
|       | cli.py::TestOptionalLoading                                                           | test_property_id_with_mortgage_file                                           | Pass   |
|       | cli.py::TestReportGeneration                                                          | test_else_branch_triggers_except                                              | Pass   |
|       | cli.py::TestReportGeneration                                                          | test_report_type_mortgage_focused_with_mortgage                               | Pass   |
|       | cli.py::TestReportGeneration                                                          | test_report_type_mortgage_focused_without_mortgage_falls_through              | Pass   |
|       | cli.py::TestReportGeneration                                                          | test_report_type_property_only                                                | Pass   |
|       | cli.py::TestReportGeneration                                                          | test_report_type_risk_focused                                                 | Pass   |
|       | cli.py::TestReportGeneration                                                          | test_success_logging_executes                                                 | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportDoesNotRaise                  | test_generate_report_generate_report_does_not_raise                           | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportFalseSkipOpen                 | test_generate_report_generate_report_false_skip_open                          | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportTrueCallOpen                  | test_generate_report_generate_report_true_call_open                           | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportFalseFunctionToGenerateReport | test_generate_report_generate_report_false_function_to_generate_report        | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportFocusedTypeWithMortgage       | test_generate_report_generate_report_focused_type_with_mortgage               | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportFocusedTypeWithoutMortgage    | test_generate_report_generate_report_focused_type_without_mortgage_falls_back | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportOnlyTypeFunction              | test_generate_report_generate_report_only_type_function                       | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportPatternFunction               | test_generate_report_generate_report_pattern_function                         | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportRiskFocusedTypeFunction       | test_generate_report_generate_report_risk_focused_type_function               | Pass   |
|       | convenience.py::TestGeneratePropertyReportGenerateReportTypeFallsBackToGenerateReport | test_generate_report_generate_report_type_falls_back_to_generate_report       | Pass   |
|       | elements.py::TestGenerateElements                                                     | test_empty_page_list                                                          | Pass   |
|       | elements.py::TestGenerateElements                                                     | test_exception_in_page_module_continues                                       | Pass   |
|       | elements.py::TestGenerateElements                                                     | test_first_page_no_page_break                                                 | Pass   |
|       | elements.py::TestGenerateElements                                                     | test_mixed_valid_and_unknown_pages                                            | Pass   |
|       | elements.py::TestGenerateElements                                                     | test_mortgage_data_passed_to_page                                             | Pass   |
|       | elements.py::TestGenerateElements                                                     | test_multiple_pages_have_page_breaks                                          | Pass   |
|       | elements.py::TestGenerateElements                                                     | test_returns_list                                                             | Pass   |
|       | elements.py::TestGenerateElements                                                     | test_unknown_page_skipped                                                     | Pass   |
|       | elements.py::TestGenerateReport                                                       | test_all_unknown_pages_still_creates_pdf                                      | Pass   |
|       | elements.py::TestGenerateReport                                                       | test_auto_select_called_when_pages_none                                       | Pass   |
|       | elements.py::TestGenerateReport                                                       | test_auto_select_not_called_when_pages_provided                               | Pass   |
|       | elements.py::TestGenerateReport                                                       | test_custom_output_filename                                                   | Pass   |
|       | elements.py::TestGenerateReport                                                       | test_custom_pages_to_include                                                  | Pass   |
|       | elements.py::TestGenerateReport                                                       | test_empty_property_data                                                      | Pass   |

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| Model | Module                                   | Test Name                                           | Status |
|-------|------------------------------------------|-----------------------------------------------------|--------|
|       | elements.py::TestGenerateReport          | test_full_data_report                               | Pass   |
|       | elements.py::TestGenerateReport          | test_pdf_exists                                     | Pass   |
|       | elements.py::TestGenerateReport          | test_pdf_non_empty                                  | Pass   |
|       | elements.py::TestGenerateReport          | test_pdf_suffix                                     | Pass   |
|       | elements.py::TestGenerateReport          | test_returns_path                                   | Pass   |
|       | elements.py::TestGenerateReport          | test_with_mortgage_data                             | Pass   |
|       | finders.py::TestEdgeCases                | test_completely_empty_property_data                 | Pass   |
|       | finders.py::TestEdgeCases                | test_negative_value_property                        | Pass   |
|       | finders.py::TestEdgeCases                | test_pdf_content_is_bytes                           | Pass   |
|       | finders.py::TestEdgeCases                | test_property_data_with_none_header                 | Pass   |
|       | finders.py::TestEdgeCases                | test_report_with_all_pages                          | Pass   |
|       | finders.py::TestEdgeCases                | test_zero_value_property                            | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_dict_with_mortgages_found                      | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_dict_with_mortgages_not_found_returns_none     | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_dict_without_mortgages_key_returns_data_itself | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_empty_mortgages_list_returns_none              | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_invalid_type_returns_data_as_is                | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_list_input_found                               | Pass   |
|       | finders.py::TestFindMortgageByPropertyId | test_list_input_not_found_returns_none              | Pass   |
|       | finders.py::TestFindPropertyById         | test_dict_without_list_key_wraps_itself             | Pass   |
|       | finders.py::TestFindPropertyById         | test_empty_properties_list_raises                   | Pass   |
|       | finders.py::TestFindPropertyById         | test_invalid_type_raises_value_error                | Pass   |
|       | finders.py::TestFindPropertyById         | test_list_input_found                               | Pass   |
|       | finders.py::TestFindPropertyById         | test_list_not_found_raises                          | Pass   |
|       | finders.py::TestFindPropertyById         | test_none_raises_value_error                        | Pass   |
|       | finders.py::TestFindPropertyById         | test_not_found_raises_value_error                   | Pass   |
|       | finders.py::TestFindPropertyById         | test_portfolio_key_found                            | Pass   |
|       | finders.py::TestFindPropertyById         | test_properties_key_found                           | Pass   |
|       | finders.py::TestOpenPdfFile              | test_darwin_success                                 | Pass   |
|       | finders.py::TestOpenPdfFile              | test_generic_exception_returns_false                | Pass   |
|       | finders.py::TestOpenPdfFile              | test_linux_success                                  | Pass   |
|       | finders.py::TestOpenPdfFile              | test_subprocess_called_process_error_returns_false  | Pass   |
|       | finders.py::TestOpenPdfFile              | test_unknown_platform_uses_webbrowser               | Pass   |
|       | finders.py::TestOpenPdfFile              | test_windows_case_insensitive                       | Pass   |
|       | finders.py::TestOpenPdfFile              | test_windows_success                                | Pass   |
|       | init.py::TestAutoSelectPages             | test_always_includes_analysis_pages                 | Pass   |
|       | init.py::TestAutoSelectPages             | test_always_includes_property_pages                 | Pass   |
|       | init.py::TestAutoSelectPages             | test_empty_property_data                            | Pass   |
|       | init.py::TestAutoSelectPages             | test_with_mortgage_includes_mortgage_pages          | Pass   |
|       | init.py::TestAutoSelectPages             | test_without_mortgage_excludes_mortgage_pages       | Pass   |
|       | init.py::TestGenerateFilename            | test_contains_property_id                           | Pass   |
|       | init.py::TestGenerateFilename            | test_ends_with_pdf                                  | Pass   |
|       | init.py::TestGenerateFilename            | test_missing_property_id_uses_unknown               | Pass   |
|       | init.py::TestGenerateFilename            | test_partial_path_missing_header                    | Pass   |

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| Model | Module                                                                                        | Test Name                                                  | Status   |
|-------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------|----------|
|       | init.py::TestGenerateFilename                                                                 | test_partial_path_missing_property_id                      | Pass     |
|       | init.py::TestGenerateFilename                                                                 | test_returns_string                                        | Pass     |
|       | init.py::TestGenerateFilename                                                                 | test_timestamp_in_filename                                 | Pass     |
|       | init.py::TestPropertyReportGeneratorCategoriesDictInitialized                                 | test_categories_dict_initialized                           | Pass     |
|       | init.py::TestPropertyReportGeneratorImplicitOutputDirPath                                     | test_implicit_output_dir_path                              | Pass     |
|       | init.py::TestPropertyReportGeneratorImplicitOutputDirString                                   | test_implicit_output_dir_string                            | Pass     |
|       | init.py::TestPropertyReportGeneratorInitOutputDirUsesConfig                                   | test_init_output_dir_uses_config                           | Pass     |
|       | init.py::TestPropertyReportGeneratorInitOutputDirCreated                                      | test_init_output_dir_created                               | Pass     |
|       | init.py::TestPropertyReportGeneratorPropertyPagesDictInitialized                              | test_property_pages_dict_initialized                       | Pass     |
|       | specialised.py::TestHeaderFooter                                                              | test_header_footer_callable                                | Pass     |
|       | specialised.py::TestHeaderFooter                                                              | test_header_footer_draws_confidential                      | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorMortgageFocusedReportCustomFilename             | test_generate_mortgage_focused_report_custom_filename      | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorMortgageFocusedReportExists                     | test_generate_mortgage_focused_report_exists               | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorPropertyOnlyReportCustomFilename                | test_generate_property_only_report_custom_filename         | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorPropertyOnlyReportExists                        | test_generate_property_only_report_exists                  | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorRiskFocusedReportCustomFilename                 | test_generate_risk_focused_report_custom_filename          | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorRiskFocusedReportWithMortgage                   | test_generate_risk_focused_report_with_mortgage            | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorRiskFocusedReportWithoutMortgage                | test_generate_risk_focused_report_without_mortgage         | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorMortgageFocusedIncludesMortgageContextPages     | test_mortgage_focused_includes_mortgage_context_pages      | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorRiskFocusedWithMortgageIncludesCurrentStatus    | test_risk_focused_with_mortgage_includes_current_status    | Pass     |
|       | specialised.py::TestSpecialisedReportGeneratorRiskFocusedWithoutMortgageExcludesCurrentStatus | test_risk_focused_without_mortgage_excludes_current_status | Pass     |
|       | specialised.py::TestUtilityMethoddest_get_page_categories_returns_copy                        | test_dest_get_page_categories_returns_copy                 | Pass     |
|       | specialised.py::TestUtilityMethoddest_get_page_categories_returns_dict                        | test_dest_get_page_categories_returns_dict                 | Pass     |
|       | specialised.py::TestUtilityMethoddest_list_available_pages_contains_all_expected              | test_dest_list_available_pages_contains_all_expected       | Expected |
|       | specialised.py::TestUtilityMethoddest_list_available_pages_returns_list                       | test_dest_list_available_pages_returns_list                | Pass     |
|       | specialised.py::TestUtilityMethoddest_validate_pages_all_invalid                              | test_validate_pages_all_invalid                            | Pass     |
|       | specialised.py::TestUtilityMethoddest_validate_pages_all_valid                                | test_validate_pages_all_valid                              | Pass     |
|       | specialised.py::TestUtilityMethoddest_validate_pages_empty_list                               | test_validate_pages_empty_list                             | Pass     |
|       | specialised.py::TestUtilityMethoddest_validate_pages_mixed                                    | test_validate_pages_mixed                                  | Pass     |
|       | test_base_page_coverage.py::TestAbstractGreenwichSubclassesReturnsEmptyList                   | test_abstract_greenwich_subclasses_returns_empty_list      | Pass     |
|       | test_base_page_coverage.py::TestFormatCurrencyNoCurrencyNotSpecified                          | test_format_currency_no_currency_not_specified             | Pass     |
|       | test_base_page_coverage.py::TestFormatCurrencyNonNymDelegates                                 | test_format_currency_non_nym_delegates                     | Pass     |
|       | test_base_page_coverage.py::TestFormatPercentageNoneReturnsNotSpecified                       | test_format_percentage_none_returns_not_specified          | Pass     |
|       | test_base_page_coverage.py::TestFormatPageNumberDelegates                                     | test_format_page_number_delegates                          | Pass     |
|       | test_base_page_coverage.py::TestFormatPlainTextStringReturnsNotSpecified                      | test_format_plain_text_string_returns_not_specified        | Pass     |
|       | test_base_page_coverage.py::TestFormatTimespanStringReturnsNotSpecified                       | test_format_timespan_string_returns_not_specified          | Pass     |
|       | test_base_page_coverage.py::TestFormatValueReturnsNotSpecified                                | test_format_value_returns_not_specified                    | Pass     |
|       | test_base_page_coverage.py::TestPropertyBasePageTitleExists                                   | test_property_base_page_title_exists                       | Pass     |
|       | test_generator_direct.py::TestGeneratorDirectExceptionDoesNotRaise                            | test_generator_direct_exception_does_not_raise             | Pass     |
|       | test_generator_direct.py::TestGeneratorDirectFalseSkips                                       | test_generator_direct_false_skips                          | Pass     |
|       | test_generator_direct.py::TestGeneratorDirectTrueTries                                        | test_generator_direct_true_tries                           | Pass     |
|       | test_generator_direct.py::TestGeneratorDirectMortgageFocusedType                              | test_generator_direct_mortgage_focused_type                | Pass     |
|       | test_generator_direct.py::TestGeneratorDirectPropertyReportType                               | test_generator_direct_property_report_type                 | Pass     |
|       | test_generator_direct.py::TestGeneratorDirectRiskReportType                                   | test_generator_direct_risk_report_type                     | Pass     |

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| Model | Module                                                                     | Test Name                                           | Status |
|-------|----------------------------------------------------------------------------|-----------------------------------------------------|--------|
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_unknown_type_falls_back                        | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_return_selected_page_direct_with_mortgage      | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_return_selected_page_direct_without_mortgage   | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_generate_report_direct_exception_continues     | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_generate_report_direct_skips_unknown           | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_generate_report_direct_unknown_id              | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_generate_report_direct_focused_report          | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_generate_report_direct_only_report             | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_generate_report_direct_report_no_mortgage      | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_generate_report_direct_report_with_mortgage    | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_report_generated_direct                        | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_report_generated_pages                         | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_report_generated_directconfig                  | Pass   |
|       | test_generator_direct.py::TestGenerateReportByProperty                     | test_report_generated_direct                        | Pass   |
|       | test_helpers.py::TestFindMortgagesByPriority                               | test_direct_with_mortgages_found                    | Pass   |
|       | test_helpers.py::TestFindMortgagesByPriority                               | test_direct_with_mortgages_not_found_returns_none   | Pass   |
|       | test_helpers.py::TestFindMortgagesByPriority                               | test_priority_type_returns_data                     | Pass   |
|       | test_helpers.py::TestFindMortgagesByPriority                               | test_priority_not_found                             | Pass   |
|       | test_helpers.py::TestFindPropertiesByDict                                  | test_dict_without_properties_wraps_in_list          | Pass   |
|       | test_helpers.py::TestFindPropertiesByDict                                  | test_invalid_type_raises                            | Pass   |
|       | test_helpers.py::TestFindPropertiesByDict                                  | test_list_input                                     | Pass   |
|       | test_helpers.py::TestFindPropertiesByDict                                  | test_not_found_raises                               | Pass   |
|       | test_helpers.py::TestFindPropertiesByDict                                  | test_portfolio_key                                  | Pass   |
|       | test_helpers.py::TestFindPropertiesByDict                                  | test_properties_key                                 | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                           | test_darwin_success                                 | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                           | test_generic_exception_returns_false                | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                           | test_linux_success                                  | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                           | test_subprocess_failure_returns_false               | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                           | test_unknown_platform_uses_webbrowser               | Pass   |
|       | test_helpers.py::TestOpenPdfFile                                           | test_windows_success                                | Pass   |
|       | test_integrator.py::TestGenerateReportByProperty                           | test_report_by_property_returns_path                | Pass   |
|       | test_integrator.py::TestGenerateReportByProperty                           | test_report_by_property_file_returns_none           | Pass   |
|       | test_integrator.py::TestGenerateReportByProperty                           | test_report_by_property_file_still_generates        | Pass   |
|       | test_integrator.py::TestGenerateReportByProperty                           | test_report_by_property_contains_property_id        | Pass   |
|       | test_integrator.py::TestGenerateReportByProperty                           | test_report_by_property_created                     | Pass   |
|       | test_integrator.py::TestGenerateReportByProperty                           | test_report_by_property_file_created                | Pass   |
|       | test_integrator.py::TestGenerateReportByProperty                           | test_report_by_property_not_found_returns_none      | Pass   |
|       | test_integrator_exception.py::TestGenerateReportByPropertyFileRequiresNone | test_generate_report_by_property_file_requires_none | Pass   |
|       | test_integrator_exception.py::TestGenerateReportByPropertyFileRequiresNone | test_generate_report_by_property_file_requires_none | Pass   |
|       | generation.py::TestGenerateRiskReport                                      | test_analysis_type                                  | Pass   |
|       | generation.py::TestGenerateRiskReport                                      | test_basic_type                                     | Pass   |
|       | generation.py::TestGenerateRiskReport                                      | test_detailed_type                                  | Pass   |
|       | generation.py::TestGenerateRiskReport                                      | test_summary_type                                   | Pass   |
|       | generation.py::TestGenerateRiskReport                                      | test_unknown_type_fallback                          | Pass   |

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| Model                        | Module                                                         | Test Name                                    | Status |
|------------------------------|----------------------------------------------------------------|----------------------------------------------|--------|
| Model 1                      | generation.py::TestRiskReportEdgeCases                         | test_select_pages_returns_list               | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_empty_gauge_data                        | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_generate_elements_exception_continues   | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_generate_elements_skips_unknown         | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_get_page_categories                     | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_high_risk_portfolio_report              | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_list_available_pages                    | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_minimal_data_report                     | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_missing_mortgage_data                   | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_output_dir_uses_config                  | Pass   |
|                              | generation.py::TestRiskReportEdgeCases                         | test_validate_pages_valid_and_invalid        | Pass   |
|                              | generation.py::TestRiskReportGeneration                        | test_analysis_report_generation              | Pass   |
|                              | generation.py::TestRiskReportGeneration                        | test_basic_report_generation                 | Pass   |
|                              | generation.py::TestRiskReportGeneration                        | test_custom_filename                         | Pass   |
|                              | generation.py::TestRiskReportGeneration                        | test_custom_page_selection                   | Pass   |
|                              | generation.py::TestRiskReportGeneration                        | test_detailed_report_generation              | Pass   |
|                              | generation.py::TestRiskReportGeneration                        | test_summary_report_generation               | Pass   |
|                              | cli.py::TestErrorHandling                                      | test_bad_flood_file_exits_with_1             | Pass   |
|                              | cli.py::TestErrorHandling                                      | test_invalid_json_exits_with_1               | Pass   |
|                              | cli.py::TestListCategories                                     | test_list_categories_iterates_all_categories | Pass   |
|                              | cli.py::TestListCategories                                     | test_list_categories_runs_without_error      | Pass   |
|                              | cli.py::TestListPages                                          | test_list_pages_does_not_open_flood_file     | Pass   |
|                              | cli.py::TestListPages                                          | test_list_pages_runs_without_error           | Pass   |
|                              | cli.py::TestReportGeneration                                   | test_logging_after_generation                | Pass   |
|                              | cli.py::TestReportGeneration                                   | test_report_type_analysis                    | Pass   |
|                              | cli.py::TestReportGeneration                                   | test_report_type_basic                       | Pass   |
|                              | cli.py::TestReportGeneration                                   | test_report_type_detailed                    | Pass   |
| cli.py::TestReportGeneration | test_report_type_else_branch                                   | Pass                                         |        |
| cli.py::TestReportGeneration | test_report_type_summary                                       | Pass                                         |        |
| Model 2                      | convenience.py::TestEdgeCases                                  | test_all_unknown_pages_still_creates_pdf     | Pass   |
|                              | convenience.py::TestEdgeCases                                  | test_completely_empty_flood_data             | Pass   |
|                              | convenience.py::TestEdgeCases                                  | test_large_property_risk_dataset             | Pass   |
|                              | convenience.py::TestEdgeCases                                  | test_missing_summary_key                     | Pass   |
|                              | convenience.py::TestEdgeCases                                  | test_negative_values_in_summary              | Pass   |
|                              | convenience.py::TestEdgeCases                                  | test_none_gauge_data_key                     | Pass   |
|                              | convenience.py::TestEdgeCases                                  | test_report_with_all_pages                   | Pass   |
|                              | convenience.py::TestEdgeCases                                  | test_single_page_report                      | Pass   |
|                              | convenience.py::TestEdgeCases                                  | test_zero_properties_at_risk                 | Pass   |
|                              | convenience.py::TestGenerateRiskReportAndOverviewTypeFunctions | test_report_and_overview_type_functions_path | Pass   |
|                              | convenience.py::TestGenerateRiskReportBasicTypeFunction        | test_report_basic_type_returns_path          | Pass   |
|                              | convenience.py::TestGenerateRiskReportDefaultTypeFunction      | test_report_default_report_type_is_basic     | Pass   |
|                              | convenience.py::TestGenerateRiskReportDetailedTypeFunction     | test_report_detailed_type_returns_path       | Pass   |
|                              | convenience.py::TestGenerateRiskReportOutputDirAsString        | test_report_output_dir_as_string             | Pass   |
|                              | convenience.py::TestGenerateRiskReportPdfIsNonEmptyBytesOnDisk | test_report_pdf_is_non_empty_bytes_on_disk   | Pass   |

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| Model | Module                                        | Test Name                                     | Status |
|-------|-----------------------------------------------|-----------------------------------------------|--------|
|       | convenience.py::TestGenerateRiskReport        | test_output_dir_type_of_output_path           | Pass   |
|       | convenience.py::TestGenerateRiskReport        | test_unknown_type_fails_to_generate_report    | Pass   |
|       | elements.py::TestGenerateElement              | test_empty_page_list                          | Pass   |
|       | elements.py::TestGenerateElement              | test_exception_in_page_continues              | Pass   |
|       | elements.py::TestGenerateElement              | test_first_page_no_page_break                 | Pass   |
|       | elements.py::TestGenerateElement              | test_mixed_valid_and_unknown_pages            | Pass   |
|       | elements.py::TestGenerateElement              | test_multiple_pages_inserts_page_breaks       | Pass   |
|       | elements.py::TestGenerateElement              | test_returns_list                             | Pass   |
|       | elements.py::TestGenerateElement              | test_unknown_page_skipped                     | Pass   |
|       | elements.py::TestGenerateReport               | test_auto_select_pages_when_none              | Pass   |
|       | elements.py::TestGenerateReport               | test_auto_select_skipped_when_provided        | Pass   |
|       | elements.py::TestGenerateReport               | test_custom_output_filename                   | Pass   |
|       | elements.py::TestGenerateReport               | test_custom_pages_to_include                  | Pass   |
|       | elements.py::TestGenerateReport               | test_empty_flood_data                         | Pass   |
|       | elements.py::TestGenerateReport               | test_full_data_report                         | Pass   |
|       | elements.py::TestGenerateReport               | test_pdf_exists                               | Pass   |
|       | elements.py::TestGenerateReport               | test_pdf_non_empty                            | Pass   |
|       | elements.py::TestGenerateReport               | test_pdf_suffix                               | Pass   |
|       | elements.py::TestGenerateReport               | test_returns_path                             | Pass   |
|       | init.py::TestAutoSelectPages                  | test_all_pages_exist                          | Pass   |
|       | init.py::TestAutoSelectPages                  | test_always_includes_appendix                 | Pass   |
|       | init.py::TestAutoSelectPages                  | test_empty_flood_data                         | Pass   |
|       | init.py::TestAutoSelectPages                  | test_includes_analysis_pages                  | Pass   |
|       | init.py::TestAutoSelectPages                  | test_includes_overview_pages                  | Pass   |
|       | init.py::TestAutoSelectPages                  | test_returns_list                             | Pass   |
|       | init.py::TestGenerateFilename                 | test_contains_flood_risk_report               | Pass   |
|       | init.py::TestGenerateFilename                 | test_contains_timestamp                       | Pass   |
|       | init.py::TestGenerateFilename                 | test_ends_with_pdf                            | Pass   |
|       | init.py::TestGenerateFilename                 | test_returns_string                           | Pass   |
|       | init.py::TestGenerateFilename                 | test_two_calls_produce_string_names           | Pass   |
|       | init.py::TestRiskReportGenerator              | test_categories_dict_initialized              | Pass   |
|       | init.py::TestRiskReportGenerator              | test_explicit_output_dir_path                 | Pass   |
|       | init.py::TestRiskReportGenerator              | test_explicit_output_dir_string               | Pass   |
|       | init.py::TestRiskReportGenerator              | test_no_output_dir_uses_config                | Pass   |
|       | init.py::TestRiskReportGenerator              | test_output_dir_created                       | Pass   |
|       | init.py::TestRiskReportGenerator              | test_pages_dict_initialized                   | Pass   |
|       | specialised.py::TestHeaderFooter              | test_header_footer_callable                   | Pass   |
|       | specialised.py::TestHeaderFooter              | test_header_footer_draws_strings              | Pass   |
|       | specialised.py::TestSpecialisedRequestManager | test_methods_detailed_page_counts             | Pass   |
|       | specialised.py::TestSpecialisedRequestManager | test_generate_analysis_report_custom_filename | Pass   |
|       | specialised.py::TestSpecialisedRequestManager | test_generate_analysis_report_exists          | Pass   |
|       | specialised.py::TestSpecialisedRequestManager | test_generate_basic_report_exists             | Pass   |
|       | specialised.py::TestSpecialisedRequestManager | test_generate_basic_report_non_empty          | Pass   |
|       | specialised.py::TestSpecialisedRequestManager | test_generate_detailed_report_exists          | Pass   |

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| Model | Module                                          | Test Name                                              | Status |
|-------|-------------------------------------------------|--------------------------------------------------------|--------|
|       | specialised.py::TestSpecialisedRequestGenerator | test_generate_detailed_report_includes_first_page_page | Pass   |
|       | specialised.py::TestSpecialisedRequestGenerator | test_generate_summary_report_custom_filename           | Pass   |
|       | specialised.py::TestSpecialisedRequestGenerator | test_generate_summary_report_exists                    | Pass   |
|       | specialised.py::TestUtilityMethod               | test_get_page_categories_contains_expected_keys        | Pass   |
|       | specialised.py::TestUtilityMethod               | test_get_page_categories_returns_copy                  | Pass   |
|       | specialised.py::TestUtilityMethod               | test_get_page_categories_returns_dict                  | Pass   |
|       | specialised.py::TestUtilityMethod               | test_list_available_pages_contains_all_expected        | Pass   |
|       | specialised.py::TestUtilityMethod               | test_list_available_pages_non_empty                    | Pass   |
|       | specialised.py::TestUtilityMethod               | test_list_available_pages_returns_list                 | Pass   |
|       | specialised.py::TestUtilityMethod               | test_validate_pages_all_invalid                        | Pass   |
|       | specialised.py::TestUtilityMethod               | test_validate_pages_all_valid                          | Pass   |
|       | specialised.py::TestUtilityMethod               | test_validate_pages_empty_list                         | Pass   |
|       | specialised.py::TestUtilityMethod               | test_validate_pages_mixed                              | Pass   |
|       | utilities.py::TestConvenienceFunction           | test_generate_risk_report_basic                        | Pass   |
|       | utilities.py::TestConvenienceFunction           | test_generate_risk_report_default                      | Pass   |
|       | utilities.py::TestConvenienceFunction           | test_generate_risk_report_detailed                     | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_count_non_numeric                          | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_count_numeric                              | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_currency_non_numeric                       | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_currency_numeric                           | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_depth_non_numeric                          | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_depth_numeric                              | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_field_name_camel_case                      | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_field_name_replacements                    | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_percentage_non_numeric                     | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_percentage_numeric                         | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_value_bool_false                           | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_value_bool_true                            | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_value_empty_string                         | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_value_float_decimal                        | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_value_float_integer                        | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_value_int                                  | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_value_non_empty_string                     | Pass   |
|       | utilities.py::TestRiskBasePageFormat            | test_format_value_none                                 | Pass   |
|       | utilities.py::TestRiskReportUtility             | test_get_page_categories                               | Pass   |
|       | utilities.py::TestRiskReportUtility             | test_list_available_pages                              | Pass   |
|       | utilities.py::TestRiskReportUtility             | test_validate_pages_invalid                            | Pass   |
|       | utilities.py::TestRiskReportUtility             | test_validate_pages_valid                              | Pass   |
|       | utilities.py::TestRiskReportWorkflow            | test_complete_workflow_all_types                       | Pass   |
|       | utilities.py::TestRiskReportWorkflow            | test_complete_workflow_basic                           | Pass   |
|       | eod_pages.py::TestEODChartsPage                 | test_insufficient_data_message                         | Pass   |
|       | eod_pages.py::TestEODChartsPage                 | test_returns_list                                      | Pass   |
|       | eod_pages.py::TestEODChartsPage                 | test_single_entry_insufficient                         | Pass   |
|       | eod_pages.py::TestEODChartsPage                 | test_sufficient_data_creates_content                   | Pass   |

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| Model | Module                                                         | Test Name                               | Status |
|-------|----------------------------------------------------------------|-----------------------------------------|--------|
|       | eod_pages.py::TestEODPnLDetailPage                             | test_many_market_movers_top_20          | Pass   |
|       | eod_pages.py::TestEODPnLDetailPage                             | test_market_move_pnl_creates_table      | Pass   |
|       | eod_pages.py::TestEODPnLDetailPage                             | test_new_trade_pnl_creates_table        | Pass   |
|       | eod_pages.py::TestEODPnLDetailPage                             | test_no_positions_shows_messages        | Pass   |
|       | eod_pages.py::TestEODPnLDetailPage                             | test_returns_list                       | Pass   |
|       | eod_pages.py::TestEODPositionsPage                             | test_long_gauge_id_truncated            | Pass   |
|       | eod_pages.py::TestEODPositionsPage                             | test_multiple_gauges_grouped            | Pass   |
|       | eod_pages.py::TestEODPositionsPage                             | test_no_positions_shows_message         | Pass   |
|       | eod_pages.py::TestEODPositionsPage                             | test_returns_list                       | Pass   |
|       | eod_pages.py::TestEODPositionsPage                             | test_single_position                    | Pass   |
|       | eod_pages.py::TestEODPositionsPage                             | test_with_positions_creates_table       | Pass   |
|       | eod_pages.py::TestEODPositionsPage                             | test_zero_pnl_renders_dash              | Pass   |
|       | eod_pages.py::TestEODReportGenerator                           | test_eod_id_in_filename                 | Pass   |
|       | eod_pages.py::TestEODReportGenerator                           | test_generate_empty_snapshot            | Pass   |
|       | eod_pages.py::TestEODReportGenerator                           | test_generate_report_returns_path       | Pass   |
|       | eod_pages.py::TestEODReportGenerator                           | test_generate_report_without_pnl_series | Pass   |
|       | eod_pages.py::TestEODReportGenerator                           | test_no_output_dir_uses_config          | Pass   |
|       | eod_pages.py::TestEODReportGenerator                           | test_output_dir_created                 | Pass   |
|       | eod_pages.py::TestEODReportGenerator                           | test_pdf_not_empty                      | Pass   |
|       | eod_pages.py::TestEODSummaryPage                               | test_date_appears_in_paragraph          | Pass   |
|       | eod_pages.py::TestEODSummaryPage                               | test_empty_summary                      | Pass   |
|       | eod_pages.py::TestEODSummaryPage                               | test_has_paragraph_and_table            | Pass   |
|       | eod_pages.py::TestEODSummaryPage                               | test_negative_pnl_snapshot              | Pass   |
|       | eod_pages.py::TestEODSummaryPage                               | test_returns_list                       | Pass   |
|       | eod_pages.py::TestPnLDetailZeroPnL                             | test_zero_pnl_returns_dash              | Pass   |
|       | test_eod_charts_coverage.py::TestDailyEmptyChartFlypyData      | test_returns_drawing                    | Pass   |
|       | test_eod_charts_coverage.py::TestDailyPnLChartPositiveNegative | test_renders                            | Pass   |
|       | test_eod_charts_coverage.py::TestDailyRegimeChartNegative      | test_renders                            | Pass   |
|       | test_eod_charts_coverage.py::TestDailyClosingPositions         | test_key_returns_drawing                | Pass   |
|       | test_eod_charts_coverage.py::TestDailyRegimeChartEmpty         | test_renders_correctly                  | Pass   |
|       | test_eod_charts_coverage.py::TestDailyClosingChart             | test_returns_drawing                    | Pass   |
|       | test_eod_charts_coverage.py::TestDailyClosingChartWithZeroPnL  | test_returns_drawing                    | Pass   |
|       | test_eod_charts_coverage.py::TestDailyPnLChartFlypyData        | test_returns_drawing                    | Pass   |
|       | open_pdf.py::TestOpenPdfFile                                   | test_called_process_error_returns_false | Pass   |
|       | open_pdf.py::TestOpenPdfFile                                   | test_darwin_calls_open                  | Pass   |
|       | open_pdf.py::TestOpenPdfFile                                   | test_generic_exception_returns_false    | Pass   |
|       | open_pdf.py::TestOpenPdfFile                                   | test_linux_calls_xdg_open               | Pass   |
|       | open_pdf.py::TestOpenPdfFile                                   | test_unknown_platform_uses_webbrowser   | Pass   |
|       | open_pdf.py::TestOpenPdfFile                                   | test_windows_calls_start                | Pass   |
|       | catchment.py::TestServeCatchmentSet                            | test_returns_404_when_file_missing      | Pass   |
|       | catchment.py::TestServeCatchmentSet                            | test_returns_html_when_file_exists      | Pass   |
|       | catchment.py::TestServeCatchmentSet                            | test_returns_json_error_or_html         | Pass   |
|       | catchment.py::TestSetCatchment                                 | test_case_insensitive_thames            | Pass   |
|       | catchment.py::TestSetCatchment                                 | test_missing_catchment_id_returns_400   | Pass   |

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| Model | Module                                  | Test Name                                  | Status |
|-------|-----------------------------------------|--------------------------------------------|--------|
|       | catchment.py::TestSetCatchment          | test_no_json_returns_400                   | Pass   |
|       | catchment.py::TestSetCatchment          | test_null_catchment_id_returns_400         | Pass   |
|       | catchment.py::TestSetCatchment          | test_options_returns_200                   | Pass   |
|       | catchment.py::TestSetCatchment          | test_unknown_catchment_returns_400         | Pass   |
|       | catchment.py::TestSetCatchment          | test_valid_thames_returns_success          | Pass   |
|       | counterparty.py::TestCounterparty       | test_get_counterparty_error_returns_500    | Pass   |
|       | counterparty.py::TestCounterparty       | test_list_counterparties_error_returns_500 | Pass   |
|       | counterparty.py::TestGetCounterparty    | test_found_returns_200                     | Pass   |
|       | counterparty.py::TestGetCounterparty    | test_found_second_counterparty             | Pass   |
|       | counterparty.py::TestGetCounterparty    | test_not_found_returns_404                 | Pass   |
|       | counterparty.py::TestListCounterparties | test_no_file_returns_empty_list            | Pass   |
|       | counterparty.py::TestListCounterparties | test_summary_fields                        | Pass   |
|       | counterparty.py::TestListCounterparties | test_with_data_returns_summary             | Pass   |
|       | cross_entity.py::TestCrossEntity        | test_list_huges_have_timeseries            | Pass   |
|       | cross_entity.py::TestCrossEntity        | test_list_huges_hazard_no_500              | Pass   |
|       | cross_entity.py::TestCrossEntity        | test_list_properties_have_flood_data       | Pass   |
|       | cross_entity.py::TestInteractivity      | test_dashboard_menus_renders               | Pass   |
|       | cross_entity.py::TestInteractivity      | test_gauge_hazard_curve_renders            | Pass   |
|       | cross_entity.py::TestInteractivity      | test_gauge_pdf_panel_renders               | Pass   |
|       | cross_entity.py::TestInteractivity      | test_gauge_storm_analysis_renders          | Pass   |
|       | cross_entity.py::TestInteractivity      | test_notifications_renders                 | Pass   |
|       | cross_entity.py::TestInteractivity      | test_property_hazard_curve_renders         | Pass   |
|       | cross_entity.py::TestInteractivity      | test_property_pdf_panel_renders            | Pass   |
|       | cross_entity.py::TestInteractivity      | test_property_storm_analysis_renders       | Pass   |
|       | cross_entity.py::TestInteractivity      | test_property_storm_js_guards_empty_events | Pass   |
|       | gauge.py::TestDataFormatValidation      | test_gaugehc_file_structure                | Pass   |
|       | gauge.py::TestDataFormatValidation      | test_gaugets_file_structure                | Pass   |
|       | gauge.py::TestDataFormatValidation      | test_propertyhc_file_structure             | Pass   |
|       | gauge.py::TestDataFormatValidation      | test_propertyts_file_structure             | Pass   |
|       | gauge.py::TestGaugeHazardRoutes         | test_gauge_hazard_existing                 | Pass   |
|       | gauge.py::TestGaugeHazardRoutes         | test_gauge_hazard_gev_params               | Pass   |
|       | gauge.py::TestGaugeHazardRoutes         | test_gauge_hazard_nonexistent_gauge        | Pass   |
|       | gauge.py::TestGaugeHazardRoutes         | test_gauge_hazard_not_found                | Pass   |
|       | gauge.py::TestGaugeHazardRoutes         | test_gauge_storms_existing                 | Pass   |
|       | gauge.py::TestGaugeIDConsistency        | test_gaugehc_ids_match_gauge_portfolio     | Pass   |
|       | gauge.py::TestGaugeIDConsistency        | test_gaugets_ids_match_gauge_portfolio     | Pass   |
|       | gauge.py::TestGaugeIDConsistency        | test_gaugets_internal_id_matches_filename  | Pass   |
|       | property.py::TestMissingDataHandling    | test_property_hazard_no_file               | Pass   |
|       | property.py::TestMissingDataHandling    | test_propertyhc_summary_no_file            | Pass   |
|       | property.py::TestMissingDataHandling    | test_propertyts_summary_no_dir             | Pass   |
|       | property.py::TestPropertyHCRoutes       | test_property_hazard_existing              | Pass   |
|       | property.py::TestPropertyHCRoutes       | test_property_hazard_nonexistent           | Pass   |
|       | property.py::TestPropertyHCRoutes       | test_property_hazard_not_in_curves         | Pass   |
|       | property.py::TestPropertyHCRoutes       | test_propertyhc_basis                      | Pass   |

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| Model | Module                                   | Test Name                                         | Status |
|-------|------------------------------------------|---------------------------------------------------|--------|
|       | property.py::TestPropertyHCRoutes        | test_propertyhc_summary                           | Pass   |
|       | property.py::TestPropertyIDConformance   | test_propertyhc_ids_match_property_ports          | Fail   |
|       | property.py::TestPropertyIDConformance   | test_propertyhc_ids_subset_of_property_ports      | Pass   |
|       | property.py::TestPropertyIDConformance   | test_propertyts_ids_match_property_ports          | Fail   |
|       | property.py::TestPropertyTSRoutes        | test_list_flood_storms                            | Pass   |
|       | property.py::TestPropertyTSRoutes        | test_list_flood_storms_includes_non_floods        | Fail   |
|       | property.py::TestPropertyTSRoutes        | test_property_floods_existing                     | Pass   |
|       | property.py::TestPropertyTSRoutes        | test_property_floods_few_events                   | Pass   |
|       | property.py::TestPropertyTSRoutes        | test_property_floods_not_found                    | Pass   |
|       | property.py::TestPropertyTSRoutes        | test_property_storms_existing                     | Pass   |
|       | property.py::TestPropertyTSRoutes        | test_property_storms_not_found                    | Pass   |
|       | property.py::TestPropertyTSRoutes        | test_property_storms_zero_floods                  | Pass   |
|       | property.py::TestPropertyTSRoutes        | test_propertyts_summary                           | Pass   |
|       | financial.py::TestPortfolioImpactData    | test_data_pts_dir_returns_404                     | Pass   |
|       | financial.py::TestPortfolioImpactOptions | test_options_returns_200                          | Pass   |
|       | financial.py::TestPortfolioImpactStorms  | test_no_found_storm_returns_404                   | Pass   |
|       | financial.py::TestPortfolioImpactStorms  | test_mortgage_enrichment_present                  | Pass   |
|       | financial.py::TestPortfolioImpactStorms  | test_property_entry_has_damage_fields             | Pass   |
|       | financial.py::TestPortfolioImpactStorms  | test_response_has_portfolio_key                   | Pass   |
|       | financial.py::TestPortfolioImpactStorms  | test_response_has_properties_list                 | Pass   |
|       | financial.py::TestPortfolioImpactStorms  | test_storm_id_in_response                         | Pass   |
|       | financial.py::TestPortfolioImpactStorms  | test_valid_storm_returns_200                      | Pass   |
|       | test_crud.py::TestCrudGetGauges          | test_get_gauge_not_found                          | Pass   |
|       | test_crud.py::TestCrudGetGauges          | test_get_gauge_success                            | Pass   |
|       | test_crud.py::TestCrudListGauges         | test_list_gauges_error_returns_500                | Pass   |
|       | test_crud.py::TestCrudListGauges         | test_list_gauges_legacy_route                     | Pass   |
|       | test_crud.py::TestCrudListGauges         | test_list_gauges_options                          | Pass   |
|       | test_crud.py::TestCrudListGauges         | test_list_gauges_returns_200                      | Pass   |
|       | test_hazard.py::TestHazardErrorPaths     | test_get_gauge_hazard_corrupted_file_returns_500  | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | test_price_prs_exception_returns_500              | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | test_price_prs_gauge_not_in_hazard                | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | test_price_prs_no_hazard_file                     | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | test_price_prs_no_json_body                       | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | test_price_prs_null_json_body                     | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | test_price_prs_severe_trigger                     | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | test_price_prs_success                            | Pass   |
|       | test_hazard.py::TestPricePrsHappyPath    | test_price_prs_with_defaults                      | Pass   |
|       | test_history.py::TestHistoryErrorPaths   | test_get_gauge_history_corrupted_file_returns_500 | Pass   |
|       | test_history.py::TestHistoryErrorPaths   | test_get_gauge_statistics_exception_returns_500   | Pass   |
|       | test_history.py::TestHistoryErrorPaths   | test_get_gaugets_exception_returns_500            | Pass   |
|       | test_reports.py::TestReportsErrorPaths   | test_gaugehd_load_error_warning                   | Pass   |
|       | test_reports.py::TestReportsErrorPaths   | test_general_exception_returns_500                | Pass   |
|       | test_reports.py::TestReportsErrorPaths   | test_hazard_load_error_warning                    | Pass   |
|       | test_reports.py::TestReportsErrorPaths   | test_import_error_returns_500                     | Pass   |

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| Model | Module                                     | Test Name                                         | Status |
|-------|--------------------------------------------|---------------------------------------------------|--------|
|       | test_reports.py::TestReportsError          | test_timeseries_load_success_with_storm_responses | Pass   |
|       | test_reports.py::TestReportsError          | test_timeseries_load_warning                      | Pass   |
|       | test_storms.py::TestStormsError            | test_gaugets_dir_corrupted_file_continues         | Pass   |
|       | test_storms.py::TestStormsError            | test_no_gaugets_dir                               | Pass   |
|       | test_storms.py::TestStormsError            | test_no_storm_sequences_file                      | Pass   |
|       | test_storms.py::TestStormsError            | test_storm_sequences_corrupted_continues          | Pass   |
|       | test_storms.py::TestStormsError            | test_storms_exception_returns_500                 | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_gauge_not_found_returns_404                  | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_no_curve_for_gauge_returns_404               | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_hazard_file_returns_404                      | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_response_has_annual_flood_probs              | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_response_has_curve_points                    | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_response_has_flood_stages                    | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_response_has_gauge_id                        | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_response_has_gev_parameters                  | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_response_has_term_structures                 | Pass   |
|       | gauges_hazard.py::TestGetGaugeHazard       | test_valid_hazard_returns_200                     | Pass   |
|       | gauges_hazard.py::TestPricePrs             | test_gauge_not_in_hazard_returns_404              | Pass   |
|       | gauges_hazard.py::TestPricePrs             | test_invalid_trigger_level_returns_400            | Pass   |
|       | gauges_hazard.py::TestPricePrs             | test_missing_gauge_id_returns_400                 | Pass   |
|       | gauges_hazard.py::TestPricePrs             | test_no_hazard_file_returns_404                   | Pass   |
|       | gauges_hazard.py::TestPricePrs             | test_no_json_returns_400                          | Pass   |
|       | gauges_hazard.py::TestPricePrs             | test_options_returns_ok                           | Pass   |
|       | gauges_history.py::TestGaugeHistory        | test_response_has_count_and_gauges_keys           | Pass   |
|       | gauges_history.py::TestGaugeHistory        | test_returns_200_with_no_gaugehd_files            | Pass   |
|       | gauges_history.py::TestGaugeHistory        | test_returns_correct_count_when_files_present     | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_days_larger_than_available_returns_all       | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_days_param_limits_observations               | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_days_param_zero_returns_all                  | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_no_history_file_returns_404                  | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_response_has_daily_observations              | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_response_has_flood_stages                    | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_response_has_gauge_metadata                  | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_response_has_statistics                      | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_valid_history_returns_200                    | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_gauge_not_found_returns_404                  | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_valid_gauge_returns_success                  | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_gauge_not_found_returns_404                  | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_response_has_count_and_readings              | Pass   |
|       | gauges_history.py::TestGetGaugeHistory     | test_valid_gauge_returns_success                  | Pass   |
|       | gauges_reports.py::TestGenerateGaugeReport | test_gauge_not_found_returns_404                  | Pass   |
|       | gauges_reports.py::TestGenerateGaugeReport | test_report_route_alias                           | Pass   |
|       | gauges_reports.py::TestGenerateGaugeReport | test_providing_gauge_id_returns_400               | Pass   |
|       | gauges_reports.py::TestGenerateGaugeReport | test_no_json_returns_400                          | Pass   |

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| Model | Module                                            | Test Name                                         | Status |
|-------|---------------------------------------------------|---------------------------------------------------|--------|
|       | gauges_reports.py::TestGenerateCentrifugeReport   | test_generate_report_on_body_returns_400          | Pass   |
|       | gauges_reports.py::TestGenerateCentrifugeReport   | test_generate_report_returns_ok                   | Pass   |
|       | gauges_reports.py::TestGenerateCentrifugeResponse | test_generate_report_response_has_file_path       | Pass   |
|       | gauges_reports.py::TestGenerateCentrifugeResponse | test_generate_report_response_has_pdf_base64      | Pass   |
|       | gauges_reports.py::TestGenerateCentrifugeReport   | test_generate_report_gauge_generates_report       | Pass   |
|       | gauges_reports.py::TestGenerateCentrifugeReport   | test_generate_report_gaugehd_file_present         | Pass   |
|       | gauges_reports.py::TestGenerateCentrifugeReport   | test_generate_report_hazard_file_present          | Pass   |
|       | gauges_storms.py::TestGetGaugeCentrifuge          | test_get_gauge_not_found_returns_404              | Pass   |
|       | gauges_storms.py::TestGetGaugeCentrifuge          | test_get_gauge_response_has_flood_simulation      | Pass   |
|       | gauges_storms.py::TestGetGaugeCentrifuge          | test_get_gauge_response_has_flood_stages          | Pass   |
|       | gauges_storms.py::TestGetGaugeCentrifuge          | test_get_gauge_response_has_gauge_id              | Pass   |
|       | gauges_storms.py::TestGetGaugeCentrifuge          | test_get_gauge_response_has_storm_responses       | Pass   |
|       | gauges_storms.py::TestGetGaugeCentrifuge          | test_get_gauge_storm_enrichment_from_storms_json  | Pass   |
|       | gauges_storms.py::TestGetGaugeCentrifuge          | test_get_gauge_invalid_gauge_returns_200          | Pass   |
|       | gauges_storms.py::TestGetGaugeCentrifuge          | test_get_gauge_with_severe_exceeded_counts_gauges | Pass   |
|       | audit.py::TestGetAuditTrail                       | test_after_log_entry_appears                      | Pass   |
|       | audit.py::TestGetAuditTrail                       | test_empty_trail_returns_200                      | Pass   |
|       | audit.py::TestGetAuditTrail                       | test_filter_by_event_type                         | Pass   |
|       | audit.py::TestGetAuditTrail                       | test_filter_by_model_id                           | Pass   |
|       | audit.py::TestGetAuditTrail                       | test_filter_by_model_id_no_match                  | Pass   |
|       | audit.py::TestGetAuditTrail                       | test_has_returned_field                           | Pass   |
|       | audit.py::TestGetAuditTrail                       | test_limit_parameter                              | Pass   |
|       | audit.py::TestGetRiskRating                       | test_model_not_found_returns_404                  | Pass   |
|       | audit.py::TestGetRiskRating                       | test_response_has_risk_rating                     | Pass   |
|       | audit.py::TestGetRiskRating                       | test_risk_rating_has_component_scores             | Pass   |
|       | audit.py::TestGetRiskRating                       | test_valid_model_returns_success                  | Pass   |
|       | audit.py::TestLogModelUsage                       | test_defaults_applied_when_no_json                | Pass   |
|       | audit.py::TestLogModelUsage                       | test_entry_has_expected_fields                    | Pass   |
|       | audit.py::TestLogModelUsage                       | test_model_not_found_returns_404                  | Pass   |
|       | audit.py::TestLogModelUsage                       | test_parameters_stored                            | Pass   |
|       | audit.py::TestLogModelUsage                       | test_valid_model_returns_200                      | Pass   |
|       | audit.py::TestOverrideRiskRating                  | test_clear_override_with_null                     | Pass   |
|       | audit.py::TestOverrideRiskRating                  | test_invalid_rating_returns_400                   | Pass   |
|       | audit.py::TestOverrideRiskRating                  | test_model_not_found_returns_404                  | Pass   |
|       | audit.py::TestOverrideRiskRating                  | test_no_reason_field_returns_400                  | Pass   |
|       | audit.py::TestOverrideRiskRating                  | test_no_reason_returns_400                        | Pass   |
|       | audit.py::TestOverrideRiskRating                  | test_override_stored_in_model                     | Pass   |
|       | audit.py::TestOverrideRiskRating                  | test_response_has_audit_entry                     | Pass   |
|       | audit.py::TestOverrideRiskRating                  | test_valid_override_returns_success               | Pass   |
|       | audit.py::TestPdfEndpoints                        | test_documentation_pdf_known_model_no_file        | Pass   |
|       | audit.py::TestPdfEndpoints                        | test_documentation_pdf_unknown_model_404          | Pass   |
|       | audit.py::TestPdfEndpoints                        | test_mrc_tor_pdf_responds                         | Pass   |
|       | audit.py::TestPdfEndpoints                        | test_parameter_inventory_pdf_responds             | Pass   |
|       | audit.py::TestPdfEndpoints                        | test_test_results_pdf_known_model_no_file         | Pass   |

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| Model | Module                                     | Test Name                                            | Status  |
|-------|--------------------------------------------|------------------------------------------------------|---------|
|       | audit.py::TestPdfEndpoints                 | test.test_results.pdf_unknown_model.404              | Pass    |
|       | audit.py::TestUpdateValidationQuestion     | test.test_valid_statuses                             | Pass    |
|       | audit.py::TestUpdateValidationQuestion     | test.test_invalid_status_returns_400                 | Pass    |
|       | audit.py::TestUpdateValidationQuestion     | test.test_model_not_found_returns_404                | Pass    |
|       | audit.py::TestUpdateValidationQuestion     | test.test_question_not_found_returns_404             | Pass    |
|       | audit.py::TestUpdateValidationQuestion     | test.test_risk_rating_recalculated                   | Pass    |
|       | audit.py::TestUpdateValidationQuestion     | test.test_valid_update_has_model_and_audit_entry     | Pass    |
|       | audit.py::TestUpdateValidationQuestion     | test.test_valid_update_returns_success               | Pass    |
|       | audit_reports.py::TestDownloadAuditReport  | test.test_report_not_found_returns_404               | Pass    |
|       | audit_reports.py::TestDownloadAuditReport  | test.test_report_pdf_served_as_attachment            | Pass    |
|       | audit_reports.py::TestDownloadAuditReport  | test.test_report_traversal_returns_403               | Pass    |
|       | audit_reports.py::TestDownloadAuditReport  | test.test_report_pdf_served_inline                   | Pass    |
|       | audit_reports.py::TestListAuditReports     | test.test_coverage_available_false_when_no_index     | Index   |
|       | audit_reports.py::TestListAuditReports     | test.test_coverage_available_true_when_index_present | Index   |
|       | audit_reports.py::TestListAuditReports     | test.test_empty_dir_returns_empty_list               | Pass    |
|       | audit_reports.py::TestListAuditReports     | test.test_file_has_size_and_modified                 | Pass    |
|       | audit_reports.py::TestListAuditReports     | test.test_file_type_label_json                       | Pass    |
|       | audit_reports.py::TestListAuditReports     | test.test_file_type_label_pdf                        | Pass    |
|       | audit_reports.py::TestListAuditReports     | test.test_file_type_label_unknown                    | Pass    |
|       | audit_reports.py::TestListAuditReports     | test.test_no_dir_returns_empty_list                  | Pass    |
|       | audit_reports.py::TestListAuditReports     | test.test_pdf_file_listed                            | Pass    |
|       | audit_reports.py::TestListAuditReports     | test.test_subdirectories_not_listed                  | Pass    |
|       | audit_reports.py::TestServeCoverageIndex   | test.test_index_coverage_dir_returns_404             | Pass    |
|       | audit_reports.py::TestServeCoverageIndex   | test.test_with_coverage_index_returns_200            | Pass    |
|       | bibliography.py::TestAddBibliographyRef    | test.test_ref_appears_in_list                        | Pass    |
|       | bibliography.py::TestAddBibliographyRef    | test.test_ref_assigns_id                             | Pass    |
|       | bibliography.py::TestAddBibliographyRef    | test.test_ref_no_json_body_uses_defaults             | Pass    |
|       | bibliography.py::TestAddBibliographyRef    | test.test_ref_returns_200                            | Pass    |
|       | bibliography.py::TestAddBibliographyRef    | test.test_ref_returns_reference                      | Pass    |
|       | bibliography.py::TestAddBibliographyRef    | test.test_ref_same_category_twice_no_duplicates      | Pass    |
|       | bibliography.py::TestAddBibliographyRef    | test.test_ref_stores_all_fields                      | Pass    |
|       | bibliography.py::TestAddBibliographyRef    | test.test_ref_with_category_updates_category         | Pass    |
|       | bibliography.py::TestDeleteBibliographyRef | test.test_ref_nonexistent_still_returns_success      | Success |
|       | bibliography.py::TestDeleteBibliographyRef | test.test_ref_returns_success                        | Pass    |
|       | bibliography.py::TestDeleteBibliographyRef | test.test_refed_ref_not_in_list                      | Pass    |
|       | bibliography.py::TestGetBibliographyRef    | test.test_empty_references                           | Pass    |
|       | bibliography.py::TestGetBibliographyRef    | test.test_has_categories_key                         | Pass    |
|       | bibliography.py::TestGetBibliographyRef    | test.test_returns_200                                | Pass    |
|       | bibliography.py::TestGetBibliographyRef    | test.test_returns_success_status                     | Pass    |
|       | bibliography.py::TestUpdateBibliographyRef | test.test_ref_not_found_returns_404                  | Pass    |
|       | bibliography.py::TestUpdateBibliographyRef | test.test_ref_updates_field                          | Pass    |
|       | bibliography.py::TestUpdateBibliographyRef | test.test_ref_updates_returns_success                | Pass    |
|       | bibliography.py::TestUpdateBibliographyRef | test.test_ref_with_new_category_adds_to_list         | Pass    |
|       | compliance.py::TestGetBcbs239              | test.test_no_data_returns_404                        | Pass    |

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| Model | Module                                   | Test Name                                         | Status |
|-------|------------------------------------------|---------------------------------------------------|--------|
|       | compliance.py::TestGetBcbs239            | test_principles_count                             | Pass   |
|       | compliance.py::TestGetBcbs239            | test_response_has_assessment                      | Pass   |
|       | compliance.py::TestGetBcbs239            | test_with_data_returns_200                        | Pass   |
|       | compliance.py::TestGetBcbs239            | test_bcbs239_pdf_responds                         | Pass   |
|       | compliance.py::TestUpdateBcbs239         | test_principles_assessment_date_updated           | Pass   |
|       | compliance.py::TestUpdateBcbs239         | test_principles_data_file_returns_404             | Pass   |
|       | compliance.py::TestUpdateBcbs239         | test_principles_not_found_returns_404             | Pass   |
|       | compliance.py::TestUpdateBcbs239         | test_principle_1_derives_non_compliant_status     | Pass   |
|       | compliance.py::TestUpdateBcbs239         | test_principle_4_derives_fully_compliant_status   | Pass   |
|       | compliance.py::TestUpdateBcbs239         | test_principle_evidence_field                     | Pass   |
|       | compliance.py::TestUpdateBcbs239         | test_principle_gaps_and_remediation               | Pass   |
|       | compliance.py::TestUpdateBcbs239         | test_principle_update_returns_200                 | Pass   |
|       | compliance.py::TestUpdateRacisActivity   | test_activity_not_found_returns_404               | Pass   |
|       | compliance.py::TestUpdateRacisActivity   | test_activity_valid_role_id_returns_400           | Pass   |
|       | compliance.py::TestUpdateRacisActivity   | test_activity_updated                             | Pass   |
|       | compliance.py::TestUpdateRacisActivity   | test_activity_raci_code_allowed                   | Pass   |
|       | compliance.py::TestUpdateRacisActivity   | test_activity_update_returns_success              | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_tier_threshold_updated            | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_response_has_audit_entry          | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_get_not_found_returns_404         | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_index_update_returns_success      | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_backup_field_updated              | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_raci_file_returns_404             | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_response_has_audit_entry          | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_role_not_found_returns_404        | Pass   |
|       | compliance.py::TestUpdateRacisEscalation | test_escalation_valid_role_update_returns_success | Pass   |
|       | documents.py::TestDeleteDocument         | test_deleted_doc_not_in_list                      | Pass   |
|       | documents.py::TestDeleteDocument         | test_not_found_returns_404                        | Pass   |
|       | documents.py::TestDeleteDocument         | test_valid_delete_returns_success                 | Pass   |
|       | documents.py::TestDeleteSaveModel        | test_delete_save_failure_returns_500              | Pass   |
|       | documents.py::TestDownloadDocument       | test_not_found_returns_404                        | Pass   |
|       | documents.py::TestDownloadDocument       | test_valid_download_returns_200                   | Pass   |
|       | documents.py::TestDownloadFile           | test_missing_file_on_disk_returns_404             | Pass   |
|       | documents.py::TestGetDocuments           | test_empty_list_returns_success                   | Pass   |
|       | documents.py::TestUploadDocument         | test_empty_filename_returns_400                   | Pass   |
|       | documents.py::TestUploadDocument         | test_no_file_returns_400                          | Pass   |
|       | documents.py::TestUploadDocument         | test_upload_appears_in_list                       | Pass   |
|       | documents.py::TestUploadDocument         | test_upload_response_has_id                       | Pass   |
|       | documents.py::TestUploadDocument         | test_upload_stores_description                    | Pass   |
|       | documents.py::TestUploadDocument         | test_valid_upload_returns_200                     | Pass   |
|       | documents.py::TestUploadSaveModel        | test_upload_save_failure_returns_500              | Pass   |
|       | inventory.py::TestInventorySummary       | test_get_models_includes_risk_fields              | Pass   |
|       | inventory.py::TestRiskRating             | test_get_risk_rating_calculation                  | Pass   |
|       | inventory.py::TestRiskRating             | test_risk_rating_clear_override                   | Pass   |

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| Model | Module                               | Test Name                                      | Status   |
|-------|--------------------------------------|------------------------------------------------|----------|
|       | inventory.py::TestRiskRating         | test_risk_rating_override                      | Pass     |
|       | inventory.py::TestRiskRating         | test_risk_rating_override_invalid_rating       | Pass     |
|       | inventory.py::TestRiskRating         | test_risk_rating_override_requires_reason      | Pass     |
|       | inventory.py::TestValidationQuestion | test_update_triggers_risk_recalculation        | Pass     |
|       | inventory.py::TestValidationQuestion | test_update_validation_question                | Pass     |
|       | inventory.py::TestValidationQuestion | test_update_validation_question_invalid_status | Pass     |
|       | inventory.py::TestValidationQuestion | test_update_validation_question_not_found      | Pass     |
|       | models.py::TestGetEditableFields     | test_has_fields                                | Pass     |
|       | models.py::TestGetEditableFields     | test_returns_200                               | Pass     |
|       | models.py::TestGetEditableFields     | test_returns_success_status                    | Pass     |
|       | models.py::TestGetModelDetail        | test_model_has_validation_questions            | Pass     |
|       | models.py::TestGetModelDetail        | test_model_not_found_returns_404               | Pass     |
|       | models.py::TestGetModelDetail        | test_response_has_audit_entries                | Pass     |
|       | models.py::TestGetModelDetail        | test_valid_model_returns_success               | Pass     |
|       | models.py::TestGetModels             | test_category_distribution                     | Pass     |
|       | models.py::TestGetModels             | test_model_has_review_status                   | Pass     |
|       | models.py::TestGetModels             | test_model_has_risk_rating                     | Pass     |
|       | models.py::TestGetModels             | test_model_has_validation_coverage             | Pass     |
|       | models.py::TestGetModels             | test_response_has_metadata                     | Pass     |
|       | models.py::TestGetModels             | test_returns_200                               | Pass     |
|       | models.py::TestGetModels             | test_returns_models_list                       | Pass     |
|       | models.py::TestGetModels             | test_returns_success_status                    | Pass     |
|       | models.py::TestGetModels             | test_tier_distribution                         | Pass     |
|       | models.py::TestUpdateModelFields     | test_empty_value_returns_400                   | Pass     |
|       | models.py::TestUpdateModelFields     | test_invalid_choice_value_returns_400          | Pass     |
|       | models.py::TestUpdateModelFields     | test_invalid_field_returns_400                 | Pass     |
|       | models.py::TestUpdateModelFields     | test_model_not_found_returns_404               | Pass     |
|       | models.py::TestUpdateModelFields     | test_no_reason_returns_400                     | Pass     |
|       | models.py::TestUpdateModelFields     | test_response_has_audit_entry                  | Pass     |
|       | models.py::TestUpdateModelFields     | test_update_lifecycle_stage                    | Pass     |
|       | models.py::TestUpdateModelFields     | test_update_validation_status                  | Pass     |
|       | models.py::TestUpdateModelFields     | test_valid_choice_field_update_returns_success | Success  |
|       | models.py::TestUpdateModelFields     | test_valid_text_field_update_returns_success   | Success  |
|       | mrc_crud.py::TestActionsCrud         | test_add_action_auto_id                        | Pass     |
|       | mrc_crud.py::TestActionsCrud         | test_add_action_missing_meeting_returns_404    | 404      |
|       | mrc_crud.py::TestActionsCrud         | test_add_action_returns_200                    | Pass     |
|       | mrc_crud.py::TestActionsCrud         | test_delete_action_missing_meeting_returns_404 | Pass 404 |
|       | mrc_crud.py::TestActionsCrud         | test_delete_action_returns_200                 | Pass     |
|       | mrc_crud.py::TestActionsCrud         | test_update_action_not_found_returns_404       | Pass 404 |
|       | mrc_crud.py::TestActionsCrud         | test_update_action_returns_200                 | Pass     |
|       | mrc_crud.py::TestAgendaCrud          | test_add_agenda_item_increments_count          | Pass     |
|       | mrc_crud.py::TestAgendaCrud          | test_add_agenda_item_returns_200               | Pass     |
|       | mrc_crud.py::TestAgendaCrud          | test_add_agenda_missing_meeting_returns_404    | 404      |
|       | mrc_crud.py::TestAgendaCrud          | test_delete_agenda_item_returns_200            | Pass     |

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| Model                                 | Module                                            | Test Name | Status |
|---------------------------------------|---------------------------------------------------|-----------|--------|
| mrc_crud.py::TestAgendaCrud           | test.delete_agenda_missing_meeting_return         | Pass      | 404    |
| mrc_crud.py::TestAgendaCrud           | test.update_agenda_item_not_found_return          | Pass      | 404    |
| mrc_crud.py::TestAgendaCrud           | test.update_agenda_item_returns_200               | Pass      |        |
| mrc_crud.py::TestDecisionsCrud        | test.add_decision_auto_ids                        | Pass      |        |
| mrc_crud.py::TestDecisionsCrud        | test.add_decision_missing_meeting_return          | Pass      | 404    |
| mrc_crud.py::TestDecisionsCrud        | test.add_decision_returns_200                     | Pass      |        |
| mrc_crud.py::TestDecisionsCrud        | test.delete_decision_missing_meeting_return       | Pass      | 404    |
| mrc_crud.py::TestDecisionsCrud        | test.delete_decision_returns_200                  | Pass      |        |
| mrc_crud.py::TestDecisionsCrud        | test.update_decision_not_found_returns_404        | Pass      |        |
| mrc_crud.py::TestDecisionsCrud        | test.update_decision_returns_200                  | Pass      |        |
| mrc_crud.py::TestMalformedAction      | test.add_action_with_malformed_existing_id        | Pass      |        |
| mrc_crud.py::TestMalformedDecision    | test.add_decision_with_malformed_existing_id      | Pass      |        |
| mrc_crud.py::TestMalformedParticipant | test.add_participant_with_malformed_existing_id   | Pass      |        |
| mrc_crud.py::TestMinutesItemsCrud     | test.add_minute_item_missing_meeting_return       | Pass      | 404    |
| mrc_crud.py::TestMinutesItemsCrud     | test.add_minute_item_returns_200                  | Pass      |        |
| mrc_crud.py::TestMinutesItemsCrud     | test.delete_minute_item_legacy_format_returns_400 | Pass      |        |
| mrc_crud.py::TestMinutesItemsCrud     | test.delete_minute_item_missing_meeting_return    | Pass      | 404    |
| mrc_crud.py::TestMinutesItemsCrud     | test.delete_minute_item_returns_200               | Pass      |        |
| mrc_crud.py::TestMinutesItemsCrud     | test.update_minute_item_legacy_format_returns_400 | Pass      |        |
| mrc_crud.py::TestMinutesItemsCrud     | test.update_minute_item_not_found_returns_404     | Pass      |        |
| mrc_crud.py::TestMinutesItemsCrud     | test.update_minute_item_returns_200               | Pass      |        |
| mrc_crud.py::TestMinutesStringTest    | test.add_minute_item_when_minutes_is_string       | Pass      |        |
| mrc_crud.py::TestMinutesUpdate        | test.update_minutes_missing_meeting_return        | Pass      | 404    |
| mrc_crud.py::TestMinutesUpdate        | test.update_minutes_returns_200                   | Pass      |        |
| mrc_crud.py::TestParticipantsCrud     | test.add_participant_auto_id                      | Pass      |        |
| mrc_crud.py::TestParticipantsCrud     | test.add_participant_missing_meeting_return       | Pass      | 404    |
| mrc_crud.py::TestParticipantsCrud     | test.add_participant_returns_200                  | Pass      |        |
| mrc_crud.py::TestParticipantsCrud     | test.delete_participant_missing_meeting_return    | Pass      | 404    |
| mrc_crud.py::TestParticipantsCrud     | test.delete_participant_returns_200               | Pass      |        |
| mrc_crud.py::TestParticipantsCrud     | test.update_participant_not_found_returns_404     | Pass      |        |
| mrc_crud.py::TestParticipantsCrud     | test.update_participant_returns_200               | Pass      |        |
| mrc_crud.py::TestUpdateMissingMeeting | test.update_action_missing_meeting_return         | Pass      | 404    |
| mrc_crud.py::TestUpdateMissingMeeting | test.update_decision_missing_meeting_return       | Pass      | 404    |
| mrc_crud.py::TestUpdateMissingMeeting | test.update_minute_item_missing_meeting_return    | Pass      | 404    |
| mrc_crud.py::TestUpdateMissingMeeting | test.update_participant_missing_meeting_return    | Pass      | 404    |
| mrc_pdf.py::TestBuildMeetingPdf       | test.all_sections                                 | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf       | test.buffer_not_empty                             | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf       | test.meeting_with_actions                         | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf       | test.meeting_with_agenda                          | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf       | test.meeting_with_decisions                       | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf       | test.meeting_with_minutes                         | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf       | test.meeting_with_no_optional_fields              | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf       | test.meeting_with_participants                    | Pass      |        |
| mrc_pdf.py::TestBuildMeetingPdf       | test.minimal_meeting                              | Pass      |        |

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| Model | Module                                                     | Test Name                              | Status |
|-------|------------------------------------------------------------|----------------------------------------|--------|
|       | mrc_pdf.py::TestBuildMeetingPdf                            | test_returns_bytesio                   | Pass   |
|       | mrc_pdf.py::TestGetMeetingPdf                              | test_meeting_not_found_returns_404     | Pass   |
|       | mrc_pdf.py::TestGetMeetingPdf                              | test_valid_meeting_content_type_pdf    | Pass   |
|       | mrc_pdf.py::TestGetMeetingPdf                              | test_valid_meeting_returns_bytes       | Pass   |
|       | mrc_pdf.py::TestGetMeetingPdf                              | test_valid_meeting_returns_pdf         | Pass   |
|       | raci.py::TestRACIMatrix                                    | test_get_raci_matrix                   | Pass   |
|       | raci.py::TestRACIMatrix                                    | test_get_raci_matrix_file_missing      | Pass   |
|       | raci.py::TestRACIMatrix                                    | test_update_raci_activity              | Pass   |
|       | raci.py::TestRACIMatrix                                    | test_update_raci_activity_invalid_role | Pass   |
|       | raci.py::TestRACIMatrix                                    | test_update_raci_escalation            | Pass   |
|       | raci.py::TestRACIMatrix                                    | test_update_raci_role                  | Pass   |
|       | raci.py::TestRACIMatrix                                    | test_update_raci_role_not_found        | Pass   |
|       | test_bibliography_coverage.py::TestBibliographySaveFailure | test_returns_500                       | Pass   |
|       | test_bibliography_coverage.py::TestBibliographySaveFailure | test_returns_500                       | Pass   |
|       | test_bibliography_coverage.py::TestBibliographySaveFailure | test_returns_500                       | Pass   |
|       | test_block2.coverage.py::TestAuditLogUsage                 | usage_audit_overflow_trim              | Pass   |
|       | test_block2.coverage.py::TestAuditLogUsage                 | usage_inventory_not_found              | Pass   |
|       | test_block2.coverage.py::TestAuditLogUsage                 | usage_save_failure                     | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_create_document_save_failure      | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_get_document_not_found            | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_get_document_success              | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_upload_document_empty_filename    | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_upload_document_meeting_not_found | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_upload_document_no_file           | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_upload_document_save_failure      | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_upload_document_success           | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_upload_document_no_documents_list | Pass   |
|       | test_block2.coverage.py::TestMrDocStore                    | test_update_document_save_failure      | Pass   |
|       | test_block2.coverage.py::TestOverwriteRating               | rating_overflow                        | Pass   |
|       | test_block2.coverage.py::TestOverwriteRating               | rating_inventory_not_found             | Pass   |
|       | test_block2.coverage.py::TestOverwriteRating               | rating_save_failure                    | Pass   |
|       | test_block2.coverage.py::TestPdfServing                    | model.doc.pdf_not_found                | Pass   |
|       | test_block2.coverage.py::TestPdfServing                    | model.doc.pdf_served                   | Pass   |
|       | test_block2.coverage.py::TestPdfServing                    | mc.tor.pdf_not_found                   | Pass   |
|       | test_block2.coverage.py::TestPdfServing                    | mc.tor.pdf_served                      | Pass   |
|       | test_block2.coverage.py::TestPdfServing                    | parameter_inventory.pdf_not_found      | Pass   |
|       | test_block2.coverage.py::TestPdfServing                    | parameter_inventory.pdf_served         | Pass   |
|       | test_block2.coverage.py::TestPdfServing                    | test_results.pdf_not_found             | Pass   |
|       | test_block2.coverage.py::TestPdfServing                    | test_results.pdf_served                | Pass   |
|       | test_block2.coverage.py::TestRiskTestingEdges              | test_inventory_not_found               | Pass   |
|       | test_block2.coverage.py::TestValidationQueue               | queue_overflow                         | Pass   |
|       | test_block2.coverage.py::TestValidationQueue               | queue_inventory_not_found              | Pass   |
|       | test_block2.coverage.py::TestValidationQueue               | queue_save_failure                     | Pass   |
|       | test_compliance_coverage.py::TestPdfGeneration             | test_returns_404                       | Pass   |

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| Model | Module                                                     | Test Name                                       | Status |
|-------|------------------------------------------------------------|-------------------------------------------------|--------|
|       | test_compliance_coverage.py::TestSaveSafeFailure           | test_save_safe_failure_returns_500              | Pass   |
|       | test_compliance_coverage.py::TestSaveActivityNoData        | test_save_activity_no_data_returns_404          | Pass   |
|       | test_compliance_coverage.py::TestSaveActivitySaveFailure   | test_save_activity_save_failure_returns_500     | Pass   |
|       | test_compliance_coverage.py::TestSaveActivityThisIsUpdated | test_save_activity_this_is_updated              | Pass   |
|       | test_compliance_coverage.py::TestSaveInteractionNoData     | test_save_interaction_no_data_returns_404       | Pass   |
|       | test_compliance_coverage.py::TestSaveEvaluationSaveFailure | test_save_evaluation_save_failure_returns_500   | Pass   |
|       | test_compliance_coverage.py::TestSaveRuleSaveFailure       | test_save_rule_save_failure_returns_500         | Pass   |
|       | test_helpers_data.py::TestLoadBibliography                 | test_loads_valid_data                           | Pass   |
|       | test_helpers_data.py::TestLoadBibliography                 | test_loads_none_when_missing                    | Pass   |
|       | test_helpers_data.py::TestLoadBibliography                 | test_loads_bibliography                         | Pass   |
|       | test_helpers_data.py::TestLoadBibliography                 | test_loads_default_when_missing                 | Pass   |
|       | test_helpers_data.py::TestLoadDocuments                    | test_loads_documents                            | Pass   |
|       | test_helpers_data.py::TestLoadDocuments                    | test_returns_empty_list_when_missing            | Pass   |
|       | test_helpers_data.py::TestLoadResources                    | test_loads_valid_data                           | Pass   |
|       | test_helpers_data.py::TestLoadResources                    | test_returns_none_when_missing                  | Pass   |
|       | test_helpers_data.py::TestSaveBibliography                 | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_data.py::TestSaveBibliography                 | test_saves_data                                 | Pass   |
|       | test_helpers_data.py::TestSaveBibliography                 | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_data.py::TestSaveBibliography                 | test_saves_bibliography                         | Pass   |
|       | test_helpers_data.py::TestSaveDocuments                    | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_data.py::TestSaveDocuments                    | test_saves_documents                            | Pass   |
|       | test_helpers_data.py::TestSaveResources                    | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_data.py::TestSaveResources                    | test_saves_data                                 | Pass   |
|       | test_helpers_io.py::TestFindMeeting                        | test_finds_existing_meeting                     | Pass   |
|       | test_helpers_io.py::TestFindMeeting                        | test_returns_error_for_missing_meeting          | Pass   |
|       | test_helpers_io.py::TestLoadAudits                         | test_loads_entries                              | Pass   |
|       | test_helpers_io.py::TestLoadAudits                         | test_returns_empty_list_on_bad_json             | Pass   |
|       | test_helpers_io.py::TestLoadAudits                         | test_returns_empty_list_when_missing            | Pass   |
|       | test_helpers_io.py::TestLoadInventory                      | test_loads_valid_file                           | Pass   |
|       | test_helpers_io.py::TestLoadInventory                      | test_returns_none_on_bad_json                   | Pass   |
|       | test_helpers_io.py::TestLoadInventory                      | test_returns_none_when_missing                  | Pass   |
|       | test_helpers_io.py::TestLoadMeetings                       | test_loads_list                                 | Pass   |
|       | test_helpers_io.py::TestLoadMeetings                       | test_returns_empty_list_when_missing            | Pass   |
|       | test_helpers_io.py::TestSaveAndRestore                     | test_saves_and_returns_response                 | Pass   |
|       | test_helpers_io.py::TestSaveAuditLog                       | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_io.py::TestSaveAuditLog                       | test_saves_entries                              | Pass   |
|       | test_helpers_io.py::TestSaveInventory                      | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_io.py::TestSaveInventory                      | test_saves_data                                 | Pass   |
|       | test_helpers_io.py::TestSaveMeetings                       | test_returns_false_on_oserror                   | Pass   |
|       | test_helpers_io.py::TestSaveMeetings                       | test_saves_meetings                             | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskConditional         | test_calculates_conditional_rating_middle_range | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskConditional         | test_calculates_conditional_rating_half_score   | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskConditional         | test_calculates_conditional_rating_full_score   | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskConditional         | test_calculates_conditional_rating_penalised    | Pass   |

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| Model | Module                                                   | Test Name                               | Status |
|-------|----------------------------------------------------------|-----------------------------------------|--------|
|       | test_helpers_risk.py::TestCalculateRiskRating            | test_coverage_acceptable                | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating            | test_limitations_penalised              | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating            | test_coverage_unacceptable              | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating            | test_review_date_gives_zero             | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating            | test_review_gives_zero_score            | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating            | test_ratings_dict_with_required_keys    | Pass   |
|       | test_helpers_risk.py::TestCalculateRiskRating            | test_rating_is_float                    | Pass   |
|       | test_helpers_risk.py::TestVqSummary                      | test_all_addressed                      | Pass   |
|       | test_helpers_risk.py::TestVqSummary                      | test_empty_questions                    | Pass   |
|       | test_helpers_risk.py::TestVqSummary                      | test_none_addressed                     | Pass   |
|       | test_helpers_risk.py::TestVqSummary                      | test_some_not_applicable                | Pass   |
|       | test_inventory_integrity.py::TestAPIEndpointAvailability | test_returns_200                        | Pass   |
|       | test_inventory_integrity.py::TestAPIEndpointAvailability | test_endpoint_succeeds                  | Pass   |
|       | test_inventory_integrity.py::TestAPIEndpointAvailability | test_model_count_matches_file           | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileExists     |                                         | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileExists     | test_valid_json                         | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileExists     | test_has_at_least_one_model             | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileExists     | test_has_file_data                      | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileExists     | test_has_model_chain                    | Pass   |
|       | test_inventory_integrity.py::TestInventoryFileExists     | test_has_file_key                       | Pass   |
|       | test_inventory_integrity.py::TestModelIDs                | test_models_have_model_id               | Pass   |
|       | test_inventory_integrity.py::TestModelIDs                | test_model_ids_are_unique               | Pass   |
|       | test_inventory_integrity.py::TestModelIDs                | test_model_ids_follow_naming_convention | Pass   |
|       | test_inventory_integrity.py::TestOverallRiskRatingString | test_is_valid                           | Pass   |
|       | test_inventory_integrity.py::TestOverallRiskRatingString | test_structuring_has_component_scores   | Pass   |
|       | test_inventory_integrity.py::TestOverallRiskRatingString | test_structuring_has_required_keys      | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[overall_risk_rating]          | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[test_coverage]                | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[assumptions]                  | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[downstream_models]            | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[limitations]                  | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[upstream_models]              | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[category]                     | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[description]                  | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[name]                         | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[owner]                        | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[status]                       | Pass   |
|       | test_inventory_integrity.py::TestRequiredFields          | test_list[version]                      | Pass   |
|       | test_inventory_integrity.py::TestTestCoverageStorage     | test_booleans_are_bool                  | Pass   |
|       | test_inventory_integrity.py::TestTestCoverageStorage     | test_has_required_keys                  | Pass   |
|       | test_inventory_integrity.py::TestTestCoverageStorage     | test_has_test_file                      | Pass   |
|       | test_models_coverage.py::TestAuditLogTruncation          | test_truncated_to_10000                 | Pass   |
|       | test_models_coverage.py::TestGetModelInventory           | test_returns_404                        | Pass   |
|       | test_models_coverage.py::TestGetModelInventory           | test_returns_404                        | Pass   |

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| Model | Module                          | Test Name                                               | Status |
|-------|---------------------------------|---------------------------------------------------------|--------|
|       | test_models_coverage.py::       | TestGetArticlesNotScheduledReviewStatus                 | Pass   |
|       | test_models_coverage.py::       | TestGetArticlesScheduledReviewStatus                    | Pass   |
|       | test_models_coverage.py::       | TestUpdateModelInventoryReturns_404                     | Pass   |
|       | test_models_coverage.py::       | TestUpdateModelSaleFutureReturns_500                    | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAgendaCRUDItem                                      | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAgendaCRUDUnknownMeetingReturns_404                 | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAgendaCRUDMultipleItemsIncrementsNumber             | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAgendaCRUDAgendaItemRenumbers                       | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAgendaCRUDAgendaUnknownMeetingReturns_404           | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAgendaCRUDAgendaItem                                | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAgendaCRUDAgendaUnknownMeetingReturns_404           | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestAgendaCRUDMissingItemReturns_404                    | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestMinutecrudItem                                      | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestMinutecrudMinuteItemLegacyFormatReturns_400         | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestMinutecrudMinuteItemLegacyFormatReturns_400         | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestMinutecrudMinuteItemNotFoundReturns_404             | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestMinutecrudMinutesText                               | Pass   |
|       | test_mrc_agenda_minutes.py::    | TestMinutecrudMinutesUnknownMeetingReturns_404          | Pass   |
|       | test_mrc_decisions_actions.py:: | TestArticleCRUDItemAutoGeneratesId                      | Pass   |
|       | test_mrc_decisions_actions.py:: | TestArticleCRUDItemUnknownMeetingReturns_404            | Pass   |
|       | test_mrc_decisions_actions.py:: | TestArticleCRUDAction                                   | Pass   |
|       | test_mrc_decisions_actions.py:: | TestArticleCRUDActionUnknownMeetingReturns_404          | Pass   |
|       | test_mrc_decisions_actions.py:: | TestArticleCRUDActionStatus                             | Pass   |
|       | test_mrc_decisions_actions.py:: | TestArticleCRUDMissingActionReturns_404                 | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecisionCRUDItemAutoGeneratesId                     | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecisionCRUDItemUnknownMeetingReturns_404           | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecisionCRUDDecision                                | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecisionCRUDDecisionUnknownMeetingReturns_404       | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecisionCRUDDecisionIncrementsId                    | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecisionCRUDDecision                                | Pass   |
|       | test_mrc_decisions_actions.py:: | TestDecisionCRUDMissingDecisionReturns_404              | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipantCRUDItemAutoGeneratesId                  | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipantCRUDItemUnknownMeetingReturns_404        | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipantCRUDParticipant                          | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipantCRUDParticipantUnknownMeetingReturns_404 | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipantCRUDBindingParticipantReturns_404        | Pass   |
|       | test_mrc_decisions_actions.py:: | TestParticipantCRUDParticipant                          | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeetingAutoGeneratesId                        | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeetingReturnsMeeting                         | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeetingAttendeesBecomeParticipants            | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeetingTitleUsed                              | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeetingDefaultChairPopulated                  | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeetingDefaultParticipantsCreated             | Pass   |
|       | test_mrc_meetings.py::          | TestCreateMeetingJsonBodyUsesDefaults                   | Pass   |

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| Model | Module | Test Name                                                                                                | Status  |
|-------|--------|----------------------------------------------------------------------------------------------------------|---------|
|       |        | test_mrc_meetings.py::TestGetMeeting::test_get_created_meeting                                           | Pass    |
|       |        | test_mrc_meetings.py::TestGetMeeting::test_get_unknown_id_returns_404                                    | Pass    |
|       |        | test_mrc_meetings.py::TestListMeetings::test_created_meeting_appears_in_list                             | Pass    |
|       |        | test_mrc_meetings.py::TestListMeetings::test_empty_list_returns_success                                  | Pass    |
|       |        | test_mrc_meetings.py::TestListMeetings::test_list_summary_has_required_keys                              | Pass    |
|       |        | test_mrc_meetings.py::TestListMeetings::test_sorted_by_date_desc                                         | Pass    |
|       |        | test_mrc_meetings.py::TestMeetingDocument::test_missing_document_returns_404                             | Pass    |
|       |        | test_mrc_meetings.py::TestMeetingDocument::test_upload_empty_filename_returns_400                        | Pass    |
|       |        | test_mrc_meetings.py::TestMeetingDocument::test_upload_no_file_returns_400                               | Pass    |
|       |        | test_mrc_meetings.py::TestMeetingDocument::test_upload_not_found_meeting_returns_404                     | Pass    |
|       |        | test_mrc_meetings.py::TestUpdateMeeting::test_update_sets_updated_at                                     | Pass    |
|       |        | test_mrc_meetings.py::TestUpdateMeeting::test_update_title                                               | Pass    |
|       |        | test_mrc_meetings.py::TestUpdateMeeting::test_update_unknown_id_returns_404                              | Pass    |
|       |        | test_report_routes.py::TestAppendOutput::test_appends_line                                               | Pass    |
|       |        | test_report_routes.py::TestAppendOutput::test_clear_then_append                                          | Pass    |
|       |        | test_report_routes.py::TestAppendOutput::test_prints_buffer_at_2000                                      | Pass    |
|       |        | test_report_routes.py::TestRunAndShow::test_appends_started_header                                       | Pass    |
|       |        | test_report_routes.py::TestRunAndShow::test_error_line_appended_on_exception                             | Pass    |
|       |        | test_report_routes.py::TestRunAndShow::test_handles_exception_gracefully                                 | Pass    |
|       |        | test_report_routes.py::TestRunAndShow::test_sets_running_false_on_completion                             | Pass    |
|       |        | test_report_routes.py::TestRunTestReport::test_ready_running_message                                     | Pass    |
|       |        | test_report_routes.py::TestRunTestReport::test_ready_running_returns_202                                 | Pass    |
|       |        | test_report_routes.py::TestRunTestReport::test_response_has_started_status                               | Pass    |
|       |        | test_report_routes.py::TestRunTestReport::test_reports_run_returns_202                                   | Pass    |
|       |        | test_report_routes.py::TestStreamProc::test_calls_proc_wait                                              | Pass    |
|       |        | test_report_routes.py::TestStreamProc::test_reads_stdout_into_buffer                                     | Pass    |
|       |        | test_report_routes.py::TestStreamProc::test_strips_newlines                                              | Pass    |
|       |        | test_report_routes.py::TestTestReportOutput::test_has_lines_field                                        | Pass    |
|       |        | test_report_routes.py::TestTestReportOutput::test_has_running_field                                      | Pass    |
|       |        | test_report_routes.py::TestTestReportOutput::test_includes_appended                                      | Pass    |
|       |        | test_report_routes.py::TestTestReportOutput::test_running_initially                                      | Pass    |
|       |        | test_report_routes.py::TestTestReportOutput::test_prints_200                                             | Pass    |
|       |        | test_report_routes.py::TestTestReportSummary::test_not_found_when_no_file                                | Pass    |
|       |        | test_report_routes.py::TestTestReportSummary::test_ok_when_file_exists                                   | Pass    |
|       |        | test_report_routes.py::TestTestReportSummary::test_summary_fields                                        | Pass    |
|       |        | test_report_routes.py::TestTestReportSummary::test_runningy_field_is_bool                                | Pass    |
|       |        | test_cluster_and_isolation.py::TestClusterIsolationStages::test_computed_per Storm                       | Storm   |
|       |        | test_cluster_and_isolation.py::TestClusterIsolationStages::test_individual_queries_enabled               | Enabled |
|       |        | test_cluster_and_isolation.py::TestClusterIsolationStages::test_longest_clusters storm has most damage   | Pass    |
|       |        | test_cluster_and_isolation.py::TestClusterIsolationStages::test_holm_isolation from sequence returns 404 | Pass    |
|       |        | test_cluster_and_isolation.py::TestNegativeEquity::test_computed_correctly                               | Pass    |
|       |        | test_cluster_and_isolation.py::TestNegativeEquity::test_flags_set_on_property                            | Pass    |
|       |        | test_cluster_and_isolation.py::TestNegativeEquity::test_damage_in_sequences_100_pct                      | Pass    |
|       |        | test_cluster_and_isolation.py::TestNegativeEquity::test_significance_negative_equity                     | Pass    |

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| Model | Module                                          | Test Name                                                  | Status |
|-------|-------------------------------------------------|------------------------------------------------------------|--------|
|       | test_cluster_and_isolation.py::TestSequence     | testSequenceSequence_no_contamination                      | Pass   |
|       | test_cluster_and_isolation.py::TestSequence     | testSequenceAIsolation storm returns only its damage       | Pass   |
|       | test_cluster_and_isolation.py::TestSequence     | testSequenceBIsolation returns only seqB damage            | Pass   |
|       | test_doublet_storms.py::TestDoublet             | testDoubletFirstStorm damage is independent of second      | Pass   |
|       | test_doublet_storms.py::TestDoublet             | testDoubletFirstStorm returns only flooded properties      | Pass   |
|       | test_doublet_storms.py::TestDoublet             | testDoubletBothStorm totals differ between storms          | Pass   |
|       | test_doublet_storms.py::TestDoublet             | testDoubletSecondStorm larger damage for properties        | Pass   |
|       | test_doublet_storms.py::TestDoublet             | testDoubletSecondStorm returns both properties             | Pass   |
|       | test_doublet_storms.py::TestDoublet             | testDoubletBothStorms response matches query               | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_max_depth                             | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_per_property_reported                 | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_in_sequence_reported                  | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_double_counted                        | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_max_depth_event                       | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_included_via_second_storm             | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_included_from_different_sequences     | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_damage_less_than_sum_of_storms        | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_options                               | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_returns_200                           | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_id_in_response                        | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_no_response_returns_404               | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_negative_Equity_from_worst_storm      | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_Negative_Equity_four                  | Pass   |
|       | test_sequence_endpoint.py::TestSequenceEndpoint | testSequenceEndpoint_Negative_Equity_storm                 | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_exception_returns_500           | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_success                         | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_missing_property_id_returns_400 | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_not_found_returns_404           | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_no_json_body_returns_400        | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_no_json_returns_error           | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_options_returns_ok              | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_property_not_found_returns_404  | Pass   |
|       | properties.py::TestGenerateMortgageReport       | testGenerateMortgageReport_import_error_returns_500        | Pass   |
|       | properties.py::TestGeneratePropertyReport       | testGeneratePropertyReport_exception_returns_500           | Pass   |
|       | properties.py::TestGeneratePropertyReport       | testGeneratePropertyReport_success                         | Pass   |
|       | properties.py::TestGeneratePropertyReport       | testGeneratePropertyReport_missing_property_id_returns_400 | Pass   |
|       | properties.py::TestGeneratePropertyReport       | testGeneratePropertyReport_no_json_body_returns_400        | Pass   |
|       | properties.py::TestGeneratePropertyReport       | testGeneratePropertyReport_no_json_returns_error           | Pass   |
|       | properties.py::TestGeneratePropertyReport       | testGeneratePropertyReport_options_returns_ok              | Pass   |
|       | properties.py::TestGeneratePropertyReport       | testGeneratePropertyReport_property_not_found_returns_404  | Pass   |
|       | properties.py::TestGeneratePropertyReport       | testGeneratePropertyReport_import_error_returns_500        | Pass   |
|       | properties.py::TestGetProperty                  | testGetProperty_test_found_returns_property                | Pass   |
|       | properties.py::TestGetProperty                  | testGetProperty_test_not_found_returns_404                 | Pass   |
|       | properties.py::TestListMortgages                | testListMortgages_empty                                    | Pass   |

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| Model | Module                              | Test Name                                  | Status |
|-------|-------------------------------------|--------------------------------------------|--------|
|       | properties.py::TestListMortgages    | test_list_mortgages_exception_returns_500  | Pass   |
|       | properties.py::TestListMortgages    | test_list_mortgages_success                | Pass   |
|       | properties.py::TestListProperties   | test_loader_exception_returns_500          | Pass   |
|       | properties.py::TestListProperties   | test_success_empty_list                    | Pass   |
|       | properties.py::TestListProperties   | test_success_with_properties               | Pass   |
|       | properties.py::TestNoJsonBody       | test_mortgage_report_null_json_returns_400 | Pass   |
|       | properties.py::TestNoJsonBody       | test_property_report_null_json_returns_400 | Pass   |
|       | properties.py::TestPropertyMortgage | test_found_returns_mortgage                | Pass   |
|       | properties.py::TestPropertyMortgage | test_loader_exception_returns_500          | Pass   |
|       | properties.py::TestPropertyMortgage | test_not_found_returns_404                 | Pass   |
|       | properties.py::TestPropertyMortgage | test_options_returns_ok                    | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_data_contains_hazard_curve            | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_data_contains_nearest_gauges          | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_data_contains_property_id             | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_data_contains_summary                 | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_known_property_returns_200            | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_known_property_status_success         | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_no_data_returns_404                   | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_no_data_status_error                  | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_options_returns_200                   | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_options_status_ok                     | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_response_has_data_key                 | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_second_property_also_works            | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_unknown_property_message_contains_id  | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_unknown_property_returns_404          | Pass   |
|       | propertyhc.py::TestPropertyHazard   | test_unknown_property_status_error         | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_entry_has_avg_basis_bps         | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_entry_has_flood_count           | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_entry_has_nearest_gauges        | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_entry_has_property_id           | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_table_count_matches             | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_table_length                    | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_basis_table_sorted_desc_by_basis      | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_nearest_gauge_has_distance_km         | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_nearest_gauge_has_gauge_id            | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_no_data_returns_404                   | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_no_data_status_error                  | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_options_returns_200                   | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_options_status_ok                     | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_response_has_basis_table              | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_response_has_count                    | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_returns_200_with_data                 | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_status_is_success                     | Pass   |
|       | propertyhc.py::TestPropertyhcBasis  | test_empty_basis_table_is_list             | Pass   |

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| Model | Module                                                                | Test Name                                            | Status |
|-------|-----------------------------------------------------------------------|------------------------------------------------------|--------|
|       | propertyhc.py::TestPropertyhcBaseEmpty                                | test_empty_count_is_zero                             | Pass   |
|       | propertyhc.py::TestPropertyhcBaseEmpty                                | test_empty_returns_200                               | Pass   |
|       | propertyhc.py::TestPropertyhcStatusDistributionAvgBasisBps            | test_distribution_avg_basis_bps                      | Pass   |
|       | propertyhc.py::TestPropertyhcStatusDistributionAvgFloodCount          | test_distribution_avg_flood_count                    | Pass   |
|       | propertyhc.py::TestPropertyhcStatusDistributionMaxDepthM              | test_distribution_max_depth_m                        | Pass   |
|       | propertyhc.py::TestPropertyhcStatusDistributionNumProperties          | test_distribution_num_properties                     | Pass   |
|       | propertyhc.py::TestPropertyhcStatusDataMessageMentionsCommand         | test_data_message_mentions_command                   | Pass   |
|       | propertyhc.py::TestPropertyhcStatusDataReturns404                     | test_data_returns_404                                | Pass   |
|       | propertyhc.py::TestPropertyhcStatusDataStatusIsError                  | test_data_status_is_error                            | Pass   |
|       | propertyhc.py::TestPropertyhcStatusOptionsReturns200                  | test_options_returns_200                             | Pass   |
|       | propertyhc.py::TestPropertyhcStatusOptionsStatusOk                    | test_options_status_ok                               | Pass   |
|       | propertyhc.py::TestPropertyhcStatusResponseHasDataKey                 | test_response_has_data_key                           | Pass   |
|       | propertyhc.py::TestPropertyhcStatusResponseHasDistribution            | test_response_has_distribution                       | Pass   |
|       | propertyhc.py::TestPropertyhcStatusResponseHasMetadata                | test_response_has_metadata                           | Pass   |
|       | propertyhc.py::TestPropertyhcStatusResponseHasSummary                 | test_response_has_summary                            | Pass   |
|       | propertyhc.py::TestPropertyhcStatusReturns200WithData                 | test_returns_200_with_data                           | Pass   |
|       | propertyhc.py::TestPropertyhcStatusStatusIsSuccess                    | test_status_is_success                               | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmptyCurvesAvgFloodCountIsZero     | test_empty_curves_avg_flood_count_is_zero            | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmptyCurvesNumPropertiesIsZero     | test_empty_curves_num_properties_is_zero             | Pass   |
|       | propertyhc.py::TestPropertyhcStatusEmptyCurvesReturns200              | test_empty_curves_returns_200                        | Pass   |
|       | claim.py::TestClaimReportGeneratorErrorHasMessage                     | test_generator_error_has_message                     | Pass   |
|       | claim.py::TestClaimReportGeneratorErrorReturns500                     | test_generator_error_returns_500                     | Pass   |
|       | claim.py::TestClaimReportGeneratorErrorStatusIsError                  | test_generator_error_status_is_error                 | Pass   |
|       | claim.py::TestClaimReportHappyPostContentLengthHeaderSet              | test_content_length_header_set                       | Pass   |
|       | claim.py::TestClaimReportHappyPostIfFilenameContainsPropId            | test_if_filename_contains_prop_id                    | Pass   |
|       | claim.py::TestClaimReportHappyPostResponseBodyIsPdfBytes              | test_response_body_is_pdf_bytes                      | Pass   |
|       | claim.py::TestClaimReportHappyPostResponseHasContentDisposition       | test_response_has_content_disposition                | Pass   |
|       | claim.py::TestClaimReportHappyPostResponseIsPdfContentType            | test_response_is_pdf_content_type                    | Pass   |
|       | claim.py::TestClaimReportHappyPostSuccessReturns200                   | test_success_returns_200                             | Pass   |
|       | claim.py::TestClaimReportHelperTestPromptSequencesFileHandled         | test_prompt_sequences_file_handled                   | Pass   |
|       | claim.py::TestClaimReportHelperTestMissingMortgageJsonStillGenerates  | test_missing_mortgage_json_still_generates           | Pass   |
|       | claim.py::TestClaimReportHelperTestMissingPropertyJsonStillGenerates  | test_missing_property_json_still_generates           | Pass   |
|       | claim.py::TestClaimReportHelperTestMissingSequencesFileStillGenerates | test_missing_sequences_file_still_generates          | Pass   |
|       | claim.py::TestClaimReportNoDirTestNoDirMessageMentionsPropertyType    | test_no_dir_message_mentions_property_type           | Pass   |
|       | claim.py::TestClaimReportNoDirTestNoDirReturns404                     | test_no_dir_returns_404                              | Pass   |
|       | claim.py::TestClaimReportNoDirTestNoDirStatusError                    | test_no_dir_status_error                             | Pass   |
|       | claim.py::TestClaimReportNoFloodEventFloodsMessageMentionsEvents      | test_no_flood_event_floods_message_mentions_events   | Pass   |
|       | claim.py::TestClaimReportNoFloodEventFloodsReturns404                 | test_no_flood_event_floods_returns_404               | Pass   |
|       | claim.py::TestClaimReportNoFloodEventFloodsStatusError                | test_no_flood_event_floods_status_error              | Pass   |
|       | claim.py::TestClaimReportNotFoundUnknownPropertyMessageContainsPdf    | test_not_found_unknown_property_message_contains_pdf | Pass   |
|       | claim.py::TestClaimReportNotFoundUnknownPropertyReturns404            | test_not_found_unknown_property_returns_404          | Pass   |
|       | claim.py::TestClaimReportNotFoundUnknownPropertyStatusError           | test_not_found_unknown_property_status_error         | Pass   |
|       | core.py::TestListFloodStorms                                          | test_no_storms_file_returns_404                      | Pass   |
|       | core.py::TestListFloodStorms                                          | test_options_returns_ok                              | Pass   |

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| Model | Module                            | Test Name                                         | Status |
|-------|-----------------------------------|---------------------------------------------------|--------|
|       | core.py::TestListFloodStorms      | test_storm_has_expected_fields                    | Pass   |
|       | core.py::TestListFloodStorms      | test_storm_name_from_intensity_category           | Pass   |
|       | core.py::TestListFloodStorms      | test_storms_list_count                            | Pass   |
|       | core.py::TestListFloodStorms      | test_storms_sorted_by_gauges_severe               | Pass   |
|       | core.py::TestListFloodStorms      | test_with_data_returns_success                    | Pass   |
|       | core.py::TestListFloodStorms      | test_no_gauges_read_error_continues               | Pass   |
|       | core.py::TestListFloodStorms      | test_no_metadata_json_read_error_continues        | Pass   |
|       | core.py::TestListFloodStorms      | test_no_property_flooding_enrichment              | Pass   |
|       | core.py::TestListFloodStorms      | test_no_property_json_read_error_continues        | Pass   |
|       | core.py::TestListFloodStorms      | test_no_property_not_in_valued_ids_skipped        | Pass   |
|       | core.py::TestListFloodStorms      | test_no_storm_with_empty_storm_id_skipped         | Pass   |
|       | core.py::TestListFloodStorms      | test_no_storms_metadata_from_storms_json          | Pass   |
|       | core.py::TestListFloodStorms      | test_no_storms_severity_classification_all_levels | Pass   |
|       | core.py::TestListFloodStorms      | test_no_zero_flood_depth_skipped                  | Pass   |
|       | core.py::TestPropertyFloods       | test_existing_property_returns_success            | Pass   |
|       | core.py::TestPropertyFloods       | test_include_readings_false_strips_readings       | Pass   |
|       | core.py::TestPropertyFloods       | test_include_readings_true_keeps_readings         | Pass   |
|       | core.py::TestPropertyFloods       | test_missing_property_returns_404                 | Pass   |
|       | core.py::TestPropertyFloods       | test_options_returns_ok                           | Pass   |
|       | core.py::TestPropertyStorms       | test_existing_property_returns_success            | Pass   |
|       | core.py::TestPropertyStorms       | test_flood_events_tagged_with_sequence_type       | Pass   |
|       | core.py::TestPropertyStorms       | test_missing_property_returns_404                 | Pass   |
|       | core.py::TestPropertyStorms       | test_nearest_gauges_enriched_with_flood_stages    | Pass   |
|       | core.py::TestPropertyStorms       | test_options_returns_ok                           | Pass   |
|       | core.py::TestPropertyStorms       | test_response_has_flood_events                    | Pass   |
|       | core.py::TestPropertyStorms       | test_response_has_nearest_gauges                  | Pass   |
|       | core.py::TestPropertyStorms       | test_response_has_property_info                   | Pass   |
|       | core.py::TestPropertyStorms       | test_no_storms_without_gauge_json                 | Pass   |
|       | core.py::TestPropertyStorms       | test_no_storms_without_storm_sequences_json       | Pass   |
|       | core.py::TestSummary              | test_no_data_returns_404                          | Pass   |
|       | core.py::TestSummary              | test_options_returns_ok                           | Pass   |
|       | core.py::TestSummary              | test_with_data_returns_success                    | Pass   |
|       | financial.py::TestPortfolioImpact | test_known_storm_returns_success                  | Pass   |
|       | financial.py::TestPortfolioImpact | test_mortgaged_property_has_ltv_fields            | Pass   |
|       | financial.py::TestPortfolioImpact | test_no_pts_dir_returns_404                       | Pass   |
|       | financial.py::TestPortfolioImpact | test_options_returns_ok                           | Pass   |
|       | financial.py::TestPortfolioImpact | test_portfolio_damage_pct_non_negative            | Pass   |
|       | financial.py::TestPortfolioImpact | test_portfolio_totals_count                       | Pass   |
|       | financial.py::TestPortfolioImpact | test_properties_sorted_by_damage_descending       | Pass   |
|       | financial.py::TestPortfolioImpact | test_property_entry_has_financial_fields          | Pass   |
|       | financial.py::TestPortfolioImpact | test_response_has_portfolio                       | Pass   |
|       | financial.py::TestPortfolioImpact | test_response_has_properties_list                 | Pass   |
|       | financial.py::TestPortfolioImpact | test_response_has_storm_id                        | Pass   |
|       | financial.py::TestPortfolioImpact | test_unknown_storm_returns_404                    | Pass   |

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| Model | Module                                  | Test Name                                           | Status |
|-------|-----------------------------------------|-----------------------------------------------------|--------|
|       | financial.py::TestSequencePortfolios    | test_known_sequence_returns_success                 | Pass   |
|       | financial.py::TestSequencePortfolios    | test_no_pts_dir_returns_404                         | Pass   |
|       | financial.py::TestSequencePortfolios    | test_options_returns_ok                             | Pass   |
|       | financial.py::TestSequencePortfolios    | test_properties_sorted_by_damage_descending         | Fail   |
|       | financial.py::TestSequencePortfolios    | test_property_entry_has_financial_field             | Pass   |
|       | financial.py::TestSequencePortfolios    | test_property_entry_has_num_sequence_flow           | Pass   |
|       | financial.py::TestSequencePortfolios    | test_property_entry_has_worst_storm_id              | Pass   |
|       | financial.py::TestSequencePortfolios    | test_response_has_damage_model                      | Pass   |
|       | financial.py::TestSequencePortfolios    | test_response_has_num_storms                        | Pass   |
|       | financial.py::TestSequencePortfolios    | test_response_has_portfolio                         | Pass   |
|       | financial.py::TestSequencePortfolios    | test_response_has_properties_list                   | Pass   |
|       | financial.py::TestSequencePortfolios    | test_response_has_sequence_id                       | Pass   |
|       | financial.py::TestSequencePortfolios    | test_unknown_sequence_returns_404                   | Pass   |
|       | risk.py::TestPortfolioVarErrors         | test_missing_sequences_message_mentions_doublet     | Pass   |
|       | risk.py::TestPortfolioVarErrors         | test_missing_sequences_returns_404                  | Pass   |
|       | risk.py::TestPortfolioVarErrors         | test_no_pts_dir_returns_404                         | Pass   |
|       | risk.py::TestPortfolioVarErrors         | test_no_pts_dir_status_error                        | Pass   |
|       | risk.py::TestPortfolioVarErrors         | test_options_returns_ok                             | Pass   |
|       | risk.py::TestPortfolioVarMissingFields  | test_missing_mortgage_json_returns_200              | Pass   |
|       | risk.py::TestPortfolioVarMissingFields  | test_missing_mortgage_json_zero_portfolio_mortgages | Pass   |
|       | risk.py::TestPortfolioVarMissingFields  | test_missing_property_json_returns_200              | Pass   |
|       | risk.py::TestPortfolioVarMissingFields  | test_missing_property_json_status_success           | Pass   |
|       | risk.py::TestPortfolioVarMissingFields  | test_missing_property_json_zero_portfolio_value     | Pass   |
|       | risk.py::TestPortfolioVarMortgages      | test_mortgage_impairment_max_is_positive            | Pass   |
|       | risk.py::TestPortfolioVarMortgages      | test_mortgage_impairment_value                      | Pass   |
|       | risk.py::TestPortfolioVarMortgages      | test_single_storm_has_mortgage_impairment           | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms | test_prob_loss_pct_is_50                            | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms | test_storm_count_is_two                             | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms | test_storms_with_damage_is_one                      | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms | test_storms_zero_damage_is_one                      | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms | test_mail_storms_contains_the_damaged_storm         | Pass   |
|       | risk.py::TestPortfolioVarMultipleStorms | test_mail_storms_limited_to_50                      | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_no_mortgage_impairment_when_above_value        | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_prob_loss_pct_is_100                           | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_property_damage_cond_mean_positive             | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_property_damage_max_is_positive                | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_storm_count_is_one                             | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_storms_with_damage_is_one                      | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_storms_zero_plus_with_damage_equals_total      | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_total_portfolio_mortgages                      | Pass   |
|       | risk.py::TestPortfolioVarNumerics       | test_total_portfolio_value                          | Pass   |
|       | risk.py::TestPortfolioVarSuccess        | test_histogram_bin_has_lo_hi_count                  | Pass   |
|       | risk.py::TestPortfolioVarSuccess        | test_mort_histogram_has_50_bins                     | Pass   |
|       | risk.py::TestPortfolioVarSuccess        | test_mortgage_impairment_sub_keys                   | Pass   |

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| Model | Module                                        | Test Name                                 | Status |
|-------|-----------------------------------------------|-------------------------------------------|--------|
|       | prs.py::TestGenerateTradePDF                  | test_generate_pdf_with_cashflows          | Pass   |
|       | prs.py::TestGenerateTradePDF                  | test_generate_pdf_with_close_out          | Pass   |
|       | prs.py::TestGenerateTradePDF                  | test_generate_pdf_with_close_out_date     | Pass   |
|       | prs.py::TestGetTradePDF                       | test_existing_pdf_served                  | Pass   |
|       | prs.py::TestGetTradePDF                       | test_missing_pdf_returns_404              | Pass   |
|       | prs.py::TestListTrades                        | test_empty_dir_returns_zero_count         | Pass   |
|       | prs.py::TestListTrades                        | test_returns_trades_list_key              | Pass   |
|       | prs.py::TestListTradesException               | test_corrupt_trade_file_returns_500       | Pass   |
|       | prs.py::TestListTradesWithFile                | test_list_returns_trades_from_json_files  | Pass   |
|       | routes.py::TestCORS                           | test_cors_headers_on_get                  | Pass   |
|       | routes.py::TestCORS                           | test_options_returns_ok                   | Pass   |
|       | routes.py::TestErrorHandling                  | test_invalid_json                         | Pass   |
|       | routes.py::TestErrorHandling                  | test_missing_content_type                 | Pass   |
|       | routes.py::TestGaugeCrudEdgeCases             | test_list_gauges_error_returns_500        | Pass   |
|       | routes.py::TestGaugeCrudEdgeCases             | test_list_gauges_options                  | Pass   |
|       | routes.py::TestGaugeRoutes                    | test_generate_report_missing_gauge_id     | Pass   |
|       | routes.py::TestGaugeRoutes                    | test_get_gauge                            | Pass   |
|       | routes.py::TestGaugeRoutes                    | test_get_gauge_not_found                  | Pass   |
|       | routes.py::TestGaugeRoutes                    | test_get_gauge_statistics                 | Pass   |
|       | routes.py::TestGaugeRoutes                    | test_get_gaugets                          | Pass   |
|       | routes.py::TestGaugeRoutes                    | test_legacy_list_gauges                   | Pass   |
|       | routes.py::TestGaugeRoutes                    | test_list_gauges                          | Pass   |
|       | routes.py::TestHealthRoutes                   | test_detailed_health_check                | Pass   |
|       | routes.py::TestHealthRoutes                   | test_health_check                         | Pass   |
|       | routes.py::TestHealthRoutes                   | test_root_redirects                       | Pass   |
|       | routes.py::TestPropertyRoutes                 | test_generate_report_missing_body         | Pass   |
|       | routes.py::TestPropertyRoutes                 | test_generate_report_missing_property_id  | Pass   |
|       | routes.py::TestPropertyRoutes                 | test_generate_report_property_not_found   | Pass   |
|       | routes.py::TestPropertyRoutes                 | test_get_property                         | Pass   |
|       | routes.py::TestPropertyRoutes                 | test_get_property_not_found               | Pass   |
|       | routes.py::TestPropertyRoutes                 | test_legacy_generate_property_report      | Pass   |
|       | routes.py::TestPropertyRoutes                 | test_list_properties                      | Pass   |
|       | routes.py::TestPropertyRoutes                 | test_options_preflight                    | Pass   |
|       | routes.py::TestServerMain                     | test_main_calls_app_run                   | Pass   |
|       | consistency.py::TestStormIdConsistency        | test_new_storm_from_list_is_runnable      | Pass   |
|       | consistency.py::TestStormIdConsistency        | test_new_storm_auto_selection_is_runnable | Pass   |
|       | consistency.py::TestStressRunInputs           | test_storm_id_exact_match_required        | Pass   |
|       | consistency.py::TestStressRunInputs           | test_storm_list_has_all_run_inputs        | Pass   |
|       | hydrograph.py::TestActualHydrographFunction   | test_actual_function_decays_after_peak    | Pass   |
|       | hydrograph.py::TestActualHydrographFunction   | test_actual_function_reaches_peak         | Pass   |
|       | hydrograph.py::TestActualHydrographFunction   | test_actual_function_returns_168_hours    | Pass   |
|       | hydrograph.py::TestActualHydrographFunction   | test_actual_function_starts_at_base       | Pass   |
|       | hydrograph.py::TestActualHydrographPeakAnchor | test_peak_anchor_matches_peak_position    | Pass   |
|       | hydrograph.py::TestGaugetsHydrograph          | test_phillips_synthesis_when_no_gaugets   | Pass   |

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| Model | Module                                       | Test Name                                                      | Status |
|-------|----------------------------------------------|----------------------------------------------------------------|--------|
|       | test_port_stress.infra.py::TestPortStress    | testSingleError_returns_500                                    | Pass   |
|       | test_port_stress.infra.py::TestPortStress    | testMissingForms_file_returns_404                              | Pass   |
|       | test_port_stress.infra.py::TestPortStress    | testForbiddenBody_returns_error                                | Pass   |
|       | test_port_stress.infra.py::TestPortStress    | testPredictorCalled_per_gauge                                  | Pass   |
|       | test_port_stress.infra.py::TestPortStress    | testStressPredictor_cache_hit                                  | Pass   |
|       | test_port_stress.infra.py::TestPortStress    | testStressPredictor_falls_back                                 | Pass   |
|       | test_port_stress.infra.py::TestStress        | testCacheCanBeReloaded_when_mtime_unchanged                    | Failed |
|       | test_port_stress.infra.py::TestStress        | testCacheCanBeReloaded_when_mtime_change                       | Pass   |
|       | test_port_stress.run.py::TestPortStress      | testSprayPplCalculation_notional                               | Pass   |
|       | test_port_stress.run.py::TestPortStress      | testSprayPplCalculation_sum_of_gauges                          | Pass   |
|       | test_port_stress.run.py::TestPortStress      | testSprayPplCalculation_notional                               | Pass   |
|       | test_port_stress.run.py::TestPortStress      | testSeverePplCalculation_matches_p_flood_to_100pct             | Pass   |
|       | test_port_stress.run.py::TestPortStress      | testSeverePplCalculation_a                                     | Pass   |
|       | test_port_stress.run.py::TestPortStress      | testSumPplCalculation_has_zero_p_flood                         | Pass   |
|       | test_port_stress.run.py::TestRunStress       | testPortfolio_gauge_results_fields                             | Pass   |
|       | test_port_stress.run.py::TestRunStress       | testPortfolio_run_gauges_severe_list                           | Pass   |
|       | test_port_stress.run.py::TestRunStress       | testPortfolio_run_gauges_warning_list                          | Pass   |
|       | test_port_stress.run.py::TestRunStress       | testPortfolio_run_has_gauge_results                            | Pass   |
|       | test_port_stress.run.py::TestRunStress       | testPortfolio_run_has_portfolio_totals                         | Pass   |
|       | test_port_stress.run.py::TestRunStress       | testPortfolio_invalid_storm                                    | Pass   |
|       | test_port_stress.run.py::TestRunStress       | testPortfolio_missing_storm_id                                 | Pass   |
|       | test_port_stress.run.py::TestRunStress       | testPortfolio_returns_success                                  | Pass   |
|       | test_port_stress.storms.py::TestGetPortfolio | testPortfolio_storms_has_storms                                | Pass   |
|       | test_port_stress.storms.py::TestGetPortfolio | testPortfolio_storms_success                                   | Pass   |
|       | test_port_stress.storms.py::TestGetPortfolio | testPortfolio_storms_has_required_fields                       | Pass   |
|       | test_port_stress.storms.py::TestGetPortfolio | testPortfolio_storms_no_data                                   | Pass   |
|       | test_port_stress.storms.py::TestGetPortfolio | testPortfolio_storms_secondary_sort                            | Pass   |
|       | test_port_stress.storms.py::TestGetPortfolio | testPortfolio_storms_sorted_by_severity                        | Pass   |
|       | test_port_stress.storms.py::TestPortfolio    | testStormNameEquals storm_id                                   | Pass   |
|       | test_port_stress.storms.py::TestPortfolio    | testStormNameDisplay_label_not_id                              | Pass   |
|       | test_port_stress.storms.py::TestPortfolio    | testGaugesSeverities_severe_before_warning                     | Pass   |
|       | test_port_stress.storms.py::TestPortfolio    | testGaugesSeverities_warning_before_alert                      | Pass   |
|       | test_port_stress.storms.py::TestPortfolio    | testGaugesSeverities_trades_have_trade_data                    | Pass   |
|       | test_port_stress.storms.py::TestPortfolio    | testStormDetails_fields                                        | Pass   |
|       | test_port_stress.storms.py::TestPortfolio    | testStormDetails_severity_for_moderate_with_equal_severe_count | Pass   |
|       | test_stress.infra.py::TestGetPredictor       | testPredictor_falls_back_to_legacy_dir                         | Pass   |
|       | test_stress.infra.py::TestGetPredictor       | testPredictor_loads_from_stressm_dir                           | Pass   |
|       | test_stress.infra.py::TestGetPredictor       | testPredictor_warns_when_no_models                             | Pass   |
|       | test_stress.infra.py::TestHydrograph         | testRemainingZeroExactly_zero                                  | Pass   |
|       | test_stress.infra.py::TestLoadStress         | testStatOSError_returns_cached                                 | Pass   |
|       | test_stress.infra.py::TestLoadStress         | testStatOSError_returns_none_when_no_data                      | Failed |
|       | test_stress.infra.py::TestRunStress          | testEdgeCase_stress_empty_gauge_id_returns_400                 | Pass   |
|       | test_stress.infra.py::TestRunStress          | testEdgeCase_stress_no_open_trades_returns_404                 | Pass   |
|       | test_stress.infra.py::TestRunStress          | testEdgeCase_stress_null_body_returns_400                      | Pass   |

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| Model | Module                                             | Test Name                                                                              | Status |
|-------|----------------------------------------------------|----------------------------------------------------------------------------------------|--------|
|       | test_stress_infra.py::TestRunStress                | test_stress_infra.py::TestRunStress:stress_predictor_not_found_returns_404             | Pass   |
|       | test_stress_infra.py::TestRunStress                | test_stress_infra.py::TestRunStress:stress_uses_gaugets_hydrograph                     | Pass   |
|       | test_stress_infra.py::TestStressCache              | test_stress_infra.py::TestStressCache:stress_predictor_cache_hit                       | Pass   |
|       | test_stress_infra.py::TestStressCache              | test_stress_infra.py::TestStressCache:stress_predictor_returns_none_when_model_missing | Pass   |
|       | test_stress_infra.py::TestStressEngine             | test_stress_infra.py::TestStressEngine:stress_engine_error_returns_500                 | Pass   |
|       | test_stress_infra.py::TestStressEngine             | test_stress_infra.py::TestStressEngine:stress_gauges_engine_error_returns_500          | Pass   |
|       | test_stress_infra.py::TestStressEngine             | test_stress_infra.py::TestStressEngine:stress_storms_engine_error_returns_500          | Pass   |
|       | test_stress_infra.py::TestStressHydrograph         | test_stress_infra.py::TestStressHydrograph:hydrograph_peak_at_zero                     | Pass   |
|       | test_stress_infra.py::TestStressHydrograph         | test_stress_infra.py::TestStressHydrograph:hydrograph_remaining_zero                   | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_hourly_fields                                 | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_invalid_storm                                 | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_knock_out                                     | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_missing_params                                | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_no_body                                       | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_portfolio_summary                             | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_probability_surface                           | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_returns_hourly                                | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:stress_trade_summary                                 | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:surface_caps_at_severe                               | Pass   |
|       | test_stress_run.py::TestRunStress                  | test_stress_run.py::TestRunStress:surface_shows_full_168h                              | Pass   |
|       | test_stress_scenario_coverage.py::TestBudget       | test_stress_scenario_coverage.py::TestBudget:budget_falls_back                         | Pass   |
|       | test_stress_scenario_coverage.py::TestHourlyGauges | test_stress_scenario_coverage.py::TestHourlyGauges:hourly_gauges_added_to_storm_hourly | Pass   |
|       | test_stress_scenario_coverage.py::TestModelAUF     | test_stress_scenario_coverage.py::TestModelAUF:from_summary_no_match                   | Pass   |
|       | test_stress_scenario_coverage.py::TestModelAUF     | test_stress_scenario_coverage.py::TestModelAUF:from_summary_raises                     | Pass   |
|       | test_stress_scenario_coverage.py::TestModelAUF     | test_stress_scenario_coverage.py::TestModelAUF:present_when_summary_has                | Pass   |
|       | test_stress_scenario_coverage.py::TestModelAUF     | test_stress_scenario_coverage.py::TestModelAUF:skip_untrained_gauge                    | Pass   |
|       | test_stress_scenario_coverage.py::TestNoGauges     | test_stress_scenario_coverage.py::TestNoGauges:with_no_gauges_returns_404              | Pass   |
|       | test_stress_scenario_coverage.py::TestPredictor    | test_stress_scenario_coverage.py::TestPredictor:not_found_returns_404                  | Pass   |
|       | test_stress_scenario_coverage.py::TestStressStorms | test_stress_scenario_coverage.py::TestStressStorms:not_found_file                      | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:has_peak_level                                    | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:has_storm_fields                                  | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:no_data                                           | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:no_gauge_id_returns_all_storms                    | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:no_matching_gauge                                 | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:returns_list                                      | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:sorted_by_peak                                    | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:gauges_has_required_fields                        | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:gauges_has_trade_counts                           | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:gauges_returns_list                               | Pass   |
|       | test_stress_storms.py::TestGetStorms               | test_stress_storms.py::TestGetStorms:gauges_sorted_by_longitude                        | Pass   |
|       | visualization.py::TestServeVisualization           | test_stress_infra.py::TestServeVisualization:storm_get_returns_json_error              | Pass   |
|       | visualization.py::TestServeVisualization           | test_stress_infra.py::TestServeVisualization:no_gauge_data_returns_error               | Pass   |
|       | visualization.py::TestServeVisualization           | test_stress_infra.py::TestServeVisualization:no_gauge_data_returns_error_status        | Pass   |
|       | visualization.py::TestServeVisualization           | test_stress_infra.py::TestServeVisualization:unknown_exception_returns_500             | Pass   |

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| Model | Module                                    | Test Name                                       | Status |
|-------|-------------------------------------------|-------------------------------------------------|--------|
|       | visualization.py::TestServeVisualizations | test_iframe_callback_insensitive                | Pass   |
|       | visualization.py::TestServeVisualizations | test_iframe_callback_redirects                  | Pass   |
|       | visualization.py::TestServeVisualizations | test_iframe_callback_attachment_returns_400     | Pass   |
|       | visualization.py::TestServeVisualizations | test_iframe_callback_attachment_message         | Pass   |
|       | visualization.py::TestServeVisualizations | test_iframe_callback_import_error_returns_503   | Pass   |
|       | visualization.py::TestServeVisualizations | test_iframe_callback_success_returns_200        | Pass   |
|       | base_loader.py::TestBaseLoader            | test_caching_prevents_reload                    | Pass   |
|       | base_loader.py::TestBaseLoader            | test_clear_cache                                | Pass   |
|       | base_loader.py::TestBaseLoader            | test_force_reload_bypasses_cache                | Pass   |
|       | base_loader.py::TestBaseLoader            | test_invalidate_cache                           | Pass   |
|       | base_loader.py::TestBaseLoader            | test_abstract_methods_enforced                  | Pass   |
|       | base_loader.py::TestBaseLoader            | test_subclass_must_implement_get_entity_ids     | Pass   |
|       | base_loader.py::TestBaseLoader            | test_subclass_must_implement_get_entity_summary | Pass   |
|       | base_loader.py::TestBaseLoader            | test_loading_exists_check                       | Skip   |
|       | base_loader.py::TestBaseLoader            | test_loading_alternative_container_key          | Pass   |
|       | base_loader.py::TestBaseLoader            | test_loading_dict_with_container_key            | Pass   |
|       | base_loader.py::TestBaseLoader            | test_loading_list_format                        | Pass   |
|       | base_loader.py::TestBaseLoader            | test_loading_missing_file                       | Pass   |
|       | base_loader.py::TestBaseLoader            | test_loading_single_object                      | Pass   |
|       | base_loader.py::TestBaseLoader            | test_entity_count                               | Pass   |
|       | base_loader.py::TestBaseLoader            | test_entity_exists                              | Pass   |
|       | base_loader.py::TestBaseLoader            | test_entity_find_by_id                          | Pass   |
|       | base_loader.py::TestBaseLoader            | test_entity_find_by_id_not_found                | Pass   |
|       | base_loader.py::TestBaseLoader            | test_entity_get_ids                             | Skip   |
|       | base_loader.py::TestBaseLoader            | test_entity_get_status                          | Skip   |
|       | base_loader.py::TestBaseLoader            | test_entity_list_all                            | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_entity_id_extraction                       | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_entity_summary                             | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_load_gauges                                | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_active_gauges                      | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_coordinates                        | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_flood_stages                       | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_gauge_name                         | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_gauge_name_not_found               | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_find_by_id                         | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_find_by_name                       | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_find_by_name_case_insensitive      | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_find_by_owner                      | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_find_by_status_active              | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_find_by_status_inactive            | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_find_by_type_river                 | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_find_by_type_tidal                 | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_gauges_in_radius                   | Pass   |
|       | gauge_loader.py::TestGaugeLoader          | test_helpers_gauges_in_radius_none_found        | Pass   |

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| Model | Module                                 | Test Name                                                                  | Status |
|-------|----------------------------------------|----------------------------------------------------------------------------|--------|
|       | loader_registry.py::TestCreateRegistry | test_create_registry                                                       | Pass   |
|       | loader_registry.py::TestCreateRegistry | test_create_registry_with_overrides                                        | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry                                                       | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_available_loaders                                     | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_clear_all_caches                                      | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_different_loaders_separate_instances                  | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_loading_instances_cached                              | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_file_overrides                                        | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_get_loader_by_name                                    | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_get_loader_unknown_type                               | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_stack_data_files                                      | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_status                                                | Skip   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_test_get_loader                                       | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_test_get_loader_by_name                               | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_test_get_loader_by_type                               | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_test_get_loader_by_type_and_name                      | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_test_get_loader_by_type_and_name_and_address          | Pass   |
|       | loader_registry.py::TestLoaderRegistry | test_loader_registry_test_get_loader_by_type_and_name_and_address_and_type | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_aggregate_mortgages                                   | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_aggregate_by_lender                                   | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_aggregate_properties_with_mortgages                   | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_aggregate_exposure                                    | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_basic_extraction                                      | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_basic_summary                                         | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_basic_mortgages                                       | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_id                                           | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_lender                                       | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_lender_case_insensitive                      | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_property_id                                  | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_property_id_not_found                        | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_status_active                                | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_status_delinquent                            | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_status_to_property                           | Pass   |
|       | mortgage_loader.py::TestMortgageLoader | test_mortgage_loader_query_by_status_to_property_with_mortgage             | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_aggregate_value                                       | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_aggregate_by_state                                    | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_basic_extraction                                      | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_basic_summary                                         | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_basic_properties                                      | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_location_coordinates                                  | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_location_coordinates_not_found                        | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_location                                              | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_location_not_found                                    | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_properties_in_radius                                  | Pass   |
|       | property_loader.py::TestPropertyLoader | test_property_loader_query_by_address                                      | Pass   |

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| Model | Module                               | Test Name                                    | Status |
|-------|--------------------------------------|----------------------------------------------|--------|
|       | property_loader.py::TestProperty     | test_dataframe_address_case_insensitive      | Pass   |
|       | property_loader.py::TestProperty     | test_dataframe_address_not_found             | Pass   |
|       | property_loader.py::TestProperty     | test_dataframe_id                            | Pass   |
|       | property_loader.py::TestProperty     | test_dataframe_state                         | Pass   |
|       | property_loader.py::TestProperty     | test_dataframe_state_case_insensitive        | Pass   |
|       | property_loader.py::TestProperty     | test_dataframe_zip                           | Pass   |
|       | property_loader.py::TestProperty     | test_dataframe_zip_multiple                  | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_change_base_gauges_for_portfolio_gauge  | Fail   |
|       | timeseries_loader.py::TestTimeseries | test_change_base_gauge_ids_match_portfolio   | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_load_basis_timestamp                    | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_load_basis_summary                      | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_load_basis_series                       | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_gauge_heading                    | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_gauge_heading                    | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_gauges_for_gauge                 | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_gauges_for_gauge_by_name         | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_gauges_for_gauge_not_found       | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_stalable_gauge_ids               | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_statistics                       | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_statistics_not_found             | Pass   |
|       | timeseries_loader.py::TestTimeseries | test_logger_end_of_range                     | Skip   |
|       | timeseries_loader.py::TestTimeseries | test_logger_end_of_range_for_gauge           | Skip   |
|       | test_b10_quick_wins.py::TestAuditee  | test_functions_truncated_above_10000         | Pass   |
|       | test_b10_quick_wins.py::TestEOD      | test_dread_not_in_new_trade                  | Pass   |
|       | test_b10_quick_wins.py::TestGauge    | test_dread_not_in_trade_exists               | Pass   |
|       | test_b10_quick_wins.py::TestInsuree  | test_area_fallthrough                        | Pass   |
|       | test_b10_quick_wins.py::TestIntere   | test_main_module_runs                        | Pass   |
|       | test_b10_quick_wins.py::TestIntere   | test_get_path_and_count_unknown_type_returns | Fail   |
|       | test_b10_quick_wins.py::TestJSON     | test_getting_gaes_on_unknown_key             | Pass   |
|       | test_b10_quick_wins.py::TestJSON     | test_getting_gaes_known_key                  | Pass   |
|       | test_b10_quick_wins.py::TestLtv      | test_layer_covers_percentage                 | Pass   |
|       | test_b10_quick_wins.py::TestProp     | test_mahatted_type_makes_typeerror           | Pass   |
|       | test_b10_quick_wins.py::TestProp     | test_empty_dataframe_encoded_as_float        | Pass   |
|       | test_b10_quick_wins.py::TestServ     | test_block_main_callable                     | Pass   |
|       | test_b10_quick_wins.py::TestStor     | test_get_block_linear_fallback               | Pass   |
|       | test_b10_quick_wins.py::TestTrad     | test_get_statistics_returns_dimensions       | Pass   |
|       | test_b10_quick_wins.py::TestVisu     | test_create_visualization_returns_instance   | Fail   |
|       | data_structures.py::TestDataVal      | test_to_result_fields                        | Pass   |
|       | data_structures.py::TestDataVal      | test_to_result_tuple                         | Pass   |
|       | data_structures.py::TestDataVal      | test_to_result_unpacking                     | Pass   |
|       | data_structures.py::TestDataVal      | test_to_result_result_fields                 | Pass   |
|       | data_structures.py::TestLoadedD      | test_can_set_gauge_data                      | Pass   |
|       | data_structures.py::TestLoadedD      | test_can_set_multiple_fields                 | Pass   |
|       | data_structures.py::TestLoadedD      | test_counts_default_to_zero                  | Pass   |

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| Model | Module                              | Test Name                                        | Status |
|-------|-------------------------------------|--------------------------------------------------|--------|
|       | data_structures.py::TestLoadedData  | test_defaults_are_none                           | Pass   |
|       | data_structures.py::TestLoadedData  | test_is_dataclass                                | Pass   |
|       | data_structures.py::TestLoadedData  | test_lookup_defaults_are_none                    | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_data_summary_contains_errors_count          | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_data_summary_contains_warnings_count        | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_data_summary_loaded_flag_false_when_no_file | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_data_summary_loaded_flag_true_for_gauge     | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_get_data_summary_empty_before_load          | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_get_data_summary_returns_dict               | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_get_validation_summary_empty_before_load    | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_get_validation_summary_is_copy              | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_get_validation_summary_returns_dict         | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_is_data_complete_depends_only_on_gauge      | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_is_data_complete_false_before_load          | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_is_data_complete_false_when_no_gauge        | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_is_data_complete_true_with_gauge            | Pass   |
|       | helpers.py::TestDataLoaderHelper    | test_validation_summary_has_gauge_key            | Pass   |
|       | helpers.py::TestPrintLoadingSummary | test_summary_with_full_data_does_not_crash       | Pass   |
|       | helpers.py::TestPrintLoadingSummary | test_summary_with_no_data_does_not_crash         | Pass   |
|       | helpers.py::TestPrintLoadingSummary | test_summary_with_partial_data_does_not_crash    | Pass   |
|       | init.py::TestDataLoaderInit         | test_init_with_input_dir                         | Pass   |
|       | init.py::TestDataLoaderInit         | test_init_with_string_path_converts_to_path      | Pass   |
|       | init.py::TestDataLoaderInit         | test_loaded_data_initialised                     | Pass   |
|       | init.py::TestDataLoaderInit         | test_private_loaders_created                     | Pass   |
|       | init.py::TestDataLoaderInit         | test_validation_results_empty_at_init            | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_counterparty_data_loaded                    | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_empty_dir_returns_loaded_data               | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_gauge_data_count_matches_records            | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_gauge_data_has_items_key                    | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_gaugehd_dir_counted                         | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_gaugets_dir_counted                         | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_gaugets_dir_multiple_files_counted          | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_hazard_curves_loaded                        | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_lookups_built_after_load                    | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_no_gaugehd_dir_count_stays_zero             | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_no_gaugets_dir_count_stays_zero             | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_no_propertyts_dir_count_stays_zero          | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_property_hazard_curves_loaded               | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_property_hazard_summary_fields_accessible   | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_propertyts_dir_counted                      | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_storm_data_list_format                      | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_storm_data_loaded_dict_format               | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_storm_data_unknown_format                   | Pass   |
|       | load_all.py::TestDataLoaderLoadAll  | test_storm_data_with_items_key                   | Pass   |

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| Model | Module                                        | Test Name                                          | Status |
|-------|-----------------------------------------------|----------------------------------------------------|--------|
|       | load.all.py::TestDataLoaderLoadAll            | test_with_gauge_data_loaded                        | Pass   |
|       | load.all.py::TestDataLoaderLoadAll            | test_with_mortgage_data_loaded                     | Pass   |
|       | load.all.py::TestDataLoaderLoadAll            | test_with_property_data_loaded                     | Pass   |
|       | loaders.py::TestLoadCounterpartyData          | test_corrupt_counterparty_file_leaves_data_none    | Pass   |
|       | loaders.py::TestLoadCounterpartyData          | test_missing_counterparty_file_leaves_data_none    | Pass   |
|       | loaders.py::TestLoadCounterpartyData          | test_valid_counterparty_file_loads                 | Pass   |
|       | loaders.py::TestLoadHazardCurveData           | test_corrupt_hazard_file_leaves_hazard_data_none   | Pass   |
|       | loaders.py::TestLoadHazardCurveData           | test_corrupt_property_hazard_file_leaves_data_none | Pass   |
|       | loaders.py::TestLoadHazardCurveData           | test_missing_hazard_file_leaves_hazard_data_none   | Pass   |
|       | loaders.py::TestLoadHazardCurveData           | test_missing_property_hazard_file_leaves_data_none | Pass   |
|       | loaders.py::TestLoadHazardCurveData           | test_valid_hazard_file_populates_hazard_data       | Pass   |
|       | loaders.py::TestLoadStormData                 | test_corrupt_storm_file_leaves_storm_data_none     | Pass   |
|       | loaders.py::TestLoadStormData                 | test_missing_storm_file_leaves_storm_data_none     | Pass   |
|       | loaders.py::TestLoadStormData                 | test_storms_dict_with_storms_key                   | Pass   |
|       | loaders.py::TestLoadStormData                 | test_storms_list_format                            | Pass   |
|       | loaders.py::TestLoadStormData                 | test_storms_other_type                             | Pass   |
|       | validation.py::TestLoadWithValidation         | test_empty_data_list_records_invalid_results       | Fail   |
|       | validation.py::TestLoadWithValidation         | test_empty_data_list_returns_none                  | Pass   |
|       | validation.py::TestLoadWithValidation         | test_exception_in_loader_returns_none              | Pass   |
|       | validation.py::TestLoadWithValidation         | test_exception_records_error_in_validation         | Pass   |
|       | validation.py::TestLoadWithValidation         | test_none_return_from_loader_returns_none          | Pass   |
|       | validation.py::TestLoadWithValidation         | test_successful_load_records_valid_results         | Pass   |
|       | validation.py::TestLoadWithValidation         | test_successful_load_returns_data_dict             | Pass   |
|       | validation.py::TestLoadWithValidation         | test_validation_summary_count_matches_items        | Pass   |
|       | validation.py::TestLoadWithValidation         | test_validation_summary_has_type_field             | Pass   |
|       | gauge_panel.py::TestGaugeGraphHistogramConfig | test_gauge_panel_height                            | Pass   |
|       | gauge_panel.py::TestGaugeGraphHistogramConfig | test_gauge_panel_width                             | Pass   |
|       | gauge_panel.py::TestGaugeGraphHistogramConfig | test_get_statistics                                | Pass   |
|       | gauge_panel.py::TestGaugeHazardCurveConfig    | test_gauge_panel_height                            | Pass   |
|       | gauge_panel.py::TestGaugeHazardCurveConfig    | test_gauge_panel_width                             | Pass   |
|       | gauge_panel.py::TestGaugeHazardCurveConfig    | test_get_statistics                                | Pass   |
|       | gauge_panel.py::TestGaugeStormAnalysisConfig  | test_gauge_panel_height                            | Pass   |
|       | gauge_panel.py::TestGaugeStormAnalysisConfig  | test_gauge_panel_width                             | Pass   |
|       | gauge_panel.py::TestGaugeStormAnalysisConfig  | test_get_statistics                                | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_fetches_gauge_history              | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_fetches_stress_storms              | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_has_storm_scenarios                | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_has_time_series_chart              | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_renders                            | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_renders_storm_list                 | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_shows_flood_stages                 | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_shows_trigger_colors               | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_storm_click_navigates_to_stress    | Pass   |
|       | gauge_panel.py::TestHistoricalTable           | test_historical_uses_event_listeners               | Pass   |

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| Model | Module                                             | Test Name                                       | Status |
|-------|----------------------------------------------------|-------------------------------------------------|--------|
|       | gauge_panel.py::TestStressTestTab                  | test_stress_fetches_storms                      | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_has_cash_price_concept              | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_has_chart_tabs                      | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_has_storm_selector                  | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_renders                             | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_runs_scenario                       | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_shows_p_flood                       | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_storm_hint_pickup                   | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_trade_table                         | Pass   |
|       | gauge_panel.py::TestStressTestTab                  | test_stress_uses_event_listeners_for_tabs       | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_alert_colour_amber                         | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_daily_observations_field                   | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_renders_historical_chart                   | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_severe_colour_red                          | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_shows_flood_alert_threshold                | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_shows_flood_warning_threshold              | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_shows_severe_flood_threshold               | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_subsamples_observations                    | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_warning_colour_orange                      | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_data_cache_keyed_by_gauge_id               | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_data_cache_history_data                    | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_data_cache_refreshed_data_on_revisit       | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_stats_bar_chart_function                   | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_stats_bar_flood_exceedances                | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_stats_bar_mean_level                       | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_stats_bar_record_years                     | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_stats_bar_renders_historical_tab           | Pass   |
|       | historical_detail.py::TestHistoricalChart          | test_stats_bar_safe_length                      | Pass   |
|       | historical_detail.py::TestHistoryEndpoint          | test_fetches_gauge_history                      | Pass   |
|       | historical_detail.py::TestHistoryEndpoint          | test_returns_gauge_id_variable                  | Pass   |
|       | historical_detail.py::TestStormChiefNavigation     | test_navigation_handler_defined                 | Pass   |
|       | historical_detail.py::TestStormChiefNavigation     | test_navigation_handler_iife_safe               | Pass   |
|       | historical_detail.py::TestStormChiefNavigation     | test_navigation_goes_to_tab_5                   | Pass   |
|       | historical_detail.py::TestStormChiefNavigation     | test_navigation_stress_storm_hint               | Pass   |
|       | historical_detail.py::TestStormChiefNavigation     | test_navigation_hint_set_before_switch_tab      | Pass   |
|       | historical_detail.py::TestStormChiefNavigation     | test_navigation_id_passed_to_navigate           | Pass   |
|       | historical_detail.py::TestStormForecastCoefficient | test_coefficient_with_stipitation_mm_consistent | Pass   |
|       | historical_detail.py::TestStormForecastCoefficient | test_gauges_with_stipitation                    | Pass   |
|       | historical_detail.py::TestStormForecastCoefficient | test_consistency_with_stipitation               | Pass   |
|       | historical_detail.py::TestStormForecastCoefficient | test_peak_level_with_stipitation                | Pass   |
|       | historical_detail.py::TestStormForecastCoefficient | test_consistency_with_stipitation               | Pass   |
|       | historical_detail.py::TestStormListOrder           | test_shows_overflow_count                       | Pass   |
|       | historical_detail.py::TestStormListOrder           | test_shows_top_30_storms                        | Pass   |
|       | historical_detail.py::TestStormListOrder           | test_sorted_by_peak_level                       | Pass   |

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| Model | Module                                      | Test Name                                    | Status |
|-------|---------------------------------------------|----------------------------------------------|--------|
|       | historical_detail.py::TestStormPresentation | test_effective_precipitation                 | Pass   |
|       | historical_detail.py::TestStormPresentation | test_gauges_severe                           | Pass   |
|       | historical_detail.py::TestStormPresentation | test_intensity_category                      | Pass   |
|       | historical_detail.py::TestStormPresentation | test_max_trigger_for_colour                  | Pass   |
|       | historical_detail.py::TestStormPresentation | test_peak_level_m                            | Pass   |
|       | historical_detail.py::TestStormPresentation | test_storm_id                                | Pass   |
|       | historical_detail.py::TestStormPresentation | test_storm_name_field                        | Pass   |
|       | historical_detail.py::TestStormScenarios    | test_endpoint_use_gauge_storms_endpoint      | Pass   |
|       | historical_detail.py::TestStormScenarios    | test_endpoint_storm_scenarios                | Pass   |
|       | historical_detail.py::TestStormScenarios    | test_storm_list_renders_in_panel             | Pass   |
|       | historical_detail.py::TestStormScenarios    | test_endpoint_trading_stress_storms_endpoint | Pass   |
|       | prs_detail.py::TestAnalyticalPrice          | test_cashflow_computation                    | Pass   |
|       | prs_detail.py::TestAnalyticalPrice          | test_survival_interpolation                  | Pass   |
|       | prs_detail.py::TestAnalyticalPrice          | test_yield_rate_interpolation                | Pass   |
|       | prs_detail.py::TestMaturityConversion       | test_compute_maturity_date                   | Pass   |
|       | prs_detail.py::TestMaturityConversion       | test_current_roll_date                       | Pass   |
|       | prs_detail.py::TestMaturityConversion       | test_last_friday_of_month                    | Pass   |
|       | prs_detail.py::TestMaturityConversion       | test_maturity_popup                          | Pass   |
|       | prs_detail.py::TestPRSControls              | test_controls_function                       | Pass   |
|       | prs_detail.py::TestPRSControls              | test_counterparty_dropdown                   | Pass   |
|       | prs_detail.py::TestPRSControls              | test_direction_toggle                        | Pass   |
|       | prs_detail.py::TestPRSControls              | test_maturity_selector                       | Pass   |
|       | prs_detail.py::TestPRSControls              | test_notional_input                          | Pass   |
|       | prs_detail.py::TestPRSControls              | test_trigger_selector                        | Pass   |
|       | prs_detail.py::TestPRSTabRender             | test_prs_tab_renders                         | Pass   |
|       | prs_detail.py::TestPRSTabRender             | test_prs_yield_curve_fetch                   | Pass   |
|       | prs_detail.py::TestTradeCommit              | test_commit_endpoint                         | Pass   |
|       | prs_detail.py::TestTradeCommit              | test_payer_flag_in_commit                    | Pass   |
|       | stress_detail.py::TestChartSubTable         | test_event_listeners_not_inline              | Pass   |
|       | stress_detail.py::TestChartSubTable         | test_flood_probability_tab                   | Pass   |
|       | stress_detail.py::TestChartSubTable         | test_stress_pnl_tab                          | Pass   |
|       | stress_detail.py::TestChartSubTable         | test_surface_tab                             | Pass   |
|       | stress_detail.py::TestKnockOutDisplay       | test_first_trigger_hour                      | Pass   |
|       | stress_detail.py::TestKnockOutDisplay       | test_ko_annotation_line                      | Pass   |
|       | stress_detail.py::TestKnockOutDisplay       | test_ko_label                                | Pass   |
|       | stress_detail.py::TestKnockOutDisplay       | test_triggered_hour_reference                | Pass   |
|       | stress_detail.py::TestStormSelector         | test_auto_selects_worst_storm                | Pass   |
|       | stress_detail.py::TestStormSelector         | test_storm_dropdown                          | Pass   |
|       | stress_detail.py::TestStressEndpoint        | test_run_endpoint                            | Pass   |
|       | stress_detail.py::TestStressEndpoint        | test_storms_endpoint                         | Pass   |
|       | stress_detail.py::TestStressTabRender       | test_stress_tab_renders                      | Pass   |
|       | stress_detail.py::TestSurfaceTable          | test_surface_ko_trim                         | Pass   |
|       | stress_detail.py::TestSurfaceTable          | test_surface_reads_probability_surface       | Pass   |
|       | stress_detail.py::TestSurfaceTable          | test_surface_renders_table                   | Pass   |

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| Model | Module                                                   | Test Name                                            | Status |
|-------|----------------------------------------------------------|------------------------------------------------------|--------|
|       | stress.detail.py::TestSurfaceTab                         | test_surface_trigger_band_shading                    | Pass   |
|       | stress.detail.py::TestSurfaceTab                         | test_surface_wrap_element                            | Pass   |
|       | js_safety.py::TestAuditReportsControl                    | test_audit_terminal_refresh_exposed_on_window        | Pass   |
|       | js_safety.py::TestAuditReportsControl                    | test_no_json_download_button                         | Pass   |
|       | js_safety.py::TestAuditReportsControl                    | test_poll_output_endpoint                            | Pass   |
|       | js_safety.py::TestAuditReportsControl                    | test_refresh_audit_tab_button                        | Pass   |
|       | js_safety.py::TestAuditReportsControl                    | test_run_test_suite_exposed_on_window                | Pass   |
|       | js_safety.py::TestAuditReportsControl                    | test_run_test_suite_function                         | Pass   |
|       | js_safety.py::TestAuditReportsControl                    | test_terminal_overlay_id                             | Pass   |
|       | js_safety.py::TestAuditReportsControl                    | test_toggle_failure_detail_exposed_on_window         | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_contains_iife                           | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_contains_regulatory_button              | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_contains_render_audit_reports           | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_contains_script_tags                    | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_contains_show_mg_panel                  | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_contains_terminal_popup                 | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_is_str                                  | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_non_empty                               | Pass   |
|       | js_safety.py::TestFullCompilation                        | test_full_js_passes_node_syntax_check                | Pass   |
|       | js_safety.py::TestGetJsReturnsString                     | test_all_modules_non_empty                           | Pass   |
|       | js_safety.py::TestGetJsReturnsString                     | test_all_modules_return_str                          | Pass   |
|       | js_safety.py::TestNoLiteralNewlineInJsStrings            | test_js_strings_no_newline_in_js_strings             | Pass   |
|       | js_safety.py::TestNoLiteralNewlineInJsStrings            | test_js_strings_reports_no_newline_in_js_strings     | Pass   |
|       | js_safety.py::TestNoLiteralNewlineInJsStrings            | test_js_strings_compiled_js_no_newline_in_js_strings | Pass   |
|       | test_backend_handler.py::TestAddToMap                    | test_add_to_map_calls_add_child                      | Pass   |
|       | test_backend_handler.py::TestBackendHandlerInit          | test_backend_handler_init                            | Pass   |
|       | test_backend_handler.py::TestBackendHandlerInit          | test_backend_handler_init_uses_config                | Pass   |
|       | test_backend_handler.py::TestBackendHandlerInit          | test_backend_handler_init_uses_copied                | Pass   |
|       | test_backend_handler.py::TestConfigureNoArgsNoChange     | test_configure_no_args_no_change                     | Pass   |
|       | test_backend_handler.py::TestConfigureUpdatesEndpoints   | test_configure_updates_endpoints                     | Pass   |
|       | test_backend_handler.py::TestConfigureUpdatesServerUrl   | test_configure_updates_server_url                    | Pass   |
|       | test_backend_handler.py::TestGetTest                     | test_contains_timeout                                | Pass   |
|       | test_backend_handler.py::TestGetTest                     | test_returns_script_tag                              | FAIL   |
|       | test_backend_handler.py::TestGetStatisticsKeys           | test_get_statistics_keys                             | Pass   |
|       | test_backend_handler.py::TestGetStatisticsSingleEndpoint | test_get_statistics_single_endpoint                  | Pass   |
|       | test_context_menus.py::TestAddToMap                      | test_add_to_map_calls_add_child                      | Pass   |
|       | test_context_menus.py::TestConfigureBothMenus            | test_configure_both_menus                            | Pass   |
|       | test_context_menus.py::TestConfigureGaugeMenu            | test_configure_gauge_menu                            | Pass   |
|       | test_context_menus.py::TestConfigureNoArgsNoChange       | test_configure_no_args_no_change                     | Pass   |
|       | test_context_menus.py::TestConfigurePropertyMenu         | test_configure_property_menu                         | Pass   |
|       | test_context_menus.py::TestContextMenusInit              | test_context_menus_init                              | Pass   |
|       | test_context_menus.py::TestContextMenusInit              | test_context_menus_init                              | Pass   |
|       | test_context_menus.py::TestGetStatisticsCustom           | test_get_statistics_custom                           | Pass   |
|       | test_context_menus.py::TestGetStatisticsDefault          | test_get_statistics_default                          | Pass   |

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| Model | Module                                              | Test Name                                               | Status |
|-------|-----------------------------------------------------|---------------------------------------------------------|--------|
|       | test_manager.py::TestConfigure                      | test_configure_menus                                    | Pass   |
|       | test_manager.py::TestConfigure                      | test_configure_notification_position                    | Pass   |
|       | test_manager.py::TestConfigure                      | test_configure_notification_timeout                     | Pass   |
|       | test_manager.py::TestConfigure                      | test_configure_returns_self                             | Pass   |
|       | test_manager.py::TestConfigure                      | test_configure_server_url                               | Pass   |
|       | test_manager.py::TestGetStatistics                  | test_returns_dict_with_expected_keys                    | Pass   |
|       | test_manager.py::TestGetStatistics                  | test_statistics_are_dicts                               | Pass   |
|       | test_manager.py::TestSetupMap                       | test_adds_all_components                                | Pass   |
|       | test_manager.py::TestSetupMap                       | test_activity_element_added                             | Pass   |
|       | test_manager.py::TestSetupMap                       | test_activities_self                                    | Pass   |
|       | test_notifications.py::TestAddNotificationToMap     | test_adds_to_map                                        | Pass   |
|       | test_notifications.py::TestAddNotificationToMap     | test_adds_to_notification_system                        | Pass   |
|       | test_notifications.py::TestAddToMap                 | test_add_to_map_calls_add_child                         | Pass   |
|       | test_notifications.py::TestBuildPosition            | test_returns_position_function                          | Pass   |
|       | test_notifications.py::TestBuildTemplate            | test_contains_all_types                                 | Pass   |
|       | test_notifications.py::TestBuildTemplate            | test_includes_template_values                           | Pass   |
|       | test_notifications.py::TestBuildTemplate            | test_returns_valid_json                                 | Pass   |
|       | test_notifications.py::TestConfigure                | test_configure_max_visible                              | Pass   |
|       | test_notifications.py::TestConfigure                | test_configure_no_args_no_change                        | Pass   |
|       | test_notifications.py::TestConfigure                | test_configure_position                                 | Pass   |
|       | test_notifications.py::TestConfigure                | test_configure_timeout                                  | Pass   |
|       | test_notifications.py::TestCreateNotificationSystem | test_create_notification_system                         | Pass   |
|       | test_notifications.py::TestCreateNotificationSystem | test_create_notification_timeout                        | Pass   |
|       | test_notifications.py::TestCreateNotificationSystem | test_create_notification_position_defaults_to_top_right | Pass   |
|       | test_notifications.py::TestCreateNotificationSystem | test_create_notification_position                       | Pass   |
|       | test_notifications.py::TestGetStatistics            | test_returns_expected_keys                              | Pass   |
|       | test_notifications.py::TestGetStatistics            | test_statistics_reflect_config                          | Pass   |
|       | test_startup.py::TestGetPreloader                   | test_get_preloader_js_matches_get_js                    | Pass   |
|       | test_startup.py::TestGetPreloader                   | test_get_preloader_js_returns_string                    | Pass   |
|       | test_startup.py::TestStartupPreloaderAddToMap       | test_add_to_map_calls_add_child                         | Pass   |
|       | test_startup.py::TestStartupPreloaderAddToMap       | test_add_to_map_output_contains_script                  | Pass   |
|       | property_panel.py::TestPropertyTreeConfigure        | test_configure_configured                               | Pass   |
|       | property_panel.py::TestPropertyTreeConfigure        | test_configure_configured_does_not_change               | Pass   |
|       | property_panel.py::TestPropertyTreeConfigure        | test_configure_configured_height                        | Pass   |
|       | property_panel.py::TestPropertyTreeConfigure        | test_configure_configured_width                         | Pass   |
|       | property_panel.py::TestPropertyTreeConfigure        | test_configure_configured_use                           | Pass   |
|       | property_panel.py::TestPropertyTreeConfigure        | test_configure_configured_returns_dict                  | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                 | test_hour_label_present                                 | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                 | test_scrubber_id_present                                | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                 | test_scrubber_max_from_config                           | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                 | test_scrubber_max_not_59                                | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                 | test_scrubber_min_is_zero                               | Pass   |
|       | sp_sim_detail.py::TestScrubberRange                 | test_scrubber_type_is_range                             | Pass   |
|       | sp_sim_detail.py::TestSimFunction                   | test_get_js_returns_nonempty                            | Pass   |

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| Model | Module                                                           | Test Name                                                         | Status |
|-------|------------------------------------------------------------------|-------------------------------------------------------------------|--------|
|       | sp_sim_detail.py::TestSimFunction                                | test_spSeekTo_defined                                             | Pass   |
|       | sp_sim_detail.py::TestSimFunction                                | test_spSetSpeed_defined                                           | Pass   |
|       | sp_sim_detail.py::TestSimFunction                                | test_spStopAnim_defined                                           | Pass   |
|       | sp_table_detail.py::TestPanelTitle                               | test_panel_title_id_present                                       | Pass   |
|       | sp_table_detail.py::TestPanelTitle                               | test_panel_title_prefix                                           | Pass   |
|       | sp_table_detail.py::TestPanelTitle                               | test_title_reads_selected_option_text                             | Pass   |
|       | sp_table_detail.py::TestPanelTitle                               | test_title_uses_label_variable_not_bare_id_concat                 | Pass   |
|       | sp_table_detail.py::TestStormList                                | test_loads_from_propertyts_storms                                 | Pass   |
|       | sp_table_detail.py::TestStormList                                | test_loads_storms_message                                         | Pass   |
|       | sp_table_detail.py::TestStormList                                | test_loads_preloaded_cache                                        | Pass   |
|       | sp_table_detail.py::TestTableStructure                           | test_reload_portfolio_impact_defined                              | Pass   |
|       | sp_table_detail.py::TestTableStructure                           | test_reload_portfolio_impact_endpoint                             | Pass   |
|       | sp_table_detail.py::TestTableStructure                           | test_resort_columns_defined                                       | Pass   |
|       | sp_table_detail.py::TestTableStructure                           | test_summary_cards_present                                        | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_refresh_main_map            | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_apply_filter                | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_click                       | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_window                      | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_01_exported                 | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_window                      | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_window_hint                 | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_show_hide                   | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_blotter_view                | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_exists                      | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_display_flex                | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_show_panel                  | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_exists                      | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_exists                      | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_preload_done                | Pass   |
|       | test_cross_iife_exports.py::TestCrossIIFEExportsPanelImports     | test_cross_iife_exports_panel_imports_open                        | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pGovAudit]     | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pGovBib]       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pStorms]       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pBlotter]      | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pEodHistory]   | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pGauges]       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pGovDocs]      | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pMarket]       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pPortStorms]   | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pStressStorms] | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_on_window[_pYieldCurve]   | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_for_cache                 | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_for_td_preload_done       | Pass   |
|       | test_cross_iife_ownership.py::TestCrossIIFEOwnershipPanelImports | test_cross_iife_ownership_panel_imports_for_td_preload_slope      | Pass   |

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| Model | Module                                                                       | Test Name                                                                    | Status |
|-------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|
|       | test_startup_preloader.py::TestGestorApplyDocumentsCacheFunctionExists       | test_startup_preloader.py::TestGestorApplyDocumentsCacheFunctionExists       | Pass   |
|       | test_startup_preloader.py::TestGestorCheckDecoratedCache                     | test_startup_preloader.py::TestGestorCheckDecoratedCache                     | Pass   |
|       | test_startup_preloader.py::TestGestorCleanDecoratedCacheAfterUse             | test_startup_preloader.py::TestGestorCleanDecoratedCacheAfterUse             | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_200_at_api_v1                    | test_startup_preloader.py::TestHealthHealth_200_at_api_v1                    | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_200_at_root                      | test_startup_preloader.py::TestHealthHealth_200_at_root                      | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_check_function_serves_both       | test_startup_preloader.py::TestHealthHealth_check_function_serves_both       | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_py_has_api_v1_health_route       | test_startup_preloader.py::TestHealthHealth_py_has_api_v1_health_route       | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_py_has_plain_health_route        | test_startup_preloader.py::TestHealthHealth_py_has_plain_health_route        | Pass   |
|       | test_startup_preloader.py::TestHealthHealth_returns_ok_status                | test_startup_preloader.py::TestHealthHealth_returns_ok_status                | Pass   |
|       | test_startup_preloader.py::TestInitCityMapAndViningLoaderBeforeNotifications | test_startup_preloader.py::TestInitCityMapAndViningLoaderBeforeNotifications | Passi  |
|       | test_startup_preloader.py::TestInitCityMapperViningImportedInInit            | test_startup_preloader.py::TestInitCityMapperViningImportedInInit            | Pass   |
|       | test_startup_preloader.py::TestInitCityMapperViningInstantiatedInInit        | test_startup_preloader.py::TestInitCityMapperViningInstantiatedInInit        | Fail   |
|       | test_startup_preloader.py::TestStartupDetailsStateKeysHaveStatHandler        | test_startup_preloader.py::TestStartupDetailsStateKeysHaveStatHandler        | Pass   |
|       | test_startup_preloader.py::TestStartupDetailsStatUsesTradesLength            | test_startup_preloader.py::TestStartupDetailsStatUsesTradesLength            | Pass   |
|       | test_startup_preloader.py::TestStartupDetailsStoryStatUsesHistoryLength      | test_startup_preloader.py::TestStartupDetailsStoryStatUsesHistoryLength      | Pass   |
|       | test_startup_preloader.py::TestStartupFangelStatsStatUsesCount               | test_startup_preloader.py::TestStartupFangelStatsStatUsesCount               | Pass   |
|       | test_startup_preloader.py::TestStartupFangelStatsStatUsesGaugesLength        | test_startup_preloader.py::TestStartupFangelStatsStatUsesGaugesLength        | Pass   |
|       | test_startup_preloader.py::TestStartupDotaailStatsStatUsesEventsLength       | test_startup_preloader.py::TestStartupDotaailStatsStatUsesEventsLength       | Pass   |
|       | test_startup_preloader.py::TestStartupDotaailStatsStatUsesReferencesLength   | test_startup_preloader.py::TestStartupDotaailStatsStatUsesReferencesLength   | Pass   |
|       | test_startup_preloader.py::TestStartupDotaailStatsStatUsesDocumentsLength    | test_startup_preloader.py::TestStartupDotaailStatsStatUsesDocumentsLength    | Pass   |
|       | test_startup_preloader.py::TestStartupDetailStatsStatUsesObjectKeysGauges    | test_startup_preloader.py::TestStartupDetailStatsStatUsesObjectKeysGauges    | Pass   |
|       | test_startup_preloader.py::TestStartupDotaillagesStatUsesMortgagesLength     | test_startup_preloader.py::TestStartupDotaillagesStatUsesMortgagesLength     | Pass   |
|       | test_startup_preloader.py::TestStartupDotaillStormsStatUsesStormsLength      | test_startup_preloader.py::TestStartupDotaillStormsStatUsesStormsLength      | Pass   |
|       | test_startup_preloader.py::TestStartupDetailStormsStatUsesStormsLength       | test_startup_preloader.py::TestStartupDetailStormsStatUsesStormsLength       | Pass   |
|       | test_startup_preloader.py::TestStartupDropelStatsStatUsesTotalPropertyTies   | test_startup_preloader.py::TestStartupDropelStatsStatUsesTotalPropertyTies   | Fais   |
|       | test_startup_preloader.py::TestStartupDotaillStormsStatUsesStormsLength      | test_startup_preloader.py::TestStartupDotaillStormsStatUsesStormsLength      | Pass   |
|       | test_startup_preloader.py::TestStartupDetailStormsStatUsesObjectKeys         | test_startup_preloader.py::TestStartupDetailStormsStatUsesObjectKeys         | Pass   |
|       | test_startup_preloader.py::TestStartupReloadBehaviorMethodDefined            | test_startup_preloader.py::TestStartupReloadBehaviorMethodDefined            | Pass   |
|       | test_startup_preloader.py::TestStartupPreloderBehavEndpointsPresent          | test_startup_preloader.py::TestStartupPreloderBehavEndpointsPresent          | Pass   |
|       | test_startup_preloader.py::TestStartupPreloadrBehavContentLoaded             | test_startup_preloader.py::TestStartupPreloadrBehavContentLoaded             | Pass   |
|       | test_startup_preloader.py::TestStartupParkletDataValueIsHazardCurves         | test_startup_preloader.py::TestStartupParkletDataValueIsHazardCurves         | Pass   |
|       | test_startup_preloader.py::TestStartupPetsoaderBehavDone                     | test_startup_preloader.py::TestStartupPetsoaderBehavDone                     | Pass   |
|       | test_startup_preloader.py::TestStartupShowsProgressionPopup                  | test_startup_preloader.py::TestStartupShowsProgressionPopup                  | Pass   |
|       | test_startup_preloader.py::TestStartupParturpBelovaderClassDefined           | test_startup_preloader.py::TestStartupParturpBelovaderClassDefined           | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresGreghVagueHist                 | test_startup_preloader.py::TestStartupDeclaresGreghVagueHist                 | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresGreghVuraudit                  | test_startup_preloader.py::TestStartupDeclaresGreghVuraudit                  | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresGreghVurbib                    | test_startup_preloader.py::TestStartupDeclaresGreghVurbib                    | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresGreghVopertyTs                 | test_startup_preloader.py::TestStartupDeclaresGreghVopertyTs                 | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresGreghVars                      | test_startup_preloader.py::TestStartupDeclaresGreghVars                      | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCachreVariotter                | test_startup_preloader.py::TestStartupDeclaresCachreVariotter                | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCachreVasdHistory              | test_startup_preloader.py::TestStartupDeclaresCachreVasdHistory              | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCachreVagauges                 | test_startup_preloader.py::TestStartupDeclaresCachreVagauges                 | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCachreVasevDocs                | test_startup_preloader.py::TestStartupDeclaresCachreVasevDocs                | Pass   |
|       | test_startup_preloader.py::TestStartupDeclaresCachreVararket                 | test_startup_preloader.py::TestStartupDeclaresCachreVararket                 | Pass   |

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| Model | Module                                                       | Test Name                                                                               | Status |
|-------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|
|       | test_startup_preloader.py::TestStartupPreloaderCacheVapor    | test_startup_preloader.py::TestStartupPreloaderCacheVapor_storms                        | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderCacheVapor    | test_startup_preloader.py::TestStartupPreloaderCacheVapor_stress_storms                 | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderCacheVapor    | test_startup_preloader.py::TestStartupPreloaderCacheVapor_yield_curve                   | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderCacheVapor    | test_startup_preloader.py::TestStartupPreloaderCacheVapor_list_function_exists          | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderCacheVapor    | test_startup_preloader.py::TestStartupPreloaderCacheVapor_storms_cache                  | Pass   |
|       | test_startup_preloader.py::TestStartupPreloaderCacheVapor    | test_startup_preloader.py::TestStartupPreloaderCacheVapor_cache_after_use               | Pass   |
|       | test_startup_preloader.py::TestTradingDeskPreloadDoneScoping | test_startup_preloader.py::TestTradingDeskPreloadDoneScoping_redeclare_td_pre_blotter   | Pass   |
|       | test_startup_preloader.py::TestTradingDeskPreloadDoneScoping | test_startup_preloader.py::TestTradingDeskPreloadDoneScoping_redeclare_td_pre_market    | Pass   |
|       | test_startup_preloader.py::TestTradingDeskPreloadDoneScoping | test_startup_preloader.py::TestTradingDeskPreloadDoneScoping_redeclare_td_pre_load_done | Pass   |
|       | blotter.js.py::TestBlotterSignedNotional                     | test_blotter_signed_notional_in_pnl_bar                                                 | Pass   |
|       | blotter.js.py::TestBlotterSignedNotional                     | test_blotter_signed_notional_payer_negative                                             | Pass   |
|       | blotter.js.py::TestMarketPnLDisplay                          | test_blotter_has_from_market_in_pnl_bar                                                 | Pass   |
|       | blotter.js.py::TestMarketPnLDisplay                          | test_blotter_has_market_pnl_column                                                      | Pass   |
|       | blotter.js.py::TestMarketPnLDisplay                          | test_commit_shows_pnl_impact                                                            | Pass   |
|       | blotter.js.py::TestMarketPnLDisplay                          | test_hazard_column_up_down_arrows                                                       | Pass   |
|       | market.js.py::TestCommitButtonPnl                            | test_commit_button_calls_commit_function                                                | Pass   |
|       | market.js.py::TestCommitButtonPnl                            | test_commit_button_in_html                                                              | Pass   |
|       | market.js.py::TestCommitButtonPnl                            | test_commit_button_not_hidden                                                           | Pass   |
|       | market.js.py::TestCommitButtonPnl                            | test_commit_function_exists                                                             | Pass   |
|       | market.js.py::TestCommitButtonPnl                            | test_commit_posts_to_endpoint                                                           | Pass   |
|       | market.js.py::TestCommitButtonPnl                            | test_commit_refreshes_blotter                                                           | Pass   |
|       | market.js.py::TestCurvesTabRender                            | test_curves_all_gauges_chart                                                            | Pass   |
|       | market.js.py::TestCurvesTabRender                            | test_curves_has_trigger_selector                                                        | Pass   |
|       | market.js.py::TestCurvesTabRender                            | test_curves_hsl_colours                                                                 | Pass   |
|       | market.js.py::TestCurvesTabRender                            | test_curves_renders                                                                     | Pass   |
|       | market.js.py::TestCurvesTabRender                            | test_curves_strips_thames_prefix                                                        | Pass   |
|       | market.js.py::TestCurvesTabRender                            | test_curves_uses_market_state                                                           | Pass   |
|       | market.js.py::TestMapClickThrough                            | test_map_hides_trading_desk                                                             | Pass   |
|       | market.js.py::TestMapClickThrough                            | test_map_new_trade_link                                                                 | Pass   |
|       | market.js.py::TestMapClickThrough                            | test_map_view_hazard_curve                                                              | Pass   |
|       | panel.py::TestInteractivityManager                           | test_interactivity_manager_exports_trading_desk                                         | Pass   |
|       | panel.py::TestInteractivityManager                           | test_interactivity_manager_has_trading_desk                                             | Pass   |
|       | panel.py::TestPreloadDoneScoping                             | test_preloader_sets_window_preload_done                                                 | Pass   |
|       | panel.py::TestPreloadDoneScoping                             | test_startup_uses_window_preload_done_in_pre                                            | Pass   |
|       | panel.py::TestPreloadDoneScoping                             | test_startup_uses_window_preload_done_set                                               | Pass   |
|       | panel.py::TestPreloadDoneScoping                             | test_tradingdesk_reads_window_preload_done                                              | Pass   |
|       | panel.py::TestTradingDeskPanel                               | test_add_to_map                                                                         | Pass   |
|       | panel.py::TestTradingDeskPanel                               | test_blotter_submodule                                                                  | Pass   |
|       | panel.py::TestTradingDeskPanel                               | test_eod_submodule                                                                      | Pass   |
|       | panel.py::TestTradingDeskPanel                               | test_map_submodule                                                                      | Pass   |
|       | panel.py::TestTradingDeskPanel                               | test_market_submodule                                                                   | Pass   |
|       | panel.py::TestTradingDeskPanel                               | test_no_unescaped_braces                                                                | Pass   |
|       | panel.py::TestTradingDeskPanel                               | test_panel_contains_all_tabs                                                            | Pass   |
|       | panel.py::TestTradingDeskPanel                               | test_panel_contains_global_export                                                       | Pass   |

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| Model | Module                            | Test Name                                       | Status |
|-------|-----------------------------------|-------------------------------------------------|--------|
|       | panel.py::TestTradingDeskPanel    | Test_panel_contains_pi_symbol                   | Pass   |
|       | panel.py::TestTradingDeskPanel    | Test_panel_renders                              | Pass   |
|       | panel.py::TestTradingDeskPanel    | Test_risk_submodule                             | Pass   |
|       | port_stress_js.py::TestPortStress | Test_portfolio_run_endpoint                     | Pass   |
|       | port_stress_js.py::TestPortStress | Test_portfolio_storms_endpoint                  | Pass   |
|       | port_stress_js.py::TestPortStress | Test_post_method_used_for_run                   | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_calls_switch_tab_stress          | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_full_detail_button               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_has_gauge_table                  | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_render_function                  | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_sets_gauge_hint                  | Pass   |
|       | port_stress_js.py::TestPortStress | Test_gauge_pnl_sets_storm_hint                  | Pass   |
|       | port_stress_js.py::TestPortStress | Test_convertion_conflicts_with_stress_tab       | Pass   |
|       | port_stress_js.py::TestPortStress | Test_suprefix_on_content                        | Pass   |
|       | port_stress_js.py::TestPortStress | Test_suprefix_on_storm_sel                      | Pass   |
|       | port_stress_js.py::TestPortStress | Test_suprefix_on_tabs                           | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_syntax_valid_with_port_stress | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_data_called            | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_artifacts                     | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_tab_in_full_js         | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_view_appended          | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_view_created           | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_in_switch_tab          | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_in_tabs_array          | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_stress_in_views_array         | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_setup_in_hide_panel           | Pass   |
|       | port_stress_js.py::TestPortStress | Test_trading_desk_hide_iife                     | Pass   |
|       | port_stress_js.py::TestPortStress | Test_pnl_did_canvas_id                          | Pass   |
|       | port_stress_js.py::TestPortStress | Test_pnl_did_render_function                    | Pass   |
|       | port_stress_js.py::TestPortStress | Test_pnl_did_threshold_colours                  | Pass   |
|       | port_stress_js.py::TestPortStress | Test_pnl_did_uses_chartjs                       | Pass   |
|       | port_stress_js.py::TestPortStress | Test_pnl_pnl_canvas_id                          | Pass   |
|       | port_stress_js.py::TestPortStress | Test_pnl_pnl_fmtgbp                             | Pass   |
|       | port_stress_js.py::TestPortStress | Test_pnl_pnl_render_function                    | Pass   |
|       | port_stress_js.py::TestPortStress | Test_pnl_pnl_uses_horizontal_bars               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_gauge_pnl_link                    | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_has_four_groups                   | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_render_function                   | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_sets_selected_gauge               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_severity_severe_colour                     | Pass   |
|       | port_stress_js.py::TestPortStress | Test_submodule_stress_has_all_sub_tabs          | Pass   |
|       | port_stress_js.py::TestPortStress | Test_submodule_stress_has_cleanup               | Pass   |
|       | port_stress_js.py::TestPortStress | Test_submodule_stress_has_create_view           | Pass   |
|       | port_stress_js.py::TestPortStress | Test_submodule_stress_has_load_data             | Pass   |

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| Model | Module                                     | Test Name                                         | Status |
|-------|--------------------------------------------|---------------------------------------------------|--------|
|       | port_stress.js.py::TestPortStressSubmodule | test_stress_has_storm_changed                     | Pass   |
|       | port_stress.js.py::TestPortStressSubmodule | test_stress_has_switch_subtab                     | Pass   |
|       | port_stress.js.py::TestPortStressSubmodule | test_stress_renders                               | Pass   |
|       | port_stress.js.py::TestPortStressSubmodule | test_stress_state_vars                            | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_full_js_contains_populate_gauge_dropdown     | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_full_js_contains_stress_gauge_changed        | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_full_js_no_unclosed_template_literals        | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_gauge_selector_element_id_present            | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_populate_gauge_dropdown_function_defined     | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_populate_storm_dropdown_function_defined     | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_storm_selector_element_id_present            | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_stress_functions_not_before_iife_closure     | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_stress_gauge_changed_binds_to_dropdown       | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_stress_gauge_changed_function_defined        | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_stress_storm_changed_binds_to_dropdown       | Pass   |
|       | stress.js.py::TestStressGaugeFunction      | test_stress_storm_changed_function_defined        | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_fallback_only_when_hint_not_matched          | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_hint_applied_before_fallback                 | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_hint_consumed_after_use                      | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_hint_fallback_to_blotter_filter              | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_load_stress_data_accepts_gauge_hint          | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_load_stress_data_stores_hint                 | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_populate_gauge_dropdown_reads_hint           | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_stress_gauge_hint_state_variable             | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_switch_tab_stress_blotter_filter_fallback    | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_switch_tab_stress_passes_gauge_hint          | Pass   |
|       | stress.js.py::TestStressGaugeHint          | test_switch_tab_stress_uses_blotter_data_fallback | Pass   |
|       | stress.js.py::TestStressKnockOut           | test_stress_ko_annotation_on_charts               | Pass   |
|       | stress.js.py::TestStressKnockOut           | test_stress_ko_status_in_table                    | Pass   |
|       | stress.js.py::TestStressKnockOut           | test_stress_stats_bar_ko_info                     | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_chart_canvas                          | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_chart_tabs                            | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_charts                                | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_cleanup                               | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_gauge_dropdown                        | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_gauges_endpoint                       | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_has_create_and_load                   | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_ko_annotation                         | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_link_property_damage                  | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_link_storm_simulation                 | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_renders                               | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_stats_bar                             | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_storm_dropdown                        | Pass   |
|       | stress.js.py::TestStressTabSubmodule       | test_stress_surface_table                         | Pass   |

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| Model | Module                              | Test Name                     | Status |
|-------|-------------------------------------|-------------------------------|--------|
|       | stress.js.py::TestStressTabSubmodel | test_stress_trade_table       | Pass   |
|       | stress.js.py::TestStressTabSubmodel | test_surface_table_scrollable | Pass   |

### 3 Flood Risk (Depth-Damage) (MKM-DD-001)

Table 4: Test Results Summary — Flood Risk (Depth-Damage) (MKM-DD-001)

|             |       |
|-------------|-------|
| Total Tests | 128   |
| Passed      | 128   |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.01s |

Table 5: Detailed Test Results — Flood Risk (Depth-Damage)

| Test Description                                                                      | Acceptance Criteria                                                       | Result | Time   |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| At distance = characteristic length, factor must equal $1/e \approx 0.368$ .          | <code>abs(f - 1.0 / math.e) &lt; 0.001</code>                             | Pass   | 0.000s |
| At 5x the decay length, attenuation should be $\leq 1\%$ ( $e^{-5} \approx 0.0067$ ). | <code>f &lt; 0.01</code>                                                  | Pass   | 0.000s |
| Negative distance is clamped to zero; must return 1.0.                                | <code>compute_attenuation(-100) == 1.0</code>                             | Pass   | 0.000s |
| At zero distance, attenuation factor must be 1.0 (no decay).                          | <code>compute_attenuation(0) == 1.0</code>                                | Pass   | 0.000s |
| Zero decay length means instant decay; must return 0.0 for any distance $\neq 0$ .    | <code>compute_attenuation(100, 0) == 0.0</code>                           | Pass   | 0.000s |
| Every hour from 0..n must be present in output, in order.                             | <code>hours == list(range(len(gauge_readings)))</code>                    | Pass   | 0.000s |
| Flood depth must be non-negative at every time step.                                  | <code>r['depth_m'] &gt;= 0.0</code>                                       | Pass   | 0.000s |
| Each hydrograph entry must contain hour, wse_m, depth_m, and flooded.                 | <code>'hour' in r; 'wse_m' in r (+2 more)</code>                          | Pass   | 0.000s |
| Empty gauge readings must return empty property hydrograph.                           | <code>build_property_hydrograph([], 8.0, 1.0, 0.9, 6.0, 0.3) == []</code> | Pass   | 0.000s |
| When floor level exceeds peak WSE, no time step should be flooded.                    | <code>r['flooded'] is False</code>                                        | Pass   | 0.000s |
| With travel time of 5 hours, first 5 hours must show zero flood depth.                | <code>r['depth_m'] == 0.0; r['flooded'] is False</code>                   | Pass   | 0.000s |
| Property hydrograph must have same number of time steps as gauge input.               | <code>len(result) == len(gauge_readings)</code>                           | Pass   | 0.000s |



| Test Description                                                           | Acceptance Criteria                                                          | Result | Time   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| Low-elevation property at gauge location must flood at peak water level.   | <code>max(peak_depths) &gt; 0</code>                                         | Pass   | 0.000s |
| Age 50 to 100                                                              | <code>mid.age &gt; young</code>                                              | Pass   | 0.000s |
| Age over 100                                                               | <code>old &gt; young</code>                                                  | Pass   | 0.000s |
| Basic returns float                                                        | <code>isinstance(score, float); 0 &lt; score &lt;= 10</code>                 | Pass   | 0.000s |
| Construction concrete lower                                                | <code>concrete &lt; brick</code>                                             | Pass   | 0.000s |
| Construction steel lowest                                                  | <code>steel &lt; concrete</code>                                             | Pass   | 0.000s |
| Construction timber higher                                                 | <code>timber &gt; brick</code>                                               | Pass   | 0.000s |
| Construction wood equals timber                                            | <code>timber == wood</code>                                                  | Pass   | 0.000s |
| High ltv multiplier                                                        | <code>high &gt; low</code>                                                   | Pass   | 0.000s |
| Ltv 60 to 80 multiplier                                                    | <code>mid &gt; base</code>                                                   | Pass   | 0.000s |
| Ltv 80 to 95 multiplier                                                    | <code>mid &gt; base</code>                                                   | Pass   | 0.000s |
| None ltv no multiplier                                                     | <code>0 &lt; score &lt;= 10</code>                                           | Pass   | 0.000s |
| Unknown construction type                                                  | <code>0 &lt; score &lt;= 10</code>                                           | Pass   | 0.000s |
| Unknown flood level                                                        | <code>0 &lt; score &lt;= 10</code>                                           | Pass   | 0.000s |
| Very high flood capped                                                     | <code>score == 10.0</code>                                                   | Pass   | 0.000s |
| Very low flood lowest score                                                | <code>vl &lt; vh</code>                                                      | Pass   | 0.000s |
| Slope from 10m to 8m over 1000m should be 0.002 (2m/1000m).                | <code>abs(s - 0.002) &lt; 1e-6</code>                                        | Pass   | 0.000s |
| Flat terrain (equal elevations) must return MIN_SLOPE, not zero.           | <code>s == MIN_SLOPE</code>                                                  | Pass   | 0.000s |
| Slope must be the same regardless of flow direction (uses absolute value). | <code>s1 == s2</code>                                                        | Pass   | 0.000s |
| Zero horizontal distance must return MIN_SLOPE to avoid division by zero.  | <code>t == 0.0; s == MIN_SLOPE</code>                                        | Pass   | 0.000s |
| At zero is max                                                             | <code>RiskAssessor.calculate_distance_risk_factor == 1.0</code>              | Pass   | 0.000s |
| Close band                                                                 | <code>RiskAssessor.calculate_distance_risk_factor == 0.8</code>              | Pass   | 0.000s |
| Far band                                                                   | <code>RiskAssessor.calculate_distance_risk_factor == 0.2</code>              | Pass   | 0.000s |
| Medium band                                                                | <code>RiskAssessor.calculate_distance_risk_factor == 0.4</code>              | Pass   | 0.000s |
| Monotone decrease                                                          | <code>all(factors[i] &gt;= factors[i + 1] for i in range(len(facto...</code> | Pass   | 0.000s |
| Near band                                                                  | <code>RiskAssessor.calculate_distance_risk_factor == 0.6</code>              | Pass   | 0.000s |

| Test Description         | Acceptance Criteria                                                                                                                                                         | Result | Time   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Negative returns default | <code>RiskAssessor.calculate_distance_risk_factor(0)</code><br><code>== 0.5</code>                                                                                          | Pass   | 0.000s |
| None returns default     | <code>RiskAssessor.calculate_distance_risk_factor(0)</code><br><code>== 0.5</code>                                                                                          | Pass   | 0.000s |
| Very close               | <code>RiskAssessor.calculate_distance_risk_factor(5)</code><br><code>== 1.0</code>                                                                                          | Pass   | 0.000s |
| Very far                 | <code>RiskAssessor.calculate_distance_risk_factor(0)</code><br><code>== 0.1</code>                                                                                          | Pass   | 0.000s |
| At high threshold        | <code>RiskAssessor.assess_flood_risk_level(1.0)</code><br><code>== "High"</code>                                                                                            | Pass   | 0.000s |
| At low threshold         | <code>RiskAssessor.assess_flood_risk_level(0.1)</code><br><code>== "Low"</code>                                                                                             | Pass   | 0.000s |
| At medium threshold      | <code>RiskAssessor.assess_flood_risk_level(0.5)</code><br><code>== "Medium"</code>                                                                                          | Pass   | 0.000s |
| Below low threshold      | <code>RiskAssessor.assess_flood_risk_level(0.05)</code><br><code>== "Low"</code>                                                                                            | Pass   | 0.000s |
| High range               | <code>RiskAssessor.assess_flood_risk_level(0.75)</code><br><code>== "High"</code>                                                                                           | Pass   | 0.000s |
| Medium range             | <code>RiskAssessor.assess_flood_risk_level(0.3)</code><br><code>== "Medium"</code>                                                                                          | Pass   | 0.000s |
| Monotone with depth      | <code>levels == order</code>                                                                                                                                                | Pass   | 0.000s |
| Negative returns unknown | <code>RiskAssessor.assess_flood_risk_level(-1.0)</code><br><code>== "Unknown"</code>                                                                                        | Pass   | 0.000s |
| None returns unknown     | <code>RiskAssessor.assess_flood_risk_level(None)</code><br><code>== "Unknown";</code><br><code>RiskAssessor.assess_ltv_risk_level(None)</code><br><code>== "Unknown"</code> | Pass   | 0.000s |
| Very high                | <code>RiskAssessor.assess_flood_risk_level(2.5)</code><br><code>== "Very High"</code>                                                                                       | Pass   | 0.000s |
| Zero is very low         | <code>RiskAssessor.assess_flood_risk_level(0.0)</code><br><code>== "Very Low"</code>                                                                                        | Pass   | 0.000s |
| Daily minus 15           | <code>r['reliability_score'] == 80</code>                                                                                                                                   | Pass   | 0.000s |
| Decommissioned           | <code>r['reliability_score'] == 0;</code><br><code>r['category'] == "Unreliable"</code>                                                                                     | Pass   | 0.000s |
| Fully operational        | <code>r['reliability_score'] == 95;</code><br><code>r['category'] == "Highly Reliable"</code>                                                                               | Pass   | 0.000s |
| Hourly minus 5           | <code>r['reliability_score'] == 90</code>                                                                                                                                   | Pass   | 0.000s |
| Maintenance required     | <code>r['reliability_score'] == 70;</code><br><code>r['category'] == "Reliable"</code>                                                                                      | Pass   | 0.000s |
| Monthly minus 50         | <code>r['reliability_score'] == 20;</code><br><code>r['category'] == "Unreliable"</code>                                                                                    | Pass   | 0.000s |

| Test Description                   | Acceptance Criteria                                                                                               | Result | Time   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Realtime no adjustment             | <code>rt['reliability_score'] ==</code><br><code>base['reliability_score']</code>                                 | Pass   | 0.000s |
| Result keys                        | <code>'reliability_score' in r;</code> <code>'category'</code><br><code>in r (+1 more)</code>                     | Pass   | 0.000s |
| Score clamped at zero              | <code>r['reliability_score'] == 0</code>                                                                          | Pass   | 0.000s |
| Temporarily offline                | <code>r['reliability_score'] == 30;</code><br><code>r['category'] == "Low Reliability"</code>                     | Pass   | 0.000s |
| Unknown status                     | <code>r['reliability_score'] == 50;</code><br><code>r['category'] == "Moderately</code><br><code>Reliable"</code> | Pass   | 0.000s |
| Weekly minus 30                    | <code>r['reliability_score'] == 65</code>                                                                         | Pass   | 0.000s |
| All items are strings              | <code>all(isinstance(r, str) for r in</code><br><code>recs)</code>                                                | Pass   | 0.000s |
| Deep flood extra items             | <code>len(deep) &gt; len(shallow)</code>                                                                          | Pass   | 0.000s |
| High risk four or more             | <code>len(recs) &gt;= 4</code>                                                                                    | Pass   | 0.000s |
| High risk mentions flood insurance | <code>any("flood insurance" in r.lower()</code><br><code>for r in recs)</code>                                    | Pass   | 0.000s |
| Low risk                           | <code>len(recs) &gt;= 2</code>                                                                                    | Pass   | 0.000s |
| Medium risk                        | <code>len(recs) &gt;= 3</code>                                                                                    | Pass   | 0.000s |
| Very low risk                      | <code>len(recs) &gt;= 1</code>                                                                                    | Pass   | 0.000s |
| Base rates by level                | <code>result == pytest.approx(rate)</code>                                                                        | Pass   | 0.000s |
| Capped at 10 pct                   | <code>result == pytest.approx(0.10)</code>                                                                        | Pass   | 0.000s |
| Depth increases premium            | <code>with_depth &gt; no_depth</code>                                                                             | Pass   | 0.000s |
| Depth multiplier formula           | <code>rate == pytest.approx(expected)</code>                                                                      | Pass   | 0.000s |
| Unknown level uses default         | <code>result == pytest.approx(0.005)</code>                                                                       | Pass   | 0.000s |
| Zero depth no effect               | <code>base == zero</code>                                                                                         | Pass   | 0.000s |
| At high boundary                   | <code>RiskAssessor.assess_ltv_risk_level(0.95)</code><br><code>== "High"</code>                                   | Pass   | 0.000s |
| At low boundary                    | <code>RiskAssessor.assess_ltv_risk_level(0.6)</code><br><code>== "Low"</code>                                     | Pass   | 0.000s |
| At moderate boundary               | <code>RiskAssessor.assess_ltv_risk_level(0.8)</code><br><code>== "Moderate"</code>                                | Pass   | 0.000s |
| Critical ltv                       | <code>RiskAssessor.assess_ltv_risk_level(0.98)</code><br><code>== "Critical"</code>                               | Pass   | 0.000s |
| High ltv                           | <code>RiskAssessor.assess_ltv_risk_level(0.88)</code><br><code>== "High"</code>                                   | Pass   | 0.000s |
| Low ltv                            | <code>RiskAssessor.assess_ltv_risk_level(0.5)</code><br><code>== "Low"</code>                                     | Pass   | 0.000s |
| Moderate ltv                       | <code>RiskAssessor.assess_ltv_risk_level(0.7)</code><br><code>== "Moderate"</code>                                | Pass   | 0.000s |

| Test Description                                                            | Acceptance Criteria                                                                                                                               | Result | Time   |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| None returns unknown                                                        | <code>RiskAssessor.assess_flood_risk_level(None)</code><br>== "Unknown";<br><code>RiskAssessor.assess_ltv_risk_level(None)</code><br>== "Unknown" | Pass   | 0.000s |
| Normalises 95 pct                                                           | <code>RiskAssessor.assess_ltv_risk_level(95)</code><br>== "High"                                                                                  | Pass   | 0.000s |
| Normalises percentage input                                                 | <code>RiskAssessor.assess_ltv_risk_level(70)</code><br>== "Moderate"                                                                              | Pass   | 0.000s |
| Monotonicity: deeper water must produce higher velocity at constant slope.  | <code>v2 &gt; v1</code>                                                                                                                           | Pass   | 0.000s |
| Negative depth is physically invalid; must return zero velocity.            | <code>compute_manning_velocity(-1.0, 0.005)</code><br>== 0.0                                                                                      | Pass   | 0.000s |
| Negative slope (downhill in opposite direction) must use absolute value.    | <code>v_pos == v_neg</code>                                                                                                                       | Pass   | 0.000s |
| Manning velocity at d=1.0m, S=0.005, n=0.04 should be ~1.77 m/s.            | <code>1.5 &lt; v &lt; 2.0</code>                                                                                                                  | Pass   | 0.000s |
| Slopes below MIN_SLOPE must be clamped to MIN_SLOPE to avoid                | <code>v == v_min</code>                                                                                                                           | Pass   | 0.000s |
| Monotonicity: steeper slope must produce higher velocity at constant depth. | <code>v2 &gt; v1</code>                                                                                                                           | Pass   | 0.000s |
| Zero depth must produce zero velocity (no flow without water).              | <code>compute_manning_velocity(0.0, 0.005)</code><br>== 0.0                                                                                       | Pass   | 0.000s |
| Zero roughness coefficient causes division by zero; must return 0.          | <code>compute_manning_velocity(1.0, 0.005,</code><br><code>roughness=0.0) == 0.0</code>                                                           | Pass   | 0.000s |
| Elevated ltv medium flood                                                   | "Moderate Risk" in r                                                                                                                              | Pass   | 0.000s |
| High flood overrides                                                        | "High Risk" in r and "flood" in<br><code>r.lower()</code>                                                                                         | Pass   | 0.000s |
| High ltv medium flood                                                       | "High Risk" in r                                                                                                                                  | Pass   | 0.000s |
| Low flood risk                                                              | "Low Risk" in r                                                                                                                                   | Pass   | 0.000s |
| Medium flood default                                                        | "Moderate Risk" in r                                                                                                                              | Pass   | 0.000s |
| Minimal risk                                                                | "Minimal" in r                                                                                                                                    | Pass   | 0.000s |
| Negative value critical                                                     | "Critical Risk" in r                                                                                                                              | Pass   | 0.000s |
| Negative value high                                                         | "High Risk" in r                                                                                                                                  | Pass   | 0.000s |
| Negative value moderate                                                     | "Moderate Risk" in r                                                                                                                              | Pass   | 0.000s |
| None ltv no crash                                                           | <code>isinstance(r, str)</code>                                                                                                                   | Pass   | 0.000s |
| Very high flood overrides                                                   | "High Risk" in r                                                                                                                                  | Pass   | 0.000s |
| Close to water increases score                                              | <code>near['vulnerability_score'] &gt;</code><br><code>far['vulnerability_score']</code>                                                          | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                                          | Result | Time   |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------|--------|
| Flooded depth computed                                                     | <code>r['flood.depth'] ==<br/>pytest.approx(1.0);<br/>r['vulnerability.score'] &gt; 0</code> | Pass   | 0.000s |
| High risk recommendations present                                          | <code>any("flood" in rec.lower() for rec<br/>in r['recommendations'])</code>                 | Pass   | 0.000s |
| No flood                                                                   | <code>r['flood.depth'] == 0.0;<br/>r['vulnerability.score'] == 0</code>                      | Pass   | 0.000s |
| None elevation returns unknown                                             | <code>r['risk.level'] == 'Unknown';<br/>r['flood.depth'] is None (+1 more)</code>            | Pass   | 0.000s |
| None flood level returns unknown                                           | <code>r['risk.level'] == 'Unknown'</code>                                                    | Pass   | 0.000s |
| Result dict keys                                                           | <code>set(r.keys()) == {'flood.depth',<br/>'risk.level', 'vulnerabi...</code>                | Pass   | 0.000s |
| Score capped at 100                                                        | <code>r['vulnerability.score'] &lt;= 100</code>                                              | Pass   | 0.000s |
| Travel time for 500m at d=1.0m,<br>S=0.005 should be well under 1<br>hour. | <code>0 &lt; t &lt; 1.0</code>                                                               | Pass   | 0.000s |
| Monotonicity: greater distance<br>must produce longer travel time.         | <code>t2 &gt; t1</code>                                                                      | Pass   | 0.000s |
| Zero depth produces zero velocity,<br>so travel time must be infinite.     | <code>t == float('inf')</code>                                                               | Pass   | 0.000s |
| Zero distance must produce zero<br>travel time.                            | <code>t == 0.0; s == MIN_SLOPE</code>                                                        | Pass   | 0.000s |
| Base pct wins when higher than<br>depth factor                             | <code>var == pytest.approx(350.000)</code>                                                   | Pass   | 0.000s |
| Depth 1 to 2m                                                              | <code>var == pytest.approx(500.000)</code>                                                   | Pass   | 0.000s |
| Depth half to 1m                                                           | <code>var == pytest.approx(250.000)</code>                                                   | Pass   | 0.000s |
| Depth over 2m                                                              | <code>var == pytest.approx(800.000)</code>                                                   | Pass   | 0.000s |
| Depth under half m                                                         | <code>var == pytest.approx(100.000)</code>                                                   | Pass   | 0.000s |
| Negative property value                                                    | <code>RiskAssessor.calculate_value_at_risk(-100.000,<br/>"High") ==...</code>                | Pass   | 0.000s |
| Risk level percentages                                                     | <code>var == pytest.approx(1.000.000 * pct)</code>                                           | Pass   | 0.000s |
| Zero depth uses base pct                                                   | <code>var == pytest.approx(150.000)</code>                                                   | Pass   | 0.000s |
| Zero property value                                                        | <code>RiskAssessor.calculate_value_at_risk(0,<br/>"High") == 0.0</code>                      | Pass   | 0.000s |



## 4 GEV Hazard Model (MKM-GH-001)

Table 6: Test Results Summary — GEV Hazard Model (MKM-GH-001)

|             |       |
|-------------|-------|
| Total Tests | 16    |
| Passed      | 16    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.30s |

Table 7: Detailed Test Results — GEV Hazard Model

| Test Description                             | Acceptance Criteria                                                                                                                                                           | Result | Time   |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Exceedance probability in range              | <code>0.0 &lt;= prob &lt;= 1.0</code>                                                                                                                                         | Pass   | 0.001s |
| Fit returns three floats                     | <code>isinstance(shape, float);</code><br><code>isinstance(loc, float) (+1 more)</code>                                                                                       | Pass   | 0.026s |
| Force gumbel sets shape zero                 | <code>shape == 0.0</code>                                                                                                                                                     | Pass   | 0.001s |
| Higher threshold lower probability           | <code>p_high &lt; p_low</code>                                                                                                                                                | Pass   | 0.000s |
| Return level increases with period           | <code>r1_100 &gt; r1_10</code>                                                                                                                                                | Pass   | 0.000s |
| Compute all responses returns dict           | <code>isinstance(responses,</code><br><code>dict); len(responses) ==</code><br><code>len(synthetic_flat_gauges) (+1</code><br><code>more)</code>                              | Pass   | 0.000s |
| Compute response returns gauge re-<br>sponse | <code>isinstance(response, GaugeResponse);</code><br><code>response.gauge_id == gauge_id (+1</code><br><code>more)</code>                                                     | Pass   | 0.000s |
| Response threshold flags                     | <code>response.exceeded_alert;</code><br><code>response.exceeded_warning</code>                                                                                               | Pass   | 0.000s |
| Annual flood probabilities in range          | <code>0.0 &lt;= curve.annual_flood_prob.alert</code><br><code>&lt;= 1.0; 0.0 &lt;=</code><br><code>curve.annual_flood_prob.warning</code><br><code>&lt;= 1.0 (+1 more)</code> | Pass   | 0.090s |
| Build returns dict of curves                 | <code>isinstance(curves, dict); len(curves)</code><br><code>== len(synthetic_flat_gauges) (+2</code><br><code>more)</code>                                                    | Pass   | 0.088s |
| Each curve is gauge hazard curve             | <code>isinstance(curve, GaugeHazardCurve);</code><br><code>curve.gauge_id == gauge_id</code>                                                                                  | Pass   | 0.097s |
| Cumulative prob increasing                   | <code>probs[i] &gt;= probs[i - 1]</code>                                                                                                                                      | Pass   | 0.000s |
| Probabilities sum to one                     | <code>abs(point.survival_prob +</code><br><code>point.prob.at_least_one - 1.0) ...</code>                                                                                     | Pass   | 0.000s |
| Returns correct length                       | <code>len(ts) == 5</code>                                                                                                                                                     | Pass   | 0.000s |

| Test Description              | Acceptance Criteria                                | Result | Time   |
|-------------------------------|----------------------------------------------------|--------|--------|
| Returns term structure points | <code>isinstance(point, TermStructurePoint)</code> | Pass   | 0.000s |
| Survival prob decreasing      | <code>survivals[i] &lt;= survivals[i - 1]</code>   | Pass   | 0.000s |

## 5 Mortgage Pricer (MKM-MP-001)

Table 8: Test Results Summary — Mortgage Pricer (MKM-MP-001)

|             |       |
|-------------|-------|
| Total Tests | 67    |
| Passed      | 67    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.30s |

Table 9: Detailed Test Results — Mortgage Pricer

| Test Description                                                | Acceptance Criteria                                                                                               | Result | Time   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Default ids assigned                                            | <code>results[0]["mortgage_id"] == "MORTGAGE_0";</code><br><code>results[0]["property_id"] == "PROPERTY_0"</code> | Pass   | 0.001s |
| Error isolation                                                 | <code>len(results) == 3;</code> "error" not in <code>results[0]</code> (+2 more)                                  | Pass   | 0.001s |
| Prices multiple mortgages                                       | <code>len(results) == 2;</code><br><code>results[0]["mortgage_id"] == "M1"</code> (+1 more)                       | Pass   | 0.001s |
| Custom tax rate                                                 | <code>MortgagePricer(tax_rate=0.40).tax_rate == 0.40</code>                                                       | Pass   | 0.000s |
| Default tax rate                                                | <code>MortgagePricer().tax_rate == 0.20</code>                                                                    | Pass   | 0.000s |
| Custom tax rate                                                 | <code>spread_high_tax &gt; spread_low_tax</code>                                                                  | Pass   | 0.000s |
| Floor at 10bps                                                  | <code>spread &gt;= 0.001</code>                                                                                   | Pass   | 0.000s |
| Higher affordability higher spread                              | <code>spread_low_income &gt; spread_high_income</code>                                                            | Pass   | 0.000s |
| Negative income fallback                                        | <code>spread == 0.15</code>                                                                                       | Pass   | 0.000s |
| Spread bounded by schedule                                      | <code>spread &gt;= 0.001; spread &lt; 0.05</code>                                                                 | Pass   | 0.000s |
| Zero income fallback                                            | <code>spread == 0.15</code>                                                                                       | Pass   | 0.000s |
| debug=True in the credit spread calculation.                    | <code>"credit_spread" in result</code>                                                                            | Pass   | 0.001s |
| debug=True exercises logging branches without error.            | <code>result["mortgage_value"] &gt; 0</code>                                                                      | Pass   | 0.001s |
| Zero income with debug=True exercises zero-income debug branch. | <code>result["credit_spread"] &gt; 0</code>                                                                       | Pass   | 0.001s |
| flood_risk_category=None with debug=True.                       | <code>result["mortgage_value"] &gt; 0</code>                                                                      | Pass   | 0.001s |

| Test Description                                                             | Acceptance Criteria                                                                                                                | Result | Time   |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| When interest_rate=0, monthly_rate==0 → loan_amount/n_periods formula.       | result["mortgage_value"] > 0                                                                                                       | Pass   | 0.001s |
| Batch passes flood risk                                                      | results[0]["flood_risk_factor"] == 1.40; results[1]["flood_risk_factor"] == 1.00 (+1 more)                                         | Pass   | 0.001s |
| Case insensitive                                                             | MortgagePricer.calculate_flood_risk_impact("high") =...;<br>MortgagePricer.calculate_flood_risk_impact("VERY LOW") ==... (+1 more) | Pass   | 0.000s |
| Flood factor in result                                                       | result["flood_risk_factor"] == 1.20                                                                                                | Pass   | 0.001s |
| Flood risk increases spread                                                  | high["credit_spread"] > low["credit_spread"];<br>high["flood_risk_factor"] > low["flood_risk_factor"] (+1 more)                    | Pass   | 0.001s |
| Flood risk multipliers[high-1.4] [category='High', expected=1.4]             | MortgagePricer.calculate_flood_risk_impact("High") == e...                                                                         | Pass   | 0.000s |
| Flood risk multipliers[low-1.05] [category='Low', expected=1.05]             | MortgagePricer.calculate_flood_risk_impact("Low") == e...                                                                          | Pass   | 0.000s |
| Flood risk multipliers[medium-1.2] [category='Medium', expected=1.2]         | MortgagePricer.calculate_flood_risk_impact("Medium") == e...                                                                       | Pass   | 0.000s |
| Flood risk multipliers[very high-1.75] [category='Very High', expected=1.75] | MortgagePricer.calculate_flood_risk_impact("Very High") == e...                                                                    | Pass   | 0.000s |
| Flood risk multipliers[very low-1.0] [category='Very Low', expected=1.0]     | MortgagePricer.calculate_flood_risk_impact("Very Low") == e...                                                                     | Pass   | 0.000s |
| No flood risk defaults to unity                                              | result["flood_risk_factor"] == 1.0                                                                                                 | Pass   | 0.001s |
| None returns unity                                                           | MortgagePricer.calculate_flood_risk_impact("None") == 1.0                                                                          | Pass   | 0.000s |
| Unknown category returns unity                                               | MortgagePricer.calculate_flood_risk_impact("Unknown") == 1.0;<br>MortgagePricer.calculate_flood_risk_impact("Extreme") == 1.0      | Pass   | 0.000s |
| Ltv thresholds[0.5-1.0] [ltv=0.5, expected=1.0]                              | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |
| Ltv thresholds[0.8-1.0] [ltv=0.8, expected=1.0]                              | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |
| Ltv thresholds[0.85-1.1] [ltv=0.85, expected=1.1]                            | pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...                                                                       | Pass   | 0.000s |

| Test Description                                  | Acceptance Criteria                                                                                 | Result | Time   |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Ltv thresholds[0.9-1.1] [ltv=0.9, expected=1.1]   | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.91-1.3] [ltv=0.91, expected=1.3] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.95-1.3] [ltv=0.95, expected=1.3] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[0.96-1.5] [ltv=0.96, expected=1.5] | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Ltv thresholds[1.0-1.5] [ltv=1.0, expected=1.5]   | <code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>                           | Pass   | 0.000s |
| Zero property value                               | <code>pricer.calculate_loan_to_value_impact(100.0, 0) == 1.5</code>                                 | Pass   | 0.000s |
| Higher rate higher payment                        | <code>high &gt; low</code>                                                                          | Pass   | 0.000s |
| Shorter term higher payment                       | <code>short &gt; long</code>                                                                        | Pass   | 0.000s |
| £200k at 3.5% over 25yr<br>£1,001.25/month.       | <code>995 &lt; payment &lt; 1010</code>                                                             | Pass   | 0.000s |
| Zero-rate degenerate case: $M = P / n$ .          | <code>payment == pytest.approx(1000.0)</code>                                                       | Pass   | 0.000s |
| Each mortgage has property id                     | <code>mapping.get("property_id") is not None</code>                                                 | Pass   | 0.019s |
| Processing stats                                  | <code>stats["successful_mortgages"] == 5;</code><br><code>stats["failed_mortgages"] == 0</code>     | Pass   | 0.029s |
| Output file exists                                | <code>result["file_path"].exists()</code>                                                           | Pass   | 0.024s |
| Output file is valid json                         | <code>isinstance(data, dict)</code>                                                                 | Pass   | 0.115s |
| Generate returns expected keys                    | <code>"data" in result;</code> <code>"file_path" in result (+1 more)</code>                         | Pass   | 0.003s |
| Mortgage count matches properties                 | <code>len(result["data"]["mortgages"]) == 5</code>                                                  | Pass   | 0.005s |
| Mortgage id format                                | <code>re.match(r"~MORT-[a-f0-9]{8}\$", mid)</code>                                                  | Pass   | 0.067s |
| Mortgage ids unique                               | <code>len(ids) == len(set(ids))</code>                                                              | Pass   | 0.017s |
| All errors returns error                          | <code>"error" in calculate_portfolio_metrics(results)</code>                                        | Pass   | 0.000s |
| High risk count                                   | <code>calculate_portfolio_metrics(results)["high_risk_mortgages"]</code>                            | Pass   | 0.000s |
| Mixed results exclude errors                      | <code>metrics["total_mortgages"] == 1;</code><br><code>metrics["error_count"] == 1</code>           | Pass   | 0.001s |
| Valid results                                     | <code>metrics["total_mortgages"] == 2;</code><br><code>metrics["error_count"] == 0 (+3 more)</code> | Pass   | 0.002s |
| Balance converges to zero                         | <code>result["outstanding_balance"][-1] &lt; 1.0</code>                                             | Pass   | 0.001s |
| Expected losses non negative                      | <code>all(e1 &gt;= 0 for e1 in result["expected_losses"])</code>                                    | Pass   | 0.001s |



| Test Description                    | Acceptance Criteria                                                                                                      | Result | Time   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Fair value less than par            | <code>result["mortgage_value"] &lt; base.mortgage_params["loan_amo...; result["discount_to_par"] &gt; 0 (+1 more)</code> | Pass   | 0.001s |
| Hazard rates positive               | <code>result["hazard_rates"][0] == 0.0; all(result["hazard_rates"][i] &gt; 0 for i in range(1, len(r...</code>           | Pass   | 0.001s |
| High ltv increases spread           | <code>high_ltv["credit_spread"] &gt; low_ltv["credit_spread"]; high_ltv["ltv_factor"] &gt; low_ltv["ltv_factor"]</code>  | Pass   | 0.001s |
| Higher haircut more loss            | <code>high["pv_losses"] &gt; low["pv_losses"]</code>                                                                     | Pass   | 0.001s |
| Input validation clamps             | <code>result["mortgage_value"] &gt; 0</code>                                                                             | Pass   | 0.000s |
| Ltv factor in result                | <code>"ltv_factor" in result; result["ltv_factor"] == 1.0</code>                                                         | Pass   | 0.001s |
| Raises on invalid types             | —                                                                                                                        | Pass   | 0.000s |
| Recovery haircut zero low ltv       | <code>all(lgd == 0 for lgd in result["lgds"])</code>                                                                     | Pass   | 0.001s |
| Survival first period less than one | <code>result["survival_probs"][1] &lt; 1.0</code>                                                                        | Pass   | 0.001s |
| Survival monotonically decreasing   | <code>surv[i] &lt;= surv[i - 1]</code>                                                                                   | Pass   | 0.001s |
| Positive rate exceeds principal     | <code>cost &gt; 200.000</code>                                                                                           | Pass   | 0.000s |
| Zero rate equals principal          | <code>cost == pytest.approx(100_000.0)</code>                                                                            | Pass   | 0.000s |

## 6 PRS Pricing Model (MKM-PR-001)

Table 10: Test Results Summary — PRS Pricing Model (MKM-PR-001)

|             |       |
|-------------|-------|
| Total Tests | 61    |
| Passed      | 61    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 7.35s |

Table 11: Detailed Test Results — PRS Pricing Model

| Test Description                                 | Acceptance Criteria                                                                                 | Result | Time   |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Basis waterfall in generated output              | <code>bw1["model.uncertainty_bp"] == 2.0;</code><br><code>bw1["terrain_bp"] == 2.0 (+6 more)</code> | Pass   | 0.561s |
| Composition flat penalty                         | <code>result["composition_bp"] ==</code><br><code>COMPOSITION_BASIS_BPS["Flat"]</code>              | Pass   | 0.000s |
| Composition pre2000 penalty                      | <code>result["composition_bp"] == 1.0</code>                                                        | Pass   | 0.000s |
| Distance basis capped                            | <code>result["distance_bp"] ==</code><br><code>DISTANCE_MAX_BPS</code>                              | Pass   | 0.000s |
| Distance basis scales linearly                   | <code>result["distance_bp"] ==</code><br><code>pytest.approx(1.0, abs=0.1)</code>                   | Pass   | 0.000s |
| Elevation above gauge reduces spread             | <code>result["elevation_bp"] ==</code><br><code>pytest.approx(-1.0, abs=0.1)</code>                 | Pass   | 0.000s |
| Elevation below gauge no benefit                 | <code>result["elevation_bp"] == 0.0</code>                                                          | Pass   | 0.000s |
| Elevation capped at 3bp                          | <code>result["elevation_bp"] ==</code><br><code>-ELEVATION_MAX_BENEFIT_BPS</code>                   | Pass   | 0.000s |
| Model uncertainty is fixed                       | <code>result["model.uncertainty_bp"] ==</code><br><code>MODEL_UNCERTAINTY_BPS</code>                | Pass   | 0.000s |
| Terrain zone1 low risk                           | <code>result["terrain_bp"] == 0.0</code>                                                            | Pass   | 0.000s |
| Terrain zone3 high risk                          | <code>result["terrain_bp"] ==</code><br><code>TERRAIN_BASIS_BPS["Zone 3a"]</code>                   | Pass   | 0.000s |
| Total basis sums components                      | <code>result["total_basis_bp"] ==</code><br><code>pytest.approx(expected_total,...)</code>          | Pass   | 0.000s |
| 1Y from Feb 2026 = last Friday May 2027.         | <code>mat == last_friday_of_month(2027, 5)</code>                                                   | Pass   | 0.000s |
| 3Y from Feb 2026 = last Friday May 2029.         | <code>mat == last_friday_of_month(2029, 5);</code><br><code>mat.month == 5 (+1 more)</code>         | Pass   | 0.000s |
| Jan 2027 uses Aug 2026 roll → Nov 2029 maturity. | <code>mat == last_friday_of_month(2029, 11)</code>                                                  | Pass   | 0.000s |

| Test Description                                                             | Acceptance Criteria                                                                                           | Result | Time   |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| 3Y from Sep 2026 = last Friday Nov 2029 (Aug roll).                          | <code>mat == last_friday_of_month(2029, 11);</code><br><code>mat.month == 11</code>                           | Pass   | 0.000s |
| 5Y from Feb 2026 = last Friday May 2031.                                     | <code>mat == last_friday_of_month(2031, 5)</code>                                                             | Pass   | 0.000s |
| 5Y from Sep 2026 = last Friday Nov 2031.                                     | <code>mat == last_friday_of_month(2031, 11)</code>                                                            | Pass   | 0.000s |
| All maturities should fall on a Friday.                                      | <code>mat.weekday() == 4</code>                                                                               | Pass   | 0.000s |
| 3Y maturity jumps from May to Nov at the Aug roll.                           | <code>may_mat.month == 5; nov_mat.month == 11 (+1 more)</code>                                                | Pass   | 0.000s |
| Aug onwards uses Aug roll.                                                   | <code>roll == last_friday_of_month(2026, 8);</code><br><code>roll.month == 8</code>                           | Pass   | 0.000s |
| December uses the same year's Aug roll.                                      | <code>roll.month == 8; roll.year == 2026</code>                                                               | Pass   | 0.000s |
| Feb-Jul uses Feb roll.                                                       | <code>roll == last_friday_of_month(2026, 2);</code><br><code>roll.month == 2</code>                           | Pass   | 0.000s |
| January uses the previous year's Aug roll.                                   | <code>roll == last_friday_of_month(2026, 8);</code><br><code>roll.month == 8 (+1 more)</code>                 | Pass   | 0.000s |
| July is the last month of the Feb roll period.                               | <code>roll.month == 2; roll.year == 2026</code>                                                               | Pass   | 0.000s |
| March is still in the Feb roll period.                                       | <code>roll.month == 2; roll.year == 2026</code>                                                               | Pass   | 0.000s |
| <code>compute_maturity_date</code> with no <code>ref_date</code> uses today. | <code>isinstance(mat, date); mat.weekday() == 4</code>                                                        | Pass   | 0.000s |
| <code>current_roll_date(None)</code> uses today and returns a valid date.    | <code>isinstance(roll, date);</code><br><code>roll.weekday() == 4</code>                                      | Pass   | 0.000s |
| <code>maturity_table</code> with no <code>ref_date</code> uses today.        | <code>len(table) == 5;</code><br><code>all(isinstance(e['maturity_date'], date) for e in table)</code>        | Pass   | 0.000s |
| Empty propertyts dir                                                         | <code>stats["total_properties"] == 0;</code><br><code>stats["properties_with_gev"] == 0</code>                | Pass   | 0.002s |
| Generator should still work without gauge hazard curves (empty basis).       | <code>stats["properties_with_gev"] == 2;</code><br><code>stats["properties_with_floor"] == 1 (+1 more)</code> | Pass   | 0.564s |
| No propertyts dir                                                            | —                                                                                                             | Pass   | 0.000s |
| Aug 2026: 31st is Monday, last Friday = 28th.                                | <code>last_friday_of_month(2026, 8) == date(2026, 8, 28)</code>                                               | Pass   | 0.000s |
| Feb 2026: 28th is Saturday, last Friday = 27th.                              | <code>last_friday_of_month(2026, 2) == date(2026, 2, 27)</code>                                               | Pass   | 0.000s |
| May 2029: 31st is Thursday, last Friday = 25th.                              | <code>last_friday_of_month(2029, 5) == date(2029, 5, 25)</code>                                               | Pass   | 0.000s |

| Test Description                                       | Acceptance Criteria                                                                                                  | Result | Time   |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Nov 2029: 30th is Friday.                              | <code>last_friday_of_month(2029, 11) == date(2029, 11, 30)</code>                                                    | Pass   | 0.000s |
| Result should always be a Friday (weekday=4).          | <code>d.weekday() == 4</code>                                                                                        | Pass   | 0.000s |
| Actual years should be strictly increasing.            | <code>years[i] &gt; years[i - 1]</code>                                                                              | Pass   | 0.000s |
| All maturities should be in May for the Feb roll.      | <code>e['maturity_date'].month == 5</code>                                                                           | Pass   | 0.000s |
| All maturities should be in November for the Aug roll. | <code>e['maturity_date'].month == 11</code>                                                                          | Pass   | 0.000s |
| Maturity string should be formatted as 'DD Mon YYYY'.  | <code>len(e['maturity_str']) &gt; 8;</code> 'May'<br><code>in e['maturity_str'] or 'Nov' in e['maturity_str']</code> | Pass   | 0.000s |
| Default table has 5 entries (1Y through 5Y).           | <code>len(table) == 5</code>                                                                                         | Pass   | 0.000s |
| Tenors should be 1, 2, 3, 4, 5.                        | <code>tenors == [1, 2, 3, 4, 5]</code>                                                                               | Pass   | 0.000s |
| Positive spread                                        | <code>spread &gt; 0; 50 &lt; spread &lt; 500</code>                                                                  | Pass   | 0.000s |
| Spread increases with hazard                           | <code>spread_high &gt; spread_low</code>                                                                             | Pass   | 0.000s |
| Spread values at different tenors                      | <code>s &gt; 0</code>                                                                                                | Pass   | 0.000s |
| Zero hazard rate                                       | <code>spread == 0.0</code>                                                                                           | Pass   | 0.000s |
| Basis calculation                                      | <code>len(nearest) &gt; 0;</code> <code>ng0["gauge_id"] == "GAUGE-001"</code> (+5 more)                              | Pass   | 0.553s |
| Depth thresholds                                       | <code>"any_flood" in thresholds;</code> <code>"moderate" in thresholds</code> (+4 more)                              | Pass   | 0.560s |
| Generate produces output                               | <code>output_path.exists();</code> <code>"metadata" in data</code> (+2 more)                                         | Pass   | 0.956s |
| Gev params structure                                   | <code>"shape" in gev;</code> <code>"loc" in gev</code> (+2 more)                                                     | Pass   | 0.527s |
| High transmission rate                                 | <code>prop3["summary"]["flood.transmission_rate"] == 1.0</code>                                                      | Pass   | 0.544s |
| Output structure                                       | <code>"PROP-001" in curves;</code> <code>"PROP-002" in curves</code> (+21 more)                                      | Pass   | 0.546s |
| Properties with gev                                    | <code>stats["properties_with_gev"] == 2;</code><br><code>stats["properties_with_floor"] == 1</code><br>(+1 more)     | Pass   | 0.703s |
| Survival decreases with tenor                          | <code>survival[i] &lt;= survival[i - 1]</code>                                                                       | Pass   | 0.598s |
| Term structure                                         | <code>ts["tenors"] == TENORS;</code> <code>threshold_name in ts</code> (+3 more)                                     | Pass   | 0.631s |
| Recovery rates by trigger                              | <code>len(spreads) &gt; 0</code>                                                                                     | Pass   | 0.603s |
| Recovery reduces spread                                | <code>spread_with_recovery &lt; spread_no_recovery</code>                                                            | Pass   | 0.000s |

| Test Description                                        | Acceptance Criteria                                           | Result | Time   |
|---------------------------------------------------------|---------------------------------------------------------------|--------|--------|
| All tenors should round-trip through maturity and back. | <code>tenor_from_maturity(mat, ref) == t</code>               | Pass   | 0.000s |
| A date not in the table returns the closest tenor.      | <code>result in [1, 2, 3, 4, 5]</code>                        | Pass   | 0.000s |
| Exact maturity date returns correct tenor.              | <code>tenor_from_maturity(mat, date(2026, 2, 24)) == 3</code> | Pass   | 0.000s |



## 7 Property Valuation (MKM-PV-001)

Table 12: Test Results Summary — Property Valuation (MKM-PV-001)

|             |        |
|-------------|--------|
| Total Tests | 55     |
| Passed      | 54     |
| Failed      | 1      |
| Skipped     | 0      |
| Duration    | 21.64s |

Table 13: Detailed Test Results — Property Valuation

| Test Description                                                 | Acceptance Criteria                                                       | Result | Time   |
|------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Line 541: non-dict schema → returns {}.                          | <code>result == {}</code>                                                 | Pass   | 0.000s |
| Line 544-545: 'type'/'options'/'description' fields are skipped. | <code>"type" not in result; "options" not in result</code>                | Pass   | 0.000s |
| Calculate insurance premium flood risk high                      | <code>premium &gt; 0</code>                                               | Pass   | 0.107s |
| Line 214: area=None → computes it first.                         | <code>isinstance(premium, float); premium &gt; 0</code>                   | Pass   | 0.101s |
| Line 211: property_value=None → computes it first.               | <code>isinstance(premium, float); premium &gt; 0</code>                   | Pass   | 0.198s |
| Calculate monthly rent all types                                 | <code>800 &lt;= rent &lt;= 15.000</code>                                  | Pass   | 0.000s |
| Lines 165-166: area=None → computes it first.                    | <code>isinstance(rent, float); rent &gt; 0</code>                         | Pass   | 0.000s |
| Lines 162-163: property_value=None → computes it first.          | <code>isinstance(rent, float); rent &gt; 0</code>                         | Pass   | 0.108s |
| Calculate monthly rent positive                                  | <code>isinstance(rent, float); 800 &lt;= rent &lt;= 15.000</code>         | Pass   | 0.000s |
| Calculate property area all types                                | <code>area &gt; 0</code>                                                  | Pass   | 0.000s |
| Lines 128-129: invalid date format → random years_ago.           | <code>isinstance(price, float)</code>                                     | Pass   | 0.000s |
| Lines 120-121: property_value=None → computes it first.          | <code>isinstance(price, float); price &gt; 0</code>                       | Pass   | 0.102s |
| Calculate sale price positive                                    | <code>isinstance(price, float); 50.000 &lt;= price &lt;= 8.000.000</code> | Pass   | 0.000s |

| Test Description                                                  | Acceptance Criteria                                                      | Result | Time   |
|-------------------------------------------------------------------|--------------------------------------------------------------------------|--------|--------|
| Lines 135-137: sale_date 1-2 years ago.                           | <code>isinstance(price, float)</code>                                    | Pass   | 0.000s |
| Lines 139-140: sale_date > 2 years ago.                           | <code>isinstance(price, float)</code>                                    | Pass   | 0.000s |
| Lines 124-128: sale_date within 1 year.                           | <code>isinstance(price, float)</code>                                    | Pass   | 0.000s |
| Line 60-61: datetime → isoformat string.                          | <code>"2024-01-01" in result</code>                                      | Pass   | 0.000s |
| Lines 64-65: np.float64 → float.                                  | <code>"3.14" in result</code>                                            | Pass   | 0.000s |
| Lines 62-63: np.integer → int.                                    | <code>result == "42"</code>                                              | Pass   | 0.011s |
| Lines 66-67: np.ndarray → list.                                   | <code>result == [1, 2, 3]</code>                                         | Pass   | 0.001s |
| Line 68: super().default() raises TypeError for unknown type.     | —                                                                        | Pass   | 0.000s |
| Lines 510-517: each fallback location has lat/lon/name/elevation. | <code>"lat" in loc; "lon" in loc (+2 more)</code>                        | Pass   | 0.000s |
| Lines 499-518: fallback generates right number of locations.      | <code>len(locs) == 5</code>                                              | Pass   | 0.001s |
| Lines 505-506: uses params.get_elevation if available.            | <code>all(loc["elevation"] == 10.0 for loc in locs)</code>               | Pass   | 0.000s |
| Lines 608-612: generate_properties returns result dict.           | <code>isinstance(result, dict); "data" in result (+1 more)</code>        | Pass   | 0.981s |
| Lines 610-612: generate_properties creates output file.           | <code>result["file_path"].exists()</code>                                | Pass   | 0.664s |
| Calculate mortgage financials returns dict                        | <code>isinstance(result, dict); "loan_amount" in result (+2 more)</code> | Pass   | 0.000s |
| Generate financial data keys                                      | <code>"mortgage_id" in data; "loan_amount" in data (+1 more)</code>      | Pass   | 0.000s |
| Interest rate range                                               | <code>0.01 &lt;= data["interest_rate"] &lt;= 0.15</code>                 | Pass   | 0.000s |
| Loan amount less than value                                       | <code>data["loan_amount"] &lt;= info["property_value"] * 1.05</code>     | Pass   | 0.000s |
| Mortgage id format                                                | <code>re.match(r"~MORT-[a-f0-9]{8}\$", data["mortgage_id"])</code>       | Pass   | 0.000s |
| Quality consistency check                                         | <code>isinstance(corrected, dict)</code>                                 | Pass   | 0.000s |
| Calculate insurance premium positive                              | <code>isinstance(premium, (int, float)); premium &gt; 0</code>           | Pass   | 0.105s |
| Calculate property area positive                                  | <code>isinstance(area, (int, float)); area &gt; 0</code>                 | Pass   | 0.000s |
| Calculate property value in range                                 | <code>50_000 &lt;= value &lt;= 10_000_000</code>                         | Pass   | 1.114s |
| Generate bedrooms detached                                        | <code>isinstance(beds, int); 1 &lt;= beds &lt;= 7</code>                 | Pass   | 0.000s |
| Generate bedrooms flat                                            | <code>isinstance(beds, int); 1 &lt;= beds &lt;= 4</code>                 | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                                   | Result | Time   |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Generate postcode format                                         | <code>isinstance(postcode, str);<br/>len(postcode) &gt;= 5</code>                     | Pass   | 0.000s |
| Generate field value decimal                                     | <code>isinstance(val, (int, float))</code>                                            | Pass   | 0.202s |
| Generate field value text                                        | <code>isinstance(val, str)</code>                                                     | FAIL   | 0.101s |
| Coordinates in thames range                                      | <code>51.0 &lt;= loc["lat"] &lt;= 52.0;    -1.0 &lt;=<br/>loc["lon"] &lt;= 1.0</code> | Pass   | 3.194s |
| Processing stats                                                 | <code>stats["successful_properties"] == 5;<br/>stats["failed_properties"] == 0</code> | Pass   | 1.525s |
| Output file exists                                               | <code>result["file_path"].exists()</code>                                             | Pass   | 1.680s |
| Output file is valid json                                        | <code>isinstance(data, dict)</code>                                                   | Pass   | 1.724s |
| Generate correct count                                           | <code>len(result["data"]["properties"]) ==<br/>5</code>                               | Pass   | 1.790s |
| Generate returns expected keys                                   | <code>"data" in result;        "file_path" in<br/>result (+1 more)</code>             | Pass   | 1.573s |
| Property id format                                               | <code>re.match(r"~PROP-[a-f0-9]{8}\$", pid)</code>                                    | Pass   | 1.758s |
| Property ids unique                                              | <code>len(ids) == len(set(ids))</code>                                                | Pass   | 4.224s |
| Construction year in range                                       | <code>1600 &lt;= year &lt;= 2026</code>                                               | Pass   | 0.000s |
| Property id format                                               | <code>re.match(r"~PROP-[a-f0-9]{8}\$",<br/>meta["property_id"])</code>                | Pass   | 0.257s |
| Returns dict with required keys                                  | <code>"property_id" in meta; "property_type"<br/>in meta</code>                       | Pass   | 0.112s |
| Lines 73-75: generate_grid_reference returns a non-empty string. | <code>isinstance(ref, str); len(ref) &gt; 4</code>                                    | Pass   | 0.000s |
| Lines 50-52: generate_owner_name returns 'Firstname Lastname'.   | <code>len(parts) == 2;    all(p[0].isupper()<br/>for p in parts)</code>               | Pass   | 0.000s |
| Lines 43-45: generate_past_date returns YYYY-MM-DD string.       | <code>isinstance(result, str);    len(parts)<br/>== 3</code>                          | Pass   | 0.000s |
| Custom days_range is respected (result is a past date).          | <code>dt &lt; datetime.now()</code>                                                   | Pass   | 0.003s |

## Failed Test Details

**test\_generate\_field\_value\_text** File: tests/port/property/rand\_property\_mortgage.py

```

1 tests/port/property/rand_property_mortgage.py:219: in test_generate_field_value_text
2     assert isinstance(val, str)
3 E   AssertionError

```

## 8 Storm-Gauge Response Model (MKM-SG-001)

Table 14: Test Results Summary — Storm-Gauge Response Model (MKM-SG-001)

|             |       |
|-------------|-------|
| Total Tests | 81    |
| Passed      | 81    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.04s |

Table 15: Detailed Test Results — Storm-Gauge Response Model

| Test Description                      | Acceptance Criteria                                                       | Result | Time   |
|---------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Gauge ids match                       | <code>{r.gauge_id for r in responses} == {gauge.gauge_id, gauge...</code> | Pass   | 0.000s |
| Near gauge higher response            | <code>near_resp.peak_level &gt;= far_resp.peak_level</code>               | Pass   | 0.000s |
| Returns one response per gauge        | <code>len(responses) == 2</code>                                          | Pass   | 0.000s |
| All kernels decrease with distance    | <code>near &gt;= far</code>                                               | Pass   | 0.000s |
| All kernels in 0 1 range              | <code>0.0 &lt;= decay &lt;= 1.0</code>                                    | Pass   | 0.000s |
| Exponential at half                   | <code>decay == pytest.approx(math.exp(-0.5), rel=1e-4)</code>             | Pass   | 0.000s |
| Exponential at one lambda             | <code>decay == pytest.approx(math.exp(-1.0), rel=1e-4)</code>             | Pass   | 0.000s |
| Gaussian at one sigma                 | <code>decay == pytest.approx(math.exp(-0.5), rel=1e-4)</code>             | Pass   | 0.000s |
| Gaussian at two sigma                 | <code>decay == pytest.approx(math.exp(-2.0), rel=1e-4)</code>             | Pass   | 0.000s |
| Linear beyond cutoff is zero          | <code>decay == 0.0</code>                                                 | Pass   | 0.000s |
| Linear midpoint                       | <code>decay == pytest.approx(0.75, abs=0.001)</code>                      | Pass   | 0.000s |
| Zero distance returns one all kernels | <code>model.compute_decay(0, 100, kernel, 1.0) == 1.0</code>              | Pass   | 0.000s |
| After end uses last point             | <code>intensity &gt;= 0</code>                                            | Pass   | 0.000s |
| At peak time is nonzero               | <code>intensity &gt; 0</code>                                             | Pass   | 0.000s |
| Before start uses first point         | <code>intensity &gt;= 0</code>                                            | Pass   | 0.000s |
| Empty track returns zero              | <code>result == 0.0</code>                                                | Pass   | 0.000s |

| Test Description                                                                 | Acceptance Criteria                                                                                                                                         | Result | Time   |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Far from track is lower                                                          | <code>near &gt; far</code>                                                                                                                                  | Pass   | 0.000s |
| Accumulation nonnegative                                                         | <code>resp.accumulation &gt;= 0</code>                                                                                                                      | Pass   | 0.000s |
| Duration nonnegative                                                             | <code>resp.duration_above_alert &gt;= 0;</code><br><code>resp.duration_above_warning &gt;= 0</code> (+1 more)                                               | Pass   | 0.000s |
| Flooded flag correct                                                             | <code>resp.flooded == (resp.peak_level &gt;= gauge.flood_alert)</code>                                                                                      | Pass   | 0.000s |
| A storm passing directly over with peak=100 should flood the gauge.              | <code>resp.peak_level &gt; gauge.base_level</code>                                                                                                          | Pass   | 0.000s |
| Peak level at least base                                                         | <code>resp.peak_level &gt;= gauge.base_level</code>                                                                                                         | Pass   | 0.000s |
| Peak level is max of series                                                      | <code>resp.peak_level ==</code><br><code>pytest.approx(max(levels))</code>                                                                                  | Pass   | 0.000s |
| Response ids match                                                               | <code>resp.gauge_id ==</code><br><code>gauge.gauge_id;</code> <code>resp.storm_id ==</code><br><code>simple_storm.storm_id</code>                           | Pass   | 0.000s |
| Timeseries length                                                                | <code>len(resp.level_timeseries)</code><br><code>== expected_steps;</code><br><code>len(resp.intensity_timeseries) ==</code><br><code>expected_steps</code> | Pass   | 0.000s |
| To dict roundtrip                                                                | <code>d['gauge_id'] == gauge.gauge_id;</code><br><code>'level_timeseries' in d</code> (+1 more)                                                             | Pass   | 0.000s |
| To dict without timeseries                                                       | <code>'level_timeseries' not in d</code>                                                                                                                    | Pass   | 0.001s |
| Encodes datetime                                                                 | <code>"2026-01-01" in result</code>                                                                                                                         | Pass   | 0.000s |
| Line 80: np.float32 is np.floating but NOT a Python float subclass in NumPy 2.x, | <code>"2.5" in result</code>                                                                                                                                | Pass   | 0.000s |
| Encodes numpy floating                                                           | <code>"3.14" in result</code>                                                                                                                               | Pass   | 0.000s |
| Encodes numpy integer                                                            | <code>"42" in result</code>                                                                                                                                 | Pass   | 0.000s |
| Encodes numpy ndarray                                                            | <code>"1.0" in result</code>                                                                                                                                | Pass   | 0.000s |
| Fallback raises for unknown                                                      | —                                                                                                                                                           | Pass   | 0.000s |
| Distance is positive                                                             | <code>dist &gt; 0</code>                                                                                                                                    | Pass   | 0.000s |
| Returns nearest of three                                                         | <code>nearest.intensity == 80.0;</code> <code>dist ==</code><br><code>pytest.approx(0.0, abs=1.0)</code>                                                    | Pass   | 0.000s |
| Single track point                                                               | <code>nearest is track[0]</code>                                                                                                                            | Pass   | 0.000s |
| Close to thames high risk                                                        | <code>result["FloodRiskScore"] &gt;= 7;</code><br><code>result["FloodRiskCategory"] in</code><br><code>["High", "Very High"]</code>                         | Pass   | 0.000s |
| Far from thames lower risk                                                       | <code>result["FloodRiskScore"] &lt;= 6;</code><br><code>result["FloodRiskCategory"] in</code><br><code>["Low", "Medium"]</code>                             | Pass   | 0.000s |
| Flood stages ordering                                                            | <code>alert &lt; warning &lt; severe</code>                                                                                                                 | Pass   | 0.001s |



| Test Description                                                                                                              | Acceptance Criteria                                                                                                                                                                     | Result | Time   |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Processing stats                                                                                                              | <code>stats["successful_gauges"] == 5;</code><br><code>stats["failed_gauges"] == 0</code>                                                                                               | Pass   | 0.001s |
| Lines 264-266: exception during <code>json.dump</code> is logged and re-raised.                                               | —                                                                                                                                                                                       | Pass   | 0.001s |
| Line 245: <code>end_time</code> is set by the first <code>generate()</code> ; the second <code>generate()</code> call         | —                                                                                                                                                                                       | Pass   | 0.001s |
| Line 290: non-dict <code>section_schema</code> returns <code>{}</code> .                                                      | <code>result == {}</code>                                                                                                                                                               | Pass   | 0.000s |
| Line 294: <code>'type'</code> , <code>'options'</code> , <code>'description'</code> , <code>'values'</code> keys are skipped. | <code>skip not in result</code>                                                                                                                                                         | Pass   | 0.000s |
| Lines 159-162: <code>count</code> <code>i</code> <code>max_gauges</code> → capped to <code>max_gauges</code> .                | <code>len(result["data"]["flood_gauges"])</code><br><code>== max_available</code>                                                                                                       | Pass   | 0.007s |
| Line 155-156: <code>count</code> <code>i</code> 1 raises <code>ValueError</code> .                                            | —                                                                                                                                                                                       | Pass   | 0.000s |
| Lines 354-355: <code>generate_gauges()</code> convenience function.                                                           | <code>"data" in result;</code><br><code>len(result["data"]["flood_gauges"])</code><br><code>== 3</code>                                                                                 | Pass   | 0.001s |
| Lines 212-215: exception in <code>_generate_single_gauge</code> increments <code>failed_gauges</code> .                       | <code>result["processing_stats"]["failed_gauges"]</code><br><code>== 1</code>                                                                                                           | Pass   | 0.000s |
| Lines 122-125: <code>log()</code> dispatches to correct logger method.                                                        | —                                                                                                                                                                                       | Pass   | 0.000s |
| Lines 309-327: missing Flood-Gauge/Header/Location/Flood-Stages are created.                                                  | <code>gauge_data["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "GAUGE-t...;</code><br><code>gauge_data["FloodGauge"]["Location"]["GaugeLatitude"]</code><br><code>== ...</code> | Pass   | 0.000s |
| Lines 99-100: <code>verbose=False</code> sets logger to WARNING.                                                              | <code>logging.getLogger("port.src.gauge.gauge")</code><br><code>== login...</code>                                                                                                      | Pass   | 0.000s |
| Output file exists                                                                                                            | <code>result["file_path"].exists()</code>                                                                                                                                               | Pass   | 0.001s |
| Output file is valid json                                                                                                     | <code>"flood_gauges" in data</code>                                                                                                                                                     | Pass   | 0.003s |
| Output json has correct count                                                                                                 | <code>len(data["flood_gauges"]) == 5</code>                                                                                                                                             | Pass   | 0.003s |
| Lines 378-381: <code>_main__</code> block calls <code>generate_gauges</code> and logs results.                                | <code>(tmp_path / 'gauge.json').exists()</code>                                                                                                                                         | Pass   | 0.005s |
| Gauge id format                                                                                                               | <code>re.match(r"^GAUGE-[a-f0-9]{8}\$",</code><br><code>gid)</code>                                                                                                                     | Pass   | 0.001s |
| Gauge ids unique                                                                                                              | <code>len(ids) == len(set(ids))</code>                                                                                                                                                  | Pass   | 0.002s |
| Generate correct count                                                                                                        | <code>len(result["data"]["flood_gauges"])</code><br><code>== 5</code>                                                                                                                   | Pass   | 0.001s |
| Generate returns expected keys                                                                                                | <code>"data" in result;</code> <code>"file_path" in</code><br><code>result (+1 more)</code>                                                                                             | Pass   | 0.004s |

| Test Description                  | Acceptance Criteria                                                                                                     | Result | Time   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Locations have coordinates        | "lat" in loc; "lon" in loc (+1 more)                                                                                    | Pass   | 0.001s |
| Date field returns date string    | isinstance(val, str);<br>re.match(r"^\{d{4}-\{d{2}-\{d{2}}\$",<br>val)                                                  | Pass   | 0.000s |
| Decimal field returns float       | isinstance(val, (int, float))                                                                                           | Pass   | 0.000s |
| Gauge name includes area          | "TestArea" in val                                                                                                       | Pass   | 0.000s |
| Menu field returns valid option   | isinstance(val, str)                                                                                                    | Pass   | 0.000s |
| Text field returns string         | isinstance(val, str); len(val) > 0                                                                                      | Pass   | 0.000s |
| Catchment name stored             | meta["catchment_name"] == "Thames"                                                                                      | Pass   | 0.000s |
| Flood stage ordering              | meta["flood_alert"] <<br>meta["flood_warning"];<br>meta["flood_warning"] <<br>meta["severe_flood_warning"] (+1<br>more) | Pass   | 0.000s |
| Gauge id format                   | re.match(r"^GAUGE-[a-f0-9]{8}\$",<br>meta["gauge_id"])                                                                  | Pass   | 0.000s |
| Historical high level range       | 5.0 <= meta["historical_high_level"]<br><= 8.0                                                                          | Pass   | 0.000s |
| Install date format               | re.match(r"^\{d{4}-\{d{2}-\{d{2}}\$",<br>meta["install_date"])                                                          | Pass   | 0.000s |
| Returns dict with required keys   | key in meta                                                                                                             | Pass   | 0.000s |
| Equatorial degree approx 111km    | 110 < d < 113                                                                                                           | Pass   | 0.000s |
| London to paris approx 340km      | 320 < d < 360                                                                                                           | Pass   | 0.000s |
| Same point is zero                | model.haversine_km(0.0, 0.0, 0.0,<br>0.0) == pytest.approx(0...                                                         | Pass   | 0.000s |
| Symmetric                         | d1 == pytest.approx(d2, rel=1e-6)                                                                                       | Pass   | 0.000s |
| Monotone increasing               | all(levels[i] < levels[i + 1] for i<br>in range(len(levels) ...                                                         | Pass   | 0.000s |
| Negative intensity returns zero   | model.intensity_to_level(-10.0,<br>gauge) == 0.0                                                                        | Pass   | 0.000s |
| Positive intensity positive level | model.intensity_to_level(50.0, gauge)<br>> 0.0                                                                          | Pass   | 0.000s |
| Sensitivity scales output         | model.intensity_to_level(50, g2) ==<br>pytest.approx( ...                                                               | Pass   | 0.000s |
| Uses historical high when set     | lv_with != lv_without                                                                                                   | Pass   | 0.000s |
| Zero intensity returns zero       | model.intensity_to_level(0.0, gauge)<br>== 0.0                                                                          | Pass   | 0.000s |

## 9 Storm Intensity Distribution (MKM-SI-001)

Table 16: Test Results Summary — Storm Intensity Distribution (MKM-SI-001)

|             |       |
|-------------|-------|
| Total Tests | 105   |
| Passed      | 105   |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.24s |

Table 17: Detailed Test Results — Storm Intensity Distribution

| Test Description                              | Acceptance Criteria                                                                                                                                                                         | Result | Time   |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Custom parameters                             | <code>d.params.base_mean == 50;</code>                                                                                                                                                      | Pass   | 0.000s |
|                                               | <code>d.params.tail_threshold == 70</code>                                                                                                                                                  |        |        |
| Default parameters                            | <code>d.params.base_mean == 45.0;</code>                                                                                                                                                    | Pass   | 0.000s |
|                                               | <code>d.params.base_std == 15.0 (+4 more)</code>                                                                                                                                            |        |        |
| From params object                            | <code>d.params.base_mean == 55</code>                                                                                                                                                       | Pass   | 0.000s |
| From scenario baseline                        | <code>d is not None;</code> <code>d.params.name ==</code><br><code>"baseline"</code>                                                                                                        | Pass   | 0.000s |
| From scenario unknown raises                  | —                                                                                                                                                                                           | Pass   | 0.000s |
| Seed gives deterministic output               | <code>s1 == s2</code>                                                                                                                                                                       | Pass   | 0.000s |
| Tail prob between 0 and 1                     | <code>0.0 &lt; d._tail_prob &lt; 1.0</code>                                                                                                                                                 | Pass   | 0.000s |
| All categories defined                        | <code>set(DURATION_PARAMS.keys()) ==</code><br><code>expected</code>                                                                                                                        | Pass   | 0.000s |
| Catastrophic extended                         | <code>high == 240</code>                                                                                                                                                                    | Pass   | 0.000s |
| Each tier's max roughly doubles the previous. | <code>1.5 &lt;= ratio &lt;= 3.0</code>                                                                                                                                                      | Pass   | 0.000s |
| Extreme extended                              | <code>DURATION_PARAMS["extreme"] == (36,</code><br><code>72, 144)</code>                                                                                                                    | Pass   | 0.000s |
| Moderate extended                             | <code>low == 12; mode == 30 (+1 more)</code>                                                                                                                                                | Pass   | 0.000s |
| Severe extended                               | <code>DURATION_PARAMS["severe"] == (24, 48,</code><br><code>96)</code>                                                                                                                      | Pass   | 0.000s |
| Gap types                                     | <code>GapType.SHORT.value == "short";</code><br><code>GapType.MEDIUM.value == "medium" (+1</code><br><code>more)</code>                                                                     | Pass   | 0.000s |
| Sequence gap mapping                          | <code>SEQUENCE_GAP_TYPE[SequenceType.DOUBLET]</code><br><code>== GapType.MEDIUM;</code><br><code>SEQUENCE_GAP_TYPE[SequenceType.CLUSTER]</code><br><code>== GapType.MEDIUM (+2 more)</code> | Pass   | 0.000s |

| Test Description            | Acceptance Criteria                                                                                                                  | Result | Time   |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Sequence types              | SequenceType.ISOLATED.value == "isolated";<br>SequenceType.DOUBLET.value == "doublet" (+2 more)                                      | Pass   | 0.000s |
| Storm count ranges          | SEQUENCE_TYPE.STORM.COUNTS[SequenceType.ISOLATED] == (1, 1);<br>SEQUENCE_TYPE.STORM.COUNTS[SequenceType.DOUBLET] == (2, 2) (+2 more) | Pass   | 0.000s |
| Above max is zero           | d.exceedance_probability(d.params.max_intensity + 10) == 0.0                                                                         | Pass   | 0.000s |
| Above threshold uses pareto | 0 < p < 1                                                                                                                            | Pass   | 0.000s |
| At max intensity is zero    | d.exceedance_probability(d.params.max_intensity) == 0.0                                                                              | Pass   | 0.000s |
| At min intensity is one     | d.exceedance_probability(d.params.min_intensity) == 1.0                                                                              | Pass   | 0.000s |
| At threshold continuous     | abs(p_just_below - p_just_above) < 0.05                                                                                              | Pass   | 0.000s |
| Below min is one            | d.exceedance_probability(d.params.min_intensity - 5) == 1.0                                                                          | Pass   | 0.000s |
| Below threshold uses normal | 0 < p < 1                                                                                                                            | Pass   | 0.000s |
| Monotonically decreasing    | all(probs[i] > probs[i + 1] for i in range(len(probs) - 1))                                                                          | Pass   | 0.000s |
| Constants                   | EVENT_WINDOW_HOURS == 168;<br>MIN_DRAINAGE_WINDOW_HOURS == 12<br>(+1 more)                                                           | Pass   | 0.000s |
| Exact boundary              | fits_in_window([78.0, 78.0], [0.0])                                                                                                  | Pass   | 0.000s |
| Exceeds window              | not fits_in_window([72.0, 72.0], [48.0])                                                                                             | Pass   | 0.000s |
| Simple doublet fits         | fits_in_window([24.0, 24.0], [48.0])                                                                                                 | Pass   | 0.000s |
| Single storm                | fits_in_window([100.0], [])                                                                                                          | Pass   | 0.000s |
| Long gaps                   | GAP_PARAMS[GapType.LONG] == (48, 96, 144)                                                                                            | Pass   | 0.000s |
| Medium gaps                 | GAP_PARAMS[GapType.MEDIUM] == (24, 48, 72)                                                                                           | Pass   | 0.000s |
| Short gaps                  | GAP_PARAMS[GapType.SHORT] == (6, 18, 36)                                                                                             | Pass   | 0.000s |
| Doublet medium              | get_gap_range(SequenceType.DOUBLET) == (24, 48, 72)                                                                                  | Pass   | 0.000s |
| Isolated zero               | get_gap_range(SequenceType.ISOLATED) == (0, 0, 0)                                                                                    | Pass   | 0.000s |

| Test Description                                                                          | Acceptance Criteria                                                                                                                   | Result | Time   |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Get duration range                                                                        | <code>get_duration_range("moderate") == (12, 30, 60)</code>                                                                           | Pass   | 0.000s |
| Get max duration                                                                          | <code>get_max_duration("catastrophic") == 240; get_max_duration("minimal") == 12</code>                                               | Pass   | 0.000s |
| Ids unique                                                                                | <code>len(ids) == 100</code>                                                                                                          | Pass   | 0.000s |
| Sequence id format                                                                        | <code>sid.startswith("STORM-"); len(sid) == 14</code>                                                                                 | Pass   | 0.000s |
| Sequence ids unique                                                                       | <code>len(ids) == 100</code>                                                                                                          | Pass   | 0.000s |
| Storm id format                                                                           | <code>sid.startswith("STORM-"); len(sid) == 14</code>                                                                                 | Pass   | 0.000s |
| One prob returns min                                                                      | <code>d.inverse_exceedance(1) == d.params.min_intensity</code>                                                                        | Pass   | 0.000s |
| Result in bounds                                                                          | <code>d.params.min_intensity &lt;= v &lt;= d.params.max_intensity; d.params.min_intensity &lt;= v &lt;= d.params.max_intensity</code> | Pass   | 0.000s |
| Roundtrip above threshold                                                                 | <code>abs(recovered - target) &lt; 3.0</code>                                                                                         | Pass   | 0.000s |
| Roundtrip below threshold                                                                 | <code>abs(recovered - target) &lt; 2.0</code>                                                                                         | Pass   | 0.000s |
| Zero prob returns max                                                                     | <code>d.inverse_exceedance(0) == d.params.max_intensity</code>                                                                        | Pass   | 0.000s |
| Fat tail higher exceedance                                                                | <code>p_fat &gt; p_thin</code>                                                                                                        | Pass   | 0.000s |
| Pareto above threshold between 0 and 1                                                    | <code>0 &lt; p &lt; 1</code>                                                                                                          | Pass   | 0.000s |
| Pareto at threshold is one                                                                | <code>d.pareto_exceedance(d.params.tail_threshold) == 1.0</code>                                                                      | Pass   | 0.000s |
| Pareto below threshold is one                                                             | <code>d.pareto_exceedance(d.params.tail_threshold - 5) == 1.0</code>                                                                  | Pass   | 0.000s |
| 100yr greater than 10yr                                                                   | <code>i100 &gt; i10</code>                                                                                                            | Pass   | 0.000s |
| Ordering 10 50 100                                                                        | <code>i10 &lt; i50 &lt; i100</code>                                                                                                   | Pass   | 0.000s |
| Result in bounds                                                                          | <code>d.params.min_intensity &lt;= v &lt;= d.params.max_intensity; d.params.min_intensity &lt;= v &lt;= d.params.max_intensity</code> | Pass   | 0.000s |
| Storms per year affects result                                                            | <code>many &gt; few</code>                                                                                                            | Pass   | 0.000s |
| Invalid category                                                                          | —                                                                                                                                     | Pass   | 0.000s |
| Mode should be near the most frequent value in a large sample.                            | <code>abs(median - mode) &lt; 15</code>                                                                                               | Pass   | 0.006s |
| Line 50: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally. | <code>low &lt;= d &lt;= high</code>                                                                                                   | Pass   | 0.000s |
| Reproducibility                                                                           | <code>d1 == d2</code>                                                                                                                 | Pass   | 0.000s |



| Test Description                                                                                                | Acceptance Criteria                                   | Result | Time   |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------|--------|
| All sampled durations fall within [min, max] for the category. [category='baseline']                            | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='catastrophic']                        | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='extreme']                             | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.001s |
| All sampled durations fall within [min, max] for the category. [category='minimal']                             | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='moderate']                            | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| All sampled durations fall within [min, max] for the category. [category='severe']                              | <code>low &lt;= d &lt;= high</code>                   | Pass   | 0.000s |
| Doublet gaps should fall in medium range (24-72h).                                                              | <code>all(24 &lt;= g &lt;= 72 for g in gaps)</code>   | Pass   | 0.000s |
| Isolated returns zero                                                                                           | <code>sample_gap(SequenceType.ISOLATED) == 0.0</code> | Pass   | 0.000s |
| Line 49: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally.                       | <code>low &lt;= g &lt;= high</code>                   | Pass   | 0.000s |
| Persistent gaps should fall in short range (6-36h).                                                             | <code>all(6 &lt;= g &lt;= 36 for g in gaps)</code>    | Pass   | 0.000s |
| Reproducibility                                                                                                 | <code>g1 == g2</code>                                 | Pass   | 0.000s |
| Within                                                                                                          | <code>low &lt;= g &lt;= high</code>                   | Pass   | 0.000s |
| <code>bounds[sequencetype.cluster]</code><br><code>[seq_type= SequenceType.CLUSTER: 'cluster' ]</code>          |                                                       |        |        |
| Within                                                                                                          | <code>low &lt;= g &lt;= high</code>                   | Pass   | 0.000s |
| <code>bounds[sequencetype.doublet]</code><br><code>[seq_type= SequenceType.DOUBLET: 'doublet' ]</code>          |                                                       |        |        |
| Within                                                                                                          | <code>low &lt;= g &lt;= high</code>                   | Pass   | 0.000s |
| <code>bounds[sequencetype.persistent]</code><br><code>[seq_type= SequenceType.PERSISTENT: 'persistent' ]</code> |                                                       |        |        |
| Different seeds different output                                                                                | <code>d1.sample(10) != d2.sample(10)</code>           | Pass   | 0.000s |

| Test Description                  | Acceptance Criteria                                                                                                                                               | Result | Time   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Sample n returns correct length   | <code>len(samples) == 100</code>                                                                                                                                  | Pass   | 0.000s |
| Sample single                     | <code>d.params.min_intensity &lt;= v &lt;=</code><br><code>d.params.max_intensity</code>                                                                          | Pass   | 0.000s |
| Samples within bounds             | <code>all(d.params.min_intensity &lt;= s &lt;=</code><br><code>d.params.max_intensity...</code>                                                                   | Pass   | 0.001s |
| Seeded reproducibility            | <code>s1 == s2</code>                                                                                                                                             | Pass   | 0.000s |
| Create basic                      | <code>storm.storm_id == "STORM-abc12345";</code><br><code>storm.duration.hours == 24.0</code>                                                                     | Pass   | 0.000s |
| From dict round trip              | <code>restored.storm_id ==</code><br><code>original.storm_id;</code><br><code>restored.duration.hours ==</code><br><code>original.duration.hours (+6 more)</code> | Pass   | 0.000s |
| Precipitation rounding            | <code>d["precipitation_mm"] == 35.12;</code><br><code>d["intensity_factor"] == 1.234568 (+1</code><br><code>more)</code>                                          | Pass   | 0.000s |
| To dict                           | <code>d["storm_id"] == "STORM-test1234";</code><br><code>d["start_time.hours"] == 48.0 (+6</code><br><code>more)</code>                                           | Pass   | 0.000s |
| Describe returns string           | <code>isinstance(desc, str); "base_mean" in</code><br><code>desc.lower() or "Base mean" in desc</code>                                                            | Pass   | 0.074s |
| Params roundtrip                  | <code>recovered.base_mean == 52;</code><br><code>recovered.name == "custom"</code>                                                                                | Pass   | 0.000s |
| Tail probability matches internal | <code>d.to_dict()['tail_probability'] ==</code><br><code>pytest.approx(d._tail...</code>                                                                          | Pass   | 0.000s |
| To dict keys                      | <code>'parameters' in result;</code><br><code>'tail_probability' in result</code>                                                                                 | Pass   | 0.000s |
| Antecedent defaults               | <code>seq.antecedent.soil_moisture ==</code><br><code>"normal"; seq.antecedent.groundwater</code><br><code>== "normal"</code>                                     | Pass   | 0.000s |
| Create doublet                    | <code>seq.num_storms == 2; len(seq.storms)</code><br><code>== 2 (+2 more)</code>                                                                                  | Pass   | 0.000s |
| From dict round trip              | <code>restored.storm_id ==</code><br><code>original.storm_id;</code><br><code>restored.duration.hours ==</code><br><code>original.duration.hours (+6 more)</code> | Pass   | 0.000s |
| Isolated no gaps                  | <code>len(seq.inter_storm_gaps_hours) == 0</code>                                                                                                                 | Pass   | 0.000s |
| Spatial correlation defaults      | <code>seq.spatial_correlation_enabled</code><br><code>is False;</code><br><code>seq.spatial_correlation_range_km</code><br><code>is None</code>                   | Pass   | 0.000s |

| Test Description              | Acceptance Criteria                                                                                          | Result | Time   |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|--------|--------|
| Spatial correlation enabled   | <code>d["spatial.correlation.enabled"]<br/>is True;<br/>d["spatial.correlation.range_km"]<br/>== 43.2</code> | Pass   | 0.000s |
| To dict                       | <code>d["storm_id"] == "STORM-test1234";<br/>d["start_time.hours"] == 48.0</code> (+6 more)                  | Pass   | 0.000s |
| Exceedance probs decreasing   | <code>stats['prob.exceed_60'] &gt;=<br/>stats['prob.exceed_70'] &gt;= ...</code>                             | Pass   | 0.000s |
| Fat tail higher p99 than thin | <code>fat['p99'] &gt; thin['p99']</code>                                                                     | Pass   | 0.148s |
| Mean in reasonable range      | <code>30 &lt; stats['mean'] &lt; 70</code>                                                                   | Pass   | 0.000s |
| Percentiles ordered           | <code>stats['p10'] &lt;= stats['p25'] &lt;=<br/>stats['p50'] &lt;= \{\}</code> ...                           | Pass   | 0.000s |
| Required keys present         | <code>k in stats</code>                                                                                      | Pass   | 0.000s |
| Duration mismatch             | <code>any("total.duration.hours" in e for e<br/>in errors)</code>                                            | Pass   | 0.000s |
| Exceeds precip window         | <code>any("MAX_PRECIP_END_HOUR" in e for e<br/>in errors)</code>                                             | Pass   | 0.000s |
| Invalid intensity category    | <code>any("invalid intensity" in e for e<br/>in errors)</code>                                               | Pass   | 0.000s |
| Invalid sequence type         | <code>any("Invalid sequence.type" in e for<br/>e in errors)</code>                                           | Pass   | 0.000s |
| Overlapping storms            | <code>any("starts at" in e and "before<br/>storm" in e for e in err...</code>                                | Pass   | 0.000s |
| Storm count mismatch          | <code>any("num.storms" in e for e in<br/>errors)</code>                                                      | Pass   | 0.000s |
| Too many storms               | <code>any("not in [1, 5]" in e for e in<br/>errors)</code>                                                   | Pass   | 0.000s |
| Valid doublet                 | <code>errors == []</code>                                                                                    | Pass   | 0.000s |

## 10 Storm Sequence Generator (MKM-SS-001)

Table 18: Test Results Summary — Storm Sequence Generator (MKM-SS-001)

|             |       |
|-------------|-------|
| Total Tests | 52    |
| Passed      | 52    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.24s |

Table 19: Detailed Test Results — Storm Sequence Generator

| Test Description                                               | Acceptance Criteria                                                                                                                | Result | Time   |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Duration must not exceed MAX_PRECIP_END_HOUR (156).            | <code>result.storms[0].duration_hours &lt;= MAX_PRECIP_END_HOUR</code>                                                             | Pass   | 0.000s |
| Isolated has one storm                                         | <code>isinstance(result, StormSequence);</code><br><code>result.num_storms == 1 (+2 more)</code>                                   | Pass   | 0.000s |
| All storm sequences                                            | <code>isinstance(seq, StormSequence)</code>                                                                                        | Pass   | 0.000s |
| Catchment propagated                                           | <code>seq.catchment_id == "rhine";</code><br><code>storm.catchment_id == "rhine"</code>                                            | Pass   | 0.001s |
| Correct count                                                  | <code>len(small.batch) == 200</code>                                                                                               | Pass   | 0.000s |
| Different seeds differ                                         | <code>types_a != types_b</code>                                                                                                    | Pass   | 0.002s |
| Force sequence type doublet                                    | <code>doublets &gt; 50; seq.sequence_type in ("doublet", "isolated")</code>                                                        | Pass   | 0.003s |
| Force sequence type isolated                                   | <code>seq.sequence_type == "isolated";</code><br><code>seq.num_storms == 1</code>                                                  | Pass   | 0.001s |
| Reproducible with same seed                                    | <code>[s.sequence_id for s in a] != [s.sequence_id for s in b];</code><br><code>sa.num_storms == sb.num_storms</code><br>(+1 more) | Pass   | 0.018s |
| Returns list                                                   | <code>isinstance(small.batch, list)</code>                                                                                         | Pass   | 0.000s |
| Custom weight heavy on extreme shifts the distribution.        | <code>all(c == "extreme" for c in cats)</code>                                                                                     | Pass   | 0.008s |
| All four default weight categories appear in the batch.        | <code>cat in categories</code>                                                                                                     | Pass   | 0.000s |
| Multi-storm sequences appear at a meaningful rate for severe+. | <code>multi_frac &gt; 0.15</code>                                                                                                  | Pass   | 0.000s |
| Creates parent dirs                                            | <code>out.exists()</code>                                                                                                          | Pass   | 0.009s |
| Load missing file raises                                       | —                                                                                                                                  | Pass   | 0.000s |
| Load returns correct count                                     | <code>len(loaded) == len(small.batch)</code>                                                                                       | Pass   | 0.021s |
| Load wrong schema raises                                       | —                                                                                                                                  | Pass   | 0.001s |

| Test Description                                   | Acceptance Criteria                                                                                                                | Result | Time   |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Loaded sequences still valid                       | <code>result["invalid"] == 0</code>                                                                                                | Pass   | 0.020s |
| Round trip num storms                              | <code>orig.num_storms == back.num_storms</code>                                                                                    | Pass   | 0.021s |
| Round trip precipitation                           | <code>abs(orig.total_precipitation_mm -<br/>back.total_precipitatio...</code>                                                      | Pass   | 0.020s |
| Round trip sequence type                           | <code>orig.sequence_type ==<br/>back.sequence_type</code>                                                                          | Pass   | 0.022s |
| Round trip storm fields                            | <code>os_.storm_index == bs_.storm_index;<br/>abs(os_.duration_hours -<br/>bs_.duration_hours) &lt; 0.01</code> (+1 more)          | Pass   | 0.024s |
| Round trip total duration                          | <code>abs(orig.total_duration_hours -<br/>back.total_duration_hours...</code>                                                      | Pass   | 0.020s |
| Save creates file                                  | <code>out.exists()</code>                                                                                                          | Pass   | 0.010s |
| Schema version in file                             | <code>raw["schema_version"] ==<br/>SCHEMA_VERSION</code>                                                                           | Pass   | 0.020s |
| Creates file                                       | <code>out.exists()</code>                                                                                                          | Pass   | 0.000s |
| Duration stats present                             | <code>d["min"] &gt; 0; d["max"] &lt;= 156</code> (+1 more)                                                                         | Pass   | 0.001s |
| Num sequences matches                              | <code>data["num_sequences"] ==<br/>len(small.batch)</code>                                                                         | Pass   | 0.001s |
| Precipitation stats present                        | <code>p["min"] &gt; 0; p["max"] &gt;= p["min"]</code> (+1 more)                                                                    | Pass   | 0.001s |
| Schema version                                     | <code>data["schema_version"] ==<br/>SCHEMA_VERSION</code>                                                                          | Pass   | 0.003s |
| Type counts sum to total                           | <code>sum(data["sequence_type_counts"].values())<br/>== len(small.b...</code>                                                      | Pass   | 0.001s |
| Type fractions sum to one                          | <code>abs(total_frac - 1.0) &lt; 0.001</code>                                                                                      | Pass   | 0.001s |
| Aggregate stats                                    | <code>result.max_intensity_factor &gt;=<br/>result.avg_intensity_factor;<br/>result.total_precipitation_mm &gt; 0</code> (+1 more) | Pass   | 0.000s |
| Catchment id in storms                             | <code>result.storms[0].catchment_id ==<br/>"rhine"</code>                                                                          | Pass   | 0.000s |
| Cluster sequence                                   | <code>result.sequence_type in<br/>(SequenceType.CLUSTER.value, Sequ...</code>                                                      | Pass   | 0.000s |
| force_sequence_type='doublet'<br>always doublet.   | → <code>result.sequence_type ==<br/>SequenceType.DOUBLET.value;<br/>result.num_storms == 2</code>                                  | Pass   | 0.000s |
| force_sequence_type='isolated'<br>always isolated. | → <code>result.sequence_type ==<br/>SequenceType.ISOLATED.value;<br/>result.num_storms == 1</code>                                 | Pass   | 0.000s |



| Test Description                                                      | Acceptance Criteria                                                                                                                  | Result | Time   |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Minimal intensity (5% sequence probability) mostly produces isolated. | <code>isolated &gt;= 10</code>                                                                                                       | Pass   | 0.001s |
| Generate multiple sequences without error.                            | <code>isinstance(result, StormSequence);</code><br><code>result.num_storms &gt;= 1</code>                                            | Pass   | 0.000s |
| Persistent sequence                                                   | <code>result.sequence_type in</code><br><code>(SequenceType.PERSISTENT.value,</code><br><code>S...</code>                            | Pass   | 0.000s |
| Returns storm sequence                                                | <code>isinstance(result, StormSequence)</code>                                                                                       | Pass   | 0.000s |
| All sequences must have total_duration   = 168h.                      | <code>result.total_duration_hours &lt;= 168.0</code>                                                                                 | Pass   | 0.000s |
| Storms have correct fields                                            | <code>storm.duration_hours &gt; 0;</code><br><code>storm.intensity_factor &gt; 0</code> (+2<br>more)                                 | Pass   | 0.000s |
| Custom catchment                                                      | <code>gen.catchment_id == "rhine";</code><br><code>gen.base_precipitation ==</code><br><code>CATCHMENT_BASE_PRECIP["rhine"]</code>   | Pass   | 0.000s |
| Default catchment                                                     | <code>gen.catchment_id == "thames";</code><br><code>gen.base_precipitation ==</code><br><code>CATCHMENT_BASE_PRECIP["thames"]</code> | Pass   | 0.000s |
| Same seed → same sequence.                                            | <code>seq1.num_storms == seq2.num_storms;</code><br><code>seq1.sequence_type ==</code><br><code>seq2.sequence_type</code>            | Pass   | 0.000s |
| Unknown catchment defaults to 35                                      | <code>gen.base_precipitation == 35.0</code>                                                                                          | Pass   | 0.000s |
| Zero validation errors across the whole batch.                        | <code>result["invalid"] == 0</code>                                                                                                  | Pass   | 0.000s |
| Medium batch all valid                                                | <code>result["invalid"] == 0</code>                                                                                                  | Pass   | 0.001s |
| No negative durations                                                 | <code>storm.duration_hours &gt; 0</code>                                                                                             | Pass   | 0.000s |
| No negative precipitation                                             | <code>seq.total_precipitation_mm &gt; 0;</code><br><code>storm.precipitation_mm &gt; 0</code>                                        | Pass   | 0.000s |
| Last storm precipitation must end by hour 156 (1e-9 FP tolerance).    | <code>last_end &lt;= MAX_PRECIP_END_HOUR +</code><br><code>_FP_TOL</code>                                                            | Pass   | 0.000s |

## 11 Trading Desk (MKM-TD-001)

Table 20: Test Results Summary — Trading Desk (MKM-TD-001)

|             |       |
|-------------|-------|
| Total Tests | 278   |
| Passed      | 278   |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 2.31s |

Table 21: Detailed Test Results — Trading Desk

| Test Description                                                                  | Acceptance Criteria                                                | Result | Time   |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------|--------|--------|
| All buttons must use font-size:11px;font-weight:600 matching header tabs.         | count >= 4                                                         | Pass   | 0.000s |
| All buttons must use padding:4px 14px matching header tab buttons.                | count >= 4                                                         | Pass   | 0.000s |
| All four buttons must sit inside a single flex container (group).                 | 'display:flex;border:1px solid #ddd;border-radius:4px;ove...       | Pass   | 0.000s |
| Commit button must be blue (#1976d2) — primary action like active header tab.     | '#1976d2' in snippet; 'color:white' in snippet                     | Pass   | 0.000s |
| History button must be grey (#f5f5f5) — inactive like unselected header tabs.     | '#f5f5f5' in snippet; '#1976d2' not in snippet                     | Pass   | 0.000s |
| Individual buttons must NOT have border-radius — container provides the rounding. | 'border-radius' not in snippet                                     | Pass   | 0.000s |
| PL Hist button must be grey (#f5f5f5) — inactive like unselected header tabs.     | '#f5f5f5' in snippet; '#1976d2' not in snippet                     | Pass   | 0.000s |
| Reset button must be grey (#f5f5f5) — inactive.                                   | '#f5f5f5' in snippet                                               | Pass   | 0.000s |
| Westminster, Chelsea and Lambeth all have trades — all must appear.               | self.WESTMINSTER_ID in active; self.CHELSEA_ID in active (+1 more) | Pass   | 0.013s |
| No trades at all → empty gauge_ids list → all right-click                         | resp.status_code == 200; data['gauge_ids'] == []                   | Pass   | 0.001s |
| A gauge that has no PRS trades must not appear in active-gauges,                  | self.NO_TRADE_ID not in data['gauge_ids']                          | Pass   | 0.014s |

| Test Description                                                           | Acceptance Criteria                                                                                                            | Result | Time   |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Lambeth Bridge (GAUGE-9042bd95) must be in active-gauges                   | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                                    | Pass   | 0.013s |
| After closing PRS-TEST-LAMBETH, Lambeth must leave active-gauges           | <code>self.LAMBETH_ID in</code><br><code>before['gauge_ids']; self.LAMBETH_ID</code><br><code>not in after['gauge_ids']</code> | Pass   | 0.034s |
| Closing Lambeth's trade must not affect Westminster or Chelsea.            | <code>self.WESTMINSTER_ID in</code><br><code>after['gauge_ids']; self.CHELSEA_ID in</code><br><code>after['gauge_ids']</code>  | Pass   | 0.053s |
| gauge_ids are returned in sorted order for stable JS dict lookup.          | <code>ids == sorted(ids)</code>                                                                                                | Pass   | 0.050s |
| Basis delta should be positive.                                            | <code>result['basis_delta_bps'] &gt; 0;</code><br><code>result['basis_dv01_gbp'] &gt; 0</code>                                 | Pass   | 0.000s |
| With constant basis, basis delta ~ gauge delta.                            | <code>abs(basis_delta['basis_delta_bps'] -</code><br><code>...</code>                                                          | Pass   | 0.000s |
| Closed trades excluded from blotter by default.                            | <code>'PRS-TEST-001' not in swap_ids</code>                                                                                    | Pass   | 0.036s |
| include_closed=true returns closed trades too.                             | <code>'PRS-TEST-001' in swap_ids</code>                                                                                        | Pass   | 0.030s |
| Hits /api/v1/trading/blotter.                                              | <code>'/api/v1/trading/blotter' in</code><br><code>blotter_js</code>                                                           | Pass   | 0.000s |
| Close-out hits /trading/close/.                                            | <code>'/trading/close/' in blotter_js</code>                                                                                   | Pass   | 0.000s |
| Corrupt JSON in prs dir is skipped via inner exception (lines 134-135).    | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                              | Pass   | 0.046s |
| Blotter returns 500 when engine raises (lines 94-96).                      | <code>resp.status_code == 500;</code><br><code>data['status'] == 'error'</code>                                                | Pass   | 0.003s |
| Close trade returns 500 when engine raises (lines 251-253).                | <code>resp.status_code == 500;</code><br><code>data['status'] == 'error'</code>                                                | Pass   | 0.001s |
| PDF regen exception is caught; trade close still succeeds (lines 236-237). | <code>resp.status_code == 200;</code><br><code>data['swap_id'] == 'PRS-TEST-001'</code><br>(+1 more)                           | Pass   | 0.022s |
| Trade with empty GaugeName gets name injected from gauge locations.        | <code>resp.status_code == 200;</code><br><code>noname[0].get('gauge_name', '')</code><br><code>!= ''</code>                    | Pass   | 0.048s |
| Close response includes final P&L.                                         | <code>'final_pnl' in data</code>                                                                                               | Pass   | 0.024s |
| After close, trade status is set to Closed in marks file.                  | <code>marks_file.exists();</code><br><code>marks['PRS-TEST-002']['trade_status']</code><br><code>== 'Closed' (+2 more)</code>  | Pass   | 0.022s |
| Closing non-existent trade returns 404.                                    | <code>resp.status_code == 404</code>                                                                                           | Pass   | 0.016s |

| Test Description                                                                     | Acceptance Criteria                                                                                      | Result | Time   |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Close-out without close-out_spread_bps returns 400.                                  | <code>resp.status_code == 400</code>                                                                     | Pass   | 0.000s |
| Settlement amount reflects the negotiated closeout spread, not fair spread.          | <code>data['status'] == 'success';</code><br><code>data['close_spread_bps'] == 400.0</code><br>(+2 more) | Pass   | 0.021s |
| Closing a trade at a negotiated spread returns success with close spread.            | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+3 more)              | Pass   | 0.025s |
| Button must call <code>tdCommitMarket()</code> .                                     | <code>'tdCommitMarket()' in market_js</code>                                                             | Pass   | 0.000s |
| td-commit-btn ID present in output.                                                  | <code>'td-commit-btn' in market_js</code>                                                                | Pass   | 0.000s |
| Commit button must NOT have <code>display:none</code> in its own style.              | <code>'display:none' not in own.style</code>                                                             | Pass   | 0.000s |
| <code>tdCommitMarket</code> function must exist.                                     | <code>'tdCommitMarket' in market_js</code>                                                               | Pass   | 0.000s |
| Hits <code>/trading/hazard-term-structure/commit</code> .                            | <code>'/trading/hazard-term-structure/commit' in market_js</code>                                        | Pass   | 0.000s |
| Hits <code>/trading/yield-curve/commit</code> .                                      | <code>'/trading/yield-curve/commit' in market_js</code>                                                  | Pass   | 0.000s |
| Commit notification shows P&L impact.                                                | <code>'total_pnl_impact' in market_js;</code><br><code>'gross_pnl_impact' in market_js</code>            | Pass   | 0.000s |
| After commit, switches to blotter tab.                                               | <code>"switchTab('blotter')" in market_js</code>                                                         | Pass   | 0.000s |
| Higher spread means higher premium                                                   | <code>pvs.high["premium_leg_pv"] &gt; pvs.low["premium_leg_pv"]</code>                                   | Pass   | 0.000s |
| Positive hazard rate                                                                 | <code>pvs["premium_leg_pv"] &gt; 0;</code><br><code>pvs["protection_leg_pv"] &gt; 0</code> (+2 more)     | Pass   | 0.000s |
| Zero hazard rate                                                                     | <code>pvs["premium_leg_pv"] == 0;</code><br><code>pvs["protection_leg_pv"] == 0</code>                   | Pass   | 0.000s |
| Returns empty history when no snapshots exist.                                       | <code>resp.status_code == 200;</code><br><code>data['history'] == []</code>                              | Pass   | 0.001s |
| Missing gauge_id returns 400.                                                        | <code>resp.status_code == 400</code>                                                                     | Pass   | 0.001s |
| After EOD, returns term structure history for that gauge.                            | <code>resp.status_code == 200;</code><br><code>data['gauge_id'] == 'GAUGE-001'</code><br>(+1 more)       | Pass   | 0.035s |
| POST <code>hazard-term-structure/commit</code> returns 500 on error (lines 341-343). | <code>resp.status_code == 500</code>                                                                     | Pass   | 0.001s |

| Test Description                                                       | Acceptance Criteria                                                                                                   | Result | Time   |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| curve-history returns 500 on engine error (lines 82-84).               | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| GET hazard-term-structure returns 500 on engine error (lines 234-236). | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| GET yield-curve returns 500 on engine error (lines 104-106).           | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| pnl-series returns 500 on engine error (lines 36-38).                  | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST hazard-term-structure/reset returns 500 on error (lines 363-365). | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST yield-curve/reset returns 500 on engine error (lines 270-272).    | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST yield-curve returns 500 on engine error (lines 211-213).          | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST hazard-term-structure/commit with null body → 400 (line 282).     | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.001s |
| POST yield-curve/commit with null body → 400 (line 145).               | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.001s |
| POST hazard-term-structure with null body → 400 (line 245).            | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.001s |
| POST yield-curve with null body → 400 (line 115).                      | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.001s |
| Closed trades excluded                                                 | <code>pnl_engine.compute_daily_pnl(trades)["num_open_trades"] == 0</code>                                             | Pass   | 0.000s |
| Daily pnl without previous eod                                         | <code>result["num_open_trades"] == 2;</code><br><code>result["total_notional"] == 15_000_000 (+1 more)</code>         | Pass   | 0.000s |
| Pnl decomposition sums                                                 | <code>abs(pos["daily_pnl"] - (pos["new_trade_pnl"] + pos["marke...))</code>                                           | Pass   | 0.000s |
| Running pnl calculation                                                | <code>abs(pos1["running_pnl"] - 43.500) &lt; 1;</code><br><code>abs(pos2["running_pnl"] - (-7.125)) &lt; 1</code>     | Pass   | 0.000s |
| Test risk grid construction.                                           | <code>len(grid['buckets']) == 3;</code><br><code>len(grid['gauges']) == 2 (+3 more)</code>                            | Pass   | 0.000s |
| Receiver trade should have negative gauge DV01.                        | <code>enriched['gauge_fs01'] &lt; 0;</code><br><code>enriched['is_payer'] is False</code>                             | Pass   | 0.003s |
| Test trade enrichment with sample data.                                | <code>enriched['swap_id'] == 'PRS-TEST001';</code><br><code>enriched['gauge_id'] == 'GAUGE-TEST-001' (+4 more)</code> | Pass   | 0.012s |



| Test Description                                                           | Acceptance Criteria                                                                                    | Result | Time   |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|--------|
| Payer and receiver DV01 should be equal magnitude, opposite sign.          | <code>abs(payer.enriched['gauge_fs01'] + rcv.enriched['gauge_fs...</code>                              | Pass   | 0.025s |
| tdHazardDirtyKeys tracks per-gauge hazard changes.                         | <code>'tdHazardDirtyKeys' in market_js</code>                                                          | Pass   | 0.000s |
| Shows UNCOMMITTED label when dirty.                                        | <code>'UNCOMMITTED' in market_js</code>                                                                | Pass   | 0.000s |
| tdYieldDirty variable tracks yield curve changes.                          | <code>'tdYieldDirty' in market_js</code>                                                               | Pass   | 0.000s |
| Also fetches blotter for live summary.                                     | <code>'/api/v1/trading/blotter' in eod_js</code>                                                       | Pass   | 0.000s |
| Fetches <code>/api/v1/trading/eod/history</code> .                         | <code>'/api/v1/trading/eod/history' in eod_js</code>                                                   | Pass   | 0.000s |
| Fetches <code>/api/v1/trading/pnl-series</code> .                          | <code>'/api/v1/trading/pnl-series' in eod_js</code>                                                    | Pass   | 0.000s |
| Hits <code>/api/v1/trading/eod</code> for submit.                          | <code>'/api/v1/trading/eod' in eod_js</code>                                                           | Pass   | 0.000s |
| <code>get_eod_history</code> returns 500 when engine raises (lines 88-90). | <code>resp.status_code == 500</code>                                                                   | Pass   | 0.001s |
| <code>get_eod_pdf</code> returns 500 on unexpected error (lines 107-109).  | <code>resp.status_code == 500</code>                                                                   | Pass   | 0.001s |
| <code>submit_eod</code> returns 500 when engine raises (lines 70-72).      | <code>resp.status_code == 500;</code><br><code>data['status'] == 'error'</code>                        | Pass   | 0.003s |
| <code>renderEodAttribution</code> function exists.                         | <code>'renderEodAttribution' in eod_js</code>                                                          | Pass   | 0.000s |
| <code>renderEodChart</code> function exists.                               | <code>'renderEodChart' in eod_js</code>                                                                | Pass   | 0.000s |
| <code>tdCleanupEodCharts</code> exists for chart cleanup.                  | <code>'tdCleanupEodCharts' in eod_js</code>                                                            | Pass   | 0.000s |
| <code>tdDownloadEodPdf</code> function exists.                             | <code>'tdDownloadEodPdf' in eod_js</code>                                                              | Pass   | 0.000s |
| <code>tdSubmitEod</code> function exists.                                  | <code>'tdSubmitEod' in eod_js</code>                                                                   | Pass   | 0.000s |
| Returns list of snapshots after submission.                                | <code>len(data['history']) == 1;</code><br><code>entry['eod_id'] == 'EOD-20260228'</code><br>(+3 more) | Pass   | 0.032s |
| Returns empty list when no snapshots exist.                                | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+2 more)            | Pass   | 0.001s |
| Can submit multiple EODs on different dates.                               | <code>data['count'] == 2</code>                                                                        | Pass   | 0.064s |
| POST with empty JSON body uses today's date (no date key).                 | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                      | Pass   | 0.033s |

| Test Description                                                          | Acceptance Criteria                                                                                                                                           | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| GET for existing PDF date serves the file.                                | <code>resp.status_code == 200;</code><br><code>resp.content_type ==</code><br><code>'application/pdf'</code>                                                  | Pass   | 0.003s |
| GET for non-existent date returns 404.                                    | <code>resp.status_code == 404</code>                                                                                                                          | Pass   | 0.000s |
| ImportError in PDF gen is caught; EOD snapshot still saved (lines 56-57). | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                                                             | Pass   | 0.026s |
| Exception in PDF gen is caught; EOD snapshot still saved (lines 58-59).   | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                                                             | Pass   | 0.039s |
| Attribution section container present.                                    | <code>'td-eod-attribution' in eod_js</code>                                                                                                                   | Pass   | 0.000s |
| P&L summary cards container present.                                      | <code>'td-eod-cards' in eod_js</code>                                                                                                                         | Pass   | 0.000s |
| Date picker for EOD date.                                                 | <code>'td-eod-date' in eod_js</code>                                                                                                                          | Pass   | 0.000s |
| EOD History section present.                                              | <code>'td-eod-history-wrap' in eod_js;</code> 'EOD History' in eod_js                                                                                         | Pass   | 0.000s |
| Submit EOD button present.                                                | <code>'td-eod-submit-btn' in eod_js;</code> 'EOD Submit' in eod_js                                                                                            | Pass   | 0.000s |
| All gauges from base should appear.                                       | <code>'GAUGE-001' in rates;</code> 'GAUGE-002' in rates                                                                                                       | Pass   | 0.001s |
| Adjusted gauges should be flagged.                                        | <code>rates['GAUGE-001']['is_adjusted']</code><br>is True;<br><code>rates['GAUGE-002']['is_adjusted']</code><br>is False                                      | Pass   | 0.022s |
| Adjusted rates should override base, non-adjusted should use base.        | <code>rates['GAUGE-001']['annual_hazard_rate_warning']</code><br>== 0.04;<br><code>rates['GAUGE-001']['annual_hazard_rate_alert']</code><br>== 0.04 (+1 more) | Pass   | 0.003s |
| Eod history                                                               | <code>len(history) == 2;</code> history[0]["date"]<br>== "2026-02-21"                                                                                         | Pass   | 0.015s |
| Generate snapshot                                                         | <code>snapshot["eod_id"] == "EOD-20260224";</code><br><code>snapshot["date"] == "2026-02-24"</code> (+3 more)                                                 | Pass   | 0.001s |
| Idempotent snapshot                                                       | <code>len(pnl_engine.get_eod_history()) ==</code><br>1                                                                                                        | Pass   | 0.003s |
| Pnl series                                                                | <code>len(series) == 2;</code> series[0]["date"]<br>== "2026-02-20" (+1 more)                                                                                 | Pass   | 0.004s |
| Snapshot persists                                                         | snapshot is not None;<br><code>snapshot["eod_id"] == "EOD-20260224"</code>                                                                                    | Pass   | 0.001s |

| Test Description                                                    | Acceptance Criteria                                                                           | Result | Time   |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| tdClearFilters function exists.                                     | 'tdClearFilters' in blotter_js                                                                | Pass   | 0.000s |
| tdApplyFilter exposed on window for external callers.               | 'tdApplyFilter' in blotter_js                                                                 | Pass   | 0.000s |
| tdFilterChanged function exists.                                    | 'tdFilterChanged' in blotter_js                                                               | Pass   | 0.000s |
| getFilteredTrades function exists.                                  | 'getFilteredTrades' in blotter_js                                                             | Pass   | 0.000s |
| tdRemoveFilter function exists.                                     | 'tdRemoveFilter' in blotter_js                                                                | Pass   | 0.000s |
| For small rates, 1bp bump in rate ~ = 1bp bump in spread.           | 0.5 < result['delta_spread.bps'] < 2.0                                                        | Pass   | 0.000s |
| DV01 scales with notional.                                          | abs(r2['dv01_gbp'] / r1['dv01_gbp'] - 2.0) < 0.01                                             | Pass   | 0.000s |
| DV01 should be positive for a buyer (spread increases with hazard). | result['dv01_gbp'] > 0;                                                                       | Pass   | 0.000s |
| At zero hazard rate, spread is at floor so delta may be minimal.    | result['delta_spread.bps'] > 0                                                                | Pass   | 0.000s |
| Balanced book                                                       | payer == rcv; len(trades) == 6                                                                | Pass   | 0.010s |
| Balanced notional                                                   | payer_not == rcv_not                                                                          | Pass   | 0.003s |
| Counterparties distributed                                          | len(ctpys) == 2                                                                               | Pass   | 0.003s |
| Files saved                                                         | len(list(output_dir.glob("PRS-*.json"))) == 6                                                 | Pass   | 0.003s |
| Payer trades below fair                                             | ps["Pricing"]["SpreadBps"] < ps["Pricing"]["FairSpreadBps"]                                   | Pass   | 0.003s |
| Receiver trades above fair                                          | ps["Pricing"]["SpreadBps"] > ps["Pricing"]["FairSpreadBps"]                                   | Pass   | 0.004s |
| Reproducible with seed                                              | t1["PhysicalSwap"]["Pricing"]["SpreadBps"] == t2["PhysicalSwap"]["Pricing"]["SpreadBps"]      | Pass   | 0.004s |
| Small mtm                                                           | abs(ps["Pricing"]["NPV"]) < ps["LegData"]["Notional"] * 0...                                  | Pass   | 0.003s |
| Valid cdm structure                                                 | ps["Header"]["SwapID"].startswith("PRS-"); isinstance(ps["LegData"]["Payer"], bool) (+2 more) | Pass   | 0.003s |
| Returns empty trades list when no PRS files exist.                  | resp.status_code == 200; data['status'] == 'success' (+1 more)                                | Pass   | 0.004s |
| Payer (long protection) trade should have positive gauge_fs01.      | len(payer_trades) > 0; t['gauge_fs01'] >= 0                                                   | Pass   | 0.021s |
| Receiver (short protection) trade should have negative gauge_fs01.  | len(rcv_trades) > 0; t['gauge_fs01'] <= 0                                                     | Pass   | 0.021s |
| Trades should have gauge_name from gauge locations.                 | 'Thames at Westminster' in names                                                              | Pass   | 0.019s |
| Each trade must have P&L decomposition fields.                      | 'daily_pnl' in trade; 'market_pnl' in trade (+2 more)                                         | Pass   | 0.019s |

| Test Description                                                             | Acceptance Criteria                                              | Result | Time   |
|------------------------------------------------------------------------------|------------------------------------------------------------------|--------|--------|
| Each trade must have prev_fair_spread_bps for curve-move indication.         | 'prev_fair_spread_bps' in trade                                  | Pass   | 0.022s |
| Each trade must have core identification and pricing fields.                 | 'swap_id' in trade; 'gauge_id' in trade (+5 more)                | Pass   | 0.018s |
| Summary must have all required aggregate fields.                             | 'num.trades' in summary; 'total_notional' in summary (+4 more)   | Pass   | 0.020s |
| Blotter returns enriched trades with status success.                         | resp.status_code == 200; data['status'] == 'success' (+1 more)   | Pass   | 0.024s |
| GET returns market_state with yield_curve and hazard_term_structure.         | resp.status_code == 200; data['status'] == 'success' (+2 more)   | Pass   | 0.003s |
| Response includes per-gauge effective rates.                                 | 'gauges' in data; len(gauge_ids) >= 1                            | Pass   | 0.003s |
| First load creates default yield curve.                                      | resp.status_code == 200; len(yc) > 0                             | Pass   | 0.003s |
| Bumping hazard curve higher should increase fair PRS spread.                 | spread_after > spread_before; spread_after - spread_before > 1.0 | Pass   | 0.012s |
| Bumping hazard curve should change trade MTM.                                | mtm_after > mtm_before                                           | Pass   | 0.002s |
| P&L from a 1bp parallel shift should approximately equal FS01.               | abs(mtm_change - fs01) < abs(fs01) * 0.20                        | Pass   | 0.005s |
| Full cycle: enrich → EOD snap → bump curve → re-enrich → daily P&L non-zero. | len(eod.snap['positions']) == 1; new_fair != prev_fair (+5 more) | Pass   | 0.005s |
| Shows down arrow for rate decrease.                                          | '\{\}\{u25bc' in blotter_js                                      | Pass   | 0.000s |
| Blotter compares to prev_fair_spread_bps.                                    | 'prev_fair_spread_bps' in blotter_js                             | Pass   | 0.000s |
| Shows up arrow for rate increase.                                            | '\{\}\{u25b2' in blotter_js                                      | Pass   | 0.000s |
| Commit returns affected_trades for the target gauge.                         | resp.status_code == 200; data['status'] == 'success' (+3 more)   | Pass   | 0.032s |
| Missing gauge_id returns 400.                                                | resp.status_code == 400                                          | Pass   | 0.000s |
| Missing rates returns 400.                                                   | resp.status_code == 400                                          | Pass   | 0.000s |
| Committing GAUGE-001 only affects its trades (2 trades).                     | data['affected_trades'] == 2                                     | Pass   | 0.028s |
| Reset all gauges.                                                            | resp.status_code == 200; 'all' in data['message']                | Pass   | 0.004s |

| Test Description                                                        | Acceptance Criteria                                                                                                   | Result | Time   |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Reset one gauge.                                                        | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                           | Pass   | 0.004s |
| GET returns per-gauge per-trigger structure.                            | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                           | Pass   | 0.002s |
| Tenor outside 1-10 returns 400.                                         | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.001s |
| Missing gauge_id returns 400.                                           | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.001s |
| POST updates one gauge/trigger/tenor point.                             | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                     | Pass   | 0.005s |
| Verify the MTM formula.                                                 | <code>abs(mtm - 80_000) &lt; 1</code>                                                                                 | Pass   | 0.000s |
| Protection seller loses when spread widens.                             | <code>mtm &lt; 0</code>                                                                                               | Pass   | 0.000s |
| Protection buyer profits when spread widens.                            | <code>mtm &gt; 0</code>                                                                                               | Pass   | 0.000s |
| MTM is zero when fair = trade spread.                                   | <code>mtm == 0</code>                                                                                                 | Pass   | 0.000s |
| Blotter has market_pnl column.                                          | <code>'market_pnl' in blotter_js</code>                                                                               | Pass   | 0.000s |
| Blotter shows Mkt P&L label.                                            | <code>'Mkt P' in blotter_js</code>                                                                                    | Pass   | 0.000s |
| GET market-state returns 500 when engine raises (lines 44-46).          | <code>resp.status_code == 500;</code><br><code>data['status'] == 'error'</code>                                       | Pass   | 0.001s |
| POST market-state/reset returns 500 when engine raises (lines 119-121). | <code>resp.status_code == 500</code>                                                                                  | Pass   | 0.001s |
| POST market-state with null JSON body → 400 (line 55).                  | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| Base rates should match what's in gaugehc.json.                         | <code>g1['annual_hazard_rate_warning'] == 0.025;</code><br><code>g1['annual_hazard_rate_severe'] == 0.01</code>       | Pass   | 0.001s |
| First load should initialise from base curves.                          | <code>'base_rates' in state;</code><br><code>'gauge_adjustments' in state</code> (+2 more)                            | Pass   | 0.002s |
| Subsequent loads should return the saved state.                         | <code>state1['base_rates'] == state2['base_rates']</code>                                                             | Pass   | 0.024s |
| Reset all adjustments.                                                  | <code>len(state['gauge_adjustments']) == 0</code>                                                                     | Pass   | 0.029s |
| Reset a single gauge's adjustments.                                     | <code>'GAUGE-001' not in state['gauge_adjustments'];</code><br><code>'GAUGE-002' in state['gauge_adjustments']</code> | Pass   | 0.049s |
| Effective rate should use adjustment if present.                        | <code>rate == 0.025; rate == 0.04</code>                                                                              | Pass   | 0.002s |



| Test Description                                                                             | Acceptance Criteria                                                               | Result | Time   |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| Gauge without adjustment should return base rate.                                            | <code>rate == 0.03</code>                                                         | Pass   | 0.002s |
| Updating a rate should persist.                                                              | <code>adj['annual_hazard_rate_warning'] == 0.035</code>                           | Pass   | 0.036s |
| <code>tdCleanupMarketCharts</code> exists.                                                   | <code>'tdCleanupMarketCharts' in market_js</code>                                 | Pass   | 0.000s |
| <code>renderGaugeList</code> function exists.                                                | <code>'renderGaugeList' in market_js</code>                                       | Pass   | 0.000s |
| Chart canvas element present.                                                                | <code>'td-market-canvas' in market_js</code>                                      | Pass   | 0.000s |
| Curve mode selector with yield/hazard options.                                               | <code>'td-curve-mode' in market_js; 'Yield Curve' in market_js (+1 more)</code>   | Pass   | 0.000s |
| <code>commit_yield_curve</code> saves all tenors at once.                                    | <code>state['yield_curve']['1'] == 0.04; state['yield_curve']['3'] == 0.05</code> | Pass   | 0.002s |
| Tenor above maximum returns last key rate.                                                   | <code>rate == r6</code>                                                           | Pass   | 0.001s |
| Tenor below minimum returns first key rate.                                                  | <code>rate == r1</code>                                                           | Pass   | 0.001s |
| Exact tenor match from yield curve.                                                          | <code>isinstance(rate, float); rate &gt; 0</code>                                 | Pass   | 0.001s |
| Non-integer tenor triggers interpolation.                                                    | <code>min(r1, r2) &lt;= rate &lt;= max(r1, r2)</code>                             | Pass   | 0.001s |
| Gauge entries with no <code>gauge_id</code> are skipped.                                     | <code>'GAUGE-B' in base; '' not in base</code>                                    | Pass   | 0.002s |
| <code>gaugehc.json</code> with <code>'gauges'</code> key is handled.                         | <code>'GAUGE-A' in state.get('base_rates', {})</code>                             | Pass   | 0.003s |
| When <code>gaugehc.json</code> is missing, <code>base curves</code> returns empty dict.      | <code>isinstance(state, dict)</code>                                              | Pass   | 0.002s |
| <code>gaugehc.json</code> with unknown format returns empty <code>base_rates</code> .        | <code>state.get('base_rates', {}) == {}</code>                                    | Pass   | 0.002s |
| <code>reset_yield_curve</code> restores defaults.                                            | <code>state['yield_curve']['1'] == MarketStateManager.DEFAULT_Y...</code>         | Pass   | 0.002s |
| Notes parameter is stored in gauge adjustments.                                              | <code>'adjustment_notes' in adj; 'override' in adj['adjustment_notes']</code>     | Pass   | 0.002s |
| New Trade button must be grey (#f5f5f5) — non-primary action.                                | <code>'#f5f5f5' in snippet; '#1976d2' not in snippet</code>                       | Pass   | 0.000s |
| Button text must be 'New Trade'.                                                             | <code>'New Trade' in market_js</code>                                             | Pass   | 0.000s |
| <code>td-newtrade-btn</code> must exist in Market tab.                                       | <code>'td-newtrade-btn' in market_js</code>                                       | Pass   | 0.000s |
| Button must call <code>tdMarketNewTrade()</code> .                                           | <code>'tdMarketNewTrade' in market_js</code>                                      | Pass   | 0.000s |
| <code>tdMarketNewTrade</code> must hide trading desk and call <code>viewHazardCurve</code> . | <code>'viewHazardCurve' in market_js; 'tdSelectedGauge' in market_js</code>       | Pass   | 0.000s |
| New Trade button must appear to the LEFT of History in the DOM.                              | <code>nt_idx &lt; hist_idx</code>                                                 | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                          | Result | Time   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| New Trade button must not have its own border-radius (container clips it). | 'border-radius' not in snippet                                               | Pass   | 0.000s |
| Market pnl nonzero with eod mark                                           | abs(result["positions"][0]["market_pnl"] - 80.000.0) < 1                     | Pass   | 0.000s |
| Market pnl zero for missing eod trade                                      | abs(pos["market_pnl"]) < 0.01;<br>abs(pos["new_trade_pnl"]) < 0.01 (+1 more) | Pass   | 0.000s |
| New trade has inception pnl                                                | abs(pos["new_trade_pnl"] - expected) < 1; abs(pos["market_pnl"]) < 0.01      | Pass   | 0.000s |
| Pnl decomposition with prev eod                                            | abs(pos["daily_pnl"] - (pos["new_trade_pnl"] + pos["marke...                 | Pass   | 0.000s |
| Shows daily P&L metric.                                                    | 'dailyPnl' in blotter_js                                                     | Pass   | 0.000s |
| Shows From Market metric.                                                  | 'From Market' in blotter_js;<br>'fromMarket' in blotter_js                   | Pass   | 0.000s |
| Shows signed net notional.                                                 | 'netNotional' in blotter_js;<br>'fmtGBP(netNotional)' in blotter_js          | Pass   | 0.000s |
| Shows Running P&L.                                                         | 'Running' in blotter_js                                                      | Pass   | 0.000s |
| After EOD snap, next blotter P&L uses snap as baseline.                    | resp.status_code == 200; 'daily_pnl' in t (+1 more)                          | Pass   | 0.038s |
| After an EOD snap, series has one entry.                                   | len(data['series']) == 1; 'date' in entry (+4 more)                          | Pass   | 0.033s |
| Returns empty series when no EOD snapshots exist.                          | resp.status_code == 200;<br>data['status'] == 'success' (+1 more)            | Pass   | 0.001s |
| Summary runs                                                               | "Market-Making Book Summary" in out;<br>"Payer (short):" in out (+1 more)    | Pass   | 0.002s |
| tdResetCurve function exists.                                              | 'tdResetCurve' in market_js                                                  | Pass   | 0.000s |
| Hits /trading/hazard-term-structure/reset.                                 | '/trading/hazard-term-structure/reset' in market_js                          | Pass   | 0.000s |
| Hits /trading/yield-curve/reset.                                           | '/trading/yield-curve/reset' in market_js                                    | Pass   | 0.000s |
| Reset clears all adjustments.                                              | resp.status_code == 200;<br>data['status'] == 'success' (+1 more)            | Pass   | 0.019s |
| Reset clears one gauge only.                                               | resp.status_code == 200; 'GAUGE-001' in data['message']                      | Pass   | 0.003s |
| risk-grid returns 500 when engine raises (lines 44-46).                    | resp.status_code == 500;<br>data['status'] == 'error'                        | Pass   | 0.012s |
| trade-map returns 500 when engine raises (lines 197-199).                  | resp.status_code == 500                                                      | Pass   | 0.001s |

| Test Description                                              | Acceptance Criteria                                                                                                                         | Result | Time   |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Grid should contain entries for traded gauges.                | <code>len(grid) &gt; 0</code>                                                                                                               | Pass   | 0.019s |
| GET returns a risk grid structure.                            | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                                                 | Pass   | 0.019s |
| Returns empty grid when no trades exist.                      | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                                           | Pass   | 0.003s |
| Longer tenor means more payments.                             | <code>a5 &gt; a1</code>                                                                                                                     | Pass   | 0.000s |
| Higher hazard rate means lower annuity (higher default prob). | <code>annuity_low &gt; annuity_high</code>                                                                                                  | Pass   | 0.000s |
| Annuity should be between 0 and tenor.                        | <code>0 &lt; annuity &lt; 5</code>                                                                                                          | Pass   | 0.000s |
| With no default risk, annuity equals risk-free annuity.       | <code>abs(annuity - expected) &lt; 1e-6</code>                                                                                              | Pass   | 0.000s |
| is_payer field used to determine sign.                        | <code>'is_payer' in blotter_js</code>                                                                                                       | Pass   | 0.000s |
| Payer notional should be negated.                             | <code>'is_payer' in blotter_js;</code> <code>'Math.abs'</code> in blotter_js                                                                | Pass   | 0.000s |
| tdToggleSort function exists.                                 | <code>'tdToggleSort' in blotter_js</code>                                                                                                   | Pass   | 0.000s |
| tdViewTrade function for click-through to PRS pricing.        | <code>'tdViewTrade' in blotter_js</code>                                                                                                    | Pass   | 0.000s |
| After knock-out, cash price must stay at full notional.       | <code>hourly[2]['per_trade'][0]['triggered'] is True;</code><br><code>hourly[2]['per_trade'][0]['cash_price'] == 1.000.000</code> (+2 more) | Pass   | 0.000s |
| Alert trade KOs before severe trade.                          | <code>'ALERT' in triggered;</code> <code>'SEVERE' in triggered</code> (+1 more)                                                             | Pass   | 0.000s |
| Trade must NOT knock out if water stays below trigger.        | <code>'T1' not in triggered</code>                                                                                                          | Pass   | 0.000s |
| Receiver (short protection) gets negative cash on KO.         | <code>hourly[1]['per_trade'][0]['cash_price'] == -1.000.000</code>                                                                          | Pass   | 0.000s |
| Trade must knock out when water $\geq$ trigger level.         | <code>'T1' in triggered;</code> <code>triggered['T1'] == 3</code>                                                                           | Pass   | 0.000s |
| Same-day snap overwrites previous.                            | <code>len(eod_files) == 1</code>                                                                                                            | Pass   | 0.048s |
| POST creates EOD snapshot.                                    | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)                                                 | Pass   | 0.041s |
| Response includes trade count and P&L summary.                | <code>'num_open_trades' in summary;</code><br><code>'total_notional' in summary</code> (+2 more)                                            | Pass   | 0.040s |

| Test Description                                                               | Acceptance Criteria                                                                          | Result | Time   |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------|--------|
| EOD-date.json created in eod/ directory.                                       | snapshot.path.exists();<br>snapshot['eod_id'] == 'EOD-20260228'<br>(+1 more)                 | Pass   | 0.035s |
| Snapshot includes yield_curve and hazard_term_structure.                       | 'yield_curve' in ms;<br>'hazard_term_structure' in ms                                        | Pass   | 0.038s |
| Snapshot includes per-trade positions.                                         | len(positions) == 16; 'swap_id' in p<br>(+2 more)                                            | Pass   | 0.034s |
| All levels should be $\geq$ base level (positive level_change).                | lvl $\geq$ base - 0.01                                                                       | Pass   | 0.000s |
| Custom num.hours should produce the right length.                              | len(levels) == 100                                                                           | Pass   | 0.000s |
| Water levels should be non-decreasing before peak.                             | levels[i] $\geq$ levels[i - 1] - 0.01                                                        | Pass   | 0.000s |
| Peak should reach base + level_change.                                         | peak $\geq$ base + change * 0.9; peak $\leq$ base + change + 0.1                             | Pass   | 0.000s |
| Water level should decrease after peak.                                        | levels[-1] < levels[peak_idx]                                                                | Pass   | 0.000s |
| Default hydrograph must return 168 hourly values (STORM_HOURS = 7-day window). | len(levels) == 168                                                                           | Pass   | 0.000s |
| Very short storm peaks early.                                                  | peak_idx $\leq$ 10                                                                           | Pass   | 0.000s |
| Hour 0 should be at base level.                                                | abs(levels[0] - 2.0) < 0.01                                                                  | Pass   | 0.000s |
| Zero duration should still produce valid levels.                               | len(levels) == 168; levels[0] > 2.0                                                          | Pass   | 0.000s |
| gauge.fs01 column present.                                                     | 'gauge.fs01' in blotter_js                                                                   | Pass   | 0.000s |
| fntMaturity function exists.                                                   | 'fntMaturity' in blotter_js                                                                  | Pass   | 0.000s |
| Files saved                                                                    | len(list(output_dir.glob("PRS-*.json"))) == 50                                               | Pass   | 0.028s |
| Has calendar spreads                                                           | calendar_spread_gauges $\geq$ 5                                                              | Pass   | 0.014s |
| Reproducible with seed                                                         | t1["PhysicalSwap"]["Pricing"]["SpreadBps"]<br>...                                            | Pass   | 0.081s |
| Trade count                                                                    | len(trades) == 50                                                                            | Pass   | 0.017s |
| Uses central gauges                                                            | gauge_ids.issubset(expected_ids)                                                             | Pass   | 0.014s |
| Valid cdm structure                                                            | ps["Header"]["SwapID"].startswith("PRS-");<br>ps["Pricing"]["RiskyAnnuity"] > 0<br>(+1 more) | Pass   | 0.015s |
| Westminster is net short                                                       | westminster_notional < 0                                                                     | Pass   | 0.021s |
| Gauge positions include FS01 broken down by tenor.                             | 'fs01.by_tenor' in g;<br>isinstance(g['fs01.by_tenor'], dict)                                | Pass   | 0.020s |

| Test Description                                                         | Acceptance Criteria                                                                           | Result | Time   |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Each gauge position has lat, lon, gauge_name.                            | 'lat' in g; 'lon' in g (+2 more)                                                              | Pass   | 0.022s |
| Each gauge has net_fs01.                                                 | 'net_fs01' in g                                                                               | Pass   | 0.021s |
| Each gauge has total and net notional.                                   | 'total_notional' in g; 'net_notional' in g (+1 more)                                          | Pass   | 0.019s |
| Gauge positions include daily P&L.                                       | 'daily_pnl' in g; 'running_pnl' in g                                                          | Pass   | 0.021s |
| Gauge positions show number of trades.                                   | gauges_by_id['GAUGE-001']['num_trades'] == 2;<br>gauges_by_id['GAUGE-002']['num_trades'] == 1 | Pass   | 0.019s |
| GET returns gauge positions.                                             | resp.status_code == 200;<br>data['status'] == 'success' (+1 more)                             | Pass   | 0.019s |
| Returns empty positions when no trades exist.                            | resp.status_code == 200;<br>len(data['gauges']) == 0                                          | Pass   | 0.004s |
| Gauge in gauge.json but not gaugehc.json gets coordinates (lines 83-85). | resp.status_code == 200                                                                       | Pass   | 0.053s |
| Property locations loaded from property.json (lines 97-101).             | resp.status_code == 200; 'gauges' in data                                                     | Pass   | 0.047s |
| Closed trades are excluded from trade map gauge aggregation (line 115).  | resp.status_code == 200;<br>'GAUGE-9042bd95' not in gauge_ids                                 | Pass   | 0.028s |
| Close trade                                                              | mark["trade_status"] == "Closed";<br>mark["close_spread_bps"] == 260.0 (+1 more)              | Pass   | 0.002s |
| Load empty marks                                                         | pnl_engine.load_trade_marks() == {}                                                           | Pass   | 0.000s |
| Update and load mark                                                     | pnl_engine.load_trade_marks()["PRS-001"]["trade_status"] == "Closed"<br>...                   | Pass   | 0.001s |
| Market making spreads are integers                                       | spread == round(spread)                                                                       | Pass   | 0.006s |
| Thames central spreads are integers                                      | spread == round(spread)                                                                       | Pass   | 0.014s |
| Updated state persists across requests.                                  | resp.status_code == 200;<br>data['market_state']['num_adjusted'] >= 1                         | Pass   | 0.018s |
| POST updates gauge rate successfully.                                    | resp.status_code == 200;<br>data['status'] == 'success' (+1 more)                             | Pass   | 0.017s |
| Missing gauge returns 400.                                               | resp.status_code == 400                                                                       | Pass   | 0.000s |
| No JSON body returns error (500 — Flask BadRequest caught).              | resp.status_code in (400, 500)                                                                | Pass   | 0.003s |



| Test Description                                      | Acceptance Criteria                                               | Result | Time   |
|-------------------------------------------------------|-------------------------------------------------------------------|--------|--------|
| Update returns affected_trades and total_pnl_impact.  | 'affected_trades' in data;<br>'total_pnl_impact' in data          | Pass   | 0.020s |
| GET returns yield_curve dict.                         | resp.status_code == 200;<br>data['status'] == 'success' (+1 more) | Pass   | 0.004s |
| Default yield curve has standard tenor points.        | len(yc) > 0                                                       | Pass   | 0.004s |
| Tenor outside 1-30 returns 400.                       | resp.status_code == 400                                           | Pass   | 0.001s |
| POST without JSON body returns error.                 | resp.status_code in (400, 500)                                    | Pass   | 0.008s |
| POST updates one tenor point.                         | resp.status_code == 200;<br>data['status'] == 'success' (+1 more) | Pass   | 0.004s |
| Commit returns affected_trades and P&L impact fields. | resp.status_code == 200;<br>data['status'] == 'success' (+4 more) | Pass   | 0.029s |
| Commit should affect all open trades.                 | data['affected_trades'] == 16                                     | Pass   | 0.025s |
| Empty rates dict returns 400.                         | resp.status_code == 400                                           | Pass   | 0.001s |
| No JSON body returns error.                           | resp.status_code in (400, 500)                                    | Pass   | 0.001s |
| Reset returns to default shape.                       | resp.status_code == 200;<br>data['status'] == 'success' (+1 more) | Pass   | 0.004s |

## 12 CDM Schema Validation (CDM-ALL)

Table 22: Test Results Summary — CDM Schema Validation (CDM-ALL)

|             |       |
|-------------|-------|
| Total Tests | 76    |
| Passed      | 76    |
| Failed      | 0     |
| Skipped     | 0     |
| Duration    | 0.01s |

Table 23: Detailed Test Results — CDM Schema Validation

| Test Description                                                                 | Acceptance Criteria                                             | Result | Time   |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------|--------|--------|
| All cdm fields present                                                           | <code>gauge_mapping_summary.fields_missing == 0</code>          | Pass   | 0.000s |
| All cdm fields present                                                           | <code>mortgage_mapping_summary.fields_missing == 0</code>       | Pass   | 0.000s |
| All cdm fields present                                                           | <code>property_mapping_summary.fields_missing == 0</code>       | Pass   | 0.000s |
| All types valid                                                                  | <code>gauge_mapping_summary.fields_type_invalid == 0</code>     | Pass   | 0.000s |
| All types valid                                                                  | <code>mortgage_mapping_summary.fields_type_invalid == 0</code>  | Pass   | 0.000s |
| All types valid                                                                  | <code>property_mapping_summary.fields_type_invalid == 0</code>  | Pass   | 0.000s |
| All values valid                                                                 | <code>gauge_mapping_summary.fields_value_invalid == 0</code>    | Pass   | 0.001s |
| All values valid                                                                 | <code>mortgage_mapping_summary.fields_value_invalid == 0</code> | Pass   | 0.000s |
| All values valid                                                                 | <code>property_mapping_summary.fields_value_invalid == 0</code> | Pass   | 0.000s |
| Get required fields returns list[gauge cdm] [cdm_name='gauge_cdm']               | <code>isinstance(cdm.get_required_fields(), list)</code>        | Pass   | 0.000s |
| Get required fields re-<br>turns list[mortgage cdm]<br>[cdm_name='mortgage_cdm'] | <code>isinstance(cdm.get_required_fields(), list)</code>        | Pass   | 0.000s |
| Get required fields re-<br>turns list[property cdm]<br>[cdm_name='property_cdm'] | <code>isinstance(cdm.get_required_fields(), list)</code>        | Pass   | 0.001s |
| Get required fields returns list[prs cdm] [cdm_name='prs_cdm']                   | <code>isinstance(cdm.get_required_fields(), list)</code>        | Pass   | 0.000s |

| Test Description                                                          | Acceptance Criteria                                 | Result | Time   |
|---------------------------------------------------------------------------|-----------------------------------------------------|--------|--------|
| Get required fields returns list[storm cdm] [cdm_name='storm_cdm']        | isinstance(cdm.get_required_fields(), list)         | Pass   | 0.000s |
| Get required fields re-returns list[stormts cdm] [cdm_name='stormts_cdm'] | isinstance(cdm.get_required_fields(), list)         | Pass   | 0.000s |
| List all fields returns paths[gauge cdm] [cdm_name='gauge_cdm']           | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[mortgage cdm] [cdm_name='mortgage_cdm']     | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[property cdm] [cdm_name='property_cdm']     | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[prs cdm] [cdm_name='prs_cdm']               | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.003s |
| List all fields returns paths[storm cdm] [cdm_name='storm_cdm']           | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| List all fields returns paths[stormts cdm] [cdm_name='stormts_cdm']       | isinstance(fields, list); len(fields) > 0 (+2 more) | Pass   | 0.000s |
| Repr returns string[gauge cdm] [cdm_name='gauge_cdm']                     | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[mortgage cdm] [cdm_name='mortgage_cdm']               | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[property cdm] [cdm_name='property_cdm']               | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[prs cdm] [cdm_name='prs_cdm']                         | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[storm cdm] [cdm_name='storm_cdm']                     | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Repr returns string[stormts cdm] [cdm_name='stormts_cdm']                 | isinstance(repr(cdm), str)                          | Pass   | 0.000s |
| Schema returns non empty dict[gauge cdm] [cdm_name='gauge_cdm']           | isinstance(schema, dict); len(schema) > 0           | Pass   | 0.000s |
| Schema returns non empty dict[mortgage cdm] [cdm_name='mortgage_cdm']     | isinstance(schema, dict); len(schema) > 0           | Pass   | 0.000s |
| Schema returns non empty dict[property cdm] [cdm_name='property_cdm']     | isinstance(schema, dict); len(schema) > 0           | Pass   | 0.000s |
| Schema returns non empty dict[prs cdm] [cdm_name='prs_cdm']               | isinstance(schema, dict); len(schema) > 0           | Pass   | 0.000s |

| Test Description                                                    | Acceptance Criteria                                                              | Result | Time   |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|--------|
| Schema returns non empty dict[storm cdm] [cdm_name='storm_cdm']     | isinstance(schema, dict); len(schema) > 0                                        | Pass   | 0.000s |
| Schema returns non empty dict[stormts cdm] [cdm_name='stormts_cdm'] | isinstance(schema, dict); len(schema) > 0                                        | Pass   | 0.000s |
| Create mapping returns flat dict                                    | isinstance(mapping, dict); mapping.get("gauge_id") == "GAUGE-test1234" (+9 more) | Pass   | 0.000s |
| Create mapping with empty input                                     | isinstance(gauge_cdm.create_mapping({}), dict)                                   | Pass   | 0.000s |
| Get field info for gauge id                                         | info is not None; info.get("type") == "text"                                     | Pass   | 0.000s |
| Required fields include gauge id                                    | any("GaugeID" in f for f in gauge_cdm.get_required_fields())                     | Pass   | 0.000s |
| Schema has flood gauge root                                         | "FloodGauge" in gauge_cdm.schema                                                 | Pass   | 0.000s |
| Schema has header section                                           | "Header" in gauge_cdm.schema["FloodGauge"]                                       | Pass   | 0.000s |
| Validate empty data returns errors                                  | isinstance(gauge_cdm.validate({}), dict)                                         | Pass   | 0.000s |
| Validate valid gauge data                                           | len(gauge_cdm.validate(data)) == 0                                               | Pass   | 0.000s |
| Create mapping returns flat dict                                    | isinstance(mapping, dict); mapping.get("gauge_id") == "GAUGE-test1234" (+9 more) | Pass   | 0.000s |
| Required fields include mortgage id                                 | any("MortgageID" in f for f in mortgage_cdm.get_required...                      | Pass   | 0.000s |
| Schema has mortgage root                                            | len(list(mortgage_cdm.schema.keys())) > 0                                        | Pass   | 0.000s |
| Validate valid mortgage                                             | isinstance(mortgage_cdm.validate(data), dict)                                    | Pass   | 0.000s |
| Create mapping returns flat dict                                    | isinstance(mapping, dict); mapping.get("gauge_id") == "GAUGE-test1234" (+9 more) | Pass   | 0.000s |
| List all fields has gauge set                                       | any("GaugeSet" in f for f in prs_cdm.list_all_fields())                          | Pass   | 0.000s |
| Required fields include swap id                                     | any("SwapID" in f for f in prs_cdm.get_required_fields())                        | Pass   | 0.000s |
| Schema has root                                                     | isinstance(stormts_cdm.schema, dict); len(stormts_cdm.schema) > 0 (+2 more)      | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                                                                                | Result | Time   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Create mapping returns flat dict                                           | <code>isinstance(mapping, dict);</code><br><code>mapping.get("gauge_id") ==</code><br><code>"GAUGE-test1234"</code> (+9 more)      | Pass   | 0.000s |
| Required fields not empty                                                  | <code>len(property_cdm.get_required_fields())</code><br><code>&gt; 0</code>                                                        | Pass   | 0.000s |
| Schema has property root                                                   | <code>len(list(property_cdm.schema.keys()))</code><br><code>&gt; 0</code>                                                          | Pass   | 0.000s |
| Validate valid property                                                    | <code>isinstance(property_cdm.validate(data), dict)</code>                                                                         | Pass   | 0.000s |
| Lines 255-256: exception in create_mapping → raises ValueError.            | —                                                                                                                                  | Pass   | 0.000s |
| Lines 213-256: create_mapping with all sections populated.                 | <code>mapping["storm_event_id"] ==</code><br><code>"STORM-001"; mapping["storm_name"] ==</code><br><code>"Eunice"</code> (+3 more) | Pass   | 0.000s |
| Create mapping returns flat dict                                           | <code>isinstance(mapping, dict);</code><br><code>mapping.get("gauge_id") ==</code><br><code>"GAUGE-test1234"</code> (+9 more)      | Pass   | 0.000s |
| Line 258-260: get_required_fields() returns list with StormEventID.        | <code>isinstance(fields, list);</code><br><code>any("StormEventID" in f for f in</code><br><code>fields)</code> (+2 more)          | Pass   | 0.000s |
| Required fields include storm id                                           | <code>any("StormEventID" in f for f in</code><br><code>storm_cdm.get_required_f...</code>                                          | Pass   | 0.000s |
| Schema has storm root                                                      | <code>len(list(storm_cdm.schema.keys())) &gt;</code><br><code>0</code>                                                             | Pass   | 0.000s |
| validate({}) returns errors for missing fields.                            | <code>isinstance(errors, dict);</code><br><code>isinstance(errors, dict)</code> (+1 more)                                          | Pass   | 0.000s |
| Lines 192-193: exception in validate → {'validation_error': [...]}.        | <code>isinstance(result, dict);</code><br><code>"validation_error" in result</code> (+2 more)                                      | Pass   | 0.000s |
| validate() with missing CatchmentID returns Header errors.                 | <code>"Header" in errors; any("CatchmentID"</code><br><code>in e for e in errors["Header"])</code> (+1 more)                       | Pass   | 0.000s |
| Lines 175-193: validate() with missing StormEventID returns Header errors. | <code>"Header" in errors;</code><br><code>any("StormEventID" in e for e in</code><br><code>errors["Header"])</code>                | Pass   | 0.000s |
| validate() with valid data returns empty dict.                             | <code>storm_cdm.validate(data) == {}</code>                                                                                        | Pass   | 0.000s |
| Lines 172-173: exception in create_mapping raises ValueError.              | —                                                                                                                                  | Pass   | 0.000s |



| Test Description                                                           | Acceptance Criteria                                                                           | Result | Time   |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Lines 150-170: create_mapping with full Parameters section.                | mapping["storm_event_id"] ==<br>"STORM-001"; mapping["storm_name"] ==<br>"Eunice" (+3 more)   | Pass   | 0.000s |
| Create mapping returns flat dict                                           | isinstance(mapping, dict);<br>mapping.get("gauge_id") ==<br>"GAUGE-test1234" (+9 more)        | Pass   | 0.000s |
| Line 185: create_reading_mapping returns flat dict.                        | mapping["timestamp"]<br>== "2022-02-18T06:00";<br>mapping["rainfall_mm"] == 12.5<br>(+1 more) | Pass   | 0.000s |
| Lines 194-199: get_required_fields() returns list.                         | isinstance(fields, list);<br>any("StormEventID" in f for f in<br>fields) (+2 more)            | Pass   | 0.000s |
| Required fields include ts id                                              | any("TimeSeriesID" in f for f in<br>stormts_cdm.get_required...                               | Pass   | 0.000s |
| Schema has root                                                            | isinstance(stormts_cdm.schema, dict);<br>len(stormts_cdm.schema) > 0 (+2 more)                | Pass   | 0.000s |
| validate({}) returns errors for both required fields.                      | isinstance(errors, dict);<br>isinstance(errors, dict) (+1 more)                               | Pass   | 0.000s |
| Lines 137-138: exception in validate → {'validation_error': [...]}.        | isinstance(result, dict);<br>"validation_error" in result (+2<br>more)                        | Pass   | 0.000s |
| validate() with missing CatchmentID returns Header errors.                 | "Header" in errors; any("CatchmentID"<br>in e for e in errors["Header"]) (+1<br>more)         | Pass   | 0.000s |
| Lines 120-138: validate() with missing TimeSeriesID returns Header errors. | "Header" in errors;<br>any("TimeSeriesID" in e for e in<br>errors["Header"])                  | Pass   | 0.000s |
| validate() with valid data returns empty dict.                             | stormts_cdm.validate(data) == {}                                                              | Pass   | 0.000s |

## A Platform Infrastructure Tests

Table 24: Test Results Summary — Platform Infrastructure (PLATFORM)

|             |         |
|-------------|---------|
| Total Tests | 4925    |
| Passed      | 4908    |
| Failed      | 2       |
| Skipped     | 15      |
| Duration    | 261.29s |

Table 25: Detailed Test Results — Platform Infrastructure

| Test Description                                                             | Acceptance Criteria                                                                             | Result | Time   |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Test that config has CATCHMENT attribute.                                    | —                                                                                               | Pass   | 0.000s |
| Test that config ENDPOINTS includes health_check for backend handler.        | <code>cfg.ENDPOINTS['health_check'] == '/api/v1/health'; ep in cfg.ENDPOINTS</code>             | Pass   | 0.000s |
| Test that select_catchment.html exists.                                      | —                                                                                               | Pass   | 0.000s |
| Test that routes register correctly.                                         | —                                                                                               | Pass   | 0.061s |
| Existing synthetic fixture — confirms endpoint logic works.                  | <code>response.status_code == 200</code>                                                        | Pass   | 0.002s |
| Mounts the real model.inventory.json and confirms the endpoint does not 500. | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+1 more) | Pass   | 0.037s |
| All models in the file are served by the endpoint.                           | <code>response.status_code == 200;</code><br><code>len(data["models"]) == expected.count</code> | Pass   | 0.039s |
| Concrete subclass returns empty list                                         | <code>result == []</code>                                                                       | Pass   | 0.000s |
| Get entity id called                                                         | <code>loader.find_by_id('A')</code> is not None                                                 | Pass   | 0.001s |
| Get entity summary called                                                    | <code>summaries[0]['id'] == 'B'</code>                                                          | Pass   | 0.001s |
| Add action auto generates id                                                 | <code>r.status_code == 200; actions[0]["id"] == "A-001"</code> (+1 more)                        | Pass   | 0.002s |
| Add action unknown meeting returns 404                                       | <code>r.status_code == 404</code>                                                               | Pass   | 0.001s |
| Delete action                                                                | <code>r.status_code == 200;</code><br><code>r.get_json()["meeting"]["actions"] == []</code>     | Pass   | 0.003s |
| Delete action unknown meeting returns 404                                    | <code>r.status_code == 404</code>                                                               | Pass   | 0.002s |

| Test Description                                                         | Acceptance Criteria                                                                                                            | Result | Time   |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Update action status                                                     | <code>r.status_code == 200;</code><br><code>r.get_json()["meeting"]["actions"][0]["status"]</code><br><code>== "Closed"</code> | Pass   | 0.002s |
| Update missing action returns 404                                        | <code>r.status_code == 404</code>                                                                                              | Pass   | 0.001s |
| Add action auto id                                                       | <code>"A-001" in ids; "A-002" in ids</code>                                                                                    | Pass   | 0.002s |
| Add action missing meeting returns 404                                   | <code>r.status_code == 404</code>                                                                                              | Pass   | 0.002s |
| Add action returns 200                                                   | <code>r.status_code == 200;</code><br><code>len(meeting["actions"]) == 2</code>                                                | Pass   | 0.002s |
| Delete action missing meeting returns 404                                | <code>r.status_code == 404</code>                                                                                              | Pass   | 0.002s |
| Delete action returns 200                                                | <code>r.status_code == 200;</code><br><code>len(r.get_json()["meeting"]["actions"])</code><br><code>== 0</code>                | Pass   | 0.002s |
| Update action not found returns 404                                      | <code>r.status_code == 404</code>                                                                                              | Pass   | 0.002s |
| Update action returns 200                                                | <code>r.status_code == 200; action["status"]</code><br><code>== "Closed"</code>                                                | Pass   | 0.002s |
| Production function: levels decay back toward base after peak.           | <code>levels[-1] &lt; max(levels)</code>                                                                                       | Pass   | 0.000s |
| Production function: peak base + level_change.                           | <code>max(levels) &gt;= base + change * 0.9</code>                                                                             | Pass   | 0.000s |
| _synthesize_hydrograph default returns 168 hours (STORM_HOURS constant). | <code>STORM_HOURS == 168; len(levels) == 168</code>                                                                            | Pass   | 0.000s |
| Production function: hour 0 = base_level.                                | <code>abs(levels[0] - 3.0) &lt; 0.01</code>                                                                                    | Pass   | 0.000s |
| Peak occurs near duration_hours * peak_position.                         | <code>abs(actual_peak_hour -</code><br><code>expected_peak_hour) &lt;= 2</code>                                                | Pass   | 0.000s |
| Add appears in list                                                      | <code>len(r.get_json()["references"]) == 1</code>                                                                              | Pass   | 0.003s |
| Add assigns id                                                           | <code>"id" in ref; len(ref["id"]) &gt; 0</code>                                                                                | Pass   | 0.002s |
| Add no json body uses defaults                                           | <code>r.status_code in (200, 400, 500)</code>                                                                                  | Pass   | 0.001s |
| Add returns 200                                                          | <code>r.status_code == 200</code>                                                                                              | Pass   | 0.002s |
| Add returns reference                                                    | <code>data["status"] == "success";</code><br><code>"reference" in data</code>                                                  | Pass   | 0.002s |
| Add same category twice no duplicate                                     | <code>r.get_json()["categories"].count("Risk")</code><br><code>== 1</code>                                                     | Pass   | 0.004s |
| Add stores all fields                                                    | <code>ref["author"] == "Jones";</code><br><code>ref["journal"] == "Nature"</code> (+1 more)                                    | Pass   | 0.002s |
| Add with category updates categories                                     | <code>"Flood Risk" in</code><br><code>r.get_json()["categories"]</code>                                                        | Pass   | 0.003s |

| Test Description                                                    | Acceptance Criteria                                                                                 | Result | Time   |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Adds to map                                                         | —                                                                                                   | Pass   | 0.001s |
| Returns notification system                                         | <code>isinstance(result, NotificationSystem)</code>                                                 | Pass   | 0.003s |
| Add to map calls add child                                          | —                                                                                                   | Pass   | 0.002s |
| Add to map calls add child                                          | —                                                                                                   | Pass   | 0.002s |
| Add to map calls add child                                          | —                                                                                                   | Pass   | 0.002s |
| Approaching retirement over 55                                      | <code>isinstance(result, list)</code>                                                               | Pass   | 0.001s |
| Early career under 25                                               | <code>isinstance(result, list)</code>                                                               | Pass   | 0.001s |
| Prime years 25 to 55                                                | <code>isinstance(result, list)</code>                                                               | Pass   | 0.001s |
| Add agenda item                                                     | <code>r.status_code == 200; len(agenda) == 1</code><br>(+2 more)                                    | Pass   | 0.001s |
| Add agenda unknown meeting returns 404                              | <code>r.status_code == 404</code>                                                                   | Pass   | 0.001s |
| Add multiple items increments number                                | <code>[a["item"] for a in agenda] == [1, 2, 3]</code>                                               | Pass   | 0.004s |
| Delete agenda item rennumbers                                       | <code>len(agenda) == 2; [a["item"] for a in agenda] == [1, 2]</code> (+1 more)                      | Pass   | 0.005s |
| Delete agenda unknown meeting returns 404                           | <code>r.status_code == 404</code>                                                                   | Pass   | 0.001s |
| Update agenda item                                                  | <code>r.status_code == 200; item["title"] == "Updated"</code> (+1 more)                             | Pass   | 0.002s |
| Update agenda unknown meeting returns 404                           | <code>r.status_code == 404</code>                                                                   | Pass   | 0.001s |
| Update missing item returns 404                                     | <code>r.status_code == 404</code>                                                                   | Pass   | 0.001s |
| Add agenda item increments count                                    | <code>len(meeting["agenda"]) == 3</code>                                                            | Pass   | 0.003s |
| Add agenda item returns 200                                         | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>              | Pass   | 0.003s |
| Add agenda missing meeting returns 404                              | <code>r.status_code == 404</code>                                                                   | Pass   | 0.002s |
| Delete agenda item returns 200                                      | <code>r.status_code == 200;</code><br><code>len(meeting["agenda"]) == 1</code> (+1 more)            | Pass   | 0.003s |
| Delete agenda missing meeting returns 404                           | <code>r.status_code == 404</code>                                                                   | Pass   | 0.002s |
| Update agenda item not found returns 404                            | <code>r.status_code == 404</code>                                                                   | Pass   | 0.001s |
| Update agenda item returns 200                                      | <code>r.status_code == 200;</code><br><code>meeting["agenda"][0]["title"] == "Updated Title"</code> | Pass   | 0.002s |
| When peak_wl is alert but warning, threshold is 'alert' (line 294). | <code>resp.status_code == 200;</code> g001 is not None (+1 more)                                    | Pass   | 0.047s |

| Test Description                                                                 | Acceptance Criteria                                                                                                                                           | Result | Time   |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Empty report                                                                     | <code>result['num_clones'] == 0;</code><br><code>result['worst'] == [] (+1 more)</code>                                                                       | Pass   | 0.000s |
| Fmt stats has python                                                             | <code>'python' in result['fmt_stats']</code>                                                                                                                  | Pass   | 0.000s |
| Line numbers extracted from 'start' field must be integers.                      | <code>isinstance(entry[1], int);</code><br><code>isinstance(entry[3], int)</code>                                                                             | Pass   | 0.000s |
| Regression: jscpd uses 'start' not 'startLine' — must not produce '?' in output. | <code>11 != '?'; 12 != '?'</code>                                                                                                                             | Pass   | 0.000s |
| Largest sorted descending                                                        | <code>lines == sorted(lines, reverse=True)</code>                                                                                                             | Pass   | 0.000s |
| Num clones correct                                                               | <code>result['num_clones'] == 2</code>                                                                                                                        | Pass   | 0.000s |
| Returns dict                                                                     | <code>isinstance(result, dict)</code>                                                                                                                         | Pass   | 0.000s |
| Total present                                                                    | <code>'total' in result;</code><br><code>result['total']['lines'] == 10000</code>                                                                             | Pass   | 0.000s |
| Worst populated                                                                  | <code>len(result['worst']) &gt; 0;</code><br><code>isinstance(fname, str) (+2 more)</code>                                                                    | Pass   | 0.000s |
| computePRSCashflows function exists.                                             | <code>'computePRSCashflows' in prs_js</code>                                                                                                                  | Pass   | 0.000s |
| interpolateSurvival function exists.                                             | <code>'interpolateSurvival' in prs_js</code>                                                                                                                  | Pass   | 0.000s |
| interpolateYieldRate function exists.                                            | <code>'interpolateYieldRate' in prs_js</code>                                                                                                                 | Pass   | 0.000s |
| Completeness percentage computed                                                 | <code>r["PropertyHeader"]["percentage"] ==</code><br><code>pytest.approx(66.67, ...)</code>                                                                   | Pass   | 0.000s |
| Empty section excluded                                                           | <code>"PropertyHeader" not in r</code>                                                                                                                        | Pass   | 0.000s |
| Mortgage data included                                                           | <code>"Mortgage Data" in r</code>                                                                                                                             | Pass   | 0.000s |
| Property header section present                                                  | <code>"PropertyHeader" in r</code>                                                                                                                            | Pass   | 0.000s |
| Returns dict                                                                     | <code>isinstance(r, dict);</code> <code>isinstance(r,</code><br><code>dict)</code>                                                                            | Pass   | 0.000s |
| Section not present excluded                                                     | <code>len(r) == 0</code>                                                                                                                                      | Pass   | 0.000s |
| All none returns zeros                                                           | <code>result["counts"]["properties"] == 0;</code><br><code>result["counts"]["mortgages"] == 0</code><br><code>(+1 more)</code>                                | Pass   | 0.000s |
| Counts match data                                                                | <code>result["counts"]["properties"] == 2;</code><br><code>result["counts"]["mortgages"] == 1</code><br><code>(+1 more)</code>                                | Pass   | 0.000s |
| Flood mortgages counted                                                          | <code>result["counts"]["flood_mortgages"]</code><br><code>== 1</code>                                                                                         | Pass   | 0.000s |
| Flood properties overlap                                                         | <code>result["counts"]["flood_properties"]</code><br><code>== 2;</code><br><code>result["overlaps"]["properties_with_flood_risk"]</code><br><code>== 1</code> | Pass   | 0.000s |

| Test Description                                                             | Acceptance Criteria                                                                           | Result | Time   |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| Returns required keys                                                        | "counts" in result; "overlaps" in result                                                      | Pass   | 0.000s |
| pts dir exists but property has no flood.events → 404.                       | client.get(_COMPOSITE_URL).status_code == 404                                                 | Pass   | 0.028s |
| pts dir exists but all flood_depths are zero → 404.                          | client.get(_COMPOSITE_URL).status_code == 404                                                 | Pass   | 0.030s |
| No pts dir returns 404                                                       | pts_client_no_data.get(_COMPOSITE_URL).status_code == 404                                     | Pass   | 0.000s |
| No pts dir status error                                                      | pts_client_no_data.get(_COMPOSITE_URL).get_json()["status"]                                   | Pass   | 0.000s |
| Options returns ok                                                           | pts_client_no_data.options(_COMPOSITE_URL).status_code == ...                                 | Pass   | 0.000s |
| Frame has gauges                                                             | "gauges" in frame0                                                                            | Pass   | 0.000s |
| Frame has hour                                                               | frame0["hour"] == 0                                                                           | Pass   | 0.000s |
| Frame has properties                                                         | "properties" in frame0                                                                        | Pass   | 0.000s |
| Frame has stats                                                              | "gauges_flooded" in stats; "properties_flooded" in stats (+1 more)                            | Pass   | 0.000s |
| Arrived true at arrival time                                                 | frames[4]["properties"][0]["arrived"] is False; frames[5]["properties"][0]["arrived"] is True | Pass   | 0.005s |
| Peak true at and after peak time                                             | frames[11]["properties"][0]["peak"] is False; frames[12]["properties"][0]["peak"] is True     | Pass   | 0.005s |
| hour i= len(readings) → flooded=False, depth_m=0, arrived=False, peak=False. | p["flooded"] is False; p["depth_m"] == 0 (+2 more)                                            | Pass   | 0.035s |
| Property does not have wse m                                                 | "wse_m" not in frames[0]["properties"][0]                                                     | Pass   | 0.000s |
| Property has peak field                                                      | "peak" in frames[0]["properties"][0]                                                          | Pass   | 0.000s |
| Property has required fields                                                 | f in p                                                                                        | Pass   | 0.000s |
| Frames list length                                                           | len(pts_env["client"].get(_COMPOSITE_URL).get_json())                                         | Pass   | 0.005s |
| N frames equals storm hours                                                  | pts_env["client"].get(_COMPOSITE_URL).get_json()["n_frames"]                                  | Pass   | 0.006s |
| N properties affected is one                                                 | pts_env["client"].get(_COMPOSITE_URL).get_json()["n_properties"]                              | Pass   | 0.006s |
| Returns 200                                                                  | pts_env["client"].get(_COMPOSITE_URL).status_code == 200                                      | Pass   | 0.005s |
| Status success                                                               | pts_env["client"].get(_COMPOSITE_URL).get_json()["status"]                                    | Pass   | 0.005s |
| Storm id is composite                                                        | pts_env["client"].get(_COMPOSITE_URL).get_json()["storm"]                                     | Pass   | 0.005s |



| Test Description                                                          | Acceptance Criteria                                                                                     | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------|--------|
| Property with two storms: composite uses the one with higher flood_depth. | data["status"] == "success";<br>data["frames"][0]["properties"][0]["depth_m"]<br>== pytest.a...         | Pass   | 0.036s |
| No pts dir returns 404                                                    | r.status_code == 404                                                                                    | Pass   | 0.001s |
| No pts dir status error                                                   | pts_client.no_data.get(<br>f"/api/v1/propertyts/...                                                     | Pass   | 0.001s |
| Options returns ok                                                        | r.status_code == 200                                                                                    | Pass   | 0.000s |
| Unknown storm message mentions storm id                                   | "STORM-GHOST" in msg                                                                                    | Pass   | 0.031s |
| Unknown storm returns 404                                                 | client.get("/api/v1/propertyts/animate/STORM-GHOST")                                                    | Pass   | 0.033s |
| Unknown storm status error                                                | client.get(<br>"/api/v1/propertyts/animate/STORM...                                                     | Pass   | 0.045s |
| Frame has gauges                                                          | "gauges" in frame0;<br>len(frame0["gauges"]) == 1                                                       | Pass   | 0.000s |
| Frame has hour field                                                      | "hour" in frame0                                                                                        | Pass   | 0.000s |
| Frame has properties                                                      | "properties" in frame0;<br>len(frame0["properties"]) == 1                                               | Pass   | 0.000s |
| Frame has stats                                                           | "gauges_flooded" in stats;<br>"properties_flooded" in stats (+1 more)                                   | Pass   | 0.000s |
| Frame hour zero is first                                                  | frame0["hour"] == 0                                                                                     | Pass   | 0.000s |
| Hour values sequential                                                    | [f["hour"] for f in frames] ==<br>list(range(STORM_HOURS))                                              | Pass   | 0.033s |
| Gauge has required fields                                                 | f in gauge0                                                                                             | Pass   | 0.000s |
| Gauge id correct                                                          | gauge0["gauge_id"] == "GAUGE-001"                                                                       | Pass   | 0.000s |
| Gauge status alert when above alert                                       | gauge0["status"] == "alert"                                                                             | Pass   | 0.000s |
| Gauge status normal when below alert                                      | g["status"] == "normal"                                                                                 | Pass   | 0.220s |
| Gauge status severe                                                       | g["status"] == "severe"                                                                                 | Pass   | 0.031s |
| Gauge status warning                                                      | g["status"] == "warning"                                                                                | Pass   | 0.033s |
| Gauge readings using water_level_m key (not waterLevel) still work.       | g["water_level_m"] == 4.3                                                                               | Pass   | 0.033s |
| Gauge with no readings defaults to zero                                   | frames[10]["gauges"][0]["water_level_m"]<br>== 0                                                        | Pass   | 0.031s |
| Water level from readings                                                 | gauge0["water_level_m"] == 4.2                                                                          | Pass   | 0.000s |
| Arrived false before arrival time                                         | frames[0]["properties"][0]["arrived"]<br>is False;<br>frames[4]["properties"][0]["arrived"]<br>is False | Pass   | 0.000s |

| Test Description                                                                       | Acceptance Criteria                                                                                    | Result | Time   |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|--------|
| Arrived true at and after arrival time                                                 | frames[5]["properties"][0]["arrived"]<br>is True;<br>frames[10]["properties"][0]["arrived"]<br>is True | Pass   | 0.000s |
| When hour $i = \text{len}(\text{readings})$ ,<br>flooded=False, depth_m=0,<br>wse_m=0. | p["flooded"] is False; p["depth_m"] ==<br>0 (+1 more)                                                  | Pass   | 0.031s |
| Property has required fields                                                           | f in p                                                                                                 | Pass   | 0.000s |
| Property id correct                                                                    | frames[0]["properties"][0]["property_id"]<br>== "PROP-001"                                             | Pass   | 0.000s |
| Gauges flooded counts non normal                                                       | stats["gauges_flooded"] == 1                                                                           | Pass   | 0.031s |
| Gauges flooded zero when normal                                                        | stats["gauges_flooded"] == 0                                                                           | Pass   | 0.034s |
| Properties flooded is positive early<br>on                                             | stats["properties_flooded"] >= 1                                                                       | Pass   | 0.031s |
| Total depth m non negative                                                             | all(f["stats"]["total_depth_m"] >= 0<br>for f in frames[:5])                                           | Pass   | 0.030s |
| Frames list length equals storm<br>hours                                               | len(data["frames"]) == STORM_HOURS                                                                     | Pass   | 0.005s |
| N frames equals storm hours                                                            | data["n_frames"] == STORM_HOURS                                                                        | Pass   | 0.007s |
| N properties affected is one                                                           | data["n_properties_affected"] == 1                                                                     | Pass   | 0.008s |
| Returns 200                                                                            | anim_client.get(f"/api/v1/propertyts/animal/STORM_ID").status_code == 200                              | Pass   | 0.004s |
| Status success                                                                         | data["status"] == "success"                                                                            | Pass   | 0.005s |
| Storm id in response                                                                   | data["storm_id"] == STORM_ID                                                                           | Pass   | 0.005s |
| Minimum spread applied for floor<br>property                                           | prob >= expected_min                                                                                   | Pass   | 0.105s |
| Return period finite for floor                                                         | rp is not None; rp > 0                                                                                 | Pass   | 0.103s |
| Appends line                                                                           | "hello world" in lines                                                                                 | Pass   | 0.000s |
| Clear then append                                                                      | lines == ["fresh line"]                                                                                | Pass   | 0.000s |
| Trims buffer at 2000                                                                   | length <= 2000                                                                                         | Pass   | 0.001s |
| All factors applied                                                                    | abs(result - round(expected, 2)) <<br>0.01                                                             | Pass   | 0.000s |
| Basic calculation                                                                      | result == 1250.0                                                                                       | Pass   | 0.000s |
| Claims factor increases premium                                                        | with_claims > base                                                                                     | Pass   | 0.000s |
| Clamped to max                                                                         | result == MAX_PREMIUM                                                                                  | Pass   | 0.000s |
| Clamped to min                                                                         | result == MIN_PREMIUM                                                                                  | Pass   | 0.000s |
| Contents premium added                                                                 | with_contents == base + 500.0                                                                          | Pass   | 0.000s |
| High flood risk increases premium                                                      | high_flood > low_flood                                                                                 | Pass   | 0.000s |
| Returns float                                                                          | isinstance(result, float)                                                                              | Pass   | 0.000s |
| Security factor reduces premium                                                        | with_security < without                                                                                | Pass   | 0.000s |
| All non negative                                                                       | np.all(result >= 0.0)                                                                                  | Pass   | 0.001s |

| Test Description                                                                  | Acceptance Criteria                                                               | Result | Time   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| For each active hour, mean(gauge-precip) approximately equals catchment_precip.   | <code>result[:, h].mean() ==<br/>pytest.approx(precip[h], rel=1e-6)</code>        | Pass   | 0.001s |
| Output shape                                                                      | <code>result.shape ==<br/>(thames_model.n_gauges, 168)</code>                     | Pass   | 0.001s |
| Different gauges get different precipitation amounts.                             | <code>col.std() &gt; 0.0</code>                                                   | Pass   | 0.001s |
| Mean across gauges of total precip approximately equals original catchment total. | <code>per_gauge_total.mean() ==<br/>pytest.approx(catchment_total,<br/>...</code> | Pass   | 0.001s |
| Zero precip hours remain zero                                                     | <code>np.all(result[:, zero_hours] == 0.0)</code>                                 | Pass   | 0.001s |
| All factors applied                                                               | <code>abs(result - round(expected, 2)) &lt;<br/>0.01</code>                       | Pass   | 0.000s |
| Basic calculation                                                                 | <code>result == 800_000.0</code>                                                  | Pass   | 0.000s |
| Clamped to max                                                                    | <code>result == MAX_PROPERTY_VALUE</code>                                         | Pass   | 0.000s |
| Clamped to min                                                                    | <code>result == MIN_PROPERTY_VALUE</code>                                         | Pass   | 0.000s |
| Flood risk discount reduces value                                                 | <code>with_risk &lt; no_risk</code>                                               | Pass   | 0.000s |
| Noise factor applied                                                              | <code>result2 &gt; result1</code>                                                 | Pass   | 0.000s |
| Returns float                                                                     | <code>isinstance(result, float)</code>                                            | Pass   | 0.000s |
| Age risk approaching retirement                                                   | <code>"Medium" in r["Age Risk"]</code>                                            | Pass   | 0.000s |
| Age risk early career                                                             | <code>"Medium" in r["Age Risk"]</code>                                            | Pass   | 0.000s |
| Age risk prime years                                                              | <code>"Age Risk" in r; "Low" in r["Age<br/>Risk"]</code>                          | Pass   | 0.000s |
| Credit risk high for poor score                                                   | <code>"High" in r["Credit Risk"]</code>                                           | Pass   | 0.000s |
| Credit risk low for high score                                                    | <code>"Credit Risk" in r; "Low" in<br/>r["Credit Risk"]</code>                    | Pass   | 0.000s |
| Credit risk low medium for good score                                             | <code>"Low-Medium" in r["Credit Risk"]</code>                                     | Pass   | 0.000s |
| Credit risk medium for fair score                                                 | <code>"Medium" in r["Credit Risk"]</code>                                         | Pass   | 0.000s |
| Employment risk established 2 to 5                                                | <code>"Low-Medium" in r["Employment Risk"]</code>                                 | Pass   | 0.000s |
| Employment risk recent under 2                                                    | <code>"Medium-High" in r["Employment<br/>Risk"]</code>                            | Pass   | 0.000s |
| Employment risk stable 5 plus                                                     | <code>"Employment Risk" in r; "Low" in<br/>r["Employment Risk"]</code>            | Pass   | 0.000s |
| Income risk low for high income                                                   | <code>"Income Risk" in r; "Low" in<br/>r["Income Risk"]</code>                    | Pass   | 0.000s |
| Income risk low medium adequate                                                   | <code>"Low-Medium" in r["Income Risk"]</code>                                     | Pass   | 0.000s |
| Income risk medium limited                                                        | <code>"Medium" in r["Income Risk"]</code>                                         | Pass   | 0.000s |
| Returns dict                                                                      | <code>isinstance(result, dict)</code>                                             | Pass   | 0.000s |
| Best and worst sections shown                                                     | <code>"Best Data Section" in r; "Weakest<br/>Data Section" in r</code>            | Pass   | 0.000s |

| Test Description                                      | Acceptance Criteria                                                              | Result | Time   |
|-------------------------------------------------------|----------------------------------------------------------------------------------|--------|--------|
| Critical completeness high                            | "Critical Data Reliability" in<br>r; "High" in r["Critical Data<br>Reliability"] | Pass   | 0.000s |
| Critical completeness low                             | "Low" in r["Critical Data<br>Reliability"]                                       | Pass   | 0.000s |
| Critical completeness medium                          | "Medium" in r["Critical Data<br>Reliability"]                                    | Pass   | 0.000s |
| Excellent over 90                                     | "Excellent" in r["Overall<br>Completeness"]                                      | Pass   | 0.000s |
| Fair 60 to 75                                         | "Fair" in r["Overall Completeness"]                                              | Pass   | 0.000s |
| Good 75 to 90                                         | "Good" in r["Overall Completeness"]                                              | Pass   | 0.000s |
| Limited 40 to 60                                      | "Limited" in r["Overall<br>Completeness"]                                        | Pass   | 0.000s |
| Poor under 40                                         | "Poor" in r["Overall Completeness"]                                              | Pass   | 0.000s |
| Building 25 years → Contemporary<br>category.         | isinstance(result, list)                                                         | Pass   | 0.001s |
| Building older than 100 years →<br>Historic category. | isinstance(result, list)                                                         | Pass   | 0.001s |
| Building 50–100 years → Mature<br>category.           | isinstance(result, list)                                                         | Pass   | 0.001s |
| Building 25–50 years → Modern<br>category.            | isinstance(result, list)                                                         | Pass   | 0.001s |
| Basic info subsection                                 | any("Basic Property Information" in<br>t for t in texts)                         | Pass   | 0.001s |
| Building age subsection                               | any("Building Age" in t for t in<br>texts)                                       | Pass   | 0.001s |
| Condition fair recommendation                         | isinstance(result, list)                                                         | Pass   | 0.001s |
| Condition good recommendation                         | isinstance(result, list)                                                         | Pass   | 0.001s |
| Condition poor recommendation                         | isinstance(result, list)                                                         | Pass   | 0.001s |
| Condition unknown falls back                          | isinstance(result, list)                                                         | Pass   | 0.002s |
| Empty property does not crash                         | isinstance(result, list)                                                         | Pass   | 0.001s |
| External features subsection                          | any("External Features" in t for t<br>in texts)                                  | Pass   | 0.001s |
| Has section header                                    | any("Property Attributes" in t for t<br>in texts)                                | Pass   | 0.001s |
| Has tables                                            | any(isinstance(e, Table) for e in<br>result)                                     | Pass   | 0.001s |
| No construction year skips age                        | isinstance(result, list)                                                         | Pass   | 0.001s |
| Physical dimensions subsection                        | any("Physical Dimensions" in t for t<br>in texts)                                | Pass   | 0.001s |

| Test Description                                                                               | Acceptance Criteria                                                                               | Result | Time   |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Returns list                                                                                   | <code>isinstance(result, list); len(result) &gt; 0</code>                                         | Pass   | 0.001s |
| Room composition subsection                                                                    | <code>any("Room Composition" in t for t in texts)</code>                                          | Pass   | 0.001s |
| With mortgage data does not crash                                                              | <code>isinstance(result, list)</code>                                                             | Pass   | 0.001s |
| Line 82: audit log $i$ 10000 entries gets trimmed.                                             | <code>r.status_code == 200; len(log) &lt;= 10000</code>                                           | Pass   | 0.068s |
| Line 57: inventory file empty/missing $\rightarrow$ 404.                                       | <code>r.status_code == 404</code>                                                                 | Pass   | 0.001s |
| Line 85: save failure $\rightarrow$ 500.                                                       | <code>r.status_code == 500</code>                                                                 | Pass   | 0.002s |
| Audit log truncated to 10000                                                                   | <code>r.status_code == 200; len(entries) == 10000 (+1 more)</code>                                | Pass   | 0.066s |
| <code>_auditTerminalRefreshNow</code> must be on window — called from onclick in innerHTML.    | <code>'window._auditTerminalRefreshNow' in audit_reports_js</code>                                | Pass   | 0.000s |
| JSON button was deliberately removed — must not reappear.                                      | <code>'test_failures_report.json' not in audit_reports_js</code>                                  | Pass   | 0.000s |
| Poll output endpoint                                                                           | <code>'test-report/output' in audit_reports_js</code>                                             | Pass   | 0.000s |
| Refresh audit tab button                                                                       | <code>'_auditTerminalRefreshNow' in audit_reports_js</code>                                       | Pass   | 0.000s |
| <code>_runTestSuite</code> must be on window — onclick attributes execute in global scope.     | <code>'window._runTestSuite' in audit_reports_js</code>                                           | Pass   | 0.000s |
| Run suite function                                                                             | <code>'_runTestSuite' in audit_reports_js</code>                                                  | Pass   | 0.000s |
| Terminal overlay id                                                                            | <code>'_audit-terminal-overlay' in audit_reports_js</code>                                        | Pass   | 0.000s |
| <code>_toggleFailureDetail</code> must be on window — called from onclick in innerHTML.        | <code>'window._toggleFailureDetail' in audit_reports_js</code>                                    | Pass   | 0.000s |
| When audit log exceeds 10 000 entries, it keeps only the last 10 000.                          | <code>len(entries) == 10_000; entries[0].get("seq") == 1 or entries[0].get("model_id") ...</code> | Pass   | 0.047s |
| <code>auto_open=True</code> enters the auto-open block (lines 324-325).                        | <code>report_path is not None; Path(report_path).exists()</code>                                  | Pass   | 0.022s |
| <code>auto_open=False</code> (default) — no open attempt, returns PDF path normally.           | <code>report_path is not None; Path(report_path).exists()</code>                                  | Pass   | 0.019s |
| <code>auto_open=True</code> with a successful <code>open_pdf_file</code> call (lines 324-325). | <code>report_path is not None; Path(report_path).exists()</code>                                  | Pass   | 0.020s |
| All pages exist                                                                                | <code>p in gen.pages</code>                                                                       | Pass   | 0.002s |

| Test Description                                                     | Acceptance Criteria                                                                         | Result | Time   |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| All pages exist                                                      | <code>p in gen.pages</code>                                                                 | Pass   | 0.001s |
| Always includes analysis pages                                       | <code>ap in pages</code>                                                                    | Pass   | 0.002s |
| Always includes appendix                                             | <code>"appendix" in pages</code>                                                            | Pass   | 0.001s |
| Always includes gauge info pages                                     | <code>gp in pages</code>                                                                    | Pass   | 0.001s |
| Always includes operational pages                                    | <code>op in pages</code>                                                                    | Pass   | 0.001s |
| Always includes property pages                                       | <code>pp in pages</code>                                                                    | Pass   | 0.002s |
| Always includes summary pages                                        | <code>sp in pages</code>                                                                    | Pass   | 0.001s |
| Empty flood data                                                     | <code>isinstance(pages, list)</code>                                                        | Pass   | 0.001s |
| Empty property data                                                  | <code>isinstance(pages, list)</code>                                                        | Pass   | 0.002s |
| Includes analysis pages                                              | <code>p in pages</code>                                                                     | Pass   | 0.001s |
| Includes overview pages                                              | <code>p in pages</code>                                                                     | Pass   | 0.001s |
| Returns list                                                         | <code>isinstance(pages, list); len(pages) &gt; 0</code>                                     | Pass   | 0.001s |
| Returns list                                                         | <code>isinstance(pages, list); len(pages) &gt; 0</code>                                     | Pass   | 0.001s |
| With hazard curve includes analysis pages                            | <code>any(p in pages for p in analysis_pages)</code>                                        | Pass   | 0.002s |
| With mortgage includes mortgage pages                                | <code>mp in pages</code>                                                                    | Pass   | 0.002s |
| With timeseries includes analysis pages                              | <code>any(p in pages for p in analysis_pages)</code>                                        | Pass   | 0.001s |
| Without mortgage excludes mortgage pages                             | <code>mp not in pages</code>                                                                | Pass   | 0.002s |
| Without timeseries or hazard no analysis pages                       | <code>not any(p in pages for p in analysis_pages)</code>                                    | Pass   | 0.002s |
| Lines 60-61: custom server_url and endpoints override config.        | <code>handler.server_url == 'http://test:9999'; handler.endpoints == {'a': '/api/a'}</code> | Pass   | 0.000s |
| Lines 60-61: default server_url and endpoints from config.           | <code>handler.server_url == config.SERVER_URL; handler.endpoints == config.ENDPOINTS</code> | Pass   | 0.000s |
| Endpoints dict is a copy, not a reference.                           | <code>'z' not in handler.endpoints</code>                                                   | Pass   | 0.000s |
| Line 154: BaseCDM.get_required_fields returns empty list by default. | <code>cdm.get_required_fields() == []</code>                                                | Pass   | 0.000s |
| get_field_info on a leaf field returns its definition.               | <code>info is not None; info["type"] == "text"</code>                                       | Pass   | 0.000s |
| Line 116: field path not found returns None.                         | <code>mortgage_cdm.get_field_info("Mortgage.Nonexistent.Field")</code>                      | Pass   | 0.000s |



| Test Description                                                            | Acceptance Criteria                                                                        | Result | Time   |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------|--------|
| Line 116: partial path that resolves to non-dict returns None.              | <code>mortgage.cdm.get_field_info("Nonexistent")</code> is None                            | Pass   | 0.000s |
| Line 118: <code>get_field_info</code> returns None when leaf is not a dict. | <code>cdm.get_field_info("Root.Field")</code> is None                                      | Pass   | 0.000s |
| Line 133: metadata keys at nested level are also skipped.                   | "Root.Sub.Name" in fields; 'options' not in path_parts (+1 more)                           | Pass   | 0.000s |
| <code>list_all_fields</code> on PRS CDM covers nested GaugeSet section.     | <code>any("GaugeSet" in f for f in fields); any("Triggers" in f for f in fields)</code>    | Pass   | 0.000s |
| Line 133: <code>list_all_fields</code> recurses nested sections.            | <code>isinstance(fields, list); len(fields) &gt; 10</code> (+2 more)                       | Pass   | 0.000s |
| Line 133: 'type', 'options', 'values', 'description', 'items' are skipped.  | <code>len(fields) == 1; fields[0] == "Section.RealField"</code>                            | Pass   | 0.000s |
| Line 157: <code>__repr__</code> returns class name.                         | "MortgageCDM" in <code>repr(mortgage.cdm)</code>                                           | Pass   | 0.000s |
| Verify data is cached after first load.                                     | <code>len(result1) == len(result2) == 3</code>                                             | Pass   | 0.001s |
| Test cache clearing.                                                        | <code>loader._cache</code> is not None; <code>loader._cache</code> is None (+1 more)       | Pass   | 0.001s |
| Verify <code>force_reload</code> bypasses cache.                            | <code>len(result1) == 3; len(result2) == 0</code>                                          | Pass   | 0.002s |
| Test cache invalidation.                                                    | <code>loader._cache_valid</code> is True; <code>loader._cache_valid</code> is False        | Pass   | 0.001s |
| Verify abstract methods must be implemented.                                | —                                                                                          | Pass   | 0.000s |
| Subclass must implement <code>get_entity_id</code> .                        | —                                                                                          | Pass   | 0.000s |
| Subclass must implement <code>get_entity_summary</code> .                   | —                                                                                          | Pass   | 0.000s |
| Test <code>file_exists</code> method.                                       | <code>loader.file_exists()</code> is True; <code>loader2.file_exists()</code> is False     | Skip   | 0.000s |
| Load data with alternative container key (portfolio).                       | <code>len(result) == 1</code>                                                              | Pass   | 0.002s |
| Load data when JSON has container key.                                      | <code>len(result) == 2</code>                                                              | Pass   | 0.001s |
| Load data when JSON is a direct list.                                       | <code>len(result) == 1; result[0]['PropertyHeader']['Header']['PropertyID'] == 'P1'</code> | Pass   | 0.001s |
| Handle missing file gracefully.                                             | <code>result == []</code>                                                                  | Pass   | 0.000s |
| Load data when JSON is single object.                                       | <code>len(result) == 1</code>                                                              | Pass   | 0.001s |
| Test count method.                                                          | <code>property_loader.count() == 3</code>                                                  | Pass   | 0.001s |

| Test Description                                                         | Acceptance Criteria                                                                                                                                     | Result | Time   |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Test exists method.                                                      | <code>property_loader.exists("PROP-001")</code><br><code>is True;</code><br><code>property_loader.exists("NONEXISTENT")</code><br><code>is False</code> | Pass   | 0.002s |
| Test finding entity by ID.                                               | <code>result is not None;</code><br><code>result['PropertyHeader']['Header']['PropertyID']</code><br><code>== "PROP..."</code>                          | Pass   | 0.002s |
| Test finding non-existent entity.                                        | <code>result is None</code>                                                                                                                             | Pass   | 0.005s |
| Test getting all IDs.                                                    | <code>len(ids) == 3; "PROP-001" in ids (+2</code><br><code>more)</code>                                                                                 | Skip   | 0.000s |
| Test get_status method.                                                  | <code>status['entity_type'] == 'property';</code><br><code>status['file.exists'] is True (+1</code><br><code>more)</code>                               | Skip   | 0.000s |
| Test listing all entities.                                               | <code>len(result) == 3; all('propertyId' in</code><br><code>item for item in result)</code>                                                             | Pass   | 0.001s |
| Line 190: gauge_elevs defaults to [prop_elevation], elev_diff = 0.       | <code>result['elevation_diff_m'] == 0.0</code>                                                                                                          | Pass   | 0.000s |
| Lines 179, 190: empty nearest_gauges defaults distances and elevations.  | <code>'total_basis_bp' in result;</code><br><code>'distance_bp' in result (+3 more)</code>                                                              | Pass   | 0.000s |
| Pdf not generated returns 404                                            | <code>r.status_code == 404;</code> "not yet<br><code>generated" in r.get_json()["message"]</code>                                                       | Pass   | 0.000s |
| Save failure returns 500                                                 | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                                                                    | Pass   | 0.002s |
| Line 74: POST add when save fails.                                       | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                                                                    | Pass   | 0.002s |
| Line 112: DELETE when save fails.                                        | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                                                                    | Pass   | 0.002s |
| Line 100: PUT update when save fails.                                    | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                                                                    | Pass   | 0.002s |
| P&L bar should display net notional (signed sum).                        | <code>'netNotional' in js;</code><br><code>'fmtGBP(netNotional)' in js</code>                                                                           | Pass   | 0.000s |
| Payer (Pay) should show negative notional.                               | <code>'is_payer' in js; 'netNotional' in js</code>                                                                                                      | Pass   | 0.000s |
| PRS-*.json glob must match PRS-{8hex} filenames (the production format). | <code>fnmatch.fnmatch(name, 'PRS-*.json');</code><br><code>fnmatch.fnmatch(name, 'PRS-*.json')</code>                                                   | Pass   | 0.001s |
| When PRS trade files are present, GET /trading/blotter returns them.     | <code>len(found) == 1</code>                                                                                                                            | Pass   | 0.002s |

| Test Description                                                   | Acceptance Criteria                                                                                      | Result | Time   |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| When QuantLib IS available, book must export its public API.       | <code>isinstance(THAMES_CENTRAL_AREAS, list); len(THAMES_CENTRAL_AREAS) &gt; 0</code>                    | Pass   | 0.000s |
| port.src.book itself can be imported without QuantLib because      | <code>hasattr(mod, "THAMES_CENTRAL_AREAS")</code>                                                        | Pass   | 0.007s |
| Thames Central book should cover exactly 10 gauge areas.           | <code>len(THAMES_CENTRAL_AREAS) == 10</code>                                                             | Pass   | 0.000s |
| models.hazard.pricing has 'import QuantLib as ql' at module level. | —                                                                                                        | Pass   | 0.002s |
| Family members shown                                               | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Has borrower header                                                | <code>any("Borrower" in t for t in texts)</code>                                                         | Pass   | 0.001s |
| Has table                                                          | <code>any(isinstance(e, Table) for e in result)</code>                                                   | Pass   | 0.001s |
| Income verification shown                                          | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Marital status shown                                               | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.001s |
| Nationality shown                                                  | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.001s |
| No borrower details returns early                                  | <code>any("No borrower" in t for t in texts)</code>                                                      | Pass   | 0.000s |
| No mortgage returns message                                        | <code>any("No mortgage" in t for t in texts)</code>                                                      | Pass   | 0.000s |
| Residency status shown                                             | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.001s |
| Secondary income shown                                             | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.001s |
| With mortgage returns list                                         | <code>isinstance(result, list) and len(result) &gt; 0</code>                                             | Pass   | 0.001s |
| All none returns empty dicts                                       | <code>result["mortgage_lookup"] == {};</code><br><code>result["gauge_flood_info"] == {} (+1 more)</code> | Pass   | 0.000s |
| All none returns four keys                                         | <code>set(result.keys()) == {"mortgage_lookup", "g...</code>                                             | Pass   | 0.000s |
| With data populates lookups                                        | <code>"P1" in result["mortgage_lookup"]</code>                                                           | Pass   | 0.000s |
| No trades returns message                                          | <code>any("No open PRS trades" in t for t in texts)</code>                                               | Pass   | 0.000s |
| Receiver trade direction                                           | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Trigger capitalized                                                | <code>any(isinstance(e, Table) for e in result)</code>                                                   | Pass   | 0.000s |
| With trades returns table                                          | <code>len(tables) &gt;= 1</code>                                                                         | Pass   | 0.000s |
| With trades shows count                                            | <code>any("Total trades: 2" in t for t in texts)</code>                                                  | Pass   | 0.000s |
| Line 186: when CDM validation returns errors, a warning is logged. | <code>record is not None; "PhysicalSwap" in record (+1 more)</code>                                      | Pass   | 0.003s |
| Diagonal is one                                                    | —                                                                                                        | Pass   | 0.000s |

| Test Description                                                          | Acceptance Criteria                                                                                            | Result | Time   |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------|--------|
| Off diagonal between zero and one                                         | <code>np.all(off_diag &gt;= 0) and<br/>np.all(off_diag &lt;= 1)</code>                                         | Pass   | 0.000s |
| Shape is n by n                                                           | <code>mat.shape == (n, n)</code>                                                                               | Pass   | 0.000s |
| Symmetric                                                                 | —                                                                                                              | Pass   | 0.000s |
| Diagonal is one                                                           | —                                                                                                              | Pass   | 0.000s |
| Higher base correlation increases off diagonal                            | <code>np.all(off_high &gt; off_low)</code>                                                                     | Pass   | 0.000s |
| Larger correlation_distance → slower exponential decay → higher off-diag. | <code>np.all(off_far &gt;= off_near)</code>                                                                    | Pass   | 0.000s |
| Off diagonal positive                                                     | <code>np.all(off &gt; 0)</code>                                                                                | Pass   | 0.000s |
| Returns square matrix                                                     | <code>mat.shape == (3, 3)</code>                                                                               | Pass   | 0.000s |
| Symmetric                                                                 | —                                                                                                              | Pass   | 0.000s |
| When flooded, arrival_time and peak_time are set.                         | <code>result["arrival_time_hrs"] is not<br/>None; result["peak_time_hrs"] is not<br/>None</code>               | Pass   | 0.001s |
| High WSE above property elevation -i flooded=True.                        | <code>result["flooded"] is True;<br/>result["flood_depth_m"] &gt; 0</code>                                     | Pass   | 0.001s |
| Gaugets with empty readings -i no readings key (not flooded).             | <code>"readings" not in result</code>                                                                          | Pass   | 0.001s |
| Low WSE -i flooded=False.                                                 | <code>result["flooded"] is False</code>                                                                        | Pass   | 0.001s |
| _build_flood_event returns dict with required keys.                       | <code>key in result; "readings" in result</code>                                                               | Pass   | 0.001s |
| Distance=0 -i travel_time=0 path.                                         | <code>result["travel_time_hrs"] == 0.0</code>                                                                  | Pass   | 0.001s |
| Both none returns empty                                                   | <code>build_gauge_flood_info(None, None) ==<br/>{}</code>                                                      | Pass   | 0.000s |
| Gauge data supplements missing fields                                     | <code>"GAUGE-002" in result;<br/>result["GAUGE-002"]["gauge_name"]<br/>== "My Gauge" (+2 more)</code>          | Pass   | 0.000s |
| Gauge without id skipped                                                  | <code>result == {}</code>                                                                                      | Pass   | 0.000s |
| Hazard curves only                                                        | <code>"GAUGE-001" in result;<br/>result["GAUGE-001"]["gauge_name"]<br/>== "Thames at Chelsea" (+2 more)</code> | Pass   | 0.001s |
| Hazard data takes precedence                                              | <code>result["GAUGE-003"]["gauge_name"] ==<br/>"From HC"</code>                                                | Pass   | 0.000s |
| Line 247: FileNotFoundError when gauge.json missing.                      | —                                                                                                              | Pass   | 0.001s |
| Line 235: FileNotFoundError when storm_sequences.json missing.            | —                                                                                                              | Pass   | 0.000s |
| Verbose=False should skip log messages.                                   | <code>result['summary']['num_gauges'] == 1;<br/>len(info_msgs) == 0</code>                                     | Pass   | 0.023s |

| Test Description                                      | Acceptance Criteria                                                                                                   | Result | Time   |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Lines 242, 254, 273, 280-281, 288: verbose logging.   | <code>result['summary']['num_gauges'] == 1;</code><br><code>result['summary']['num_storms'] == 20</code><br>(+3 more) | Pass   | 0.025s |
| No market state returns message                       | <code>any("No market hazard term structure" in t for t in texts)</code>                                               | Pass   | 0.000s |
| Some tenor slots missing → show 'N/A'.                | <code>any(isinstance(e, Table) for e in result)</code>                                                                | Pass   | 0.000s |
| With adjustments returns extra table                  | <code>len(tables) &gt;= 2</code>                                                                                      | Pass   | 0.000s |
| With hts returns table                                | <code>len(tables) &gt;= 1</code>                                                                                      | Pass   | 0.000s |
| All sections                                          | <code>len(content) &gt; 1000</code>                                                                                   | Pass   | 0.003s |
| Buffer not empty                                      | <code>len(content) &gt; 100</code>                                                                                    | Pass   | 0.003s |
| Meeting with actions                                  | <code>isinstance(result, io.BytesIO)</code>                                                                           | Pass   | 0.001s |
| Meeting with agenda                                   | <code>isinstance(result, io.BytesIO)</code>                                                                           | Pass   | 0.001s |
| Meeting with decisions                                | <code>isinstance(result, io.BytesIO)</code>                                                                           | Pass   | 0.001s |
| Meeting with minutes                                  | <code>isinstance(result, io.BytesIO)</code>                                                                           | Pass   | 0.001s |
| Meeting with no date/time/location/chair still works. | <code>isinstance(result, io.BytesIO)</code>                                                                           | Pass   | 0.001s |
| Meeting with participants                             | <code>isinstance(result, io.BytesIO)</code>                                                                           | Pass   | 0.003s |
| Minimal meeting                                       | <code>isinstance(result, io.BytesIO)</code>                                                                           | Pass   | 0.001s |
| Returns bytesio                                       | <code>isinstance(result, io.BytesIO)</code>                                                                           | Pass   | 0.003s |
| Empty dict returns empty                              | <code>build_mortgage_lookup({}) == {}</code>                                                                          | Pass   | 0.000s |
| Loan from loan details fallback                       | <code>result["PROP-002"]["OriginalLoan"] == 300.000</code>                                                            | Pass   | 0.000s |
| Mortgage without property id skipped                  | <code>build_mortgage_lookup(data) == {}</code>                                                                        | Pass   | 0.000s |
| Multiple mortgages                                    | <code>set(result.keys()) == {"P1", "P2"}</code>                                                                       | Pass   | 0.000s |
| No items key returns empty                            | <code>build_mortgage_lookup({"other": []}) == {}</code>                                                               | Pass   | 0.000s |
| None returns empty                                    | <code>build_mortgage_lookup(None) == {}</code>                                                                        | Pass   | 0.000s |
| Single mortgage                                       | <code>"PROP-001" in result;</code><br><code>result["PROP-001"]["mortgage_id"] == "MORT-001" (+3 more)</code>          | Pass   | 0.000s |
| Basic risk info                                       | <code>"M1" in result["by_mortgage_id"]; "P1" in result["by_property_id"] (+3 more)</code>                             | Pass   | 0.000s |
| Missing mortgage or property id skipped               | <code>result == {"by_mortgage_id": {}, "by_property_id": {}}</code>                                                   | Pass   | 0.000s |
| No property flood info uses unknown                   | <code>result["by_mortgage_id"]["M2"]["flood_risk_level"] == "Un..."</code>                                            | Pass   | 0.000s |
| None mortgage returns empty structure                 | <code>result == {"by_mortgage_id": {}, "by_property_id": {}}</code>                                                   | Pass   | 0.000s |

| Test Description                                                  | Acceptance Criteria                                                                                 | Result | Time   |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Value at risk calculation                                         | result["by.mortgage.id"]["M4"]["mortgage.value.at.risk"]<br>...                                     | Pass   | 0.000s |
| 60 i= percentile i 80 -i 'Elevated'.                              | any('Elevated' in t for t in texts)                                                                 | Pass   | 0.106s |
| FloodAlert and FloodWarning in metadata -i no crash.              | len(elements) > 0                                                                                   | Pass   | 0.113s |
| 80 i= percentile i 95 -i 'High'.                                  | any('High' in t for t in texts)                                                                     | Pass   | 0.105s |
| percentile i 40 -i 'Low'.                                         | any('Low' in t for t in texts)                                                                      | Pass   | 0.106s |
| 40 i= percentile i 60 -i 'Normal'.                                | any('Normal' in t for t in texts)                                                                   | Pass   | 0.106s |
| Single same_month_levels entry -i stdev guard -i no crash.        | len(elements) > 0                                                                                   | Pass   | 0.110s |
| Stats table and distribution statistics header present.           | any('Distribution Statistics' in t for t in texts)                                                  | Pass   | 0.132s |
| percentile i= 95 -i 'Very High'.                                  | any('Very High' in t for t in texts)                                                                | Pass   | 0.106s |
| Returns position function                                         | 'getPositionStyles' in css;<br>'top-right' in css (+1 more)                                         | Pass   | 0.000s |
| All none returns empty                                            | build_property_flood_info(None, None, None) == {}                                                   | Pass   | 0.000s |
| From property data when not in phc                                | "PROP-002" in result;<br>result["PROP-002"]["flood_zone"]<br>== "Zone 2" (+1 more)                  | Pass   | 0.000s |
| From property hazard data                                         | "PROP-001" in result;<br>result["PROP-001"]["annual_flood_prob"]<br>== pytest.approx(...) (+2 more) | Pass   | 0.000s |
| Phc takes precedence over property data                           | result["PROP-003"]["annual_flood_prob"]<br>== pytest.approx(...)                                    | Pass   | 0.000s |
| alert i= peak i warning -i 'Alert'.                               | len(elements) > 1                                                                                   | Pass   | 0.000s |
| ValueError/TypeError in date parsing (lines 179-180) -i continue. | len(elements) > 0                                                                                   | Pass   | 0.000s |
| Days within 3 days grouped; higher level updates peak.            | not any('No flood events' in t for t in texts)                                                      | Pass   | 0.000s |
| Consecutive days -i single event with range format.               | len(elements) > 1                                                                                   | Pass   | 0.000s |
| No days exceeding alert -i 'No flood events recorded'.            | any('No flood events' in t for t in texts)                                                          | Pass   | 0.000s |
| Peak i= severe.level -i 'Severe' row in table.                    | len(elements) > 1                                                                                   | Pass   | 0.000s |
| Single-day flood: date_str = start.strftime(...) only.            | len(elements) > 1                                                                                   | Pass   | 0.000s |
| Days i 3 days apart -i two separate flood events.                 | len(elements) > 1                                                                                   | Pass   | 0.000s |



| Test Description                                                 | Acceptance Criteria                                                 | Result | Time   |
|------------------------------------------------------------------|---------------------------------------------------------------------|--------|--------|
| warning j= peak j severe -j 'Warning'.                           | len(elements) > 1                                                   | Pass   | 0.000s |
| Empty schema returns empty dict                                  | make_generator(tmp_path)..build_section({}, 0, {}) == {}            | Pass   | 0.000s |
| Nested dict without type recurses                                | "SubSection" in result;<br>isinstance(result["SubSection"], dict)   | Pass   | 0.000s |
| Non dict schema returns empty dict                               | make_generator(tmp_path)..build_section("not a dict", 0, ...)       | Pass   | 0.000s |
| None schema returns empty dict                                   | make_generator(tmp_path)..build_section(None, 0, {}) == {}          | Pass   | 0.000s |
| None value from generate field value excluded                    | "AField" not in result                                              | Pass   | 0.000s |
| Reserved key description skipped                                 | "description" not in<br>make_generator(tmp_path)..build.sect...     | Pass   | 0.000s |
| Reserved key options skipped                                     | "options" not in<br>make_generator(tmp_path)..build.section(...     | Pass   | 0.000s |
| Reserved key type skipped                                        | "type" not in result                                                | Pass   | 0.000s |
| Reserved key values skipped                                      | "values" not in<br>make_generator(tmp_path)..build.section({...     | Pass   | 0.000s |
| Empty dataframe returns none                                     | build_spatial_index(df) is None                                     | Pass   | 0.000s |
| None gauge data returns none                                     | build_spatial_index(None) is None                                   | Pass   | 0.000s |
| With gauges returns kdtree                                       | isinstance(result, cKDTree)                                         | Pass   | 0.000s |
| Lines 43-45: coordinates are converted to radians.               | tree.data[0][1] ==<br>pytest.approx(np.pi, rel=1e-6)                | Pass   | 0.000s |
| Single gauge should still produce a valid KDTree.                | isinstance(tree, cKDTree); tree.n == 1                              | Pass   | 0.000s |
| Lines 42-46: non-empty DataFrame builds a cKDTree.               | isinstance(tree, cKDTree); tree.n == 3                              | Pass   | 0.000s |
| Counts default to zero                                           | s["alert_sequences"] == 0;<br>s["warning_sequences"] == 0 (+1 more) | Pass   | 0.000s |
| Num gauges                                                       | s["num_gauges"] == 4                                                | Pass   | 0.000s |
| Num sequences                                                    | s["num_sequences"] == 7                                             | Pass   | 0.001s |
| Returns required keys                                            | key in s                                                            | Pass   | 0.001s |
| Contains all types                                               | t.key in parsed; 'icon' in<br>parsed[t.key] (+2 more)               | Pass   | 0.000s |
| Info template values                                             | parsed['info']['color'] == '#2196F3'                                | Pass   | 0.000s |
| Returns valid json                                               | isinstance(parsed, dict)                                            | Pass   | 0.000s |
| Line 88: exceedance probs between 0 and 0.01 are capped to 0.01. | pt.annual.exceedance_prob >= 0.01 or<br>pt.annual.exceedance...     | Pass   | 0.030s |

| Test Description                                                  | Acceptance Criteria                                                     | Result | Time   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|
| Line 61: log() does nothing when verbose=False.                   | "should not appear" not in caplog.text                                  | Pass   | 0.000s |
| Line 61: log() calls logger.info when verbose=True.               | "test message" in caplog.text                                           | Pass   | 0.003s |
| Line 152: progress logged every 10 gauges.                        | len(curves) == 11; "10/11" in caplog.text or "Processed" in caplog.text | Pass   | 0.226s |
| All days before start                                             | d < today                                                               | Pass   | 0.000s |
| Chronological order                                               | days[i] > days[i - 1]                                                   | Pass   | 0.000s |
| Ends before start                                                 | d < start                                                               | Pass   | 0.000s |
| Large count                                                       | len(days) == 252; d.weekday() < 5                                       | Pass   | 0.000s |
| Last element should be the business day immediately before start. | last < start; last.weekday() < 5                                        | Pass   | 0.000s |
| No weekends                                                       | d.weekday() < 5                                                         | Pass   | 0.000s |
| Returns ascending order                                           | result == sorted(result)                                                | Pass   | 0.000s |
| Returns business days only                                        | d.weekday() < 5                                                         | Pass   | 0.000s |
| Returns correct count                                             | len(days) == 63                                                         | Pass   | 0.000s |
| Returns correct count                                             | len(result) == 5                                                        | Pass   | 0.000s |
| Single day                                                        | len(days) == 1; days[0].weekday() < 5                                   | Pass   | 0.000s |
| Single day                                                        | len(result) == 1; result[0].weekday() < 5                               | Pass   | 0.000s |
| Skips weekends                                                    | wd not in returned_set                                                  | Pass   | 0.000s |
| The day before Monday is Sunday — should skip back to Friday.     | days[0].weekday() == 4                                                  | Pass   | 0.000s |
| 5 days back from Wednesday should contain no Saturday/Sunday.     | len(days) == 5; d.weekday() < 5                                         | Pass   | 0.000s |
| Zero days                                                         | result == []                                                            | Pass   | 0.000s |
| Test CORS headers on GET request.                                 | response.status_code == 200                                             | Pass   | 0.000s |
| Test OPTIONS requests return 200.                                 | response.status_code == 200                                             | Pass   | 0.001s |
| Larger area gives larger energy                                   | sum(large) > sum(small)                                                 | Pass   | 0.000s |
| Returns positive value                                            | result > 0; result > 0 (+2 more)                                        | Pass   | 0.000s |
| Uses default area when missing                                    | result > 0; result > 0                                                  | Pass   | 0.000s |
| Is float                                                          | isinstance(result, float);<br>isinstance(result, float)                 | Pass   | 0.000s |
| Larger area tends to give larger emissions                        | sum(large) > sum(small)                                                 | Pass   | 0.000s |
| Returns positive value                                            | result > 0; result > 0 (+2 more)                                        | Pass   | 0.000s |
| Uses default area when missing                                    | result > 0; result > 0                                                  | Pass   | 0.000s |
| Correct calculation                                               | abs(result - 2430.0) < 0.01                                             | Pass   | 0.000s |

| Test Description                                                                 | Acceptance Criteria                                                                       | Result | Time   |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Is float                                                                         | <code>isinstance(result, float);</code><br><code>isinstance(result, float)</code>         | Pass   | 0.000s |
| Returns positive value                                                           | <code>result &gt; 0; result &gt; 0 (+2 more)</code>                                       | Pass   | 0.000s |
| Uses defaults when missing                                                       | <code>result &gt;= 0; abs(result - 2430.0) &lt;</code><br><code>0.01</code>               | Pass   | 0.000s |
| Zero usage returns zero                                                          | <code>result == 0.0</code>                                                                | Pass   | 0.000s |
| Above severe level some positive                                                 | <code>np.any(depths &gt; 0)</code>                                                        | Pass   | 0.001s |
| Below severe level returns zeros                                                 | —                                                                                         | Pass   | 0.000s |
| Depths capped at 5m                                                              | <code>np.all(depths &lt;= 5.0)</code>                                                     | Pass   | 0.000s |
| Empty gauge data returns zeros                                                   | —                                                                                         | Pass   | 0.000s |
| No gauge data returns zeros                                                      | —                                                                                         | Pass   | 0.000s |
| Returns correct length                                                           | <code>len(depths) == len(properties)</code>                                               | Pass   | 0.000s |
| Lines 50-57: non-None<br>gauge_data calls calcu-<br>late_synthetic_flood_depths. | <code>len(depths) == 3; isinstance(depths,</code><br><code>np.ndarray)</code>             | Pass   | 0.000s |
| Key remapping: 'flow_cumecs' →<br>'flow'.                                        | <code>"mean_flow" in result;</code><br><code>"mean_flow_cumecs" not in result</code>      | Pass   | 0.000s |
| Flow stats should include q5_flow<br>and q95_flow.                               | <code>"q5_flow" in result; "q95_flow" in</code><br><code>result</code>                    | Pass   | 0.000s |
| Returns dict                                                                     | <code>isinstance(result, dict);</code><br><code>isinstance(result, dict) (+1 more)</code> | Pass   | 0.000s |
| Less than remaining energy                                                       | <code>result &lt;= 9001</code>                                                            | Pass   | 0.000s |
| No negative when electricity exceeds<br>total                                    | <code>result == 0</code>                                                                  | Pass   | 0.000s |
| Returns non negative                                                             | <code>result &gt;= 0</code>                                                               | Pass   | 0.000s |
| Uses defaults when missing                                                       | <code>result &gt;= 0; abs(result - 2430.0) &lt;</code><br><code>0.01</code>               | Pass   | 0.000s |
| Fraction of total                                                                | <code>all(4500 &lt;= r &lt;= 12001 for r in</code><br><code>results)</code>               | Pass   | 0.000s |
| Returns positive value                                                           | <code>result &gt; 0; result &gt; 0 (+2 more)</code>                                       | Pass   | 0.000s |
| Uses default energy when missing                                                 | <code>result &gt; 0</code>                                                                | Pass   | 0.000s |
| Key remapping: 'level_meters' →<br>'level'.                                      | <code>"mean_level" in result;</code><br><code>"mean_level_meters" not in result</code>    | Pass   | 0.001s |
| Has min max                                                                      | <code>"min_level" in result; "max_level" in</code><br><code>result</code>                 | Pass   | 0.001s |
| Returns dict                                                                     | <code>isinstance(result, dict);</code><br><code>isinstance(result, dict) (+1 more)</code> | Pass   | 0.001s |
| Buy to let type                                                                  | <code>result["mortgage_type"] ==</code><br><code>"Buy-to-Let"</code>                      | Pass   | 0.000s |
| High flood risk reduces ltv                                                      | <code>high_risk["ltv_ratio"] &lt;=</code><br><code>normal["ltv_ratio"] + 0.01</code>      | Pass   | 0.000s |

| Test Description                     | Acceptance Criteria                                                                                | Result | Time   |
|--------------------------------------|----------------------------------------------------------------------------------------------------|--------|--------|
| Loan amount positive                 | <code>result["loan.amount"] &gt; 0</code>                                                          | Pass   | 0.000s |
| Monthly payment positive             | <code>result["monthly_payment"] &gt; 0</code>                                                      | Pass   | 0.000s |
| Required keys present                | <code>key in result</code>                                                                         | Pass   | 0.000s |
| Returns dict                         | <code>isinstance(result, dict)</code>                                                              | Pass   | 0.000s |
| Second home type                     | <code>result["mortgage_type"] == "Second Home"</code>                                              | Pass   | 0.000s |
| Conditional rating middle range      | <code>result["calculated_rating"] in ("Conditional", "Unaccepta...</code>                          | Pass   | 0.000s |
| Due soon gives half score            | <code>result["component_scores"]["review_currency"] == 0.5</code>                                  | Pass   | 0.000s |
| Far review gives full score          | <code>result["component_scores"]["review_currency"] == 1.0</code>                                  | Pass   | 0.000s |
| High assumptions penalised           | <code>result["component_scores"]["assumption_risk"] == 0.0</code>                                  | Pass   | 0.000s |
| High coverage acceptable             | <code>result["calculated_rating"] == "Acceptable"</code>                                           | Pass   | 0.000s |
| High limitations penalised           | <code>result["component_scores"]["limitation_risk"] == 0.0</code>                                  | Pass   | 0.000s |
| No coverage unacceptable             | <code>result["calculated_rating"] == "Unacceptable"</code>                                         | Pass   | 0.000s |
| No review date gives zero            | <code>result["component_scores"]["review_currency"] == 0.0</code>                                  | Pass   | 0.000s |
| Overdue review gives zero score      | <code>result["component_scores"]["review_currency"] == 0.0</code>                                  | Pass   | 0.000s |
| Returns dict with required keys      | <code>"calculated_rating" in result;</code><br><code>"calculated_score" in result</code> (+1 more) | Pass   | 0.000s |
| Score is float                       | <code>isinstance(result["calculated_score"], float)</code>                                         | Pass   | 0.000s |
| Heat pump returns zero               | <code>result == 0</code>                                                                           | Pass   | 0.000s |
| Multiple returns positive            | <code>result &gt; 0</code>                                                                         | Pass   | 0.000s |
| No solar returns zero                | <code>result == 0</code>                                                                           | Pass   | 0.000s |
| No system returns zero               | <code>result == 0</code>                                                                           | Pass   | 0.000s |
| Solar pv returns positive            | <code>result &gt; 0</code>                                                                         | Pass   | 0.000s |
| Solar uses default area when missing | <code>result &gt; 0</code>                                                                         | Pass   | 0.000s |
| Annual maxima present                | <code>"annual_maxima" in stats;</code><br><code>stats["annual_maxima"]["years_count"] == 2</code>  | Pass   | 0.000s |
| Basic stats keys                     | <code>key in stats</code>                                                                          | Pass   | 0.000s |
| Empty returns empty dict             | <code>GaugeHistoricalDaily().calculate_statistics()</code><br><code>== {}</code>                   | Pass   | 0.000s |

| Test Description                       | Acceptance Criteria                                                                         | Result | Time   |
|----------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Extreme events above p99               | "extreme_events" in stats;<br>stats["extreme_events"]["count"]<br>>= 1                      | Pass   | 0.000s |
| Mean correct                           | gen.calculate_statistics(flows)["mean_flow"]<br>== pytest.ap...                             | Pass   | 0.000s |
| Min max correct                        | stats["min_flow"] ==<br>pytest.approx(5.0); stats["max_flow"]<br>== pytest.approx(100.0)    | Pass   | 0.000s |
| Monthly means present                  | "01" in stats["monthly_means"]; "02"<br>in stats["monthly_means"]                           | Pass   | 0.000s |
| Percentiles present                    | "percentiles" in stats; p in<br>stats["percentiles"]                                        | Pass   | 0.001s |
| Single flow no std dev error           | gen.calculate_statistics(flows)["std_dev"]<br>== 0                                          | Pass   | 0.001s |
| All percentile keys present            | p in stats["percentiles"]                                                                   | Pass   | 0.000s |
| Annual maxima events capped at ten     | len(stats["annual_maxima"]["events"])<br><= 10                                              | Pass   | 0.000s |
| Extreme events top 10 capped           | len(stats["extreme_events"]["top_10"])<br><= 10                                             | Pass   | 0.000s |
| Max flow date correctly identified     | stats["max_flow_date"] ==<br>"2020-01-02"; stats["max_flow"]<br>== pytest.approx(200.0)     | Pass   | 0.000s |
| Q5 flow equals p95                     | stats["q5_flow"] ==<br>stats["percentiles"]["p95"]                                          | Pass   | 0.000s |
| Q95 flow equals p5                     | stats["q95_flow"] ==<br>stats["percentiles"]["p5"]                                          | Pass   | 0.000s |
| Record start and end match data        | stats["record_start"] ==<br>"2018-05-01"; stats["record_end"]<br>== "2021-08-15"            | Pass   | 0.000s |
| Single year annual maxima std dev zero | stats["annual_maxima"]["years_count"]<br>== 1;<br>stats["annual_maxima"]["std_dev"]<br>== 0 | Pass   | 0.000s |
| Total years approximate                | stats["total_days"] == 20;<br>stats["total_years"] == round(20<br>/ 365.25, 1)              | Pass   | 0.000s |
| Two flow std dev non zero              | gen.calculate_statistics(flows)["std_dev"]<br>> 0                                           | Pass   | 0.000s |
| All same values                        | result["mean_level_meters"] == 3.0;<br>result["std_dev"] == 0.0                             | Pass   | 0.000s |

| Test Description                                 | Acceptance Criteria                                                                                                        | Result | Time   |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Empty returns empty dict                         | <code>calculate.timeseries_statistics([], "level_meters") == {}</code>                                                     | Pass   | 0.000s |
| Flood exceedances zero when all below            | <code>result["flood_exceedances"]["flood_alert"] == 0</code>                                                               | Pass   | 0.000s |
| Flood stages adds exceedances                    | <code>"flood_exceedances" in result;</code><br><code>result["flood_exceedances"]["severe_warning"]["count"] &gt;= 1</code> | Pass   | 0.001s |
| Flow cumecs adds q flows                         | <code>"q95_flow" in result;</code> <code>"q5_flow" in result</code>                                                        | Pass   | 0.000s |
| Has annual maxima                                | <code>"annual_maxima" in result;</code> <code>"years_count" in result["annual_maxima"]</code>                              | Pass   | 0.001s |
| Has basic keys                                   | <code>"total_days" in result;</code> <code>"mean_level_meters" in result</code> (+2 more)                                  | Pass   | 0.001s |
| Has extreme events                               | <code>"extreme_events" in result</code>                                                                                    | Pass   | 0.001s |
| Has monthly means                                | <code>"monthly_means" in result</code>                                                                                     | Pass   | 0.001s |
| Has percentiles                                  | <code>"percentiles" in result;</code> <code>"p5" in result["percentiles"]</code> (+1 more)                                 | Pass   | 0.001s |
| Max date correct                                 | <code>result["max_level_meters.date"] == "2026-01-02"</code>                                                               | Pass   | 0.000s |
| No flood stages no exceedances key               | <code>"flood_exceedances" not in result</code>                                                                             | Pass   | 0.001s |
| Record start end                                 | <code>result["record_start"] == "2020-01-01";</code> <code>result["record_end"] == "2025-12-31"</code>                     | Pass   | 0.000s |
| Returns dict                                     | <code>isinstance(result, dict);</code> <code>isinstance(result, dict)</code> (+1 more)                                     | Pass   | 0.001s |
| Single observation no stdev error                | <code>result["std_dev"] == 0</code>                                                                                        | Pass   | 0.000s |
| Total days correct                               | <code>result["total_days"] == 100</code>                                                                                   | Pass   | 0.000s |
| Explicit none params                             | <code>isinstance(level, float)</code>                                                                                      | Pass   | 0.000s |
| None params uses defaults                        | <code>isinstance(level, float); level &gt; 0</code>                                                                        | Pass   | 0.000s |
| Uses addEventListener, not onclick for sub-tabs. | <code>'addEventListener' in stress_js;</code> <code>'stress-ctab-0' in stress_js</code> (+2 more)                          | Pass   | 0.000s |
| Flood Probability sub-tab present.               | <code>'Flood Probability' in stress_js</code>                                                                              | Pass   | 0.000s |
| Stress P&L sub-tab present.                      | <code>'Stress P' in stress_js</code>                                                                                       | Pass   | 0.000s |
| Surface sub-tab present.                         | <code>'Surface' in stress_js</code>                                                                                        | Pass   | 0.000s |
| Nearby ranges round to same cache key.           | <code>L1 is L2</code>                                                                                                      | Pass   | 0.000s |
| Second call at same range returns cached object. | <code>L1 is L2</code>                                                                                                      | Pass   | 0.000s |



| Test Description                                                                                             | Acceptance Criteria                                                          | Result | Time   |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| Cholesky lower triangular                                                                                    | <code>np.allclose(np.triu(L, k=1), 0.0)</code>                               | Pass   | 0.001s |
| <code>L @ L.T == C</code> within numerical tolerance.                                                        | <code>np.allclose(L @ L.T, C, atol=1e-10)</code>                             | Pass   | 0.002s |
| Thames cholesky reconstruction                                                                               | <code>np.allclose(L @ L.T, C, atol=1e-8)</code>                              | Pass   | 0.014s |
| When <code>np.linalg.cholesky</code> raises <code>LinAlgError</code> on first call,                          | <code>L.shape == (model.n_gauges, model.n_gauges); call_count[0] == 2</code> | Pass   | 0.001s |
| Generator error has message                                                                                  | <code>r.get_json().get("message")</code>                                     | Pass   | 0.005s |
| Generator error returns 500                                                                                  | <code>r.status_code == 500</code>                                            | Pass   | 0.021s |
| Generator error status is error                                                                              | <code>r.get_json()["status"] == "error"</code>                               | Pass   | 0.005s |
| Claim ref is built from last 8 chars of <code>property_id</code> .                                           | <code>len(result) &gt; 1000</code>                                           | Pass   | 0.034s |
| Generate empty sequence lookup                                                                               | <code>isinstance(result, bytes)</code>                                       | Pass   | 0.031s |
| Generate returns bytes                                                                                       | <code>isinstance(result, bytes); len(result) &gt; 0</code>                   | Pass   | 0.024s |
| Generate without mortgage                                                                                    | <code>isinstance(result, bytes); result[:4] == b'%PDF'</code>                | Pass   | 0.019s |
| Minimal <code>prop_data</code> with a single flood event should not raise.                                   | <code>isinstance(result, bytes)</code>                                       | Pass   | 0.016s |
| Output is valid pdf                                                                                          | <code>result[:4] == b'%PDF'</code>                                           | Pass   | 0.025s |
| Pdf size reasonable                                                                                          | <code>len(result) &gt; 8.000</code>                                          | Pass   | 0.028s |
| Content length header set                                                                                    | <code>r.headers.get("Content-Length") == str(len(fake_pdf))</code>           | Pass   | 0.003s |
| Filename contains prop id                                                                                    | <code>PROP_ID in cd</code>                                                   | Pass   | 0.005s |
| Response body is pdf bytes                                                                                   | <code>r.data == fake_pdf</code>                                              | Pass   | 0.004s |
| Response has content disposition                                                                             | <code>"attachment" in cd; PROP_ID in cd</code>                               | Pass   | 0.007s |
| Response is pdf content type                                                                                 | <code>r.content_type == "application/pdf"</code>                             | Pass   | 0.005s |
| Success returns 200                                                                                          | <code>r.status_code == 200</code>                                            | Pass   | 0.016s |
| Corrupt <code>storm_sequences.json</code> triggers the except in <code>load_sequence_lookup</code> (L56-57). | <code>r.status_code == 200; seq_lookup == {}</code>                          | Pass   | 0.037s |
| <code>load_mortgage_record</code> exception (L89-91) returns <code>None</code> ; report still generated.     | <code>r.status_code == 200; mort_record is None</code>                       | Pass   | 0.002s |
| <code>load_prop_record</code> exception (L71-73) returns <code>{}</code> ; report still generated.           | <code>r.status_code == 200; prop_record == {}</code>                         | Pass   | 0.002s |
| <code>load_sequence_lookup</code> exception (L56-57) returns empty dict; report still generated.             | <code>r.status_code == 200</code>                                            | Pass   | 0.002s |

| Test Description                                           | Acceptance Criteria                                                                                   | Result | Time   |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| No dir message mentions propertyts                         | "propertyts" in msg or "property" in msg or "not found" i...                                          | Pass   | 0.001s |
| No dir returns 404                                         | r.status_code == 404                                                                                  | Pass   | 0.001s |
| No dir status error                                        | r.get_json()["status"] == "error"                                                                     | Pass   | 0.001s |
| No floods message mentions events                          | "flood" in msg or "event" in msg                                                                      | Pass   | 0.002s |
| No floods returns 404                                      | r.status_code == 404                                                                                  | Pass   | 0.004s |
| No floods status error                                     | r.get_json()["status"] == "error"                                                                     | Pass   | 0.002s |
| Unknown property message contains id                       | "PROP-GHOST" in r.get_json()["message"]                                                               | Pass   | 0.001s |
| Unknown property returns 404                               | r.status_code == 404                                                                                  | Pass   | 0.001s |
| Unknown property status error                              | r.get_json()["status"] == "error"                                                                     | Pass   | 0.001s |
| Metadata categories contains expected keys                 | key in cats                                                                                           | Pass   | 0.000s |
| Metadata categories station has required subkeys           | key in station_keys                                                                                   | Pass   | 0.000s |
| Schema version is string                                   | isinstance(GaugeHistoricalDaily.SCHEMA_VERSION, str);<br>len(GaugeHistoricalDaily.SCHEMA_VERSION) > 0 | Pass   | 0.000s |
| Classifiers that exist should reference current gauge IDs. | len(valid) > 0                                                                                        | Pass   | 0.001s |
| At least one trained classifier must exist.                | len(classifiers) > 0                                                                                  | Pass   | 0.002s |
| training_summary.json must exist alongside classifiers.    | —                                                                                                     | Pass   | 0.001s |
| Above high threshold is high                               | _classify_property_risk(0.5) == "High"                                                                | Pass   | 0.000s |
| High threshold                                             | _classify_property_risk(0.03) == "High"                                                               | Pass   | 0.000s |
| Just below medium is low                                   | _classify_property_risk(0.009) == "Low"                                                               | Pass   | 0.000s |
| Medium threshold                                           | _classify_property_risk(0.01) == "Medium"                                                             | Pass   | 0.000s |
| Tiny prob is low                                           | _classify_property_risk(0.005) == "Low"                                                               | Pass   | 0.000s |
| Zero is negligible                                         | _classify_property_risk(0.0) == "Negligible"                                                          | Pass   | 0.000s |
| Closed trade should not appear in stress results.          | resp.status_code == 200; t['swap_id'] != 'PRS-CLOSED-001'                                             | Pass   | 0.052s |

| Test Description                                                     | Acceptance Criteria                                                                               | Result | Time   |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| post_damage.ltv is recomputed independently for each storm.          | ltvs["STORM-cl2"] ><br>ltvs["STORM-cl1"]; ltvs["STORM-cl2"]<br>> ltvs["STORM-cl3"]                | Pass   | 0.014s |
| Each cluster storm individually queryable                            | r.status_code == 200; data["status"]<br>== "success"                                              | Pass   | 0.013s |
| STORM-cl2 (damage_ratio=0.15) causes more damage than cl1 or cl3.    | damages["STORM-cl2"] ><br>damages["STORM-cl1"];<br>damages["STORM-cl2"] ><br>damages["STORM-cl3"] | Pass   | 0.005s |
| Querying a storm_id that didn't breach alert returns 404.            | r.status_code == 404                                                                              | Pass   | 0.003s |
| Summary max_depth_m reflects the worst storm in the cluster.         | result["summary"]["max_depth_m"] >=<br>0.0                                                        | Pass   | 0.001s |
| Cluster produces three flood events                                  | len(data["flood_events"]) == 3                                                                    | Pass   | 0.003s |
| Cluster three distinct storm ids in events                           | storm_ids == {"STORM-c001a",<br>"STORM-c001b", "STORM-c001c"}                                     | Pass   | 0.003s |
| The 'from port.src.book import ...' statement in cmd.port must be    | blotter_idx != -1; book_idx != -1 (+1<br>more)                                                    | Pass   | 0.001s |
| Importing app.commands.port must not trigger QuantLib.               | hasattr(mod, "cmd.port"); hasattr(mod,<br>"register_parser")                                      | Pass   | 0.002s |
| All sections together                                                | isinstance(result, list); len(tables)<br>>= 2                                                     | Pass   | 0.000s |
| Feeding bad data (non-dict regulatory) exercises the except block.   | isinstance(result, list)                                                                          | Pass   | 0.000s |
| mortgage_data without 'Mortgage' key → uses top-level.               | isinstance(result, list)                                                                          | Pass   | 0.000s |
| Commit button must call tdCommitMarket().                            | 'tdCommitMarket()' in js or<br>'tdCommitMarket(' in js                                            | Pass   | 0.000s |
| Market tab must contain a visible commit button with correct ID.     | 'td-commit-btn' in js                                                                             | Pass   | 0.000s |
| Commit button must NOT have display:none in its own style.           | 'display:none' not in own.style                                                                   | Pass   | 0.000s |
| tdCommitMarket function must be defined.                             | 'tdCommitMarket = function' in js or<br>\{\} 'fu...                                               | Pass   | 0.000s |
| Commit function must POST to hazard-term-structure/commit.           | '/trading/hazard-term-structure/commit'<br>in js                                                  | Pass   | 0.000s |
| After commit, must switch to blotter tab (which loads blotter data). | "switchTab('blotter')" in js                                                                      | Pass   | 0.000s |
| Close out path exercised                                             | r.status_code in (200, 400)                                                                       | Pass   | 0.037s |
| Commit with cashflows — exercises PDF cashflow schedule section.     | r.status_code in (200, 400)                                                                       | Pass   | 0.008s |

| Test Description                                                                                | Acceptance Criteria                                                                                      | Result | Time   |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Commit with maturity_date provided explicitly.                                                  | <code>r.status_code in (200, 400)</code>                                                                 | Pass   | 0.006s |
| Empty json returns 400                                                                          | <code>r.status_code == 400;</code><br><code>r.get_json()["status"] == "error"</code>                     | Pass   | 0.001s |
| Commit with enough fields to pass CDM validation.                                               | <code>r.status_code in (200, 400);</code><br><code>r.get_json()["status"] in ("success", "error")</code> | Pass   | 0.047s |
| No json returns error                                                                           | <code>r.status_code in (400, 500);</code><br><code>r.get_json()["status"] == "error"</code>              | Pass   | 0.015s |
| Validation fails returns 400                                                                    | <code>r.status_code == 400; data["status"] == "error" (+1 more)</code>                                   | Pass   | 0.026s |
| commit_yield_curve creates yield_curve from default when missing.                               | <code>updated['yield_curve']['1'] == 0.04;</code><br><code>updated['yield_curve']['2'] == 0.045</code>   | Pass   | 0.002s |
| Advised flag                                                                                    | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Business purpose false                                                                          | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Business purpose true                                                                           | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Common section returns elements                                                                 | <code>isinstance(result, list); len(result) &gt; 2</code>                                                | Pass   | 0.000s |
| Fca reference number                                                                            | <code>len(tables) &gt;= 1</code>                                                                         | Pass   | 0.000s |
| Multiple regulatory flags                                                                       | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Contains header                                                                                 | <code>"Scenario Comparison" in result</code>                                                             | Pass   | 0.380s |
| Contains scenario names                                                                         | <code>name in result</code>                                                                              | Pass   | 0.382s |
| None uses all scenarios                                                                         | <code>len(result) &gt; 50</code>                                                                         | Pass   | 0.397s |
| Returns string                                                                                  | <code>isinstance(result, str)</code>                                                                     | Pass   | 0.373s |
| Single scenario                                                                                 | <code>first_name in result</code>                                                                        | Pass   | 0.078s |
| Two scenarios                                                                                   | <code>name in result</code>                                                                              | Pass   | 0.156s |
| Shim class is same                                                                              | <code>New is Shim</code>                                                                                 | Pass   | 0.000s |
| Shim imports generator                                                                          | <code>ClaimReportGenerator is not None</code>                                                            | Pass   | 0.002s |
| Line 159: _create_summary calls get_entity_summary.                                             | <code>summary == {'id': 'Y', 'name': 'Ypsilon'}</code>                                                   | Pass   | 0.000s |
| Line 155: _extract_id calls get_entity_id.                                                      | <code>loader._extract_id(entity) == 'X'</code>                                                           | Pass   | 0.000s |
| Line 163: _get_container_keys returns CONTAINER_KEYS.                                           | <code>loader._get_container_keys() == ['items']</code>                                                   | Pass   | 0.000s |
| Line 167: get_file_status is alias for get_status.                                              | <code>status['entity_name'] == 'stub';</code><br><code>status['exists'] is True (+1 more)</code>         | Pass   | 0.000s |
| Persistent sequences (SHORT gaps) should show stronger compounding than doublets (MEDIUM gaps). | <code>persistent_excess &gt;= doublet_excess - 0.1</code>                                                | Pass   | 0.050s |

| Test Description                                                                    | Acceptance Criteria                                                                  | Result | Time   |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| During the gap, level should drop from Storm 1 peak.                                | <code>level.at.gap.end &lt; s1_peak</code>                                           | Pass   | 0.001s |
| For a short-gap persistent sequence, level does not return to baseline during gap.  | <code>level.at.gap.end &gt; default_gauge.base_level + 0.05</code>                   | Pass   | 0.001s |
| Spec requirement: Storm 2 peak $\geq$ Storm 1 peak in $\geq 60\%$ of doublets.      | <code>fraction &gt;= 0.60</code>                                                     | Pass   | 0.036s |
| dist $\geq 1.0$ - $\geq$ short-circuit to <code>_build_flood_event</code> directly. | <code>result is not None; result["storm_id"] == "STORM-001"</code>                   | Pass   | 0.001s |
| Multiple gauges in nearest - $\geq$ IDW path used.                                  | <code>result is not None</code>                                                      | Pass   | 0.001s |
| <code>gauge_responses</code> has no entry for nearest gauge - $\geq$ None.          | <code>result is None</code>                                                          | Pass   | 0.001s |
| Normal IDW path returns dict with <code>storm_id</code> .                           | <code>result is not None; result["storm_id"] == "STORM-001"</code>                   | Pass   | 0.001s |
| Line 126: annuity $j=0$ returns <code>annual_hazard_rate * 10000</code> .           | <code>result &gt; 0</code>                                                           | Pass   | 0.000s |
| Negative hazard returns zero                                                        | <code>result == 0.0</code>                                                           | Pass   | 0.000s |
| Verify <code>compute_prs_spread</code> uses <code>yield_curve</code> when provided. | <code>result &gt; 0</code>                                                           | Pass   | 0.000s |
| Zero hazard returns zero                                                            | <code>result == 0.0</code>                                                           | Pass   | 0.000s |
| Pre-wetting the catchment before the same storm raises the peak.                    | <code>np.max(wet) &gt; np.max(dry)</code>                                            | Pass   | 0.001s |
| Level should be declining in the later part of the drainage window.                 | <code>np.all(np.diff(tail) &lt;= 0.01)</code>                                        | Pass   | 0.004s |
| Extreme storm should produce higher peak than moderate.                             | <code>np.max(l_ext) &gt; np.max(l_mod)</code>                                        | Pass   | 0.001s |
| With identical first storm parameters, doublet produces $\geq$ isolated peak.       | <code>np.max(l_dbl) &gt;= np.max(l_iso)</code>                                       | Pass   | 0.001s |
| Severe storm should raise level above baseline.                                     | <code>np.max(levels) &gt; default_gauge.base_level + 0.5</code>                      | Pass   | 0.000s |
| River level must always be $\geq$ <code>base_level</code> .                         | <code>np.all(levels &gt;= default_gauge.base_level - 1e-6)</code>                    | Pass   | 0.005s |
| Spec constraint: no precipitation after hour 156.                                   | <code>np.all(p[156:] == pytest.approx(0.0))</code>                                   | Pass   | 0.001s |
| Output shape                                                                        | <code>p.shape == (EVENT_WINDOW_HOURS,); levels.shape == (EVENT_WINDOW_HOURS,)</code> | Pass   | 0.000s |
| Level at hour 167 should be below the peak level (recovering, not diverging).       | <code>final &lt;= peak + 0.01; final_excess &lt; 1.5</code>                          | Pass   | 0.006s |

| Test Description                                                      | Acceptance Criteria                                                                                                                                   | Result | Time   |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Peak above base                                                       | <code>peak &gt;= default_gauge.base_level</code>                                                                                                      | Pass   | 0.004s |
| Peak indices match storm indices                                      | <code>idx == storm.storm_index</code>                                                                                                                 | Pass   | 0.002s |
| Returns one peak per storm                                            | <code>len(peaks) == seq.num_storms</code>                                                                                                             | Pass   | 0.004s |
| Spatial correlation constants in config.port must be self-consistent. | <code>SPATIAL_CORR_ENABLED is True;</code><br><code>SPATIAL_CORR_MODEL_TYPE ==</code><br><code>"exponential" (+9 more)</code>                         | Pass   | 0.000s |
| Config structure                                                      | <code>sc["enabled"] is True;</code><br><code>sc["model.type"] == "exponential"</code><br><code>(+3 more)</code>                                       | Pass   | 0.002s |
| Round trip params                                                     | <code>loaded.base_range_km == 35.0;</code><br><code>loaded.nugget == 0.08</code>                                                                      | Pass   | 0.001s |
| Save creates file                                                     | <code>path.exists()</code>                                                                                                                            | Pass   | 0.000s |
| Configure both menus                                                  | <code>handler.property_menu == pm;</code><br><code>handler.gauge_menu == gm</code>                                                                    | Pass   | 0.000s |
| Configure gauge menu                                                  | <code>handler.gauge_menu == new_gm</code>                                                                                                             | Pass   | 0.000s |
| Configure max visible                                                 | <code>ns.max_visible == 10</code>                                                                                                                     | Pass   | 0.000s |
| Configure menus                                                       | <code>mgr.context_menus.property_menu ==</code><br><code>custom_prop</code>                                                                           | Pass   | 0.000s |
| Configure no args no change                                           | <code>handler.server_url ==</code><br><code>'http://test:1234'</code>                                                                                 | Pass   | 0.000s |
| Configure no args no change                                           | <code>handler.property_menu ==</code><br><code>DEFAULT_PROPERTY_MENU;</code><br><code>handler.gauge_menu ==</code><br><code>DEFAULT_GAUGE_MENU</code> | Pass   | 0.000s |
| Configure no args no change                                           | <code>ns.position ==</code><br><code>NotificationPosition.TOP_LEFT;</code><br><code>ns.timeout == 7000 (+1 more)</code>                               | Pass   | 0.000s |
| Configure notification position                                       | <code>mgr.notifications.position ==</code><br><code>NotificationPosition.BOTTOM...</code>                                                             | Pass   | 0.000s |
| Configure notification timeout                                        | <code>mgr.notifications.timeout == 2000</code>                                                                                                        | Pass   | 0.000s |
| Configure position                                                    | <code>ns.position ==</code><br><code>NotificationPosition.BOTTOM_LEFT</code>                                                                          | Pass   | 0.000s |
| Configure property menu                                               | <code>handler.property_menu == new_pm</code>                                                                                                          | Pass   | 0.000s |
| Configure returns self                                                | <code>result is mgr</code>                                                                                                                            | Pass   | 0.000s |
| Configure server url                                                  | <code>result is mgr; mgr.backend.server_url</code><br><code>== 'http://new:9999'</code>                                                               | Pass   | 0.000s |
| Configure timeout                                                     | <code>ns.timeout == 3000</code>                                                                                                                       | Pass   | 0.000s |
| Configure updates endpoints                                           | <code>handler.endpoints == {'a': '/b',</code><br><code>'c': '/d'}</code>                                                                              | Pass   | 0.000s |



| Test Description                                                                         | Acceptance Criteria                                                                                   | Result | Time   |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| Configure updates server url                                                             | <code>handler.server_url == 'http://new:2222'; handler.endpoints == {'a': '/b'}</code>                | Pass   | 0.000s |
| Margin positive                                                                          | <code>MARGIN &gt; 0</code>                                                                            | Pass   | 0.000s |
| Page dimensions                                                                          | <code>PAGE.W &gt; 0; PAGE.H &gt; 0 (+1 more)</code>                                                   | Pass   | 0.000s |
| Construction year shown                                                                  | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Empty property does not crash                                                            | <code>isinstance(self._page().generate_elements(prop, list))</code>                                   | Pass   | 0.000s |
| Floor level shown                                                                        | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Has section header                                                                       | <code>any("Construction" in t for t in texts)</code>                                                  | Pass   | 0.001s |
| Has table                                                                                | <code>any(isinstance(e, Table) for e in result)</code>                                                | Pass   | 0.000s |
| No construction data returns message                                                     | <code>any("No construction" in t for t in texts)</code>                                               | Pass   | 0.000s |
| No construction year                                                                     | <code>isinstance(self._page().generate_elements(prop, list))</code>                                   | Pass   | 0.000s |
| Returns list with data                                                                   | <code>isinstance(result, list) and len(result) &gt; 0</code>                                          | Pass   | 0.001s |
| Site height shown                                                                        | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Brick construction                                                                       | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.001s |
| Concrete construction                                                                    | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.001s |
| Empty construction type                                                                  | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Masonry construction                                                                     | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Steel construction                                                                       | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Timber construction                                                                      | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Unknown construction type                                                                | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Custom menus                                                                             | <code>handler.property_menu == pm; handler.gauge_menu == gm</code>                                    | Pass   | 0.000s |
| Default init                                                                             | <code>handler.property_menu == DEFAULT_PROPERTY_MENU; handler.gauge_menu == DEFAULT_GAUGE_MENU</code> | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null. [var='_preGovAudit'] | <code>f'window.{var}' in src</code>                                                                   | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null. [var='_preGovBib']   | <code>f'window.{var}' in src</code>                                                                   | Pass   | 0.000s |

| Test Description                                                                                  | Acceptance Criteria                    | Result | Time   |
|---------------------------------------------------------------------------------------------------|----------------------------------------|--------|--------|
| Each startup cache variable must be initialised as window.X = null.<br>[var='_preStorms']         | f'window.{var}' in src                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null.<br>[var='_tdPreBlotter']      | f'window.{var}' in src                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null.<br>[var='_tdPreEodHistory']   | f'window.{var}' in src                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null.<br>[var='_tdPreGauges']       | f'window.{var}' in src                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null.<br>[var='_tdPreGovDocs']      | f'window.{var}' in src                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null.<br>[var='_tdPreMarket']       | f'window.{var}' in src                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null.<br>[var='_tdPrePortStorms']   | f'window.{var}' in src                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null.<br>[var='_tdPreStressStorms'] | f'window.{var}' in src                 | Pass   | 0.000s |
| Each startup cache variable must be initialised as window.X = null.<br>[var='_tdPreYieldCurve']   | f'window.{var}' in src                 | Pass   | 0.000s |
| No startup cache var should have a bare var declaration.                                          | f'var {var}' not in src                | Pass   | 0.000s |
| The preload-done flag must NOT be a var (regression: 2026-03-12 incident).                        | 'var _tdPreloadDone' not in src        | Pass   | 0.003s |
| Window td preload done initialised                                                                | 'window._tdPreloadDone = false' in src | Pass   | 0.000s |
| Window td preload done set after settle                                                           | 'window._tdPreloadDone = true' in src  | Pass   | 0.000s |
| Blotter nulls cache after consume                                                                 | 'window._tdPreBlotter = null' in src   | Pass   | 0.000s |
| Blotter reads window pre blotter                                                                  | 'window._tdPreBlotter' in src          | Pass   | 0.000s |
| Gov docs nulls cache after consume                                                                | 'window._tdPreGovDocs = null' in src   | Pass   | 0.000s |
| Gov docs reads window pre gov docs                                                                | 'window._tdPreGovDocs' in src          | Pass   | 0.002s |

| Test Description                                                              | Acceptance Criteria                     | Result | Time   |
|-------------------------------------------------------------------------------|-----------------------------------------|--------|--------|
| Port stress nulls cache after consume                                         | 'window._tdPrePortStorms = null' in src | Pass   | 0.000s |
| Port stress reads window pre port storms                                      | 'window._tdPrePortStorms' in src        | Pass   | 0.002s |
| preloader.py must not assign bare _tdPreloadDone (creates unreliable global). | not bare_assigns                        | Pass   | 0.000s |
| trading/preloader.py fallback must also set window._tdPreloadDone.            | 'window._tdPreloadDone = true' in src   | Pass   | 0.002s |
| Storm portfolio nulls cache after consume                                     | 'window._preStorms = null' in src       | Pass   | 0.000s |
| Storm portfolio reads window pre storms                                       | 'window._preStorms' in src              | Pass   | 0.002s |
| showPanel() must read window._tdPreloadDone, not bare _tdPreloadDone.         | 'window._tdPreloadDone' in src          | Pass   | 0.000s |
| Blotter setup no bare pre blotter                                             | —                                       | Pass   | 0.000s |
| Gov docs no bare pre gov docs                                                 | —                                       | Pass   | 0.001s |
| Port stress no bare pre port storms                                           | —                                       | Pass   | 0.001s |
| preloader.py checks should all use window._tdPreloadDone.                     | not bad_lines                           | Pass   | 0.001s |
| Storm table no bare pre storms                                                | —                                       | Pass   | 0.001s |
| Tradingdesk no bare preload done                                              | —                                       | Pass   | 0.001s |
| Governance JS sub-modules must be syntactically valid.                        | —                                       | Pass   | 0.173s |
| startup.py JS must be syntactically valid.                                    | —                                       | Pass   | 0.061s |
| Storm portfolio syntax                                                        | —                                       | Pass   | 0.060s |
| Trading aggregate submodule syntax                                            | —                                       | Pass   | 0.065s |
| Trading blotter submodule syntax                                              | —                                       | Pass   | 0.058s |
| Trading curves submodule syntax                                               | —                                       | Pass   | 0.083s |
| Trading desk js syntax                                                        | —                                       | Pass   | 0.076s |
| Trading eod submodule syntax                                                  | —                                       | Pass   | 0.058s |
| Trading fs01 submodule syntax                                                 | —                                       | Pass   | 0.062s |
| Trading market submodule syntax                                               | —                                       | Pass   | 0.061s |
| Trading port stress submodule syntax                                          | —                                       | Pass   | 0.108s |
| Trading stress submodule syntax                                               | —                                       | Pass   | 0.062s |
| Aggregate generates                                                           | len(js) > 1000                          | Pass   | 0.000s |
| Blotter generates                                                             | len(js) > 1000                          | Pass   | 0.000s |

| Test Description                                                                                | Acceptance Criteria                                                                                                  | Result | Time   |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Curves generates                                                                                | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Eod generates                                                                                   | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Fs01 generates                                                                                  | <code>len(js) &gt; 500</code>                                                                                        | Pass   | 0.000s |
| Governance bibliography generates                                                               | <code>len(js) &gt; 500</code>                                                                                        | Pass   | 0.000s |
| Governance docs generates                                                                       | <code>len(js) &gt; 500</code>                                                                                        | Pass   | 0.000s |
| Market generates                                                                                | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Port stress generates                                                                           | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Preloader generates                                                                             | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Startup generates                                                                               | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Storm sp table generates                                                                        | <code>len(js) &gt; 500</code>                                                                                        | Pass   | 0.000s |
| Stress generates                                                                                | <code>len(js) &gt; 1000</code>                                                                                       | Pass   | 0.000s |
| Trading desk generates                                                                          | <code>len(js) &gt; 50.000</code>                                                                                     | Pass   | 0.000s |
| blotter/actions.py must call <code>window.refreshMainMapFS01</code> after close-out.            | <code>'window.refreshMainMapFS01'</code> in <code>src</code>                                                         | Pass   | 0.008s |
| <code>tdApplyFilter</code> must be on window for FS01 cell click to work.                       | <code>'window.tdApplyFilter'</code> in <code>src</code>                                                              | Pass   | 0.003s |
| <code>tdRiskCellClick</code> must be on window for blotter filter drill-through.                | <code>'window.tdRiskCellClick'</code> in <code>src</code>                                                            | Pass   | 0.002s |
| <code>_tdPendingFilter</code> must be on window for context-menu → blotter filter.              | <code>'window._tdPendingFilter'</code> in <code>blotter_src</code>                                                   | Pass   | 0.001s |
| <code>refreshMainMapFS01</code> must be on window for blotter close-out to call it.             | <code>'window.refreshMainMapFS01'</code> in <code>td_map_src</code>                                                  | Pass   | 0.002s |
| <code>_stressStormHint</code> must be set via <code>window.*</code> for cross-panel navigation. | <code>'window._stressStormHint'</code> in <code>hist_src</code>                                                      | Pass   | 0.003s |
| Stress setup reads window hint                                                                  | <code>'window._stressStormHint'</code> in <code>stress_src</code>                                                    | Pass   | 0.002s |
| TradingDeskPanel must export <code>TradingDesk</code> and <code>showTradingDesk</code> .        | <code>'window.TradingDesk'</code> in <code>src</code> ;<br><code>'window.showTradingDesk'</code> in <code>src</code> | Pass   | 0.018s |
| Create panel creates blotter view                                                               | <code>'createBlotterView'</code> in <code>js</code>                                                                  | Pass   | 0.002s |
| Open panel function exists                                                                      | <code>'_tdOpenPanel'</code> in <code>js</code>                                                                       | Pass   | 0.003s |
| Open panel sets display flex                                                                    | <code>"style.display = 'flex'"</code> in <code>js</code>                                                             | Pass   | 0.003s |
| Pi button calls show panel                                                                      | <code>'showPanel()'</code> in <code>js</code>                                                                        | Pass   | 0.002s |
| Fallback preloader function must exist for first-open race condition.                           | <code>'_tdRunPreload'</code> in <code>js</code>                                                                      | Pass   | 0.002s |
| Show panel function exists                                                                      | <code>'function showPanel()'</code> in <code>js</code>                                                               | Pass   | 0.003s |
| Show panel reads window preload done                                                            | <code>'window._tdPreloadDone'</code> in <code>js</code>                                                              | Pass   | 0.003s |

| Test Description                                                    | Acceptance Criteria                                                                                                       | Result | Time   |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Panel must switch to blotter tab when opened.                       | <code>"switchTab('blotter')" in js</code>                                                                                 | Pass   | 0.002s |
| Generate risk report basic                                          | <code>generate_risk_report(sample_portfolio_data,</code><br><code>output_dir=ou...</code>                                 | Pass   | 0.018s |
| Generate risk report default                                        | <code>generate_risk_report(sample_portfolio_data,</code><br><code>output_dir=ou...</code>                                 | Pass   | 0.018s |
| Generate risk report detailed                                       | <code>generate_risk_report(sample_portfolio_data,</code><br><code>output_dir=ou...</code>                                 | Pass   | 0.024s |
| Diagonal is one                                                     | <code>np.allclose(np.diag(C), 1.0)</code>                                                                                 | Pass   | 0.000s |
| Longer range -i higher off-diagonal correlations.                   | <code>C.long[mask].mean() &gt;</code><br><code>C.short[mask].mean()</code>                                                | Pass   | 0.000s |
| Higher nugget -i lower off-diagonal values.                         | <code>C.high[mask].mean() &lt;</code><br><code>C.low[mask].mean()</code>                                                  | Pass   | 0.000s |
| Off diagonal in open interval                                       | <code>0.0 &lt; C[i, j] &lt; 1.0</code>                                                                                    | Pass   | 0.000s |
| Positive definite                                                   | <code>np.all(eigenvalues &gt; 0)</code>                                                                                   | Pass   | 0.011s |
| Symmetric                                                           | <code>np.allclose(C, C.T)</code>                                                                                          | Pass   | 0.000s |
| Thames positive definite                                            | <code>np.all(eigenvalues &gt; 0)</code>                                                                                   | Pass   | 0.004s |
| All gauge files corrupt → no storms (but file still written).       | <code>result["total_storms"] == 0</code>                                                                                  | Pass   | 0.003s |
| Lines 151-153: unreadable gauge file is skipped with warning.       | <code>out.exists() and (out /</code><br><code>"_index.json").exists();</code><br><code>result["total_storms"] == 1</code> | Pass   | 0.002s |
| Lines 133-134: invalid JSON in storms.json logs warning, continues. | <code>out.exists() and (out /</code><br><code>"_index.json").exists();</code><br><code>result["total_storms"] == 1</code> | Pass   | 0.006s |
| Exactly at 119999 should be band 'A'.                               | <code>band == "A"</code>                                                                                                  | Pass   | 0.000s |
| Line 159: property value i 120000 returns band 'A'.                 | <code>band == "A"</code>                                                                                                  | Pass   | 0.000s |
| Line 161: property value = 120000 returns band 'B'.                 | <code>band == "B"</code>                                                                                                  | Pass   | 0.000s |
| Property value 140000 returns band 'B'.                             | <code>band == "B"</code>                                                                                                  | Pass   | 0.000s |
| Exactly at 155999 should still be band 'B'.                         | <code>band == "B"</code>                                                                                                  | Pass   | 0.000s |
| All counterparties have party id                                    | <code>len(counterparties) &gt; 0;</code><br><code>ctpy["CounterpartySet"]["Party"]["PartyID"].startswith("C...</code>     | Pass   | 0.003s |
| Cdm structure                                                       | <code>"Party" in ctpy;</code> <code>"PartyID" in</code><br><code>ctpy["Party"] (+1 more)</code>                           | Pass   | 0.001s |
| Custom count                                                        | <code>len(result["data"]) == 3</code>                                                                                     | Pass   | 0.001s |

| Test Description                                             | Acceptance Criteria                                                                              | Result | Time   |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------|--------|
| Generates counterparty file                                  | (pipeline_dir /<br>"counterparty.json").exists()                                                 | Pass   | 0.000s |
| Lines 263-269: ContactInformation fields mapped.             | mapping.get("primary_email") ==<br>"test@bank.com"; mapping.get("city")<br>== "London"           | Pass   | 0.000s |
| Lines 280-285: count fields in mapping.                      | "counterparty_count" in mapping;<br>mapping["counterparty_count"] == 2                           | Pass   | 0.000s |
| Minimal data (empty Counterparty-Set) returns dict.          | isinstance(mapping, dict)                                                                        | Pass   | 0.000s |
| Lines 245-289: create_mapping with full data.                | isinstance(mapping, dict);<br>mapping["party_id"] == "CTPY-001"<br>(+3 more)                     | Pass   | 0.000s |
| Lines 291-292: exception → raises ValueError.                | —                                                                                                | Pass   | 0.000s |
| Lines 288-289: None values are stripped from result.         | None not in mapping.values()                                                                     | Pass   | 0.000s |
| Lines 294-301: get_required_fields() returns non-empty list. | isinstance(fields, list); len(fields)<br>> 0 (+2 more)                                           | Pass   | 0.000s |
| Schema is dict                                               | isinstance(cdm.schema, dict);<br>len(cdm.schema) > 0                                             | Pass   | 0.000s |
| Lines 201-202: empty Counterparties array → error.           | "Counterparties" in errors                                                                       | Pass   | 0.000s |
| validate({}) returns errors for all required fields.         | isinstance(errors, dict)                                                                         | Pass   | 0.000s |
| Lines 211-214: invalid role (not Party1/Party2) → error.     | "Counterparties" in errors;<br>any("Invalid" in e for e in<br>errors["Counterparties"])          | Pass   | 0.000s |
| Lines 209-210: missing CounterpartyRole → error.             | "Counterparties" in errors;<br>any("CounterpartyRole" in e for<br>e in errors["Counterparti...]) | Pass   | 0.002s |
| Lines 188-189: missing PartyID → Party errors.               | "Party" in errors; any("PartyID" in e<br>for e in errors["Party"])                               | Pass   | 0.000s |
| Lines 191-192: missing PartyName → Party errors.             | "Party" in errors; any("PartyName" in<br>e for e in errors["Party"])                             | Pass   | 0.000s |
| Lines 218-219: missing PartyRef → error.                     | "Counterparties" in errors;<br>any("PartyRef" in e for e in<br>errors["Counterparties"])         | Pass   | 0.000s |
| Lines 222-225: two CPs with same role → error.               | "Counterparties" in errors                                                                       | Pass   | 0.000s |
| Valid counterparty data produces empty error dict.           | errors == {}                                                                                     | Pass   | 0.000s |



| Test Description                                                   | Acceptance Criteria                                            | Result | Time   |
|--------------------------------------------------------------------|----------------------------------------------------------------|--------|--------|
| Lines 74-75:<br>json.JSONDecodeError in<br>get_counterparty → 500. | r.status_code == 500; data["status"]<br>== "error"             | Pass   | 0.002s |
| Lines 58-59:<br>json.JSONDecodeError → 500.                        | r.status_code == 500; data["status"]<br>== "error"             | Pass   | 0.002s |
| Creates output file                                                | output_file.exists()                                           | Pass   | 0.001s |
| Data is list                                                       | isinstance(result["data"], list);<br>len(result["data"]) == 3  | Pass   | 0.001s |
| Default count uses all counterparties                              | len(result["data"]) ==<br>len(_ALL.COUNTERPARTIES)             | Pass   | 0.005s |
| File path is pathlib path                                          | isinstance(result["file_path"], Path)                          | Pass   | 0.001s |
| Metadata has total generated                                       | result["metadata"]["total_generated"]<br>== 4                  | Pass   | 0.001s |
| Output file is valid json                                          | "counterparties" in data;<br>"generation_metadata" in data     | Pass   | 0.009s |
| Returns dict with required keys                                    | "data" in result; "file_path" in<br>result (+1 more)           | Pass   | 0.001s |
| Zero count returns empty                                           | result["data"] == []                                           | Pass   | 0.000s |
| Engines not none                                                   | engine is not None                                             | Pass   | 0.000s |
| Returns dict with engines                                          | isinstance(engines, dict); "ISDA" in<br>engines (+2 more)      | Pass   | 0.007s |
| Default day counter                                                | curve is not None; curve is not None                           | Pass   | 0.000s |
| Returns curve and quote                                            | curve is not None; quote is not None                           | Pass   | 0.000s |
| Actual360 day counter accepted                                     | curve is not None                                              | Pass   | 0.000s |
| Quote is simple quote                                              | isinstance(quote, ql.SimpleQuote)                              | Pass   | 0.000s |
| Quote value matches input                                          | quote.value() == pytest.approx(rate,<br>rel=1e-10)             | Pass   | 0.000s |
| Survival probability formula                                       | sp == pytest.approx(math.exp(-lam *<br>3), rel=0.02)           | Pass   | 0.000s |
| Very high hazard rate near zero survival                           | s1 == pytest.approx(math.exp(-2.0),<br>rel=0.01)               | Pass   | 0.000s |
| Returns hazard curve                                               | curve is not None                                              | Pass   | 0.000s |
| Survival probability decreases                                     | sp2 < sp1                                                      | Pass   | 0.000s |
| Zero hazard near full survival                                     | sp > 0.99                                                      | Pass   | 0.000s |
| Auto generates id                                                  | "id" in data["meeting"];<br>len(data["meeting"]["id"]) > 0     | Pass   | 0.001s |
| Create returns meeting                                             | r.status_code == 200; data["status"]<br>== "success" (+1 more) | Pass   | 0.001s |
| Custom attendees become participants                               | "Alice Smith" in names                                         | Pass   | 0.002s |

| Test Description                          | Acceptance Criteria                                                                               | Result | Time   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Custom title used                         | <code>data["meeting"]["title"] == "Quarterly Review"</code>                                       | Pass   | 0.001s |
| Default chair populated                   | <code>data["meeting"]["chair"] == "Johnny Mattimore"</code>                                       | Pass   | 0.001s |
| Default participants created              | <code>len(participants) &gt;= 1</code>                                                            | Pass   | 0.001s |
| No json body uses defaults                | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>            | Pass   | 0.001s |
| Custom max notifications                  | <code>ns.max_visible == 8</code>                                                                  | Pass   | 0.000s |
| Custom timeout                            | <code>ns.timeout == 1000</code>                                                                   | Pass   | 0.000s |
| Invalid position defaults to top right    | <code>ns.position.value == "top-right"</code>                                                     | Pass   | 0.000s |
| Valid position                            | <code>ns.position.value == "bottom-left"</code>                                                   | Pass   | 0.000s |
| Creates registry                          | <code>create_registry(str(tmp_path)) is not None</code>                                           | Pass   | 0.000s |
| With file overrides                       | <code>registry.get_gauge_loader().count() == 2</code>                                             | Pass   | 0.001s |
| Test create_registry helper.              | <code>registry.data_dir == populated_data_dir;</code> <code>registry is not None (+1 more)</code> | Pass   | 0.002s |
| Test create_registry with file overrides. | <code>loader.count() == 0</code>                                                                  | Pass   | 0.002s |
| Auto generated id starts with storm       | <code>storm.storm_id.startswith("STORM-")</code>                                                  | Pass   | 0.000s |
| Decay kernel stored                       | <code>storm.decay_kernel == DecayKernel.EXPONENTIAL</code>                                        | Pass   | 0.000s |
| Decay parameter stored                    | <code>storm.decay_parameter == pytest.approx(1.2)</code>                                          | Pass   | 0.000s |
| Default num track points                  | <code>len(storm.track) == 25</code>                                                               | Pass   | 0.000s |
| Default start time is recent              | <code>before &lt;= storm.start_time &lt;= after</code>                                            | Pass   | 0.000s |
| Duration hours stored                     | <code>storm.duration_hours == pytest.approx(36.0)</code>                                          | Pass   | 0.000s |
| Explicit id preserved                     | <code>storm.storm_id == "MY-STORM-001"</code>                                                     | Pass   | 0.000s |
| Explicit name                             | <code>storm.name == "Ciara"</code>                                                                | Pass   | 0.000s |
| Explicit start time                       | <code>storm.start_time == t</code>                                                                | Pass   | 0.000s |
| Footprint km stored                       | <code>storm.footprint_km == pytest.approx(60.0)</code>                                            | Pass   | 0.000s |
| Num track points                          | <code>len(storm.track) == 10</code>                                                               | Pass   | 0.000s |
| Peak intensity stored                     | <code>storm.peak_intensity == pytest.approx(75.0)</code>                                          | Pass   | 0.000s |
| Returns storm object                      | <code>isinstance(storm, Storm)</code>                                                             | Pass   | 0.000s |
| Speed is positive                         | <code>storm.speed_kmh &gt; 0</code>                                                               | Pass   | 0.000s |
| Track end time equals duration            | <code>storm.track[-1].time_hours == pytest.approx(24.0)</code>                                    | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                                     | Result | Time   |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|--------|
| Track longitude moves from start to end                                    | <code>lons[0] &lt; lons[-1]</code>                                                      | Pass   | 0.000s |
| Track points are trackpoint instances                                      | <code>isinstance(tp, TrackPoint)</code>                                                 | Pass   | 0.000s |
| Track start time is zero                                                   | <code>storm.track[0].time.hours ==<br/>pytest.approx(0.0)</code>                        | Pass   | 0.000s |
| Track time increases                                                       | <code>all(times[i] &lt;= times[i + 1] for i<br/>in range(len(times) - 1))</code>        | Pass   | 0.000s |
| Actual360 day counter produces curve                                       | <code>0.0 &lt; sp &lt;= 1.0</code>                                                      | Pass   | 0.000s |
| Implied hazard rate formula                                                | <code>sp5 == pytest.approx(expected_5yr,<br/>rel=0.01)</code>                           | Pass   | 0.000s |
| Near zero survival positive hazard                                         | <code>s1 &lt; 0.9</code>                                                                | Pass   | 0.000s |
| Single point high risk ts                                                  | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Single point term structure                                                | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Survival strictly positive                                                 | <code>sp &gt; 0.0</code>                                                                | Pass   | 0.000s |
| Custom day counter                                                         | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Default day counter                                                        | <code>curve is not None; curve is not None</code>                                       | Pass   | 0.000s |
| Full survival at start                                                     | <code>abs(sp - 1.0) &lt; 1e-6</code>                                                    | Pass   | 0.001s |
| Returns curve                                                              | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Line 83: <code>final_survival j= 0</code> → uses 0.05 default hazard rate. | <code>curve is not None</code>                                                          | Pass   | 0.000s |
| Line 83: <code>final_survival j= 1</code> → uses 0.05 default hazard rate. | <code>curve is not None</code>                                                          | Pass   | 0.007s |
| Excellent credit 800 plus                                                  | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Fair credit 580 to 669                                                     | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Good credit 670 to 739                                                     | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Poor credit under 580                                                      | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Very good credit 740 to 799                                                | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Excellent 800 plus                                                         | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Fair 580 to 669                                                            | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Good 670 to 739                                                            | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Poor under 580                                                             | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Very good 740 to 799                                                       | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Gauges returned by list endpoint should have valid timeseries responses.   | <code>response.status_code == 200;<br/>ts_response.status_code in [200,<br/>404]</code> | Pass   | 0.002s |
| Gauge hazard endpoint should never return 500 for listed gauges.           | <code>hz_response.status_code != 500</code>                                             | Pass   | 0.002s |

| Test Description                                                                | Acceptance Criteria                                                                                | Result | Time   |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------|--------|
| Properties returned by list endpoint should have flood data if property exists. | <code>response.status_code == 200;</code><br><code>flood_response.status_code in [200, 404]</code> | Pass   | 0.002s |
| crud.py lines 63-67: gauge not found -i 404.                                    | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code>               | Pass   | 0.031s |
| crud.py lines 59-72: valid gauge retrieval.                                     | <code>r.status_code == 200;</code> <code>data["status"] == "success" (+1 more)</code>              | Pass   | 0.031s |
| crud.py lines 51-53: loader error -i 500.                                       | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>               | Pass   | 0.032s |
| crud.py line 34: /list_gauges alias works.                                      | <code>r.status_code == 200;</code><br><code>r.get_json()["count"] == 1</code>                      | Pass   | 0.030s |
| crud.py lines 37-38: OPTIONS returns ok.                                        | <code>r.status_code == 200</code>                                                                  | Pass   | 0.029s |
| crud.py lines 40-50: successful gauge listing.                                  | <code>r.status_code == 200;</code> <code>data["status"] == "success" (+2 more)</code>              | Pass   | 0.046s |
| hd present but daily_observations missing -i same paragraph.                    | <code>any('No historical daily data' in t for t in texts)</code>                                   | Pass   | 0.000s |
| daily_observations = [] -i falsy, caught by first guard.                        | <code>any('No historical daily data' in t for t in texts)</code>                                   | Pass   | 0.000s |
| Normal flow: valid obs -i chart + table elements returned.                      | <code>len(elements) &gt; 3</code>                                                                  | Pass   | 0.104s |
| ValueError/KeyError in date parse loop -i skipped.                              | <code>len(elements) &gt; 0</code>                                                                  | Pass   | 0.104s |
| Invalid date in latest observation -i current_month = None.                     | <code>len(elements) &gt; 0</code>                                                                  | Pass   | 0.225s |
| No obs in same month -i same_month_levels = all_levels.                         | <code>len(elements) &gt; 0</code>                                                                  | Pass   | 0.100s |
| No hd data -i 'No historical daily data' paragraph.                             | <code>any('No historical daily data' in t for t in texts)</code>                                   | Pass   | 0.000s |
| level is None -i skipped, does not crash.                                       | <code>len(elements) &gt; 0</code>                                                                  | Pass   | 0.102s |
| Boe base shown                                                                  | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Current payment shown                                                           | <code>isinstance(result, list)</code>                                                              | Pass   | 0.001s |
| Has status header                                                               | <code>any("Status" in t or "Mortgage" in t for t in texts)</code>                                  | Pass   | 0.000s |
| Has table                                                                       | <code>any(isinstance(e, Table) for e in result)</code>                                             | Pass   | 0.000s |
| Interest paid shown                                                             | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Last payment date shown                                                         | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Last status shown                                                               | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |
| Lending rate shown                                                              | <code>isinstance(result, list)</code>                                                              | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                                                      | Result | Time   |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| No current status returns message                             | <code>any("No current status" in t for t in texts)</code>                                | Pass   | 0.000s |
| No mortgage returns message                                   | <code>any("No mortgage" in t for t in texts)</code>                                      | Pass   | 0.000s |
| Outstanding balance shown                                     | <code>isinstance(result, list)</code>                                                    | Pass   | 0.001s |
| Principal paid shown                                          | <code>isinstance(result, list)</code>                                                    | Pass   | 0.000s |
| Total payments shown                                          | <code>isinstance(result, list)</code>                                                    | Pass   | 0.000s |
| With mortgage returns list                                    | <code>isinstance(result, list) and len(result) &gt; 0</code>                             | Pass   | 0.000s |
| Curves tab should show all gauges on one chart.               | <code>'tdRenderAllCurves' in js;</code><br><code>'tdCurveAllData' in js (+1 more)</code> | Pass   | 0.000s |
| Curves tab should have alert/warning/severe trigger selector. | <code>'td-curve-trigger' in js; 'severe' in js (+2 more)</code>                          | Pass   | 0.000s |
| Curves tab should use HSL for distinct line colours.          | <code>'hsl(' in js</code>                                                                | Pass   | 0.000s |
| Curves sub-module should render without errors.               | <code>len(js) &gt; 0</code>                                                              | Pass   | 0.000s |
| Legend should strip 'Thames' prefix for brevity.              | <code>'Thames' in js</code>                                                              | Pass   | 0.000s |
| Curves tab should fetch from market-state endpoint.           | <code>'/api/v1/trading/market-state' in js</code>                                        | Pass   | 0.000s |
| Good dti under 28                                             | <code>isinstance(result, list)</code>                                                    | Pass   | 0.001s |
| High dti 36 to 43                                             | <code>isinstance(result, list)</code>                                                    | Pass   | 0.001s |
| Moderate dti 28 to 36                                         | <code>isinstance(result, list)</code>                                                    | Pass   | 0.001s |
| No payment dti unknown                                        | <code>isinstance(result, list)</code>                                                    | Pass   | 0.001s |
| Very high dti over 43                                         | <code>isinstance(result, list)</code>                                                    | Pass   | 0.001s |
| Empty daily pnl returns drawing                               | <code>isinstance(result, Drawing)</code>                                                 | Pass   | 0.000s |
| Mixed positive negative renders                               | <code>isinstance(result, Drawing)</code>                                                 | Pass   | 0.000s |
| Negative daily pnl renders                                    | <code>isinstance(result, Drawing)</code>                                                 | Pass   | 0.001s |
| Commercial higher damage                                      | <code>com_ratio &gt;= res_ratio</code>                                                   | Pass   | 0.000s |
| Damage capped                                                 | <code>calculate_damage_ratio(10.0) &lt;= 1.0</code>                                      | Pass   | 0.000s |
| Damage increases with depth                                   | <code>ratios == sorted(ratios)</code>                                                    | Pass   | 0.000s |
| No damage no flood                                            | <code>calculate_damage_ratio(0.0) == 0.0</code>                                          | Pass   | 0.000s |
| GaugeHC file must have required structure.                    | <code>'hazard_curves' in data; not missing</code>                                        | Pass   | 0.001s |
| Gaugets files must have required structure.                   | <code>'flood_simulation' in data; 'readings' in sim (+1 more)</code>                     | Pass   | 0.002s |
| PropertyHC file must have required structure.                 | <code>'property_hazard_curves' in data; 'metadata' in data (+2 more)</code>              | Pass   | 0.001s |
| PropertyTS files must have required fields.                   | <code>not missing; not missing_loc (+3 more)</code>                                      | Pass   | 0.002s |

| Test Description                                              | Acceptance Criteria                                                                  | Result | Time    |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|---------|
| Every step in manifest must appear in dependency graph.       | <code>len(missing) == 0</code>                                                       | Skip   | 0.002s  |
| <code>data_lineage.json</code> must exist after any port run. | <code>"steps" in data; len(data["steps"]) &gt; 0</code>                              | Pass   | 0.001s  |
| No step should have stale inputs.                             | <code>len(stale) == 0</code>                                                         | Pass   | 0.003s  |
| Recorded output hashes must match current files on disk.      | —                                                                                    | Pass   | 70.117s |
| Data summary contains errors count                            | <code>"errors" in dl.get_data_summary()["gauge"]</code>                              | Pass   | 0.001s  |
| Data summary contains warnings count                          | <code>"warnings" in dl.get_data_summary()["gauge"]</code>                            | Pass   | 0.001s  |
| Data summary loaded flag false when no file                   | <code>summary["gauge"]["loaded"] is False</code>                                     | Pass   | 0.000s  |
| Data summary loaded flag true for gauge                       | <code>dl.get_data_summary()["gauge"]["loaded"] is True</code>                        | Pass   | 0.001s  |
| Get data summary empty before load                            | <code>DataLoader(input_dir=tmp_path).get_data_summary() == {}</code>                 | Pass   | 0.000s  |
| Get data summary returns dict                                 | <code>isinstance(dl.get_data_summary(), dict)</code>                                 | Pass   | 0.001s  |
| Get validation summary empty before load                      | <code>DataLoader(input_dir=tmp_path).get_validation_summary() == ...</code>          | Pass   | 0.000s  |
| Get validation summary is copy                                | <code>s1 is not s2</code>                                                            | Pass   | 0.002s  |
| Get validation summary returns dict                           | <code>isinstance(dl.get_validation_summary(), dict)</code>                           | Pass   | 0.001s  |
| Is data complete depends only on gauge                        | <code>dl.is_data_complete() is False</code>                                          | Pass   | 0.001s  |
| Is data complete false before load                            | <code>DataLoader(input_dir=tmp_path).is_data_complete() is False</code>              | Pass   | 0.000s  |
| Is data complete false when no gauge                          | <code>dl.is_data_complete() is False</code>                                          | Pass   | 0.001s  |
| Is data complete true with gauge                              | <code>dl.is_data_complete() is True</code>                                           | Pass   | 0.002s  |
| Validation summary has gauge key                              | <code>"gauge" in dl.get_validation_summary()</code>                                  | Pass   | 0.001s  |
| Init with input dir                                           | <code>dl.input_dir == tmp_path</code>                                                | Pass   | 0.000s  |
| Init with string path converts to path                        | <code>isinstance(dl.input_dir, Path); dl.input_dir == tmp_path</code>                | Pass   | 0.000s  |
| Loaded data initialised                                       | <code>isinstance(dl.loaded_data, LoadedData)</code>                                  | Pass   | 0.000s  |
| Private loaders created                                       | <code>dl._gauge_loader is not None; dl._property_loader is not None (+1 more)</code> | Pass   | 0.000s  |



| Test Description                          | Acceptance Criteria                                                                                  | Result | Time    |
|-------------------------------------------|------------------------------------------------------------------------------------------------------|--------|---------|
| Validation results empty at init          | <code>dl._validation.results == {}</code>                                                            | Pass   | 0.000s  |
| Counterparty data loaded                  | <code>result.counterparty_data is not None</code>                                                    | Pass   | 0.002s  |
| Empty dir returns loaded data             | <code>isinstance(result, LoadedData)</code>                                                          | Pass   | 0.000s  |
| Gauge data count matches records          | <code>result.gauge_data["count"] == len(result.gauge_data["item...</code>                            | Pass   | 0.011s  |
| Gauge data has items key                  | <code>"items" in result.gauge_data</code>                                                            | Pass   | 0.001s  |
| Gaugehd dir counted                       | <code>result.gaugehd_count == 1</code>                                                               | Pass   | 0.002s  |
| Gaugets dir counted                       | <code>result.gaugets_count == 1</code>                                                               | Pass   | 0.001s  |
| Gaugets dir multiple files counted        | <code>result.gaugets_count == 3</code>                                                               | Pass   | 0.001s  |
| Hazard curves loaded                      | <code>result.hazard_data is not None</code>                                                          | Pass   | 0.002s  |
| Lookups built after load                  | <code>result.mortgage_lookup is not None;</code><br><code>result.gauge_flood_info is not None</code> | Pass   | 0.001s  |
| No gaugehd dir count stays zero           | <code>result.gaugehd_count == 0</code>                                                               | Pass   | 0.002s  |
| No gaugets dir count stays zero           | <code>result.gaugets_count == 0</code>                                                               | Pass   | 0.001s  |
| No propertyts dir count stays zero        | <code>result.propertyts_count == 0</code>                                                            | Pass   | 0.001s  |
| Property hazard curves loaded             | <code>result.property_hazard_data is not None</code>                                                 | Pass   | 0.002s  |
| Property hazard summary fields accessible | <code>"summary" in result.property_hazard_data</code>                                                | Pass   | 0.011s  |
| Propertyts dir counted                    | <code>result.propertyts_count &gt;= 1</code>                                                         | Pass   | 0.002s  |
| Storm data list format                    | <code>result.storm_data is not None</code>                                                           | Pass   | 0.003s  |
| Storm data loaded dict format             | <code>result.storm_data is not None</code>                                                           | Pass   | 0.002s  |
| Storm data unknown format                 | <code>result.storm_data is not None</code>                                                           | Pass   | 0.004s  |
| Storm data with items key                 | <code>result.storm_data is not None</code>                                                           | Pass   | 0.002s  |
| With gauge data loaded                    | <code>result.gauge_data is not None</code>                                                           | Pass   | 0.003s  |
| With mortgage data loaded                 | <code>result.mortgage_data is not None</code>                                                        | Pass   | 0.002s  |
| With property data loaded                 | <code>result.property_data is not None</code>                                                        | Pass   | 0.002s  |
| Both empty                                | <code>isinstance(self._page().generate_elements({}), list)</code>                                    | Pass   | 0.001s  |
| Empty property does not crash             | <code>isinstance(self._page().generate_elements({}), list)</code>                                    | Pass   | 0.000s  |
| Has data header                           | <code>any("Data" in t for t in texts)</code>                                                         | Pass   | 0.001s  |
| Has table                                 | <code>any(isinstance(e, Table) for e in result)</code>                                               | Pass   | 0.001s  |
| No sections in property                   | <code>isinstance(result, list)</code>                                                                | Pass   | 0.001s  |
| Returns list                              | <code>isinstance(self._page().generate_elements(_make_property()), list)</code>                      | Pass   | 0.001s; |
| Returns non empty                         | <code>len(self._page().generate_elements(_make_property())) &gt; 0</code>                            | Pass   | 0.001s  |
| With mortgage data                        | <code>isinstance(result, list) and len(result) &gt; 0</code>                                         | Pass   | 0.001s  |

| Test Description                                      | Acceptance Criteria                                                                        | Result | Time   |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------|--------|--------|
| Invalid result fields                                 | result.is_valid is False; "file not found" in result.errors                                | Pass   | 0.000s |
| Is named tuple                                        | isinstance(result, tuple)                                                                  | Pass   | 0.000s |
| Tuple unpacking                                       | is_valid is True                                                                           | Pass   | 0.000s |
| Valid result fields                                   | result.is_valid is True;<br>result.warnings == ["minor issue"]<br>(+2 more)                | Pass   | 0.000s |
| Datetime encoded as isoformat                         | "2024-06-15" in result                                                                     | Pass   | 0.000s |
| Nested numpy in dict                                  | result["a"] == 1; abs(result["b"] - 2.5) < 1e-6 (+1 more)                                  | Pass   | 0.000s |
| Numpy float32 encoded                                 | abs(result - 1.5) < 0.01                                                                   | Pass   | 0.001s |
| Numpy float64 encoded as float                        | abs(result - 3.14) < 1e-6                                                                  | Pass   | 0.000s |
| Numpy int32 encoded                                   | result == 7                                                                                | Pass   | 0.000s |
| Numpy int64 encoded as int                            | result == 99; isinstance(result, int)                                                      | Pass   | 0.002s |
| Numpy ndarray encoded as list                         | result == [1, 2, 3]                                                                        | Pass   | 0.000s |
| Unknown type raises type error                        | —                                                                                          | Pass   | 0.000s |
| Line 59-60: datetime -i isoformat.                    | "2024-06-01" in result                                                                     | Pass   | 0.000s |
| Line 63-64: np.floating -i float.                     | "2.71" in result                                                                           | Pass   | 0.000s |
| Line 61-62: np.integer -i int.                        | result == "99"                                                                             | Pass   | 0.000s |
| Line 65-66: np.ndarray -i list.                       | result == [10, 20]                                                                         | Pass   | 0.000s |
| Line 67: super().default() -i Type-Error for unknown. | —                                                                                          | Pass   | 0.000s |
| Exponential value                                     | DecayKernel.EXPONENTIAL.value == "exponential"                                             | Pass   | 0.000s |
| Gaussian value                                        | IntensityProfile.GAUSSIAN.value == "gaussian";<br>DecayKernel.GAUSSIAN.value == "gaussian" | Pass   | 0.000s |
| Linear value                                          | DecayKernel.LINEAR.value == "linear"                                                       | Pass   | 0.000s |
| Add decision auto generates id                        | r.status_code == 200;<br>decisions[0]["id"] == "D-001"                                     | Pass   | 0.001s |
| Add decision unknown meeting returns 404              | r.status_code == 404                                                                       | Pass   | 0.001s |
| Delete decision                                       | r.status_code == 200;<br>len(r.get_json()["meeting"]["decisions"]) == 1                    | Pass   | 0.004s |
| Delete decision unknown meeting returns 404           | r.status_code == 404                                                                       | Pass   | 0.002s |
| Second decision increments id                         | ids == ["D-001", "D-002"]                                                                  | Pass   | 0.002s |

| Test Description                                                | Acceptance Criteria                                                                                                             | Result | Time   |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Update decision                                                 | <code>r.status_code == 200;</code><br><code>r.get_json()["meeting"]["decisions"][0]["description"]</code><br><code>==...</code> | Pass   | 0.002s |
| Update missing decision returns 404                             | <code>r.status_code == 404</code>                                                                                               | Pass   | 0.001s |
| Add decision auto ids                                           | <code>"D-001" in ids; "D-002" in ids</code>                                                                                     | Pass   | 0.002s |
| Add decision missing meeting returns 404                        | <code>r.status_code == 404</code>                                                                                               | Pass   | 0.002s |
| Add decision returns 200                                        | <code>r.status_code == 200;</code><br><code>len(meeting["decisions"]) == 2</code>                                               | Pass   | 0.002s |
| Delete decision missing meeting returns 404                     | <code>r.status_code == 404</code>                                                                                               | Pass   | 0.002s |
| Delete decision returns 200                                     | <code>r.status_code == 200;</code><br><code>len(r.get_json()["meeting"]["decisions"])</code><br><code>== 0</code>               | Pass   | 0.002s |
| Update decision not found returns 404                           | <code>r.status_code == 404</code>                                                                                               | Pass   | 0.002s |
| Update decision returns 200                                     | <code>r.status_code == 200;</code><br><code>decisions[0]["description"] ==</code><br><code>"Updated decision"</code>            | Pass   | 0.002s |
| Delete is idempotent — filtering out a missing ID is still 200. | <code>r.status_code == 200</code>                                                                                               | Pass   | 0.002s |
| Delete returns success                                          | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>                                          | Pass   | 0.003s |
| Deleted ref not in list                                         | <code>ref_id not in ids</code>                                                                                                  | Pass   | 0.004s |
| Deleted doc not in list                                         | <code>doc_id not in ids</code>                                                                                                  | Pass   | 0.003s |
| Not found returns 404                                           | <code>r.status_code == 404; r.status_code ==</code><br><code>404</code>                                                         | Pass   | 0.001s |
| Valid delete returns success                                    | <code>r.status_code == 200; data["status"]</code><br><code>== "success"</code>                                                  | Pass   | 0.003s |
| Delete save failure returns 500                                 | <code>r.status_code == 500; data["status"]</code><br><code>== "error"</code>                                                    | Pass   | 0.027s |
| Gaugehd references gauges                                       | <code>gid in gauge_ids</code>                                                                                                   | Pass   | 0.004s |
| Gaugets reads gauge                                             | <code>gid in gauge_ids</code>                                                                                                   | Pass   | 0.002s |
| Hazard reads gauge and storms                                   | <code>len(hc_curves) &gt; 0; gid in gauge_ids</code>                                                                            | Pass   | 0.001s |
| Mortgage reads property                                         | <code>mid in prop_ids</code>                                                                                                    | Pass   | 0.000s |
| Propertyhc references properties                                | <code>pid in prop_ids</code>                                                                                                    | Pass   | 0.001s |
| Propertyts uses nearest gauges                                  | <code>ng["gauge_id"] in gauge_ids</code>                                                                                        | Pass   | 0.001s |
| Commercial higher damage than residential                       | <code>com_damage &gt; res_damage</code>                                                                                         | Pass   | 0.000s |
| Floor level reduces damage                                      | <code>damages[0] &gt;= damages[3]</code>                                                                                        | Pass   | 0.000s |
| Large depth all near one                                        | <code>np.all(damages &gt; 0.5)</code>                                                                                           | Pass   | 0.000s |

| Test Description                                                                | Acceptance Criteria                                                  | Result | Time   |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------|--------|--------|
| Output clipped to 0 1                                                           | <code>np.all((damages &gt;= 0) &amp; (damages &lt;= 1))</code>       | Pass   | 0.000s |
| Returns array same length                                                       | <code>len(damages) == len(properties)</code>                         | Pass   | 0.000s |
| Zero depths all zero damage                                                     | —                                                                    | Pass   | 0.022s |
| Line 150: commercial factor = 1.2.                                              | <code>result[1] &gt; result[0]</code>                                | Pass   | 0.000s |
| Lines 159-160: adjusted_depths = max(depths - floor_levels, 0).                 | <code>result[3] == pytest.approx(0.0)</code>                         | Pass   | 0.000s |
| Full function returns correct length                                            | <code>len(result) == len(properties)</code>                          | Pass   | 0.000s |
| Line 151: industrial factor = 0.9.                                              | <code>result[2] &lt; result[0]</code>                                | Pass   | 0.000s |
| Line 165: <code>np.clip(final_damage, 0, 1)</code> .                            | <code>np.all(result &lt;= 1.0); np.all(result &gt;= 0.0)</code>      | Pass   | 0.000s |
| Line 155: <code>.get(str(pt).lower(), 1.0)</code><br>— 'other' defaults to 1.0. | <code>result[4] == pytest.approx(result[0])</code>                   | Pass   | 0.000s |
| Zero depths all zero                                                            | —                                                                    | Pass   | 0.000s |
| Deep flood -i damage ratio approaches 1.                                        | <code>result &gt; 0.5</code>                                         | Pass   | 0.001s |
| Positive depth -i positive damage ratio.                                        | <code>0.0 &lt; result &lt;= 1.0</code>                               | Pass   | 0.001s |
| 0m depth -i 0 damage.                                                           | <code>gen..depth_to_damage(0.0) == 0.0</code>                        | Pass   | 0.002s |
| Line 80: after loop exhausted, gauges_warning i 3 → 'baseline'.                 | <code>.derive_intensity_category(-1, 2) == "baseline"</code>         | Pass   | 0.000s |
| Negative severe with five warnings returns moderate                             | <code>.derive_intensity_category(-1, 5) == "moderate"</code>         | Pass   | 0.000s |
| Lines 78-79: gauges_severe i 0 (doesn't match any threshold),                   | <code>.derive_intensity_category(-1, 3) == "moderate"</code>         | Pass   | 0.000s |
| Negative severe zero warnings returns baseline                                  | <code>.derive_intensity_category(-1, 0) == "baseline"</code>         | Pass   | 0.000s |
| When severe i= 1, warning count is irrelevant.                                  | <code>.derive_intensity_category(1, 10) == "moderate"</code>         | Pass   | 0.000s |
| gauges_severe=0 matches (0, 'baseline') in the severity table.                  | <code>.derive_intensity_category(0, 0) == "baseline"</code>          | Pass   | 0.000s |
| Both missing returns na                                                         | <code>page..derive_tenor("", "") == "N/A"</code>                     | Pass   | 0.000s |
| Five year tenor                                                                 | <code>page..derive_tenor("2024-01-01", "2029-01-01") == "5Y"</code>  | Pass   | 0.000s |
| Invalid date format returns na                                                  | <code>page..derive_tenor("not-a-date", "2026-01-01") == "N/A"</code> | Pass   | 0.000s |
| Missing end returns na                                                          | <code>page..derive_tenor("2024-01-01", "") == "N/A"</code>           | Pass   | 0.000s |
| Missing start returns na                                                        | <code>page..derive_tenor("", "2026-01-01") == "N/A"</code>           | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                         | Result | Time   |
|------------------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------|
| Valid dates returns years                                        | <code>page..derive_tenor("2024-01-01", "2026-01-01") == "2Y"</code>         | Pass   | 0.000s |
| Empty property info returns valid                                | <code>result in ["Residential", "Buy-to-Let", "Second Home", "H...]</code>  | Pass   | 0.000s |
| Multi family returns btl or residential                          | <code>result in ["Buy-to-Let", "Residential"]</code>                        | Pass   | 0.000s |
| Rental history returns btl or residential                        | <code>result in ["Buy-to-Let", "Residential"]</code>                        | Pass   | 0.000s |
| Rental income returns btl or residential                         | <code>result in ["Buy-to-Let", "Residential"]</code>                        | Pass   | 0.000s |
| Vacant property                                                  | <code>result in ["Buy-to-Let", "Second Home", "Residential"]</code>         | Pass   | 0.000s |
| Buy to let returns investment or primary                         | <code>result in ["Investment", "PrimaryResidence"]</code>                   | Pass   | 0.000s |
| Holiday home returns second residence                            | <code>..determine_occupancy_type("Holiday Home") == "SecondResid..."</code> | Pass   | 0.000s |
| Residential returns primary or second                            | <code>result in ["PrimaryResidence", "SecondResidence"]</code>              | Pass   | 0.000s |
| Second home returns second residence                             | <code>..determine_occupancy_type("Second Home") == "SecondReside..."</code> | Pass   | 0.000s |
| Gauge IDs should be derived from location, not random UUIDs.     | —                                                                           | Skip   | 0.001s |
| Haversine known distance                                         | <code>6.0 &lt; distance &lt; 8.0</code>                                     | Pass   | 0.000s |
| Haversine same point                                             | <code>calculate_distance_km(lat, lon, lat, lon) == pytest.appro...</code>   | Pass   | 0.000s |
| Haversine teddington to richmond                                 | <code>2.5 &lt; distance &lt; 4.0</code>                                     | Pass   | 0.000s |
| Diagonal zero                                                    | <code>np.allclose(np.diag(D), 0.0)</code>                                   | Pass   | 0.000s |
| 51.5N 0.0 -i 51.5N 1.0E is approximately 69km.                   | <code>65.0 &lt; d &lt; 75.0</code>                                          | Pass   | 0.000s |
| Positive off diagonal                                            | <code>D[i, j] &gt; 0.0</code>                                               | Pass   | 0.000s |
| Shape                                                            | <code>D.shape == (n, n)</code>                                              | Pass   | 0.000s |
| Symmetric                                                        | <code>np.allclose(D, D.T)</code>                                            | Pass   | 0.000s |
| Thames gauges span ~80km corridor (Reading to Purfleet).         | <code>10.0 &lt; max_d &lt; 120.0</code>                                     | Pass   | 0.000s |
| Thames n gauges                                                  | <code>thames_model.n_gauges == 52</code>                                    | Pass   | 0.000s |
| Close distance label                                             | <code>"Very close to Thames" in str(elements)</code>                        | Pass   | 0.001s |
| Zero distance highest risk                                       | <code>"On Thames River" in str(elements)</code>                             | Pass   | 0.001s |
| STORM-d1a damage for PROP-A uses only that storm's damage_ratio. | <code>prop.a["damage_amount"] == pytest.approx(expected.damage)</code>      | Pass   | 0.003s |

| Test Description                                                         | Acceptance Criteria                                                                             | Result | Time   |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Query STORM-d1a: only PROP-A (depth=0) appears; PROP-B depth=0 excluded. | <code>r.status_code == 200;</code> <code>"PROP-A" in prop_ids (+1 more)</code>                  | Pass   | 0.004s |
| Total damage is different for each storm in the doublet.                 | <code>total2 &gt; total1</code>                                                                 | Pass   | 0.004s |
| STORM-d1b causes more damage to PROP-A (deeper flood).                   | <code>dmg2 &gt; dm1</code>                                                                      | Pass   | 0.005s |
| Query STORM-d1b: both PROP-A and PROP-B have positive depth.             | <code>r.status_code == 200;</code> <code>"PROP-A" in prop_ids (+1 more)</code>                  | Pass   | 0.003s |
| Response storm_id matches the queried storm.                             | <code>data["storm_id"] == "STORM-d1b"</code>                                                    | Pass   | 0.004s |
| Each storm computes its own damage_ratio from its own flood depth.       | <code>"damage_ratio" in event;</code> <code>0.0 &lt;= event["damage_ratio"] &lt;= 1.0</code>    | Pass   | 0.002s |
| Doublet events have distinct storm ids                                   | <code>"STORM-d001a" in event_storm_ids;</code><br><code>"STORM-d001b" in event_storm_ids</code> | Pass   | 0.002s |
| Doublet produces two flood events                                        | <code>result is not None; prop_file.exists()</code><br><code>(+1 more)</code>                   | Pass   | 0.002s |
| <code>floods_at_nearest_gauge == 2</code> for doublet sequence.          | <code>result["summary"]["floods_at_nearest_gauge"] == 2</code>                                  | Pass   | 0.001s |
| File not found returns 404                                               | <code>r.status_code == 404</code>                                                               | Pass   | 0.001s |
| Non pdf served as attachment                                             | <code>r.status_code == 200; "attachment" in r.headers.get("Content-Disposition", "")</code>     | Pass   | 0.001s |
| Path traversal returns 403                                               | <code>r.status_code in (403, 404)</code>                                                        | Pass   | 0.002s |
| Valid pdf served inline                                                  | <code>r.status_code == 200</code>                                                               | Pass   | 0.009s |
| Not found returns 404                                                    | <code>r.status_code == 404; r.status_code == 404</code>                                         | Pass   | 0.002s |
| Valid download returns 200                                               | <code>r.status_code == 200</code>                                                               | Pass   | 0.004s |
| File missing on disk returns 404                                         | <code>r.status_code == 404; "not found" in data["message"].lower()</code>                       | Pass   | 0.028s |
| Insufficient data message                                                | <code>any("Insufficient" in t for t in texts)</code>                                            | Pass   | 0.000s |
| Returns list                                                             | <code>isinstance(result, list); len(result) &gt; 0 (+6 more)</code>                             | Pass   | 0.001s |
| Single entry insufficient                                                | <code>any("Insufficient" in t for t in texts)</code>                                            | Pass   | 0.000s |
| Sufficient data creates content                                          | <code>len(result) &gt; 2</code>                                                                 | Pass   | 0.001s |
| More than 20 positions → only top 20 movers shown.                       | <code>isinstance(result, list)</code>                                                           | Pass   | 0.001s |
| Market move pnl creates table                                            | <code>len(tables) &gt;= 2</code>                                                                | Pass   | 0.001s |
| New trade pnl creates table                                              | <code>len(tables) &gt;= 1</code>                                                                | Pass   | 0.000s |



| Test Description                                                                                          | Acceptance Criteria                                                                                   | Result | Time   |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| No positions shows messages                                                                               | <code>any("No new trades" in t for t in texts); any("No market move" in t for t in texts)</code>      | Pass   | 0.001s |
| Returns list                                                                                              | <code>isinstance(result, list); len(result) &gt; 0 (+6 more)</code>                                   | Pass   | 0.000s |
| Position with notional=0 triggers <code>fmt_gbp(0) → em-dash</code> path.                                 | <code>isinstance(result, list); len(result) &gt; 0</code>                                             | Pass   | 0.001s |
| Gauge IDs longer than 12 chars should not crash.                                                          | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.001s |
| Positions from 2 gauges should produce group subheaders.                                                  | <code>len(tables) &gt;= 1</code>                                                                      | Pass   | 0.001s |
| No positions shows message                                                                                | <code>any("No open positions" in t for t in texts)</code>                                             | Pass   | 0.001s |
| Returns list                                                                                              | <code>isinstance(result, list); len(result) &gt; 0 (+6 more)</code>                                   | Pass   | 0.001s |
| Single position                                                                                           | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.001s |
| With positions creates table                                                                              | <code>any(isinstance(e, Table) for e in result)</code>                                                | Pass   | 0.001s |
| Line 48: <code>daily_pnl=0</code> and <code>running_pnl=0 → fmt_gbp</code> returns <code>em-dash</code> . | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.000s |
| Eod id in filename                                                                                        | <code>"EOD-2026-03-08" in path.name</code>                                                            | Pass   | 0.005s |
| Generate empty snapshot                                                                                   | <code>path.exists()</code>                                                                            | Pass   | 0.004s |
| Generate report returns path                                                                              | <code>path.exists(); path.suffix == ".pdf"</code>                                                     | Pass   | 0.012s |
| Generate report without pnl series                                                                        | <code>path.exists()</code>                                                                            | Pass   | 0.005s |
| Lines 49-50: <code>output_dir=None → uses config.get_eod_dir()</code> .                                   | <code>gen.output_dir == eod_dir</code>                                                                | Pass   | 0.001s |
| Output dir created                                                                                        | <code>new_dir.exists()</code>                                                                         | Pass   | 0.001s |
| Pdf not empty                                                                                             | <code>path.stat().st_size &gt; 1000</code>                                                            | Pass   | 0.010s |
| Date appears in paragraph                                                                                 | <code>any("2026-03-08" in t for t in texts)</code>                                                    | Pass   | 0.001s |
| Empty summary                                                                                             | <code>isinstance(result, list); len(result) &gt; 0</code>                                             | Pass   | 0.001s |
| Has paragraph and table                                                                                   | <code>any(isinstance(e, Paragraph) for e in result); any(isinstance(e, Table) for e in result)</code> | Pass   | 0.001s |
| Negative pnl snapshot                                                                                     | <code>isinstance(result, list)</code>                                                                 | Pass   | 0.001s |
| Returns list                                                                                              | <code>isinstance(result, list); len(result) &gt; 0 (+6 more)</code>                                   | Pass   | 0.376s |
| All unknown pages still creates pdf                                                                       | <code>path.exists()</code>                                                                            | Pass   | 0.001s |
| Completely empty flood data                                                                               | <code>path.exists()</code>                                                                            | Pass   | 0.011s |

| Test Description                                                | Acceptance Criteria                                                   | Result | Time   |
|-----------------------------------------------------------------|-----------------------------------------------------------------------|--------|--------|
| Completely empty gauge data                                     | <code>path.exists()</code>                                            | Pass   | 0.016s |
| Completely empty property data                                  | <code>path.exists()</code>                                            | Pass   | 0.015s |
| Gauge data with hazard curve and timeseries                     | <code>path.exists()</code>                                            | Pass   | 0.019s |
| Verify no crash or timeout with 500 property entries.           | <code>path.exists()</code>                                            | Pass   | 0.013s |
| Missing summary key                                             | <code>path.exists()</code>                                            | Pass   | 0.011s |
| Negative value property                                         | <code>path.exists()</code>                                            | Pass   | 0.018s |
| Negative/nonsensical values should not crash the generator.     | <code>path.exists()</code>                                            | Pass   | 0.011s |
| None flood gauge header                                         | <code>path.exists()</code>                                            | Pass   | 0.012s |
| None gauge data key                                             | <code>path.exists()</code>                                            | Pass   | 0.006s |
| Output is non empty bytes                                       | <code>isinstance(content, bytes);<br/>len(content) &gt; 0</code>      | Pass   | 0.104s |
| Pdf content is bytes                                            | <code>isinstance(content, bytes);<br/>len(content) &gt; 0</code>      | Pass   | 0.020s |
| Property data with none header                                  | <code>path.exists()</code>                                            | Pass   | 0.008s |
| Report with all pages                                           | <code>path.exists(); path.stat().st_size &gt; 0</code>                | Pass   | 0.017s |
| Report with all pages                                           | <code>path.exists(); path.stat().st_size &gt; 0</code>                | Pass   | 0.022s |
| Report with all pages                                           | <code>path.exists(); path.stat().st_size &gt; 0</code>                | Pass   | 0.047s |
| Single page report                                              | <code>path.exists()</code>                                            | Pass   | 0.003s |
| Single page report                                              | <code>path.exists()</code>                                            | Pass   | 0.003s |
| Zero properties at risk                                         | <code>path.exists()</code>                                            | Pass   | 0.011s |
| Zero value property                                             | <code>path.exists()</code>                                            | Pass   | 0.017s |
| Spec requirement: stronger storms -¿ broader spatial footprint. | <code>r_moderate &lt; r_severe &lt; r_extreme</code>                  | Pass   | 0.000s |
| intensity_factor ¿ 1 still gives at least base range.           | <code>r ==<br/>pytest.approx(small_model.params.base_range_km)</code> | Pass   | 0.000s |
| intensity_factor = 1.0 -¿ range = base_range.                   | <code>r ==<br/>pytest.approx(small_model.params.base_range_km)</code> | Pass   | 0.000s |
| range = base * (1 + rho * (factor - 1)).                        | <code>m.effective_range_km(2.0) ==<br/>pytest.approx(56.0)</code>     | Pass   | 0.000s |
| Good elevation over 20m                                         | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Low elevation 5 to 10m                                          | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Moderate elevation 10 to 20m                                    | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| No ground level                                                 | <code>isinstance(result, list)</code>                                 | Pass   | 0.000s |
| Very low elevation under 5m                                     | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Concerning under 1 year                                         | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Excellent 5 plus years                                          | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Fair 1 to 2 years                                               | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Good 2 to 5 years                                               | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |

| Test Description                                                                    | Acceptance Criteria                                                                                                   | Result | Time   |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| No curves raises value error                                                        | —                                                                                                                     | Pass   | 0.002s |
| Empty storm_id returns 400.                                                         | <code>resp.status_code == 400</code>                                                                                  | Pass   | 0.000s |
| Line 160-161: responses with empty storm_id are skipped.                            | <code>result["total_storms"] == 1;</code><br><code>data["storms"][0]["storm_id"] == "STORM-valid001"</code>           | Pass   | 0.002s |
| storm_id key missing entirely → <code>resp.get('storm_id', "") == ""</code> → skip. | <code>result["total_storms"] == 1</code>                                                                              | Pass   | 0.002s |
| Full pipeline generates all outputs                                                 | <code>(pipeline_dir / "gaugets").is_dir();</code><br><code>(pipeline_dir / "gaugehd").is_dir()</code><br>(+3 more)    | Pass   | 0.000s |
| Gauge count propagates                                                              | <code>n == N_GAUGES;</code><br><code>len(list((pipeline_dir / "gaugets").glob("GAUGE-*.json"))...</code><br>(+2 more) | Pass   | 0.001s |
| Pipeline output file integrity                                                      | <code>expected_key in data</code>                                                                                     | Pass   | 0.002s |
| Property count propagates                                                           | <code>n == N_PROPERTIES;</code><br><code>json.load(f)["summary"]["total_properties"]</code><br><code>== n</code>      | Pass   | 0.000s |
| Analyze epc rating all bands                                                        | <code>expected_word in result</code>                                                                                  | Pass   | 0.000s |
| Analyze epc rating case insensitive                                                 | <code>"Excellent" in</code><br><code>page._analyze_epc_rating("a")</code>                                             | Pass   | 0.000s |
| Analyze epc rating unknown                                                          | <code>"Unknown" in result</code>                                                                                      | Pass   | 0.000s |
| Exercises energy intensity branch when area is known.                               | <code>len(tables) &gt;= 1</code>                                                                                      | Pass   | 0.000s |
| Full data set exercises all branches without exception.                             | <code>isinstance(result, list); len(result)</code><br><code>&gt; 2</code>                                             | Pass   | 0.001s |
| No energy data fallback message                                                     | <code>any("No energy performance data" in</code><br><code>t for t in texts)</code>                                    | Pass   | 0.000s |
| Returns list                                                                        | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0</code>                                             | Pass   | 0.003s |
| With building fabric                                                                | <code>isinstance(result, list)</code>                                                                                 | Pass   | 0.000s |
| With energy usage                                                                   | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0</code>                                             | Pass   | 0.000s |
| With ratings data                                                                   | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0</code>                                             | Pass   | 0.000s |
| With ratings includes table                                                         | <code>len(tables) &gt;= 1</code>                                                                                      | Pass   | 0.000s |
| Close point pushed away                                                             | <code>dist &gt;=</code><br><code>PropertyPortfolioGenerator.MIN_RIVER_DISTANCE_M</code><br><code>*...</code>          | Pass   | 0.000s |
| Far point unchanged                                                                 | <code>new_lat == pytest.approx(lat); new_lon</code><br><code>== pytest.approx(lon)</code>                             | Pass   | 0.000s |

| Test Description                                                                          | Acceptance Criteria                                                                                           | Result | Time   |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| Returns tuple of two floats                                                               | <code>len(result) == 2; all(isinstance(v, float) for v in result)</code>                                      | Pass   | 0.000s |
| Lines 353-355: <code>FileNotFoundError</code> → <code>logger.error + sys.exit(1)</code> . | <code>1 in exited_with</code>                                                                                 | Pass   | 0.001s |
| Lines 451-453: <code>FileNotFoundError</code> → <code>logger.error + sys.exit(1)</code> . | <code>1 in exited</code>                                                                                      | Pass   | 0.001s |
| Lines 510-512: <code>FileNotFoundError</code> → <code>logger.error + sys.exit(1)</code> . | <code>1 in exited</code>                                                                                      | Pass   | 0.001s |
| Test handling of invalid JSON.                                                            | <code>response.status_code == 400</code>                                                                      | Pass   | 0.000s |
| Lines 451-453: bad JSON → <code>json.JSONDecodeError</code> → <code>sys.exit(1)</code> .  | <code>1 in exited</code>                                                                                      | Pass   | 0.046s |
| Lines 510-512: bad JSON → <code>json.JSONDecodeError</code> → <code>sys.exit(1)</code> .  | <code>1 in exited</code>                                                                                      | Pass   | 0.001s |
| Lines 353-355: bad JSON → <code>json.JSONDecodeError</code> → <code>sys.exit(1)</code> .  | <code>1 in exited_with</code>                                                                                 | Pass   | 0.002s |
| Test handling of missing content type.                                                    | <code>response.status_code in [400, 415, 500]</code>                                                          | Pass   | 0.000s |
| Kent multiplier                                                                           | <code>result &gt; 0</code>                                                                                    | Pass   | 0.000s |
| London multiplier                                                                         | <code>london &gt; 0</code>                                                                                    | Pass   | 0.000s |
| New property                                                                              | <code>result &gt; 0</code>                                                                                    | Pass   | 0.000s |
| Old property                                                                              | <code>result &gt; 0</code>                                                                                    | Pass   | 0.000s |
| Returns float                                                                             | <code>isinstance(result, float)</code>                                                                        | Pass   | 0.000s |
| Surrey multiplier                                                                         | <code>result &gt; 0</code>                                                                                    | Pass   | 0.000s |
| With area                                                                                 | <code>result &gt; 0</code>                                                                                    | Pass   | 0.000s |
| With bedrooms                                                                             | <code>result &gt; 0</code>                                                                                    | Pass   | 0.000s |
| Zero property value falls through                                                         | <code>result &gt; 0</code>                                                                                    | Pass   | 0.000s |
| Severe floods should be rare.                                                             | <code>prob &lt;= 0.05 + 1e-9</code>                                                                           | Pass   | 0.000s |
| Normal levels should be common.                                                           | <code>prob &gt; 0.5</code>                                                                                    | Pass   | 0.000s |
| Return period should be inverse of probability.                                           | <code>return_period &gt; 1</code>                                                                             | Pass   | 0.000s |
| Get gauge id missing key                                                                  | <code>page._get_gauge_id({}) == "Unknown Gauge"; page._get_gauge_id(None) == "Unknown Gauge"</code>           | Pass   | 0.000s |
| Malformed gauge data returns error element                                                | <code>isinstance(elements, list); len(elements) &gt; 0</code>                                                 | Pass   | 0.000s |
| Extracts ids                                                                              | <code>extract_property_ids(data) == {"P1", "P2"}; extract_mortgage_ids(data) == {"M1", "M2"} (+1 more)</code> | Pass   | 0.000s |

| Test Description                              | Acceptance Criteria                                                                                                        | Result | Time   |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Missing gauge id skipped                      | <code>extract_gauge_ids(data) == set()</code>                                                                              | Pass   | 0.000s |
| None returns empty set                        | <code>extract_property_ids(None) == set();</code><br><code>extract_mortgage_ids(None) == set()</code><br>(+2 more)         | Pass   | 0.000s |
| Dict format                                   | <code>len(result) == 2</code>                                                                                              | Pass   | 0.000s |
| Empty list                                    | <code>_extract_gauges({"flood_gauges": []}) == []</code>                                                                   | Pass   | 0.000s |
| List format                                   | <code>result == [_G1, _G2]</code>                                                                                          | Pass   | 0.000s |
| Missing key returns empty                     | <code>_extract_gauges({}) == []</code>                                                                                     | Pass   | 0.000s |
| Extracts ids                                  | <code>extract_property_ids(data) == {"P1", "P2"};</code> <code>extract_mortgage_ids(data) == {"M1", "M2"} (+1 more)</code> | Pass   | 0.000s |
| None returns empty set                        | <code>extract_property_ids(None) == set();</code><br><code>extract_mortgage_ids(None) == set()</code><br>(+2 more)         | Pass   | 0.000s |
| Extracts property ids from mortgages          | <code>extract_mortgage_property_ids(data) == {"P1", "P2"}</code>                                                           | Pass   | 0.000s |
| None returns empty set                        | <code>extract_property_ids(None) == set();</code><br><code>extract_mortgage_ids(None) == set()</code><br>(+2 more)         | Pass   | 0.000s |
| Empty items returns empty set                 | <code>extract_property_ids({"items": []}) == set()</code>                                                                  | Pass   | 0.000s |
| Extracts ids                                  | <code>extract_property_ids(data) == {"P1", "P2"};</code> <code>extract_mortgage_ids(data) == {"M1", "M2"} (+1 more)</code> | Pass   | 0.000s |
| Missing property id skipped                   | <code>extract_property_ids(data) == set()</code>                                                                           | Pass   | 0.000s |
| None returns empty set                        | <code>extract_property_ids(None) == set();</code><br><code>extract_mortgage_ids(None) == set()</code><br>(+2 more)         | Pass   | 0.000s |
| Construction year extracted                   | <code>gen._extract_property_info(self._full_record["construction_year"])</code>                                            | Pass   | 0.000s |
| Flood risk extracted                          | <code>gen._extract_property_info(self._full_record["flood_risk"])</code>                                                   | Pass   | 0.000s |
| Lat lon extracted                             | <code>abs(info["latitude"] - 51.45) &lt; 0.001;</code> <code>abs(info["longitude"] + 0.30) &lt; 0.001</code>               | Pass   | 0.000s |
| Missing attrs default construction year       | <code>gen._extract_property_info({"PropertyHeader": {"Header": ...}})</code>                                               | Pass   | 0.000s |
| Missing header defaults property id           | <code>gen._extract_property_info({"PropertyHeader": {"prope...}})</code>                                                   | Pass   | 0.000s |
| Missing location defaults postcode and coords | <code>info["postcode"] == "SW1A 1AA";</code><br><code>abs(info["latitude"] - 51.5) &lt; 0.01</code><br>(+1 more)           | Pass   | 0.000s |

| Test Description                            | Acceptance Criteria                                                                                               | Result | Time   |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Missing property header returns defaults    | <code>info["property_id"] == "";</code><br><code>info["property_value"] == 500000</code><br>(+1 more)             | Pass   | 0.000s |
| Missing risk assessment defaults flood risk | <code>gen._extract_property_info({"PropertyHeader": ...</code><br><code>{"Header": ...</code>                     | Pass   | 0.000s |
| Missing valuation defaults property value   | <code>gen._extract_property_info(record)["property_value"]</code><br><code>== 5...</code>                         | Pass   | 0.000s |
| Postcode extracted                          | <code>gen._extract_property_info(self._full_record["postcode"]</code>                                             | Pass   | 0.000s |
| Property id extracted from header           | <code>gen._extract_property_info(self._full_record["property_id"]</code>                                          | Pass   | 0.000s |
| Property type extracted                     | <code>gen._extract_property_info(self._full_record["property_type"]</code>                                        | Pass   | 0.000s |
| Property value extracted                    | <code>gen._extract_property_info(self._full_record["property_value"]</code>                                       | Pass   | 0.000s |
| Age band factors within reasonable bounds   | <code>lo &gt; 0; hi &gt;= lo</code>                                                                               | Pass   | 0.000s |
| Base area ranges all positive               | <code>lo &gt; 0; hi &gt; lo</code>                                                                                | Pass   | 0.000s |
| Base price per sqm all positive             | <code>lo &gt; 0; hi &gt; lo</code>                                                                                | Pass   | 0.001s |
| Condition factors ordered by severity       | <code>excellent_lo &gt; poor_hi</code>                                                                            | Pass   | 0.000s |
| Epc factors a highest                       | <code>a_lo &gt; g_hi</code>                                                                                       | Pass   | 0.000s |
| Flood risk factors decrease with risk       | <code>very_low_lo &gt; very_high_hi</code>                                                                        | Pass   | 0.000s |
| Min max value bounds                        | <code>MIN_PROPERTY_VALUE == 150.000;</code><br><code>MAX_PROPERTY_VALUE == 5.000.000</code>                       | Pass   | 0.000s |
| Proximity zones have range key              | <code>'range' in data</code>                                                                                      | Pass   | 0.000s |
| Rent per sqm all present                    | <code>set(RENT_PER_SQM.keys()) ==</code><br><code>expected_types</code>                                           | Pass   | 0.000s |
| Rental yield rates all present              | <code>set(RENTAL_YIELD_RATES.keys()) ==</code><br><code>expected_types</code>                                     | Pass   | 0.000s |
| Empty counterparty file uses fallback       | <code>len(trades) &gt; 0;</code><br><code>all(cid.startswith("CTPY-") for</code><br><code>cid in ctpy_ids)</code> | Pass   | 0.005s |
| Missing counterparty file uses fallback     | <code>len(trades) &gt; 0</code>                                                                                   | Pass   | 0.005s |
| None field def returns empty string         | <code>result == ""</code>                                                                                         | Pass   | 0.000s |
| Numeric field def returns empty string      | <code>result == ""</code>                                                                                         | Pass   | 0.000s |
| String field def returned as is             | <code>result == "literal_value"</code>                                                                            | Pass   | 0.000s |
| No type defaults to text                    | <code>"SomeField" in result</code>                                                                                | Pass   | 0.000s |
| Unknown type returns empty                  | <code>result == ""</code>                                                                                         | Pass   | 0.000s |
| 5 year filter                               | <code>len(result) &lt; len(flows); len(result)</code><br><code>&gt; 0</code>                                      | Pass   | 0.002s |
| Empty flows returns empty                   | <code>GaugeHistoricalDaily().filter_by_years([],</code><br><code>years=10) == []</code>                           | Pass   | 0.000s |



| Test Description                            | Acceptance Criteria                                                                         | Result | Time   |
|---------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Empty list returns empty                    | <code>result == []</code>                                                                   | Pass   | 0.000s |
| Filter 1 year keeps only recent             | <code>"2010-01-01" not in [f["date"] for f in result]</code>                                | Pass   | 0.000s |
| Filter keeps recent                         | <code>"2000-06-01" not in dates;</code><br><code>"2020-06-01" in dates</code>               | Pass   | 0.000s |
| Large years returns all                     | <code>len(result) == len(flows)</code>                                                      | Pass   | 0.001s |
| None years returns all                      | <code>len(gen.filter_by_years(flows, years=None)) == 2</code>                               | Pass   | 0.000s |
| None years returns all                      | <code>len(result) == len(flows)</code>                                                      | Pass   | 0.000s |
| All dates within range kept                 | <code>len(gen.filter_by_years(flows, years=5)) == 3</code>                                  | Pass   | 0.000s |
| Zero years returns only most recent year    | <code>"2020-01-01" in [f["date"] for f in result]</code>                                    | Pass   | 0.000s |
| Empty property does not crash               | <code>isinstance(self._page().generate_elements(_make_property(), list))</code>             | Pass   | 0.000s |
| Has financial header                        | <code>any("Financial" in t for t in texts)</code>                                           | Pass   | 0.000s |
| Has table                                   | <code>any(isinstance(e, Table) for e in result)</code>                                      | Pass   | 0.000s |
| No area skips value per sqm                 | <code>isinstance(result, list)</code>                                                       | Pass   | 0.000s |
| No property value                           | <code>isinstance(result, list)</code>                                                       | Pass   | 0.000s |
| No valuation data                           | <code>isinstance(result, list)</code>                                                       | Pass   | 0.000s |
| Other valuation field with value in name    | <code>isinstance(result, list)</code>                                                       | Pass   | 0.000s |
| Other valuation field without value in name | <code>isinstance(result, list)</code>                                                       | Pass   | 0.000s |
| Returns list                                | <code>isinstance(self._page().generate_elements(_make_property(), list))</code>             | Pass   | 0.000s |
| Returns non empty list                      | <code>len(self._page().generate_elements(_make_property(), list)) &gt; 0</code>             | Pass   | 0.001s |
| Value per sqm shown when area present       | <code>isinstance(result, list)</code>                                                       | Pass   | 0.000s |
| Atl basin                                   | <code>"EVT-001" in ids; "EVT-002" in ids</code><br><code>(+1 more)</code>                   | Pass   | 0.001s |
| Case insensitive                            | <code>result is not None; len(result) &gt;= 1</code>                                        | Pass   | 0.001s |
| Unknown basin returns empty                 | <code>loader.find_by_basin("XYZ") == []</code>                                              | Pass   | 0.000s |
| Wpac basin                                  | <code>len(result) == 1;</code><br><code>loader.get_entity_id(result[0]) == "EVT-003"</code> | Pass   | 0.001s |
| All category 4                              | <code>all(s["TCEvent"]["Header"]["Category"] == 4 ...)</code>                               | Pass   | 0.001s |
| Category 3                                  | <code>"EVT-003" in ids</code>                                                               | Pass   | 0.001s |

| Test Description                                                           | Acceptance Criteria                                                                                                           | Result | Time   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Category 5                                                                 | <code>len(result) == 1;</code><br><code>loader.get_entity_id(result[0])</code><br><code>== "EVT-002"</code>                   | Pass   | 0.000s |
| No matches returns empty                                                   | <code>loader.find.by_category(99) == []</code>                                                                                | Pass   | 0.001s |
| Find by county case insensitive                                            | <code>len(results) == 1</code>                                                                                                | Pass   | 0.001s |
| Find by county empty portfolio                                             | <code>results == []</code>                                                                                                    | Pass   | 0.001s |
| Find by county exact match                                                 | <code>len(results) == 1;</code><br><code>results[0]["PropertyHeader"]["Header"]["PropertyID"]</code><br><code>== "..."</code> | Pass   | 0.001s |
| Property with no County field should not crash.                            | <code>results == []</code>                                                                                                    | Pass   | 0.001s |
| Find by county no match                                                    | <code>results == []</code>                                                                                                    | Pass   | 0.001s |
| Find by county partial match                                               | <code>len(results) == 2;</code> <code>ids ==</code><br><code>{"PROP-001", "PROP-003"}</code>                                  | Pass   | 0.001s |
| Case insensitive                                                           | <code>result is not None; len(result) &gt;= 1</code>                                                                          | Pass   | 0.000s |
| Exact match                                                                | <code>result is not None;</code><br><code>loader.get_entity_id(result) ==</code><br><code>"EVT-001"</code>                    | Pass   | 0.000s |
| No match returns none                                                      | <code>loader.find.by_name("Nonsense Storm")</code><br><code>is None</code>                                                    | Pass   | 0.000s |
| Partial match                                                              | <code>result is not None;</code><br><code>loader.get_entity_id(result) ==</code><br><code>"EVT-003"</code>                    | Pass   | 0.001s |
| Missing season returns empty                                               | <code>loader.find.by_season(1900) == []</code>                                                                                | Pass   | 0.001s |
| Season 2017                                                                | <code>len(result) == 1;</code><br><code>loader.get_entity_id(result[0])</code><br><code>== "EVT-001"</code>                   | Pass   | 0.001s |
| Season with multiple                                                       | <code>len(result) == 1</code>                                                                                                 | Pass   | 0.001s |
| Dict with 'flood_gauges' list → finds first gauge correctly.               | <code>result["FloodGauge"]["Header"]["GaugeName"]</code><br><code>== "Gauge One"</code>                                       | Pass   | 0.000s |
| Dict with 'flood_gauges' list → finds correct gauge by ID.                 | <code>result["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "GAUGE-002"</code>                                         | Pass   | 0.000s |
| Plain list of gauge dicts → finds the correct one.                         | <code>result["FloodGauge"]["Header"]["GaugeID"]</code><br><code>== "GAUGE-B"</code>                                           | Pass   | 0.000s |
| Dict with 'FloodGauge' key (single object) → returns the dict itself.      | <code>result is sample_gauge_data</code>                                                                                      | Pass   | 0.000s |
| Flood gauges key found                                                     | <code>result == data["flood_gauges"][0]</code>                                                                                | Pass   | 0.000s |
| Dict with neither 'flood_gauges' nor 'FloodGauge' key → ValueError raised. | —                                                                                                                             | Pass   | 0.000s |
| Invalid dict structure raises                                              | —                                                                                                                             | Pass   | 0.000s |

| Test Description                                                                | Acceptance Criteria                                      | Result | Time   |
|---------------------------------------------------------------------------------|----------------------------------------------------------|--------|--------|
| Invalid type raises                                                             | —                                                        | Pass   | 0.000s |
| Passing a string instead of dict/list<br>→ ValueError raised.                   | —                                                        | Pass   | 0.000s |
| List input found                                                                | result == gauges[0]                                      | Pass   | 0.000s |
| Malformed gauge entry skipped                                                   | result == data["flood_gauges"][1]                        | Pass   | 0.000s |
| Gauge ID not in plain list → Val-<br>ueError raised.                            | —                                                        | Pass   | 0.000s |
| Gauge ID not present in data →<br>ValueError raised.                            | —                                                        | Pass   | 0.000s |
| Not found raises value error                                                    | —                                                        | Pass   | 0.000s |
| Single gauge dict returns self                                                  | result is data                                           | Pass   | 0.000s |
| Line 361: dict with neither<br>'flood_gauges' nor 'FloodGauge' →<br>ValueError. | —                                                        | Pass   | 0.000s |
| Flood gauges key first gauge                                                    | result["FloodGauge"]["Header"]["GaugeID"]<br>== "G-A"    | Pass   | 0.000s |
| Flood gauges key last gauge                                                     | result["FloodGauge"]["Header"]["GaugeID"]<br>== "G-C"    | Pass   | 0.000s |
| Line 375: gauge_id not present in<br>any entry → ValueError.                    | —                                                        | Pass   | 0.000s |
| Integer input raises                                                            | —                                                        | Pass   | 0.000s |
| Line 363: list input → list used di-<br>rectly as gauge collection.             | result["FloodGauge"]["Header"]["GaugeID"]<br>== "G-LIST" | Pass   | 0.000s |
| Malformed entry missing flood<br>gauge key skipped                              | result["FloodGauge"]["Header"]["GaugeID"]<br>== "G-REAL" | Pass   | 0.000s |
| Malformed entry with none skipped                                               | result["FloodGauge"]["Header"]["GaugeID"]<br>== "G-OK"   | Pass   | 0.000s |
| Single flood gauge dict returns itself                                          | result is data                                           | Pass   | 0.000s |
| String input raises                                                             | —                                                        | Pass   | 0.000s |
| Finds existing meeting                                                          | meeting is not None; meeting["id"] ==<br>"M1" (+1 more)  | Pass   | 0.030s |
| Returns error for missing meeting                                               | meeting is None; err is not None                         | Pass   | 0.026s |
| Dict with mortgages found                                                       | result["amount"] == 100                                  | Pass   | 0.000s |
| Dict with mortgages found                                                       | result["amount"] == 100                                  | Pass   | 0.000s |
| Dict with mortgages not found re-<br>turns none                                 | result is None                                           | Pass   | 0.000s |
| Dict with mortgages not found re-<br>turns none                                 | result is None                                           | Pass   | 0.000s |
| Dict without mortgages key returns<br>data itself                               | result is not None                                       | Pass   | 0.000s |
| Empty mortgages list returns none                                               | result is None                                           | Pass   | 0.000s |

| Test Description                                           | Acceptance Criteria                                                  | Result | Time   |
|------------------------------------------------------------|----------------------------------------------------------------------|--------|--------|
| Lines 406-407: non-dict, non-list -j<br>return data as-is. | result == "some_string"                                              | Pass   | 0.000s |
| Invalid type returns data as is                            | result == "some_string"                                              | Pass   | 0.000s |
| List input found                                           | result == data[0];    result["amount"]<br>== 200                     | Pass   | 0.000s |
| List input found                                           | result["amount"] == 200                                              | Pass   | 0.000s |
| List input not found returns none                          | result is None                                                       | Pass   | 0.000s |
| Dict without list key wraps itself                         | result is data                                                       | Pass   | 0.000s |
| Dict without properties wraps in list                      | result == data                                                       | Pass   | 0.000s |
| Empty properties list raises                               | —                                                                    | Pass   | 0.000s |
| Invalid type raises                                        | —                                                                    | Pass   | 0.000s |
| Invalid type raises value error                            | —                                                                    | Pass   | 0.000s |
| List input                                                 | result == data[0]                                                    | Pass   | 0.000s |
| List input found                                           | result == data[0];    result["amount"]<br>== 200                     | Pass   | 0.000s |
| List not found raises                                      | —                                                                    | Pass   | 0.000s |
| None raises value error                                    | —                                                                    | Pass   | 0.000s |
| Not found raises                                           | —                                                                    | Pass   | 0.000s |
| Not found raises value error                               | —                                                                    | Pass   | 0.000s |
| Portfolio key                                              | result == data["portfolio"][0]                                       | Pass   | 0.000s |
| Portfolio key found                                        | result["PropertyHeader"]["Header"]["Property"]<br>== "PA"            | Pass   | 0.000s |
| Properties key                                             | result == data["properties"][0]                                      | Pass   | 0.000s |
| Properties key found                                       | result["PropertyHeader"]["Header"]["Property"]<br>== "P-002"         | Pass   | 0.000s |
| Train all avg auc in range                                 | 0.0 <= summary["avg_auc_roc"] <= 1.0                                 | Pass   | 0.110s |
| Train all creates summary file                             | (model_dir /<br>"training_summary.json").exists()                    | Pass   | 0.313s |
| Train all returns summary                                  | "num_gauges" in summary; "num_trained"<br>in summary (+1 more)       | Pass   | 0.351s |
| Some gauges trained, some skipped.                         | summary["num_trained"] >= 1;<br>summary["num_skipped"] >= 1          | Pass   | 0.150s |
| Train gauge creates joblib file                            | (model_dir /<br>"GAUGE-001.joblib").exists()                         | Pass   | 0.287s |
| Train gauge has metrics                                    | "metrics" in result;    "auc_roc" in<br>result["metrics"] (+1 more)  | Pass   | 0.120s |
| Train gauge returns dict                                   | isinstance(result, dict)                                             | Pass   | 3.054s |
| Train gauge single class skipped                           | result["status"] == "skipped";<br>result["reason"] == "single_class" | Pass   | 0.002s |
| Train gauge status trained                                 | result["status"] == "trained"                                        | Pass   | 0.150s |
| Flood above trigger                                        | depth > 0.0                                                          | Pass   | 0.000s |

| Test Description                                                                                | Acceptance Criteria                                                                                                       | Result | Time   |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Flood attenuates with distance                                                                  | <code>depth_close &gt; depth_far</code>                                                                                   | Pass   | 0.000s |
| High elevation no flood                                                                         | <code>depth == 0.0</code>                                                                                                 | Pass   | 0.000s |
| No flood below trigger                                                                          | <code>depth == 0.0</code>                                                                                                 | Pass   | 0.000s |
| Calculate time above warning level.                                                             | <code>duration &gt; 0</code>                                                                                              | Pass   | 0.000s |
| Duration should be in reasonable range.                                                         | <code>5 &lt; duration &lt; 30</code>                                                                                      | Pass   | 0.000s |
| No time above threshold for normal readings.                                                    | <code>duration == 0</code>                                                                                                | Pass   | 0.000s |
| Available gauges                                                                                | <code>"GAUGE-001" in gauges</code>                                                                                        | Pass   | 0.000s |
| Available gauges empty dir                                                                      | <code>predictor.available_gauges() == []</code>                                                                           | Pass   | 0.000s |
| Lines 188-192: <code>_load_summary()</code> loads <code>training_summary.json</code> .          | <code>'num_gauges' in summary; 'num_trained' in summary</code>                                                            | Pass   | 0.002s |
| <code>_load_summary</code> caches on second call.                                               | <code>s1 is s2</code>                                                                                                     | Pass   | 0.001s |
| Predict caches model                                                                            | <code>"GAUGE-001" in predictor._models</code>                                                                             | Pass   | 0.008s |
| Predict high level higher probability                                                           | <code>p_high &gt; p_low</code>                                                                                            | Pass   | 0.008s |
| Predict in range                                                                                | <code>0.0 &lt;= prob &lt;= 1.0</code>                                                                                     | Pass   | 0.020s |
| Predict missing model raises                                                                    | —                                                                                                                         | Pass   | 0.000s |
| Predict returns float                                                                           | <code>isinstance(prob, float)</code>                                                                                      | Pass   | 0.012s |
| Predict series all in range                                                                     | <code>all(0.0 &lt;= p &lt;= 1.0 for p in result)</code>                                                                   | Pass   | 0.007s |
| Predict series default derivatives                                                              | <code>len(result) == 2</code>                                                                                             | Pass   | 0.007s |
| Predict series returns list                                                                     | <code>isinstance(result, list); len(result) == 4</code>                                                                   | Pass   | 0.007s |
| Predict series with derivatives                                                                 | <code>len(result) == 3</code>                                                                                             | Pass   | 0.008s |
| <code>predict_surface</code> returns a 2D probability grid (hours × levels).                    | <code>'water_levels' in result; 'hours' in result (+3 more)</code>                                                        | Pass   | 0.010s |
| Lines 59-74: model builds without error given valid inputs.                                     | <code>model.properties is not None; model.flood_event == {"GAUGE-000": 1.5}</code>                                        | Pass   | 8.737s |
| Line 72-74: <code>correlation_matrix</code> is set after <code>__init__</code> .                | <code>model.correlation_matrix is not None</code>                                                                         | Pass   | 0.001s |
| Lines 62-63: <code>custom_correlation_distance</code> and <code>base_correlation</code> stored. | <code>model.correlation_distance == 500; model.base_correlation == 0.6</code>                                             | Pass   | 0.001s |
| Lines 86-88: missing elevation → warning + default 20.0.                                        | <code>any("elevation" in str(warning.message).lower() for warni...; (model.properties["elevation"] == 20.0).all())</code> | Pass   | 0.001s |

| Test Description                                                              | Acceptance Criteria                                                                                                            | Result | Time   |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Lines 94-95: missing floor_level_metres → default 0.0.                        | "floor_level_metres" in model.properties.columns;<br>(model.properties["floor_level_metres"] == 0.0).all()                     | Pass   | 0.001s |
| Line 81: missing 'geometry' column raises ValueError.                         | —                                                                                                                              | Pass   | 0.000s |
| Lines 83-84: property_id auto-assigned when absent.                           | "property_id" in model.properties.columns;<br>model.properties["property_id"].iloc[0].startswith("PROP.")                      | Pass   | 0.001s |
| Lines 90-92: missing property_type → warning + default 'residential'.         | any("property_type" in str(warning.message).lower()<br>for w...;<br>(model.properties["property_type"] == "residential").all() | Pass   | 0.001s |
| Line 81: missing 'value' column raises ValueError.                            | —                                                                                                                              | Pass   | 0.000s |
| Line 59: properties is a copy, original not mutated.                          | set(props.columns) == original.cols                                                                                            | Pass   | 0.001s |
| Line 71: kdtree built when gauge_data provided.                               | model.gauge_data is not None                                                                                                   | Pass   | 0.005s |
| Lines 131-139: analyze_risk_clusters() returns DataFrame, sets risk_clusters. | isinstance(result, pd.DataFrame);<br>model.risk_clusters is not None                                                           | Pass   | 0.076s |
| Lines 114-120: analyze_spatial_concentration() returns DataFrame.             | isinstance(result, pd.DataFrame)                                                                                               | Pass   | 0.007s |
| Lines 122-129: calculate_advanced_metrics() returns dict.                     | isinstance(result, dict)                                                                                                       | Pass   | 0.002s |
| Line 103: calculate_direct_impacts() returns ndarray.                         | isinstance(impacts, np.ndarray); len(impacts) == len(model.properties)                                                         | Pass   | 0.001s |
| Line 99: calculate_flood_depths() returns ndarray.                            | isinstance(depths, np.ndarray); len(depths) == len(model.properties)                                                           | Pass   | 0.001s |
| Line 151: _haversine_distance() delegates to haversine_distance.              | isinstance(d, float); d > 0                                                                                                    | Pass   | 0.000s |
| Lines 107-112: simulate_portfolio_impact() returns dict.                      | isinstance(result, dict)                                                                                                       | Pass   | 0.016s |
| Default params produces data                                                  | len(result) > 0; "readings" in result[0]                                                                                       | Pass   | 0.001s |



| Test Description                                                                | Acceptance Criteria                                                                                        | Result | Time   |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------|--------|
| String start time is parsed                                                     | <code>len(result) == 50</code>                                                                             | Pass   | 0.000s |
| Unknown zone                                                                    | <code>"Unknown" in r or "Zone X" in r</code>                                                               | Pass   | 0.000s |
| Zone 1 low risk                                                                 | <code>"Low" in r</code>                                                                                    | Pass   | 0.000s |
| Zone 2 medium risk                                                              | <code>"Medium" in r</code>                                                                                 | Pass   | 0.000s |
| Zone 3a high risk                                                               | <code>"High" in r</code>                                                                                   | Pass   | 0.000s |
| Zone 3b floodplain                                                              | <code>"floodplain" in r.lower() or<br/>"Functional" in r</code>                                            | Pass   | 0.000s |
| Zone in page output                                                             | <code>isinstance(result, list)</code>                                                                      | Pass   | 0.002s |
| Large value                                                                     | <code>'1,000,000' in result;<br/>result.startswith('£')</code>                                             | Pass   | 0.000s |
| Negative                                                                        | <code>'5,000' in result</code>                                                                             | Pass   | 0.000s |
| Non numeric fallback                                                            | <code>fmt_gbp('N/A') == 'N/A'</code>                                                                       | Pass   | 0.000s |
| None fallback                                                                   | <code>result == 'None'</code>                                                                              | Pass   | 0.000s |
| Positive float                                                                  | <code>fmt_gbp(1234.56) == '£1,235'</code>                                                                  | Pass   | 0.021s |
| Zero                                                                            | <code>fmt_gbp(0) == '£0'</code>                                                                            | Pass   | 0.000s |
| line 286: <code>f'{value:.6f}'</code> .                                         | <code>result == '51.500000'</code>                                                                         | Pass   | 0.000s |
| Negative coordinate                                                             | <code>result == '-0.100000'</code>                                                                         | Pass   | 0.000s |
| None delegates to format value                                                  | <code>page._format.coordinate(None) == 'Not<br/>specified'</code>                                          | Pass   | 0.000s |
| None currency returns not specified                                             | <code>result == 'Not specified'</code>                                                                     | Pass   | 0.000s |
| String currency delegates                                                       | <code>result == "N/A"</code>                                                                               | Pass   | 0.000s |
| Abbreviation replacements                                                       | <code>'GPS' in result</code>                                                                               | Pass   | 0.000s |
| Camel case split                                                                | <code>'Gauge' in result</code>                                                                             | Pass   | 0.000s |
| Non numeric delegates                                                           | <code>page._format.frequency(None) == 'Not<br/>specified'</code>                                           | Pass   | 0.000s |
| Positive returns times                                                          | <code>page._format.frequency(3) == '3<br/>times'</code>                                                    | Pass   | 0.000s |
| line 294: <code>'Never'</code> .                                                | <code>page._format.frequency(0) == 'Never'</code>                                                          | Pass   | 0.000s |
| Integer value                                                                   | <code>result == '5.000 m'</code>                                                                           | Pass   | 0.000s |
| Non numeric delegates to format value                                           | <code>result == 'Not specified'</code>                                                                     | Pass   | 0.000s |
| Numeric value with custom unit                                                  | <code>result == '1.234 km'</code>                                                                          | Pass   | 0.000s |
| line 280: <code>f'{value:.3f} {unit}'</code> .                                  | <code>result == '3.500 m'</code>                                                                           | Pass   | 0.000s |
| None percentage returns not specified                                           | <code>result == 'Not specified'</code>                                                                     | Pass   | 0.000s |
| String percentage delegates                                                     | <code>result == "N/A"</code>                                                                               | Pass   | 0.000s |
| Blank/whitespace-only string (line 272) - <code>i</code> 'Not specified'.       | <code>page._format.value('') == 'Not<br/>specified'; page._format.value(' ') ==<br/>'Not specified'</code> | Pass   | 0.000s |
| False returns no                                                                | <code>page._format.value(False) == 'No'</code>                                                             | Pass   | 0.000s |
| <code>float.is_integer()</code> - <code>i</code> <code>str(int(value))</code> . | <code>result == '3'</code>                                                                                 | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                                                      | Result | Time   |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| float with fractional part -j 2 decimal places.                      | result == '3.14'                                                                         | Pass   | 0.000s |
| int branch (line 270): str(value).                                   | result == '42'                                                                           | Pass   | 0.000s |
| None returns not specified                                           | page._format_value(None) == 'Not specified'                                              | Pass   | 0.000s |
| Normal string returned                                               | page._format_value('hello') == 'hello'                                                   | Pass   | 0.000s |
| True returns yes                                                     | page._format_value(True) == 'Yes'                                                        | Pass   | 0.000s |
| Empty string returns not specified                                   | page._format_value('') == 'Not specified'                                                | Pass   | 0.000s |
| Whitespace string returns not specified                              | page._format_value(' ') == 'Not specified'                                               | Pass   | 0.000s |
| None returns not specified                                           | page._format_value(None) == 'Not specified'                                              | Pass   | 0.000s |
| Pad foundation                                                       | isinstance(result, list)                                                                 | Pass   | 0.000s |
| Pile foundation                                                      | isinstance(result, list)                                                                 | Pass   | 0.000s |
| Raft foundation                                                      | isinstance(result, list)                                                                 | Pass   | 0.000s |
| Strip foundation                                                     | isinstance(result, list)                                                                 | Pass   | 0.000s |
| Unknown foundation                                                   | isinstance(result, list)                                                                 | Pass   | 0.000s |
| Four exceedances high                                                | "High frequency" in str(elements)                                                        | Pass   | 0.001s |
| One exceedance                                                       | "One severe level exceedance" in str(elements)                                           | Pass   | 0.000s |
| Two exceedances moderate                                             | "Moderate frequency" in str(elements)                                                    | Pass   | 0.001s |
| Zero exceedances                                                     | "No severe level exceedances recorded" in str(elements)                                  | Pass   | 0.000s |
| Distance matrix built                                                | thames_model.dist_matrix.shape == (52, 52)                                               | Pass   | 0.000s |
| From dict                                                            | p.base_range_km == 50.0; p.nugget == 0.10 (+1 more)                                      | Pass   | 0.123s |
| Lines 323-325: construct model from a gauge.json file.               | model.n_gauges == 3; model.dist_matrix.shape == (3, 3)                                   | Pass   | 0.023s |
| Loads 52 gauges                                                      | thames_model.n_gauges == 52                                                              | Pass   | 0.000s |
| Full js contains iife                                                | '(function()'' in full_governance_js or '(function ()'' in ...                           | Pass   | 0.000s |
| The Leaflet control that adds the Regulatory shield must be present. | 'GovernanceControl' in full_governance_js; 'Regulatory Compliance' in full_governance_js | Pass   | 0.000s |
| Full js contains render audit reports                                | 'renderAuditReports' in full_governance_js                                               | Pass   | 0.000s |

| Test Description                                          | Acceptance Criteria                                   | Result | Time   |
|-----------------------------------------------------------|-------------------------------------------------------|--------|--------|
| Full js contains script tags                              | '<script>' in full_governance_js;                     | Pass   | 0.000s |
|                                                           | '</script>' in full_governance_js                     |        |        |
| Full js contains show mg panel                            | 'showMgPanel' in full_governance_js                   | Pass   | 0.000s |
| Terminal popup added in last session must be present.     | '_openTerminalPopup' in full_governance_js;           | Pass   | 0.000s |
|                                                           | '_auditTerminalRefreshNow' in full_governance_js      |        |        |
| Full js is str                                            | isinstance(full_governance_js, str)                   | Pass   | 0.000s |
| Full js non empty                                         | len(full_governance_js) > 10.000                      | Pass   | 0.000s |
| Run node -check on the compiled governance script.        | result.returncode == 0                                | Pass   | 1.278s |
| Base level calculated                                     | abs(gc.base_level - 1.4) < 0.01                       | Pass   | 0.000s |
| Creates gauge config                                      | isinstance(gc, GaugeConfig)                           | Pass   | 0.000s |
| Defaults when no flood stages                             | gc.flood_alert == 4.0                                 | Pass   | 0.000s |
| Handles case where FloodGauge key is missing (flat data). | gc.flood_alert == 3.5                                 | Pass   | 0.000s |
| Flood alert set                                           | gc.flood_alert == 4.0                                 | Pass   | 0.000s |
| Gauge id set                                              | gc.gauge_id == "GAUGE-099"                            | Pass   | 0.000s |
| Historical high set                                       | gc.historical_high == 6.0                             | Pass   | 0.000s |
| Lines 32-34: loader.list_all() raises → 500 response.     | response.status_code == 500;                          | Pass   | 0.000s |
|                                                           | data['status'] == 'error'                             |        |        |
| Line 19: OPTIONS on /api/v1/gauges returns 200.           | response.status_code == 200;                          | Pass   | 0.000s |
|                                                           | data['status'] == 'ok'                                |        |        |
| Empty historical daily shows fallback                     | any(t for t in texts)                                 | Pass   | 0.000s |
| No timeseries data shows fallback                         | any("No historical" in t or "No" in t for t in texts) | Pass   | 0.000s |
| Returns list                                              | isinstance(result, list); len(result) > 0             | Pass   | 0.000s |
| With observations includes content                        | len(result) >= 2                                      | Skip   | 0.108s |
| With observations returns elements                        | len(result) > 2                                       | Skip   | 0.122s |
| Certified conclusion                                      | "certification standards" in str(result)              | Pass   | 0.001s |
| Empty data does not crash                                 | isinstance(result, list)                              | Pass   | 0.000s |
| Freq high significance                                    | "Regular severe events" in str(result)                | Pass   | 0.001s |
| Freq low significance                                     | "Low frequency events" in str(result)                 | Pass   | 0.001s |
| Freq zero no exceedances conclusion                       | "No historical severe flood events" in str(result)    | Pass   | 0.001s |

| Test Description                                                     | Acceptance Criteria                                        | Result | Time   |
|----------------------------------------------------------------------|------------------------------------------------------------|--------|--------|
| Freq zero significance                                               | "No severe exceedances" in<br>str(result)                  | Pass   | 0.001s |
| Gauge overview summary section                                       | any("Gauge Overview Summary" in t<br>for t in texts)       | Pass   | 0.001s |
| Get gauge id missing                                                 | page._get_gauge_id({}) == "Unknown<br>Gauge"               | Pass   | 0.000s |
| Get gauge id present                                                 | page._get_gauge_id(_make_gauge()) ==<br>"GAUGE-001"        | Pass   | 0.000s |
| Get gauge name missing                                               | page._get_gauge_name({}) == "Unknown<br>Gauge Name"        | Pass   | 0.000s |
| Get gauge name present                                               | page._get_gauge_name(_make_gauge()) ==<br>"Chelsea Bridge" | Pass   | 0.000s |
| Has tables                                                           | any(isinstance(e, Table) for e in<br>result)               | Pass   | 0.001s |
| High risk conclusion                                                 | "high flood risk" in str(result)                           | Pass   | 0.001s |
| hist_high < severe_level → 'Exceeds<br>severe warning' significance. | "Exceeds severe warning" in<br>str(result)                 | Pass   | 0.001s |
| hist_high ≥ severe_level → 'Within<br>expected range' significance.  | "Within expected range" in<br>str(result)                  | Pass   | 0.001s |
| Key metrics summary section                                          | any("Key Metrics Summary" in t for t<br>in texts)          | Pass   | 0.001s |
| Low risk conclusion                                                  | "Low flood risk" in str(result)                            | Pass   | 0.001s |
| Moderate risk conclusion                                             | "Moderate flood risk" in str(result)                       | Pass   | 0.001s |
| Operational status conclusion                                        | "fully operational" in str(result)                         | Pass   | 0.001s |
| Operational status summary section                                   | any("Operational Status Summary" in<br>t for t in texts)   | Pass   | 0.001s |
| Report conclusions section                                           | any("Report Conclusions" in t for t<br>in texts)           | Pass   | 0.001s |
| Returns list                                                         | isinstance(result, list); len(result)<br>> 0               | Pass   | 0.001s |
| Timeseries alert conclusion                                          | "ALERT" in str(result)                                     | Pass   | 0.001s |
| Timeseries normal conclusion                                         | "normal water levels" in str(result)                       | Pass   | 0.001s |
| Timeseries severe conclusion                                         | "CRITICAL" in str(result)                                  | Pass   | 0.001s |
| Timeseries warning conclusion                                        | "WARNING" in str(result)                                   | Pass   | 0.001s |
| Title present                                                        | any("Comprehensive Data Summary" in<br>t for t in texts)   | Pass   | 0.001s |
| Historical graph no dates fallback                                   | isinstance(result, list)                                   | Pass   | 0.000s |
| Historical graph with thresholds                                     | isinstance(result, list); len(result)<br>> 0               | Pass   | 0.044s |
| No storm responses shows message                                     | any("No storm" in t for t in texts)                        | Pass   | 0.000s |

| Test Description                                                           | Acceptance Criteria                                                      | Result | Time   |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------|--------|--------|
| No timeseries data shows no data message                                   | <code>any("No historical" in t or "No storm" in t for t in texts)</code> | Pass   | 0.000s |
| Realised floods no exceedances                                             | <code>any("No flood events" in t for t in texts)</code>                  | Pass   | 0.000s |
| Realised floods with exceedances                                           | <code>len(tables) &gt;= 1</code>                                         | Pass   | 0.000s |
| Returns list                                                               | <code>isinstance(result, list); len(result) &gt; 0</code>                | Pass   | 0.000s |
| With historical daily returns more elements                                | <code>len(result) &gt; 2</code>                                          | Pass   | 0.774s |
| With storm responses includes table                                        | <code>len(tables) &gt;= 1</code>                                         | Pass   | 0.001s |
| Worst storms severity classification                                       | <code>isinstance(result, list)</code>                                    | Pass   | 0.000s |
| Worst storms table top 10                                                  | <code>len(tables) == 1</code>                                            | Pass   | 0.000s |
| Current level at alert                                                     | <code>"FLOOD ALERT" in str(result)</code>                                | Pass   | 0.001s |
| Current level at severe                                                    | <code>"SEVERE FLOOD WARNING" in str(result)</code>                       | Pass   | 0.001s |
| Current level at warning                                                   | <code>"FLOOD WARNING" in str(result)</code>                              | Pass   | 0.001s |
| Current level below alert is normal                                        | <code>"Normal" in str(result)</code>                                     | Pass   | 0.001s |
| Empty data does not crash                                                  | <code>isinstance(result, list)</code>                                    | Pass   | 0.000s |
| Get gauge id missing                                                       | <code>page._get_gauge_id({}) == "Unknown Gauge"</code>                   | Pass   | 0.000s |
| Get gauge id present                                                       | <code>page._get_gauge_id(_make_gauge()) == "GAUGE-001"</code>            | Pass   | 0.000s |
| Has tables                                                                 | <code>any(isinstance(e, Table) for e in result)</code>                   | Pass   | 0.000s |
| Historical exceedance section                                              | <code>any("Historical Threshold Exceedance" in t for t in texts)</code>  | Pass   | 0.000s |
| No timeseries shows message                                                | <code>any("No current timeseries data" in t for t in texts)</code>       | Pass   | 0.000s |
| Returns list                                                               | <code>isinstance(result, list); len(result) &gt; 0</code>                | Pass   | 0.001s |
| Threshold analysis section                                                 | <code>any("Threshold Analysis" in t for t in texts)</code>               | Pass   | 0.000s |
| Alert-to-warning and warning-to-severe ranges are added to analysis table. | <code>isinstance(result, list)</code>                                    | Pass   | 0.000s |
| Title present                                                              | <code>any("Flood Stages" in t for t in texts)</code>                     | Pass   | 0.001s |
| Uk thresholds section                                                      | <code>any("UK Flood Stage Thresholds" in t for t in texts)</code>        | Pass   | 0.000s |
| With timeseries shows current status section                               | <code>any("Current Status vs Thresholds" in t for t in texts)</code>     | Pass   | 0.001s |

| Test Description                                                           | Acceptance Criteria                                                            | Result | Time   |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|--------|
| Lines 251-254: generate_basic_report() body on real generator.             | path.exists(); path.suffix == ".pdf"                                           | Pass   | 0.014s |
| Line 315: analysis branch in gauge_generator.generate_report.              | path.exists()                                                                  | Pass   | 0.004s |
| Lines 321-328: auto_open=True — open attempt caught, does not raise.       | path.exists()                                                                  | Pass   | 0.015s |
| Lines 307-311: generate_report() in gauge_generator.py, basic type.        | path.exists()                                                                  | Pass   | 0.015s |
| Lines 316-317: unknown type falls back to generate_report.                 | path.exists()                                                                  | Pass   | 0.010s |
| Line 313: monitoring branch in gauge_generator.generate_report.            | path.exists()                                                                  | Pass   | 0.003s |
| Line 119: generate_report(pages_to_include=None) calls _auto_select_pages. | path.exists()                                                                  | Pass   | 0.011s |
| Lines 102-103 in __init__.py: auto_open=True, open returns False.          | result == fake_pdf                                                             | Pass   | 0.015s |
| Lines 285-286: configure(panel_height=...) updates panel_height.           | obj.panel_height == "600px";<br>obj.panel_height == "700px" (+1 more)          | Pass   | 0.000s |
| Lines 283-284: configure(panel_width=...) updates panel_width.             | obj.panel_width == "800px";<br>obj.panel_width == "900px" (+1 more)            | Pass   | 0.000s |
| Line 290: get_statistics() returns dict with panel_width and panel_height. | stats["panel_width"] == "700px";<br>stats["panel_height"] == "500px" (+4 more) | Pass   | 0.000s |
| Lines 480-481: configure(panel_height=...) updates panel_height.           | obj.panel_height == "600px";<br>obj.panel_height == "700px" (+1 more)          | Pass   | 0.000s |
| Lines 478-479: configure(panel_width=...) updates panel_width.             | obj.panel_width == "800px";<br>obj.panel_width == "900px" (+1 more)            | Pass   | 0.000s |
| Line 485: get_statistics() returns dict.                                   | stats["panel_width"] == "700px";<br>stats["panel_height"] == "500px" (+4 more) | Pass   | 0.000s |
| Empty return period levels no extra table                                  | len(tables) >= 1                                                               | Pass   | 0.000s |



| Test Description                                                          | Acceptance Criteria                                                                                                                     | Result | Time   |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| No hazard curve shows fallback                                            | <code>any("No hazard" in t or "hazard" in t.lower() for t in te...</code>                                                               | Pass   | 0.000s |
| Returns list                                                              | <code>isinstance(result, list)</code>                                                                                                   | Pass   | 0.000s |
| With hazard curve basic                                                   | <code>len(result) &gt; 2</code>                                                                                                         | Pass   | 0.000s |
| With return period levels renders table                                   | <code>len(tables) &gt;= 2</code>                                                                                                        | Pass   | 0.000s |
| GET /api/v1/gauges/{id}/hazard returns hazard data for known gauge.       | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+6 more)                                         | Pass   | 0.003s |
| GEV parameters should be present and numeric.                             | <code>gev['location'] is not None;</code><br><code>gev['scale'] is not None</code> (+3 more)                                            | Pass   | 0.003s |
| GET /api/v1/gauges/{id}/hazard returns 404 for nonexistent gauge.         | <code>response.status_code == 404</code>                                                                                                | Pass   | 0.002s |
| GET /api/v1/gauges/{id}/hazard returns 404 for gauge without hazard data. | <code>response.status_code == 404</code>                                                                                                | Pass   | 0.003s |
| GET /api/v1/gauges/{id}/storms returns storm data for known gauge.        | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+2 more)                                         | Pass   | 0.004s |
| calculate_statistics_from_levels via class.                               | <code>isinstance(stats, dict)</code>                                                                                                    | Pass   | 0.002s |
| Calculate statistics via class                                            | <code>isinstance(stats, dict)</code>                                                                                                    | Pass   | 0.000s |
| Custom years                                                              | <code>ghd.years_of_history == 30</code>                                                                                                 | Pass   | 0.000s |
| Default years                                                             | <code>ghd.years_of_history == 50</code>                                                                                                 | Pass   | 0.000s |
| Filter by years explicit                                                  | <code>isinstance(result, list)</code>                                                                                                   | Pass   | 0.000s |
| Filter by years none                                                      | <code>result == flows</code>                                                                                                            | Pass   | 0.000s |
| generate_from_gauge_portfolio with explicit years overrides default.      | <code>result["years_included"] == 3</code>                                                                                              | Pass   | 0.255s |
| generate_from_gauge_portfolio via class method.                           | <code>isinstance(result, dict);</code><br><code>result["years_included"] == 5</code>                                                    | Pass   | 0.217s |
| When years=None, uses years_of_history.                                   | <code>result["years_included"] == 10</code>                                                                                             | Pass   | 0.001s |
| Generate from nrfa via class                                              | <code>output.exists();</code> <code>isinstance(result, dict)</code>                                                                     | Pass   | 0.001s |
| generate_synthetic_timeseries via class method.                           | <code>isinstance(result, list); len(result) &gt; 0</code>                                                                               | Pass   | 0.003s |
| Generates per gauge files                                                 | <code>gaugets_dir.exists();</code><br><code>len(list(gaugets_dir.glob("GAUGE-*.json")))</code><br><code>&gt;= N_GAUGES</code> (+2 more) | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                   | Result | Time   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------|--------|--------|
| Output has daily observations                                         | "daily_observations" in data;<br>len(data["daily_observations"]) > 0  | Pass   | 0.016s |
| Parse nrfa csv via class                                              | len(flows) == 10                                                      | Pass   | 0.002s |
| Statistics computed                                                   | "statistics" in data; "mean_level" in data["statistics"] (+1 more)    | Pass   | 0.002s |
| Cdm available flag is bool                                            | hasattr(mod, "CDM_AVAILABLE");<br>isinstance(mod.CDM_AVAILABLE, bool) | Pass   | 0.000s |
| Response has count and gauges keys                                    | "count" in data; "gauges" in data (+1 more)                           | Pass   | 0.001s |
| Returns 200 with no gaugehd files                                     | r.status_code == 200; data["status"] == "success" (+1 more)           | Pass   | 0.001s |
| Returns correct count when files present                              | r.status_code == 200; data["status"] == "success" (+2 more)           | Pass   | 0.001s |
| gauge.json must contain at least one gauge.                           | len(ids) > 0                                                          | Pass   | 0.001s |
| Gauge IDs in gaugehc.json must exist in gauge.json.                   | not orphaned                                                          | Pass   | 0.002s |
| Gauge IDs in gaugets/ files must exist in gauge.json.                 | not orphaned                                                          | Pass   | 0.001s |
| Each gaugets file's internal gauge_id must match its filename.        | internal_id == gf.stem                                                | Pass   | 0.002s |
| gaugets/ files must reference gauge IDs from gauge.json.              | len(missing) == 0                                                     | Pass   | 3.550s |
| Every gauge in gauge.json must have a hazard curve.                   | len(missing) == 0                                                     | Pass   | 0.003s |
| gaugehc.json must not contain IDs absent from gauge.json.             | len(orphan) == 0                                                      | Pass   | 0.003s |
| Every gauge referenced by a trade must exist in gauge.json.           | len(missing) == 0                                                     | Pass   | 0.005s |
| Every gauge with trades must have a hazard curve (for PRS pricing).   | len(missing) == 0                                                     | Pass   | 0.007s |
| gauge_random.py must NOT use GAUGE-{i:03d} style sequential IDs.      | 'GAUGE-{i:}' not in src;<br>'GAUGE-{index}' not in src (+3 more)      | Pass   | 0.000s |
| gauge_random.py contains the GAUGE- prefix.                           | 'GAUGE-' in src                                                       | Pass   | 0.087s |
| gauge_random.py uses uuid for ID generation (not sequential counter). | 'uuid' in src; 'uuid' in src (+1 more)                                | Pass   | 0.000s |
| uuid4()[8] always produces 8 lower-case hex chars — 100 samples.      | GAUGE_ID.RE.match(gauge_id);<br>PROP_ID.RE.match(prop_id) (+2 more)   | Pass   | 0.000s |

| Test Description                                                    | Acceptance Criteria                                                                                                        | Result | Time   |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Accepts valid uuid derived ids                                      | <code>GAUGE_ID.RE.match('GAUGE-9042bd95');</code><br><code>GAUGE_ID.RE.match('GAUGE-a1b2c3d4')</code><br>(+2 more)         | Pass   | 0.000s |
| Rejects non hex chars                                               | <code>not GAUGE_ID.RE.match('GAUGE-9042xd95');</code><br><code>not GAUGE_ID.RE.match('GAUGE-9042gd95')</code>              | Pass   | 0.000s |
| GAUGE-001, GAUGE-002 are NOT protocol-compliant.                    | <code>not GAUGE_ID.RE.match('GAUGE-001');</code><br><code>not GAUGE_ID.RE.match('GAUGE-002')</code><br>(+5 more)           | Pass   | 0.000s |
| Gauge IDs must be lowercase; GAUGE-9042BD95 is invalid.             | <code>not GAUGE_ID.RE.match('GAUGE-9042BD95');</code><br><code>not GAUGE_ID.RE.match('GAUGE-A1B2C3D4')</code>              | Pass   | 0.000s |
| Rejects wrong length                                                | <code>not GAUGE_ID.RE.match('GAUGE-9042bd9');</code><br><code>not GAUGE_ID.RE.match('GAUGE-9042bd951')</code><br>(+1 more) | Pass   | 0.000s |
| Gauge id present in all responses                                   | <code>"gauge_id" not in resp;</code><br><code>gr.get("gauge_id") ==</code><br><code>"GAUGE-deadbeef"</code>                | Pass   | 0.003s |
| Multiple gauges have correct ids                                    | <code>found_ids == {"GAUGE-00000001",</code><br><code>"GAUGE-00000002", "GAUGE-...</code>                                  | Pass   | 0.003s |
| Corrupted gauge json does not crash                                 | <code>result["processing_stats"]["successful_properties"]</code><br><code>== 2</code>                                      | Pass   | 0.628s |
| Gauge id map empty when no gauge json                               | <code>isinstance(gen._gauge_id_map, dict)</code>                                                                           | Pass   | 0.789s |
| When gauge.json exists with proper structure, gauge IDs are loaded. | <code>0 in gen._gauge_id_map;</code><br><code>gen._gauge_id_map[0] == "GAUGE-111"</code>                                   | Pass   | 0.614s |
| Also supports 'gauges' as the top-level key.                        | <code>gen._gauge_id_map.get(0) == "G-AAA"</code>                                                                           | Pass   | 0.608s |
| Get most recent reading.                                            | <code>current ==</code><br><code>flood_event_readings[-1]['level']</code>                                                  | Pass   | 0.000s |
| Detect falling water levels.                                        | <code>trend == "FALLING"</code>                                                                                            | Pass   | 0.000s |
| Detect rising water levels.                                         | <code>trend == "RISING"</code>                                                                                             | Pass   | 0.000s |
| Detect stable water levels.                                         | <code>trend == "STABLE"</code>                                                                                             | Pass   | 0.000s |
| Detect peak level in time series.                                   | <code>peak['level'] &gt; 6.0; 'timestamp' in</code><br><code>peak</code>                                                   | Pass   | 0.000s |
| Peak should occur around hour 24.                                   | <code>20 &lt; hours_to_peak &lt; 28</code>                                                                                 | Pass   | 0.000s |
| Count                                                               | <code>GaugeLoader(tmp_path).count() == 4</code>                                                                            | Pass   | 0.001s |
| Floodgauges key also works                                          | <code>GaugeLoader(tmp_path).count() == 2</code>                                                                            | Pass   | 0.001s |
| Load all returns list                                               | <code>len(GaugeLoader(tmp_path).load_all())</code><br><code>== 3</code>                                                    | Pass   | 0.001s |
| Load missing file returns empty                                     | <code>GaugeLoader(tmp_path).load_all() ==</code><br><code>[]</code>                                                        | Pass   | 0.000s |

| Test Description                               | Acceptance Criteria                                                                                             | Result | Time   |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------|--------|
| Test GaugeID extraction from nested structure. | <code>gauge.id is not None;</code><br><code>gauge.id.startswith("GAUGE-")</code>                                | Pass   | 0.001s |
| Test gauge summary generation.                 | <code>len(summaries) == 3;</code> <code>'gaugeId' in summary (+5 more)</code>                                   | Pass   | 0.001s |
| Test loading gauge portfolio.                  | <code>len(gauges) == 3</code>                                                                                   | Pass   | 0.001s |
| Caching works                                  | <code>loader.load_all() is loader.load_all()</code>                                                             | Pass   | 0.001s |
| Clear cache                                    | <code>not loader..cache_valid</code>                                                                            | Pass   | 0.001s |
| Force reload                                   | <code>len(loader.load_all(force_reload=True)) == 5</code>                                                       | Pass   | 0.001s |
| Test getting active gauges.                    | <code>len(active) == 2; status == "Active"</code>                                                               | Pass   | 0.001s |
| Test getting gauge coordinates.                | <code>coords is not None;</code> <code>lat == pytest.approx(51.4309, rel=1e-4)</code><br><code>(+1 more)</code> | Pass   | 0.004s |
| Test getting flood stage thresholds.           | <code>stages is not None;</code><br><code>stages['ActionStage'] == 4.5</code> <code>(+3 more)</code>            | Pass   | 0.001s |
| Test getting gauge name by ID.                 | <code>name == "Thames at Teddington"</code>                                                                     | Pass   | 0.001s |
| Test getting name for non-existent gauge.      | <code>name is None</code>                                                                                       | Pass   | 0.001s |
| Exists false                                   | <code>GaugeLoader(tmp_path).exists("GAUGE-999") is False</code>                                                 | Pass   | 0.001s |
| Exists true                                    | <code>GaugeLoader(tmp_path).exists("GAUGE-000") is True</code>                                                  | Pass   | 0.001s |
| Find by id missing returns none                | <code>GaugeLoader(tmp_path).find_by_id("GAUGE-999") is None</code>                                              | Pass   | 0.001s |
| Find by id returns entity                      | <code>result is not None;</code><br><code>result["FloodGauge"]["Header"]["GaugeID"] == "GAUGE-001"</code>       | Pass   | 0.001s |
| List all returns summaries                     | <code>len(summaries) == 2;</code> <code>"gaugeId" in summaries[0]</code>                                        | Pass   | 0.001s |
| Test finding gauge by ID.                      | <code>gauge is not None;</code><br><code>gauge['FloodGauge']['Header']['GaugeName'] == "Thames at ..."</code>   | Pass   | 0.002s |
| Test finding gauge by name.                    | <code>gauge is not None;</code><br><code>gauge_loader.get_entity_id(gauge) == "GAUGE-001"</code>                | Pass   | 0.001s |
| Test name search is case insensitive.          | <code>gauge is not None</code>                                                                                  | Pass   | 0.001s |
| Test finding gauges by owner.                  | <code>len(ea_gauges) == 2; len(usgs_gauges) == 1</code>                                                         | Pass   | 0.001s |
| Test finding active gauges.                    | <code>len(active) == 2</code>                                                                                   | Pass   | 0.001s |

| Test Description                                                                             | Acceptance Criteria                                                                                                     | Result | Time   |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Test finding inactive gauges.                                                                | <code>len(inactive) == 1;</code><br><code>gauge_loader.get_entity_id(inactive[0])</code><br><code>== "GAUGE-003"</code> | Pass   | 0.001s |
| Test finding river gauges.                                                                   | <code>len(river) == 2</code>                                                                                            | Pass   | 0.001s |
| Test finding tidal gauges.                                                                   | <code>len(tidal) == 1</code>                                                                                            | Pass   | 0.003s |
| Test finding gauges within radius.                                                           | <code>len(nearby) &gt;= 1; len(nearby) &gt;= 2</code>                                                                   | Pass   | 0.002s |
| Test radius search with no matches.                                                          | <code>len(nearby) == 0</code>                                                                                           | Pass   | 0.001s |
| Get status returns dict                                                                      | <code>"entity_name" in status;</code> <code>"path" in</code><br><code>status (+1 more)</code>                           | Pass   | 0.000s |
| Distance <code>j 100</code> → 'Very close to Thames'.                                        | <code>"Very close to Thames" in</code><br><code>str(result)</code>                                                      | Pass   | 0.001s |
| Distance <code>== 0</code> → 'On Thames River' and 'Directly on Thames'.                     | <code>"On Thames River" in str(result)</code>                                                                           | Pass   | 0.001s |
| Empty data does not crash                                                                    | <code>isinstance(result, list)</code>                                                                                   | Pass   | 0.000s |
| Flood risk section                                                                           | <code>any("Flood Risk" in t for t in</code><br><code>texts)</code>                                                      | Pass   | 0.000s |
| Geographic coordinates section                                                               | <code>any("Geographic Coordinates" in t</code><br><code>for t in texts)</code>                                          | Pass   | 0.000s |
| Has tables                                                                                   | <code>any(isinstance(e, Table) for e in</code><br><code>result)</code>                                                  | Pass   | 0.000s |
| Latitude <code>51.4–51.5</code> → 'Central London area'.                                     | <code>"Central London area" in str(result)</code>                                                                       | Pass   | 0.001s |
| Latitude <code>j 51.5</code> → 'North London area'.                                          | <code>"North London area" in str(result)</code>                                                                         | Pass   | 0.001s |
| Latitude <code>j 51.4</code> → 'South London area'.                                          | <code>"South London area" in str(result)</code>                                                                         | Pass   | 0.001s |
| No thames info does not crash                                                                | <code>isinstance(result, list)</code>                                                                                   | Pass   | 0.000s |
| Positioning summary section                                                                  | <code>any("Positioning Summary" in t for t</code><br><code>in texts)</code>                                             | Pass   | 0.000s |
| Returns list                                                                                 | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0</code>                                               | Pass   | 0.000s |
| Risk category in location profile                                                            | <code>"High flood risk location" in</code><br><code>str(result)</code>                                                  | Pass   | 0.001s |
| Risk score shown as 'X/10'.                                                                  | <code>"7.5/10" in str(result)</code>                                                                                    | Pass   | 0.001s |
| Thames context section                                                                       | <code>any("Thames River Context" in t for</code><br><code>t in texts)</code>                                            | Pass   | 0.000s |
| Title present                                                                                | <code>any("Geographic Location" in t for t</code><br><code>in texts)</code>                                             | Pass   | 0.000s |
| Lines 89-90: <code>exception in create_gauge_mapping</code> raises <code>ValueError</code> . | —                                                                                                                       | Pass   | 0.000s |

| Test Description                                            | Acceptance Criteria                                           | Result | Time   |
|-------------------------------------------------------------|---------------------------------------------------------------|--------|--------|
| Line 100: get_nrfa_fields returns list of NRFA field names. | isinstance(fields, list); len(fields) == 15 (+3 more)         | Pass   | 0.000s |
| Empty data does not crash                                   | isinstance(result, list)                                      | Pass   | 0.000s |
| Empty readings list does not crash                          | isinstance(result, list)                                      | Pass   | 0.000s |
| Get gauge id missing                                        | page._get_gauge_id({}) == "Unknown Gauge"                     | Pass   | 0.000s |
| Get gauge id present                                        | page._get_gauge_id(_make_gauge()) == "GAUGE-001"              | Pass   | 0.000s |
| Has tables                                                  | any(isinstance(e, Table) for e in result)                     | Pass   | 0.001s |
| Historical statistics section                               | any("Historical Measurement Statistics" in t for t in texts)  | Pass   | 0.000s |
| Measurement configuration section                           | any("Measurement Configuration" in t for t in texts)          | Pass   | 0.000s |
| No timeseries shows message                                 | any("No current timeseries data" in t for t in texts)         | Pass   | 0.000s |
| Operational status section                                  | any("Operational Status" in t for t in texts)                 | Pass   | 0.000s |
| Returns list                                                | isinstance(result, list); len(result) > 0                     | Pass   | 0.000s |
| Title present                                               | any("Current Measurements" in t for t in texts)               | Pass   | 0.000s |
| Current readings table is added when timeseries provided.   | len(tables_with) >= len(tables_without)                       | Pass   | 0.001s |
| With timeseries shows current readings                      | any("Current Readings" in t for t in texts)                   | Pass   | 0.001s |
| Calculate level range.                                      | level_range > 2.0                                             | Pass   | 0.000s |
| Calculate maximum level.                                    | max_level > 6.0                                               | Pass   | 0.000s |
| Calculate mean water level.                                 | 2.5 < mean < 3.5                                              | Pass   | 0.000s |
| Calculate minimum level.                                    | min_level > 0                                                 | Pass   | 0.000s |
| Calculate maximum rate of rise.                             | rate > 0                                                      | Pass   | 0.000s |
| When name is empty, uses area_name.                         | "Westminster" in result; "Thames" in result                   | Pass   | 0.000s |
| No name key falls back to area                              | "Putney" in result                                            | Pass   | 0.000s |
| No hazard curve shows fallback                              | any("No hazard" in t or "hazard" in t.lower() for t in te...) | Pass   | 0.000s |
| Returns list                                                | isinstance(result, list)                                      | Pass   | 0.000s |
| With nonzero rates renders spreads                          | len(tables) >= 1                                              | Pass   | 0.000s |
| With zero rates renders na                                  | len(tables) >= 1                                              | Pass   | 0.000s |



| Test Description                                                          | Acceptance Criteria                                                                         | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Basic gauge report produces a valid PDF file.                             | <code>report_path.exists();</code><br><code>report_path.suffix == '.pdf'</code> (+1 more)   | Pass   | 0.018s |
| Full gauge report produces a valid PDF.                                   | <code>report_path.exists();</code><br><code>report_path.suffix == '.pdf'</code>             | Pass   | 0.014s |
| Output directory is created automatically.                                | <code>new_dir.exists(); report_path.exists()</code>                                         | Pass   | 0.018s |
| Generated filename includes the gauge ID.                                 | <code>'GAUGE-test0001' in report_path.name</code>                                           | Pass   | 0.017s |
| Gauge report PDF should be non-trivial in size.                           | <code>report_path.stat().st_size &gt; 5000</code>                                           | Pass   | 0.018s |
| Generated file is a valid PDF.                                            | <code>header == b'%PDF-'</code>                                                             | Pass   | 0.088s |
| Report handles minimal gauge data without crashing.                       | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 0</code>        | Pass   | 0.016s |
| Generator initialises with output directory.                              | <code>gen.output_dir == tmp_path</code>                                                     | Pass   | 0.001s |
| Generator lists all registered gauge page modules.                        | <code>'title_overview' in pages;</code><br><code>'sensor_details' in pages</code> (+7 more) | Pass   | 0.001s |
| Validates known gauge page names.                                         | <code>'title_overview' in valid;</code> <code>'bogus' in invalid</code>                     | Pass   | 0.001s |
| Line 163 (else branch): no time-series or hazard -i no analysis pages.    | <code>not any(p in pages for p in analysis_pages)</code>                                    | Pass   | 0.001s |
| Line 163 (if branch): gauge_data.get('hazard_curve') -i include analysis. | <code>"risk.assessment" in pages</code>                                                     | Pass   | 0.001s |
| Lines 269-270: generate_analysis_report creates PDF.                      | <code>path.exists()</code>                                                                  | Pass   | 0.004s |
| Lines 208-210: exception in page -i logged, continue to next page.        | <code>isinstance(elements, list)</code>                                                     | Pass   | 0.002s |
| Lines 190-191: unknown page -i warning logged, skipped.                   | <code>isinstance(elements, list)</code>                                                     | Pass   | 0.001s |
| Lines 176-177: missing GaugeID -i uses 'unknown' in filename.             | <code>"unknown" in name;</code><br><code>name.endswith(".pdf")</code>                       | Pass   | 0.001s |
| Line 175: gauge_id extracted for filename.                                | <code>"GAUGE-001" in name</code>                                                            | Pass   | 0.001s |
| Lines 261-263: generate_monitoring_report creates PDF.                    | <code>path.exists()</code>                                                                  | Pass   | 0.003s |
| Line 313: get_page_categories returns dict.                               | <code>isinstance(cats, dict);</code> <code>"gauge_info" in cats</code>                      | Pass   | 0.001s |

| Test Description                                                     | Acceptance Criteria                                                          | Result | Time   |
|----------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| Line 279: list_available_pages returns list of strings.              | isinstance(pages, list);<br>"title_overview" in pages                        | Pass   | 0.001s |
| Lines 59-60: output_dir=None -i uses config.get_gauge_reports_dir(). | gen.output_dir == gauge_dir                                                  | Pass   | 0.001s |
| Line 315: validate_pages with all valid pages.                       | "title_overview" in valid; invalid == []                                     | Pass   | 0.001s |
| validate_pages with some invalid page names.                         | "title_overview" in valid;<br>"nonexistent_page" in invalid                  | Pass   | 0.001s |
| Categories dict initialized                                          | "gauge_info" in gen.categories;<br>"operational" in gen.categories (+2 more) | Pass   | 0.001s |
| Explicit output dir path                                             | isinstance(gen.output_dir, Path)                                             | Pass   | 0.002s |
| Explicit output dir string                                           | gen.output_dir == tmp_path                                                   | Pass   | 0.002s |
| No output dir uses config                                            | gen.output_dir == gauge_dir                                                  | Pass   | 0.002s |
| Output dir created                                                   | new_dir.exists()                                                             | Pass   | 0.001s |
| Pages dict initialized                                               | expected <= set(gen.pages.keys())                                            | Pass   | 0.002s |
| Excludes timeseries when requested                                   | "intensity_timeseries" not in d;<br>"level_timeseries" not in d              | Pass   | 0.000s |
| Flooded is bool                                                      | isinstance(d["flooded"], bool);<br>d["flooded"] is True                      | Pass   | 0.000s |
| Has required keys                                                    | "gauge_id" in d; "storm_id" in d (+2 more)                                   | Pass   | 0.000s |
| Includes timeseries by default                                       | "intensity_timeseries" in d;<br>"level_timeseries" in d (+1 more)            | Pass   | 0.000s |
| Returns dict                                                         | isinstance(d, dict);<br>isinstance(d, dict)                                  | Pass   | 0.000s |
| Values rounded                                                       | d["peak_level"] == 5.2;<br>d["accumulation"] == 4.5                          | Pass   | 0.000s |
| Generates elements list                                              | isinstance(elements, list);<br>len(elements) > 0                             | Pass   | 0.000s |
| No timeseries shows placeholder                                      | any("No current timeseries" in t for t in texts)                             | Pass   | 0.000s |
| With timeseries data                                                 | isinstance(elements, list);<br>len(elements) > 0                             | Pass   | 0.000s |
| Test POST /api/v1/gauges/report without gaugeId.                     | response.status_code == 400; 'Gauge ID is required' in data['message']       | Pass   | 0.000s |
| Test GET /api/v1/gauges/idi.                                         | response.status_code == 200;<br>data['status'] == 'success' (+1 more)        | Pass   | 0.001s |
| Test GET /api/v1/gauges/idi with invalid ID.                         | response.status_code == 404                                                  | Pass   | 0.001s |

| Test Description                                    |  | Acceptance Criteria                                                      | Result | Time   |
|-----------------------------------------------------|--|--------------------------------------------------------------------------|--------|--------|
| Test GET /api/v1/gauges/{id}/statistics.            |  | response.status_code == 200;<br>data['status'] == 'success' (+1<br>more) | Pass   | 0.002s |
| Test GET /api/v1/gauges/{id}/timeseries.            |  | response.status_code == 200;<br>data['status'] == 'success' (+1<br>more) | Pass   | 0.002s |
| Test legacy GET /api/v1/list_gauges endpoint.       |  | response.status_code == 200;<br>data['count'] == 3                       | Pass   | 0.002s |
| Test GET /api/v1/gauges.                            |  | response.status_code == 200;<br>data['status'] == 'success' (+2<br>more) | Pass   | 0.002s |
| Empty data does not crash                           |  | isinstance(result, list)                                                 | Pass   | 0.000s |
| Gauge id in subtitle                                |  | any("GAUGE-001" in t for t in texts)                                     | Pass   | 0.001s |
| Gauge information section                           |  | any("Gauge Information" in t for t<br>in texts)                          | Pass   | 0.001s |
| Get gauge id missing                                |  | page._get_gauge_id({}) == "Unknown<br>Gauge"                             | Pass   | 0.000s |
| Get gauge id present                                |  | page._get_gauge_id(_make_gauge()) ==<br>"GAUGE-001"                      | Pass   | 0.000s |
| Has tables                                          |  | any(isinstance(e, Table) for e in<br>result)                             | Pass   | 0.001s |
| Historical sensor statistics section                |  | any("Historical Sensor Statistics"<br>in t for t in texts)               | Pass   | 0.001s |
| Location information section                        |  | any("Location Information" in t for<br>t in texts)                       | Pass   | 0.001s |
| Measurement configuration section                   |  | any("Measurement Configuration" in t<br>for t in texts)                  | Pass   | 0.001s |
| Missing SensorDetails → empty tables, no exception. |  | isinstance(result, list)                                                 | Pass   | 0.000s |
| Returns list                                        |  | isinstance(result, list); len(result)<br>> 0                             | Pass   | 0.001s |
| Title present                                       |  | any("Sensor Details" in t for t in<br>texts)                             | Pass   | 0.001s |
| With timeseries does not crash                      |  | isinstance(result, list)                                                 | Pass   | 0.001s |
| Level at alert threshold.                           |  | status == "ALERT"                                                        | Pass   | 0.000s |
| Level between alert and warning.                    |  | status == "ALERT"                                                        | Pass   | 0.000s |
| Level below alert = normal.                         |  | status == "NORMAL"                                                       | Pass   | 0.000s |
| Level at severe threshold.                          |  | status == "SEVERE"                                                       | Pass   | 0.000s |
| Level at warning threshold.                         |  | status == "WARNING"                                                      | Pass   | 0.000s |

| Test Description                                                       | Acceptance Criteria                                                                         | Result | Time   |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Lines 270-271: configure(panel_height=...) panel_height.               | config-<br>updates<br>obj.panel_height == "600px";<br>obj.panel_height == "700px" (+1 more) | Pass   | 0.000s |
| Lines 268-269: configure(panel_width=...) panel_width.                 | config-<br>updates<br>obj.panel_width == "800px";<br>obj.panel_width == "900px" (+1 more)   | Pass   | 0.000s |
| Line 275: get_statistics() returns dict.                               | stats["panel_width"] == "700px";<br>stats["panel_height"] == "500px" (+4 more)              | Pass   | 0.000s |
| Empty timeseries returns short list                                    | isinstance(result, list); len(result) > 0                                                   | Pass   | 0.000s |
| Gauge name in title                                                    | any("Chelsea Gauge" in t for t in texts)                                                    | Pass   | 0.001s |
| Generates per gauge files                                              | gaugets_dir.exists();<br>len(list(gaugets_dir.glob("GAUGE-*.json")) >= N_GAUGES (+2 more)   | Pass   | 0.000s |
| Missing gauge name does not crash                                      | isinstance(result, list)                                                                    | Pass   | 0.001s |
| No timeseries message included                                         | any("No time series data" in t for t in texts)                                              | Pass   | 0.000s |
| No timeseries returns short list                                       | len(result) > 0                                                                             | Pass   | 0.000s |
| Output file has readings                                               | len(gauge_files) > 0;<br>"flood_simulation" in data (+1 more)                               | Pass   | 0.001s |
| Requires gauge portfolio                                               | —                                                                                           | Pass   | 0.000s |
| Returns list                                                           | isinstance(result, list)                                                                    | Pass   | 0.025s |
| Timeseries mixed statuses                                              | isinstance(result, list)                                                                    | Pass   | 0.106s |
| Readings with non-zero thresholds exercising fill_between paths.       | isinstance(result, list)                                                                    | Pass   | 0.108s |
| With timeseries returns more elements                                  | len(result) >= len(empty_result)                                                            | Pass   | 0.147s |
| With timeseries table included                                         | len(tables) >= 1                                                                            | Pass   | 0.126s |
| DEFAULT_PARAMS and generate() default must both be 168 hours (7 days). | GaugeTimeSeriesGenerator.DEFAULT_PARAMS["simulation_hours"] == 168                          | Pass   | 0.000s |
| generate(simulation_hours) default parameter must be 168.              | default == 168                                                                              | Pass   | 0.000s |
| Convenience function generate_gaugets() default must be 168.           | default == 168                                                                              | Pass   | 0.000s |
| peak_hour_max must be strictly less than simulation_hours.             | p["peak_hour_max"] < p["simulation_hours"]                                                  | Pass   | 0.000s |

| Test Description                                                                                     | Acceptance Criteria                                                          | Result | Time   |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| peak_hour_min must be less than peak_hour_max.                                                       | <code>p["peak_hour_min"] &lt; p["peak_hour_max"]</code>                      | Pass   | 0.000s |
| 168-hour simulation produces significantly more readings than 60-hour.                               | <code>files60 and files168; count168 &gt; count60</code>                     | Pass   | 0.297s |
| Lines 127-130: <code>configure()</code> updates a known <code>sim_param</code> key.                  | <code>gen.sim_params['simulation_hours'] == 48</code>                        | Pass   | 0.000s |
| Lines 127-129: unknown key is silently ignored (if branch is False).                                 | <code>'nonexistent_param' not in gen.sim_params</code>                       | Pass   | 0.000s |
| <code>generate(168)</code> should produce ~168 readings per gauge.                                   | <code>len(files) &gt; 0; sim.get("simulation_hours") == 168 (+1 more)</code> | Pass   | 0.284s |
| Explicit override stores the passed value in <code>simulation_hours</code> metadata.                 | <code>sim.get("simulation_hours") == 60</code>                               | Pass   | 0.180s |
| Lines 157-159: <code>gauge.json</code> absent → <code>FileNotFoundException</code> re-raised.        | —                                                                            | Pass   | 0.000s |
| <code>simulation_hours</code> is written into each gaugets JSON.                                     | <code>"simulation_hours" in data["flood_simulation"]</code>                  | Pass   | 0.175s |
| Line 175: stale <code>GAUGE-*.json</code> files in <code>gaugets/</code> are deleted before writing. | <code>stale.exists(); not stale.exists()</code>                              | Pass   | 0.050s |
| Unparseable install date is silently skipped.                                                        | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Empty data does not crash                                                                            | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Gauge id in output                                                                                   | <code>any("GAUGE-001" in t for t in texts)</code>                            | Pass   | 0.000s |
| Gauge overview section header                                                                        | <code>any("Gauge Overview" in t for t in texts)</code>                       | Pass   | 0.000s |
| Gauge summary section header                                                                         | <code>any("Gauge Summary" in t for t in texts)</code>                        | Pass   | 0.000s |
| Get gauge id missing                                                                                 | <code>page._get_gauge_id({}) == "Unknown Gauge"</code>                       | Pass   | 0.000s |
| Get gauge id present                                                                                 | <code>page._get_gauge_id(_make_gauge()) == "GAUGE-001"</code>                | Pass   | 0.000s |
| Has tables                                                                                           | <code>any(isinstance(e, Table) for e in result)</code>                       | Pass   | 0.000s |
| Valid install date triggers gauge age row in summary.                                                | <code>isinstance(result, list); len(result) &gt; 0</code>                    | Pass   | 0.000s |
| No install date does not crash                                                                       | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Returns list                                                                                         | <code>isinstance(result, list); len(result) &gt; 0</code>                    | Pass   | 0.001s |
| Title paragraph present                                                                              | <code>any("Comprehensive Flood Gauge Analysis Report" in t for ...</code>    | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                                                 | Result | Time   |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Unknown gauge fallback                                           | <code>any("Unknown Gauge" in t for t in texts)</code>                                               | Pass   | 0.000s |
| With timeseries does not crash                                   | <code>isinstance(result, list)</code>                                                               | Pass   | 0.000s |
| Lines 68-69: exception returns validation_error dict.            | <code>"validation_error" in result; isinstance(result["validation_error"], list) (+2 more)</code>   | Pass   | 0.000s |
| Lines 61,64: invalid menu field value appended to sensor_errors. | <code>"SensorDetails" in errors; any("DataSourceType" in e for e in errors["SensorDetails"])</code> | Pass   | 0.000s |
| Multiple invalid menu fields all collected.                      | <code>"SensorDetails" in errors; len(errors["SensorDetails"]) == 3</code>                           | Pass   | 0.000s |
| Valid menu field values produce no SensorDetails errors.         | <code>"SensorDetails" not in errors</code>                                                          | Pass   | 0.000s |
| Curve points present                                             | <code>not missing</code>                                                                            | Pass   | 0.000s |
| Each gauge has required fields                                   | <code>not missing</code>                                                                            | Pass   | 0.000s |
| File exists                                                      | <code>GAUGEHC_PATH.exists(); MARKET_STATE.exists()</code>                                           | Pass   | 0.001s |
| alert ; warning ; severe for every gauge.                        | <code>not violations</code>                                                                         | Pass   | 0.000s |
| Gauge count at least 40                                          | <code>count &gt;= 40</code>                                                                         | Pass   | 0.000s |
| Gauge ids are well formed                                        | <code>not bad</code>                                                                                | Pass   | 0.000s |
| Has hazard curves key                                            | <code>"hazard_curves" in gaugehc</code>                                                             | Pass   | 0.000s |
| alert ; warning ; severe (higher trigger = lower probability).   | <code>not violations</code>                                                                         | Pass   | 0.000s |
| Hazard rates positive                                            | <code>not violations</code>                                                                         | Pass   | 0.000s |
| Term structure present                                           | <code>not missing</code>                                                                            | Pass   | 0.000s |
| Lines 73-77: main() with no GDF files logs info.                 | —                                                                                                   | Pass   | 0.000s |
| Lines 70-71: main() processes files, handles errors.             | —                                                                                                   | Pass   | 0.006s |
| Line 37: output_dir defaults to input_dir when None.             | <code>result == []</code>                                                                           | Pass   | 0.000s |
| Lines 46-47: error processing a CSV file is logged and skipped.  | <code>result == []</code>                                                                           | Pass   | 0.003s |
| directory.main() is callable.                                    | <code>callable(main)</code>                                                                         | Pass   | 0.000s |
| Each file has required keys                                      | <code>not bad</code>                                                                                | Pass   | 2.827s |
| Flood simulation has readings                                    | <code>not bad</code>                                                                                | Pass   | 2.316s |
| Gauge id field matches filename                                  | <code>not bad</code>                                                                                | Pass   | 2.353s |
| Gaugets dir exists                                               | <code>GAUGETS_DIR.exists()</code>                                                                   | Pass   | 0.000s |
| Gaugets gauge id format                                          | <code>not bad</code>                                                                                | Pass   | 0.000s |



| Test Description                                                                                                       | Acceptance Criteria                                                                                         | Result | Time   |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| Gaugets not empty                                                                                                      | <code>gaugets_files</code>                                                                                  | Pass   | 0.000s |
| simulation_hours should match number of reading entries.                                                               | not mismatches                                                                                              | Pass   | 2.337s |
| Line 72: <code>datetime</code> → <code>isoformat</code> string.                                                        | "2026-01-01" in result                                                                                      | Pass   | 0.000s |
| Lines 75-76: <code>np.float32</code> is <code>np.floating</code> but not Python float → <code>default()</code> called. | "2.5" in result                                                                                             | Pass   | 0.000s |
| Lines 73-74: <code>np.integer</code> → <code>int</code> .                                                              | "42" in result                                                                                              | Pass   | 0.000s |
| Lines 77-78: <code>np.ndarray</code> → list via <code>tolist()</code> .                                                | "1.0" in result                                                                                             | Pass   | 0.000s |
| Line 79: unknown type → <code>super().default()</code> → <code>TypeError</code> .                                      | —                                                                                                           | Pass   | 0.000s |
| When no <code>gaugets</code> file exists, hydrograph falls back to synthesis.                                          | <code>resp.status_code == 200;</code><br><code>max(water_levels) &gt; min(water_levels)</code>              | Pass   | 0.007s |
| <code>gaugets</code> readings using 'level' key (alternate field name) are extracted.                                  | <code>resp.status_code == 200;</code><br><code>max(water_levels) &gt; min(water_levels)</code>              | Pass   | 0.009s |
| <code>gaugets</code> readings using 'waterLevel' key are correctly extracted.                                          | <code>resp.status_code == 200;</code><br><code>max(water_levels) &gt; min(water_levels)</code><br>(+1 more) | Pass   | 0.008s |
| Lines 241-244: <code>__main__</code> block runs <code>generate_gaugets(168)</code> and logs results.                   | <code>(tmp_path / 'gaugets').exists()</code>                                                                | Pass   | 0.004s |
| Random module <code>peak_hour_max</code> must be <code>simulation_hours</code> .                                       | <code>DEFAULT_PARAMS["peak_hour_max"] &lt;</code><br><code>DEFAULT_PARAMS["simulat...</code>                | Pass   | 0.000s |
| <code>gaugets_random.DEFAULT_PARAMS</code> <code>simulation_hours</code> must be 168.                                  | <code>DEFAULT_PARAMS["simulation_hours"] ==</code><br>168                                                   | Pass   | 0.000s |
| Daily flows filtered by years                                                                                          | <code>len(result_3yr["daily_flows"]) &gt;</code><br><code>len(result_1yr["daily_fl...</code>                | Pass   | 0.036s |
| Explicit gauge id overrides metadata                                                                                   | <code>result["station_metadata"]["station_id"]</code><br><code>== "OVERRIDE-ID"</code>                      | Pass   | 0.019s |
| Output file written                                                                                                    | <code>output_path.exists();</code><br><code>json.load(f)["data_type"] ==</code><br>"GaugeHistoricalDaily"   | Pass   | 0.023s |
| Returns dict with required keys                                                                                        | <code>key in result</code>                                                                                  | Pass   | 0.019s |
| Avg basis is float                                                                                                     | <code>isinstance(stats["avg_basis_bps"],</code><br><code>float)</code>                                      | Pass   | 0.383s |
| Avg transmission rate between 0 and 1                                                                                  | <code>0.0 &lt;= stats["avg-transmission_rate"]</code><br><code>&lt;= 1.0</code>                             | Pass   | 0.267s |
| Min spread bps in stats                                                                                                | <code>stats["min_spread_bps"] ==</code><br><code>MIN_PRS_SPREAD_BPS</code>                                  | Pass   | 0.112s |

| Test Description                                                 | Acceptance Criteria                                                                                                                                 | Result | Time   |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Total flood events counted                                       | <code>stats["total_flood_events"] == 7</code>                                                                                                       | Pass   | 0.553s |
| No gauges in portfolio returns empty list.                       | <code>result == []</code>                                                                                                                           | Pass   | 0.002s |
| Line 50: error path extracts GaugeID from FloodGauge.Header.     | <code>len(result) == 0; "GAUGE-BADONE" in caplog.text</code>                                                                                        | Pass   | 0.001s |
| Line 50: completely empty entry uses UNKNOWN as gauge_id.        | <code>len(result) == 0; "UNKNOWN" in caplog.text</code>                                                                                             | Pass   | 0.001s |
| Good gauges succeed even when bad ones fail.                     | <code>len(result) == 1</code>                                                                                                                       | Pass   | 0.262s |
| Lines 83-110: generate_all_gauge_histories processes all gauges. | <code>isinstance(result, list); len(result) == 1</code>                                                                                             | Pass   | 0.256s |
| Lines 105-107: error on one gauge -i logged, continues.          | <code>isinstance(result, list)</code>                                                                                                               | Pass   | 0.496s |
| Line 60-61: house with 3 bedrooms returns 1 or 2 bathrooms.      | <code>results.issubset({1, 2}); len(results) == 2</code>                                                                                            | Pass   | 0.000s |
| Line 62-63: house with 4+ bedrooms returns 2 or 3 bathrooms.     | <code>results.issubset({2, 3}); len(results) == 2</code>                                                                                            | Pass   | 0.000s |
| 5 bedrooms also takes the 4+ path (line 62-63).                  | <code>val in (2, 3)</code>                                                                                                                          | Pass   | 0.000s |
| Advised flag                                                     | <code>isinstance(generate_boolean_value("AdvisedFlag", self._fd...)</code>                                                                          | Pass   | 0.000s |
| Btl commercial purpose                                           | <code>generate_boolean_value("BusinessOrCommercialPurpose", self._fd...); generate_boolean_value("BusinessOrCommercialPurpose", self._fd...)</code> | Pass   | 0.000s |
| Default flag from data                                           | <code>generate_boolean_value("DefaultFlag", self._fd(is_default...); generate_boolean_value("DefaultFlag", self._fd(is_default...)</code>           | Pass   | 0.000s |
| Execution only flag                                              | <code>isinstance(generate_boolean_value("ExecutionOnlyFlag", self._fd...)</code>                                                                    | Pass   | 0.000s |
| First time buyer btl always false                                | <code>generate_boolean_value("FirstTimeBuyerFlag", self._fd(mor...)</code>                                                                          | Pass   | 0.000s |
| First time buyer expensive property                              | <code>isinstance(result, bool)</code>                                                                                                               | Pass   | 0.000s |
| First time buyer normal property                                 | <code>isinstance(result, bool)</code>                                                                                                               | Pass   | 0.000s |
| In arrears flag                                                  | <code>generate_boolean_value("InArrearsFlag", self._fd(is_in.ar...)</code>                                                                          | Pass   | 0.000s |
| Mmr compliant                                                    | <code>generate_boolean_value("MMRCompliantFlag", self._fd()) is...</code>                                                                           | Pass   | 0.000s |

| Test Description                                | Acceptance Criteria                                                                                         | Result | Time    |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|---------|
| Stress compliant                                | <code>generate_boolean_value("StressTestCompliant",</code><br><code>self.f...</code>                        | Pass   | 0.000s  |
| Unknown field returns bool                      | <code>isinstance(generate_boolean_value("Unknown",</code><br><code>self.fd()),...</code>                    | Pass   | 0.000s  |
| Count respected                                 | <code>len(result["data"]) == 3</code>                                                                       | Pass   | 0.000s  |
| Returns dict                                    | <code>isinstance(result, dict); "data" in</code><br><code>result</code>                                     | Pass   | 0.000s  |
| Fair section 50 to 75 gets enhance-<br>ment rec | <code>any("OtherSection" in rec for rec in</code><br><code>r)</code>                                        | Pass   | 0.000s  |
| Good sections maintain recommen-<br>dation      | <code>any("Maintain" in rec or "maintain"</code><br><code>in rec for rec in r)</code>                       | Pass   | 0.000s  |
| Max 5 recommendations                           | <code>len(r) &lt;= 5</code>                                                                                 | Pass   | 0.000s  |
| Poor section under 50 gets recom-<br>mendation  | <code>any("SomeSection" in rec for rec in</code><br><code>r)</code>                                         | Pass   | 0.000s  |
| Returns list                                    | <code>isinstance(self._page().generate_elements("page-property",</code><br><code>isinstance(r, list)</code> | Pass   | 0.000s; |
| Application date is string                      | <code>isinstance(result, str); len(result)</code><br><code>== 10</code>                                     | Pass   | 0.000s  |
| Default date not defaulted                      | <code>result is None</code>                                                                                 | Pass   | 0.000s  |
| Default date when defaulted                     | <code>isinstance(result, str)</code>                                                                        | Pass   | 0.000s  |
| Disbursal date                                  | <code>isinstance(result, str)</code>                                                                        | Pass   | 0.000s  |
| Maturity date                                   | <code>isinstance(result, str)</code>                                                                        | Pass   | 0.000s  |
| Unknown field                                   | <code>1 &lt;= result &lt;= 10; isinstance(result,</code><br><code>str) (+1 more)</code>                     | Pass   | 0.000s  |
| Application property valuation                  | <code>generate_decimal_value("ApplicationPropertyValuation",</code><br><code>se...</code>                   | Pass   | 0.000s  |
| Aprc initial rate                               | <code>isinstance(result, float)</code>                                                                      | Pass   | 0.000s  |
| Aprc secondary rate                             | <code>isinstance(result, float)</code>                                                                      | Pass   | 0.000s  |
| Borrower income                                 | <code>generate_decimal_value("BorrowerIncome",</code><br><code>self.fd()) == 7...</code>                    | Pass   | 0.000s  |
| Current balance                                 | <code>generate_decimal_value("CurrentBalance",</code><br><code>self.fd()) == 2...</code>                    | Pass   | 0.000s  |
| Current lending rate                            | <code>abs(result - 4.0) &lt; 0.01</code>                                                                    | Pass   | 0.000s  |
| Current ltv                                     | <code>abs(result - 75.0) &lt; 0.01</code>                                                                   | Pass   | 0.000s  |
| Current payment                                 | <code>generate_decimal_value("CurrentPayment",</code><br><code>self.fd()) == 1500</code>                    | Pass   | 0.000s  |
| Debt to income ratio                            | <code>isinstance(result, float)</code>                                                                      | Pass   | 0.000s  |
| Debt to income zero income                      | <code>result == 0.3</code>                                                                                  | Pass   | 0.000s  |
| Default field                                   | <code>isinstance(result, float)</code>                                                                      | Pass   | 0.000s  |
| Interest rate                                   | <code>isinstance(result, float)</code>                                                                      | Pass   | 0.000s  |
| Loan to value ratio                             | <code>abs(result - 80.0) &lt; 0.01</code>                                                                   | Pass   | 0.000s  |

| Test Description                                       | Acceptance Criteria                                                           | Result | Time   |
|--------------------------------------------------------|-------------------------------------------------------------------------------|--------|--------|
| Ltv field                                              | <code>abs(result - 75.0) &lt; 0.01</code>                                     | Pass   | 0.000s |
| Original lending rate                                  | <code>abs(result - 4.0) &lt; 0.01</code>                                      | Pass   | 0.000s |
| Original loan                                          | <code>generate_decimal_value("OriginalLoan",<br/>self._fd()) == 320000</code> | Pass   | 0.000s |
| Original loan amount                                   | <code>generate_decimal_value("OriginalLoanAmount",<br/>self._fd()) ...</code> | Pass   | 0.000s |
| Original ltv                                           | <code>abs(result - 80.0) &lt; 0.01</code>                                     | Pass   | 0.000s |
| Outstanding balance                                    | <code>generate_decimal_value("OutstandingBalance",<br/>self._fd()) ...</code> | Pass   | 0.000s |
| Purchase value                                         | <code>generate_decimal_value("PurchaseValue",<br/>self._fd()) == 40...</code> | Pass   | 0.000s |
| Empty page list                                        | <code>elements == []</code>                                                   | Pass   | 0.002s |
| Empty page list                                        | <code>elements == []</code>                                                   | Pass   | 0.001s |
| Empty page list returns empty                          | <code>elements == []</code>                                                   | Pass   | 0.001s |
| Exception in page continues                            | <code>isinstance(elements, list);<br/>len(elements) &gt; 0</code>             | Pass   | 0.016s |
| If one page raises, the remainder are still processed. | <code>isinstance(elements, list);<br/>len(elements) &gt; 0</code>             | Pass   | 0.001s |
| Exception in page module continues                     | <code>isinstance(elements, list);<br/>len(elements) &gt; 0</code>             | Pass   | 0.002s |
| First page no page break                               | <code>not any(isinstance(e, PageBreak) for<br/>e in elements)</code>          | Pass   | 0.002s |
| First page no page break                               | <code>not any(isinstance(e, PageBreak) for<br/>e in elements)</code>          | Pass   | 0.002s |
| First page no page break                               | <code>not any(isinstance(e, PageBreak) for<br/>e in elements)</code>          | Pass   | 0.001s |
| Mixed valid and unknown pages                          | <code>len(elements) &gt; 0</code>                                             | Pass   | 0.002s |
| Mixed valid and unknown pages                          | <code>len(elements) &gt; 0</code>                                             | Pass   | 0.002s |
| Mixed valid and unknown pages                          | <code>len(elements) &gt; 0</code>                                             | Pass   | 0.003s |
| Mortgage data passed to page                           | —                                                                             | Pass   | 0.003s |
| Multiple pages have page break                         | <code>any(isinstance(e, PageBreak) for e<br/>in elements)</code>              | Pass   | 0.002s |
| Multiple pages have page breaks                        | <code>any(isinstance(e, PageBreak) for e<br/>in elements)</code>              | Pass   | 0.002s |
| Multiple pages inserts page breaks                     | <code>has_break</code>                                                        | Pass   | 0.009s |
| No gauge id key returns message                        | <code>any("No gauge ID" in t for t in<br/>texts)</code>                       | Pass   | 0.000s |
| None timeseries passed to page                         | —                                                                             | Pass   | 0.002s |
| Returns list                                           | <code>isinstance(elements, list)</code>                                       | Pass   | 0.002s |
| Returns list                                           | <code>isinstance(result, list); len(result)<br/>&gt; 0</code>                 | Pass   | 0.001s |

| Test Description                            | Acceptance Criteria                                                                                                | Result | Time   |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------|--------|
| Returns list                                | <code>isinstance(elements, list)</code>                                                                            | Pass   | 0.002s |
| Returns list                                | <code>isinstance(elements, list)</code>                                                                            | Pass   | 0.001s |
| Timeseries passed to page                   | —                                                                                                                  | Pass   | 0.002s |
| Unknown page skipped                        | <code>len(elements) == 0</code>                                                                                    | Pass   | 0.001s |
| Unknown page skipped                        | <code>len(elements) == 0</code>                                                                                    | Pass   | 0.002s |
| Unknown page skipped                        | <code>isinstance(elements, list);</code><br><code>len(elements) == 0</code>                                        | Pass   | 0.001s |
| With gauge id returns trading header        | <code>any("Trading Activity" in t for t in texts)</code>                                                           | Pass   | 0.001s |
| When eod_date is None, uses today's date.   | <code>snapshot['date'] == date.today().isoformat();</code><br><code>snapshot['eod_id'] == expected_id</code>       | Pass   | 0.001s |
| Empty property list returns zero mortgages  | <code>result["data"]["mortgages"] == [];</code><br><code>result["processing_stats"]["total_mortgages"] == 0</code> | Pass   | 0.019s |
| Missing property file raises file not found | —                                                                                                                  | Pass   | 0.000s |
| Boolean type                                | <code>isinstance(result, bool)</code>                                                                              | Pass   | 0.000s |
| Date type                                   | <code>isinstance(result, str)</code>                                                                               | Pass   | 0.000s |
| Decimal type                                | <code>isinstance(result, (int, float))</code>                                                                      | Pass   | 0.000s |
| Enum type                                   | <code>result == "Monthly"</code>                                                                                   | Pass   | 0.000s |
| Integer type                                | <code>result == 300</code>                                                                                         | Pass   | 0.000s |
| Menu type                                   | <code>result == "Monthly"</code>                                                                                   | Pass   | 0.000s |
| Non dict field def returns string           | <code>result == "literal"</code>                                                                                   | Pass   | 0.000s |
| Non dict non string returns empty           | <code>result == ""</code>                                                                                          | Pass   | 0.000s |
| Text type                                   | <code>result == "thames"</code>                                                                                    | Pass   | 0.000s |
| Unknown type returns empty                  | <code>result == ""</code>                                                                                          | Pass   | 0.000s |
| Contains flood risk report                  | <code>"flood_risk_report" in gen.generate_filename()</code>                                                        | Pass   | 0.002s |
| Contains gauge id                           | <code>"GAUGE-001" in name</code>                                                                                   | Pass   | 0.001s |
| Contains property id                        | <code>"PROP-001" in name</code>                                                                                    | Pass   | 0.002s |
| Contains timestamp                          | <code>digits &gt;= 8</code>                                                                                        | Pass   | 0.001s |
| Ends with pdf                               | <code>gen.generate_filename(gauge_data).endswith(".pdf")</code>                                                    | Pass   | 0.001s |
| Ends with pdf                               | <code>gen.generate_filename(prop_data).endswith(".pdf")</code>                                                     | Pass   | 0.002s |
| Ends with pdf                               | <code>gen.generate_filename().endswith(".pdf")</code>                                                              | Pass   | 0.001s |
| Missing gauge id uses unknown               | <code>"unknown" in name;</code><br><code>name.endswith(".pdf")</code>                                              | Pass   | 0.001s |
| Missing property id uses unknown            | <code>"unknown" in name;</code><br><code>name.endswith(".pdf")</code>                                              | Pass   | 0.002s |
| Partial path missing gauge id               | <code>"unknown" in name</code>                                                                                     | Pass   | 0.002s |
| Partial path missing header                 | <code>"unknown" in name</code>                                                                                     | Pass   | 0.001s |

| Test Description                                      | Acceptance Criteria                                                                       | Result | Time   |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Partial path missing header                           | "unknown" in name                                                                         | Pass   | 0.002s |
| Partial path missing property id                      | "unknown" in name                                                                         | Pass   | 0.002s |
| Returns string                                        | isinstance(name, str)                                                                     | Pass   | 0.001s |
| Returns string                                        | isinstance(name, str)                                                                     | Pass   | 0.002s |
| Returns string                                        | isinstance(name, str)                                                                     | Pass   | 0.001s |
| Timestamp in filename                                 | digits >= 8                                                                               | Pass   | 0.001s |
| Timestamp in filename                                 | digits >= 8                                                                               | Pass   | 0.002s |
| Two calls produce string names                        | isinstance(n1, str) and<br>isinstance(n2, str)                                            | Pass   | 1.105s |
| Estimates value when zero                             | result["loan_amount"] > 0                                                                 | Pass   | 0.000s |
| Flood risk passed through                             | result["flood_risk"] == "High"                                                            | Pass   | 0.000s |
| Returns dict with mortgage id                         | "mortgage_id" in result;<br>result["mortgage_id"].startswith("MORT-")                     | Pass   | 0.000s |
| Uses provided property value                          | result["property_value"] == 600000                                                        | Pass   | 0.000s |
| When output_dir=None, uses config.get_gaugehd_dir().  | isinstance(result, dict)                                                                  | Pass   | 0.253s |
| Daily observations generated                          | len(result["daily_observations"]) > 0                                                     | Pass   | 0.223s |
| Gauge metadata extracted                              | result["gauge_metadata"]["gauge_id"]<br>== "GAUGE-TEST01"                                 | Pass   | 0.246s |
| Gauge data without FloodGauge wrapper.                | result["gauge_metadata"]["gauge_id"]<br>== "GAUGE-FLAT"                                   | Pass   | 0.228s |
| Output file created                                   | (tmp_path /<br>"gauge_GAUGE-TEST01_hd.json").exists()                                     | Pass   | 0.267s |
| Returns dict                                          | isinstance(result, dict)                                                                  | Pass   | 0.238s |
| Schema version                                        | result["schema_version"] == "1.0"                                                         | Pass   | 0.295s |
| When output_path=None, uses config.get_gaugehd_dir(). | isinstance(result, dict);<br>(tmp_path / "gaugehd" /<br>"gauge_AUTO-01_hd.json").exists() | Pass   | 0.002s |
| Creates json file                                     | output.exists()                                                                           | Pass   | 0.002s |
| Gauge id override                                     | result["station_metadata"]["station_id"]<br>== "GAUGE-CUSTOM"                             | Pass   | 0.002s |
| Returns dict                                          | isinstance(result, dict)                                                                  | Pass   | 0.002s |
| Schema version in output                              | result["schema_version"] ==<br>"1.0"; result["data_type"] ==<br>"GaugeHistoricalDaily"    | Pass   | 0.002s |
| Station metadata extracted                            | result["station_metadata"]["station_name"]<br>== "Thames at ..."                          | Pass   | 0.002s |
| Current balance does not exceed original loan         | loan["CurrentBalance"] <=<br>loan["OriginalLoanAmount"]                                   | Pass   | 0.010s |
| Loan details values are positive                      | loan["OriginalLoanAmount"] > 0;<br>loan["CurrentBalance"] > 0 (+3 more)                   | Pass   | 0.002s |



| Test Description                                                           | Acceptance Criteria                                               | Result | Time   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------|--------|--------|
| Ltv within valid range                                                     | 10 <=<br>m["Mortgage"]["LoanDetails"]["LTV"]<br><= 100            | Pass   | 0.002s |
| Mortgage data key is list                                                  | isinstance(result["data"]["mortgages"],<br>list)                  | Pass   | 0.002s |
| Mortgage data structure has mortgage key                                   | "Mortgage" in m                                                   | Pass   | 0.002s |
| Mortgage header has required fields                                        | "MortgageID" in header; "PropertyID"<br>in header (+1 more)       | Pass   | 0.002s |
| Mortgage ids key is list                                                   | isinstance(result["data"]["mortgage_ids"],<br>list)               | Pass   | 0.002s |
| Output file is valid json                                                  | isinstance(json.load(f), dict)                                    | Pass   | 0.003s |
| Output json contains generation metadata                                   | "generated_at" in meta; "catchment" in<br>meta (+1 more)          | Pass   | 0.015s |
| Output json contains mortgages key                                         | "mortgages" in json.load(f)                                       | Pass   | 0.010s |
| Processing stats contain timing                                            | stats["start_time"] is not None;<br>stats["end_time"] is not None | Pass   | 0.002s |
| Gauge reading returns all fields                                           | "gaugeId" in reading; "waterLevel" in<br>reading (+1 more)        | Pass   | 0.000s |
| Gauge reading with string timestamp                                        | reading["timestamp"] ==<br>"2024-01-01T00:00:00"                  | Pass   | 0.000s |
| report_type='analysis' calls generate_analysis_report.                     | —                                                                 | Pass   | 0.001s |
| auto_open=True with open_pdf_file raising -i logs warning, does not raise. | result is not None                                                | Pass   | 0.001s |
| auto_open=True tries to open the PDF (lines 95-106).                       | mock_open.call_count >= 0                                         | Pass   | 0.001s |
| report_type='basic' calls generate_basic_report.                           | —                                                                 | Pass   | 0.001s |
| Gauge not found returns 404                                                | r.status_code == 404; data["status"]<br>== "error"                | Pass   | 0.002s |
| The /generate_gauge_report alias also works.                               | r.status_code == 200                                              | Pass   | 0.023s |
| Missing gauge id returns 400                                               | r.status_code == 400; data["status"]<br>== "error"                | Pass   | 0.001s |
| report_type='monitoring' with timeseries calls generate_monitoring_report. | —                                                                 | Pass   | 0.001s |
| No json returns 400                                                        | r.status_code in (400, 415, 500)                                  | Pass   | 0.001s |

| Test Description                                                              | Acceptance Criteria                                                                         | Result | Time   |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Sending JSON null body makes get_json() return None → line 29.                | <code>r.status_code == 400</code>                                                           | Pass   | 0.001s |
| Options returns ok                                                            | <code>r.status_code == 200</code>                                                           | Pass   | 0.001s |
| Response has file path                                                        | <code>"file_path" in data;</code><br><code>data["file_path"].endswith(".pdf")</code>        | Pass   | 0.024s |
| Response has pdf base64                                                       | <code>"pdf_base64" in data;</code><br><code>len(data["pdf_base64"]) &gt; 0</code>           | Pass   | 0.023s |
| Return value is the PDF path.                                                 | <code>result == tmp_path /</code><br><code>"gauge_report.pdf"</code>                        | Pass   | 0.001s |
| Unknown report_type calls generate_report.                                    | —                                                                                           | Pass   | 0.001s |
| Valid gauge generates report                                                  | <code>r.status_code == 200;</code> <code>data["status"]</code><br><code>== "success"</code> | Pass   | 0.022s |
| gaugehd file exists → exercises lines 62-69.                                  | <code>r.status_code == 200</code>                                                           | Pass   | 0.119s |
| gaugehc.json exists → exercises lines 76-82.                                  | <code>r.status_code == 200</code>                                                           | Pass   | 0.050s |
| Analysis type calls generate analysis                                         | —                                                                                           | Pass   | 0.001s |
| Auto open false returns path                                                  | <code>result == fake_pdf</code>                                                             | Pass   | 0.001s |
| Auto open true handles exception                                              | <code>result is not None</code>                                                             | Pass   | 0.001s |
| Lines 343-344: auto_open=True + open_pdf_file succeeds → both lines covered.  | <code>result == fake_pdf</code>                                                             | Pass   | 0.002s |
| Basic type calls generate basic                                               | —                                                                                           | Pass   | 0.002s |
| Monitoring type with timeseries calls monitoring                              | —                                                                                           | Pass   | 0.001s |
| Monitoring without timeseries falls back                                      | —                                                                                           | Pass   | 0.001s |
| Returns path                                                                  | <code>result == fake_pdf</code>                                                             | Pass   | 0.001s |
| Unknown type calls generate report                                            | —                                                                                           | Pass   | 0.001s |
| Lines 236-237: generate_gauges() instantiates generator and calls generate(). | <code>result['data']['num_gauges'] == 5</code>                                              | Pass   | 0.000s |
| Creates output file                                                           | <code>output.exists()</code>                                                                | Pass   | 0.002s |
| History entries have date                                                     | <code>'date' in entry</code>                                                                | Pass   | 0.003s |
| Metadata lists gauges                                                         | <code>'GAUGE-005' in</code><br><code>data['metadata']['gauges']</code>                      | Pass   | 0.003s |
| Multiple gauges                                                               | <code>'GAUGE-001' in data['history'];</code><br><code>'GAUGE-002' in data['history']</code> | Pass   | 0.002s |
| No eod files returns early                                                    | <code>not output.exists()</code>                                                            | Pass   | 0.001s |

| Test Description                                                       | Acceptance Criteria                                                            | Result | Time   |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|--------|
| Output has history                                                     | 'history' in data;                                                             | Pass   | 0.005s |
| Output has metadata                                                    | 'GAUGE-001' in data['history']                                                 | Pass   | 0.002s |
| Tenors sorted numerically                                              | 'metadata' in data;                                                            | Pass   | 0.002s |
| Two eod files two history entries                                      | data['metadata']['num_snapshots'] == 2                                         | Pass   | 0.005s |
| Empty dir produces no file                                             | len(entries) == 1; tenor_keys == sorted(tenor_keys, key=int)                   | Pass   | 0.005s |
| History entries have date                                              | len(entries) == 2                                                              | Pass   | 0.005s |
| History entries have tenors                                            | not output.exists(); not output.exists()                                       | Pass   | 0.000s |
| History keyed by gauge                                                 | entry["date"] == "2026-01-02"                                                  | Pass   | 0.002s |
| History keyed by trigger                                               | "1" in entry and "2" in entry and "3" in entry                                 | Pass   | 0.004s |
| Metadata has generated at                                              | "GAUGE-042" in data["history"]                                                 | Pass   | 0.002s |
| Metadata num snapshots                                                 | "alert" in gauge_hist; "warning" in gauge_hist                                 | Pass   | 0.002s |
| Multiple gauges in metadata                                            | "generated_at" in data["metadata"];                                            | Pass   | 0.005s |
| Multiple snapshots produce multiple entries                            | "generated_at" in data["metadata"]                                             | Pass   | 0.005s |
| Output has metadata and history                                        | data["metadata"]["num_snapshots"] == 3; data["metadata"]["num_snapshots"] == 3 | Pass   | 0.004s |
| Single snapshot creates file                                           | set(data["metadata"]["gauges"]) == {"GAUGE-001", "GAUGE-0...                   | Pass   | 0.002s |
| Standalone check: _business_days always returns Mon-Fri dates.         | len(alert_entries) == 3                                                        | Pass   | 0.007s |
| Pre-existing EOD-*.json files are deleted before simulation.           | "metadata" in data; "history" in data                                          | Pass   | 0.004s |
| Creates eod dir                                                        | output.exists(); output.exists()                                               | Pass   | 0.005s |
| Empty trades returns zero                                              | len(result) == 63; d.weekday() < 5                                             | Pass   | 0.000s |
| A trade that fails enrich_trade is skipped gracefully (except branch). | not stale.exists()                                                             | Pass   | 0.004s |
| More trades should generally yield more (or equal) snapshots.          | (trading_dir / 'eod').exists()                                                 | Pass   | 0.002s |
| Returns integer                                                        | count == 0                                                                     | Pass   | 0.003s |
| Same seed should produce same random walk (deterministic).             | count == 0                                                                     | Pass   | 0.015s |
|                                                                        | count > 0                                                                      | Pass   | 0.236s |
|                                                                        | isinstance(result, int)                                                        | Pass   | 0.002s |
|                                                                        | count1 == count2                                                               | Pass   | 0.004s |

| Test Description                                                    | Acceptance Criteria                                                                                                                  | Result | Time   |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| With a real trade + real engine, we expect 0 snapshots.             | <code>count &gt; 0</code>                                                                                                            | Pass   | 0.151s |
| EOD snapshot JSON files are created in eod/ directory.              | <code>len(eod_files) == count</code>                                                                                                 | Pass   | 0.180s |
| hazard_curve_history.json and trade_pnl_history.json are created.   | <code>(trading_dir / 'hazard_curve_history.json').exists();</code><br><code>(trading_dir / 'trade_pnl_history.json').exists()</code> | Pass   | 0.361s |
| Arrears months current                                              | <code>generate_integer_value("ArrearsMonths", self._fd()) == 0</code>                                                                | Pass   | 0.000s |
| Arrears months in arrears                                           | <code>1 &lt;= result &lt;= 6</code>                                                                                                  | Pass   | 0.000s |
| Behavioral score                                                    | <code>30 &lt;= result &lt;= 100</code>                                                                                               | Pass   | 0.000s |
| Borrower age btl                                                    | <code>35 &lt;= result &lt;= 65</code>                                                                                                | Pass   | 0.000s |
| Borrower age expensive                                              | <code>35 &lt;= result &lt;= 65</code>                                                                                                | Pass   | 0.000s |
| Borrower age normal                                                 | <code>25 &lt;= result &lt;= 60</code>                                                                                                | Pass   | 0.000s |
| Credit score expensive property                                     | <code>700 &lt;= result &lt;= 800</code>                                                                                              | Pass   | 0.000s |
| Credit score low rate                                               | <code>750 &lt;= result &lt;= 850</code>                                                                                              | Pass   | 0.000s |
| Credit score normal                                                 | <code>650 &lt;= result &lt;= 750</code>                                                                                              | Pass   | 0.000s |
| Days in arrears current                                             | <code>generate_integer_value("DaysInArrears", self._fd()) == 0</code>                                                                | Pass   | 0.000s |
| Days in arrears defaulted                                           | <code>90 &lt;= result &lt;= 365</code>                                                                                               | Pass   | 0.000s |
| Days in arrears in arrears                                          | <code>1 &lt;= result &lt;= 89</code>                                                                                                 | Pass   | 0.000s |
| Family members                                                      | <code>result in [1, 2, 3, 4]</code>                                                                                                  | Pass   | 0.000s |
| Loan term                                                           | <code>generate_integer_value("LoanTerm", self._fd()) == 300</code>                                                                   | Pass   | 0.000s |
| Number of borrowers                                                 | <code>result in [1, 2]</code>                                                                                                        | Pass   | 0.000s |
| Original term                                                       | <code>generate_integer_value("OriginalTerm", self._fd()) == 300</code>                                                               | Pass   | 0.000s |
| Prepayment risk high rate                                           | <code>7 &lt;= result &lt;= 10</code>                                                                                                 | Pass   | 0.000s |
| Prepayment risk low rate                                            | <code>1 &lt;= result &lt;= 5</code>                                                                                                  | Pass   | 0.000s |
| Remaining term                                                      | <code>result == 240</code>                                                                                                           | Pass   | 0.000s |
| Total payments                                                      | <code>result == 60</code>                                                                                                            | Pass   | 0.000s |
| Unknown field                                                       | <code>1 &lt;= result &lt;= 10; isinstance(result, str) (+1 more)</code>                                                              | Pass   | 0.000s |
| Each location has required keys                                     | <code>"lat" in loc; "lon" in loc (+4 more)</code>                                                                                    | Pass   | 0.000s |
| Elevation is non negative                                           | <code>loc["elevation"] &gt;= 0</code>                                                                                                | Pass   | 0.000s |
| With 3 gauge points, fallback generator is called.                  | <code>len(locs) == 3</code>                                                                                                          | Pass   | 0.000s |
| Fallback used when no gauge points                                  | <code>len(locs) == 3</code>                                                                                                          | Pass   | 0.000s |
| When primary_idx == 0, triangle is gauges 0, 1, 2 - sorted [0,1,2]. | <code>set(ref) == {0, 1, 2}</code>                                                                                                   | Pass   | 0.000s |

| Test Description                                                        | Acceptance Criteria                                                                   | Result | Time   |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Reference gauge indices sorted                                          | <code>idxs == sorted(idx)</code>                                                      | Pass   | 0.001s |
| Reference gauge indices three elements                                  | <code>len(loc["reference-gauge-indices"]) == 3</code>                                 | Pass   | 0.000s |
| Returns correct count                                                   | <code>len(locs) == 10</code>                                                          | Pass   | 0.001s |
| When params has no get_elevation, elevation is sampled from             | <code>len(locs) == 5; 0 &lt;= loc["elevation"] &lt;= 40</code>                        | Pass   | 0.000s |
| Application channel                                                     | <code>result in ["Retail", "Broker", "Correspondent"]</code>                          | Pass   | 0.000s |
| Loan purpose btl                                                        | <code>result in ["Purchase", "Refinancing"]</code>                                    | Pass   | 0.000s |
| Loan purpose residential                                                | <code>result in ["Purchase", "Refinancing", "Home Improvement"]</code>                | Pass   | 0.000s |
| Mortgage type                                                           | <code>result == "Buy-to-Let"</code>                                                   | Pass   | 0.000s |
| Occupancy type                                                          | <code>result == "Investment"</code>                                                   | Pass   | 0.000s |
| Original rate type                                                      | <code>isinstance(result, str) and len(result) &gt; 0</code>                           | Pass   | 0.000s |
| Payment frequency                                                       | <code>generate_menu_value("PaymentFrequency", {}, 0, self._fd())...</code>            | Pass   | 0.000s |
| Repayment type btl                                                      | <code>result in ["Interest only", "Repayment", "Part and part"]</code>                | Pass   | 0.000s |
| Repayment type residential                                              | <code>result in ["Repayment", "Interest only", "Part and part"]</code>                | Pass   | 0.000s |
| Unknown field no options                                                | <code>result == ""</code>                                                             | Pass   | 0.000s |
| Unknown field with options                                              | <code>result in ["X", "Y"]</code>                                                     | Pass   | 0.000s |
| Convenience function creates pdf                                        | <code>output.exists()</code>                                                          | Pass   | 0.011s |
| Convenience function returns path                                       | <code>isinstance(result, Path)</code>                                                 | Pass   | 0.012s |
| Lines 203-208: exception → 500.                                         | <code>r.status_code in (200, 500)</code>                                              | Pass   | 0.001s |
| Lines 174-193: full success path with mocked mortgage report generator. | <code>r.status_code in (200, 500)</code>                                              | Pass   | 0.002s |
| Missing property id returns 400                                         | <code>r.status_code == 400; "Property ID" in r.get_json()["message"] (+2 more)</code> | Pass   | 0.001s |
| Mortgage not found returns 404                                          | <code>r.status_code == 404; "mortgage" in r.get_json()["message"].lower()</code>      | Pass   | 0.001s |
| Line 149: get_json() returns None → 400.                                | <code>r.status_code in (400, 415, 500); r.status_code in (400, 415, 500)</code>       | Pass   | 0.001s |
| No json returns error                                                   | <code>r.status_code in (400, 415); r.status_code in (400, 415)</code>                 | Pass   | 0.001s |
| Options returns ok                                                      | <code>r.status_code == 200; r.get_json()["status"] == "ok" (+2 more)</code>           | Pass   | 0.001s |

| Test Description                                                                                               | Acceptance Criteria                                                                   | Result | Time   |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Property not found returns 404                                                                                 | <code>r.status_code == 404; r.status_code == 404</code>                               | Pass   | 0.001s |
| Mortgage import error returns 500                                                                              | <code>r.status_code == 500; data["status"] == "error"</code>                          | Pass   | 0.003s |
| Raises without property file                                                                                   | —                                                                                     | Pass   | 0.001s |
| Returns dict when property file exists                                                                         | <code>isinstance(result, dict); "data" in result (+1 more)</code>                     | Pass   | 0.024s |
| Lines 332-334: <code>open_pdf_file</code> raises - <code>Warning</code> logged, no raise.                      | <code>result is not None</code>                                                       | Pass   | 0.001s |
| Lines 335-336: <code>auto_open=False</code> - <code>Warning</code> skips opening.                              | —                                                                                     | Pass   | 0.001s |
| Lines 327-334: <code>auto_open=True</code> - <code>Warning</code> calls <code>open_pdf_file</code> .           | <code>mock_open.called</code>                                                         | Pass   | 0.001s |
| Lines 128-133: exception in report generation → 500.                                                           | <code>r.status_code in (500, 404, 400)</code>                                         | Pass   | 0.006s |
| Lines 94-119: full success path with mocked report generator.                                                  | <code>r.status_code in (200, 500); data["status"] == "success" (+1 more)</code>       | Pass   | 0.003s |
| Missing property id returns 400                                                                                | <code>r.status_code == 400; "Property ID" in r.get_json()["message"] (+2 more)</code> | Pass   | 0.001s |
| Lines 319-320: <code>'mortgage-focused'</code> with <code>mortgage_data</code> .                               | —                                                                                     | Pass   | 0.001s |
| Line 74: <code>get_json()</code> returns <code>None</code> → 400.                                              | <code>r.status_code in (400, 415, 500); r.status_code in (400, 415, 500)</code>       | Pass   | 0.001s |
| No json returns error                                                                                          | <code>r.status_code in (400, 415); r.status_code in (400, 415)</code>                 | Pass   | 0.001s |
| Options returns ok                                                                                             | <code>r.status_code == 200; r.get_json()["status"] == "ok" (+2 more)</code>           | Pass   | 0.001s |
| Property not found returns 404                                                                                 | <code>r.status_code == 404; r.status_code == 404</code>                               | Pass   | 0.001s |
| Line 317-318: <code>'property-only'</code> - <code>Warning</code> generate <code>property_only_report</code> . | —                                                                                     | Pass   | 0.001s |
| Lines 321-322: <code>'risk-focused'</code> .                                                                   | —                                                                                     | Pass   | 0.001s |
| Lines 323-324: <code>unknown type</code> - <code>Warning</code> generate <code>report</code> .                 | —                                                                                     | Pass   | 0.001s |
| Auto open exception does not raise                                                                             | <code>result is not None</code>                                                       | Pass   | 0.001s |
| Auto open false skips open                                                                                     | —                                                                                     | Pass   | 0.001s |
| Auto open true calls open                                                                                      | —                                                                                     | Pass   | 0.001s |
| Full type falls back to generate report                                                                        | —                                                                                     | Pass   | 0.001s |



| Test Description                                            | Acceptance Criteria                                                                                           | Result | Time   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| Mortgage focused type with mortgage                         | —                                                                                                             | Pass   | 0.001s |
| Mortgage focused without mortgage falls back                | —                                                                                                             | Pass   | 0.001s |
| Property only type                                          | —                                                                                                             | Pass   | 0.001s |
| Returns path                                                | <code>result == fake_pdf</code>                                                                               | Pass   | 0.001s |
| Risk focused type                                           | —                                                                                                             | Pass   | 0.001s |
| Unknown type falls back to generate report                  | —                                                                                                             | Pass   | 0.001s |
| Line 167: log milestone every 50 properties.                | <code>any("50" in m for m in log_messages)</code>                                                             | Pass   | 0.017s |
| Line 171: ratio branch when total_storms_at_gauge $\neq$ 0. | <code>"gauge_to_property_ratio" in result;</code><br><code>result["gauge_to_property_ratio"] &gt;= 0.0</code> | Pass   | 0.219s |
| Line 176: ratio=0 branch when total_storms_at_gauge == 0.   | <code>result["gauge_to_property_ratio"] == 0.0</code>                                                         | Pass   | 0.006s |
| All unknown pages still creates pdf                         | <code>path.exists()</code>                                                                                    | Pass   | 0.002s |
| All unknown pages still creates pdf                         | <code>path.exists()</code>                                                                                    | Pass   | 0.002s |
| Auto select called when pages none                          | —                                                                                                             | Pass   | 0.024s |
| Auto select called when pages none                          | —                                                                                                             | Pass   | 0.017s |
| Auto select not called when pages provided                  | —                                                                                                             | Pass   | 0.005s |
| Auto select not called when pages provided                  | —                                                                                                             | Pass   | 0.003s |
| Auto select pages when none                                 | —                                                                                                             | Pass   | 0.017s |
| Auto select skipped when provided                           | —                                                                                                             | Pass   | 0.004s |
| Custom output filename                                      | <code>path.name == "my_gauge.pdf";</code><br><code>path.exists()</code>                                       | Pass   | 0.012s |
| Custom output filename                                      | <code>path.name == "my_prop.pdf";</code><br><code>path.exists()</code>                                        | Pass   | 0.017s |
| Custom output filename                                      | <code>path.name == "my_report.pdf";</code><br><code>path.exists()</code>                                      | Pass   | 0.023s |
| Custom pages to include                                     | <code>path.exists()</code>                                                                                    | Pass   | 0.005s |
| Custom pages to include                                     | <code>path.exists()</code>                                                                                    | Pass   | 0.005s |
| Custom pages to include                                     | <code>path.exists()</code>                                                                                    | Pass   | 0.016s |
| Empty flood data                                            | <code>path.exists()</code>                                                                                    | Pass   | 0.017s |
| Empty gauge data                                            | <code>path.exists()</code>                                                                                    | Pass   | 0.011s |
| Empty property data                                         | <code>path.exists()</code>                                                                                    | Pass   | 0.015s |
| Full data report                                            | <code>path.exists(); path.stat().st_size &gt; 0</code>                                                        | Pass   | 0.022s |
| Full data report                                            | <code>path.exists(); path.stat().st_size &gt; 0</code>                                                        | Pass   | 0.023s |
| Pdf exists                                                  | <code>gen.generate_report(gauge_data).exists()</code>                                                         | Pass   | 0.011s |

| Test Description                                                  | Acceptance Criteria                                               | Result | Time   |
|-------------------------------------------------------------------|-------------------------------------------------------------------|--------|--------|
| Pdf exists                                                        | <code>gen.generate_report(prop_data).exists()</code>              | Pass   | 0.017s |
| Pdf exists                                                        | <code>path.exists()</code>                                        | Pass   | 0.017s |
| Pdf is bytes                                                      | <code>isinstance(content, bytes);<br/>len(content) &gt; 0</code>  | Pass   | 0.033s |
| Pdf non empty                                                     | <code>path.stat().st_size &gt; 0</code>                           | Pass   | 0.012s |
| Pdf non empty                                                     | <code>path.stat().st_size &gt; 0</code>                           | Pass   | 0.017s |
| Pdf non empty                                                     | <code>path.stat().st_size &gt; 0</code>                           | Pass   | 0.021s |
| Pdf suffix                                                        | <code>gen.generate_report(gauge_data).suffix<br/>== ".pdf"</code> | Pass   | 0.011s |
| Pdf suffix                                                        | <code>gen.generate_report(prop_data).suffix<br/>== ".pdf"</code>  | Pass   | 0.020s |
| Pdf suffix                                                        | <code>path.suffix == ".pdf"</code>                                | Pass   | 0.021s |
| Returns path                                                      | <code>isinstance(result, Path)</code>                             | Pass   | 0.012s |
| Returns path                                                      | <code>isinstance(result, Path)</code>                             | Pass   | 0.017s |
| Returns path                                                      | <code>isinstance(result, Path)</code>                             | Pass   | 0.015s |
| With mortgage data                                                | <code>path.exists()</code>                                        | Pass   | 0.023s |
| With timeseries data                                              | <code>path.exists()</code>                                        | Pass   | 0.044s |
| gauge_integrator should handle<br>flood_gauges key too.           | <code>result is not None</code>                                   | Pass   | 0.002s |
| Found gauge returns path                                          | <code>result is not None;<br/>isinstance(result, Path)</code>     | Pass   | 0.002s |
| Gauge not found returns none                                      | <code>result is None</code>                                       | Pass   | 0.023s |
| Missing file returns none                                         | <code>result is None</code>                                       | Pass   | 0.000s |
| Output contains gauge id                                          | <code>"GAUGE-001" in content</code>                               | Pass   | 0.002s |
| Output dir created                                                | <code>output_dir.exists()</code>                                  | Pass   | 0.001s |
| Output file created                                               | <code>result.exists()</code>                                      | Pass   | 0.002s |
| Corrupt json returns none                                         | <code>result is None</code>                                       | Pass   | 0.043s |
| Non json file returns none                                        | <code>result is None</code>                                       | Pass   | 0.001s |
| Found property returns path                                       | <code>result is not None;<br/>isinstance(result, Path)</code>     | Pass   | 0.003s |
| Missing property file returns none                                | <code>result is None</code>                                       | Pass   | 0.025s |
| No mortgage file still generates                                  | <code>result is not None</code>                                   | Pass   | 0.001s |
| Output contains property id                                       | <code>"PROP-001" in content</code>                                | Pass   | 0.002s |
| Output dir created                                                | <code>output_dir.exists()</code>                                  | Pass   | 0.001s |
| Output file created                                               | <code>result.exists()</code>                                      | Pass   | 0.001s |
| Property not found returns none                                   | <code>result is None</code>                                       | Pass   | 0.021s |
| A file with invalid JSON triggers an<br>exception -i return None. | <code>result is None</code>                                       | Pass   | 0.001s |
| Simulated write permission error re-<br>turns None.               | <code>result is None</code>                                       | Pass   | 0.001s |

| Test Description                                                          | Acceptance Criteria                                                                   | Result | Time   |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Import error returns 500                                                  | <code>r.status_code == 500; data["status"] == "error"</code>                          | Pass   | 0.089s |
| Has report generated timestamp                                            | <code>"Report Generated" in r</code>                                                  | Pass   | 0.000s |
| Last updated extracted                                                    | <code>r["Data Last Updated"] == "2025-01-01"</code>                                   | Pass   | 0.000s |
| Mortgage id extracted                                                     | <code>r.get("Mortgage ID") == "MORT-001"</code>                                       | Pass   | 0.000s |
| Property id extracted                                                     | <code>r["Property ID"] == "PROP-999"</code>                                           | Pass   | 0.000s |
| Returns dict                                                              | <code>isinstance(r, dict); isinstance(r, dict)</code>                                 | Pass   | 0.000s |
| Scope combined with mortgage                                              | <code>"Mortgage" in r["Analysis Scope"]</code>                                        | Pass   | 0.000s |
| Scope property only                                                       | <code>"Property-Only" in r["Analysis Scope"]</code>                                   | Pass   | 0.000s |
| Unknown property id when missing                                          | <code>r["Property ID"] == "Unknown"</code>                                            | Pass   | 0.000s |
| Upnrn extracted                                                           | <code>r["UPRN"] == "123456789"</code>                                                 | Pass   | 0.000s |
| Lines 263-264: 'analysis' type.                                           | <code>path.exists()</code>                                                            | Pass   | 0.018s |
| Line 257-258: 'basic' type.                                               | <code>path.exists()</code>                                                            | Pass   | 0.018s |
| Lines 259-260: 'detailed' type.                                           | <code>path.exists()</code>                                                            | Pass   | 0.025s |
| Lines 261-262: 'summary' type.                                            | <code>path.exists()</code>                                                            | Pass   | 0.010s |
| Lines 265-266: unknown type → generate_report.                            | <code>path.exists()</code>                                                            | Pass   | 0.024s |
| Analysis type returns path                                                | <code>path.exists()</code>                                                            | Pass   | 0.011s |
| Basic type returns path                                                   | <code>isinstance(path, Path); path.exists()</code>                                    | Pass   | 0.011s |
| Omitting report_type should default to 'basic'.                           | —                                                                                     | Pass   | 0.011s |
| Detailed type returns path                                                | <code>path.exists()</code>                                                            | Pass   | 0.013s |
| Output dir as string                                                      | <code>path.exists()</code>                                                            | Pass   | 0.011s |
| Pdf is non empty bytes on disk                                            | <code>isinstance(content, bytes); len(content) &gt; 0</code>                          | Pass   | 0.015s |
| Summary type returns path                                                 | <code>path.exists()</code>                                                            | Pass   | 0.009s |
| Unknown type falls back to generate report                                | <code>path.exists()</code>                                                            | Pass   | 0.013s |
| Empty gaugets directory must raise FileNotFoundError.                     | —                                                                                     | Pass   | 0.005s |
| Each index entry must have gauge_ids_alert for directory-based filtering. | <code>"gauge_ids_alert" in entry; "GAUGE-00000001" in entry["gauge_ids_alert"]</code> | Pass   | 0.005s |
| Index has required top level keys                                         | <code>key in data</code>                                                              | Pass   | 0.003s |
| Individual storm files must contain the full gauge_responses array.       | <code>"gauge_responses" in storm; len(storm["gauge_responses"]) == 1</code>           | Pass   | 0.002s |

| Test Description                                                            | Acceptance Criteria                                                                 | Result | Time   |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|--------|
| Storm id prefix uses config constant                                        | <code>data["storm_id_prefix"] == STORM_ID.PREFIX</code>                             | Pass   | 0.003s |
| Storms where no gauge exceeded alert must NOT appear in output.             | <code>result["total_storms"] == 0</code>                                            | Pass   | 0.002s |
| Total storms matches list length                                            | <code>data["total_storms"] == len(data["storms"])</code>                            | Pass   | 0.004s |
| Output directory must contain <code>_index.json</code> and per-storm files. | <code>(out_dir / "_index.json").exists(); (out_dir / f"{sid}.json").exists()</code> | Pass   | 0.003s |
| Gauge summary gauge ids                                                     | <code>len(d["gauge_ids"]) == 3</code>                                               | Pass   | 0.000s |
| Gauge summary json written                                                  | <code>(gauge_dir / GAUGE_SUMMARY_FILENAME).exists()</code>                          | Pass   | 0.000s |
| Gauge summary num gauges                                                    | <code>d["num_gauges"] == 3</code>                                                   | Pass   | 0.001s |
| Gauge summary schema version                                                | <code>d["schema_version"] == SCHEMA_VERSION.SPATIAL</code>                          | Pass   | 0.046s |
| Gauge summary sequence count                                                | <code>len(d["sequences"]) == 40</code>                                              | Pass   | 0.000s |
| Sequences file loadable                                                     | <code>len(seqs) == 40</code>                                                        | Pass   | 0.001s |
| Sequences json written                                                      | <code>(gauge_dir / SEQUENCES_FILENAME).exists()</code>                              | Pass   | 0.000s |
| Summary json written                                                        | <code>(gauge_dir / SUMMARY_FILENAME).exists()</code>                                | Pass   | 0.000s |
| Alert count non negative                                                    | <code>full_run["alert_sequences"] &gt;= 0</code>                                    | Pass   | 0.000s |
| Elapsed positive                                                            | <code>full_run["elapsed_seconds"] &gt; 0</code>                                     | Pass   | 0.000s |
| Num gauges                                                                  | <code>full_run["num_gauges"] == 3</code>                                            | Pass   | 0.000s |
| Num sequences                                                               | <code>full_run["num_sequences"] == 40</code>                                        | Pass   | 0.000s |
| Returns dict                                                                | <code>isinstance(full_run, dict)</code>                                             | Pass   | 0.000s |
| Severe lte warning                                                          | <code>full_run["severe_sequences"] &lt;= full_run["warning_sequenc...</code>        | Pass   | 0.000s |
| Type counts present                                                         | <code>"type_counts" in full_run; sum(full_run["type_counts"].values()) == 40</code> | Pass   | 0.000s |
| Warning lte alert                                                           | <code>full_run["warning_sequences"] &lt;= full_run["alert_sequences"]</code>        | Pass   | 0.000s |
| All levels positive                                                         | <code>all(obs["level_meters"] &gt; 0 for obs in result)</code>                      | Pass   | 0.006s |
| Correct length approximately                                                | <code>728 &lt;= len(result) &lt;= 732</code>                                        | Pass   | 0.003s |
| Dates are sequential                                                        | <code>(d_curr - d_prev).days == 1</code>                                            | Pass   | 0.004s |
| Different seeds different output                                            | <code>levels1 != levels2</code>                                                     | Pass   | 0.013s |
| Each obs has date and level                                                 | <code>"date" in obs; "level_meters" in obs (+1 more)</code>                         | Pass   | 0.002s |
| Gauge data without FloodGauge wrapper uses defaults.                        | <code>isinstance(result, list)</code>                                               | Pass   | 0.001s |

| Test Description                                                                | Acceptance Criteria                                                        | Result | Time   |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| When historical high date is within range, that day gets historical_high level. | <code>match[0]["level_meters"] == 7.5</code>                               | Pass   | 0.001s |
| Invalid date format doesn't crash.                                              | <code>isinstance(result, list)</code>                                      | Pass   | 0.001s |
| Historical high date before start_date is ignored.                              | <code>isinstance(result, list)</code>                                      | Pass   | 0.001s |
| Gauge data with no FloodStages uses sensible defaults.                          | <code>isinstance(result, list); len(result) &gt; 0</code>                  | Pass   | 0.001s |
| Returns list                                                                    | <code>isinstance(result, list)</code>                                      | Pass   | 0.002s |
| Seed reproducible                                                               | <code>r1 == r2</code>                                                      | Pass   | 0.003s |
| Account status defaulted                                                        | <code>result == "Default"</code>                                           | Pass   | 0.000s |
| Account status normal                                                           | <code>result in ["Current", "Arrears", "Closed"]</code>                    | Pass   | 0.000s |
| Borrower employment btl                                                         | <code>isinstance(result, str)</code>                                       | Pass   | 0.000s |
| Catchment id                                                                    | <code>generate_text_value("CatchmentID", 0, self._fd()) == "tha..."</code> | Pass   | 0.000s |
| Currency                                                                        | <code>generate_text_value("currency", 0, self._fd()) == "GBP"</code>       | Pass   | 0.000s |
| Employment type residential                                                     | <code>isinstance(result, str)</code>                                       | Pass   | 0.000s |
| Flood risk category                                                             | <code>result == "Medium"</code>                                            | Pass   | 0.000s |
| Lastatus defaulted                                                              | <code>result == "Defaulted"</code>                                         | Pass   | 0.000s |
| Lastatus normal                                                                 | <code>result in ["Current", "Completed", "Redeemed"]</code>                | Pass   | 0.000s |
| Marital status older                                                            | <code>isinstance(result, str)</code>                                       | Pass   | 0.000s |
| Marital status young                                                            | <code>result in ["Single", "Married", "Civil Partnership"]</code>          | Pass   | 0.000s |
| Member id                                                                       | <code>result.startswith("MEMBER-")</code>                                  | Pass   | 0.000s |
| Mortgage id                                                                     | <code>generate_text_value("MortgageID", 0, self._fd()) == "MORT..."</code> | Pass   | 0.000s |
| Mortgage provider                                                               | <code>isinstance(result, str)</code>                                       | Pass   | 0.000s |
| Property id                                                                     | <code>generate_text_value("PropertyID", 0, self._fd()) == "PROP..."</code> | Pass   | 0.000s |
| Unknown field                                                                   | <code>1 &lt;= result &lt;= 10; isinstance(result, str) (+1 more)</code>    | Pass   | 0.000s |
| Uprn no existing                                                                | <code>result.startswith("UPRN-") or isinstance(result, str)</code>         | Pass   | 0.000s |
| Uprn with existing                                                              | <code>result == "UPRN-12345"</code>                                        | Pass   | 0.000s |
| Alias returns same structure                                                    | <code>len(result) == 50; "readings" in result[0]</code>                    | Pass   | 0.000s |
| CloseOutDate with Payable SettlementDirection.                                  | <code>pdf_path.exists()</code>                                             | Pass   | 0.003s |

| Test Description                                                        | Acceptance Criteria                                                         | Result | Time   |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------|
| _generate_trade_pdf creates a PDF without cashflows.                    | pdf_path.exists();<br>pdf_path.stat().st_size > 1000                        | Pass   | 0.002s |
| _generate_trade_pdf with cashflows exercises CF schedule section.       | pdf_path.exists()                                                           | Pass   | 0.003s |
| _generate_trade_pdf with CloseOutOf exercises close-out section.        | pdf_path.exists()                                                           | Pass   | 0.003s |
| _generate_trade_pdf with CloseOutDate (original trade that was closed). | pdf_path.exists()                                                           | Pass   | 0.003s |
| Pdf failure returns empty list                                          | pdfs == []                                                                  | Pass   | 0.001s |
| Pdf partial failure returns successful only                             | len(pdfs) == 1                                                              | Pass   | 0.001s |
| Creates output file                                                     | output.exists()                                                             | Pass   | 0.007s |
| History entry has required keys                                         | 'date' in entry; 'mark' in entry (+2 more)                                  | Pass   | 0.002s |
| Multiple eods accumulate                                                | len(trade['history']) == 3                                                  | Pass   | 0.003s |
| Multiple trades in same eod                                             | 'PRS-A' in data['trades']; 'PRS-B' in data['trades'] (+1 more)              | Pass   | 0.003s |
| No eod files returns early                                              | not output.exists()                                                         | Pass   | 0.000s |
| Num snapshots in metadata                                               | data['metadata']['num_snapshots'] == 2                                      | Pass   | 0.004s |
| Output has metadata                                                     | 'metadata' in data; 'num_trades' in data['metadata']                        | Pass   | 0.004s |
| Output has trades                                                       | 'trades' in data; 'PRS-001' in data['trades']                               | Pass   | 0.003s |
| Position without swap id skipped                                        | data['trades'] == {};<br>data['metadata']['num_trades'] == 0                | Pass   | 0.003s |
| Trade has history                                                       | 'history' in trade;<br>len(trade['history']) == 1                           | Pass   | 0.002s |
| Trade metadata preserved                                                | trade['gauge_id'] == 'GAUGE-001';<br>trade['trigger'] == 'severe' (+3 more) | Pass   | 0.003s |
| Empty dir produces no file                                              | not output.exists();<br>not output.exists()                                 | Pass   | 0.000s |
| History entry structure                                                 | entry["date"] == "2026-01-02";<br>entry["mark"] == 215.0 (+2 more)          | Pass   | 0.002s |
| Metadata has generated at                                               | "generated_at" in data["metadata"];<br>"generated_at" in data["metadata"]   | Pass   | 0.002s |



| Test Description                                                  | Acceptance Criteria                                                                         | Result | Time   |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Metadata num snapshots                                            | <code>data["metadata"]["num_snapshots"] == 3; data["metadata"]["num_snapshots"] == 3</code> | Pass   | 0.003s |
| Multiple days builds history vector                               | <code>len(data["trades"]["PRS-001"]["history"]) == 3</code>                                 | Pass   | 0.004s |
| Multiple trades in snapshot                                       | <code>data["metadata"]["num_trades"] == 3</code>                                            | Pass   | 0.002s |
| Output has metadata and trades                                    | <code>"metadata" in data; "trades" in data</code>                                           | Pass   | 0.003s |
| Position missing swap id is skipped                               | <code>"PRS-001" in data["trades"]; len(data["trades"]) == 1</code>                          | Pass   | 0.002s |
| Single snapshot creates file                                      | <code>output.exists(); output.exists()</code>                                               | Pass   | 0.002s |
| Trade keyed by swap id                                            | <code>"PRS-001" in data["trades"]</code>                                                    | Pass   | 0.002s |
| Trade metadata fields                                             | <code>"gauge_id" in t; "trigger" in t (+3 more)</code>                                      | Pass   | 0.002s |
| Explicit years overrides instance default                         | <code>result["years_included"] == 1</code>                                                  | Pass   | 0.026s |
| Instance years used when no years arg                             | <code>result["years_included"] == 2</code>                                                  | Pass   | 0.051s |
| If all sequence storms are sub-alert, portfolio flood count is 0. | <code>result["total_storms_at_gauge"] == 0; result["properties_with_floods"] == 0</code>    | Pass   | 0.004s |
| Portfolio summary total_storms_at_gauge counts all alert storms.  | <code>result["total_storms_at_gauge"] == 2</code>                                           | Pass   | 0.005s |
| portfolio_flood_summary.json is written with per-property data.   | <code>summary_path.exists(); summary["summary"]["total_gauges"] == 1 (+1 more)</code>       | Pass   | 0.005s |
| Generate output path none uses config input dir                   | <code>isinstance(result, dict); (tmp_path / "gaugehd.39001.json").exists()</code>           | Pass   | 0.021s |
| Banks always have CSA Agreement = True.                           | <code>platform["CSA Agreement"] is True</code>                                              | Pass   | 0.001s |
| Counterparties list has two parties                               | <code>len(counterparties) == 2</code>                                                       | Pass   | 0.000s |
| Counterparty has counterparty set                                 | <code>"CounterpartySet" in ctpy</code>                                                      | Pass   | 0.000s |
| Isda master agreement is true                                     | <code>platform["ISDAMasterAgreement"] is True</code>                                        | Pass   | 0.000s |
| Party has accounts                                                | <code>"Accounts" in party; len(party["Accounts"]) &gt;= 1</code>                            | Pass   | 0.000s |
| Party has contact info                                            | <code>"ContactInformation" in party; "PrimaryEmail" in party["ContactInformation"]</code>   | Pass   | 0.000s |
| Party has id                                                      | <code>"PartyID" in party; party["PartyID"].startswith("CTPY-")</code>                       | Pass   | 0.001s |

| Test Description                                                                 | Acceptance Criteria                                                             | Result | Time   |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|--------|
| Party has name                                                                   | "PartyName" in party;<br>isinstance(party["PartyName"],<br>str)                 | Pass   | 0.000s |
| Party has natural person                                                         | "NaturalPersons" in party;<br>len(party["NaturalPersons"]) ==<br>1              | Pass   | 0.000s |
| Platform has credit rating                                                       | "CreditRating" in platform;<br>platform["CreditRating"] in [ "AAA",<br>"AA+..." | Pass   | 0.001s |
| Unique party ids                                                                 | len(set(ids)) == 5                                                              | Pass   | 0.001s |
| Generator exception falls through to<br>type                                     | isinstance(result, float);      0 <=<br>result <= 1000                          | Pass   | 0.000s |
| Gauge importable without quantlib                                                | hasattr(mod,<br>"GaugePortfolioGenerator")                                      | Pass   | 0.030s |
| models.hazard.builder must not<br>drag in QuantLib pricing at import.            | mod is not None                                                                 | Pass   | 0.015s |
| Mortgage importable without<br>quantlib                                          | hasattr(mod,<br>"MortgagePortfolioGenerator")                                   | Pass   | 0.002s |
| Importing port.src (package __init__)<br>must not require QuantLib.              | hasattr(mod,<br>"GaugePortfolioGenerator")                                      | Pass   | 0.004s |
| Storm multi importable without<br>quantlib                                       | mod is not None                                                                 | Pass   | 0.043s |
| Depth damage importable without<br>geopandas                                     | hasattr(mod, 'scalar_depth_damage')                                             | Pass   | 0.004s |
| __init__.py must not have an ex-<br>ecutable FloodRiskModel import<br>statement. | 'FloodRiskModel' not in executable                                              | Pass   | 0.002s |
| propertyts imports<br>scalar_depth_damage — must<br>work without geopandas.      | hasattr(mod,<br>'PropertyTimeSeriesGenerator')                                  | Pass   | 0.003s |
| Scalar depth damage works without<br>geopandas                                   | 0.0 < result <= 1.0                                                             | Pass   | 0.002s |
| Spatial importable without geopan-<br>das                                        | hasattr(mod, 'haversine.distance')                                              | Pass   | 0.007s |
| 100 years is heritage                                                            | get_age.band(1925, current_year=2025)<br>== 'heritage'                          | Pass   | 0.000s |
| 10 years is modern                                                               | get_age.band(2015, current_year=2025)<br>== 'modern'                            | Pass   | 0.000s |
| 25 years is established                                                          | get_age.band(2000, current_year=2025)<br>== 'established'                       | Pass   | 0.000s |

| Test Description                | Acceptance Criteria                                             | Result | Time   |
|---------------------------------|-----------------------------------------------------------------|--------|--------|
| 50 years is period              | get_age.band(1975, current_year=2025)<br>== 'period'            | Pass   | 0.000s |
| Default current year            | result in AGE_BAND_FACTORS                                      | Pass   | 0.000s |
| Established                     | get_age.band(1985, current_year=2025)<br>== 'established'       | Pass   | 0.000s |
| Exactly 24 years is modern      | get_age.band(2001, current_year=2025)<br>== 'modern'            | Pass   | 0.000s |
| Exactly 49 years is established | get_age.band(1976, current_year=2025)<br>== 'established'       | Pass   | 0.000s |
| Exactly 99 years is period      | get_age.band(1926, current_year=2025)<br>== 'period'            | Pass   | 0.000s |
| Exactly 9 years is new build    | get_age.band(2016, current_year=2025)<br>== 'new_build'         | Pass   | 0.000s |
| Heritage old                    | get_age.band(1800, current_year=2025)<br>== 'heritage'          | Pass   | 0.000s |
| Modern                          | get_age.band(2005, current_year=2025)<br>== 'modern'            | Pass   | 0.000s |
| New build                       | get_age.band(2020, current_year=2025)<br>== 'new_build'         | Pass   | 0.000s |
| Period                          | get_age.band(1950, current_year=2025)<br>== 'period'            | Pass   | 0.000s |
| 100 years is heritage           | get_age.premium.band(1925,<br>current_year=2025) == 'heritage'  | Pass   | 0.000s |
| 10 years is modern              | get_age.premium.band(2015,<br>current_year=2025) == 'modern'    | Pass   | 0.000s |
| 30 years is established         | get_age.premium.band(1995,<br>current_year=2025) == 'establi... | Pass   | 0.000s |
| Default current year            | result in AGE_PREMIUM_FACTORS                                   | Pass   | 0.000s |
| Established                     | get_age.premium.band(1960,<br>current_year=2025) == 'establi... | Pass   | 0.000s |
| Exactly 29 years modern         | get_age.premium.band(1996,<br>current_year=2025) == 'modern'    | Pass   | 0.000s |
| Exactly 99 years established    | get_age.premium.band(1926,<br>current_year=2025) == 'establi... | Pass   | 0.000s |
| Exactly 9 years new build       | get_age.premium.band(2016,<br>current_year=2025) == 'new_build' | Pass   | 0.000s |
| Heritage old                    | get_age.premium.band(1850,<br>current_year=2025) == 'heritage'  | Pass   | 0.000s |
| Modern                          | get_age.premium.band(2000,<br>current_year=2025) == 'modern'    | Pass   | 0.000s |

| Test Description                                     | Acceptance Criteria                                                       | Result | Time   |
|------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| New build                                            | <code>get_age_premium_band(2020, current_year=2025) == 'new_build'</code> | Pass   | 0.000s |
| Exactly 100 third band                               | <code>(lo, hi) == AREA_PREMIUM_BANDS[2][1]</code>                         | Pass   | 0.000s |
| Exactly 50 second band                               | <code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>                         | Pass   | 0.000s |
| Just below 100 second band                           | <code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>                         | Pass   | 0.000s |
| Just below 50 first band                             | <code>(lo, hi) == AREA_PREMIUM_BANDS[0][1]</code>                         | Pass   | 0.000s |
| Large area fourth band                               | <code>(lo, hi) == AREA_PREMIUM_BANDS[3][1]</code>                         | Pass   | 0.000s |
| Medium area second band                              | <code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>                         | Pass   | 0.000s |
| Returns tuple                                        | <code>isinstance(result, tuple); len(result) == 2</code>                  | Pass   | 0.001s |
| Small area first band                                | <code>(lo, hi) == AREA_PREMIUM_BANDS[0][1]</code>                         | Pass   | 0.000s |
| After log entry appears                              | <code>data["total_entries"] == 1; len(data["entries"]) == 1</code>        | Pass   | 0.004s |
| Empty trail returns 200                              | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>  | Pass   | 0.001s |
| Filter by event type                                 | <code>all(e["event_type"] == "calibration" for e in entries)</code>       | Pass   | 0.004s |
| Filter by model id                                   | <code>r.get_json()["total_entries"] == 1</code>                           | Pass   | 0.005s |
| Filter by model id no match                          | <code>r.get_json()["total_entries"] == 0</code>                           | Pass   | 0.004s |
| Has returned field                                   | <code>"returned" in r.get_json()</code>                                   | Pass   | 0.001s |
| Limit parameter                                      | <code>r.get_json()["returned"] == 2</code>                                | Pass   | 0.008s |
| Missing file returns empty list                      | <code>result == []</code>                                                 | Pass   | 0.000s |
| File with two gauges should return both IDs.         | <code>set(result) == {"GAUGE-A", "GAUGE-B"}</code>                        | Pass   | 0.001s |
| Returns correct id                                   | <code>"GAUGE-001" in result</code>                                        | Pass   | 0.001s |
| Returns list                                         | <code>isinstance(result, list)</code>                                     | Pass   | 0.001s |
| Binary file returns empty                            | <code>result == []</code>                                                 | Pass   | 0.001s |
| Corrupt json returns empty                           | <code>result == []</code>                                                 | Pass   | 0.001s |
| No data returns 404                                  | <code>r.status_code == 404; r.get_json()["status"] == "error"</code>      | Pass   | 0.001s |
| Principles count                                     | <code>len(r.get_json()["assessment"]["principles"]) == 2</code>           | Pass   | 0.002s |
| Response has assessment                              | <code>"assessment" in data; "principles" in data["assessment"]</code>     | Pass   | 0.002s |
| With data returns 200                                | <code>r.status_code == 200; r.get_json()["status"] == "success"</code>    | Pass   | 0.001s |
| Returns 200 if PDF exists, 404 if not yet generated. | <code>r.status_code in (200, 404)</code>                                  | Pass   | 0.020s |
| Empty references                                     | <code>r.get_json()["references"] == []</code>                             | Pass   | 0.001s |
| Has categories key                                   | <code>"categories" in r.get_json()</code>                                 | Pass   | 0.001s |
| Returns 200                                          | <code>r.status_code == 200</code>                                         | Pass   | 0.001s |

| Test Description                                                    | Acceptance Criteria                                                                                      | Result | Time   |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Returns success status                                              | <code>r.get_json()["status"] == "success"</code>                                                         | Pass   | 0.001s |
| Found returns 200                                                   | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                 | Pass   | 0.002s |
| Found second counterparty                                           | <code>r.status_code == 200</code>                                                                        | Pass   | 0.002s |
| Not found returns 404                                               | <code>r.status_code == 404; data["status"] == "error"</code>                                             | Pass   | 0.002s |
| Empty list returns success                                          | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                 | Pass   | 0.001s |
| Has fields                                                          | <code>"owner" in fields; "rag_rating" in fields</code>                                                   | Pass   | 0.001s |
| Returns 200                                                         | <code>r.status_code == 200; r.status_code == 200</code>                                                  | Pass   | 0.001s |
| Returns success status                                              | <code>r.get_json()["status"] == "success"; r.get_json()["status"] == "success"</code>                    | Pass   | 0.000s |
| Empty entity returns none                                           | <code>loader.get_entity_id({}) is None</code>                                                            | Pass   | 0.000s |
| Extracts event id                                                   | <code>loader.get_entity_id(storms[0]) == "EVT-001"</code>                                                | Pass   | 0.001s |
| Missing id returns none                                             | <code>loader.get_entity_id(entity) is None; loader.get_track("EVT-NONEXISTENT") is None (+1 more)</code> | Pass   | 0.000s |
| Missing fields return none                                          | <code>summary["eventId"] is None; summary["name"] is None</code>                                         | Pass   | 0.000s |
| Returns summary dict                                                | <code>summary["eventId"] == "EVT-001"; summary["name"] == "Harvey" (+3 more)</code>                      | Pass   | 0.002s |
| get_eod_snapshot returns snapshot data for existing date.           | <code>result is not None; result['date'] == '2026-03-15'</code>                                          | Pass   | 0.002s |
| get_eod_snapshot returns None when no snapshot exists for the date. | <code>result is None</code>                                                                              | Pass   | 0.000s |
| Gauge not found returns 404                                         | <code>r.status_code == 404; r.get_json()["status"] == "error"</code>                                     | Pass   | 0.002s |
| No curve for gauge returns 404                                      | <code>r.status_code == 404</code>                                                                        | Pass   | 0.003s |
| No hazard file returns 404                                          | <code>r.status_code == 404; "not available" in r.get_json()["message"].lower() (+1 more)</code>          | Pass   | 0.002s |
| Response has annual flood probs                                     | <code>"alert" in probs; "warning" in probs (+1 more)</code>                                              | Pass   | 0.003s |
| Response has curve points                                           | <code>len(r.get_json()["curve_points"]) == 3</code>                                                      | Pass   | 0.003s |
| Response has flood stages                                           | <code>"FloodAlert" in stages; "FloodWarning" in stages (+1 more)</code>                                  | Pass   | 0.003s |

| Test Description                       | Acceptance Criteria                                                                                  | Result | Time   |
|----------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Response has gauge id                  | <code>r.get_json()["gauge_id"] == GAUGE_ID</code>                                                    | Pass   | 0.003s |
| Response has gev parameters            | <code>"location" in gev; "scale" in gev (+1 more)</code>                                             | Pass   | 0.002s |
| Response has term structures           | <code>"alert" in ts; "warning" in ts (+1 more)</code>                                                | Pass   | 0.002s |
| Valid hazard returns 200               | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>               | Pass   | 0.003s |
| Days larger than available returns all | <code>len(data["daily_observations"]) == 12</code>                                                   | Pass   | 0.002s |
| Days param limits observations         | <code>len(data["daily_observations"]) == 5</code>                                                    | Pass   | 0.002s |
| Days param zero returns all            | <code>len(data["daily_observations"]) == 12</code>                                                   | Pass   | 0.001s |
| No history file returns 404            | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code>                 | Pass   | 0.001s |
| Response has daily observations        | <code>"daily_observations" in data;</code><br><code>len(data["daily_observations"]) == 12</code>     | Pass   | 0.002s |
| Response has flood stages              | <code>"flood_stages" in data</code>                                                                  | Pass   | 0.002s |
| Response has gauge metadata            | <code>"gauge_metadata" in data</code>                                                                | Pass   | 0.002s |
| Response has statistics                | <code>"statistics" in data;</code><br><code>data["statistics"]["mean_level"] == 2.5</code>           | Pass   | 0.002s |
| Valid history returns 200              | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>               | Pass   | 0.025s |
| Missing key defaults to zero rate      | <code>result == [0.0] * len(TENORS)</code>                                                           | Pass   | 0.000s |
| Negative rate returns all zeros        | <code>result == [0.0] * len(TENORS)</code>                                                           | Pass   | 0.000s |
| Positive rate returns positive spreads | <code>len(result) == len(TENORS); all(s &gt; 0 for s in result)</code>                               | Pass   | 0.000s |
| Spread list has one entry per tenor    | <code>len(result) == len(TENORS)</code>                                                              | Pass   | 0.000s |
| Zero rate returns all zeros            | <code>result == [0.0] * len(TENORS)</code>                                                           | Pass   | 0.000s |
| Gauge not found returns 404            | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+2 more)    | Pass   | 0.002s |
| Valid gauge returns success            | <code>r.status_code == 200; data["status"] == "success" (+4 more)</code>                             | Pass   | 0.003s |
| Gauge not found returns 404            | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code>                 | Pass   | 0.007s |
| Response has flood simulation          | <code>"flood_simulation" in data;</code><br><code>"num_timesteps" in data["flood_simulation"]</code> | Pass   | 0.008s |
| Response has flood stages              | <code>"flood_stages" in r.get_json()</code>                                                          | Pass   | 0.004s |
| Response has gauge id                  | <code>r.get_json()["gauge_id"] == GAUGE_ID</code>                                                    | Pass   | 0.003s |



| Test Description                                                           | Acceptance Criteria                                                                         | Result | Time   |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------|
| Response has storm responses                                               | "storm_responses" in data; "num_storms" in data["storm_responses"]                          | Pass   | 0.004s |
| Storm responses enriched with metadata from storm_sequences.json.          | "name" in responses[0] or responses[0].get("name") == ""                                    | Pass   | 0.003s |
| Valid gauge returns 200                                                    | r.status_code == 200; r.get_json()["status"] == "success"                                   | Pass   | 0.004s |
| When exceeded_severe=True, severe count computed from gauges dir.          | r.status_code == 200                                                                        | Pass   | 0.003s |
| Gauge not found returns 404                                                | r.status_code == 404; r.get_json()["status"] == "error" (+2 more)                           | Pass   | 0.003s |
| Response has count and readings                                            | "count" in data; "readings" in data (+1 more)                                               | Pass   | 0.002s |
| Valid gauge returns success                                                | r.status_code == 200; data["status"] == "success" (+4 more)                                 | Pass   | 0.024s |
| When term structure doesn't have the requested tenor, falls back           | rate == base_rate                                                                           | Pass   | 0.001s |
| Unknown gauge with no term structure returns 0.0 from flat rate fall-back. | rate == 0.0                                                                                 | Pass   | 0.014s |
| Missing id returns none                                                    | loader.get_entity_id(entity) is None; loader.get_track("EVT-NONEXISTENT") is None (+1 more) | Pass   | 0.000s |
| Returns impact dict                                                        | isinstance(impact, dict); "TotalDamage" in impact                                           | Pass   | 0.001s |
| Contains timeout                                                           | '30000' in js                                                                               | Pass   | 0.000s |
| Returns script tag                                                         | '<script>' in js; '</script>' in js (+2 more)                                               | FAIL   | 0.002s |
| All modules non empty                                                      | len(js) > 50                                                                                | Pass   | 0.000s |
| All modules return str                                                     | isinstance(js, str)                                                                         | Pass   | 0.000s |
| Excludes category 1 and 2                                                  | "EVT-004" not in ids                                                                        | Pass   | 0.001s |
| Includes cat 3 4 5                                                         | "EVT-001" in ids; "EVT-002" in ids (+1 more)                                                | Pass   | 0.001s |
| Returns category 3 plus                                                    | all(c >= 3 for c in categories)                                                             | Pass   | 0.001s |
| Get created meeting                                                        | r.status_code == 200; data["status"] == "success" (+1 more)                                 | Pass   | 0.002s |
| Unknown id returns 404                                                     | r.status_code == 404; r.get_json()["status"] == "error"                                     | Pass   | 0.001s |

| Test Description               | Acceptance Criteria                                                                                                   | Result | Time   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Meeting not found returns 404  | <code>r.status_code == 404;</code>                                                                                    | Pass   | 0.001s |
|                                | <code>r.get_json()["status"] == "error"</code>                                                                        |        |        |
| Valid meeting content type pdf | <code>"application/pdf" in r.content_type</code>                                                                      | Pass   | 0.005s |
| Valid meeting returns bytes    | <code>len(r.data) &gt; 100</code>                                                                                     | Pass   | 0.005s |
| Valid meeting returns pdf      | <code>r.status_code == 200</code>                                                                                     | Pass   | 0.004s |
| Model has validation questions | <code>"validation_questions" in model;</code><br><code>len(model["validation_questions"]) == 9</code>                 | Pass   | 0.003s |
| Model not found returns 404    | <code>r.status_code == 404; r.status_code == 404 (+1 more)</code>                                                     | Pass   | 0.002s |
| Response has audit entries     | <code>"audit_entries" in r.get_json()</code>                                                                          | Pass   | 0.002s |
| Valid model returns success    | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                              | Pass   | 0.003s |
| No inventory returns 404       | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br><code>(+3 more)</code>        | Pass   | 0.001s |
| Category distribution          | <code>"category_distribution" in data</code>                                                                          | Pass   | 0.002s |
| Model has review status        | <code>"review_status" in model;</code><br><code>model["review_status"] in ("On Track", "Upcoming", "Due S...</code>   | Pass   | 0.002s |
| Model has risk rating          | <code>"risk_rating" in model</code>                                                                                   | Pass   | 0.002s |
| Model has validation coverage  | <code>"validation_coverage" in model</code>                                                                           | Pass   | 0.002s |
| Response has metadata          | <code>"metadata" in data</code>                                                                                       | Pass   | 0.002s |
| Returns 200                    | <code>r.status_code == 200; r.status_code == 200</code>                                                               | Pass   | 0.002s |
| Returns models list            | <code>"models" in data; data["total_models"] &gt;= 1</code>                                                           | Pass   | 0.002s |
| Returns success status         | <code>r.get_json()["status"] == "success";</code><br><code>r.get_json()["status"] == "success"</code>                 | Pass   | 0.002s |
| Tier distribution              | <code>"tier_distribution" in data</code>                                                                              | Pass   | 0.002s |
| No inventory returns 404       | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br><code>(+3 more)</code>        | Pass   | 0.001s |
| Not scheduled review status    | <code>r.status_code == 200;</code><br><code>model["review_status"] == "Not Scheduled"</code>                          | Pass   | 0.002s |
| Overdue review status          | <code>r.status_code == 200;</code><br><code>model["review_status"] == "Overdue"</code>                                | Pass   | 0.002s |
| Returns expected keys          | <code>result["NRFASStationID"] == "39001";</code><br><code>result["MeanFlow"] == pytest.approx(55.0) (+2 more)</code> | Pass   | 0.001s |

| Test Description                                                         | Acceptance Criteria                                                                    | Result | Time   |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Truncates data quality notes                                             | <code>len(result["DataQualityNotes"]) &lt;= 500</code>                                 | Pass   | 0.000s |
| All expected keys present                                                | <code>key in result</code>                                                             | Pass   | 0.000s |
| Catchment area always none                                               | <code>result["CatchmentArea"] is None</code>                                           | Pass   | 0.000s |
| Custom flow units                                                        | <code>gen.get_nrfa_metadata_for_cdm({}, {"units": "ft3/s"}, "f....</code>              | Pass   | 0.000s |
| Database source from station meta-data                                   | <code>result["DatabaseSource"] == "TestDB"</code>                                      | Pass   | 0.001s |
| Default flow units m3 per s                                              | <code>gen.get_nrfa_metadata_for_cdm({}, {}, "f.json")["FlowUnit...</code>              | Pass   | 0.000s |
| Empty data quality notes when no description                             | <code>gen.get_nrfa_metadata_for_cdm({}, {}, "f.json")["DataQual...</code>              | Pass   | 0.000s |
| Missing database source returns none                                     | <code>gen.get_nrfa_metadata_for_cdm({}, {}, "f.json")["Database...</code>              | Pass   | 0.000s |
| Missing stats keys return none                                           | <code>result[key] is None</code>                                                       | Pass   | 0.000s |
| Response has 'storms' list with count $\geq 0$ .                         | <code>'storms' in data; data['count'] &gt; 0</code>                                    | Pass   | 0.002s |
| Returns 200 with status=success.                                         | <code>resp.status_code == 200; data['status'] == 'success'</code>                      | Pass   | 0.002s |
| Each storm has storm_id, name, intensity_category, gauges_severe,        | <code>'storm_id' in s; 'name' in s (+5 more)</code>                                    | Pass   | 0.002s |
| Returns 404 when stress_storms not found.                                | <code>resp.status_code == 404</code>                                                   | Pass   | 0.001s |
| When gauges_severe is equal, storms sorted by gauges_warning desc.       | <code>s1['gauges_warning'] &gt;= s2['gauges_warning']</code>                           | Pass   | 0.003s |
| Storms sorted by gauges_severe desc — STORM_PORT_SEVERE must come first. | <code>len(storms) &gt;= 2; storms[0]['storm_id'] == STORM_PORT_SEVERE (+1 more)</code> | Pass   | 0.002s |
| Lines 91-96: empty stressm dir falls back to legacy stress/ dir.         | <code>result is mock_predictor</code>                                                  | Pass   | 0.001s |
| Lines 86-90: stressm dir with joblib files loads predictor.              | <code>result is mock_predictor</code>                                                  | Pass   | 0.008s |
| Line 97: no models in either directory logs warning, returns None.       | <code>result is None</code>                                                            | Pass   | 0.001s |
| Get preloader js matches get js                                          | <code>_get_preloader_js() == get_js()</code>                                           | Pass   | 0.000s |
| Get preloader js returns string                                          | <code>isinstance(result, str); 'DOMContentLoaded' in result</code>                     | Pass   | 0.000s |
| Found returns property                                                   | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>               | Pass   | 0.001s |

| Test Description                       | Acceptance Criteria                                                                                                                                                                     | Result | Time   |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Not found returns 404                  | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+2 more)                                                                                       | Pass   | 0.001s |
| Close high risk high                   | <code>get_proximity_zone(50, 'High') ==</code><br><code>'close_high_risk'</code>                                                                                                        | Pass   | 0.000s |
| Close high risk medium                 | <code>get_proximity_zone(100, 'Medium') ==</code><br><code>'close_high_risk'</code>                                                                                                     | Pass   | 0.000s |
| Close high risk very high              | <code>get_proximity_zone(10, 'Very High')</code><br><code>== 'close_high_risk'</code>                                                                                                   | Pass   | 0.000s |
| Close low risk low                     | <code>get_proximity_zone(150, 'Low') ==</code><br><code>'close_low_risk'</code>                                                                                                         | Pass   | 0.000s |
| Close low risk very low                | <code>get_proximity_zone(100, 'Very Low')</code><br><code>== 'close_low_risk'</code>                                                                                                    | Pass   | 0.000s |
| Exactly 200m is medium zone            | <code>get_proximity_zone(200, 'Low') ==</code><br><code>'medium'</code>                                                                                                                 | Pass   | 0.000s |
| Exactly 500m is far                    | <code>get_proximity_zone(500, 'Low') ==</code><br><code>'far'</code>                                                                                                                    | Pass   | 0.000s |
| Far zone                               | <code>get_proximity_zone(600, 'Medium') ==</code><br><code>'far'</code>                                                                                                                 | Pass   | 0.000s |
| Medium zone                            | <code>get_proximity_zone(300, 'Very Low')</code><br><code>== 'medium'</code>                                                                                                            | Pass   | 0.000s |
| Medium zone upper bound                | <code>get_proximity_zone(499, 'Very High')</code><br><code>== 'medium'</code>                                                                                                           | Pass   | 0.000s |
| Returns none when quantlib unavailable | <code>result is None</code>                                                                                                                                                             | Pass   | 0.004s |
| Model not found returns 404            | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+3 more)                                                                                       | Pass   | 0.002s |
| Response has risk rating               | <code>"risk_rating" in data;</code><br><code>"calculated_rating" in rr</code> (+1 more)                                                                                                 | Pass   | 0.002s |
| Risk rating has component scores       | <code>"component_scores" in rr</code>                                                                                                                                                   | Pass   | 0.002s |
| Valid model returns success            | <code>r.status_code == 200;</code> <code>data["status"]</code><br><code>== "success"</code> (+1 more)                                                                                   | Pass   | 0.002s |
| Custom statistics                      | <code>stats['property_menu_items'] == 2;</code><br><code>stats['gauge_menu_items'] == 1</code> (+1 more)                                                                                | Pass   | 0.000s |
| Default statistics                     | <code>stats['property_menu_items']</code><br><code>== len(DEFAULT_PROPERTY_MENU);</code><br><code>stats['gauge_menu_items'] ==</code><br><code>len(DEFAULT_GAUGE_MENU)</code> (+1 more) | Pass   | 0.000s |

| Test Description                                                  | Acceptance Criteria                                                                                 | Result | Time   |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------|--------|
| Get statistics keys                                               | stats['server_url']<br>== 'http://test:1234';<br>stats['total_endpoints'] == 2 (+1 more)            | Pass   | 0.000s |
| Get statistics single endpoint                                    | stats['total_endpoints'] == 1;<br>stats['endpoints'] == ['only']                                    | Pass   | 0.000s |
| Returns dict with expected keys                                   | 'notifications' in stats;<br>'backend_handler' in stats (+1 more)                                   | Pass   | 0.000s |
| Returns expected keys                                             | 'default_position' in stats;<br>'auto.dismiss_timeout' in stats<br>(+3 more)                        | Pass   | 0.000s |
| Statistics are dicts                                              | isinstance(stats['notifications'], dict);<br>isinstance(stats['backend_handler'], dict) (+1 more)   | Pass   | 0.000s |
| Statistics reflect config                                         | stats['default_position']<br>== 'bottom-center';<br>stats['auto.dismiss_timeout'] == 9000 (+1 more) | Pass   | 0.000s |
| Cluster returns 3 or 4                                            | count in (3, 4)                                                                                     | Pass   | 0.000s |
| Doublet returns 2                                                 | get_storm_count(SequenceType.DOUBLET, rng) == 2                                                     | Pass   | 0.000s |
| Isolated returns 1                                                | get_storm_count(SequenceType.ISOLATED, rng) == 1                                                    | Pass   | 0.000s |
| Persistent returns 4 or 5                                         | count in (4, 5)                                                                                     | Pass   | 0.000s |
| Each storm has peak_level.m.                                      | 'peak_level.m' in s; s['peak_level.m'] > 0                                                          | Pass   | 0.009s |
| Storms have required identification fields.                       | 'storm_id' in s; 'name' in s (+3 more)                                                              | Pass   | 0.003s |
| Returns 404 when stress_storms not found.                         | resp.status_code == 404                                                                             | Pass   | 0.000s |
| Missing gauge_id returns total count (used by startup preloader). | resp.status_code == 200;<br>data['status'] == 'success' (+2 more)                                   | Pass   | 0.002s |
| Returns empty list for gauge with no storms.                      | resp.status_code == 200;<br>data['storms'] == []                                                    | Pass   | 0.001s |
| Returns storm list for specified gauge.                           | resp.status_code == 200;<br>data['status'] == 'success' (+2 more)                                   | Pass   | 0.006s |

| Test Description                                                                | Acceptance Criteria                                                                                                      | Result | Time   |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Storms sorted by peak level descending (worst first).                           | <code>peaks == sorted(peaks, reverse=True)</code>                                                                        | Pass   | 0.002s |
| Each storm returned at most once                                                | <code>len(result) == 1</code>                                                                                            | Pass   | 0.001s |
| Empty for remote location                                                       | <code>result == []</code>                                                                                                | Pass   | 0.000s |
| Finds nearby storm                                                              | <code>"EVT-001" in ids</code>                                                                                            | Pass   | 0.001s |
| Large radius finds more                                                         | <code>len(large) &gt;= len(small)</code>                                                                                 | Pass   | 0.001s |
| Each gauge has gauge_id, gauge_name, lon, lat, trade_count.                     | <code>'gauge_id' in g; 'gauge_name' in g (+3 more)</code>                                                                | Pass   | 0.022s |
| Gauges with trades show non-zero trade_count.                                   | <code>sum(counts) &gt; 0</code>                                                                                          | Pass   | 0.017s |
| Returns gauge list with trade counts.                                           | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success' (+2 more)</code>                              | Pass   | 0.021s |
| Gauges sorted west to east (ascending longitude).                               | <code>lons == sorted(lons)</code>                                                                                        | Pass   | 0.014s |
| Missing id returns none                                                         | <code>loader.get_entity_id(entity) is None;</code><br><code>loader.get_track("EVT-NONEXISTENT") is None (+1 more)</code> | Pass   | 0.001s |
| Returns track for existing storm                                                | <code>track is not None; isinstance(track, list) (+1 more)</code>                                                        | Pass   | 0.001s |
| Track has lat lon                                                               | <code>"Latitude" in track[0]; "Longitude" in track[0]</code>                                                             | Pass   | 0.001s |
| get_trade_mark returns mark data for an existing trade.                         | <code>mark['fair_spread_bps'] == 250.0</code>                                                                            | Pass   | 0.013s |
| get_trade_mark returns empty dict for unknown trade.                            | <code>mark == {}</code>                                                                                                  | Pass   | 0.000s |
| get_trade_mark returns merged data after multiple updates.                      | <code>mark['fair_spread_bps'] == 200.0;</code><br><code>mark['mtm'] == 5000.0</code>                                     | Pass   | 0.007s |
| If the PDF exists, it should be served with 200.                                | <code>r.status_code == 200; r.content_type == "application/pdf"</code>                                                   | Pass   | 0.030s |
| Missing pdf returns 404                                                         | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code>                                     | Pass   | 0.000s |
| When yield_curve is empty dict, falls back to risk_free_rate.                   | <code>rate == 0.042</code>                                                                                               | Pass   | 0.000s |
| When yield_curve key is missing, falls back to risk_free_rate.                  | <code>rate == 0.05</code>                                                                                                | Pass   | 0.000s |
| When tenor falls exactly between integer keys but loop exits, returns fallback. | <code>0.035 &lt;= rate &lt;= 0.045</code>                                                                                | Pass   | 0.000s |
| Apply documents function exists                                                 | <code>'applyDocuments' in src</code>                                                                                     | Pass   | 0.000s |
| Checks gov docs cache                                                           | <code>'window..tdPreGovDocs' in src</code>                                                                               | Pass   | 0.000s |



| Test Description                                               | Acceptance Criteria                                                                    | Result | Time   |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Clears cache after use                                         | 'window._preStorms = null' in src;<br>'window._tdPreGovDocs = null' in src             | Pass   | 0.000s |
| Approximate one degree lat                                     | 110.000 < d < 112.000                                                                  | Pass   | 0.000s |
| Approximate one degree lon at equator                          | 110.000 < d < 112.000                                                                  | Pass   | 0.000s |
| Same point returns zero                                        | d == pytest.approx(0.0, abs=1e-6)                                                      | Pass   | 0.000s |
| Symmetry                                                       | d1 == pytest.approx(d2, rel=1e-9)                                                      | Pass   | 0.000s |
| Equatorial degree approx 111km                                 | 111.000 < d < 112.000                                                                  | Pass   | 0.000s |
| London paris approx 340km                                      | 330.000 < d < 350.000                                                                  | Pass   | 0.000s |
| North south degree approx 111km                                | 110.000 < d < 112.000                                                                  | Pass   | 0.000s |
| Returns float                                                  | isinstance(haversine_distance(51.0, -0.1, 52.0, 0.0), float)                           | Pass   | 0.000s |
| Same point is zero                                             | haversine_distance(51.5, -0.1, 51.5, -0.1) == pytest.approx(0.0, abs=1e-6)             | Pass   | 0.000s |
| Symmetry                                                       | d1 == pytest.approx(d2, rel=1e-9)                                                      | Pass   | 0.000s |
| hazard.py lines 100-102: exception reading hazard file -i 500. | r.status_code == 500;<br>r.get_json()["status"] == "error"                             | Pass   | 0.030s |
| Custom day count                                               | curve is not None                                                                      | Pass   | 0.000s |
| High hazard gives low survival                                 | sp < 0.1                                                                               | Pass   | 0.000s |
| Returns credit curve                                           | curve is not None                                                                      | Pass   | 0.033s |
| Survival prob decreases                                        | sp2 < sp1                                                                              | Pass   | 0.002s |
| Zero hazard gives survival near one                            | sp >= 0.999                                                                            | Pass   | 0.000s |
| Header footer callable                                         | —                                                                                      | Pass   | 0.003s |
| Header footer callable                                         | —                                                                                      | Pass   | 0.004s |
| Header footer callable                                         | —                                                                                      | Pass   | 0.002s |
| Header footer draws confidential                               | mock_canvas.drawCentredString.called                                                   | Pass   | 0.003s |
| Header footer draws confidential footer                        | mock_canvas.drawCentredString.called                                                   | Pass   | 0.003s |
| Header footer draws header strings                             | mock_canvas.drawString.call_count >= 1;<br>mock_canvas.drawRightString.call_count >= 2 | Pass   | 0.003s |
| Header footer draws strings                                    | mock_canvas.drawString.call_count >= 1;<br>mock_canvas.drawRightString.call_count >= 2 | Pass   | 0.002s |
| Health 200 at api v1                                           | r.status_code == 200                                                                   | Pass   | 0.001s |
| Health 200 at root                                             | r.status_code == 200                                                                   | Pass   | 0.000s |
| A single function must handle both route decorators.           | api_idx != -1 and plain_idx != -1 and fn_idx != -1; api_idx < fn_idx (+1 more)         | Pass   | 0.000s |

| Test Description                                         | Acceptance Criteria                                                                                                  | Result | Time   |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|
| health.py must declare /api/v1/health as a route alias.  | <code>'/api/v1/health' in src</code>                                                                                 | Pass   | 0.004s |
| Backward-compat: /health must still exist.               | <code>"/health" in src</code>                                                                                        | Pass   | 0.000s |
| Health returns ok status                                 | <code>data['status'] == 'ok'</code>                                                                                  | Pass   | 0.000s |
| Test GET /health/detailed returns detailed status.       | <code>response.status_code in [200, 503];</code><br><code>data['status'] in ['healthy', 'degraded'] (+2 more)</code> | Pass   | 0.000s |
| Test GET /health returns health status.                  | <code>response.status_code == 200;</code><br><code>data['status'] == 'ok' (+1 more)</code>                           | Pass   | 0.000s |
| Test GET / redirects to catchment selector.              | <code>response.status_code == 302</code>                                                                             | Pass   | 0.001s |
| Medium height 10 to 15m                                  | <code>isinstance(result, list)</code>                                                                                | Pass   | 0.000s |
| No height skips classification                           | <code>isinstance(result, list)</code>                                                                                | Pass   | 0.000s |
| Standard height under 10m                                | <code>isinstance(result, list)</code>                                                                                | Pass   | 0.000s |
| Tall structure over 15m                                  | <code>isinstance(result, list)</code>                                                                                | Pass   | 0.000s |
| Generate monitoring high score                           | <code>"Weekly" in r.values()</code>                                                                                  | Pass   | 0.000s |
| Generate monitoring low score                            | <code>"Annually" in r.values()</code>                                                                                | Pass   | 0.000s |
| Generate monitoring medium score                         | <code>"Monthly" in r.values()</code>                                                                                 | Pass   | 0.000s |
| Generate recommendations high score                      | <code>any("URGENT" in rec for rec in r)</code>                                                                       | Pass   | 0.000s |
| Generate recommendations low score                       | <code>any("LOW" in rec for rec in r)</code>                                                                          | Pass   | 0.000s |
| Generate recommendations medium score                    | <code>any("HIGH PRIORITY" in rec or "MODERATE" in rec for rec i...</code>                                            | Pass   | 0.000s |
| Identify key factors high risk                           | <code>any("High Risk Factor" in k for k in r.keys())</code>                                                          | Pass   | 0.000s |
| Identify key factors medium risk                         | <code>any("Medium Risk Factor" in k for k in r.keys())</code>                                                        | Pass   | 0.000s |
| Alert line uses amber (#FFC107).                         | <code>'#FFC107' in hist_js</code>                                                                                    | Pass   | 0.000s |
| Chart uses daily_observations from history API response. | <code>'daily_observations' in hist_js</code>                                                                         | Pass   | 0.000s |
| Historical chart canvas element defined.                 | <code>'hist-timeseries-chart' in hist_js</code>                                                                      | Pass   | 0.000s |
| Severe line uses red (#F44336).                          | <code>'#F44336' in hist_js</code>                                                                                    | Pass   | 0.000s |
| Alert threshold line on the 50yr chart.                  | <code>'FloodAlert' in hist_js</code>                                                                                 | Pass   | 0.000s |
| Warning threshold line on the 50yr chart.                | <code>'FloodWarning' in hist_js</code>                                                                               | Pass   | 0.000s |
| Severe threshold line on the 50yr chart.                 | <code>'SevereFloodWarning' in hist_js</code>                                                                         | Pass   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                              | Result | Time   |
|--------------------------------------------------------------------------|------------------------------------------------------------------|--------|--------|
| Subsamples every 7th day for performance.                                | '7' in hist_js; 'sampleObs' in hist_js                           | Pass   | 0.000s |
| Warning line uses orange (#FF9800).                                      | '#FF9800' in hist_js                                             | Pass   | 0.000s |
| Cache is keyed by gauge ID (_gaugeId).                                   | '_gaugeId' in hist_js                                            | Pass   | 0.000s |
| Historical data is cached to avoid repeated fetches.                     | '_histData' in hist_js                                           | Pass   | 0.000s |
| If cache exists for same gauge, skip API fetch.                          | '_histData._gaugeId' in hist_js or<br>{ 'his...                  | Pass   | 0.000s |
| Exceeded severe label                                                    | "Exceeded severe warning level" in<br>str(elements)              | Pass   | 0.001s |
| Within severe level                                                      | "4.000m" in str(elements)                                        | Pass   | 0.001s |
| Lines 138-141: historical high injected after main loop.                 | len(match) == 1;<br>match[0]["level_meters"] == 9.99             | Pass   | 0.008s |
| When historical_high_dt is None (out of range), post-loop block skipped. | all(o["level_meters"] != 8.0 for o in<br>result)                 | Pass   | 0.001s |
| When historical_high_date is None, both branches skipped.                | isinstance(result, list); len(result)<br>> 0                     | Pass   | 0.001s |
| Chart cleanup function defined.                                          | '_cleanupHistCharts' in hist_js                                  | Pass   | 0.000s |
| Stats bar shows flood exceedances.                                       | 'flood_exceedances' in hist_js                                   | Pass   | 0.000s |
| Stats bar shows mean level.                                              | 'mean_level' in hist_js or 'Mean' in<br>hist_js                  | Pass   | 0.000s |
| Stats bar shows record years.                                            | 'record_years' in hist_js or 'Record'<br>in hist_js              | Pass   | 0.000s |
| Should fetch from /api/v1/gauges/{gauge_id}/history.                     | '/api/v1/gauges/' in js; '/history'<br>in js                     | Pass   | 0.000s |
| Should fetch from /api/v1/trading/stress/storms.                         | '/api/v1/trading/stress/storms' in<br>js                         | Pass   | 0.000s |
| Right panel should show storm scenarios list.                            | 'hist-storms-panel' in js;<br>'hist-storms-list' in js (+1 more) | Pass   | 0.000s |
| Should render a time series chart.                                       | 'hist-timeseries-chart' in js;<br>'renderHistorical' in js       | Pass   | 0.000s |
| Historical tab JS should render without errors.                          | len(js) > 0                                                      | Pass   | 0.000s |
| Should have function to render storm list.                               | '_renderStormList' in js;<br>'hist-storm-row' in js              | Pass   | 0.000s |
| Chart should show flood stage threshold lines.                           | 'FloodAlert' in js; 'FloodWarning' in<br>js (+1 more)            | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                                          | Result | Time   |
|---------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| Storm rows should show trigger-colored dots.                  | '#FFC107' in js; '#FF9800' in js (+1 more)                                   | Pass   | 0.000s |
| Storm row click should navigate to Stress Test tab.           | '_navigateToStress' in js; '_stressStormHint' in js (+1 more)                | Pass   | 0.000s |
| Should use addEventListener (not inline onclick) for storms.  | 'addEventListener' in js                                                     | Pass   | 0.000s |
| ghc_historical.get_js() returns substantial JS output.        | len(hist_js) > 1000                                                          | Pass   | 0.000s |
| Output long enough to contain real function definitions.      | len(hist_js) > 3000                                                          | Pass   | 0.000s |
| Historical tab fetches from /gauges/{gauge_id}/history.       | '/api/v1/gauges/' in hist_js; '/history' in hist_js                          | Pass   | 0.000s |
| Fetch URL is built using gaugeId variable.                    | 'gaugeId' in hist_js                                                         | Pass   | 0.000s |
| history.py lines 88-90: corrupted gaugehd file -i 500.        | r.status_code == 500; r.get_json()["status"] == "error"                      | Pass   | 0.032s |
| history.py lines 144-146: statistics loader exception -i 500. | r.status_code == 500                                                         | Pass   | 0.030s |
| history.py lines 119-121: timeseries loader exception -i 500. | r.status_code == 500                                                         | Pass   | 0.029s |
| Assess historical risks active subsidence                     | "Ground Stability Risk" in result; "High" in result["Ground Stability Risk"] | Pass   | 0.000s |
| Assess historical risks high env                              | "Environmental Risk" in result; "High" in result["Environmental Risk"]       | Pass   | 0.000s |
| Assess historical risks low env                               | "Low" in result["Environmental Risk"]                                        | Pass   | 0.000s |
| Assess historical risks minor subsidence                      | "Medium" in result["Ground Stability Risk"]                                  | Pass   | 0.000s |
| Assess historical risks moderate env                          | "Medium" in result["Environmental Risk"]                                     | Pass   | 0.000s |
| Assess historical risks moderate flood                        | "Medium" in result["Historical Flood Risk"]                                  | Pass   | 0.000s |
| Assess historical risks no flood damage                       | "Low" in result["Historical Flood Risk"]                                     | Pass   | 0.000s |
| Assess historical risks severe flood                          | "Historical Flood Risk" in result; "High" in result["Historical Flood Risk"] | Pass   | 0.000s |
| Assess historical risks stable ground                         | "Low" in result["Ground Stability Risk"]                                     | Pass   | 0.000s |

| Test Description                                                                                   | Acceptance Criteria                                                         | Result | Time   |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------|
| Build flood history bad json                                                                       | <code>any("Error" in t or "error" in t.lower() for t in texts)</code>       | Pass   | 0.001s |
| Build flood history no flood days                                                                  | <code>any("No flood events" in t for t in texts)</code>                     | Pass   | 0.022s |
| Build flood history no hd file                                                                     | <code>any("No historical" in t or "historical" in t.lower() for...</code>   | Pass   | 0.000s |
| Build flood history no reference gauges                                                            | <code>any("No reference gauges" in t for t in texts)</code>                 | Pass   | 0.000s |
| Build flood history with flood data                                                                | <code>isinstance(result, list); len(tables) &gt;= 1</code>                  | Pass   | 0.025s |
| Empty flood events                                                                                 | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Empty property does not crash                                                                      | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Has history header                                                                                 | <code>any("History" in t or "Flood" in t for t in texts)</code>             | Pass   | 0.000s |
| No flood history                                                                                   | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Returns list                                                                                       | <code>isinstance(result, list); len(result) &gt; 0</code>                   | Pass   | 0.000s |
| With environmental issues                                                                          | <code>isinstance(result, list); len(tables) &gt;= 1</code>                  | Pass   | 0.000s |
| With fire incidents                                                                                | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| With ground conditions                                                                             | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| With mortgage data                                                                                 | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Hmda minimal fields                                                                                | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Hmda rate spread formatted                                                                         | <code>isinstance(result, list)</code>                                       | Pass   | 0.000s |
| Hmda section creates table                                                                         | <code>len(tables) &gt;= 1</code>                                            | Pass   | 0.000s |
| When gauges file has invalid JSON, exception is caught (lines 249-250)                             | <code>resp.status_code == 200; data['status'] == 'success'</code>           | Pass   | 0.049s |
| When gauges file has <code>!STORM_HOURS</code> readings, hydrograph is truncated (line 245).       | <code>resp.status_code == 200</code>                                        | Pass   | 0.050s |
| When gauges JSON is corrupt, exception caught (lines 109-110),                                     | <code>resp.status_code == 200; data['status'] == 'success' (+1 more)</code> | Pass   | 0.034s |
| When gauges has fewer than <code>STORM_HOURS</code> readings, pads with last value (line 106-108). | <code>resp.status_code == 200; len(data['hourly']) == 168</code>            | Pass   | 0.040s |
| When <code>peak_hour == num_hours</code> (remaining=0), <code>frac=1.0</code> for post-peak hours. | <code>len(levels) == 5; abs(levels[0] - 3.0) &lt; 0.01</code>               | Pass   | 0.000s |
| <code>run_series</code> result matches manual step-by-step.                                        | <code>qf == pytest.approx(qf_series[h], rel=1e-6)</code>                    | Pass   | 0.002s |

| Test Description                                                 | Acceptance Criteria                                                                            | Result | Time   |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------|--------|
| Wet initial state produces higher early quickflow.               | <code>qf_wet[:6].mean() &gt; qf_dry[:6].mean()</code>                                          | Pass   | 0.000s |
| No rain no quickflow                                             | <code>np.all(qf == pytest.approx(0.0))</code>                                                  | Pass   | 0.000s |
| Quickflow positive                                               | <code>np.all(qf &gt;= 0.0)</code>                                                              | Pass   | 0.002s |
| Shape                                                            | <code>qf.shape == (168,); len(states) == 168</code>                                            | Pass   | 0.000s |
| Soil moisture drains to near zero after long dry period.         | <code>states[-1].soil_moisture &lt; 1.0</code>                                                 | Pass   | 0.000s |
| Dry start no precip no quickflow                                 | <code>qf == pytest.approx(0.0); new_state.soil_moisture == pytest.approx(0.0) (+1 more)</code> | Pass   | 0.000s |
| Groundwater accumulates from soil                                | <code>new_state.groundwater &gt; 0.0</code>                                                    | Pass   | 0.000s |
| Negative precip treated as zero                                  | <code>qf &gt;= 0.0; new_state.soil_moisture &gt;= 0.0</code>                                   | Pass   | 0.000s |
| Precip increases soil moisture                                   | <code>new_state.soil_moisture &gt; 0.0</code>                                                  | Pass   | 0.000s |
| Precip produces quickflow                                        | <code>qf &gt; 0.0</code>                                                                       | Pass   | 0.000s |
| Soil moisture drains without rain                                | <code>new_state.soil_moisture &lt; 50.0</code>                                                 | Pass   | 0.000s |
| Elevated antecedent state produces more quickflow per mm.        | <code>qf_wet &gt; qf_dry</code>                                                                | Pass   | 0.000s |
| All zero weights returns zero                                    | <code>result == pytest.approx(7.5)</code>                                                      | Pass   | 0.000s |
| Closer gauge has higher weight                                   | <code>result &lt; 5.0</code>                                                                   | Pass   | 0.000s |
| Empty list returns zero                                          | <code>idw_interpolate_from_gauges([]) == 0.0</code>                                            | Pass   | 0.000s |
| Equal distance gauges averages wse                               | <code>result == pytest.approx(3.0)</code>                                                      | Pass   | 0.000s |
| Near gauge returns wse directly                                  | <code>result == pytest.approx(5.0)</code>                                                      | Pass   | 0.000s |
| Power one changes result                                         | <code>r2 != pytest.approx(r1, rel=0.01)</code>                                                 | Pass   | 0.000s |
| Single far gauge                                                 | <code>result == pytest.approx(3.0)</code>                                                      | Pass   | 0.000s |
| Line 158: power parameter affects weights.                       | <code>r3 &lt; r1</code>                                                                        | Pass   | 0.000s |
| Line 149-150: empty input → 0.0.                                 | <code>idw_interpolate_from_gauges([]) == 0.0</code>                                            | Pass   | 0.000s |
| Gauge at exactly min_distance=1.0 is NOT less than, so IDW used. | <code>result == pytest.approx(5.0)</code>                                                      | Pass   | 0.000s |
| Multiple gauges produce weighted result.                         | <code>2.0 &lt; result &lt; 6.0; result &lt; 4.0</code>                                         | Pass   | 0.000s |
| Line 156-157: dist ; min_distance → return wse.                  | <code>result == pytest.approx(42.0)</code>                                                     | Pass   | 0.000s |
| Idw gauge spreads computed with two gauges                       | <code>"any_flood" in idw; len(idw["any_flood"]) == len(TENORS)</code>                          | Pass   | 0.258s |
| With empty nearest_gauges, idw_gauge_spreads is still a dict.    | <code>isinstance(result.get("idw_gauge_spreads"), dict)</code>                                 | Pass   | 0.267s |



| Test Description                                               | Acceptance Criteria                                                                                                            | Result | Time   |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Fair income 30k to 50k                                         | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.000s |
| Good income 50k to 100k                                        | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.000s |
| High income over 100k                                          | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.001s |
| Limited income under 30k                                       | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.000s |
| Total income with secondary                                    | <code>isinstance(result, list)</code>                                                                                          | Pass   | 0.001s |
| Area = float('inf') skips all threshold checks, hits fallback. | <code>isinstance(result, tuple);</code><br><code>len(result) == 2</code>                                                       | Pass   | 0.000s |
| Age premium factors all positive                               | <code>lo &gt; 0; hi &gt;= lo</code>                                                                                            | Pass   | 0.000s |
| Area bands are sorted                                          | <code>thresholds == sorted(thresholds)</code>                                                                                  | Pass   | 0.000s |
| Base rate range                                                | <code>lo &gt; 0; hi &gt; lo</code>                                                                                             | Pass   | 0.000s |
| Construction type factors present                              | <code>'Brick' in CONSTRUCTION_TYPE_FACTORS;</code><br><code>'Timber Frame' in</code><br><code>CONSTRUCTION_TYPE_FACTORS</code> | Pass   | 0.000s |
| Flood risk factors increase with risk                          | <code>very_high_hi &gt; very_low_lo</code>                                                                                     | Pass   | 0.000s |
| Flood risk factors present                                     | <code>set(FLOOD_RISK_PREMIUM_FACTORS.keys())</code><br><code>== expected</code>                                                | Pass   | 0.000s |
| Min max premium bounds                                         | <code>MIN_PREMIUM == 200;      MAX_PREMIUM ==</code><br><code>20.000</code>                                                    | Pass   | 0.000s |
| Property type factors present                                  | <code>set(PROPERTY_TYPE_PREMIUM_FACTORS.keys())</code><br><code>== expected</code>                                             | Pass   | 0.000s |
| Unknown integer field returns int                              | <code>isinstance(result, int); 1 &lt;= result</code><br><code>&lt;= 10</code>                                                  | Pass   | 0.000s |
| Integer type returns int                                       | <code>isinstance(result, int); 0 &lt;= result</code><br><code>&lt;= 100</code>                                                 | Pass   | 0.000s |
| Baseline alert only                                            | <code>self.run(tmp_path, n_severe=0,</code><br><code>n_warning=0) == "baseline"</code>                                         | Pass   | 0.003s |
| Catastrophic 10 severe                                         | <code>self.run(tmp_path, n_severe=10) ==</code><br><code>"catastrophic"</code>                                                 | Pass   | 0.011s |
| Extreme 6 severe                                               | <code>self.run(tmp_path, n_severe=6) ==</code><br><code>"extreme"</code>                                                       | Pass   | 0.005s |
| Moderate 1 severe                                              | <code>self.run(tmp_path, n_severe=1) ==</code><br><code>"moderate"</code>                                                      | Pass   | 0.004s |
| Severe 3 severe                                                | <code>self.run(tmp_path, n_severe=3) ==</code><br><code>"severe"</code>                                                        | Pass   | 0.004s |
| Batch generator populates category                             | <code>all(s.intensity_category in valid for</code><br><code>s in seqs)</code>                                                  | Pass   | 0.001s |
| From dict missing category defaults empty                      | <code>reconstructed.intensity_category ==</code><br><code>""</code>                                                            | Pass   | 0.000s |
| From dict round trip                                           | <code>reconstructed.intensity_category ==</code><br><code>s.intensity_category</code>                                          | Pass   | 0.000s |

| Test Description                                                        | Acceptance Criteria                                                                                  | Result | Time   |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Serialized sequences have category                                      | <code>all(s.intensity_category in valid for s in seqs)</code>                                        | Pass   | 0.001s |
| To dict includes intensity category                                     | <code>"intensity_category" in d; d["intensity_category"] == s.intensity_category</code>              | Pass   | 0.001s |
| Running 'python -m models.intensity' executes <code>cli.main()</code> . | <code>result.returncode == 0; "Storm Intensity" in result.stdout or "usage" in result.s...</code>    | Pass   | 0.722s |
| Beta value                                                              | <code>IntensityProfile.BETA.value == "beta"</code>                                                   | Pass   | 0.000s |
| Gaussian value                                                          | <code>IntensityProfile.GAUSSIAN.value == "gaussian"; DecayKernel.GAUSSIAN.value == "gaussian"</code> | Pass   | 0.000s |
| Triangular value                                                        | <code>IntensityProfile.TRIANGULAR.value == "triangular"</code>                                       | Pass   | 0.000s |
| All profiles produce valid storm                                        | <code>isinstance(storm, Storm); len(storm.track) &gt; 0</code>                                       | Pass   | 0.000s |
| Beta profile creates storm                                              | <code>storm is not None; len(storm.track) &gt; 0</code>                                              | Pass   | 0.000s |
| Gaussian profile peak near midpoint                                     | <code>8 &lt;= peak_idx &lt;= 16</code>                                                               | Pass   | 0.000s |
| Triangular max equals peak intensity                                    | <code>max(tp.intensity for tp in storm.track) == pytest.approx(...</code>                            | Pass   | 0.000s |
| Triangular profile peak near midpoint                                   | <code>8 &lt;= peak_idx &lt;= 16</code>                                                               | Pass   | 0.000s |
| Unknown sequence type falls back to 1 storm.                            | <code>count == 1</code>                                                                              | Pass   | 0.003s |
| ContextMenuHandler JS renders without f-string errors.                  | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                                  | Pass   | 0.005s |
| GaugeHazardCurve JS renders without f-string errors.                    | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                                  | Pass   | 0.001s |
| GaugePDFPanel JS renders without f-string errors.                       | <code>len(js) &gt; 0; 'GaugePDFPanel' in js</code>                                                   | Pass   | 0.000s |
| GaugeStormAnalysis JS renders without f-string errors.                  | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                                  | Pass   | 0.001s |
| NotificationSystem JS renders without f-string errors.                  | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                                  | Pass   | 0.003s |
| PropertyHazardCurvePanel JS renders without f-string errors.            | <code>len(js) &gt; 0; '&lt;script&gt;' in js</code>                                                  | Pass   | 0.000s |
| PropertyPDFPanel JS renders without f-string errors.                    | <code>len(js) &gt; 0; 'PropertyPDFPanel' in js</code>                                                | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                                                       | Result | Time   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------|--------|
| PropertyStormAnalysis JS renders without f-string errors.             | len(js) > 0; ' <script&gt;' (+1="" in="" js="" more)<="" td=""><td>Pass</td><td>1.369s</td></script&gt;'> | Pass   | 1.369s |
| PropertyStormAnalysis JS must guard against empty flood_events array. | 'flood_events.length === 0' in js                                                                         | Pass   | 0.000s |
| TradingDeskPanel should be in __all__.                                | 'TradingDeskPanel' in __all__                                                                             | Pass   | 0.000s |
| InteractivityManager should have trading_desk attribute.              | hasattr(mgr, 'trading_desk')                                                                              | Pass   | 0.000s |
| Setup map adds preloader before notifications                         | preloader_idx != -1; notifications_idx != -1 (+1 more)                                                    | Pass   | 0.000s |
| Startup preloader imported in init                                    | 'from .startup import StartupPreloader' in src                                                            | Pass   | 0.003s |
| Startup preloader instantiated in init                                | 'self.startup_preloader = StartupPreloader()' in src                                                      | Pass   | 0.000s |
| Elevated rate 5 to 7                                                  | isinstance(result, list)                                                                                  | Pass   | 0.001s |
| High rate over 7                                                      | isinstance(result, list)                                                                                  | Pass   | 0.001s |
| Low rate under 3                                                      | isinstance(result, list)                                                                                  | Pass   | 0.001s |
| Moderate rate 3 to 5                                                  | isinstance(result, list)                                                                                  | Pass   | 0.001s |
| No rate unknown                                                       | isinstance(result, list)                                                                                  | Pass   | 0.001s |
| Line 64: empty yield_curve returns default fallback 0.04.             | result == 0.04                                                                                            | Pass   | 0.000s |
| Line 64: empty yield_curve returns fallback.                          | result == 0.05                                                                                            | Pass   | 0.000s |
| Line 77: fallthrough at end of for loop returns last key rate.        | interpolate_yield_rate(curve, 4.0) == 0.04; result == pytest.approx(0.035)                                | Pass   | 0.000s |
| Line 71: t != keys[-1] returns the last key's rate.                   | result == 0.045                                                                                           | Pass   | 0.000s |
| Line 71: t == keys[-1] returns the last key's rate.                   | result == 0.045                                                                                           | Pass   | 0.000s |
| File exists                                                           | INVENTORY_PATH.exists()                                                                                   | Pass   | 0.000s |
| File is valid json                                                    | isinstance(data, dict)                                                                                    | Pass   | 0.008s |
| Has at least one model                                                | len(models) > 0                                                                                           | Pass   | 0.000s |
| Has metadata                                                          | "metadata" in inventory; "framework" in inventory["metadata"]                                             | Pass   | 0.000s |
| Has model chain                                                       | "model_chain" in inventory                                                                                | Pass   | 0.000s |
| Has models key                                                        | "models" in inventory                                                                                     | Pass   | 0.000s |
| Get models includes risk fields                                       | response.status_code == 200; data["status"] == "success" (+4 more)                                        | Pass   | 0.002s |

| Test Description                                                                                            | Acceptance Criteria                                   | Result | Time   |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------|--------|
| Line 125: <code>_inverse_normal(1.5)</code> returns <code>+inf</code> .                                     | <code>result == float('inf')</code>                   | Pass   | 0.000s |
| Line 123: <code>_inverse_normal(-0.5)</code> returns <code>-inf</code> .                                    | <code>result == float('-inf')</code>                  | Pass   | 0.000s |
| Line 125: <code>_inverse_normal(1.0)</code> returns <code>+inf</code> .                                     | <code>result == float('inf')</code>                   | Pass   | 0.000s |
| Line 123: <code>_inverse_normal(0)</code> returns <code>-inf</code> .                                       | <code>result == float('-inf')</code>                  | Pass   | 0.000s |
| Lines 170-171: <code>p &lt; p_high (0.97575)</code> uses the upper tail formula.                            | <code>result == pytest.approx(2.326, abs=0.02)</code> | Pass   | 0.000s |
| Lines 170-171: <code>p=0.98</code> also triggers high-p branch.                                             | <code>result == pytest.approx(2.054, abs=0.02)</code> | Pass   | 0.000s |
| Lines 170-171: high-p branch returns positive value.                                                        | <code>result &gt; 0</code>                            | Pass   | 0.000s |
| Lines 170-171: verify roundtrip CDF( <code>inverse.CDF(p)</code> ) <code>p</code> for high <code>p</code> . | <code>recovered == pytest.approx(p, abs=0.001)</code> | Pass   | 0.000s |
| Below average 2 to 4pct                                                                                     | <code>isinstance(result, list)</code>                 | Pass   | 0.000s |
| Excellent over 8pct                                                                                         | <code>isinstance(result, list)</code>                 | Pass   | 0.000s |
| Fair 4 to 6pct                                                                                              | <code>isinstance(result, list)</code>                 | Pass   | 0.000s |
| Good 6 to 8pct                                                                                              | <code>isinstance(result, list)</code>                 | Pass   | 0.000s |
| Poor under 2pct                                                                                             | <code>isinstance(result, list)</code>                 | Pass   | 0.000s |
| Get raises on unknown key                                                                                   | —                                                     | Pass   | 0.000s |
| <code>JSONFileConfig.get('hazard_curves')</code> returns the hazard curves filename.                        | <code>result == JSONFileConfig.HAZARD_CURVES</code>   | Pass   | 0.000s |
| Datetime encoded as isoformat                                                                               | <code>"2024-01-15" in result</code>                   | Pass   | 0.000s |
| Numpy float32 encoded                                                                                       | <code>abs(result - 1.5) &lt; 0.01</code>              | Pass   | 0.000s |
| Numpy float encoded                                                                                         | <code>abs(result - 3.14) &lt; 1e-6</code>             | Pass   | 0.000s |
| Numpy int32 encoded                                                                                         | <code>result == 7</code>                              | Pass   | 0.000s |
| Numpy int encoded                                                                                           | <code>result == 42; isinstance(result, int)</code>    | Pass   | 0.000s |
| Numpy ndarray encoded as list                                                                               | <code>result == [1, 2, 3]</code>                      | Pass   | 0.000s |
| Unknown type raises type error                                                                              | —                                                     | Pass   | 0.000s |
| Charts reference <code>first_trigger_hour</code> for KO line.                                               | <code>'first_trigger_hour' in stress_js</code>        | Pass   | 0.000s |
| Charts have KO annotation line.                                                                             | <code>'koLine' in stress_js</code>                    | Pass   | 0.000s |
| Shows KO $H\{n\}$ label for triggered trades.                                                               | <code>'KO H' in stress_js</code>                      | Pass   | 0.000s |
| References <code>triggered_hour</code> for KO status.                                                       | <code>'triggered_hour' in stress_js</code>            | Pass   | 0.000s |
| LTV $< 1$ (e.g. 75 instead of 0.75) should still work.                                                      | <code>isinstance(result, list)</code>                 | Pass   | 0.000s |
| Ltv over 60 low medium                                                                                      | <code>isinstance(result, list)</code>                 | Pass   | 0.000s |

| Test Description                           | Acceptance Criteria                                                                    | Result | Time   |
|--------------------------------------------|----------------------------------------------------------------------------------------|--------|--------|
| Ltv over 70 medium                         | <code>isinstance(result, list)</code>                                                  | Pass   | 0.000s |
| Ltv over 80 medium high                    | <code>isinstance(result, list)</code>                                                  | Pass   | 0.000s |
| Ltv over 90 high risk                      | <code>isinstance(result, list)</code>                                                  | Pass   | 0.000s |
| Ltv under 60 low risk                      | <code>isinstance(result, list)</code>                                                  | Pass   | 0.000s |
| High ltv 90 to 95                          | <code>isinstance(result, list)</code>                                                  | Pass   | 0.001s |
| Low ltv under 70                           | <code>isinstance(result, list)</code>                                                  | Pass   | 0.001s |
| Low medium ltv 70 to 80                    | <code>isinstance(result, list)</code>                                                  | Pass   | 0.001s |
| Medium ltv 80 to 90                        | <code>isinstance(result, list)</code>                                                  | Pass   | 0.001s |
| Very high ltv over 95                      | <code>isinstance(result, list)</code>                                                  | Pass   | 0.001s |
| Bad date string gracefully skipped         | <code>isinstance(elements, list)</code>                                                | Pass   | 0.000s |
| Moderate recent activity                   | <code>"Moderate recent activity" in str(elements)</code>                               | Pass   | 0.001s |
| No recent activity                         | <code>"No recent severe activity" in str(elements)</code>                              | Pass   | 0.001s |
| Recent activity within 1 year              | <code>"Recent severe level activity" in str(elements)</code>                           | Pass   | 0.001s |
| Body style alignment                       | <code>s.alignment == TA_CENTER</code>                                                  | Pass   | 0.000s |
| Body style bold                            | <code>s.fontName == 'Helvetica-Bold'</code>                                            | Pass   | 0.000s |
| Body style default                         | <code>isinstance(s, ParagraphStyle); s.fontName == 'Helvetica'</code>                  | Pass   | 0.000s |
| Build header footer callable               | <code>callable(cb)</code>                                                              | Pass   | 0.000s |
| White hdr style returns paragraph style    | <code>isinstance(s, ParagraphStyle); s.fontName == 'Helvetica-Bold'</code>             | Pass   | 0.000s |
| Coverage available false when no index     | <code>r.get_json()["coverage_available"] is False</code>                               | Pass   | 0.001s |
| Coverage available true when index present | <code>r.get_json()["coverage_available"] is True</code>                                | Pass   | 0.001s |
| Empty dir returns empty list               | <code>data["reports"] == []</code>                                                     | Pass   | 0.001s |
| File has size and modified                 | <code>reports[0]["size"] &gt; 0; "modified" in reports[0]</code>                       | Pass   | 0.001s |
| File type label json                       | <code>r.get_json()["reports"][0]["type"] == "JSON"</code>                              | Pass   | 0.001s |
| File type label pdf                        | <code>r.get_json()["reports"][0]["type"] == "PDF"</code>                               | Pass   | 0.001s |
| File type label unknown                    | <code>r.get_json()["reports"][0]["type"] == "XYZ"</code>                               | Pass   | 0.001s |
| No dir returns empty list                  | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>               | Pass   | 0.000s |
| Pdf file listed                            | <code>len(data["reports"]) == 1; data["reports"][0]["filename"] == "report.pdf"</code> | Pass   | 0.001s |

| Test Description                                                                                         | Acceptance Criteria                                                                                          | Result | Time   |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------|--------|
| Subdirectories not listed                                                                                | <code>r.get_json()["reports"] == []</code>                                                                   | Pass   | 0.001s |
| sys.exit(0) fires before the gauge file is opened.                                                       | —                                                                                                            | Pass   | 0.002s |
| sys.exit(0) fires before the property file is opened.                                                    | —                                                                                                            | Pass   | 0.003s |
| Lines 324-327: all category/page pairs are logged.                                                       | —                                                                                                            | Pass   | 0.002s |
| Lines 412-419: <code>-list-categories</code> iterates category dict, sys.exit(0).                        | —                                                                                                            | Pass   | 0.002s |
| Lines 469-476: <code>-list-categories</code> iterates category dict, sys.exit(0).                        | —                                                                                                            | Pass   | 0.002s |
| Lines 321-328: <code>-list-categories</code> → <code>get_page_categories()</code> iterated, sys.exit(0). | —                                                                                                            | Pass   | 0.001s |
| No file returns empty list                                                                               | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                                     | Pass   | 0.001s |
| Summary fields                                                                                           | <code>first["counterparty_id"] == "CTPY-001"; first["name"] == "Test Bank PLC" (+2 more)</code>              | Pass   | 0.002s |
| With data returns summary                                                                                | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                     | Pass   | 0.002s |
| No storms file returns 404                                                                               | <code>r.status_code == 404</code>                                                                            | Pass   | 0.001s |
| Options returns ok                                                                                       | <code>r.status_code == 200; r.status_code == 200 (+2 more)</code>                                            | Pass   | 0.001s |
| Storm has expected fields                                                                                | <code>len(storms) &gt;= 1; "storm_id" in s (+3 more)</code>                                                  | Pass   | 0.004s |
| Storm name is capitalised intensity_category, not empty.                                                 | <code>s["name"] == s["intensity_category"].capitalize()</code>                                               | Pass   | 0.006s |
| Storms list count                                                                                        | <code>data["count"] &gt;= 1</code>                                                                           | Pass   | 0.005s |
| Storms sorted by gauges_severe descending (most severe first).                                           | <code>counts == sorted(counts, reverse=True)</code>                                                          | Pass   | 0.005s |
| With data returns success                                                                                | <code>r.status_code == 200; r.get_json()["status"] == "success" (+3 more)</code>                             | Pass   | 0.021s |
| Lines 218-219: invalid JSON in gaugets file triggers except continue.                                    | <code>data["status"] == "success"; data["count"] &gt;= 1</code>                                              | Pass   | 0.055s |
| Lines 235-239: invalid metadata JSON triggers except continue.                                           | <code>r.status_code == 200</code>                                                                            | Pass   | 0.032s |
| Lines 280-292: property flooding enriches storm with depth/damage.                                       | <code>storms["STORM-D"]["properties_flooded"] == 1; storms["STORM-D"]["max_depth_m"] == 1.2 (+1 more)</code> | Pass   | 0.079s |



| Test Description                                                                                                   | Acceptance Criteria                                                                                                       | Result | Time   |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Lines 280-281: property.json read error triggers except pass.                                                      | <code>r.status_code == 200; data["status"] == "success"</code>                                                            | Pass   | 0.046s |
| Line 288: property not in valued_ids is skipped.                                                                   | <code>storms["STORM-E"]["properties_flooded"] == 0</code>                                                                 | Pass   | 0.069s |
| Line 204: storm responses with empty storm_id are skipped.                                                         | <code>data["status"] == "success"; "" not in storm_ids (+1 more)</code>                                                   | Pass   | 0.046s |
| Lines 235-237: storms.json enrichment path (not sequences).                                                        | <code>s["intensity_category"] == "extreme"; s["effective_precipitation_mm"] == 80.0</code>                                | Pass   | 0.038s |
| Lines 254-266: storms without metadata get classified by gauges_severe count.                                      | <code>storms["S-CAT"]["intensity_category"] == "catastrophic"; storms["S-CAT"]["name"] == "Catastrophic" (+4 more)</code> | Pass   | 0.333s |
| Line 292: flood events with depth $\neq 0$ are skipped.                                                            | <code>storms["STORM-F"]["properties_flooded"] == 0</code>                                                                 | Pass   | 0.139s |
| Created meeting appears in list                                                                                    | <code>len(data["meetings"]) == 1; data["meetings"][0]["title"] == "Board Meeting"</code>                                  | Pass   | 0.002s |
| Empty list returns success                                                                                         | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                                  | Pass   | 0.001s |
| List summary has required keys                                                                                     | <code>key in m</code>                                                                                                     | Pass   | 0.002s |
| Sorted by date desc                                                                                                | <code>dates == sorted(dates, reverse=True)</code>                                                                         | Pass   | 0.003s |
| List mortgages empty                                                                                               | <code>r.status_code == 200; r.get_json()["count"] == 0</code>                                                             | Pass   | 0.004s |
| List mortgages exception returns 500                                                                               | <code>r.status_code == 500; r.get_json()["status"] == "error"</code>                                                      | Pass   | 0.001s |
| List mortgages success                                                                                             | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                                                  | Pass   | 0.002s |
| <code>sys.exit(0)</code> fires before the flood file is opened.                                                    | —                                                                                                                         | Pass   | 0.001s |
| <code>sys.exit(0)</code> fires before the gauge file is opened.                                                    | —                                                                                                                         | Pass   | 0.002s |
| <code>sys.exit(0)</code> fires before the property file is opened.                                                 | —                                                                                                                         | Pass   | 0.002s |
| Lines 405-410: <code>-list-pages</code> iterates available pages, <code>sys.exit(0)</code> .                       | —                                                                                                                         | Pass   | 0.056s |
| Lines 462-467: <code>-list-pages</code> iterates available pages, <code>sys.exit(0)</code> .                       | —                                                                                                                         | Pass   | 0.007s |
| Lines 314-319: <code>-list-pages</code> → <code>list_available_pages()</code> iterated, <code>sys.exit(0)</code> . | —                                                                                                                         | Pass   | 0.002s |

| Test Description                                                 | Acceptance Criteria                                                                                                                             | Result | Time   |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Loader exception returns 500                                     | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code><br>(+1 more)                                               | Pass   | 0.001s |
| Success empty list                                               | <code>r.status_code == 200; data["status"]</code><br><code>== "success" (+2 more)</code>                                                        | Pass   | 0.001s |
| Success with properties                                          | <code>r.status_code == 200;</code><br><code>r.get_json()["count"] == 2</code>                                                                   | Pass   | 0.001s |
| Empty dir returns zero count                                     | <code>r.status_code == 200; data["status"]</code><br><code>== "success" (+2 more)</code>                                                        | Pass   | 0.001s |
| Returns trades list key                                          | <code>"trades" in r.get_json()</code>                                                                                                           | Pass   | 0.001s |
| Corrupt trade file returns 500                                   | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                                                            | Pass   | 0.029s |
| list_prs_trades reads PRS-*.json files and returns them.         | <code>r.status_code == 200; data["count"] ==</code><br><code>1 (+1 more)</code>                                                                 | Pass   | 0.029s |
| Loads entries                                                    | <code>result[0]["event"] == "test"</code>                                                                                                       | Pass   | 0.001s |
| Returns empty list on bad json                                   | <code>_load_audit_log() == []</code>                                                                                                            | Pass   | 0.001s |
| Returns empty list when missing                                  | <code>_load_audit_log() == [];</code><br><code>_load_meetings() == []</code>                                                                    | Pass   | 0.000s |
| Loads valid data                                                 | <code>"principles" in result; "roles" in</code><br><code>result</code>                                                                          | Pass   | 0.001s |
| Returns none when missing                                        | <code>_load_bcbs239() is None; _load_raci()</code><br><code>is None</code>                                                                      | Pass   | 0.000s |
| Loads bibliography                                               | <code>result["references"][0]["id"] ==</code><br><code>"R1"</code>                                                                              | Pass   | 0.001s |
| Returns default when missing                                     | <code>result == {"references": [],</code><br><code>"categories": []}</code>                                                                     | Pass   | 0.000s |
| Line 208: file exists with empty counterparties → fallback list. | <code>len(result) == 21; result[0]["id"] ==</code><br><code>"CTPY-001" (+1 more)</code>                                                         | Pass   | 0.026s |
| Line 207-209: file doesn't exist → fallback list.                | <code>len(result) == 21</code>                                                                                                                  | Pass   | 0.002s |
| When file has entries, those are returned (not fallback).        | <code>len(result) == 1; result[0]["id"] ==</code><br><code>"CTPY-REAL" (+1 more)</code>                                                         | Pass   | 0.003s |
| Corrupt counterparty file leaves data none                       | <code>dl.loaded_data.counterparty_data is</code><br><code>None</code>                                                                           | Pass   | 0.001s |
| Missing counterparty file leaves data none                       | <code>dl.loaded_data.counterparty_data is</code><br><code>None</code>                                                                           | Pass   | 0.000s |
| Valid counterparty file loads                                    | <code>dl.loaded_data.counterparty_data</code><br><code>is not None; "counterparties" in</code><br><code>dl.loaded_data.counterparty_data</code> | Pass   | 0.001s |
| Missing gaugehc returns empty and default storms                 | <code>curves == {}; n_storms == 1000</code>                                                                                                     | Pass   | 0.000s |

| Test Description                                                   | Acceptance Criteria                                                  | Result | Time   |
|--------------------------------------------------------------------|----------------------------------------------------------------------|--------|--------|
| Present gaugehc returns curves                                     | "GAUGE-001" in curves; n.storms == 500                               | Pass   | 0.001s |
| Present gaugehc returns num storms                                 | n.storms == 250                                                      | Pass   | 0.001s |
| Lines 74-75: missing file raises FileNotFoundError.                | —                                                                    | Pass   | 0.000s |
| Lines 72-80: load_gauge_portfolio reads gauge.json.                | isinstance(result, list); len(result) == 1                           | Pass   | 0.021s |
| Lines 59-64: get_reports_dir raises → fallback to get_output_path. | result == []                                                         | Pass   | 0.000s |
| Line 67: prs_dir doesn't exist → return empty list.                | result == []                                                         | Pass   | 0.000s |
| Lines 75-76: trade with matching gauge_id → appended to list.      | len(result) == 1                                                     | Pass   | 0.025s |
| Trade for different gauge not appended.                            | result == []                                                         | Pass   | 0.001s |
| Empty gauge list raises                                            | —                                                                    | Pass   | 0.001s |
| File not found raises                                              | —                                                                    | Pass   | 0.000s |
| Gauge has id                                                       | result[0]["gauge_id"] == "GAUGE-999"                                 | Pass   | 0.001s |
| Gauge that maps to no gauge_id raises ValueError.                  | —                                                                    | Pass   | 0.001s |
| Loads gauges                                                       | isinstance(result, list); len(result) == 1                           | Pass   | 0.001s |
| Empty portfolio returns empty list                                 | result == []                                                         | Pass   | 0.001s |
| Loads gauge from portfolio                                         | len(result) == 1;<br>isinstance(result[0], GaugeConfig)<br>(+1 more) | Pass   | 0.023s |
| Loads documents                                                    | _load_gov_documents()[0]["doc_id"] == "D1"                           | Pass   | 0.001s |
| Returns empty list when missing                                    | _load_gov_documents() == []                                          | Pass   | 0.000s |
| Corrupt hazard file leaves hazard data none                        | dl.loaded_data.hazard_data is None                                   | Pass   | 0.001s |
| Corrupt property hazard file leaves phc none                       | dl.loaded_data.property_hazard_data is None                          | Pass   | 0.001s |
| Loaded data has gauge                                              | "TEST-G001" in<br>result["hazard_curves"]                            | Pass   | 0.001s |
| Loads from file                                                    | "hazard_curves" in result                                            | Pass   | 0.006s |
| Missing hazard file leaves hazard data none                        | dl.loaded_data.hazard_data is None                                   | Pass   | 0.000s |
| Missing property hazard file leaves phc none                       | dl.loaded_data.property_hazard_data is None                          | Pass   | 0.000s |

| Test Description                                             | Acceptance Criteria                                                                                | Result | Time   |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------|--------|
| Valid hazard file populates hazard data                      | <code>dl.loaded_data.hazard_data is not None; "hazard_curves" in dl.loaded_data.hazard_data</code> | Pass   | 0.001s |
| Empty json object returns empty dict                         | <code>result == {}</code>                                                                          | Pass   | 0.000s |
| Invalid json raises json decode error                        | —                                                                                                  | Pass   | 0.001s |
| Missing file raises file not found                           | —                                                                                                  | Pass   | 0.000s |
| Returns full structure                                       | <code>result["meta"]["version"] == 2; "G-001" in result["hazard_curves"]</code>                    | Pass   | 0.001s |
| Loads valid file                                             | <code>result["models"][0]["model_id"] == "M1"</code>                                               | Pass   | 0.001s |
| Returns none on bad json                                     | <code>_load_inventory() is None</code>                                                             | Pass   | 0.001s |
| Returns none when missing                                    | <code>_load_inventory() is None</code>                                                             | Pass   | 0.000s |
| Lines 89-91: exception → returns {}.                         | <code>result == {}</code>                                                                          | Pass   | 0.000s |
| Loads list                                                   | <code>_load_meetings()[0]["id"] == "M1"</code>                                                     | Pass   | 0.001s |
| Returns empty list when missing                              | <code>_load_audit_log() == [];</code><br><code>_load_meetings() == []</code>                       | Pass   | 0.000s |
| When mortgage.json has invalid JSON, the exception is caught | <code>r.status_code == 200; len(props) == 1</code><br>(+1 more)                                    | Pass   | 0.029s |
| Missing property json returns 404                            | <code>r.status_code == 404</code>                                                                  | Pass   | 0.002s |
| Missing property json sequence returns 404                   | <code>r.status_code == 404</code>                                                                  | Pass   | 0.002s |
| Loads valid data                                             | <code>"principles" in result; "roles" in result</code>                                             | Pass   | 0.001s |
| Returns none when missing                                    | <code>_load_bcbs239() is None; _load_raci() is None</code>                                         | Pass   | 0.000s |
| Corrupt storm file leaves storm data none                    | <code>dl.loaded_data.storm_data is None</code>                                                     | Pass   | 0.003s |
| Missing storm file leaves storm data none                    | <code>dl.loaded_data.storm_data is None</code>                                                     | Pass   | 0.000s |
| Storms dict with storms key                                  | <code>dl.loaded_data.storm_data is not None</code>                                                 | Pass   | 0.001s |
| Storms list format                                           | <code>dl.loaded_data.storm_data is not None</code>                                                 | Pass   | 0.006s |
| Storms other type                                            | <code>dl.loaded_data.storm_data is not None</code>                                                 | Pass   | 0.001s |
| File not found raises                                        | —                                                                                                  | Pass   | 0.000s |
| Loads list                                                   | <code>isinstance(result, list); len(result) == 5</code>                                            | Pass   | 0.002s |
| Storm has id                                                 | <code>result[0]["storm_id"] == "STORM-0000"</code>                                                 | Pass   | 0.002s |
| When path.stat() raises OSError, return existing cache.      | <code>result == {"storms": [{"storm_id": "cached"}]}</code>                                        | Pass   | 0.002s |

| Test Description                                            | Acceptance Criteria                                                                                                  | Result | Time   |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|
| When path.stat() raises OSError and no cache, returns None. | result is None                                                                                                       | Pass   | 0.000s |
| Empty data list records invalid result                      | dl._validation_results["empty_type"].is_valid is False; "No data loaded" in dl._validation_results["empty_type"].... | Pass   | 0.000s |
| Empty data list returns none                                | dl._load_with_validation(mock_loader, "empty_type") is None                                                          | Pass   | 0.000s |
| Exception in loader returns none                            | dl._load_with_validation(mock_loader, "error_type") is None                                                          | Pass   | 0.004s |
| Exception records error in validation                       | dl._validation_results["error_type"].is_valid is False; "disk error" in dl._validation_results["error_type"].erro... | Pass   | 0.001s |
| None return from loader returns none                        | dl._load_with_validation(mock_loader, "none_type") is None                                                           | Pass   | 0.000s |
| Successful load records valid result                        | "gauge_test" in dl._validation_results; dl._validation_results["gauge_test"].is_valid is True                        | Pass   | 0.001s |
| Successful load returns data dict                           | result is not None; result["count"] == 2 (+1 more)                                                                   | Pass   | 0.000s |
| Validation summary count matches items                      | dl._validation_results["three_items"].summary["count"] == 3                                                          | Pass   | 0.000s |
| Validation summary has type field                           | dl._validation_results["xtype"].summary["type"] == "xtype"                                                           | Pass   | 0.000s |
| Can set gauge data                                          | d.gauge_data is not None                                                                                             | Pass   | 0.000s |
| Can set multiple fields                                     | d.gaugets_count == 3; d.gaugehd_count == 5                                                                           | Pass   | 0.000s |
| Counts default to zero                                      | d.gaugets_count == 0; d.gaugehd_count == 0 (+1 more)                                                                 | Pass   | 0.000s |
| Defaults are none                                           | getattr(d, field) is None                                                                                            | Pass   | 0.000s |
| Is dataclass                                                | dataclasses.is_dataclass(LoadedData)                                                                                 | Pass   | 0.000s |
| Lookup defaults are none                                    | getattr(d, field) is None                                                                                            | Pass   | 0.000s |
| Test creating a registry.                                   | registry.data_dir == populated.data_dir; registry is not None (+1 more)                                              | Pass   | 0.000s |
| Test getting list of available loader types.                | 'property' in loaders; 'mortgage' in loaders (+3 more)                                                               | Pass   | 0.000s |
| Clear all caches                                            | not loader._cache_valid                                                                                              | Pass   | 0.001s |
| Get all status                                              | "gauge" in status; "entity_name" in status["gauge"]                                                                  | Pass   | 0.001s |
| Invalidate all caches                                       | —                                                                                                                    | Pass   | 0.000s |

| Test Description                               | Acceptance Criteria                                                                      | Result | Time   |
|------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| Test clearing all loader caches.               | prop_loader._cache is None;<br>gauge_loader._cache is None                               | Pass   | 0.003s |
| Test that different loader types are separate. | prop_loader is not gauge_loader                                                          | Pass   | 0.000s |
| Test that loader instances are cached.         | loader1 is loader2                                                                       | Pass   | 0.000s |
| Check data files all missing                   | all(v is False for v in<br>result.values())                                              | Pass   | 0.000s |
| Check data files present                       | result["property"] is True;<br>result["gauge"] is False                                  | Pass   | 0.000s |
| File overrides                                 | registry.get_gauge_loader().count()<br>== 3                                              | Pass   | 0.003s |
| Get data dir                                   | LoaderRegistry(data_dir=tmp_path).get_data_dir()<br>== tmp_path                          | Pass   | 0.000s |
| Test providing custom filenames.               | len(props) == 1;<br>loader.get_entity_id(props[0]) ==<br>"CUSTOM-001"                    | Pass   | 0.002s |
| Test getting loader by name string.            | loader is not None; loader.count() ==<br>3                                               | Pass   | 0.001s |
| Test getting unknown loader type.              | loader is None                                                                           | Pass   | 0.000s |
| Get available loaders                          | "property" in available; "gauge" in<br>available (+1 more)                               | Pass   | 0.000s |
| Get gauge loader                               | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("gauge"), GaugeLoader)           | Pass   | 0.000s |
| Get loader by name                             | LoaderRegistry(data_dir=tmp_path).get_loader("gauge")<br>is ...                          | Pass   | 0.000s |
| Get loader unknown name returns none           | LoaderRegistry(data_dir=tmp_path).get_loader("nonexistent")                              | Pass   | 0.000s |
| Get mortgage loader                            | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("mortgage"), MortgageLoader)     | Pass   | 0.000s |
| Get property loader                            | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("property"), PropertyLoader)     | Pass   | 0.000s |
| Get storm loader                               | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("storm"), StormLoader)           | Pass   | 0.000s |
| Get timeseries loader                          | isinstance(LoaderRegistry(data_dir=tmp_path).get_loader("timeseries"), TimeseriesLoader) | Pass   | 0.000s |
| Loader cached on repeated access               | registry.get_property_loader() is<br>registry.get_property_loader()                      | Pass   | 0.000s |
| Test checking data file existence.             | status['property'] is True;<br>status['mortgage'] is True (+3<br>more)                   | Pass   | 0.000s |
| Test getting status for all loaders.           | 'property' in status; 'gauge' in<br>status (+1 more)                                     | Skip   | 0.000s |
| Test getting GaugeLoader.                      | isinstance(loader, GaugeLoader);<br>loader.count() == 3                                  | Pass   | 0.002s |



| Test Description                                               | Acceptance Criteria                                                                  | Result | Time   |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| Test getting MortgageLoader.                                   | <code>isinstance(loader, MortgageLoader);</code><br><code>loader.count() == 3</code> | Pass   | 0.002s |
| Test getting PropertyLoader.                                   | <code>isinstance(loader, PropertyLoader);</code><br><code>loader.count() == 3</code> | Pass   | 0.002s |
| Test getting StormLoader.                                      | <code>isinstance(loader, StormLoader)</code>                                         | Pass   | 0.000s |
| Test getting TimeseriesLoader.                                 | <code>isinstance(loader, TimeseriesLoader)</code>                                    | Pass   | 0.000s |
| Address subsection header                                      | <code>any("Primary Address" in t for t in texts)</code>                              | Pass   | 0.001s |
| Administrative subsection header                               | <code>any("Administrative Areas" in t for t in texts)</code>                         | Pass   | 0.001s |
| Area classification subsection header                          | <code>any("Area Classification" in t for t in texts)</code>                          | Pass   | 0.001s |
| Density high                                                   | <code>isinstance(result, list)</code>                                                | Pass   | 0.000s |
| Density low                                                    | <code>isinstance(result, list)</code>                                                | Pass   | 0.000s |
| Density low medium                                             | <code>isinstance(result, list)</code>                                                | Pass   | 0.000s |
| Density medium                                                 | <code>isinstance(result, list)</code>                                                | Pass   | 0.001s |
| Density very high                                              | <code>isinstance(result, list)</code>                                                | Pass   | 0.001s |
| Empty property does not crash                                  | <code>isinstance(result, list)</code>                                                | Pass   | 0.000s |
| Geographic subsection header                                   | <code>any("Geographic Information" in t for t in texts)</code>                       | Pass   | 0.001s |
| Has section header                                             | <code>any("Location Details" in t for t in texts)</code>                             | Pass   | 0.001s |
| Has tables                                                     | <code>any(isinstance(e, Table) for e in result)</code>                               | Pass   | 0.001s |
| Missing lat/lon means coordinates rows are omitted — no crash. | <code>isinstance(result, list)</code>                                                | Pass   | 0.000s |
| Returns list                                                   | <code>isinstance(result, list); len(result) &gt; 0</code>                            | Pass   | 0.001s |
| With mortgage data does not crash                              | <code>isinstance(result, list)</code>                                                | Pass   | 0.001s |
| Debug does not raise                                           | —                                                                                    | Pass   | 0.000s |
| Debug level does not raise                                     | —                                                                                    | Pass   | 0.000s |
| Error does not raise                                           | —                                                                                    | Pass   | 0.000s |
| Error level does not raise                                     | —                                                                                    | Pass   | 0.000s |
| Info does not raise                                            | —                                                                                    | Pass   | 0.000s |
| Info level does not raise                                      | —                                                                                    | Pass   | 0.000s |
| Log does not raise                                             | —                                                                                    | Pass   | 0.000s |
| Log with empty string                                          | —                                                                                    | Pass   | 0.000s |
| Success falls back gracefully                                  | —                                                                                    | Pass   | 0.000s |
| Success level falls back gracefully                            | —                                                                                    | Pass   | 0.000s |
| Unknown level falls back to info                               | —                                                                                    | Pass   | 0.000s |
| Warning does not raise                                         | —                                                                                    | Pass   | 0.001s |

| Test Description                                                      | Acceptance Criteria                                                                       | Result | Time   |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Warning level does not raise                                          | —                                                                                         | Pass   | 0.000s |
| Appends entry                                                         | len(entries) == 1;<br>entries[0]["action"] == "test_action"<br>(+1 more)                  | Pass   | 0.001s |
| Failures in audit logging must not propagate.                         | —                                                                                         | Pass   | 0.000s |
| Corrupted audit log → start fresh, don't raise.                       | len(entries) == 1                                                                         | Pass   | 0.001s |
| Creates audit file                                                    | os.path.exists(audit_path)                                                                | Pass   | 0.000s |
| Defaults applied when no json                                         | r.status_code == 200;<br>entry["event_type"] == "usage"<br>(+1 more)                      | Pass   | 0.002s |
| Entry has expected fields                                             | entry["model_id"] == "MKM-TEST-001";<br>entry["event_type"] == "calibration"<br>(+2 more) | Pass   | 0.002s |
| Entry has required keys                                               | "timestamp" in entry; "model_id" in entry (+3 more)                                       | Pass   | 0.001s |
| Existing file appended                                                | len(entries) == 2; "existing" in entries[0]                                               | Pass   | 0.002s |
| Model not found returns 404                                           | r.status_code == 404;<br>r.get_json()["status"] == "error"<br>(+3 more)                   | Pass   | 0.002s |
| Multiple calls accumulate                                             | len(entries) == 2                                                                         | Pass   | 0.001s |
| Parameters stored                                                     | r.get_json()["entry"]["parameters"]<br>== {"key": "value"}                                | Pass   | 0.002s |
| Valid model returns 200                                               | r.status_code == 200;<br>r.get_json()["status"] == "success"                              | Pass   | 0.002s |
| ltv_ratio $\div$ 1 is interpreted as a percentage and divided by 100. | score > 0                                                                                 | Pass   | 0.000s |
| Compare mode                                                          | "Scenario Comparison" in captured.out                                                     | Pass   | 0.386s |
| Custom notional and spread                                            | rc == 0                                                                                   | Pass   | 0.002s |
| Default describe mode                                                 | len(captured.out) > 0                                                                     | Pass   | 0.079s |
| Default gauge selection returns zero                                  | rc == 0                                                                                   | Pass   | 0.002s |
| Exceedance json mode                                                  | "exceedance_probability" in data                                                          | Pass   | 0.000s |
| Exceedance mode                                                       | "70" in captured.out                                                                      | Pass   | 0.000s |
| Explicit gauge returns zero                                           | rc == 0                                                                                   | Pass   | 0.002s |
| Flat flag dispatches flat mode                                        | rc == 0                                                                                   | Pass   | 0.002s |
| Json default mode                                                     | "summary_statistics" in data                                                              | Pass   | 0.294s |
| List flag logs gauge names                                            | "THAMES-TEST-99" in combined                                                              | Pass   | 0.002s |
| List flag returns zero                                                | rc == 0                                                                                   | Pass   | 0.002s |

| Test Description                                                          | Acceptance Criteria                                                                                                  | Result | Time   |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------|--------|
| Lines 120-122: default (no -nrfa-dir) calls generate_all_gauge_histories. | —                                                                                                                    | Pass   | 0.384s |
| main() exits cleanly if jscpd fails.                                      | —                                                                                                                    | Pass   | 0.000s |
| Lines 107-108: without -catchment, config.CATCHMENT is untouched.         | config.CATCHMENT == original                                                                                         | Pass   | 0.352s |
| main() prints file/clone/pct stats to stdout.                             | 'Clones' in out; 'PDF saved' in out                                                                                  | Pass   | 0.233s |
| main() writes code_duplication_report.pdf to audit dir.                   | (tmp_path / 'code_duplication_report.pdf').exists();<br>(tmp_path / 'code_duplication_report.pdf').stat().st_size... | Pass   | 0.237s |
| Line 102: -c short flag for catchment.                                    | —                                                                                                                    | Pass   | 0.371s |
| Line 101: -y short flag for years.                                        | —                                                                                                                    | Pass   | 0.280s |
| main() logs errors for unreadable files and continues.                    | —                                                                                                                    | Pass   | 0.003s |
| Lines 107-108: -catchment sets config.CATCHMENT.                          | config.CATCHMENT == "severn"                                                                                         | Pass   | 0.360s |
| main() processes any *_gdf.csv files found in config.input_dir.           | —                                                                                                                    | Pass   | 0.029s |
| main() handles an empty directory gracefully.                             | —                                                                                                                    | Pass   | 0.001s |
| Lines 117-119: -nrfa-dir calls process_nrfa_directory.                    | list(gaugehd_dir.glob("gauge.*_hd.json"))                                                                            | Pass   | 0.003s |
| Line 101: -years flag is passed through.                                  | —                                                                                                                    | Pass   | 0.248s |
| Missing gauge logs available gauges                                       | "THAMES-G042" in combined                                                                                            | Pass   | 0.002s |
| Missing gauge logs error                                                  | any("GHOST-GAUGE" in r.message for r in caplog.records)                                                              | Pass   | 0.001s |
| Missing gauge returns one                                                 | rc == 1                                                                                                              | Pass   | 0.001s |
| Return period json mode                                                   | "return_period_years" in data;<br>"intensity" in data                                                                | Pass   | 0.000s |
| Return period mode                                                        | "100" in captured.out                                                                                                | Pass   | 0.000s |
| Samples json mode                                                         | "samples" in data;<br>len(data["samples"]) == 3                                                                      | Pass   | 0.000s |
| Samples mode                                                              | "5 samples" in captured.out                                                                                          | Pass   | 0.000s |
| Scenario mode                                                             | len(captured.out) > 0                                                                                                | Pass   | 0.078s |
| Seed reproducible                                                         | out1 == out2                                                                                                         | Pass   | 0.001s |
| Trigger alert explicit                                                    | rc == 0                                                                                                              | Pass   | 0.002s |
| Trigger severe explicit                                                   | rc == 0                                                                                                              | Pass   | 0.001s |

| Test Description                                                           | Acceptance Criteria                                                                                  | Result | Time   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Trigger warning explicit                                                   | <code>rc == 0</code>                                                                                 | Pass   | 0.002s |
| Report with 2% duplication should render without error.                    | <code>analysis['total']['percentage'] &lt; 3;</code><br><code>len(pdf) &gt; 0</code>                 | Pass   | 0.230s |
| Report with 8% duplication renders correctly.                              | <code>pdf[:4] == b'PDF'</code>                                                                       | Pass   | 0.234s |
| Pdf header                                                                 | <code>pdf[:4] == b'PDF'</code>                                                                       | Pass   | 0.237s |
| Pdf nonempty                                                               | <code>len(pdf) &gt; 2000</code>                                                                      | Pass   | 0.232s |
| Returns bytes                                                              | <code>isinstance(result, bytes)</code>                                                               | Pass   | 1.500s |
| Peak precipitation is in the storm's active window.                        | <code>peak_in_storm &gt;= peak_outside</code>                                                        | Pass   | 0.000s |
| Drainage window (hours 156-167) must be precipitation-free.                | <code>np.all(p[156:] ==</code><br><code>pytest.approx(0.0))</code>                                   | Pass   | 0.001s |
| Non negative                                                               | <code>np.all(p &gt;= 0.0)</code>                                                                     | Pass   | 0.000s |
| Output shape                                                               | <code>p.shape == (EVENT_WINDOW_HOURS,);</code><br><code>levels.shape == (EVENT_WINDOW_HOURS,)</code> | Pass   | 0.000s |
| Sum of hourly series == sum of storm precipitation_mm values.              | <code>p.sum() == pytest.approx(expected,</code><br><code>rel=1e-4)</code>                            | Pass   | 0.001s |
| Add action with malformed existing id                                      | <code>r.status_code == 200; "A-002" in ids</code>                                                    | Pass   | 0.002s |
| Malformed decision ID is skipped; new decision still gets a valid auto-ID. | <code>r.status_code == 200; "D-002" in ids</code>                                                    | Pass   | 0.002s |
| Add participant with malformed existing id                                 | <code>r.status_code == 200; "P-002" in ids</code>                                                    | Pass   | 0.002s |
| Click-through should hide trading desk before opening gauge.               | <code>'TradingDesk.hide' in js</code>                                                                | Pass   | 0.000s |
| Map should have New Trade link.                                            | <code>'window.tdNewTrade' in js;</code> <code>'New</code><br><code>Trade' in js</code>               | Pass   | 0.000s |
| Map should have tdViewHazard-Curve navigating to gauge panel.              | <code>'window.tdViewHazardCurve' in js;</code><br><code>'GaugeHazardCurve' in js</code>              | Pass   | 0.000s |
| Any flood maps to alert                                                    | <code>gen._map.threshold.to_gauge.trigger("any_flood</code><br><code>== "alert"</code>               | Pass   | 0.000s |
| Moderate maps to warning                                                   | <code>gen._map.threshold.to_gauge.trigger("moderate</code><br><code>== "warning"</code>              | Pass   | 0.000s |
| Severe maps to severe                                                      | <code>gen._map.threshold.to_gauge.trigger("severe</code><br><code>== "severe"</code>                 | Pass   | 0.000s |
| Unknown key returns warning                                                | <code>gen._map.threshold.to_gauge.trigger("nonexistent</code><br><code>== "wa..."</code>             | Pass   | 0.000s |
| P&L bar must show 'From Market' metric.                                    | <code>'From Market' in js;</code> <code>'fromMarket' in</code><br><code>js</code>                    | Pass   | 0.000s |

| Test Description                                               | Acceptance Criteria                                                  | Result | Time   |
|----------------------------------------------------------------|----------------------------------------------------------------------|--------|--------|
| Blotter must have a Mkt P&L column.                            | "'market.pnl'" in js or<br>'"market.pnl"' in js;<br>js<br>'Mkt P' in | Pass   | 0.000s |
| Market tab commit notification must show P&L impact.           | 'total.pnl.impact' in js;<br>'gross.pnl.impact' in js                | Pass   | 0.000s |
| Hazard column must show up/down arrows when fair != prev_fair. | 'prev.fair.spread.bps' in js;<br>'\{\}\u25b2' in js (+1 more)        | Pass   | 0.000s |
| Entry level under 250k                                         | isinstance(result, list)                                             | Pass   | 0.000s |
| Middle 250k to 500k                                            | isinstance(result, list)                                             | Pass   | 0.000s |
| Premium over 1m                                                | isinstance(result, list)                                             | Pass   | 0.000s |
| Upper middle 500k to 1m                                        | isinstance(result, list)                                             | Pass   | 0.000s |
| Base rates not empty                                           | br                                                                   | Pass   | 0.000s |
| File exists                                                    | GAUGEHC_PATH.exists();<br>MARKET_STATE.exists()                      | Pass   | 0.000s |
| Has required keys                                              | not missing                                                          | Pass   | 0.000s |
| Hazard term structure has triggers                             | "alert" in triggers                                                  | Pass   | 0.000s |
| Hazard term structure not empty                                | hts                                                                  | Pass   | 0.000s |
| Last updated present                                           | "last.updated" in ms                                                 | Pass   | 0.000s |
| Yield curve has all tenors                                     | not missing                                                          | Pass   | 0.000s |
| Yield curve rates in range                                     | not bad                                                              | Pass   | 0.000s |
| computeMaturityDate function exists.                           | 'computeMaturityDate' in prs-js                                      | Pass   | 0.000s |
| currentRollDate function exists.                               | 'currentRollDate' in prs-js                                          | Pass   | 0.000s |
| lastFridayOfMonth function exists.                             | 'lastFridayOfMonth' in prs-js                                        | Pass   | 0.000s |
| showMaturityPopup function exists.                             | 'showMaturityPopup' in prs-js                                        | Pass   | 0.000s |
| Mcob advice rejection reason                                   | isinstance(result, list)                                             | Pass   | 0.000s |
| Mcob all fields returns list                                   | isinstance(result, list); len(result)<br>> 5                         | Pass   | 0.000s |
| Mcob section creates tables                                    | len(tables) >= 2                                                     | Pass   | 0.000s |
| Mcob with minimal fields                                       | isinstance(result, list)                                             | Pass   | 0.000s |
| Serve missing document returns 404                             | r.status_code == 404                                                 | Pass   | 0.002s |
| Upload empty filename returns 400                              | r.status_code == 400                                                 | Pass   | 0.004s |
| Upload no file returns 400                                     | r.status_code == 400;<br>r.get_json()["status"] == "error"           | Pass   | 0.002s |
| Upload not found meeting returns 404                           | r.status_code == 404                                                 | Pass   | 0.003s |
| Unknown menu no options returns empty                          | result == ""                                                         | Pass   | 0.000s |
| Unknown menu with options cycles                               | result == "OptionB"                                                  | Pass   | 0.000s |
| Unknown menu with options wraps                                | result == "X"                                                        | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                                                                                                                     | Result | Time   |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Duration from storms json                                            | <code>storm["duration_hours"] == 72;</code><br><code>storm["peak_position"] == pytest.approx(0.3) (+1 more)</code>                                      | Pass   | 0.003s |
| Fallback to config defaults                                          | <code>storm["duration_hours"] == STRESS_STORM_DEFAULT_DURATION....;</code><br><code>storm["peak_position"] == pytest.approx(STRESS_STORM_DEFA...</code> | Pass   | 0.002s |
| Passing a non-existent <code>storms_json_path</code> must not crash. | <code>out.exists()</code> and <code>(out / "_index.json").exists()</code>                                                                               | Pass   | 0.001s |
| Distant point returns larger distance                                | <code>d_far &gt; d_near</code>                                                                                                                          | Pass   | 0.000s |
| Point on gauge returns small distance                                | <code>d &lt; 10.000</code>                                                                                                                              | Pass   | 0.000s |
| Returns float                                                        | <code>isinstance(d, float); d &gt;= 0</code>                                                                                                            | Pass   | 0.000s |
| Add minute item                                                      | <code>r.status_code == 200;</code><br><code>isinstance(minutes, list) (+1 more)</code>                                                                  | Pass   | 0.001s |
| Delete minute item legacy format returns 400                         | <code>r.status_code == 400</code>                                                                                                                       | Pass   | 0.003s |
| Update minute item legacy format returns 400                         | <code>r.status_code == 400</code>                                                                                                                       | Pass   | 0.005s |
| Update minute item not found returns 404                             | <code>r.status_code == 404</code>                                                                                                                       | Pass   | 0.003s |
| Update minutes text                                                  | <code>r.status_code == 200;</code><br><code>r.get_json()["meeting"]["minutes"] == "# Minutes\{\}n\{\}nAll a...</code>                                   | Pass   | 0.002s |
| Update minutes unknown meeting returns 404                           | <code>r.status_code == 404</code>                                                                                                                       | Pass   | 0.002s |
| Add minute item missing meeting returns 404                          | <code>r.status_code == 404</code>                                                                                                                       | Pass   | 0.001s |
| Add minute item returns 200                                          | <code>r.status_code == 200;</code><br><code>isinstance(meeting["minutes"], list) (+1 more)</code>                                                       | Pass   | 0.002s |
| When minutes is a string, deleting returns 400.                      | <code>r.status_code == 400</code>                                                                                                                       | Pass   | 0.003s |
| Delete minute item missing meeting returns 404                       | <code>r.status_code == 404</code>                                                                                                                       | Pass   | 0.002s |
| Delete minute item returns 200                                       | <code>r.status_code == 200;</code><br><code>len(meeting["minutes"]) == 1 (+1 more)</code>                                                               | Pass   | 0.003s |
| When minutes is a string (legacy), updating an item returns 400.     | <code>r.status_code == 400</code>                                                                                                                       | Pass   | 0.003s |



| Test Description                                                       | Acceptance Criteria                                                                               | Result | Time   |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Update minute item not found returns 404                               | <code>r.status_code == 404</code>                                                                 | Pass   | 0.001s |
| Update minute item returns 200                                         | <code>r.status_code == 200; item["text"] == "Updated text."</code>                                | Pass   | 0.002s |
| Set minutes to a string via update, then add a minutes-item.           | <code>r.status_code == 200; isinstance(meeting["minutes"], list) (+2 more)</code>                 | Pass   | 0.004s |
| Update minutes missing meeting returns 404                             | <code>r.status_code == 404</code>                                                                 | Pass   | 0.001s |
| Update minutes returns 200                                             | <code>r.status_code == 200; r.get_json()["meeting"]["minutes"] == "New minutes text"</code>       | Pass   | 0.002s |
| Property hazard returns 404 when propertyhc.json missing.              | <code>response.status_code == 404; data['status'] == 'error'</code>                               | Pass   | 0.001s |
| PropertyHC summary returns 404 with helpful message when file missing. | <code>response.status_code == 404; data['status'] == 'error' (+1 more)</code>                     | Pass   | 0.001s |
| PropertyTS summary returns 404 when directory missing.                 | <code>response.status_code == 404; data['status'] == 'error'</code>                               | Pass   | 0.001s |
| No crash when gauge json absent                                        | <code>isinstance(result, dict)</code>                                                             | Pass   | 0.003s |
| Num gauges zero without gauge json                                     | <code>result["num-gauges"] == 0</code>                                                            | Pass   | 0.002s |
| Sequences still written without gauge json                             | <code>(tmp_path / SEQUENCES_FILENAME).exists()</code>                                             | Pass   | 0.002s |
| Only alert storms create flood events                                  | <code>result["summary"]["floods_at_nearest_gauge"] == 2</code>                                    | Pass   | 0.002s |
| Sub alert storms not in flood events                                   | <code>"STORM-m001b" not in event_storm_ids; "STORM-m001d" not in event_storm_ids (+2 more)</code> | Pass   | 0.002s |
| When predictor.load_summary has no matching gauge, model_auc is None.  | <code>resp.status_code == 200; data['model_auc'] is None</code>                                   | Pass   | 0.047s |
| When predictor.load_summary raises, model_auc is None (lines 136-137). | <code>resp.status_code == 200; data['model_auc'] is None</code>                                   | Pass   | 0.030s |
| When predictor.load_summary has matching gauge, model_auc is set.      | <code>resp.status_code == 200; data['model_auc'] == 0.95</code>                                   | Pass   | 0.026s |
| When gauge status is not 'trained', model_auc remains None.            | <code>resp.status_code == 200; data['model_auc'] is None</code>                                   | Pass   | 0.033s |
| All models have model id                                               | <code>"model_id" in m; isinstance(m["model_id"], str) (+1 more)</code>                            | Pass   | 0.000s |

| Test Description                                                               | Acceptance Criteria                                                                                                                 | Result | Time   |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Model ids are unique                                                           | <code>len(ids) == len(set(ids))</code>                                                                                              | Pass   | 0.000s |
| Model ids follow naming convention                                             | <code>mid.startswith("MKM-")</code>                                                                                                 | Pass   | 0.000s |
| Lines 362-363: exception in create_mapping raises ValueError.                  | —                                                                                                                                   | Pass   | 0.000s |
| Line 280: valid data returns empty dict.                                       | <code>mortgage_cdm.validate(data) == {}</code>                                                                                      | Pass   | 0.000s |
| Lines 282-283: exception in validate returns validation_error.                 | <code>"validation_error" in result;</code><br><code>isinstance(result["validation_error"],</code><br><code>list) (+1 more)</code>   | Pass   | 0.000s |
| Line 268: header_errors list appended to errors dict when non-empty.           | <code>"Header" in errors;</code><br><code>len(errors["Header"]) &gt;= 2</code>                                                      | Pass   | 0.000s |
| Line 262: Missing CatchmentID produces header error.                           | <code>"Header" in errors; any("CatchmentID"</code><br><code>in e for e in errors["Header"]) (+2</code><br><code>more)</code>        | Pass   | 0.000s |
| Line 259: Missing MortgageID produces header error.                            | <code>"Header" in errors; any("MortgageID"</code><br><code>in e for e in errors["Header"])</code>                                   | Pass   | 0.000s |
| Line 275: Missing OriginalLoan produces FinancialTerms error.                  | <code>"FinancialTerms" in errors;</code><br><code>any("OriginalLoan" in e for e in</code><br><code>errors["FinancialTerms"])</code> | Pass   | 0.000s |
| Line 265: Missing PropertyID produces header error.                            | <code>"Header" in errors; any("PropertyID"</code><br><code>in e for e in errors["Header"])</code>                                   | Pass   | 0.000s |
| Line 278: terms_errors appended when non-empty.                                | <code>"FinancialTerms" in errors</code>                                                                                             | Pass   | 0.000s |
| Cost analysis subsection                                                       | <code>any("Cost Analysis" in t for t in</code><br><code>texts)</code>                                                               | Pass   | 0.000s |
| Net upfront $\geq$ 5% of loan $\rightarrow$ High assessment.                   | <code>isinstance(result, list)</code>                                                                                               | Pass   | 0.000s |
| Net upfront $\geq$ 0.5% $\rightarrow$ Low assessment.                          | <code>isinstance(result, list)</code>                                                                                               | Pass   | 0.000s |
| Net upfront 0.5–1.5% $\rightarrow$ Low-Medium assessment.                      | <code>isinstance(result, list)</code>                                                                                               | Pass   | 0.000s |
| Net upfront 1.5–3% $\rightarrow$ Medium assessment.                            | <code>isinstance(result, list)</code>                                                                                               | Pass   | 0.000s |
| Net upfront 3–5% $\rightarrow$ Medium-High assessment.                         | <code>isinstance(result, list)</code>                                                                                               | Pass   | 0.000s |
| Discount points + original rate triggers estimated points/rate reduction rows. | <code>isinstance(result, list)</code>                                                                                               | Pass   | 0.000s |
| Empty mortgage dict does not crash                                             | <code>isinstance(result, list)</code>                                                                                               | Pass   | 0.000s |

| Test Description                                                      | Acceptance Criteria                                                   | Result | Time   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------|--------|--------|
| Has tables                                                            | <code>any(isinstance(e, Table) for e in result)</code>                | Pass   | 0.000s |
| Lender credits reduces net cost                                       | <code>isinstance(result, list)</code>                                 | Pass   | 0.000s |
| No financial terms returns message                                    | <code>any("No cost information available" in t for t in texts)</code> | Pass   | 0.000s |
| No mortgage data returns message                                      | <code>any("No mortgage data available" in t for t in texts)</code>    | Pass   | 0.000s |
| Origination setup subsection                                          | <code>any("Origination" in t for t in texts)</code>                   | Pass   | 0.000s |
| Penalties subsection                                                  | <code>any("Penalties" in t for t in texts)</code>                     | Pass   | 0.000s |
| Returns list with data                                                | <code>isinstance(result, list); len(result) &gt; 0</code>             | Pass   | 0.000s |
| Section header present                                                | <code>any("Mortgage Costs" in t for t in texts)</code>                | Pass   | 0.000s |
| TotalLoanCosts + OriginalLoan triggers percentage row.                | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Total costs overview subsection                                       | <code>any("Total Loan Costs" in t for t in texts)</code>              | Pass   | 0.000s |
| LoanToValueRatio shown when different from OriginalLTV.               | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Core loan terms subsection                                            | <code>any("Core Loan Terms" in t for t in texts)</code>               | Pass   | 0.001s |
| Dti high risk                                                         | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Dti low risk                                                          | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Dti medium high risk                                                  | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Dti medium risk                                                       | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Dti subsection                                                        | <code>any("Debt-to-Income" in t for t in texts)</code>                | Pass   | 0.000s |
| Has tables                                                            | <code>any(isinstance(e, Table) for e in result)</code>                | Pass   | 0.001s |
| Interest rates subsection                                             | <code>any("Interest Rates" in t for t in texts)</code>                | Pass   | 0.000s |
| Loan as % of purchase value row is computed when both fields present. | <code>isinstance(result, list)</code>                                 | Pass   | 0.000s |
| Ltv high risk                                                         | <code>isinstance(result, list)</code>                                 | Pass   | 0.000s |
| Ltv low medium risk                                                   | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Ltv low risk                                                          | <code>isinstance(result, list)</code>                                 | Pass   | 0.001s |
| Ltv medium risk                                                       | <code>isinstance(result, list)</code>                                 | Pass   | 0.000s |
| Ltv subsection                                                        | <code>any("Loan-to-Value" in t for t in texts)</code>                 | Pass   | 0.000s |
| Ltv very high risk                                                    | <code>isinstance(result, list)</code>                                 | Pass   | 0.000s |

| Test Description                                  | Acceptance Criteria                                                                           | Result | Time   |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------|--------|--------|
| No financial terms returns message                | <code>any("No detailed financial terms available" in t for t in ...)</code>                   | Pass   | 0.000s |
| No mortgage data returns message                  | <code>any("No mortgage data available" in t for t in texts)</code>                            | Pass   | 0.000s |
| Returns list with data                            | <code>isinstance(result, list); len(result) &gt; 0</code>                                     | Pass   | 0.001s |
| Section header present                            | <code>any("Detailed Mortgage Financial Terms" in t for t in texts)</code>                     | Pass   | 0.001s |
| Init no output dir uses config                    | <code>gen.output_dir == reports_dir; reports_dir.exists()</code>                              | Pass   | 0.001s |
| Generate filename non dict raises attribute error | <code>"unknown" in fname</code>                                                               | Pass   | 0.001s |
| Generate filename none mortgage data              | <code>"unknown" in fname; fname.endswith(".pdf")</code>                                       | Pass   | 0.001s |
| Page exception is caught                          | <code>isinstance(elements, list); len(elements) &gt; 0</code>                                 | Pass   | 0.002s |
| Unknown page in order is skipped                  | <code>isinstance(elements, list); len(elements) &gt; 0</code>                                 | Pass   | 0.002s |
| Generator does not use sequential format          | <code>'GAUGE-{i:}' not in src; 'GAUGE-{index}' not in src (+3 more)</code>                    | Pass   | 0.000s |
| Generator source uses mort prefix                 | <code>'MORT-' in src</code>                                                                   | Pass   | 0.052s |
| Generator source uses uuid                        | <code>'uuid' in src; 'uuid' in src (+1 more)</code>                                           | Pass   | 0.000s |
| Uuid4 produces protocol matching ids              | <code>GAUGE_ID_RE.match(gauge_id); PROP_ID_RE.match(prop_id) (+2 more)</code>                 | Pass   | 0.000s |
| Accepts valid                                     | <code>PROP_ID_RE.match('PROP-a3f7bc12'); PROP_ID_RE.match('PROP-00000000') (+6 more)</code>   | Pass   | 0.000s |
| Rejects sequential                                | <code>not PROP_ID_RE.match('PROP-001'); not PROP_ID_RE.match('PROP-1234') (+3 more)</code>    | Pass   | 0.000s |
| Test getting delinquent mortgages.                | <code>len(delinquent) == 1; delinquent[0]['MortgageID'] == "MORT-003"</code>                  | Pass   | 0.002s |
| Test exposure aggregation by lender.              | <code>"First National Bank" in exposure; "Regional Credit Union" in exposure (+2 more)</code> | Pass   | 0.001s |
| Test getting property IDs that have mortgages.    | <code>len(prop_ids) == 3; "PROP-001" in prop_ids (+2 more)</code>                             | Pass   | 0.001s |
| Test total loan exposure calculation.             | <code>total == 1640000</code>                                                                 | Pass   | 0.002s |
| List all returns summaries                        | <code>len(MortgageLoader(tmp_path).list_all()) == 2</code>                                    | Pass   | 0.001s |

| Test Description                                                                       | Acceptance Criteria                                                                  | Result | Time   |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| Load all returns list                                                                  | <code>len(MortgageLoader(tmp_path).load_all()) == 3</code>                           | Pass   | 0.001s |
| Test MortgageID extraction.                                                            | <code>mort_id is not None;</code><br><code>mort_id.startswith("MORT-")</code>        | Pass   | 0.002s |
| Test mortgage summary generation.                                                      | <code>len(summaries) == 3; 'mortgageId' in summary (+3 more)</code>                  | Pass   | 0.001s |
| Test loading mortgage portfolio.                                                       | <code>len(mortgages) == 3</code>                                                     | Pass   | 0.002s |
| Find by id                                                                             | <code>MortgageLoader(tmp_path).find_by_id("MORT-001") is not None</code>             | Pass   | 0.001s |
| Mortgage without top-level PropertyID falls back to nested Mortgage.Header.PropertyID. | <code>MortgageLoader(tmp_path).find_by_property_id("PROP-NESTED") is not None</code> | Pass   | 0.001s |
| Missing returns none                                                                   | <code>MortgageLoader(tmp_path).find_by_id("MORT-999") is None</code>                 | Pass   | 0.001s |
| Test finding mortgage by ID.                                                           | <code>mort is not None; mort['PropertyID'] == "PROP-001" (+1 more)</code>            | Pass   | 0.002s |
| Test finding mortgages by lender.                                                      | <code>len(morts) == 2</code>                                                         | Pass   | 0.001s |
| Test lender search is case insensitive.                                                | <code>len(morts) == 2</code>                                                         | Pass   | 0.002s |
| Test finding mortgage by property ID.                                                  | <code>mort is not None; mort['MortgageID'] == "MORT-001"</code>                      | Pass   | 0.002s |
| Test finding mortgage for property without mortgage.                                   | <code>mort is None</code>                                                            | Pass   | 0.001s |
| Test finding active mortgages.                                                         | <code>len(morts) == 2</code>                                                         | Pass   | 0.002s |
| Test finding delinquent mortgages.                                                     | <code>len(morts) == 1;</code><br><code>morts[0]['MortgageID'] == "MORT-003"</code>   | Pass   | 0.002s |
| Application subsection                                                                 | <code>any("Application Information" in t for t in texts)</code>                      | Pass   | 0.001s |
| Denial reason hidden when default                                                      | <code>isinstance(result, list)</code>                                                | Pass   | 0.001s |
| Denial reason shown when not default                                                   | <code>isinstance(result, list)</code>                                                | Pass   | 0.001s |
| Empty mortgage dict does not crash                                                     | <code>isinstance(result, list)</code>                                                | Pass   | 0.000s |
| mortgage_data without outer 'Mortgage' key is handled via fallback.                    | <code>isinstance(result, list)</code>                                                | Pass   | 0.000s |
| Has tables                                                                             | <code>any(isinstance(e, Table) for e in result)</code>                               | Pass   | 0.001s |
| Key financial terms subsection                                                         | <code>any("Key Financial Terms" in t for t in texts)</code>                          | Pass   | 0.001s |
| LTV row is computed when both PurchaseValue and OriginalLoan are present.              | <code>isinstance(result, list)</code>                                                | Pass   | 0.001s |

| Test Description                                       | Acceptance Criteria                                                    | Result | Time   |
|--------------------------------------------------------|------------------------------------------------------------------------|--------|--------|
| Mortgage features subsection                           | <code>any("Mortgage Features" in t for t in texts)</code>              | Pass   | 0.001s |
| Mortgage identification subsection                     | <code>any("Mortgage Identification" in t for t in texts)</code>        | Pass   | 0.001s |
| No mortgage data returns message                       | <code>any("No mortgage data available" in t for t in texts)</code>     | Pass   | 0.000s |
| None mortgage data returns message                     | <code>any("No mortgage data available" in t for t in texts)</code>     | Pass   | 0.000s |
| Returns list with data                                 | <code>isinstance(result, list); len(result) &gt; 0</code>              | Pass   | 0.001s |
| Section header present                                 | <code>any("Mortgage Overview" in t for t in texts)</code>              | Pass   | 0.001s |
| Test that mortgage PropertyID links to valid property. | <code>prop is not None</code>                                          | Pass   | 0.003s |
| Test finding mortgage for each property.               | <code>prop is not None; mort is not None</code>                        | Pass   | 0.002s |
| Generate elements returns list                         | <code>isinstance(elements, list); len(elements) &gt; 0</code>          | Pass   | 0.003s |
| Generate filename fallback                             | <code>fname.endswith(".pdf"); "unknown" in fname</code>                | Pass   | 0.001s |
| Generate filename with id                              | <code>"MORT-test0001" in fname; fname.endswith(".pdf")</code>          | Pass   | 0.001s |
| Generate report creates pdf                            | <code>output.exists(); output.suffix == ".pdf"</code>                  | Pass   | 0.012s |
| Generate report returns path                           | <code>isinstance(result, Path)</code>                                  | Pass   | 0.012s |
| Init creates output dir                                | <code>output_dir.exists()</code>                                       | Pass   | 0.001s |
| Page order set                                         | <code>"title" in gen.page_order; gen.page_order[0] == "title"</code>   | Pass   | 0.001s |
| Pages initialised                                      | <code>"title" in gen.pages; "mortgage_overview" in gen.pages</code>    | Pass   | 0.001s |
| Elements non empty                                     | <code>len(elements) &gt; 0</code>                                      | Pass   | 0.001s |
| Empty property data does not crash                     | <code>isinstance(elements, list)</code>                                | Pass   | 0.000s |
| Extracts mortgage id                                   | <code>"MORT-test0001" in full_text</code>                              | Pass   | 0.000s |
| No mortgage data returns elements                      | <code>isinstance(elements, list); len(elements) &gt; 0</code>          | Pass   | 0.000s |
| Returns list                                           | <code>isinstance(elements, list)</code>                                | Pass   | 0.430s |
| Table elements included                                | <code>len(tables) &gt;= 1</code>                                       | Pass   | 0.000s |
| Line 108: <code>_save_meetings</code> fails → 500.     | <code>r.status_code == 500; "Failed" in r.get_json()["message"]</code> | Pass   | 0.001s |
| Line 195: document file not found → 404.               | <code>r.status_code == 404</code>                                      | Pass   | 0.002s |



| Test Description                                                      | Acceptance Criteria                                                                       | Result | Time   |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Line 196: serve an uploaded document.                                 | <code>r.status_code == 200</code>                                                         | Pass   | 0.004s |
| Line 162: file with empty filename → 400.                             | <code>r.status_code == 400</code>                                                         | Pass   | 0.003s |
| Line 155: meeting not found → 404.                                    | <code>r.status_code == 404</code>                                                         | Pass   | 0.002s |
| Line 158: no file in request → 400.                                   | <code>r.status_code == 400</code>                                                         | Pass   | 0.003s |
| Line 184: save failure after upload → 500.                            | <code>r.status_code == 500</code>                                                         | Pass   | 0.004s |
| Lines 164-186: upload a file to a meeting.                            | <code>r.status_code == 200; doc["filename"] == "report.pdf"</code>                        | Pass   | 0.003s |
| Line 179: meeting without 'documents' key gets it created.            | <code>r.status_code == 200</code>                                                         | Pass   | 0.005s |
| Line 144: <code>_save_meetings</code> fails on update → 500.          | <code>r.status_code == 500</code>                                                         | Pass   | 0.003s |
| Count gauges exceeding warning level.                                 | <code>count &gt;= 0; count &lt;= len(sample_gauges_df)</code>                             | Pass   | 0.003s |
| Find gauge with highest current level.                                | <code>highest is not None; 'gauge_id' in highest (+1 more)</code>                         | Pass   | 0.024s |
| First storm gets greek name                                           | <code>name; name[0].isupper()</code>                                                      | Pass   | 0.002s |
| Name from storms json takes priority                                  | <code>data["storms"][0]["name"] == "CustomStormName"</code>                               | Pass   | 0.003s |
| When segment is a point (seg2 j 1e-18), returns start point distance. | <code>abs(nx - 51.5) &lt; 1e-9; abs(ny + 0.1) &lt; 1e-9 (+1 more)</code>                  | Pass   | 0.000s |
| Distance is non negative                                              | <code>d &gt;= 0</code>                                                                    | Pass   | 0.000s |
| Midpoint projects to midpoint                                         | <code>abs(nx - 51.5) &lt; 0.01; abs(ny - 0.0) &lt; 0.01 (+1 more)</code>                  | Pass   | 0.000s |
| Point before start clamps to start                                    | <code>abs(nx - 51.0) &lt; 0.01; abs(ny - 0.0) &lt; 0.01</code>                            | Pass   | 0.000s |
| Point past end clamps to end                                          | <code>abs(nx - 52.0) &lt; 0.01; abs(ny - 0.0) &lt; 0.01</code>                            | Pass   | 0.000s |
| Damage amount computed correctly                                      | <code>prop["damage_amount"] == pytest.approx(expected)</code>                             | Pass   | 0.009s |
| Negative equity flag set on property                                  | <code>data["properties"][0]["negative_equity"] is True</code>                             | Pass   | 0.003s |
| LTV j 100% when outstanding j post-damage value.                      | <code>prop["post_damage_ltv"] &gt; 100.0</code>                                           | Pass   | 0.003s |
| Severe storm triggers negative equity                                 | <code>r.status_code == 200; data["portfolio"]["mortgages_in_negative_equity"] == 1</code> | Pass   | 0.009s |
| Same for mortgage-report: null JSON body → line 168 → 400.            | <code>r.status_code in (400, 500)</code>                                                  | Pass   | 0.001s |

| Test Description                                                                           | Acceptance Criteria                                                               | Result | Time   |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| Sending content-type application/json with null body → get_json() = None → 400.            | <code>r.status_code in (400, 500)</code>                                          | Pass   | 0.001s |
| All modules no newline in js strings                                                       | <code>failures == {}</code>                                                       | Pass   | 0.004s |
| Audit reports no newline in js strings                                                     | <code>offenders == []</code>                                                      | Pass   | 0.001s |
| Full compiled js no newline in js strings                                                  | <code>offenders == []</code>                                                      | Pass   | 0.004s |
| No mortgage message                                                                        | <code>any("No mortgage data" in t for t in texts)</code>                          | Pass   | 0.000s |
| No mortgage returns list                                                                   | <code>isinstance(result, list); len(result) &gt; 0</code>                         | Pass   | 0.000s |
| Gauge that exists in storm data but has no trades returns 404.                             | <code>resp.status_code == 404; 'No open trades' in data.get('message', '')</code> | Pass   | 0.049s |
| Empty regulatory returns message                                                           | <code>any("No regulatory information" in t for t in texts)</code>                 | Pass   | 0.000s |
| Returns list                                                                               | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| gauge.json present but every record missing gauge.id → n_all == 0.                         | <code>isinstance(result, dict); result["num_gauges"] == 0 (+1 more)</code>        | Pass   | 0.055s |
| Sequences still written when all gauges invalid                                            | <code>(tmp_path / SEQUENCES.FILENAME).exists()</code>                             | Pass   | 0.026s |
| Line 95: <code>_normal_pdf(0)</code> should return $1/\sqrt{2\pi}$ .                       | <code>result == pytest.approx(expected, rel=1e-10)</code>                         | Pass   | 0.000s |
| Line 95: PDF decreases as $-x$ increases.                                                  | <code>d._normal_pdf(0) &gt; d._normal_pdf(1) &gt; d._normal_pdf(2)</code>         | Pass   | 0.000s |
| Line 95: <code>_normal_pdf</code> at positive value.                                       | <code>result == pytest.approx(expected, rel=1e-10)</code>                         | Pass   | 0.000s |
| Line 95: <code>_normal_pdf</code> is symmetric.                                            | <code>d._normal_pdf(2.0) == pytest.approx(d._normal_pdf(-2.0))</code>             | Pass   | 0.000s |
| Normalize float returns empty                                                              | <code>loader._normalize_to_list(3.14) == []</code>                                | Pass   | 0.000s |
| Normalize integer returns empty                                                            | <code>result == []</code>                                                         | Pass   | 0.001s |
| Normalize string returns empty                                                             | <code>result == []</code>                                                         | Pass   | 0.000s |
| Normalize true returns empty                                                               | <code>loader._normalize_to_list(True) == []</code>                                | Pass   | 0.000s |
| <code>-stressm</code> alone must always run stressm regardless of <code>-nostress</code> . | —                                                                                 | Pass   | 0.005s |
| <code>-nostress</code> must be a registered argument on the port subparser.                | <code>args.nostress is True</code>                                                | Pass   | 0.009s |
| With <code>-all -nostress</code> , generate_stressm must not be called.                    | —                                                                                 | Pass   | 0.044s |
| Number type returns float                                                                  | <code>isinstance(result, float); 0 &lt;= result &lt;= 1000</code>                 | Pass   | 0.000s |

| Test Description                                                                 | Acceptance Criteria                                          | Result | Time   |
|----------------------------------------------------------------------------------|--------------------------------------------------------------|--------|--------|
| CalledProcessError → returns False.                                              | result is False                                              | Pass   | 0.001s |
| macOS: subprocess.run(['open', ...]) is called.                                  | result is True;<br>mock_run.call_args[0][0][0] == "open"     | Pass   | 0.002s |
| Darwin success                                                                   | result is True;<br>mock_run.call_args[0][0][0] == "open"     | Pass   | 0.000s |
| Darwin success                                                                   | result is True                                               | Pass   | 0.000s |
| Generic exception returns false                                                  | result is False                                              | Pass   | 0.000s |
| Generic exception returns false                                                  | result is False                                              | Pass   | 0.000s |
| Any other exception → returns False.                                             | result is False                                              | Pass   | 0.000s |
| Linux: subprocess.run(['xdg-open', ...]) is called.                              | result is True;<br>mock_run.call_args[0][0][0] == "xdg-open" | Pass   | 0.000s |
| Linux success                                                                    | result is True; "xdg-open" in args                           | Pass   | 0.000s |
| Linux success                                                                    | result is True                                               | Pass   | 0.000s |
| Subprocess called process error returns false                                    | result is False                                              | Pass   | 0.000s |
| Subprocess failure returns false                                                 | result is False                                              | Pass   | 0.000s |
| Unknown platform uses webbrowser                                                 | result is True                                               | Pass   | 0.000s |
| Unknown platform uses webbrowser                                                 | result is True                                               | Pass   | 0.001s |
| Unknown platform: falls back to webbrowser.open().                               | result is True                                               | Pass   | 0.000s |
| Windows: subprocess.run(['start', ...]) is called.                               | result is True;<br>mock_run.call_args[0][0][0] == "start"    | Pass   | 0.001s |
| Windows case insensitive                                                         | isinstance(result, bool)                                     | Pass   | 0.000s |
| Windows success                                                                  | result is True                                               | Pass   | 0.000s |
| Windows success                                                                  | result is True                                               | Pass   | 0.000s |
| Lines 432-433: -gauge-id → _find_gauge_by_id called.                             | —                                                            | Pass   | 0.054s |
| Lines 484-486: -mortgage-file → mortgage JSON is opened and parsed.              | —                                                            | Pass   | 0.007s |
| Lines 489-490: -property-id with no mortgage file → _find_property_by_id called. | —                                                            | Pass   | 0.006s |

| Test Description                                                                                     | Acceptance Criteria                                                                                             | Result | Time   |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------|--------|
| Lines 491-492: both <code>-property-id</code> and <code>-mortgage-file</code> → both finders called. | —                                                                                                               | Pass   | 0.006s |
| Lines 427-429: <code>-timeseries-file</code> → <code>timeseries</code> JSON opened and parsed.       | —                                                                                                               | Pass   | 0.008s |
| Very high flood risk + arrears + very high LTV + poor credit → CRITICAL.                             | <code>isinstance(result, list)</code>                                                                           | Pass   | 0.001s |
| Very low flood risk + good payment + low LTV + excellent credit → LOW.                               | <code>isinstance(result, list)</code>                                                                           | Pass   | 0.001s |
| Property only high flood risk                                                                        | <code>isinstance(result, list)</code>                                                                           | Pass   | 0.000s |
| Property only very low flood risk                                                                    | <code>isinstance(result, list)</code>                                                                           | Pass   | 0.000s |
| Effective rating is valid                                                                            | <code>rating in valid_ratings</code>                                                                            | Pass   | 0.000s |
| Overall risk rating has component scores                                                             | <code>isinstance(cs, dict)</code>                                                                               | Pass   | 0.000s |
| Overall risk rating has required keys                                                                | <code>key in orr</code>                                                                                         | Pass   | 0.000s |
| Clear override with null                                                                             | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>                          | Pass   | 0.006s |
| Invalid rating returns 400                                                                           | <code>r.status_code == 400;</code><br><code>r.get_json()["status"] == "error"</code>                            | Pass   | 0.002s |
| Model not found returns 404                                                                          | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+3 more)               | Pass   | 0.002s |
| No reason field returns 400                                                                          | <code>r.status_code == 400</code>                                                                               | Pass   | 0.002s |
| No reason returns 400                                                                                | <code>r.status_code == 400</code>                                                                               | Pass   | 0.002s |
| Override stored in model                                                                             | <code>rr["mrc_override"] == "Conditional";</code><br><code>rr["effective_rating"] == "Conditional"</code>       | Pass   | 0.004s |
| Response has audit entry                                                                             | <code>"audit_entry" in data;</code><br><code>data["audit_entry"]["event_type"] == "risk_rating_override"</code> | Pass   | 0.003s |
| Valid override returns success                                                                       | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>                          | Pass   | 0.003s |
| Line 301: audit overflow during override.                                                            | <code>r.status_code == 200</code>                                                                               | Pass   | 0.034s |
| Line 248: inventory not found → 404.                                                                 | <code>r.status_code == 404</code>                                                                               | Pass   | 0.002s |
| Line 282: save failure → 500.                                                                        | <code>r.status_code == 500</code>                                                                               | Pass   | 0.002s |
| Custom gauge basket size                                                                             | <code>cdm.gauge.basket_size == 10</code>                                                                        | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                                                                               | Result | Time   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Default gauge basket size                                     | <code>cdm.gauge.basket.size == 20</code>                                                                          | Pass   | 0.000s |
| Lines 333-334: exception in create_mapping raises ValueError. | —                                                                                                                 | Pass   | 0.000s |
| Verify all sections of create_mapping produce correct keys.   | <code>mapping["swap_id"] == "PRS-1";</code><br><code>mapping["notional"] == 1000000</code> (+3 more)              | Pass   | 0.000s |
| <code>validate({})</code> returns errors.                     | <code>isinstance(errors, dict)</code>                                                                             | Pass   | 0.000s |
| Lines 255-256: Empty GaugeBasket.                             | <code>"GaugeSet" in errors;</code><br><code>any("GaugeBasket" in e for e in errors["GaugeSet"])</code>            | Pass   | 0.000s |
| Lines 263-264: exception returns validation_error.            | <code>"validation_error" in result;</code><br><code>isinstance(result["validation_error"], list)</code> (+1 more) | Pass   | 0.000s |
| Lines 258-259: gauge errors collected.                        | <code>"GaugeSet" in errors</code>                                                                                 | Pass   | 0.000s |
| Lines 244-245: header errors collected.                       | <code>"Header" in errors;</code><br><code>len(errors["Header"]) &gt;= 2</code>                                    | Pass   | 0.000s |
| Line 241-242: Missing CatchmentID.                            | <code>"Header" in errors; any("CatchmentID" in e for e in errors["Header"])</code> (+2 more)                      | Pass   | 0.000s |
| Line 251-252: Missing GaugeSetID.                             | <code>"GaugeSet" in errors;</code><br><code>any("GaugeSetID" in e for e in errors["GaugeSet"])</code>             | Pass   | 0.000s |
| Line 238-239: Missing SwapID.                                 | <code>"Header" in errors; any("SwapID" in e for e in errors["Header"])</code>                                     | Pass   | 0.000s |
| Line 261: valid data returns empty dict.                      | <code>prs_cdm.validate(data) == {}</code>                                                                         | Pass   | 0.000s |
| buildPRSControls function exists.                             | <code>'buildPRSControls' in prs_js</code>                                                                         | Pass   | 0.000s |
| Counterparty selector present.                                | <code>'prs-counterparty' in prs_js</code>                                                                         | Pass   | 0.000s |
| Payer/Receiver direction selector present.                    | <code>'payer' in prs_js.lower(); 'receiver' in prs_js.lower()</code> (+1 more)                                    | Pass   | 0.000s |
| Maturity/tenor selector present.                              | <code>'prs-maturity' in prs_js</code>                                                                             | Pass   | 0.000s |
| Notional input field present.                                 | <code>'prs-notional' in prs_js</code>                                                                             | Pass   | 0.000s |
| Trigger level selector present.                               | <code>'prs-trigger' in prs_js</code>                                                                              | Pass   | 0.000s |
| <code>ghc_prs.get_js()</code> returns valid JS.               | <code>len(prs_js) &gt; 1000</code>                                                                                | Pass   | 0.000s |
| Fetches yield curve from /api/v1/trading/yield-curve.         | <code>'/api/v1/trading/yield-curve' in prs_js</code>                                                              | Pass   | 0.000s |
| Each trade has physical swap root                             | <code>not bad</code>                                                                                              | Pass   | 0.109s |
| Each trade has swap id                                        | <code>not bad</code>                                                                                              | Pass   | 0.004s |
| Prs directory exists                                          | <code>PRS_DIR.exists()</code>                                                                                     | Pass   | 0.000s |
| Swap id matches filename                                      | <code>not bad</code>                                                                                              | Pass   | 0.004s |

| Test Description                                                                             | Acceptance Criteria                                                                                     | Result | Time   |
|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------|--------|
| Thames Central book has 16 trades.                                                           | <code>len(prs_files) &gt;= 16</code>                                                                    | Pass   | 0.000s |
| Trade statuses are valid                                                                     | <code>not bad</code>                                                                                    | Pass   | 0.004s |
| Trades are present                                                                           | <code>len(prs_files) &gt;= 1</code>                                                                     | Pass   | 0.000s |
| Trades have gauge set                                                                        | <code>not bad</code>                                                                                    | Pass   | 0.004s |
| Book thames uses prs prefix                                                                  | <code>'PRS-' in src</code>                                                                              | Pass   | 0.042s |
| Book thames uses uuid                                                                        | <code>'uuid' in src</code>                                                                              | Pass   | 0.000s |
| <code>uuid4().hex[:8].upper()</code> — the book generator pattern — always matches.          | <code>PRS_ID_RE.match(swap_id)</code>                                                                   | Pass   | 0.001s |
| Some paths may produce lowercase — also accepted.                                            | <code>PRS_ID_RE.match('PRS-a8b3c4d5');</code><br><code>PRS_ID_RE.match('PRS-ffffffff')</code>           | Pass   | 0.000s |
| <code>book.py</code> uses <code>uuid4().hex[:8].upper()</code> → uppercase hex.              | <code>PRS_ID_RE.match('PRS-A8B3C4D5');</code><br><code>PRS_ID_RE.match('PRS-FFFFFFFF') (+1 more)</code> | Pass   | 0.000s |
| PRS-TEST-001 style IDs used in test fixtures are NOT production format.                      | <code>not PRS_ID_RE.match('PRS-TEST-001');</code><br><code>not PRS_ID_RE.match('PRS-VAUXHALL')</code>   | Pass   | 0.000s |
| Port src all does not include book symbols                                                   | <code>not overlap</code>                                                                                | Pass   | 0.000s |
| The package <code>__init__.py</code> must not contain 'from .book import'.                   | <code>"from .book import" not in src</code>                                                             | Pass   | 0.003s |
| Package <code>__init__.py</code> must not expose <code>generate_thames_central_book</code> . | <code>not hasattr(pkg,</code><br><code>"generate_thames_central_book")</code>                           | Pass   | 0.000s |
| Contains claim ref paragraph                                                                 | <code>len(tables) &gt; 0</code>                                                                         | Pass   | 0.001s |
| No flood events still renders                                                                | <code>len(result) &gt; 0</code>                                                                         | Pass   | 0.001s |
| Coordinates that are not floats should not raise.                                            | <code>isinstance(result, list)</code>                                                                   | Pass   | 0.001s |
| Returns list                                                                                 | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0 (+4 more)</code>                     | Pass   | 0.001s |
| Stats reflect events                                                                         | <code>any(isinstance(e, Paragraph) for e</code><br><code>in result)</code>                              | Pass   | 0.001s |
| Without mortgage                                                                             | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0</code>                               | Pass   | 0.003s |
| Events with no <code>sequence_id</code> (None) should render without error.                  | <code>isinstance(result, list)</code>                                                                   | Pass   | 0.001s |
| No flood events                                                                              | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0 (+2 more)</code>                     | Pass   | 0.000s |
| Returns list                                                                                 | <code>isinstance(result, list); len(result)</code><br><code>&gt; 0 (+4 more)</code>                     | Pass   | 0.001s |
| Stats bar present                                                                            | <code>len(paragraphs) &gt; 0</code>                                                                     | Pass   | 0.001s |



| Test Description                                                    | Acceptance Criteria                                                                     | Result | Time   |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|--------|
| Table included                                                      | <code>any(isinstance(e, Table) for e in result)</code>                                  | Pass   | 0.001s |
| Events with sequence_id not in lookup default to 'isolated'.        | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Contains tables                                                     | <code>len(tables) &gt;= 2</code>                                                        | Pass   | 0.002s |
| Events with damage ratio $\geq$ 5% trigger red background rows.     | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Multiple sequences grouped                                          | <code>len(tables) &gt;= 3</code>                                                        | Pass   | 0.002s |
| No flood events                                                     | <code>isinstance(result, list); len(result) &gt; 0 (+2 more)</code>                     | Pass   | 0.000s |
| Returns list                                                        | <code>isinstance(result, list); len(result) &gt; 0 (+4 more)</code>                     | Pass   | 0.002s |
| Zero property value should not raise a ZeroDivisionError.           | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Contains tables                                                     | <code>any(isinstance(e, Table) for e in result)</code>                                  | Pass   | 0.001s |
| High damage + large mortgage triggers negative equity highlighting. | <code>isinstance(result, list); any(isinstance(e, Table) for e in result)</code>        | Pass   | 0.001s |
| No flood events with mortgage                                       | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| With mortgage                                                       | <code>isinstance(result, list); len(result) &gt; 0</code>                               | Pass   | 0.001s |
| Without mortgage                                                    | <code>isinstance(result, list); any('No mortgage' in t for t in texts) (+2 more)</code> | Pass   | 0.000s |
| Contains reference table                                            | <code>len(tables) &gt;= 2</code>                                                        | Pass   | 0.001s |
| Assessed damage = prop_value * max damage_ratio per sequence.       | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| When damage exceeds property value, LTV is very high.               | <code>isinstance(result, list)</code>                                                   | Pass   | 0.001s |
| Returns list                                                        | <code>isinstance(result, list); len(result) &gt; 0</code>                               | Pass   | 0.001s |
| Signature block present                                             | <code>len(paragraphs) &gt; 3</code>                                                     | Pass   | 0.001s |
| Without mortgage                                                    | <code>isinstance(result, list); any('No mortgage' in t for t in texts) (+2 more)</code> | Pass   | 0.001s |
| sp-panel-title element ID must be referenced in the JS.             | <code>'sp-panel-title' in table_js</code>                                               | Pass   | 0.000s |
| Title must start with 'Portfolio Storm Impact'.                     | <code>'Portfolio Storm Impact' in table_js</code>                                       | Pass   | 0.000s |

| Test Description                                             | Acceptance Criteria                                                         | Result | Time   |
|--------------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------|
| Panel title must use options[selectedIndex].textContent.     | 'selectedIndex' in table.js;                                                | Pass   | 0.000s |
| Title assignment must NOT directly concatenate bare stormId. | 'textContent' in table.js                                                   | Pass   | 0.000s |
| Base level derived from alert                                | "Portfolio Storm Impact \{\}\u2014<br>' + stormId" not in table.js          | Pass   | 0.000s |
| Flood alert                                                  | r["base_level"] == pytest.approx(3.5 * 0.35)                                | Pass   | 0.000s |
| Flood warning                                                | _parse_gauge(_G1)["flood.alert"] ==<br>pytest.approx(3.5)                   | Pass   | 0.000s |
| Gauge id                                                     | _parse_gauge(_G1)["flood.warning"] ==<br>pytest.approx(4.6)                 | Pass   | 0.000s |
| Lat lon                                                      | _parse_gauge(_G1)["gauge_id"] ==<br>"GAUGE-test001"                         | Pass   | 0.000s |
| Missing alert returns none                                   | r["lat"] == pytest.approx(51.46);<br>r["lon"] == pytest.approx(-0.30)       | Pass   | 0.000s |
| Missing gauge id returns none                                | _parse_gauge(bad) is None                                                   | Pass   | 0.000s |
| Missing lat returns none                                     | _parse_gauge(bad) is None                                                   | Pass   | 0.000s |
| Severe warning                                               | _parse_gauge(bad) is None                                                   | Pass   | 0.000s |
| Valid record returns dict                                    | _parse_gauge(_G1)["severe.warning"]<br>== pytest.approx(5.5)                | Pass   | 0.000s |
| Warning defaults when absent                                 | result is not None                                                          | Pass   | 0.000s |
| Empty file                                                   | r is not None; r["flood.warning"] ==<br>pytest.approx(4.0 * 1.33) (+1 more) | Pass   | 0.000s |
| Flow values are floats                                       | metadata == {}; flows == []                                                 | Pass   | 0.000s |
| Flows have date and flow                                     | isinstance(f["flow.cumecs"], float)                                         | Pass   | 0.007s |
| Flows have date and flow cumecs                              | len(flows) == 5; "date" in f (+1 more)                                      | Pass   | 0.002s |
| Invalid rows skipped                                         | len(flows) > 0; "date" in flows[0]<br>(+1 more)                             | Pass   | 0.007s |
| Metadata contains station id                                 | len(flows) == 2                                                             | Pass   | 0.001s |
| Metadata contains station name                               | "station_id" in metadata;<br>metadata["station_id"] == "39001"              | Pass   | 0.002s |
| Metadata contains station name                               | metadata.get("station_name") == "My<br>Test Gauge"                          | Pass   | 0.007s |
| Metadata contains station name                               | metadata.get("station_name") == "Test<br>River"                             | Pass   | 0.003s |
| Metadata station id                                          | metadata.get("station_id") == "40007"                                       | Pass   | 0.008s |
| Returns metadata and flows                                   | isinstance(metadata, dict);<br>isinstance(flows, list)                      | Pass   | 0.007s |
| Returns tuple                                                | isinstance(metadata, dict);<br>isinstance(flows, list)                      | Pass   | 0.001s |
| Short row skipped                                            | len(flows) == 2                                                             | Pass   | 0.005s |

| Test Description                               | Acceptance Criteria                                                                                                       | Result | Time   |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Lines with fewer than 2 columns are skipped.   | <code>len(flows) == 1</code>                                                                                              | Pass   | 0.001s |
| All metadata categories parsed                 | <code>metadata.get("file_timestamp") == "2024-01-01";</code><br><code>metadata.get("database_id") == "7" (+4 more)</code> | Pass   | 0.001s |
| Flow row with non numeric value skipped        | <code>len(flows) == 2</code>                                                                                              | Pass   | 0.007s |
| Metadata row with no value uses empty string   | <code>metadata.get("station_name") == "";</code><br><code>metadata.get("station_id") == "39001" (+1 more)</code>          | Pass   | 0.001s |
| Metadata row with three columns stores value   | <code>metadata.get("dataType_units") == "m3/s"</code>                                                                     | Pass   | 0.002s |
| Add participant auto generates id              | <code>r.status_code == 200;</code><br><code>new_p["id"].startswith("P-")</code>                                           | Pass   | 0.001s |
| Add participant unknown meeting returns 404    | <code>r.status_code == 404</code>                                                                                         | Pass   | 0.001s |
| Delete participant                             | <code>r.status_code == 200;</code> <code>pid not in remaining</code>                                                      | Pass   | 0.002s |
| Delete participant unknown meeting returns 404 | <code>r.status_code == 404</code>                                                                                         | Pass   | 0.001s |
| Update missing participant returns 404         | <code>r.status_code == 404</code>                                                                                         | Pass   | 0.005s |
| Update participant                             | <code>r.status_code == 200;</code><br><code>updated["status"] == "Attended"</code>                                        | Pass   | 0.002s |
| Add participant auto id                        | <code>"P-001" in ids; "P-002" in ids</code>                                                                               | Pass   | 0.003s |
| Add participant missing meeting returns 404    | <code>r.status_code == 404</code>                                                                                         | Pass   | 0.002s |
| Add participant returns 200                    | <code>r.status_code == 200;</code><br><code>len(meeting["participants"]) == 2</code>                                      | Pass   | 0.002s |
| Delete participant missing meeting returns 404 | <code>r.status_code == 404</code>                                                                                         | Pass   | 0.002s |
| Delete participant returns 200                 | <code>r.status_code == 200;</code><br><code>len(r.get_json()["meeting"]["participants"]) == 0</code>                      | Pass   | 0.002s |
| Update participant not found returns 404       | <code>r.status_code == 404</code>                                                                                         | Pass   | 0.002s |
| Update participant returns 200                 | <code>r.status_code == 200;</code> <code>p["role"] == "CF0"</code>                                                        | Pass   | 0.003s |
| All holiday fields shown                       | <code>isinstance(result, list)</code>                                                                                     | Pass   | 0.001s |

| Test Description                                              | Acceptance Criteria                                                               | Result | Time   |
|---------------------------------------------------------------|-----------------------------------------------------------------------------------|--------|--------|
| Holidays taken shown                                          | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Last holiday date shown                                       | <code>isinstance(result, list)</code>                                             | Pass   | 0.001s |
| Without holiday fields, section should not appear.            | <code>not any("Holiday" in t for t in texts)</code>                               | Pass   | 0.000s |
| Total holidays shown                                          | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Arrears balance shown                                         | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Five missed payments poor                                     | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Four missed payments concerning                               | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Highest arrears shown                                         | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| In arrears flag true                                          | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| In arrears score 5                                            | <code>isinstance(result, list)</code>                                             | Pass   | 0.001s |
| Many missed payments over 3                                   | <code>isinstance(result, list)</code>                                             | Pass   | 0.001s |
| No missed payments good                                       | <code>isinstance(result, list)</code>                                             | Pass   | 0.001s |
| One missed payment good                                       | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Some missed payments 1 to 3                                   | <code>isinstance(result, list)</code>                                             | Pass   | 0.001s |
| Three missed payments concerning                              | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Two missed payments good                                      | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Zero missed payments excellent                                | <code>isinstance(result, list)</code>                                             | Pass   | 0.000s |
| Documentation pdf known model no file 404                     | <code>r.status_code == 404</code>                                                 | Pass   | 0.001s |
| Documentation pdf unknown model 404                           | <code>r.status_code == 404</code>                                                 | Pass   | 0.001s |
| Mrc tor pdf responds                                          | <code>r.status_code in (200, 404)</code>                                          | Pass   | 0.008s |
| Parameter inventory pdf responds                              | <code>r.status_code in (200, 404)</code>                                          | Pass   | 0.011s |
| Results pdf known model no file 404                           | <code>r.status_code == 404</code>                                                 | Pass   | 0.001s |
| Results pdf unknown model 404                                 | <code>r.status_code == 404</code>                                                 | Pass   | 0.001s |
| Line 323: valid model dir but PDF file missing → 404.         | <code>r.status_code == 404; "not found" in r.get_json()["message"].lower()</code> | Pass   | 0.001s |
| Line 325: model documentation PDF served when file exists.    | <code>r.status_code == 200</code>                                                 | Pass   | 0.002s |
| Line 357: MRC ToR PDF missing → 404.                          | <code>r.status_code == 404</code>                                                 | Pass   | 0.001s |
| Line 359: MRC ToR PDF served.                                 | <code>r.status_code == 200</code>                                                 | Pass   | 0.002s |
| Line 347: parameter inventory PDF missing → 404.              | <code>r.status_code == 404</code>                                                 | Pass   | 0.000s |
| Line 349: parameter inventory PDF served.                     | <code>r.status_code == 200</code>                                                 | Pass   | 0.002s |
| Line 337: valid model dir but test_results.pdf missing → 404. | <code>r.status_code == 404</code>                                                 | Pass   | 0.000s |
| Line 339: test results PDF served when file exists.           | <code>r.status_code == 200</code>                                                 | Pass   | 0.002s |

| Test Description                                                           | Acceptance Criteria                                                                      | Result | Time   |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| STORM-SEVERE-001 (peak 6.5m) should come before STORM-MODERATE-002 (4.8m). | <code>severe_idx &lt; moderate_idx</code>                                                | Pass   | 0.003s |
| GET /trading/stress/storms returns storms sorted by peak_level_m DESC.     | <code>len(storms) &gt;= 2;</code><br><code>sorted(peaks, reverse=True)</code>            | Pass   | 0.004s |
| storms[0] must have the highest peak_level_m.                              | <code>storms[0]['peak_level_m'] &gt;= storms[-1]['peak_level_m']</code>                  | Pass   | 0.003s |
| When seg_start == seg_end, norm ~ 0 -i degenerate fallback path.           | <code>abs(lon - (-0.30)) &lt; 1e-9</code>                                                | Pass   | 0.000s |
| Offset moves point                                                         | <code>dist &gt;= PropertyPortfolioGenerator.MIN_OFFSET_M * 0.5</code>                    | Pass   | 0.000s |
| Returns two floats                                                         | <code>isinstance(lat, float);</code><br><code>isinstance(lon, float)</code>              | Pass   | 0.000s |
| Persistent all events have readings                                        | <code>"flood_depth_m" in event;</code><br><code>"damage_ratio" in event (+2 more)</code> | Pass   | 0.003s |
| Persistent four events recorded                                            | <code>result["summary"]["floods_at_nearest_gauge"] == 4</code>                           | Pass   | 0.001s |
| Historical EOD snapshots must be generated (63 business days).             | <code>len(eod_files) &gt;= 50</code>                                                     | FAIL   | 0.001s |
| gaugets/ must have one file per gauge in gauge.json.                       | <code>n_files == n_gauges</code>                                                         | Pass   | 0.001s |
| gaugehc.json must have one curve per gauge in gauge.json.                  | <code>n_curves == n_gauges</code>                                                        | Pass   | 0.017s |
| propertyhc.json must have one curve per property in property.json.         | <code>n_curves == n_props</code>                                                         | Pass   | 0.034s |
| Every gauge referenced by PRS trades must exist in gauge.json.             | <code>not missing</code>                                                                 | Pass   | 0.059s |
| PRS trade files must exist in the blotter directory.                       | <code>len(trade_files) &gt;= 10</code>                                                   | Pass   | 0.001s |
| Line 48: fnt_gbp(0) returns em-dash.                                       | <code>isinstance(result, list)</code>                                                    | Pass   | 0.000s |
| Missing positions key returns drawing                                      | <code>isinstance(result, Drawing)</code>                                                 | Pass   | 0.000s |
| Negative gauge P&L -i bar drawn to the left.                               | <code>isinstance(result, Drawing)</code>                                                 | Pass   | 0.000s |
| No positions returns drawing                                               | <code>isinstance(result, Drawing)</code>                                                 | Pass   | 0.000s |
| Positions exist but all running_pnl = 0 -i gauge_pnl has values.           | <code>isinstance(result, Drawing)</code>                                                 | Pass   | 0.000s |

| Test Description                                                                                                                                                                  | Acceptance Criteria                                                                                                                                                      | Result | Time   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| JS must POST to <code>/api/v1/trading/stress/portfolio-run</code> .                                                                                                               | <code>'/api/v1/trading/stress/portfolio-run'</code> in <code>ps_js</code>                                                                                                | Pass   | 0.000s |
| JS must fetch from <code>/api/v1/trading/stress/portfolio-storms</code> .                                                                                                         | <code>'/api/v1/trading/stress/portfolio-storms'</code> in <code>ps_js</code>                                                                                             | Pass   | 0.000s |
| <code>portfolio-run</code> endpoint must use POST method.                                                                                                                         | <code>'POST'</code> in <code>ps_js</code>                                                                                                                                | Pass   | 0.000s |
| Full Detail button must navigate to Tab 7 via <code>switchTab('stress')</code> .                                                                                                  | <code>"switchTab('stress')"</code> in <code>ps_js</code>                                                                                                                 | Pass   | 0.000s |
| 'Full Detail' link button must be present — navigates to Tab 7 gauge view.                                                                                                        | <code>'Full Detail'</code> in <code>ps_js</code>                                                                                                                         | Pass   | 0.000s |
| <code>_psRenderGaugeTable</code> must be defined — renders per-gauge trade rows.                                                                                                  | <code>'_psRenderGaugeTable'</code> in <code>ps_js</code>                                                                                                                 | Pass   | 0.000s |
| <code>_psRenderGaugePnlTab</code> must be defined.                                                                                                                                | <code>'_psRenderGaugePnlTab'</code> in <code>ps_js</code>                                                                                                                | Pass   | 0.000s |
| <code>tdStressGaugeHint</code> must be set — passes selected gauge to Tab 7 pre-selection.                                                                                        | <code>'tdStressGaugeHint'</code> in <code>ps_js</code>                                                                                                                   | Pass   | 0.000s |
| <code>_stressStormHint</code> must be set — passes selected storm to Tab 7 auto-selection.                                                                                        | <code>'_stressStormHint'</code> in <code>ps_js</code>                                                                                                                    | Pass   | 0.000s |
| <code>ps-</code> IDs must not collide with Tab 7's <code>td-stress-gauge</code> ID.                                                                                               | <code>'td-stress-gauge'</code> not in <code>ps_js</code>                                                                                                                 | Pass   | 0.000s |
| Content area uses <code>ps-content</code> ID.                                                                                                                                     | <code>'ps-content'</code> in <code>ps_js</code>                                                                                                                          | Pass   | 0.000s |
| Storm selector uses <code>ps-</code> prefix (not <code>td-stress-*</code> ).                                                                                                      | <code>'ps-storm-sel'</code> in <code>ps_js</code>                                                                                                                        | Pass   | 0.000s |
| Sub-tab button IDs are built using <code>ps-tab-</code> prefix.                                                                                                                   | <code>"ps-tab-"</code> in <code>ps_js</code> or <code>"'ps-tab-' +"</code> in <code>ps_js</code> or <code>'ps-tab-...; "'pfflood'</code> in <code>ps_js</code> (+3 more) | Pass   | 0.000s |
| Rendered full JS must be of substantial length — sanity check that <code>loadPortStressData()</code> must be called in the <code>portstress</code> <code>switchTab</code> branch. | <code>len(full_js) &gt; 200000</code>                                                                                                                                    | Pass   | 0.000s |
|                                                                                                                                                                                   | <code>'loadPortStressData'</code> in <code>full_js</code>                                                                                                                | Pass   | 0.000s |
| <code>psResult</code> must survive f-string rendering — absent if brace escaping broke.                                                                                           | <code>'psResult'</code> in <code>full_js</code>                                                                                                                          | Pass   | 0.000s |



| Test Description                                                                | Acceptance Criteria                                 | Result | Time   |
|---------------------------------------------------------------------------------|-----------------------------------------------------|--------|--------|
| 'Port Stress' tab label must appear in the rendered trading desk.               | 'Port Stress' in full_js                            | Pass   | 0.000s |
| portStressView variable must appear — panel appends it to DOM.                  | 'portStressView' in full_js                         | Pass   | 0.000s |
| createPortStressView() call must appear in panel construction.                  | 'createPortStressView' in full_js                   | Pass   | 0.000s |
| switchTab must handle tab === 'portstress' branch.                              | "tab === 'portstress'" in full_js                   | Pass   | 0.000s |
| tabs array must include the port-stress entry.                                  | "id: 'portstress'" in full_js                       | Pass   | 0.000s |
| switchTab views array must include 'portstress'.                                | "'portstress'" in full_js                           | Pass   | 0.000s |
| hidePanel must call psCleanupCharts() to destroy chart instances.               | 'psCleanupCharts' in full_js                        | Pass   | 0.000s |
| psResult state variable must appear inside the IIFE, not after it.              | idx > 0; idx < iife_close                           | Pass   | 0.000s |
| Canvas element uses ps-pflood-canvas ID.                                        | 'ps-pflood-canvas' in ps_js                         | Pass   | 0.000s |
| _psRenderPFloodTab must be defined.                                             | '_psRenderPFloodTab' in ps_js                       | Pass   | 0.000s |
| Threshold bar colours must be present: severe red, warning orange, alert amber. | '#c62828' in ps_js; '#e65100' in ps_js<br>(+1 more) | Pass   | 0.000s |
| P(flood) tab creates a Chart.js instance.                                       | 'Chart(' in ps_js                                   | Pass   | 0.000s |
| Canvas element uses ps-portpnl-canvas ID.                                       | 'ps-portpnl-canvas' in ps_js                        | Pass   | 0.000s |
| Portfolio P&L tab uses the shared fmtGBP formatter.                             | 'fmtGBP' in ps_js                                   | Pass   | 0.000s |
| _psRenderPortPnlTab must be defined.                                            | '_psRenderPortPnlTab' in ps_js                      | Pass   | 0.000s |
| Horizontal bar chart needs indexAxis: 'y' in Chart.js options.                  | 'indexAxis' in ps_js                                | Pass   | 0.000s |
| Severity rows must navigate to Gauge P&L sub-tab via psSwitchSubTab.            | 'psSwitchSubTab' in ps_js                           | Pass   | 0.000s |
| All four severity group labels must be present in the JS output.                | 'SEVERE' in ps_js; 'WARNING' in ps_js<br>(+2 more)  | Pass   | 0.000s |
| _psRenderSeverityTab must be defined.                                           | '_psRenderSeverityTab' in ps_js                     | Pass   | 0.000s |

| Test Description                                                        | Acceptance Criteria                                                           | Result | Time    |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|---------|
| Severity row click must set psSelectedGaugeId before switching sub-tab. | 'psSelectedGaugeId' in ps_js                                                  | Pass   | 0.000s  |
| Severe group must use the #ffebee background colour.                    | ps_js.lower().find('#ffebee') >= 0                                            | Pass   | 0.000s  |
| All four sub-tab render functions must be present.                      | '_psRenderPFloodTab' in ps_js;<br>'_psRenderPortPnlTab' in ps_js<br>(+2 more) | Pass   | 0.000s  |
| 'psCleanupCharts' must be defined — destroys Chart.js instances.        | 'psCleanupCharts' in ps_js                                                    | Pass   | 0.000s  |
| 'createPortStressView' must be defined — builds the Tab 8 DOM.          | 'createPortStressView' in ps_js                                               | Pass   | 0.000s  |
| 'loadPortStressData' must be defined — fetches storm list on tab open.  | 'loadPortStressData' in ps_js                                                 | Pass   | 0.000s  |
| 'psStormChanged' must be defined — handles storm dropdown selection.    | 'psStormChanged' in ps_js                                                     | Pass   | 0.000s  |
| 'psSwitchSubTab' must be defined — controls the 4 sub-tabs.             | 'psSwitchSubTab' in ps_js                                                     | Pass   | 0.000s  |
| port_stress.get_js() returns non-empty string without errors.           | len(ps_js) > 0                                                                | Pass   | 0.000s  |
| Module-level state variables must be declared.                          | 'psResult' in ps_js; 'psActiveSubTab' in ps_js (+1 more)                      | Pass   | 0.000s  |
| Return value still has the correct structure after classifier training. | result["num_sequences"] == 20;<br>"type_counts" in result                     | Pass   | 18.555s |
| train_classifier=True without gauge_id covers lines 364-384.            | result["num_gauges"] == 3;<br>stressm_dir.exists() (+1 more)                  | Pass   | 19.965s |
| Known storm returns success                                             | r.status_code == 200;<br>r.get_json()["status"] == "success"                  | Pass   | 0.003s  |
| Mortgaged property has ltv fields                                       | len(mortgaged) >= 1;<br>"outstanding_balance" in p (+3 more)                  | Pass   | 0.003s  |
| No pts dir returns 404                                                  | r.status_code == 404;<br>r.get_json()["status"] == "error" (+2 more)          | Pass   | 0.001s  |
| Options returns ok                                                      | r.status_code == 200; r.status_code == 200                                    | Pass   | 0.001s  |
| Portfolio damage pct non negative                                       | pct >= 0                                                                      | Pass   | 0.004s  |
| Portfolio totals count                                                  | port["properties_affected"] ==<br>len(r.get_json()["propterti...              | Pass   | 0.004s  |

| Test Description                                                             | Acceptance Criteria                                                                                                          | Result | Time   |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Properties sorted by damage descending                                       | <code>amounts == sorted(amounts, reverse=True);</code><br><code>amounts == sorted(amounts, reverse=True)</code>              | Pass   | 0.003s |
| Property entry has financial fields                                          | <code>"property_id" in prop;</code><br><code>"property_value" in prop (+9 more)</code>                                       | Pass   | 0.003s |
| Response has portfolio                                                       | <code>"portfolio" in data;</code><br><code>"total_properties" in port (+4 more)</code>                                       | Pass   | 0.003s |
| Response has properties list                                                 | <code>"properties" in data;</code><br><code>len(data["properties"]) &gt;= 1 (+2 more)</code>                                 | Pass   | 0.003s |
| Response has storm id                                                        | <code>r.get_json()["storm_id"] == sid</code>                                                                                 | Pass   | 0.004s |
| Unknown storm returns 404                                                    | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code>                                         | Pass   | 0.008s |
| No pts dir returns 404                                                       | <code>r.status_code == 404; data["status"] == "error"</code>                                                                 | Pass   | 0.001s |
| Options returns 200                                                          | <code>r.status_code == 200</code>                                                                                            | Pass   | 0.001s |
| Unknown storm returns 404                                                    | <code>r.status_code == 404; data["status"] == "error"</code>                                                                 | Pass   | 0.024s |
| Mortgage enrichment present                                                  | <code>entry["has_mortgage"] is True;</code><br><code>entry["outstanding_balance"] == pytest.approx(280_000) (+2 more)</code> | Pass   | 0.003s |
| Property entry has damage fields                                             | <code>entry["property_id"] == "PROP-001";</code><br><code>entry["flood_depth_m"] == pytest.approx(0.5) (+2 more)</code>      | Pass   | 0.002s |
| Response has portfolio key                                                   | <code>"portfolio" in data; key in portfolio</code>                                                                           | Pass   | 0.003s |
| Response has properties list                                                 | <code>"properties" in data; len(props) &gt;= 1</code>                                                                        | Pass   | 0.003s |
| Storm id in response                                                         | <code>data["storm_id"] == "STORM-001"</code>                                                                                 | Pass   | 0.004s |
| Valid storm returns 200                                                      | <code>r.status_code == 200; data["status"] == "success"</code>                                                               | Pass   | 0.003s |
| API must not return name == storm_id (produces 'ID (ID)' in UI).             | <code>s['name']</code>                                                                                                       | Pass   | 0.002s |
| At least one storm must have a human-readable name (not a raw STORM-xxx id). | <code>named</code>                                                                                                           | Pass   | 0.002s |
| In gauge_results, severe gauges appear before warning gauges.                | <code>levels == sorted(levels)</code>                                                                                        | Pass   | 0.033s |
| Warning gauges appear before alert gauges in results.                        | <code>levels[i] &lt;= levels[i + 1]</code>                                                                                   | Pass   | 0.035s |
| Gauges with num_trades > 0 have non-empty trades list.                       | <code>len(g['trades']) &gt; 0</code>                                                                                         | Pass   | 0.034s |

| Test Description                                                                                                    | Acceptance Criteria                                                                                     | Result | Time   |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------|--------|
| Each trade has swap_id, trigger, notional, mtm, cash_price, stress_pnl, Exception in load_stress_storms causes 500. | <code>found_trade; 'swap_id' in t (+8 more)</code>                                                      | Pass   | 0.032s |
| When two storms have equal gauges_severe, the one with patch_get_engines to raise RuntimeError → 500.               | <code>resp.status_code == 500</code>                                                                    | Pass   | 0.003s |
| No stress_storms/ + cache cleared → portfolio-run returns 404.                                                      | <code>storms[0]['intensity_category'] == 'catastrophic'</code>                                          | Pass   | 0.000s |
| POST with no body returns 400 or 500.                                                                               | <code>resp.status_code == 500</code>                                                                    | Pass   | 0.003s |
| Payer (is_payer=True) has positive notional in trade details.                                                       | <code>resp.status_code == 404</code>                                                                    | Pass   | 0.001s |
| portfolio_stress_pnl == sum(g['stress_pnl'] for g in gauges).                                                       | <code>resp.status_code in (400, 500)</code>                                                             | Pass   | 0.001s |
| Receiver (is_payer=False) has negative notional in trade details.                                                   | <code>t['notional'] &gt; 0</code>                                                                       | Pass   | 0.024s |
| When peak_level_m is severe_level, p_flood should be 1.0.                                                           | <code>abs(data['portfolio_stress_pnl'] - expected_total) &lt; 0.01</code>                               | Pass   | 0.030s |
| stress_pnl = cash_price - mtm; cash_price = signed_notional * p_flood.                                              | <code>t['notional'] &lt; 0</code>                                                                       | Pass   | 0.024s |
| Gauge not in storm's gauge_responses has p_flood=0.0 and impacted=False.                                            | <code>gauge_001 is not None; gauge_001['p_flood'] == 1.0</code>                                         | Pass   | 0.033s |
| With 3 impacted gauges in STORM_PORT_SEVERE, predictor.predict is called                                            | <code>data['status'] == 'success'; abs(t['stress_pnl'] - (t['cash_price'] - t['mtm'])) &lt; 0.01</code> | Pass   | 0.028s |
| _stressm_predictor_cache is reused on second call.                                                                  | <code>lambeth['p_flood'] == 0.0; lambeth['impacted'] is False</code>                                    | Pass   | 0.023s |
| If stressm/ dir missing, falls back gracefully — no crash.                                                          | <code>pred.predict.call_count &gt;= 1</code>                                                            | Pass   | 0.024s |
| Missing sequences message mentions doublet                                                                          | <code>result is sentinel</code>                                                                         | Pass   | 0.003s |
| Missing sequences returns 404                                                                                       | <code>result is None or callable(getattr(result, 'predict', None))</code>                               | Pass   | 0.000s |
| No pts dir returns 404                                                                                              | <code>"storm_sequences" in msg or "not found" in msg</code>                                             | Pass   | 0.030s |
| No pts dir status error                                                                                             | <code>r.status_code == 404</code>                                                                       | Pass   | 0.027s |
|                                                                                                                     | <code>r.status_code == 404</code>                                                                       | Pass   | 0.001s |
|                                                                                                                     | <code>pts_client_no_data.get(URL).get_json()["status"] == "error"</code>                                | Pass   | 0.001s |

| Test Description                               | Acceptance Criteria                                                                                                                                    | Result | Time    |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| Options returns ok                             | <code>r.status_code == 200</code>                                                                                                                      | Pass   | 0.001s  |
| Missing mortgage json returns 200              | <code>client.get(URL).status_code == 200</code>                                                                                                        | Pass   | 0.152s  |
| Missing mortgage json zero portfolio mortgages | <code>client.get(URL).get_json()["total_portfolio_mortgages"] == ...</code>                                                                            | Pass   | 0.155s  |
| Missing property json returns 200              | <code>client.get(URL).status_code == 200</code>                                                                                                        | Pass   | 0.140s  |
| Missing property json status success           | <code>client.get(URL).get_json()["status"] == "success"</code>                                                                                         | Pass   | 0.394s  |
| Missing property json zero portfolio value     | <code>client.get(URL).get_json()["total_portfolio_value"] == 0.0</code>                                                                                | Pass   | 0.141s  |
| Mortgage impairment max is positive            | <code>mi["max"] &gt; 0</code>                                                                                                                          | Pass   | 0.116s  |
| Mortgage impairment value                      | <code>mi["max"] == pytest.approx(140000.0, rel=1e-3)</code>                                                                                            | Pass   | 0.116s  |
| Tail storm has mortgage impairment             | <code>len(tail) &gt;= 1;</code><br><code>tail[0]["mortgage_impairment"] &gt; 0</code>                                                                  | Pass   | 0.119s  |
| Prob loss pct is 50                            | <code>client_two_storms.get(URL).get_json()["prob_loss_pct"] == ...</code>                                                                             | Pass   | 0.106s  |
| Storm count is two                             | <code>client_two_storms.get(URL).get_json()["storm_count"] == 2</code>                                                                                 | Pass   | 0.106s  |
| Storms with damage is one                      | <code>pts_env["client"].get(URL).get_json()["storms_with_damage"]</code><br><code>client_two_storms.get(URL).get_json()["storms_with_damage...]</code> | Pass   | 0.110s; |
| Storms zero damage is one                      | <code>client_two_storms.get(URL).get_json()["storms_zero_damage"]</code>                                                                               | Pass   | 0.109s  |
| Tail storms contains the damaged storm         | <code>"STORM-0001" in storm_ids</code>                                                                                                                 | Pass   | 0.104s  |
| Tail storms limited to 50                      | <code>len(client_two_storms.get(URL).get_json()["tail_storms"])</code>                                                                                 | Pass   | 0.111s  |
| No mortgage impairment when above water        | <code>mi["max"] == 0.0</code>                                                                                                                          | Pass   | 0.126s  |
| Prob loss pct is 100                           | <code>pts_env["client"].get(URL).get_json()["prob_loss_pct"] == ...</code>                                                                             | Pass   | 0.142s  |
| Property damage cond mean positive             | <code>pts_env["client"].get(URL).get_json()["property_damage"]</code>                                                                                  | Pass   | 0.120s. |
| Property damage max is positive                | <code>pts_env["client"].get(URL).get_json()["property_damage"]</code>                                                                                  | Pass   | 0.111s. |
| Storm count is one                             | <code>pts_env["client"].get(URL).get_json()["storm_count"] == 1</code>                                                                                 | Pass   | 0.115s  |
| Storms with damage is one                      | <code>pts_env["client"].get(URL).get_json()["storms_with_damage"]</code><br><code>client_two_storms.get(URL).get_json()["storms_with_damage...]</code> | Pass   | 0.110s; |
| Storms zero plus with damage equals total      | <code>data["storms_with_damage"] +</code><br><code>data["storms_zero_damage"] == ...</code>                                                            | Pass   | 0.120s  |
| Total portfolio mortgages                      | <code>pts_env["client"].get(URL).get_json()["total_portfolio_mortgages"]</code>                                                                        | Pass   | 0.106s  |
| Total portfolio value                          | <code>pts_env["client"].get(URL).get_json()["total_portfolio_value"]</code>                                                                            | Pass   | 0.108s  |

| Test Description                                                                     | Acceptance Criteria                                                                   | Result | Time   |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Histogram bin has lo hi count                                                        | "lo" in bin0 and "hi" in bin0 and<br>"count" in bin0                                  | Pass   | 0.146s |
| Mort histogram has 50 bins                                                           | len(pts_env["client"].get(URL).get_json()["mort_histogram"]) == 50                    | Pass   | 0.113s |
| Mortgage impairment sub keys                                                         | key in mi                                                                             | Pass   | 0.115s |
| Prop histogram has 50 bins                                                           | len(pts_env["client"].get(URL).get_json()["prop_histogram"]) == 50                    | Pass   | 0.114s |
| Property damage sub keys                                                             | key in pd                                                                             | Pass   | 0.123s |
| Returns 200                                                                          | pts_env["client"].get(URL).status_code == 200                                         | Pass   | 0.136s |
| Status success                                                                       | pts_env["client"].get(URL).get_json()["status"] == "success"                          | Pass   | 0.121s |
| Tail storm has required fields                                                       | len(storms) >= 1; f in s                                                              | Pass   | 0.114s |
| Tail storms is list                                                                  | isinstance(pts_env["client"].get(URL).get_json()["tail_storms"], list)                | Pass   | 0.136s |
| Tail storms sorted descending                                                        | damages == sorted(damages,<br>reverse=True)                                           | Pass   | 0.115s |
| Top level keys present                                                               | key in data                                                                           | Pass   | 0.121s |
| Portfolio value from known property                                                  | client_unknown_prop.get(URL).get_json()["total_portfolio_value"] == 0                 | Pass   | 0.115s |
| Unknown prop id is skipped                                                           | data["storms_with_damage"] == 0                                                       | Pass   | 0.110s |
| Zero depth cond mean is zero                                                         | client_zero_depth.get(URL).get_json()["property_damage"] == 0                         | Pass   | 0.128s |
| Zero depth prob loss is zero                                                         | client_zero_depth.get(URL).get_json()["probability_loss_pct"] == ...                  | Pass   | 0.142s |
| Zero depth property damage max is zero                                               | client_zero_depth.get(URL).get_json()["property_damage"] == 0                         | Pass   | 0.130s |
| Zero depth storms with damage is zero                                                | client_zero_depth.get(URL).get_json()["storms_with_damage"] == 0                      | Pass   | 0.127s |
| If predictor.predict raises, p_flood defaults to 0.0.                                | resp.status_code == 200;<br>g001['p_flood'] == 0.0                                    | Pass   | 0.022s |
| When _get_predictor returns None, returns 404.                                       | resp.status_code == 404; 'models<br>not found' in data.get('message',<br>'').lower()  | Pass   | 0.025s |
| trading/preloader.py must set window._tdPreloadDone (not bare assignment).           | 'window._tdPreloadDone = true' in src                                                 | Pass   | 0.000s |
| startup.py must initialise via window._tdPreloadDone (not bare var).                 | 'window._tdPreloadDone = false' in<br>src; 'var _tdPreloadDone = false' not<br>in src | Pass   | 0.001s |
| startup.py must set window._tdPreloadDone = true after all fetches settle.           | 'window._tdPreloadDone = true' in src                                                 | Pass   | 0.000s |
| tradingdesk.py showPanel() must read window._tdPreloadDone, not bare _tdPreloadDone. | 'window._tdPreloadDone' in js                                                         | Pass   | 0.000s |



| Test Description                                                       | Acceptance Criteria                                                                      | Result | Time   |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| All three tabs must fetch from /trading/stress/storms.                 | '/trading/stress/storms' in js                                                           | Pass   | 0.000s |
| Gauge Stress tab auto-selects storms[0] (worst case).                  | 'storms[0]' in ghc_stress_js or<br>'storms.length > 0' in gh...                          | Pass   | 0.000s |
| Gauge Stress tab storm dropdown uses all required storm fields.        | field in ghc_stress_js                                                                   | Pass   | 0.000s |
| Historical tab storm list uses all required storm fields.              | field in hist_js                                                                         | Pass   | 0.000s |
| Trading Desk storm dropdown includes peak_level_m in the label.        | 'peak_level_m' in td_stress_js                                                           | Pass   | 0.000s |
| Trading Desk Stress tab also auto-selects worst case storm.            | 'withTrades' in td_stress_js or \{\}<br>'storms....                                      | Pass   | 0.000s |
| Trading Desk Stress tab storm dropdown uses all required storm fields. | field in td_stress_js                                                                    | Pass   | 0.000s |
| Alert trigger                                                          | result["trigger_level"] == "alert";<br>result["trigger_threshold_m"] == 4.0              | Pass   | 0.000s |
| Custom notional                                                        | result["notional"] == 5.000.000                                                          | Pass   | 0.000s |
| Flat hazard curve                                                      | "fair_spread_bps" in result;<br>result["use_term_structure"] is<br>False                 | Pass   | 0.000s |
| Gauge not in hazard returns 404                                        | r.status_code == 404                                                                     | Pass   | 0.002s |
| Invalid trigger level returns 400                                      | r.status_code == 400;<br>r.get_json()["status"] == "error"                               | Pass   | 0.001s |
| Missing gauge id returns 400                                           | r.status_code == 400                                                                     | Pass   | 0.001s |
| No hazard file returns 404                                             | r.status_code == 404; "not available"<br>in r.get_json()["message"].lower()<br>(+1 more) | Pass   | 0.001s |
| No json returns 400                                                    | r.status_code in (400, 500)                                                              | Pass   | 0.001s |
| Options returns ok                                                     | r.status_code == 200                                                                     | Pass   | 0.001s |
| Result has fair spread bps                                             | "fair_spread_bps" in result;<br>result["fair_spread_bps"] > 0                            | Pass   | 0.000s |
| Result has npv                                                         | "npv" in result;<br>isinstance(result["npv"], float)                                     | Pass   | 0.000s |
| Result has survival probabilities                                      | len(sp) == 3; "1yr" in sp                                                                | Pass   | 0.000s |
| Returns dict                                                           | isinstance(result, dict)                                                                 | Pass   | 0.016s |
| Severe trigger                                                         | result["trigger_level"] == "severe";<br>result["trigger_threshold_m"] == 5.0             | Pass   | 0.000s |
| Term structure true                                                    | result["use_term_structure"] is True                                                     | Pass   | 0.000s |
| Annual hazard rate matches input                                       | r["annual_hazard_rate"] ==<br>gauge["annual_hazard_rate.warn...                          | Pass   | 0.000s |
| Gauge id in result                                                     | r["gauge_id"] == gauge["gauge_id"]                                                       | Pass   | 0.000s |

| Test Description                                          | Acceptance Criteria                                                                                      | Result | Time   |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Gauge name in result                                      | <code>r["gauge_name"] == gauge["gauge_name"]</code>                                                      | Pass   | 0.000s |
| Severe annual hazard rate matches input                   | <code>r["annual_hazard_rate"] == gauge["annual_hazard_rate_seve...</code>                                | Pass   | 0.000s |
| hazard.py lines 174-176: exception during pricing -j 500. | <code>r.status_code == 500</code>                                                                        | Pass   | 0.032s |
| hazard.py lines 146-147: gauge not in hazard data -j 404. | <code>r.status_code == 404</code>                                                                        | Pass   | 0.030s |
| hazard.py lines 139-140: no gaugehc.json -j 404.          | <code>r.status_code == 404</code>                                                                        | Pass   | 0.030s |
| hazard.py line 123: no JSON body -j 400.                  | <code>r.status_code in (400, 500)</code>                                                                 | Pass   | 0.029s |
| hazard.py lines 121-123: JSON null body -j 400.           | <code>r.status_code == 400;</code><br><code>r.get_json()["status"] == "error"</code>                     | Pass   | 0.028s |
| All trigger levels work: alert, warning, severe.          | <code>r.status_code == 200</code>                                                                        | Pass   | 0.193s |
| hazard.py lines 133-172: successful PRS pricing.          | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                                 | Pass   | 0.291s |
| hazard.py: PRS pricing with default params.               | <code>r.status_code == 200</code>                                                                        | Pass   | 0.155s |
| Npv is sum of legs                                        | <code>r["npv"] == pytest.approx(r["protection_leg...</code>                                              | Pass   | 0.000s |
| Premium leg negative                                      | <code>r["premium_leg_npv"] &lt; 0</code>                                                                 | Pass   | 0.000s |
| Protection leg positive                                   | <code>r["protection_leg_npv"] &gt; 0</code>                                                              | Pass   | 0.000s |
| Nonzero recovery rate stored                              | <code>r["recovery_rate"] == pytest.approx(0.4)</code>                                                    | Pass   | 0.000s |
| Recovery rate affects fair spread                         | <code>r0["fair_spread_bps"] &gt; r40["fair_spread_bps"]</code>                                           | Pass   | 0.000s |
| Higher risk free rate changes npv                         | <code>r_low["npv"] != pytest.approx(r_high["npv"], rel=1e-4)</code>                                      | Pass   | 0.000s |
| Risk free rate not in output                              | <code>"risk_free_rate" not in r</code>                                                                   | Pass   | 0.000s |
| Fair spread is float                                      | <code>isinstance(r["fair_spread"], float)</code>                                                         | Pass   | 0.000s |
| Fair upfront is float                                     | <code>isinstance(r["fair_upfront"], float)</code>                                                        | Pass   | 0.000s |
| Npv is float                                              | <code>isinstance(r["npv"], float)</code>                                                                 | Pass   | 0.000s |
| Running spread bps identity                               | <code>r["running_spread_bps"] == pytest.approx(r["running_sprea...</code>                                | Pass   | 0.000s |
| Survival probs all in zero one                            | <code>0.0 &lt; val &lt;= 1.0</code>                                                                      | Pass   | 0.000s |
| Tenor 1yr gives one survival entry                        | <code>len(r["survival_probabilities"]) == 1;</code><br><code>"1yr" in r["survival_probabilities"]</code> | Pass   | 0.000s |

| Test Description                                                    | Acceptance Criteria                                         | Result | Time   |
|---------------------------------------------------------------------|-------------------------------------------------------------|--------|--------|
| Use term structure stored false                                     | r["use_term_structure"] is False                            | Pass   | 0.000s |
| Use term structure stored true                                      | r["use_term_structure"] is True                             | Pass   | 0.000s |
| Not trained no metrics printed                                      | "AUC" not in out                                            | Pass   | 0.000s |
| Not trained status                                                  | "insufficient_data" in out;<br>"GAUGE-bad" in out (+1 more) | Pass   | 0.000s |
| Trained shows feature importance                                    | "log_h_s" in out; "log_t_end" in out                        | Pass   | 0.000s |
| Trained shows label info                                            | "severe_warning" in out; "5.50" in out                      | Pass   | 0.000s |
| Trained status prints metrics                                       | "trained" in out; "0.9200" in out (+1 more)                 | Pass   | 0.000s |
| Percentile between alert and warning → '← ALERT' marker (line 480). | "ALERT" in out                                              | Pass   | 0.001s |
| All sequence types reported                                         | "isolated" in out; "doublet" in out<br>(+1 more)            | Pass   | 0.000s |
| Categories with zero peaks are skipped via 'continue' (line 493).   | "moderate" in out; "catastrophic" not in out                | Pass   | 0.000s |
| Output contains gauge id                                            | "GAUGE-test-pgp" in out                                     | Pass   | 0.000s |
| Percentile $i$ = severe → '← SEVERE' marker.                        | "SEVERE" in out                                             | Pass   | 0.000s |
| Percentile between warning and severe → '← WARNING' marker.         | "WARNING" in out                                            | Pass   | 0.000s |
| Summary with full data does not crash                               | —                                                           | Pass   | 0.004s |
| Summary with no data does not crash                                 | —                                                           | Pass   | 0.000s |
| Summary with partial data does not crash                            | —                                                           | Pass   | 0.000s |
| Logs gauge name                                                     | isinstance(result["gauge_name"], str)                       | Pass   | 0.000s |
| Logs without error                                                  | len(caplog.records) >= 0                                    | Pass   | 0.001s |
| Alert trigger in log                                                | "ALERT" in combined                                         | Pass   | 0.000s |
| All survival years in log                                           | key in combined                                             | Pass   | 0.000s |
| Contract terms header in log                                        | "Contract Terms" in combined                                | Pass   | 0.000s |
| Fair spread in log                                                  | "Fair Spread" in combined                                   | Pass   | 0.000s |
| Gauge id in log                                                     | result["gauge_id"] in combined                              | Pass   | 0.000s |
| Gauge name in log                                                   | result["gauge_name"] in combined                            | Pass   | 0.000s |
| Npv value in log                                                    | "NPV" in combined                                           | Pass   | 0.000s |
| Premium leg in log                                                  | "Premium Leg" in combined                                   | Pass   | 0.000s |
| Protection leg in log                                               | "Protection Leg" in combined                                | Pass   | 0.000s |
| Recovery rate in log                                                | "Recovery Rate" in combined                                 | Pass   | 0.000s |
| Severe trigger in log                                               | "SEVERE" in combined                                        | Pass   | 0.000s |
| Survival curve label in log                                         | "Survival Curve" in combined                                | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                                             | Result | Time   |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Trigger level in log                                             | "WARNING" in combined                                                                           | Pass   | 0.000s |
| Empty directory returns empty list                               | result == []                                                                                    | Pass   | 0.000s |
| Non matching files skipped                                       | result == []                                                                                    | Pass   | 0.001s |
| Output dir none uses input dir                                   | isinstance(result, list);<br>any("gaugehd.39999" in r for r in<br>result)                       | Pass   | 0.015s |
| Output files created                                             | all(Path(p).exists() for p in<br>result)                                                        | Pass   | 0.016s |
| Process directory error continues                                | isinstance(result, list)                                                                        | Pass   | 0.016s |
| Processes matching files                                         | len(result) == 2                                                                                | Pass   | 0.032s |
| Lines 143-144: bad CSV -i error<br>logged, continues.            | isinstance(result, list)                                                                        | Pass   | 0.004s |
| Lines 113-147: process_nrf_directory scans CSV<br>files.         | isinstance(result, list); len(result)<br>== 1                                                   | Pass   | 0.002s |
| Line 134: output_dir=None uses<br>config.get_gaugehd_dir().      | isinstance(result, list)                                                                        | Pass   | 0.004s |
| Lines 131-147: no matching files -i<br>empty list.               | result == []                                                                                    | Pass   | 0.000s |
| Line 61/80: custom glob pattern se-<br>lects different files.    | len(result) == 1                                                                                | Pass   | 0.009s |
| Line 87-88: error on one file doesn't<br>stop processing others. | len(result) >= 1                                                                                | Pass   | 0.003s |
| Multiple CSV files in directory are<br>all processed.            | len(result) == 2                                                                                | Pass   | 0.004s |
| Line 81: station_id extracted by<br>stripping _gdf from stem.    | len(result) == 1;<br>result[0]                                                                  | Pass   | 0.002s |
| Line 288: empty gauge_lookup -i no<br>nearest -i None.           | result is None                                                                                  | Pass   | 0.001s |
| Line 283: empty prop_id -i None.                                 | result is None                                                                                  | Pass   | 0.001s |
| Line 283: prop_lat == 0 -i None.                                 | result is None                                                                                  | Pass   | 0.001s |
| Non flooded events excluded from<br>depth count                  | result["flood_count"] == 4                                                                      | Pass   | 0.255s |
| Zero flooded events uses floor pric-<br>ing                      | result["has_gev"] is False;<br>result["pricing_method"] == "floor"<br>(+1 more)                 | Pass   | 0.103s |
| Exactly min events triggers gev                                  | result["flood_count"] ==<br>MIN_EVENTS_FOR_GEV                                                  | Pass   | 0.252s |
| Many events enables gev                                          | result["flood_count"] == 10;<br>result["gev_params"] is not None<br>or result["has_gev"] is ... | Pass   | 0.233s |

| Test Description                                                     | Acceptance Criteria                                                                                               | Result | Time   |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| One below min events uses floor                                      | result["has_gev"] is False;<br>result["pricing_method"] == "floor"                                                | Pass   | 0.112s |
| Max depth positive for flooded property                              | result["summary"]["max_depth_m"] > 0;<br>result["summary"]["mean_depth_m"] > 0                                    | Pass   | 0.268s |
| Transmission rate zero when no gauge floods                          | result["nearest_gauges"][0]["flood_transmission_rate"] == ...                                                     | Pass   | 0.110s |
| Zero depths for non flooded                                          | result["summary"]["max_depth_m"] == 0.0; result["summary"]["mean_depth_m"] == 0.0                                 | Pass   | 0.105s |
| Property with elevation > gauge peak -> depth 0.                     | result["summary"]["floods_at_property"] == 0                                                                      | Pass   | 0.001s |
| Gauge response with exceeded_alert=False -> 0 gauge storms.          | result["summary"]["floods_at_nearest_gauge"] == 0                                                                 | Pass   | 0.001s |
| Summary dict has floods_at_nearest_gauge etc.                        | "property_id" in s;<br>"floods_at_nearest_gauge" in s (+2 more)                                                   | Pass   | 0.002s |
| Full processing writes a JSON file for the property.                 | result is not None; (pts_dir / "PROP-001.json").exists()                                                          | Pass   | 0.001s |
| If a single property raises during generation, failed_properties     | result["processing_stats"]["failed_properties"] >= 1;<br>result["processing_stats"]["successful_properties"] >= 2 | Pass   | 0.603s |
| getSampleStyleSheet() includes 'Title' by default, so line 65 branch | 'Title' in page.styles                                                                                            | Pass   | 0.000s |
| Trade with property_id computes basis_delta_bps and basis_dv01.      | enriched['property_id'] == 'PROP-001'; 'basis_delta_bps' in enriched (+2 more)                                    | Pass   | 0.002s |
| Existing property returns success                                    | r.status_code == 200;<br>r.get_json()["status"] == "success" (+3 more)                                            | Pass   | 0.009s |
| Include readings false strips readings                               | "readings" not in event                                                                                           | Pass   | 0.002s |
| Include readings true keeps readings                                 | "readings" in data["flood_events"][0];<br>len(data["flood_events"][0]["readings"]) == 168                         | Pass   | 0.002s |
| Missing property returns 404                                         | r.status_code == 404;<br>r.get_json()["status"] == "error" (+1 more)                                              | Pass   | 0.000s |

| Test Description                                                                 | Acceptance Criteria                                                                   | Result | Time   |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Options returns ok                                                               | <code>r.status_code == 200; r.status_code == 200 (+2 more)</code>                     | Pass   | 0.001s |
| GET /api/v1/properties/{id}/hazard returns hazard data for property with curves. | <code>response.status_code == 200; data['status'] == 'success' (+2 more)</code>       | Pass   | 0.002s |
| GET /api/v1/properties/{id}/hazard returns 404 for unknown property.             | <code>response.status_code == 404</code>                                              | Pass   | 0.002s |
| GET /api/v1/properties/{id}/hazard returns 404 for property without curves.      | <code>response.status_code == 404; data['status'] == 'error' (+1 more)</code>         | Pass   | 0.002s |
| GET /api/v1/propertyhc/basis returns basis table.                                | <code>response.status_code == 200; data['status'] == 'success' (+5 more)</code>       | Pass   | 0.002s |
| GET /api/v1/propertyhc/summary returns portfolio hazard summary.                 | <code>response.status_code == 200; data['status'] == 'success' (+3 more)</code>       | Pass   | 0.002s |
| Data contains hazard curve                                                       | <code>"hazard_curve" in r.get_json()["data"]</code>                                   | Pass   | 0.003s |
| Data contains nearest gauges                                                     | <code>"nearest_gauges" in r.get_json()["data"]</code>                                 | Pass   | 0.002s |
| Data contains property id                                                        | <code>r.get_json()["data"]["property_id"] == "PROP-001"</code>                        | Pass   | 0.002s |
| Data contains summary                                                            | <code>"summary" in r.get_json()["data"]</code>                                        | Pass   | 0.003s |
| Known property returns 200                                                       | <code>r.status_code == 200</code>                                                     | Pass   | 0.001s |
| Known property status success                                                    | <code>r.get_json()["status"] == "success"</code>                                      | Pass   | 0.002s |
| No data returns 404                                                              | <code>r.status_code == 404; r.status_code == 404 (+1 more)</code>                     | Pass   | 0.001s |
| No data status error                                                             | <code>r.get_json()["status"] == "error"; r.get_json()["status"] == "error"</code>     | Pass   | 0.001s |
| Options returns 200                                                              | <code>r.status_code == 200; r.status_code == 200 (+1 more)</code>                     | Pass   | 0.001s |
| Options status ok                                                                | <code>r.get_json()["status"] == "ok"; r.get_json()["status"] == "ok" (+1 more)</code> | Pass   | 0.001s |
| Response has data key                                                            | <code>"data" in r.get_json(); r.get_json()</code> "data" in                           | Pass   | 0.002s |
| Second property also works                                                       | <code>r.status_code == 200; r.get_json()["data"]["property_id"] == "PROP-002"</code>  | Pass   | 0.004s |
| Unknown property message contains id                                             | <code>"PROP-GHOST" in r.get_json()["message"]</code>                                  | Pass   | 0.002s |

| Test Description                                                                                                   | Acceptance Criteria                                                                                         | Result | Time   |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| Unknown property returns 404                                                                                       | <code>r.status_code == 404</code>                                                                           | Pass   | 0.006s |
| Unknown property status error                                                                                      | <code>r.get_json()["status"] == "error"</code>                                                              | Pass   | 0.005s |
| <code>configure(panel_width,</code><br><code>panel_height)</code> updates both.                                    | <code>obj.panel_width == "1000px";</code><br><code>obj.panel_height == "800px"</code>                       | Pass   | 0.000s |
| <code>configure()</code> with no args leaves values unchanged.                                                     | <code>obj.panel_width == original_width;</code><br><code>obj.panel_height == original_height</code>         | Pass   | 0.000s |
| Lines 303-304: <code>configure(panel_height=...)</code> updates <code>panel_height</code> .                        | <code>obj.panel_height == "700px"</code>                                                                    | Pass   | 0.000s |
| Lines 301-302: <code>configure(panel_width=...)</code> updates <code>panel_width</code> .                          | <code>obj.panel_width == "900px"</code>                                                                     | Pass   | 0.000s |
| Line 308: <code>get_statistics()</code> returns dict with <code>panel_width</code> and <code>panel_height</code> . | <code>stats["panel_width"] == "750px";</code><br><code>stats["panel_height"] == "550px"</code>              | Pass   | 0.000s |
| <code>get_statistics()</code> always returns a dict.                                                               | <code>isinstance(stats, dict); "panel_width"</code><br><code>in stats (+1 more)</code>                      | Pass   | 0.000s |
| PropertyID must be populated in Header (not just PropertyAttributes).                                              | <code>len(empty_header_ids) == 0</code>                                                                     | Pass   | 0.003s |
| <code>property.json</code> must contain at least one property.                                                     | <code>len(ids) &gt; 0</code>                                                                                | Pass   | 0.003s |
| Property IDs in <code>propertyhc.json</code> must exist in <code>property.json</code> .                            | <code>not orphaned</code>                                                                                   | Pass   | 0.024s |
| Properties with hazard curves should also have timeseries data.                                                    | <code>not missing</code>                                                                                    | Pass   | 0.001s |
| <code>propertyts/</code> files must reference property IDs from <code>property.json</code> .                       | <code>len(missing) == 0</code>                                                                              | Pass   | 6.474s |
| Property IDs in <code>propertyts/</code> files must exist in <code>property.json</code> .                          | <code>not orphaned</code>                                                                                   | Pass   | 0.011s |
| Generator does not use sequential format                                                                           | <code>'GAUGE-{i:}' not in src;</code><br><code>'GAUGE-{index}' not in src (+3 more)</code>                  | Pass   | 0.000s |
| Generator source uses prop prefix                                                                                  | <code>'PROP-' in src</code>                                                                                 | Pass   | 0.013s |
| Generator source uses uuid                                                                                         | <code>'uuid' in src; 'uuid' in src (+1 more)</code>                                                         | Pass   | 0.000s |
| Uuid4 produces protocol matching ids                                                                               | <code>GAUGE_ID.RE.match(gauge_id);</code><br><code>PROP_ID.RE.match(prop_id) (+2 more)</code>               | Pass   | 0.000s |
| Accepts valid                                                                                                      | <code>PROP_ID.RE.match('PROP-a3f7bc12');</code><br><code>PROP_ID.RE.match('PROP-00000000') (+6 more)</code> | Pass   | 0.000s |



| Test Description                                    | Acceptance Criteria                                                                                      | Result | Time   |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Rejects sequential                                  | <code>not PROP_ID_RE.match('PROP-001');</code> <code>not PROP_ID_RE.match('PROP-1234')</code> (+3 more)  | Pass   | 0.000s |
| Rejects uppercase                                   | <code>not PROP_ID_RE.match('PROP-A3F7BC12');</code> <code>not STORM_ID_RE.match('STORM-C7F4B8E2')</code> | Pass   | 0.000s |
| Test total portfolio value calculation.             | <code>total == 2050000</code>                                                                            | Pass   | 0.002s |
| Test value aggregation by state.                    | <code>'FL' in values;</code> <code>'TX' in values</code> (+2 more)                                       | Pass   | 0.001s |
| Custom filename                                     | <code>PropertyLoader(tmp_path, filename="my_props.json").count(...</code>                                | Pass   | 0.001s |
| Load all returns list                               | <code>len(PropertyLoader(tmp_path).load_all()) == 3</code>                                               | Pass   | 0.001s |
| Missing file empty                                  | <code>PropertyLoader(tmp_path).load_all() == []</code>                                                   | Pass   | 0.000s |
| Test PropertyID extraction from nested structure.   | <code>prop_id is not None;</code> <code>prop_id.startswith("PROP-")</code>                               | Pass   | 0.001s |
| Test property summary generation.                   | <code>len(summaries) == 3;</code> <code>'propertyId' in summary</code> (+4 more)                         | Pass   | 0.001s |
| Test loading property portfolio.                    | <code>len(properties) == 3</code>                                                                        | Pass   | 0.002s |
| Test getting coordinates.                           | <code>coords is not None;</code> <code>lat == pytest.approx(25.7617, rel=1e-4)</code> (+1 more)          | Pass   | 0.001s |
| Test getting coordinates for non-existent property. | <code>coords is None</code>                                                                              | Pass   | 0.002s |
| Test getting location data.                         | <code>location is not None;</code> <code>location['City'] == "Miami"</code> (+1 more)                    | Pass   | 0.001s |
| Test getting location for non-existent property.    | <code>location is None</code>                                                                            | Pass   | 0.001s |
| Test finding properties within radius.              | <code>len(nearby) &gt;= 1;</code> <code>len(nearby) == 2</code> (+1 more)                                | Pass   | 0.002s |
| Find by id                                          | <code>PropertyLoader(tmp_path).find_by_id("PROP-001") is not None</code>                                 | Pass   | 0.001s |
| List all has property id                            | <code>"propertyId" in summaries[0]</code>                                                                | Pass   | 0.001s |
| Test finding property by address.                   | <code>prop is not None;</code> <code>"PROP-001" == property_loader.get_entity_id(prop)</code>            | Pass   | 0.001s |
| Test address search is case insensitive.            | <code>prop is not None</code>                                                                            | Pass   | 0.002s |
| Test address search with no match.                  | <code>prop is None</code>                                                                                | Pass   | 0.002s |

| Test Description                                                          | Acceptance Criteria                                                           | Result | Time   |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|--------|
| Test finding property by ID.                                              | prop is not None;<br>prop['PropertyHeader']['Location']['City']<br>== "Miami" | Pass   | 0.002s |
| Test finding properties by state.                                         | len(fl_props) == 2; len(tx_props) == 1                                        | Pass   | 0.002s |
| Test state search is case insensitive.                                    | len(props) == 2                                                               | Pass   | 0.001s |
| Test finding properties by zip code.                                      | len(props) == 1;<br>property_loader.get_entity_id(props[0])<br>== "PROP-001"  | Pass   | 0.001s |
| Test finding multiple properties in same zip.                             | len(props) == 1                                                               | Pass   | 0.002s |
| Passing an unhandled type triggers super().default() → TypeError.         | —                                                                             | Pass   | 0.000s |
| Lines 62-64: ground_level falls back to default_elevation.                | mapping["ground_level_meters"] == 10.0; mapping["elevation"] == 10.0          | Pass   | 0.000s |
| Lines 136-137: exception in create_mapping raises ValueError.             | —                                                                             | Pass   | 0.001s |
| Found returns mortgage                                                    | r.status_code == 200; data["status"]<br>== "success" (+2 more)                | Pass   | 0.001s |
| Loader exception returns 500                                              | r.status_code == 500;<br>r.get_json()["status"] == "error"<br>(+1 more)       | Pass   | 0.001s |
| Not found returns 404                                                     | r.status_code == 404;<br>r.get_json()["status"] == "error"<br>(+2 more)       | Pass   | 0.001s |
| Options returns ok                                                        | r.status_code == 200;<br>r.get_json()["status"] == "ok"<br>(+2 more)          | Pass   | 0.001s |
| Line 258: continue when prop_id not in prop_values → no properties → 404. | r.status_code == 404                                                          | Pass   | 0.003s |
| Line 193: break when prop_id not in prop_values → no properties → 404.    | r.status_code == 404                                                          | Pass   | 0.003s |
| Output json contains generation metadata                                  | "generated_at" in meta; "catchment" in meta (+1 more)                         | Pass   | 0.654s |
| Output json contains properties key                                       | "properties" in data                                                          | Pass   | 0.974s |
| Property header has property id                                           | "PropertyHeader" in prop;<br>"PropertyAttributes" in prop["PropertyHeader"]   | Pass   | 0.911s |
| Result data contains locations                                            | len(result["data"]["locations"]) == 3                                         | Pass   | 1.106s |

| Test Description                                                     | Acceptance Criteria                                                                                | Result | Time   |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------|--------|
| Full property+mortgage report produces a valid PDF file.             | <code>report_path.exists();</code><br><code>report_path.suffix == '.pdf'</code> (+1 more)          | Pass   | 0.032s |
| Mortgage-focused report produces a valid PDF.                        | <code>report_path.exists();</code><br><code>report_path.suffix == '.pdf'</code> (+1 more)          | Pass   | 0.024s |
| Output directory is created automatically if it doesn't exist.       | <code>new_dir.exists(); report_path.exists()</code>                                                | Pass   | 0.027s |
| Generated filename includes the property ID.                         | <code>'PROP-test0001' in report_path.name</code>                                                   | Pass   | 0.022s |
| Full report PDF should be non-trivial in size (multi-page).          | <code>report_path.stat().st_size &gt; 10000</code>                                                 | Pass   | 0.034s |
| Generated file is a valid PDF (starts with %PDF).                    | <code>header == b'%PDF-'</code>                                                                    | Pass   | 0.052s |
| Property-only report (no mortgage) produces a valid PDF.             | <code>report_path.exists();</code><br><code>report_path.suffix == '.pdf'</code> (+1 more)          | Pass   | 0.022s |
| Report handles minimal/sparse property data without crashing.        | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 0</code>               | Pass   | 0.018s |
| Risk-focused report produces a valid PDF.                            | <code>report_path.exists();</code><br><code>report_path.suffix == '.pdf'</code> (+1 more)          | Pass   | 0.018s |
| Auto-select includes mortgage pages when mortgage data present.      | <code>'mortgage_overview' in pages;</code><br><code>'mortgage_details' in pages</code>             | Pass   | 0.002s |
| Auto-select excludes mortgage pages when no mortgage data.           | <code>'title_overview' in pages;</code><br><code>'mortgage_overview' not in pages</code> (+1 more) | Pass   | 0.002s |
| Generator initialises with output directory.                         | <code>gen.output_dir == tmp_path</code>                                                            | Pass   | 0.002s |
| Generator lists all registered page modules.                         | <code>'title_overview' in pages;</code> <code>'location' in pages</code> (+2 more)                 | Pass   | 0.002s |
| Validates known page names.                                          | <code>'title_overview' in valid;</code> <code>'location' in valid</code> (+1 more)                 | Pass   | 0.002s |
| Line 174-175: <code>mortgage_data -i</code> includes mortgage pages. | <code>"mortgage_overview" in pages</code>                                                          | Pass   | 0.002s |
| Line 174 (else): <code>no mortgage_data -i</code> no mortgage pages. | <code>"mortgage_overview" not in pages</code>                                                      | Pass   | 0.002s |
| Lines 214-216: exception in page <code>-i</code> logged, continue.   | <code>isinstance(elements, list)</code>                                                            | Pass   | 0.002s |
| Lines 198-199: unknown page <code>-i</code> skip.                    | <code>isinstance(elements, list)</code>                                                            | Pass   | 0.002s |

| Test Description                                                        | Acceptance Criteria                                                       | Result | Time   |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Lines 184-185: missing PropertyID -¿ 'unknown'.                         | "unknown" in name                                                         | Pass   | 0.002s |
| Lines 257-264: mortgage-focused report.                                 | path.exists()                                                             | Pass   | 0.018s |
| Lines 251-255: property-only report.                                    | path.exists()                                                             | Pass   | 0.016s |
| Lines 266-276: risk-focused report without mortgage data.               | path.exists()                                                             | Pass   | 0.011s |
| Lines 266-276: risk-focused report with mortgage data.                  | path.exists()                                                             | Pass   | 0.013s |
| Line 313: get_page_categories.                                          | "property" in cats; "mortgage" in cats                                    | Pass   | 0.002s |
| Line 285: list_available_pages.                                         | isinstance(pages, list); "title_overview" in pages                        | Pass   | 0.002s |
| Lines 69-70: output_dir=None -¿ uses config.get_property_reports_dir(). | gen.output_dir == prop_dir                                                | Pass   | 0.002s |
| Line 315: validate_pages.                                               | "title_overview" in valid; "bogus_page" in invalid                        | Pass   | 0.002s |
| Categories dict initialized                                             | "property" in gen.categories; "mortgage" in gen.categories (+1 more)      | Pass   | 0.002s |
| Explicit output dir path                                                | isinstance(gen.output_dir, Path)                                          | Pass   | 0.002s |
| Explicit output dir string                                              | gen.output_dir == tmp_path                                                | Pass   | 0.002s |
| No output dir uses config                                               | gen.output_dir == prop_dir                                                | Pass   | 0.002s |
| Output dir created                                                      | new_dir.exists()                                                          | Pass   | 0.002s |
| Property pages dict initialized                                         | expected <= set(gen.property_pages.keys())                                | Pass   | 0.002s |
| Test POST /api/v1/properties/report without body.                       | response.status_code == 400                                               | Pass   | 0.001s |
| Test POST /api/v1/properties/report without propertyId.                 | response.status_code == 400; 'Property ID is required' in data['message'] | Pass   | 0.001s |
| Test POST /api/v1/properties/report with invalid property.              | response.status_code == 404                                               | Pass   | 0.002s |
| Test GET /api/v1/properties/{id}.                                       | response.status_code == 200; data['status'] == 'success' (+1 more)        | Pass   | 0.002s |

| Test Description                                                                         | Acceptance Criteria                                                                                             | Result | Time   |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------|--------|
| Test GET /api/v1/properties/jid <sub>i</sub> with invalid ID.                            | response.status_code == 404;<br>data['status'] == 'error'                                                       | Pass   | 0.002s |
| Test legacy POST /api/v1/generate_property_report endpoint.                              | response.status_code == 404                                                                                     | Pass   | 0.002s |
| Test GET /api/v1/properties.                                                             | response.status_code == 200;<br>data['status'] == 'success' (+2 more)                                           | Pass   | 0.001s |
| Test OPTIONS preflight request.                                                          | response.status_code == 200                                                                                     | Pass   | 0.000s |
| Existing property returns success                                                        | r.status_code == 200;<br>r.get_json()["status"] == "success"<br>(+3 more)                                       | Pass   | 0.012s |
| Flood events tagged with sequence type                                                   | len(events) >= 1; "sequence_type" in events[0]                                                                  | Pass   | 0.005s |
| Missing property returns 404                                                             | r.status_code == 404;<br>r.get_json()["status"] == "error"<br>(+1 more)                                         | Pass   | 0.001s |
| Nearest gauges enriched with flood stages                                                | len(gauges) >= 1; "flood_stages" in gauges[0]                                                                   | Pass   | 0.005s |
| Options returns ok                                                                       | r.status_code == 200; r.status_code == 200 (+2 more)                                                            | Pass   | 0.001s |
| Response has flood events                                                                | "flood_events" in data;<br>len(data["flood_events"]) >= 1                                                       | Pass   | 0.006s |
| Response has nearest gauges                                                              | "nearest_gauges" in data                                                                                        | Pass   | 0.004s |
| Response has property info                                                               | "property_info" in data; "elevation_m" in data["property_info"]                                                 | Pass   | 0.004s |
| Lines 157-158: gauge.json missing triggers except pass.                                  | r.status_code == 200;<br>ng["flood_stages"] == {}                                                               | Pass   | 0.035s |
| Lines 134-135: storm_sequences.json missing triggers except pass.                        | r.status_code == 200; data["status"] == "success" (+1 more)                                                     | Pass   | 0.035s |
| GET /api/v1/propertyts/storms returns ALL storms from storm_sequences.json.              | response.status_code == 200;<br>data['status'] == 'success' (+9 more)                                           | Pass   | 0.006s |
| Non-flooding storms must appear in the list with properties_flooded=0.                   | storm_map['STORM-001']['properties_flooded'] > 0;<br>storm_map['STORM-002']['properties_flooded'] > 0 (+2 more) | Pass   | 0.005s |
| GET /api/v1/properties/jid <sub>i</sub> /floods returns flood events for known property. | response.status_code == 200;<br>data['status'] == 'success' (+2 more)                                           | Pass   | 0.002s |

| Test Description                                                         | Acceptance Criteria                                                                                  | Result | Time   |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------|--------|
| Property with few flood events should still return data from propertyts. | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)      | Pass   | 0.003s |
| GET /api/v1/properties/{id}/floods returns 404 for unknown property.     | <code>response.status_code == 404;</code><br><code>data['status'] == 'error'</code>                  | Pass   | 0.001s |
| GET /api/v1/properties/{id}/storms returns storm analysis data.          | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+4 more)      | Pass   | 0.004s |
| GET /api/v1/properties/{id}/storms returns 404 for unknown property.     | <code>response.status_code == 404</code>                                                             | Pass   | 0.001s |
| Property with zero flood events should return empty array, not error.    | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+4 more)      | Pass   | 0.026s |
| GET /api/v1/propertyts/summary returns portfolio flood summary.          | <code>response.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)      | Pass   | 0.002s |
| Generates per property files                                             | <code>pts_dir.exists();</code> (pts_dir / "portfolio_flood_summary.json").exists()                   | Pass   | 0.000s |
| Per property files have flood data                                       | "property_id" in data;<br>"nearest_gauges" in data (+1 more)                                         | Pass   | 0.001s |
| Portfolio summary has statistics                                         | "total_properties" in s;<br>s["total_properties"] == N_PROPERTIES (+1 more)                          | Pass   | 0.006s |
| np.float64 is converted to Python float by encoder.                      | <code>isinstance(result, float); abs(result - 3.14) &lt; 1e-10</code>                                | Pass   | 0.000s |
| Lines 69-70: exception returns validation_error dict.                    | "validation_error" in result;<br><code>isinstance(result["validation_error"], list)</code> (+2 more) | Pass   | 0.000s |
| get_required_fields returns expected paths.                              | <code>len(fields) == 4;</code><br>"PropertyHeader.Header.PropertyID" in fields (+1 more)             | Pass   | 0.000s |
| Line 65: location_errors appended when non-empty.                        | "Location" in errors;<br><code>len(errors["Location"]) == 2</code>                                   | Pass   | 0.000s |
| Line 60: Missing LatitudeDegrees.                                        | "Location" in errors;<br><code>any("LatitudeDegrees" in e for e in errors["Location"])</code>        | Pass   | 0.000s |
| Line 62: Missing LongitudeDegrees.                                       | "Location" in errors;<br><code>any("LongitudeDegrees" in e for e in errors["Location"])</code>       | Pass   | 0.000s |
| Valid data returns no errors.                                            | <code>property_validate(data) == {}</code>                                                           | Pass   | 0.000s |

| Test Description                                    | Acceptance Criteria                                                                                   | Result | Time   |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| No mortgage returns zero fill                       | <code>r.status_code == 200;</code><br><code>prop["has.mortgage"] is False</code> (+4 more)            | Pass   | 0.003s |
| No mortgage sequence                                | <code>r.status_code == 200;</code><br><code>prop["has.mortgage"] is False</code>                      | Pass   | 0.003s |
| Basis entry has avg basis bps                       | <code>"avg.basis.bps" in entry</code>                                                                 | Pass   | 0.002s |
| Basis entry has flood count                         | <code>"flood.count" in entry</code>                                                                   | Pass   | 0.002s |
| Basis entry has nearest gauges                      | <code>"nearest.gauges" in entry</code>                                                                | Pass   | 0.009s |
| Basis entry has property id                         | <code>"property.id" in entry</code>                                                                   | Pass   | 0.004s |
| Basis table count matches                           | <code>data["count"] ==</code><br><code>len(data["basis.table"])</code>                                | Pass   | 0.002s |
| Basis table length                                  | <code>r.get_json()["count"] == 2</code>                                                               | Pass   | 0.002s |
| Results must be sorted highest avg_basis_bps first. | <code>bases == sorted(bases, reverse=True)</code>                                                     | Pass   | 0.002s |
| Nearest gauge has distance km                       | <code>"distance.km" in</code><br><code>entry["nearest_gauges"][0]</code>                              | Pass   | 0.002s |
| Nearest gauge has gauge id                          | <code>"gauge.id" in</code><br><code>entry["nearest_gauges"][0]</code>                                 | Pass   | 0.002s |
| No data returns 404                                 | <code>r.status_code == 404; r.status_code ==</code><br><code>404 (+1 more)</code>                     | Pass   | 0.003s |
| No data status error                                | <code>r.get_json()["status"] == "error";</code><br><code>r.get_json()["status"] == "error"</code>     | Pass   | 0.001s |
| Options returns 200                                 | <code>r.status_code == 200; r.status_code ==</code><br><code>200 (+1 more)</code>                     | Pass   | 0.001s |
| Options status ok                                   | <code>r.get_json()["status"] == "ok";</code><br><code>r.get_json()["status"] == "ok" (+1 more)</code> | Pass   | 0.001s |
| Response has basis table                            | <code>"basis.table" in r.get_json()</code>                                                            | Pass   | 0.003s |
| Response has count                                  | <code>"count" in r.get_json()</code>                                                                  | Pass   | 0.002s |
| Returns 200 with data                               | <code>r.status_code == 200; r.status_code ==</code><br><code>200</code>                               | Pass   | 0.026s |
| Status is success                                   | <code>r.get_json()["status"] == "success";</code><br><code>r.get_json()["status"] == "success"</code> | Pass   | 0.002s |
| Empty basis table is list                           | <code>isinstance(r.get_json()["basis.table"],</code><br><code>list)</code>                            | Pass   | 0.001s |
| Empty count is zero                                 | <code>r.get_json()["count"] == 0</code>                                                               | Pass   | 0.002s |
| Empty returns 200                                   | <code>r.status_code == 200</code>                                                                     | Pass   | 0.002s |
| Distribution avg basis bps                          | <code>"avg.basis.bps" in dist;</code><br><code>dist["avg.basis.bps"] &gt;= 0</code>                   | Pass   | 0.002s |
| Distribution avg flood count                        | <code>"avg.flood.count" in dist;</code><br><code>dist["avg.flood.count"] &gt; 0</code>                | Pass   | 0.003s |



| Test Description                                        | Acceptance Criteria                                                         | Result | Time   |
|---------------------------------------------------------|-----------------------------------------------------------------------------|--------|--------|
| Distribution max depth m                                | "max_depth_m" in dist;<br>dist["max_depth_m"] > 0                           | Pass   | 0.004s |
| Distribution num properties                             | dist["num_properties"] == 2                                                 | Pass   | 0.002s |
| No data message mentions command                        | "port" in<br>r.get_json()["message"].lower()<br>or "propertyhc"...          | Pass   | 0.001s |
| No data returns 404                                     | r.status_code == 404; r.status_code == 404 (+1 more)                        | Pass   | 0.001s |
| No data status is error                                 | r.get_json()["status"] == "error"                                           | Pass   | 0.001s |
| Options returns 200                                     | r.status_code == 200; r.status_code == 200 (+1 more)                        | Pass   | 0.001s |
| Options status ok                                       | r.get_json()["status"] == "ok";<br>r.get_json()["status"] == "ok" (+1 more) | Pass   | 0.001s |
| Response has data key                                   | "data" in r.get_json(); "data" in r.get_json()                              | Pass   | 0.002s |
| Response has distribution                               | "distribution" in<br>r.get_json()["data"]                                   | Pass   | 0.002s |
| Response has metadata                                   | "metadata" in r.get_json()["data"]                                          | Pass   | 0.001s |
| Response has summary                                    | "summary" in r.get_json()["data"]                                           | Pass   | 0.002s |
| Returns 200 with data                                   | r.status_code == 200; r.status_code == 200                                  | Pass   | 0.025s |
| Status is success                                       | r.get_json()["status"] == "success";<br>r.get_json()["status"] == "success" | Pass   | 0.002s |
| Empty curves avg flood count is zero                    | r.get_json()["data"]["distribution"]["avg_flood_count"] == 0                | Pass   | 0.003s |
| Empty curves num properties is zero                     | r.get_json()["data"]["distribution"]["num_properties"] == 0                 | Pass   | 0.002s |
| Empty curves returns 200                                | r.status_code == 200                                                        | Pass   | 0.002s |
| When no natural measures, recommendation should appear. | isinstance(result, list)                                                    | Pass   | 0.000s |
| Natural measures section rendered                       | any("Natural" in t for t in texts)                                          | Pass   | 0.000s |
| Empty property returns list                             | isinstance(result, list); len(result) > 0                                   | Pass   | 0.000s |
| Empty protection measures dict shows fallback           | any("No protection" in t for t in texts)                                    | Pass   | 0.000s |
| No protection measures shows fallback                   | any("No protection" in t or "Protection" in t for t in te...)               | Pass   | 0.000s |
| Full protection fewer recommendations                   | isinstance(result, list)                                                    | Pass   | 0.001s |

| Test Description                                                                            | Acceptance Criteria                                                          | Result | Time   |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| Generate protection recommendations returns dict                                            | <code>isinstance(result, dict)</code>                                        | Pass   | 0.000s |
| Missing critical measures in recommendations                                                | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Recommendations all natural false                                                           | <code>"Natural Solutions" in result</code>                                   | Pass   | 0.000s |
| Recommendations flood re eligible                                                           | <code>"Insurance" in result</code>                                           | Pass   | 0.000s |
| Recommendations missing critical                                                            | <code>"Priority Installations" in result</code>                              | Pass   | 0.000s |
| Recommendations section always rendered                                                     | <code>any("Recommendation" in t for t in texts)</code>                       | Pass   | 0.000s |
| All installed excellent rating                                                              | <code>any("Resilience" in t or "Protection" in t for t in texts)</code>      | Pass   | 0.000s |
| Building measures listed                                                                    | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| 8 of 10 = 80% → Excellent.                                                                  | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| 4 of 10 = 40% → Fair.                                                                       | <code>isinstance(result, list)</code>                                        | Pass   | 0.001s |
| 6 of 10 = 60% → Good.                                                                       | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| 2 of 10 = 20% → Limited.                                                                    | <code>isinstance(result, list)</code>                                        | Pass   | 0.001s |
| 0 of 10 = 0% → Poor.                                                                        | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Critical measures listed                                                                    | <code>any(isinstance(e, Table) for e in result)</code>                       | Pass   | 0.000s |
| Emergency measures listed                                                                   | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Uncategorised measure still shown                                                           | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Excess amount shown                                                                         | <code>any(isinstance(e, Table) for e in result)</code>                       | Pass   | 0.000s |
| Insurance premium shown                                                                     | <code>any(isinstance(e, Table) for e in result)</code>                       | Pass   | 0.000s |
| Other risk fields shown                                                                     | <code>isinstance(result, list); len(result) &gt; 1</code>                    | Pass   | 0.000s |
| Recommendations include flood re                                                            | <code>isinstance(result, list)</code>                                        | Pass   | 0.000s |
| Risk assessment section rendered                                                            | <code>any("Insurance" in t for t in texts)</code>                            | Pass   | 0.000s |
| All entity ID suffixes are exactly 8 characters.                                            | <code>all(v == 8 for v in suffix_lengths.values())</code>                    | Pass   | 0.000s |
| <code>str(uuid.uuid4())[8]</code> always yields lowercase hex.                              | <code>suffix == suffix.lower(); re.match(r'^[0-9a-f]{8}\$', suffix)</code>   | Pass   | 0.001s |
| All four entity types use <code>str(uuid.uuid4())[8]</code> — same derivation, same format. | <code>pattern.match(entity_id)</code>                                        | Pass   | 0.000s |
| All exports                                                                                 | <code>'ClaimReportGenerator' in pkg.__all__</code>                           | Pass   | 0.000s |
| Import from package                                                                         | <code>ClaimReportGenerator is not None</code>                                | Pass   | 0.000s |
| Current exceeds original is capped                                                          | <code>result["Mortgage"]["LoanDetails"]["CurrentBalance"] &lt;= 40...</code> | Pass   | 0.000s |

| Test Description                               | Acceptance Criteria                                                                                                             | Result | Time   |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Interest over 15 capped                        | result["Mortgage"]["LoanDetails"]["InterestRate"]<br>== 12.0                                                                    | Pass   | 0.000s |
| Interest under 1 raised                        | result["Mortgage"]["LoanDetails"]["InterestRate"]<br>== 2.0                                                                     | Pass   | 0.000s |
| Ltv over 100 capped at 95                      | result["Mortgage"]["LoanDetails"]["LTV"]<br>== 95                                                                               | Pass   | 0.000s |
| Ltv under 10 set to 60                         | result["Mortgage"]["LoanDetails"]["LTV"]<br>== 60                                                                               | Pass   | 0.000s |
| Valid data unchanged                           | result["Mortgage"]["LoanDetails"]["CurrentBalance"]<br>== 35...;<br>result["Mortgage"]["LoanDetails"]["LTV"]<br>== 80 (+1 more) | Pass   | 0.000s |
| Current exceeds original is capped to 95pct    | result["Mortgage"]["LoanDetails"]["CurrentBalance"]<br><= 30...                                                                 | Pass   | 0.000s |
| Current within original unchanged              | result["Mortgage"]["LoanDetails"]["CurrentBalance"]<br>== 35...                                                                 | Pass   | 0.000s |
| Delegates to random module when method present | result is sentinel                                                                                                              | Pass   | 0.000s |
| Ltv over 100 set to 95                         | result["Mortgage"]["LoanDetails"]["LTV"]<br>== 95                                                                               | Pass   | 0.000s |
| Ltv under 10 set to 60                         | result["Mortgage"]["LoanDetails"]["LTV"]<br>== 60                                                                               | Pass   | 0.000s |
| Valid ltv unchanged                            | result["Mortgage"]["LoanDetails"]["LTV"]<br>== 75                                                                               | Pass   | 0.000s |
| Get raci matrix                                | response.status_code == 200;<br>data["status"] == "success" (+3 more)                                                           | Pass   | 0.020s |
| Get raci matrix file missing                   | response.status_code == 404                                                                                                     | Pass   | 0.001s |
| Update raci activity                           | response.status_code == 200;<br>data["status"] == "success" (+3 more)                                                           | Pass   | 0.003s |
| Update raci activity invalid role              | response.status_code == 400                                                                                                     | Pass   | 0.002s |
| Update raci escalation                         | response.status_code == 200;<br>data["status"] == "success" (+3 more)                                                           | Pass   | 0.003s |
| Update raci role                               | response.status_code == 200;<br>data["status"] == "success" (+3 more)                                                           | Pass   | 0.003s |
| Update raci role not found                     | response.status_code == 404                                                                                                     | Pass   | 0.002s |
| No raci data returns 404                       | r.status_code == 404; r.status_code == 404                                                                                      | Pass   | 0.000s |

| Test Description                                                               | Acceptance Criteria                                                                                                 | Result | Time   |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------|--------|
| Save raci failure returns 500                                                  | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code><br>(+4 more)                   | Pass   | 0.002s |
| Tier emphasis updated                                                          | <code>r.status_code == 200;</code><br><code>activities[0]["tier_emphasis"] ==</code><br><code>"Tier 1 focus"</code> | Pass   | 0.003s |
| No raci data returns 404                                                       | <code>r.status_code == 404; r.status_code ==</code><br><code>404</code>                                             | Pass   | 0.000s |
| Save raci failure returns 500                                                  | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code><br>(+4 more)                   | Pass   | 0.002s |
| Save raci failure returns 500                                                  | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code><br>(+4 more)                   | Pass   | 0.002s |
| Remaining zero uses base amplitude                                             | <code>isinstance(level, float)</code>                                                                               | Pass   | 0.000s |
| Empty EndDate returns default tenor of 5.                                      | <code>result == 5</code>                                                                                            | Pass   | 0.000s |
| Invalid date format returns default tenor of 5 (ValueError path).              | <code>result == 5</code>                                                                                            | Pass   | 0.000s |
| Missing EndDate key returns default tenor of 5.                                | <code>result == 5</code>                                                                                            | Pass   | 0.000s |
| None EndDate returns default tenor of 5 (TypeError path).                      | <code>result == 5</code>                                                                                            | Pass   | 0.000s |
| Valid far-future EndDate returns positive tenor.                               | <code>result &gt;= 1</code>                                                                                         | Pass   | 0.000s |
| Lines 503-504: default 'full' report_type → args.property_pages AttributeError | <code>1 in exited</code>                                                                                            | Pass   | 0.004s |
| Lines 349-351: success-logging block executes without error.                   | —                                                                                                                   | Pass   | 0.004s |
| Lines 444-445: monitoring without timeseries_data → condition False → else.    | —                                                                                                                   | Pass   | 0.004s |
| Lines 442-443: -report-type analysis → generate_analysis_report path.          | —                                                                                                                   | Pass   | 0.003s |
| Lines 344-345: -report-type analysis → generate_analysis_report path executed. | —                                                                                                                   | Pass   | 0.004s |
| Lines 438-439: -report-type basic → generate_basic_report path.                | —                                                                                                                   | Pass   | 0.010s |

| Test Description                                                                                                            | Acceptance Criteria                                                       | Result | Time   |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Lines 338-339: <code>-report-type basic</code> → <code>generate_basic_report</code> path executed.                          | —                                                                         | Pass   | 0.005s |
| Lines 340-341: <code>-report-type detailed</code> → <code>generate_detailed_report</code> path executed.                    | —                                                                         | Pass   | 0.004s |
| Lines 346-347: <code>unmapped</code> <code>report_type</code> → <code>generate_report(pages_to_include=args.pages)</code> . | —                                                                         | Pass   | 0.006s |
| Lines 444-445: <code>-report-type full</code> (not <code>basic/monitoring/analysis</code> ) → <code>else</code> branch.     | —                                                                         | Pass   | 0.004s |
| Lines 440-441: <code>-report-type monitoring + timeseries</code> → <code>monitoring</code> path.                            | —                                                                         | Pass   | 0.005s |
| Lines 499-500: <code>-report-type mortgage-focused + mortgage</code> file → <code>focused</code> report.                    | —                                                                         | Pass   | 0.007s |
| Lines 499 condition <code>False</code> (no <code>mortgage_data</code> ) → falls to <code>elif risk-focused / else</code> .  | <code>1 in exited</code>                                                  | Pass   | 0.009s |
| Lines 497-498: <code>-report-type property-only</code> → <code>generate_property_only_report</code> .                       | —                                                                         | Pass   | 0.006s |
| Lines 501-502: <code>-report-type risk-focused</code> → <code>generate_risk_focused_report</code> .                         | —                                                                         | Pass   | 0.004s |
| Lines 342-343: <code>-report-type summary</code> → <code>generate_summary_report</code> path executed.                      | —                                                                         | Pass   | 0.004s |
| Lines 447-449: <code>success-logging</code> block runs after generation.                                                    | —                                                                         | Pass   | 0.004s |
| Lines 506-508: <code>success-logging</code> block runs after a successful generation.                                       | —                                                                         | Pass   | 0.005s |
| <code>reports.py</code> lines 87-88: <code>gaugehd</code> file read error - <code>i</code> , warning, continues.            | <code>r.status_code == 200</code>                                         | Pass   | 0.037s |
| <code>reports.py</code> lines 135-140: <code>general</code> Exception - <code>i</code> , 500.                               | <code>r.status_code == 500; "disk full" in r.get_json()["message"]</code> | Pass   | 0.032s |
| <code>reports.py</code> lines 100-101: <code>hazard</code> file read error - <code>i</code> , warning, continues.           | <code>r.status_code == 200</code>                                         | Pass   | 0.036s |

| Test Description                                                        | Acceptance Criteria                                        | Result | Time   |
|-------------------------------------------------------------------------|------------------------------------------------------------|--------|--------|
| reports.py lines 128-133: ImportError -i 500.                           | r.status_code == 500; "not available"                      | Pass   | 0.035s |
| reports.py lines 70-72: storm_responses loaded -i time-series_data set. | in r.get_json()["message"].lower()<br>r.status_code == 200 | Pass   | 0.030s |
| reports.py lines 73-74: timeseries load failure -i warning, continues.  | r.status_code == 200                                       | Pass   | 0.032s |
| Field is dict[overall risk rating] [field='overall_risk_rating']        | field in m; isinstance(m[field], dict)                     | Pass   | 0.000s |
| Field is dict[coverage] [field='test_coverage']                         | field in m; isinstance(m[field], dict)                     | Pass   | 0.000s |
| Field is list[assumptions] [field='assumptions']                        | field in m; isinstance(m[field], list)                     | Pass   | 0.000s |
| Field is list[downstream models] [field='downstream_models']            | field in m; isinstance(m[field], list)                     | Pass   | 0.000s |
| Field is list[limitations] [field='limitations']                        | field in m; isinstance(m[field], list)                     | Pass   | 0.000s |
| Field is list[upstream models] [field='upstream_models']                | field in m; isinstance(m[field], list)                     | Pass   | 0.000s |
| Field is string[category] [field='category']                            | field in m; isinstance(m[field], str)<br>(+1 more)         | Pass   | 0.000s |
| Field is string[description] [field='description']                      | field in m; isinstance(m[field], str)<br>(+1 more)         | Pass   | 0.000s |
| Field is string[name] [field='name']                                    | field in m; isinstance(m[field], str)<br>(+1 more)         | Pass   | 0.000s |
| Field is string[owner] [field='owner']                                  | field in m; isinstance(m[field], str)<br>(+1 more)         | Pass   | 0.000s |
| Field is string[status] [field='status']                                | field in m; isinstance(m[field], str)<br>(+1 more)         | Pass   | 0.000s |
| Field is string[version] [field='version']                              | field in m; isinstance(m[field], str)<br>(+1 more)         | Pass   | 0.000s |
| Case insensitive                                                        | resolve_model_id("GEV") == "MKM-GH-001"                    | Pass   | 0.000s |
| Depth damage maps to dd                                                 | resolve_model_id("depth_damage") == "MKM-DD-001"           | Pass   | 0.000s |
| Distribution maps to si                                                 | resolve_model_id("distribution") == "MKM-SI-001"           | Pass   | 0.000s |
| Mkm prefix returned as is                                               | resolve_model_id("MKM-GH-001") == "MKM-GH-001"             | Pass   | 0.003s |
| Shorthand gev resolved                                                  | resolve_model_id("gev") == "MKM-GH-001"                    | Pass   | 0.000s |

| Test Description                                         | Acceptance Criteria                                                                | Result | Time   |
|----------------------------------------------------------|------------------------------------------------------------------------------------|--------|--------|
| Shorthand intensity resolved                             | <code>resolve_model_id("intensity") == "MKM-SI-001"</code>                         | Pass   | 0.000s |
| Shorthand mortgage resolved                              | <code>resolve_model_id("mortgage") == "MKM-MP-001"</code>                          | Pass   | 0.000s |
| Shorthand prs resolved                                   | <code>resolve_model_id("prs") == "MKM-PR-001"</code>                               | Pass   | 0.000s |
| Spatial maps to sp                                       | <code>resolve_model_id("spatial") == "MKM-SP-001"</code>                           | Pass   | 0.000s |
| Unknown shorthand returned as is                         | <code>resolve_model_id("unknown_model") == "unknown_model"</code>                  | Pass   | 0.000s |
| Returns dict with total storms                           | <code>isinstance(result, dict); "total_storms" in result (+1 more)</code>          | Pass   | 0.002s |
| Returns elapsed s                                        | <code>"elapsed_s" in result; result["elapsed_s"] &gt;= 0</code>                    | Pass   | 0.002s |
| revalue_all(trades, None) loads market state internally. | <code>len(enriched) == 1; enriched[0]['swap_id'] == 'PRS-TEST001' (+1 more)</code> | Pass   | 0.002s |
| Combined mode with mortgage                              | <code>isinstance(result, list) and len(result) &gt; 0</code>                       | Pass   | 0.001s |
| Empty property does not crash                            | <code>isinstance(self._page().generate_elements({}), list)</code>                  | Pass   | 0.000s |
| Has risk header                                          | <code>any("Risk" in t for t in texts)</code>                                       | Pass   | 0.000s |
| Has table                                                | <code>any(isinstance(e, Table) for e in result)</code>                             | Pass   | 0.001s |
| Property only mode                                       | <code>isinstance(result, list) and len(result) &gt; 0</code>                       | Pass   | 0.001s |
| Returns list                                             | <code>isinstance(self._page().generate_elements(Lake_property), list)</code>       | Pass   | 0.001s |
| Canal distance shown                                     | <code>isinstance(result, list)</code>                                              | Pass   | 0.001s |
| Empty property does not crash                            | <code>isinstance(self._page().generate_elements({}), list)</code>                  | Pass   | 0.000s |
| Flood risk type shown                                    | <code>isinstance(result, list)</code>                                              | Pass   | 0.001s |
| Has risk header                                          | <code>any("Risk" in t for t in texts)</code>                                       | Pass   | 0.001s |
| Has table                                                | <code>any(isinstance(e, Table) for e in result)</code>                             | Pass   | 0.001s |
| Lake distance shown                                      | <code>isinstance(result, list)</code>                                              | Pass   | 0.001s |
| Last flood date shown                                    | <code>isinstance(result, list)</code>                                              | Pass   | 0.001s |
| No defence scheme                                        | <code>isinstance(result, list)</code>                                              | Pass   | 0.001s |
| No ea zone                                               | <code>isinstance(result, list)</code>                                              | Pass   | 0.000s |
| No last flood date                                       | <code>isinstance(result, list)</code>                                              | Pass   | 0.001s |
| Returns list                                             | <code>isinstance(self._page().generate_elements(Lake_property), list)</code>       | Pass   | 0.001s |



| Test Description                                                  | Acceptance Criteria                                                                             | Result | Time   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Line 257: non-numeric → delegates to <code>_format_value</code> . | <code>page._format_count(None) == "Not specified"</code>                                        | Pass   | 0.000s |
| Line 256: numeric → comma-formatted.                              | <code>"1,234,567" in result</code>                                                              | Pass   | 0.000s |
| Line 239: non-numeric → delegates to <code>_format_value</code> . | <code>result == "Not specified"</code>                                                          | Pass   | 0.000s |
| Line 238: numeric → £ formatted.                                  | <code>"£" in result; "1,234,567" in result</code>                                               | Pass   | 0.000s |
| Line 251: non-numeric → delegates to <code>_format_value</code> . | <code>page._format_depth("N/A") == "N/A"</code>                                                 | Pass   | 0.000s |
| Line 250: numeric → formatted with 'm'.                           | <code>"1.50m" in result</code>                                                                  | Pass   | 0.000s |
| Lines 204-215: camelCase → 'Camel Case'.                          | <code>"Risk" in result</code>                                                                   | Pass   | 0.000s |
| Lines 207-213: abbreviation replacements applied.                 | <code>"ID" in result</code>                                                                     | Pass   | 0.000s |
| Line 245: non-numeric → delegates to <code>_format_value</code> . | <code>page._format_percentage(None) == "Not specified"</code>                                   | Pass   | 0.000s |
| Line 244: numeric → formatted with %.                             | <code>"%" in result; "12.3" in result</code>                                                    | Pass   | 0.000s |
| Line 222: False → 'No'.                                           | <code>page._format_value(False) == "No"</code>                                                  | Pass   | 0.000s |
| Line 222: True → 'Yes'.                                           | <code>page._format_value(True) == "Yes"</code>                                                  | Pass   | 0.000s |
| Lines 230-231: blank string → 'Not specified'.                    | <code>page._format_value(" ") == "Not specified"</code>                                         | Pass   | 0.000s |
| Lines 226-227: float with fractional part → 2dp string.           | <code>result == "3.14"</code>                                                                   | Pass   | 0.000s |
| Lines 224-225: float with no fractional part → int string.        | <code>page._format_value(5.0) == "5"</code>                                                     | Pass   | 0.000s |
| Lines 228-229: integer → plain string.                            | <code>page._format_value(42) == "42"</code>                                                     | Pass   | 0.000s |
| Lines 232-233: non-empty string → passed through.                 | <code>page._format_value("hello") == "hello"</code>                                             | Pass   | 0.000s |
| Line 220: None → 'Not specified'.                                 | <code>page._format_value(None) == "Not specified"</code>                                        | Pass   | 0.000s |
| High risk                                                         | <code>classify_risk_level(3.0) == "HIGH"</code>                                                 | Pass   | 0.000s |
| Low risk                                                          | <code>classify_risk_level(0.3) == "LOW"</code>                                                  | Pass   | 0.000s |
| Medium risk                                                       | <code>classify_risk_level(1.0) == "MEDIUM"</code>                                               | Pass   | 0.000s |
| Minimal risk                                                      | <code>classify_risk_level(0.0) == "MINIMAL"</code>                                              | Pass   | 0.000s |
| Get risk rating calculation                                       | <code>response.status_code == 200;</code><br><code>data["status"] == "success"</code> (+5 more) | Pass   | 0.002s |

| Test Description                                                                              | Acceptance Criteria                                                                                         | Result | Time   |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| Risk rating clear override                                                                    | <code>response.status_code == 200;</code><br><code>rr["mrc_override"] is None</code> (+1 more)              | Pass   | 0.005s |
| Risk rating override                                                                          | <code>response.status_code == 200;</code><br><code>rr["mrc_override"] == "Unacceptable"</code><br>(+3 more) | Pass   | 0.004s |
| Risk rating override invalid rating                                                           | <code>response.status_code == 400</code>                                                                    | Pass   | 0.002s |
| Risk rating override requires reason                                                          | <code>response.status_code == 400</code>                                                                    | Pass   | 0.003s |
| Line 210: inventory not found → 404.                                                          | <code>r.status_code == 404</code>                                                                           | Pass   | 0.001s |
| Lines 126-134: <code>_auto_select_pages</code> returns known pages.                           | <code>isinstance(pages, list); len(pages) &gt; 0</code>                                                     | Pass   | 0.001s |
| Empty gauge data                                                                              | <code>RiskReportGenerator(output_dir).generate_basic_report()</code>                                        | Pass   | 0.017s |
| Lines 162-164: page raises → continue to next.                                                | <code>isinstance(elements, list)</code>                                                                     | Pass   | 0.001s |
| Lines 148-149: unknown page → warning, skipped.                                               | <code>isinstance(elements, list)</code>                                                                     | Pass   | 0.001s |
| Line 265: <code>get_page_categories</code> .                                                  | <code>isinstance(cats, dict)</code>                                                                         | Pass   | 0.001s |
| High risk portfolio report                                                                    | <code>RiskReportGenerator(output_dir).generate_detailed_report()</code>                                     | Pass   | 0.025s |
| Line 261: <code>list_available_pages</code> .                                                 | <code>isinstance(pages, list); len(pages) &gt; 0</code>                                                     | Pass   | 0.001s |
| Minimal data report                                                                           | <code>RiskReportGenerator(output_dir).generate_basic_report()</code>                                        | Pass   | 0.016s |
| Missing mortgage data                                                                         | <code>RiskReportGenerator(output_dir).generate_detailed_report()</code>                                     | Pass   | 0.023s |
| Lines 60-62: <code>output_dir=None</code> → uses <code>config.get_risk_reports_dir()</code> . | <code>gen.output_dir == risk_dir</code>                                                                     | Pass   | 0.000s |
| Lines 270-336: <code>validate_pages</code> .                                                  | <code>"title" in valid;</code> <code>"nonexistent" in invalid</code>                                        | Pass   | 0.001s |
| Analysis report generation                                                                    | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 0</code>                        | Pass   | 0.018s |
| Basic report generation                                                                       | <code>report_path.exists();</code><br><code>report_path.suffix == ".pdf"</code> (+1 more)                   | Pass   | 0.036s |
| Custom filename                                                                               | <code>report_path.name == custom_name;</code><br><code>report_path.exists()</code>                          | Pass   | 0.017s |
| Custom page selection                                                                         | <code>report_path.exists()</code>                                                                           | Pass   | 0.013s |
| Detailed report generation                                                                    | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 0</code>                        | Pass   | 0.027s |
| Summary report generation                                                                     | <code>report_path.exists();</code><br><code>report_path.stat().st_size &gt; 0</code>                        | Pass   | 0.010s |

| Test Description                                                   | Acceptance Criteria                                                        | Result | Time   |
|--------------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Categories dict initialized                                        | "overview" in gen.categories;<br>"analysis" in gen.categories (+1<br>more) | Pass   | 0.001s |
| Explicit output dir path                                           | isinstance(gen.output_dir, Path)                                           | Pass   | 0.001s |
| Explicit output dir string                                         | gen.output_dir == tmp_path                                                 | Pass   | 0.001s |
| output_dir=None → falls back to<br>config.get_reports_dir('risk'). | gen.output_dir == risk_dir                                                 | Pass   | 0.001s |
| Output dir created                                                 | new_dir.exists()                                                           | Pass   | 0.001s |
| Pages dict initialized                                             | expected <= set(gen.pages.keys())                                          | Pass   | 0.001s |
| Get page categories                                                | isinstance(categories, dict); key in<br>categories                         | Pass   | 0.001s |
| List available pages                                               | isinstance(pages, list); key in pages                                      | Pass   | 0.001s |
| Validate pages invalid                                             | "title" in valid; "nonexistent_page"<br>in invalid                         | Pass   | 0.001s |
| Validate pages valid                                               | len(valid) == 2; len(invalid) == 0                                         | Pass   | 0.001s |
| Complete workflow all types                                        | p.exists(); p.stat().st_size > 0                                           | Pass   | 0.070s |
| Complete workflow basic                                            | report_path.exists();<br>report_path.suffix == ".pdf" (+1<br>more)         | Pass   | 0.019s |
| High risk label                                                    | "High Risk" in str(elements)                                               | Pass   | 0.001s |
| Low risk label                                                     | "Low Risk" in str(elements)                                                | Pass   | 0.001s |
| Moderate risk label                                                | "Moderate Risk" in str(elements)                                           | Pass   | 0.001s |
| Non numeric score formatted                                        | isinstance(elements, list)                                                 | Pass   | 0.000s |
| Very high risk label                                               | "Very High Risk" in str(elements)                                          | Pass   | 0.001s |
| Clay soil adds 1                                                   | result["score"] >= 1                                                       | Pass   | 0.000s |
| Close river adds 1                                                 | result["score"] >= 1                                                       | Pass   | 0.000s |
| High flood risk adds 3                                             | result["score"] >= 3                                                       | Pass   | 0.000s |
| High score level                                                   | result["level"] == "High Risk"                                             | Pass   | 0.000s |
| Low elevation adds 2                                               | result["score"] >= 2                                                       | Pass   | 0.000s |
| Low flood risk adds 1                                              | result["score"] >= 1                                                       | Pass   | 0.000s |
| Low medium score level                                             | result["level"] in ("Medium Risk",<br>"Low-Medium Risk")                   | Pass   | 0.000s |
| Low risk level                                                     | result["level"] == "Low Risk"                                              | Pass   | 0.000s |
| Medium flood risk adds 2                                           | result["score"] >= 2                                                       | Pass   | 0.000s |
| Medium score level                                                 | "Medium" in result["level"] or<br>result["score"] >= 4                     | Pass   | 0.000s |
| Moderate elevation adds 1                                          | result["score"] >= 1                                                       | Pass   | 0.000s |
| No risk factors message                                            | "No major" in<br>result["primary_factors"]                                 | Pass   | 0.000s |
| Peat soil adds 2                                                   | result["score"] >= 2                                                       | Pass   | 0.000s |
| Returns all keys                                                   | k in result                                                                | Pass   | 0.000s |

| Test Description                                                                                                                            | Acceptance Criteria                                                                                                               | Result | Time   |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Very close river adds 2                                                                                                                     | <code>result["score"] &gt;= 2</code>                                                                                              | Pass   | 0.000s |
| Very high flood risk adds 4                                                                                                                 | <code>result["score"] &gt;= 4</code>                                                                                              | Pass   | 0.000s |
| Very high score level                                                                                                                       | <code>result["level"] == "Very High Risk"</code>                                                                                  | Pass   | 0.000s |
| Very low elevation adds 3                                                                                                                   | <code>result["score"] &gt;= 3</code>                                                                                              | Pass   | 0.000s |
| Close 100 to 500m                                                                                                                           | <code>isinstance(result, list)</code>                                                                                             | Pass   | 0.001s |
| Good distance over 1000m                                                                                                                    | <code>isinstance(result, list)</code>                                                                                             | Pass   | 0.001s |
| Moderate 500 to 1000m                                                                                                                       | <code>isinstance(result, list)</code>                                                                                             | Pass   | 0.001s |
| Very close under 100m                                                                                                                       | <code>isinstance(result, list)</code>                                                                                             | Pass   | 0.001s |
| <code>_run_and_save</code> should append startup header lines to the output buffer.                                                         | <code>any("=" * 10 in ln for ln in lines);</code><br><code>any("Audit run complete" in ln for</code><br><code>ln in lines)</code> | Pass   | 0.004s |
| When an exception is raised, an ERROR line should appear in output.                                                                         | <code>any("ERROR" in ln for ln in lines)</code>                                                                                   | Pass   | 0.002s |
| <code>_run_and_save</code> should catch exceptions and still set <code>_running=False</code> .                                              | <code>tr_mod._running is False</code>                                                                                             | Pass   | 0.002s |
| <code>_run_and_save</code> should set <code>_running=False</code> in the finally block.                                                     | <code>tr_mod._running is False</code>                                                                                             | Pass   | 0.022s |
| Each gauge has <code>gauge_id</code> , <code>gauge_name</code> , <code>p_flood</code> , <code>p_flood_pct</code> , <code>threshold</code> , | <code>'gauge_id' in g; 'gauge_name' in g (+8</code><br><code>more)</code>                                                         | Pass   | 0.035s |
| Response has <code>gauges_severe</code> as list of <code>gauge_ids</code> ; for STORM_PORT_SEVERE                                           | <code>'gauges_severe' in data;</code><br><code>isinstance(data['gauges_severe'],</code><br><code>list) (+2 more)</code>           | Pass   | 0.031s |
| <code>gauges_warning</code> list includes GAUGE-002.                                                                                        | <code>'gauges_warning' in data; 'GAUGE-002'</code><br><code>in data['gauges_warning']</code>                                      | Pass   | 0.068s |
| Response has <code>'gauges'</code> list.                                                                                                    | <code>'gauges' in data;</code><br><code>isinstance(data['gauges'], list)</code>                                                   | Pass   | 0.037s |
| Response has <code>portfolio_stress_pnl</code> , <code>portfolio_mtm</code> , <code>num_gauges</code> .                                     | <code>'portfolio_stress_pnl' in data;</code><br><code>'portfolio_mtm' in data (+1 more)</code>                                    | Pass   | 0.032s |
| POST with non-existent <code>storm_id</code> returns 404.                                                                                   | <code>resp.status_code == 404</code>                                                                                              | Pass   | 0.002s |
| POST with no body or empty <code>storm_id</code> returns 400.                                                                               | <code>resp.status_code == 400</code>                                                                                              | Pass   | 0.000s |
| POST STORM-PORT-001 returns 200 with <code>status=success</code> .                                                                          | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>                                                 | Pass   | 0.025s |
| Each hour has <code>water_level</code> , <code>p_flood</code> , <code>portfolio</code> fields.                                              | <code>'hour' in h; 'water_level' in h (+4</code><br><code>more)</code>                                                            | Pass   | 0.038s |
| Non-existent storm returns 404.                                                                                                             | <code>resp.status_code == 404</code>                                                                                              | Pass   | 0.002s |
| Severe storm triggers knock-out ( $P(\text{flood})=100\%$ ).                                                                                | <code>data['hourly'][first_breach]['p_flood']</code><br><code>== 1.0; h['p_flood'] is None</code>                                 | Pass   | 0.031s |

| Test Description                                                                          | Acceptance Criteria                                                                             | Result | Time   |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------|--------|
| Missing gauge_id or storm_id returns 400.                                                 | <code>resp.status_code == 400</code>                                                            | Pass   | 0.001s |
| No JSON body returns error.                                                               | <code>resp.status_code in (400, 500)</code>                                                     | Pass   | 0.002s |
| Response includes portfolio-level stress summary.                                         | <code>'num_trades' in summary;</code><br><code>'total_notional' in summary</code> (+5 more)     | Pass   | 0.038s |
| Response includes probability surface grid.                                               | <code>'probability_surface' in data;</code><br><code>'water_levels' in surface</code> (+7 more) | Pass   | 0.051s |
| POST returns 60-hour forecast array.                                                      | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code> (+1 more)     | Pass   | 0.032s |
| Response includes per-trade summary.                                                      | <code>'trades' in data; len(data['trades']) == 2</code> (+4 more)                               | Pass   | 0.023s |
| Surface water levels cap at severe – no rows above severe.                                | <code>max(surface['water_levels']) &lt;= severe</code>                                          | Pass   | 0.042s |
| Surface must cover full 168h storm horizon regardless of KO.                              | <code>max(surface['hours']) &gt;= 164</code>                                                    | Pass   | 0.073s |
| Empty gauge_id returns 400.                                                               | <code>resp.status_code == 400</code>                                                            | Pass   | 0.001s |
| No open trades at gauge returns 404.                                                      | <code>resp.status_code == 404</code>                                                            | Pass   | 0.003s |
| Null JSON body returns 400.                                                               | <code>resp.status_code == 400</code>                                                            | Pass   | 0.001s |
| Predictor None (no model dir) returns 404.                                                | <code>resp.status_code == 404</code>                                                            | Pass   | 0.226s |
| Gaugets file present uses hydrograph built from real readings.                            | <code>resp.status_code == 200;</code><br><code>data['status'] == 'success'</code>               | Pass   | 0.037s |
| Already running message                                                                   | <code>data["status"] == "running";</code><br><code>"message" in data</code> (+1 more)           | Pass   | 0.001s |
| Already running returns 202                                                               | <code>resp.status_code == 202</code>                                                            | Pass   | 0.001s |
| Response has started status                                                               | <code>data["status"] == "started"</code>                                                        | Pass   | 0.001s |
| Starts run returns 202                                                                    | <code>resp.status_code == 202</code>                                                            | Pass   | 0.001s |
| Empty running pnl returns drawing                                                         | <code>isinstance(result, Drawing)</code>                                                        | Pass   | 0.000s |
| All positive                                                                              | <code>np.all(M &gt; 0.0)</code>                                                                 | Pass   | 0.000s |
| Empirical pairwise correlation approximately matches model correlation for large samples. | <code>abs(emp_corr[i, j] - model_corr[i, j]) &lt; 0.15</code>                                   | Pass   | 0.008s |
| Higher intensity -i longer range -i more spatially coherent (lower variance).             | <code>np.mean(vars_extreme) &lt; np.mean(vars_moderate)</code>                                  | Pass   | 0.009s |
| Normalisation ensures catchment mean is preserved.                                        | <code>M.mean() == pytest.approx(1.0, abs=1e-10)</code>                                          | Pass   | 0.000s |

| Test Description                                                                         | Acceptance Criteria                                                                                         | Result | Time   |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|--------|
| Pre-normalisation, $E[M] = \frac{\exp(\sigma^2/2)}{\exp(\sigma^2/2)} = 1$ theoretically. | <code>np.mean(pre_norms) == pytest.approx(1.0, abs=0.05)</code>                                             | Pass   | 0.003s |
| Reproducible with seed                                                                   | <code>np.allclose(M1, M2)</code>                                                                            | Pass   | 0.000s |
| Shape                                                                                    | <code>M.shape == (small_model.n_gauges,)</code>                                                             | Pass   | 0.000s |
| When rng is None, falls back to <code>np.random.randn</code> (line 251).                 | <code>M.shape == (model.n_gauges,); M.mean() == pytest.approx(1.0, abs=1e-10) (+1 more)</code>              | Pass   | 0.000s |
| Catastrophic weights: 15% doublet, 45% cluster, 40% persistent.                          | <code>doublets &lt; 150</code>                                                                              | Pass   | 0.003s |
| Defined categories[catastrophic] [cat='catastrophic']                                    | <code>result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code>                                   | Pass   | 0.000s |
| Defined categories[extreme] [cat='extreme']                                              | <code>result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code>                                   | Pass   | 0.000s |
| Defined categories[moderate] [cat='moderate']                                            | <code>result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code>                                   | Pass   | 0.000s |
| Defined categories[severe] [cat='severe']                                                | <code>result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code>                                   | Pass   | 0.000s |
| Returns sequence type                                                                    | <code>isinstance(result, SequenceType); result in [SequenceType.DOUBLET, SequenceType.CLUSTER, Se...</code> | Pass   | 0.000s |
| Unknown category uses DEFAULT_TYPE_WEIGHTS.                                              | <code>isinstance(result, SequenceType)</code>                                                               | Pass   | 0.000s |
| Lines 122-123: 30% chance subsequent storm goes down.                                    | <code>below_base &gt; 10</code>                                                                             | Pass   | 0.001s |
| Storm index 0 has downward variation (up to -20%).                                       | <code>all(r &lt;= base + 0.001 for r in results)</code>                                                     | Pass   | 0.001s |
| First storm returns positive                                                             | <code>result &gt; 0</code>                                                                                  | Pass   | 0.000s |
| Result should be within +/-20% of base for first storm.                                  | <code>all(r &gt;= lower - 0.001 for r in results); all(r &lt;= upper + 0.001 for r in results)</code>       | Pass   | 0.001s |
| 70% chance of maintaining/increasing subsequent storm intensity.                         | <code>above_base &gt; 100</code>                                                                            | Pass   | 0.000s |
| Subsequent storm positive                                                                | <code>result &gt; 0</code>                                                                                  | Pass   | 0.000s |
| <code>max(0.1, ...)</code> prevents zero or negative intensity.                          | <code>result &gt;= 0.1</code>                                                                               | Pass   | 0.000s |
| Bool passed through                                                                      | <code>result["flag"] is True</code>                                                                         | Pass   | 0.000s |
| Empty params                                                                             | <code>result == {}</code>                                                                                   | Pass   | 0.000s |
| Float passed through                                                                     | <code>result["rate"] == 0.05</code>                                                                         | Pass   | 0.000s |
| Int passed through                                                                       | <code>result["n"] == 42</code>                                                                              | Pass   | 0.000s |

| Test Description                                             | Acceptance Criteria                                                                               | Result | Time   |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Long list summarised                                         | <code>isinstance(result["items"], str); "15 items" in result["items"]</code>                      | Pass   | 0.000s |
| None passed through                                          | <code>result["x"] is None</code>                                                                  | Pass   | 0.000s |
| Object converted to string                                   | <code>result["obj"] == "my_object"</code>                                                         | Pass   | 0.000s |
| Short list converted to str                                  | <code>isinstance(result["items"], list); len(result["items"]) == 3</code>                         | Pass   | 0.000s |
| String passed through                                        | <code>result["name"] == "test"</code>                                                             | Pass   | 0.000s |
| Tuple treated like list                                      | <code>isinstance(result["t"], list)</code>                                                        | Pass   | 0.000s |
| Saves and returns response                                   | <code>resp is not None</code>                                                                     | Pass   | 0.026s |
| Returns false on oserror                                     | <code>result is False; _save_audit_log([]) is False (+1 more)</code>                              | Pass   | 0.000s |
| Saves entries                                                | <code>result is True; json.loads(p.read_text())[0]["event"] == "created"</code>                   | Pass   | 0.001s |
| Returns false on oserror                                     | <code>_save_bcbs239({}) is False; _save_raci({}) is False (+2 more)</code>                        | Pass   | 0.000s |
| Saves data                                                   | <code>_save_bcbs239({"principles": []}) is True; _save_raci({"roles": []}) is True</code>         | Pass   | 0.000s |
| Returns false on oserror                                     | <code>_save_bcbs239({}) is False; _save_raci({}) is False (+2 more)</code>                        | Pass   | 0.000s |
| Saves bibliography                                           | <code>_save_bibliography({"references": [], "categories": []}) ...</code>                         | Pass   | 0.000s |
| Save error is reraised                                       | —                                                                                                 | Pass   | 0.011s |
| Creates gauge file                                           | <code>(tmp_path / "gaugets" / "GAUGE-001.json").exists()</code>                                   | Pass   | 0.000s |
| If gauge file already exists, merge storm_responses into it. | <code>"flood_simulation" in data; "storm_responses" in data</code>                                | Pass   | 0.002s |
| Multiple gauges                                              | <code>(gaugets_dir / "GAUGE-001.json").exists(); (gaugets_dir / "GAUGE-002.json").exists()</code> | Pass   | 0.000s |
| Saved file has responses                                     | <code>"storm_responses" in data; len(data["storm_responses"]["responses"]) == 3</code>            | Pass   | 0.002s |
| Returns false on oserror                                     | <code>_save_bcbs239({}) is False; _save_raci({}) is False (+2 more)</code>                        | Pass   | 0.000s |
| Saves documents                                              | <code>_save_gov_documents([{"doc_id": "D1"}]) is True</code>                                      | Pass   | 0.000s |
| Creates file                                                 | <code>output_path.exists(); output_path</code>                                                    | Pass   | 0.000s |



| Test Description                    | Acceptance Criteria                                                                                                       | Result | Time   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Creates parent dirs                 | <code>output_path.exists()</code>                                                                                         | Pass   | 0.000s |
| File has hazard curves              | <code>"GAUGE-001" in data["hazard_curves"]</code>                                                                         | Pass   | 0.001s |
| File has metadata                   | <code>"metadata" in data;</code><br><code>data["metadata"]["catchment_id"]</code><br><code>== "thames"</code>             | Pass   | 0.002s |
| Metadata merged                     | <code>data["metadata"]["num_storms"] == 100</code>                                                                        | Pass   | 0.002s |
| Returns false on oserror            | <code>result is False; _save_audit_log([])</code><br><code>is False (+1 more)</code>                                      | Pass   | 0.000s |
| Saves data                          | <code>result is True;</code><br><code>json.loads(p.read_text())["models"]</code><br><code>== []</code>                    | Pass   | 0.001s |
| Returns false on oserror            | <code>result is False; _save_audit_log([])</code><br><code>is False (+1 more)</code>                                      | Pass   | 0.000s |
| Saves meetings                      | <code>result is True;</code><br><code>json.loads(p.read_text())[0]["id"] ==</code><br><code>"M1"</code>                   | Pass   | 0.001s |
| Returns false on oserror            | <code>_save_bcbs239({}) is False;</code><br><code>_save_raci({}) is False (+2 more)</code>                                | Pass   | 0.000s |
| Saves data                          | <code>_save_bcbs239({"principles": []}) is</code><br><code>True; _save_raci({"roles": []}) is</code><br><code>True</code> | Pass   | 0.000s |
| Above max depth returns one         | <code>scalar_depth_damage(100.0) ==</code><br><code>pytest.approx(1.0)</code>                                             | Pass   | 0.000s |
| Between control points interpolated | <code>0.25 &lt; d &lt; 0.40</code>                                                                                        | Pass   | 0.000s |
| Damage between 0 and 1              | <code>0 &lt;= scalar_depth_damage(depth) &lt;=</code><br><code>1.0</code>                                                 | Pass   | 0.000s |
| Half meter damage                   | <code>d == pytest.approx(DAMAGE_POINTS[2],</code><br><code>abs=1e-6)</code>                                               | Pass   | 0.000s |
| Max depth returns one               | <code>scalar_depth_damage(DEPTH_POINTS[-1])</code><br><code>== pytest.approx(1.0)</code>                                  | Pass   | 0.000s |
| Monotonically increasing            | <code>all(damages[i] &lt;= damages[i + 1] for</code><br><code>i in range(len(damag...</code>                              | Pass   | 0.000s |
| Negative depth returns zero         | <code>scalar_depth_damage(-1.0) == 0.0;</code><br><code>scalar_depth_damage(-100.0) == 0.0</code>                         | Pass   | 0.000s |
| One meter damage                    | <code>d == pytest.approx(DAMAGE_POINTS[3],</code><br><code>abs=1e-6)</code>                                               | Pass   | 0.000s |
| Returns float                       | <code>isinstance(scalar_depth_damage(0.75),</code><br><code>float)</code>                                                 | Pass   | 0.000s |
| Small depth returns small damage    | <code>d == pytest.approx(DAMAGE_POINTS[1],</code><br><code>abs=1e-9)</code>                                               | Pass   | 0.000s |
| Zero depth returns zero             | <code>scalar_depth_damage(0.0) == 0.0</code>                                                                              | Pass   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                          | Result | Time   |
|--------------------------------------------------------------------------|--------------------------------------------------------------|--------|--------|
| Every control point must return a value in [0, 1].                       | <code>0.0 &lt;= val &lt;= 1.0</code>                         | Pass   | 0.000s |
| Very small positive depth (between 0 and first control point).           | <code>0.0 &lt; d &lt; 0.05</code>                            | Pass   | 0.000s |
| 'base.level_m' field — used in Stress tab storm info bar.                | <code>'base.level_m' in s</code>                             | Pass   | 0.004s |
| 'duration_hours' field — used in Stress tab storm info bar.              | <code>'duration_hours' in s</code>                           | Pass   | 0.004s |
| 'effective_precipitation_mm' field — shown in storm row.                 | <code>'effective_precipitation_mm' in s</code>               | Pass   | 0.003s |
| 'gauges_severe' field — severity indicator in storm row.                 | <code>'gauges_severe' in s</code>                            | Pass   | 0.004s |
| 'intensity_category' field — colour-coded in storm row.                  | <code>'intensity_category' in s</code>                       | Pass   | 0.003s |
| 'level_change_m' field — used in Stress tab storm info bar.              | <code>'level_change_m' in s</code>                           | Pass   | 0.003s |
| 'max_trigger' field — used for trigger colour coding in Historical list. | <code>'max_trigger' in s</code>                              | Pass   | 0.003s |
| 'name' field — shown in storm row title.                                 | <code>'name' in s</code>                                     | Pass   | 0.003s |
| 'peak_level_m' field — primary sort key and shown in Trading Desk label. | <code>'peak_level_m' in s</code>                             | Pass   | 0.004s |
| 'peak_position' field — used in hydrograph synthesis.                    | <code>'peak_position' in s</code>                            | Pass   | 0.003s |
| Hour label element must be present to display current hour.              | <code>'sp-sim-hour-label' in sim_js</code>                   | Pass   | 0.000s |
| Scrubber element must have the expected DOM ID.                          | <code>'sp-sim-scrubber' in sim_js</code>                     | Pass   | 0.000s |
| scrubber.max must equal STORM_SIMULATION_HOURS - 1 (0-indexed).          | <code>f"scrubber.max = '{expected}'" in sim_js</code>        | Pass   | 0.000s |
| Regression guard: the old wrong value '59' must not be present.          | <code>"scrubber.max = '59'" not in sim_js</code>             | Pass   | 0.000s |
| scrubber.min must be '0' so the timeline starts at hour 0.               | <code>"scrubber.min = '0'" in sim_js</code>                  | Pass   | 0.000s |
| Scrubber element must be an input[type=range].                           | <code>"scrubber.type = 'range'" in sim_js</code>             | Pass   | 0.000s |
| Case insensitive                                                         | <code>seq.type_color('DOUBLET') == colors.lightyellow</code> | Pass   | 0.000s |

| Test Description                                                                 | Acceptance Criteria                                                                                                                            | Result | Time   |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Cluster                                                                          | <code>seq_type.color('cluster') == colors.lightsalmon</code>                                                                                   | Pass   | 0.000s |
| Doublet                                                                          | <code>seq_type.color('doublet') == colors.lightyellow</code>                                                                                   | Pass   | 0.000s |
| Isolated                                                                         | <code>seq_type.color('isolated') == colors.lightblue</code>                                                                                    | Pass   | 0.000s |
| None returns white                                                               | <code>seq_type.color(None) == colors.white</code>                                                                                              | Pass   | 0.000s |
| Persistent                                                                       | <code>seq_type.color('persistent') == colors.mistyrose</code>                                                                                  | Pass   | 0.000s |
| Unknown returns white                                                            | <code>seq_type.color('unknown') == colors.white</code>                                                                                         | Pass   | 0.000s |
| Response declares the max_depth damage model.                                    | <code>data["damage_model"] == "max_depth"</code>                                                                                               | Pass   | 0.006s |
| Each property entry includes num_sequence_floods.                                | <code>prop_a["num_sequence_floods"] &gt;= 1</code>                                                                                             | Pass   | 0.004s |
| Response includes num_storms_in_sequence.                                        | <code>"num_storms_in_sequence" in data;</code><br><code>data["num_storms_in_sequence"] &gt;= 1</code>                                          | Pass   | 0.008s |
| Damage is max depth, not $0.10 + 0.18 = 0.28$ .                                  | <code>prop_a["damage_ratio"] &lt; 0.20</code>                                                                                                  | Pass   | 0.005s |
| PROP-A: second storm (0.65m) is worst — sequence damage uses 0.18 ratio.         | <code>prop_a["damage_ratio"] == pytest.approx(0.18);</code><br><code>prop_a["damage_amount"] == pytest.approx(350_000 * 0.18)</code> (+1 more) | Pass   | 0.006s |
| PROP-B: first storm depth=0 so excluded, second storm 0.55m -i included.         | <code>"PROP-B" in prop_ids</code>                                                                                                              | Pass   | 0.004s |
| PROP-C is flooded by STORM-other, not STORM-test1 — must not appear.             | <code>"PROP-C" not in prop_ids</code>                                                                                                          | Pass   | 0.004s |
| Sequence total damage i sum of individual storm damages (not additive).          | <code>seq.damage &lt; dmg.d1a + dmg.d1b</code>                                                                                                 | Pass   | 0.005s |
| Sequence endpoint options                                                        | <code>r.status_code == 200</code>                                                                                                              | Pass   | 0.001s |
| Sequence endpoint returns 200                                                    | <code>r.status_code == 200;</code> <code>data["status"] == "success"</code>                                                                    | Pass   | 0.006s |
| Sequence id in response                                                          | <code>data["sequence_id"] == "STORM-test1"</code>                                                                                              | Pass   | 0.005s |
| Unknown sequence returns 404                                                     | <code>r.status_code == 404</code>                                                                                                              | Pass   | 0.006s |
| Worst storm (17% damage on 300k -i, 249k i 280k outstanding -i, negative equity. | <code>data["portfolio"]["mortgages_in_negative_equity"] == 1;</code> <code>prop["negative_equity"] is True</code>                              | Pass   | 0.003s |

| Test Description                                                | Acceptance Criteria                                                                                   | Result | Time   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|--------|
| Sequence num floods is four                                     | <code>data["properties"][0]["num.sequence_floods"] == 4</code>                                        | Pass   | 0.005s |
| Sequence damage uses storm p3 (1.8m, ratio=0.17), not p1 or p2. | <code>prop["worst_storm_id"] == "STORM-p3"; prop["damage_ratio"] == pytest.approx(0.17)</code>        | Pass   | 0.005s |
| Querying seqA1 does not return seqB damage.                     | <code>dmg_a != dmg_b; dmg_a != dmg_a + dmg_b</code>                                                   | Pass   | 0.004s |
| Seqa first storm returns only its damage                        | <code>prop["damage.amount"] == pytest.approx(400_000 * 0.12)</code>                                   | Pass   | 0.026s |
| Seqb storm returns only seqb damage                             | <code>prop["damage.amount"] == pytest.approx(400_000 * 0.06)</code>                                   | Pass   | 0.003s |
| Known sequence returns success                                  | <code>r.status_code == 200; r.get_json()["status"] == "success"</code>                                | Pass   | 0.015s |
| No pts dir returns 404                                          | <code>r.status_code == 404; r.get_json()["status"] == "error"</code><br>(+2 more)                     | Pass   | 0.001s |
| Options returns ok                                              | <code>r.status_code == 200; r.status_code == 200</code>                                               | Pass   | 0.000s |
| Properties sorted by damage descending                          | <code>amounts == sorted(amounts, reverse=True); amounts == sorted(amounts, reverse=True)</code>       | Pass   | 0.004s |
| Property entry has financial fields                             | <code>"property_id" in prop; "property_value" in prop</code> (+9 more)                                | Pass   | 0.003s |
| Property entry has num sequence floods                          | <code>"num.sequence_floods" in prop; prop["num.sequence_floods"] &gt;= 1</code>                       | Pass   | 0.005s |
| Property entry has worst storm id                               | <code>"worst_storm_id" in prop</code>                                                                 | Pass   | 0.004s |
| Response has damage model                                       | <code>r.get_json()["damage_model"] == "max_depth"</code>                                              | Pass   | 0.004s |
| Response has num storms                                         | <code>"num.storms.in.sequence" in r.get_json(); r.get_json()["num.storms.in.sequence"] &gt;= 1</code> | Pass   | 0.004s |
| Response has portfolio                                          | <code>"portfolio" in data; "total_properties" in port</code> (+4 more)                                | Pass   | 0.005s |
| Response has properties list                                    | <code>"properties" in data; len(data["properties"]) &gt;= 1</code> (+2 more)                          | Pass   | 0.005s |
| Response has sequence id                                        | <code>r.get_json()["sequence_id"] == seq_id</code>                                                    | Pass   | 0.004s |

| Test Description                                                    | Acceptance Criteria                                                                                          | Result | Time   |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------|--------|
| Unknown sequence returns 404                                        | <code>r.status_code == 404;</code>                                                                           | Pass   | 0.003s |
| Alert flags length matches num gauges                               | <code>r.get_json()["status"] == "error"</code><br><code>all(len(r["alert"]) == 3 for r in records)</code>    | Pass   | 0.000s |
| All have intensity category                                         | <code>all(r["intensity_category"] in valid for r in records)</code>                                          | Pass   | 0.000s |
| All have sequence id                                                | <code>all("sequence_id" in r for r in records)</code>                                                        | Pass   | 0.000s |
| All have sequence type                                              | <code>all(r["sequence_type"] in valid for r in records)</code>                                               | Pass   | 0.000s |
| All have total precip                                               | <code>all("total_precip_mm" in r for r in records); all(r["total_precip_mm"] &gt; 0 for r in records)</code> | Pass   | 0.000s |
| Peak hours in valid range                                           | <code>all(0 &lt;= h &lt;= 167 for h in r["peak_hours"])</code>                                               | Pass   | 0.000s |
| Peak hours length matches num gauges                                | <code>all(len(r["peak_hours"]) == 3 for r in records)</code>                                                 | Pass   | 0.000s |
| Peaks length matches num gauges                                     | <code>all(len(r["peaks_m"]) == 3 for r in records)</code>                                                    | Pass   | 0.000s |
| Peaks positive                                                      | <code>all(p &gt; 0 for p in r["peaks_m"])</code>                                                             | Pass   | 0.000s |
| Severe flags length matches num gauges                              | <code>all(len(r["severe"]) == 3 for r in records)</code>                                                     | Pass   | 0.000s |
| Severe implies warning                                              | <code>r["warning"][i]</code>                                                                                 | Pass   | 0.000s |
| Warning flags length matches num gauges                             | <code>all(len(r["warning"]) == 3 for r in records)</code>                                                    | Pass   | 0.000s |
| Every sequence that breached warning must also have breached alert. | <code>r["alert"][i]</code>                                                                                   | Pass   | 0.000s |
| Returns 404 when file missing                                       | <code>r.status_code in (200, 404, 500)</code>                                                                | Pass   | 0.004s |
| Lines 33-35: send_file when HTML file actually exists.              | <code>r.status_code in (200, 404, 500)</code>                                                                | Pass   | 0.059s |
| Returns json error or html                                          | <code>r.status_code in (200, 404, 500)</code>                                                                | Pass   | 0.001s |
| No coverage dir returns 404                                         | <code>r.status_code == 404</code>                                                                            | Pass   | 0.001s |
| With coverage index returns 200                                     | <code>r.status_code == 200</code>                                                                            | Pass   | 0.002s |
| The response body is valid JSON with an error status.               | <code>r.status_code in (500, 503); data is not None (+1 more)</code>                                         | Pass   | 0.001s |
| Without gauge.json, visualization fails with 500.                   | <code>r.status_code in (500, 503)</code>                                                                     | Pass   | 0.001s |
| No gauge data returns error status                                  | <code>data["status"] == "error"</code>                                                                       | Pass   | 0.001s |
| Lines 97-99: exception in serve_visualization_for_catchment → 500.  | <code>r.status_code in (302, 500)</code>                                                                     | Pass   | 0.025s |

| Test Description                                                                  | Acceptance Criteria                                                                                                          | Result | Time   |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Matching is case-insensitive.                                                     | <code>r.status_code == 302</code>                                                                                            | Pass   | 0.000s |
| 'thames' catchment redirects to main visualization route.                         | <code>r.status_code == 302</code>                                                                                            | Pass   | 0.000s |
| Unknown catchment returns 400                                                     | <code>r.status_code == 400; data["status"] == "error" (+1 more)</code>                                                       | Pass   | 0.000s |
| Unsupported catchment message                                                     | <code>r.status_code == 400; "thames" in r.get_json()["message"].lower()</code>                                               | Pass   | 0.000s |
| Lines 59-65: ImportError → 503.                                                   | <code>r.status_code in (500, 503)</code>                                                                                     | Pass   | 0.028s |
| Lines 50-57: mock TCEventVisualization to return existing file.                   | <code>r.status_code in (200, 500)</code>                                                                                     | Pass   | 0.034s |
| <code>main()</code> calls <code>app.run()</code> with configured host/port/debug. | <code>len(run_calls) == 1; run_calls[0]["host"] == config.SERVER_HOST (+1 more)</code>                                       | Pass   | 0.027s |
| <code>server.main()</code> is callable (test by importing, not running).          | <code>callable(main)</code>                                                                                                  | Pass   | 0.000s |
| Simulate what routes/ <code>gauges.py:generate_report()</code> does.              | <code>report_path.exists(); report_path.suffix == '.pdf' (+1 more)</code>                                                    | Pass   | 0.166s |
| Test with actual Thames gauge data if available.                                  | <code>report_path.exists(); report_path.stat().st_size &gt; 5000</code>                                                      | Pass   | 0.019s |
| Simulate what routes/ <code>properties.py:generate_report()</code> does.          | <code>report_path.exists(); report_path.suffix == '.pdf' (+1 more)</code>                                                    | Pass   | 0.035s |
| Test with actual Thames portfolio data if available.                              | <code>report_path.exists(); report_path.stat().st_size &gt; 10000</code>                                                     | Pass   | 0.370s |
| Case insensitive thames                                                           | <code>r.status_code == 200</code>                                                                                            | Pass   | 0.001s |
| Missing catchment id returns 400                                                  | <code>r.status_code == 400</code>                                                                                            | Pass   | 0.001s |
| No json returns 400                                                               | <code>r.status_code in (400, 415, 500)</code>                                                                                | Pass   | 0.002s |
| Line 64: non-empty body but null/empty catchment → 400.                           | <code>r.status_code == 400; "Catchment ID is required" in data["message"]</code>                                             | Pass   | 0.001s |
| Options returns 200                                                               | <code>r.status_code == 200</code>                                                                                            | Pass   | 0.001s |
| Unknown catchment returns 400                                                     | <code>r.status_code == 400; data["status"] == "error"</code>                                                                 | Pass   | 0.001s |
| Valid thames returns success                                                      | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                                                     | Pass   | 0.001s |
| Existing mortgage dict updated                                                    | <code>data["Mortgage"]["Header"]["ExistingFieldID"] == "keep"; data["Mortgage"]["Header"]["MortgageID"] == "MORT-new"</code> | Pass   | 0.000s |
| Initialises header key when absent                                                | <code>"Header" in data["Mortgage"]</code>                                                                                    | Pass   | 0.000s |

| Test Description                                   | Acceptance Criteria                                                                                           | Result | Time   |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| Initialises mortgage key when absent               | "Mortgage" in data                                                                                            | Pass   | 0.000s |
| Missing property id key defaults empty string      | data["Mortgage"]["Header"]["PropertyID"] == ""                                                                | Pass   | 0.000s |
| Sets catchment id                                  | data["Mortgage"]["Header"]["CatchmentID"] == "thames"                                                         | Pass   | 0.000s |
| Sets loan details fields                           | loan["OriginalLoanAmount"] == fd["loan.amount"];<br>loan["CurrentBalance"] == fd["current.balance"] (+3 more) | Pass   | 0.000s |
| Sets mortgage id                                   | data["Mortgage"]["Header"]["MortgageID"] == "MORT-abc12"                                                      | Pass   | 0.000s |
| Sets property id in header                         | data["Mortgage"]["Header"]["PropertyID"] == "PROP-xyz"                                                        | Pass   | 0.000s |
| Catchment id set in attrs                          | data["PropertyHeader"]["PropertyAttributes"]["CatchmentID"] == "thames"                                       | Pass   | 0.000s |
| Elevation sets ground level                        | loc["RiskAssessment"]["GroundLevelMeters"] == pytest.approx(51.45)                                            | Pass   | 0.000s |
| Empty reference gauge indices no ref key           | "ReferenceGauges" not in data.get("PropertyHeader", {})                                                       | Pass   | 0.000s |
| Lat lon set in location                            | loc["LatitudeDegrees"] == pytest.approx(51.45);<br>loc["LongitudeDegrees"] == pytest.approx(-0.31)            | Pass   | 0.000s |
| No elevation in location does not set ground level | "RiskAssessment" not in loc                                                                                   | Pass   | 0.000s |
| No reference gauge indices no ref key              | "ReferenceGauges" not in data.get("PropertyHeader", {})                                                       | Pass   | 0.000s |
| Property id set in attrs                           | data["PropertyHeader"]["PropertyAttributes"]["PropertyID"] == "PROP-xyz"                                      | Pass   | 0.000s |
| Reference gauges fallback when not in map          | ref == ["GAUGE-001", "GAUGE-002", "GAUGE-003"]                                                                | Pass   | 0.000s |
| Reference gauges set from gauge id map             | ref == ["GAUGE-abc", "GAUGE-def", "GAUGE-ghi"]                                                                | Pass   | 0.000s |
| Adds all components                                | mock.html.add_child.call_count > 0                                                                            | Pass   | 0.068s |
| Copyright element added                            | len(calls) > 0                                                                                                | Pass   | 0.068s |
| Returns self                                       | result is mgr                                                                                                 | Pass   | 0.069s |
| Required keys                                      | key in styles                                                                                                 | Pass   | 0.000s |
| Returns stylesheet                                 | isinstance(styles, StyleSheet1)                                                                               | Pass   | 0.000s |
| Styles are paragraph styles                        | isinstance(styles[key], ParagraphStyle)                                                                       | Pass   | 0.000s |
| All defined categories                             | isinstance(result, bool)                                                                                      | Pass   | 0.000s |



| Test Description                                                          | Acceptance Criteria                                                                                               | Result | Time   |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|--------|
| Catastrophic has 85% chance — over many trials, majority should be True.  | <code>sum(results) &gt; 140</code>                                                                                | Pass   | 0.000s |
| Minimal has 5% chance — most should be False.                             | <code>sum(results) &lt; 30</code>                                                                                 | Pass   | 0.000s |
| Returns bool                                                              | <code>isinstance(result, bool)</code>                                                                             | Pass   | 0.000s |
| Unknown category uses default 0.15 probability.                           | <code>50 &lt; sum(results) &lt; 350</code>                                                                        | Pass   | 0.001s |
| <code>sp_sim.get_js()</code> must return a non-empty string.              | <code>len(sim_js) &gt; 100</code>                                                                                 | Pass   | 0.000s |
| <code>spSeekTo()</code> must be defined — called by scrubber oninput.     | <code>'spSeekTo' in sim_js</code>                                                                                 | Pass   | 0.000s |
| <code>spSetSpeed()</code> must be defined — called by speed buttons.      | <code>'spSetSpeed' in sim_js</code>                                                                               | Pass   | 0.000s |
| <code>spStopAnim()</code> must be defined — cleans up animation interval. | <code>'spStopAnim' in sim_js</code>                                                                               | Pass   | 0.000s |
| Business days is three months                                             | <code>60 &lt;= NUM_BUSINESS_DAYS &lt;= 66</code>                                                                  | Pass   | 0.000s |
| Daily vol is small                                                        | <code>0.0001 &lt;= DAILY_HAZARD_VOL &lt;= 0.001</code>                                                            | Pass   | 0.000s |
| Random walk floors at 0.0001 — vol must not exceed that floor.            | <code>DAILY_HAZARD_VOL &gt; 0; MAX_TOTAL_MOVE &gt; 0</code>                                                       | Pass   | 0.000s |
| Cumulative cap should allow meaningful drift.                             | <code>MAX_TOTAL_MOVE &gt;= 5 * DAILY_HAZARD_VOL</code>                                                            | Pass   | 0.000s |
| Max move bounds vol                                                       | <code>MAX_TOTAL_MOVE &gt; DAILY_HAZARD_VOL; MAX_TOTAL_MOVE &lt;= 0.01</code>                                      | Pass   | 0.000s |
| Test complete gauge analysis workflow.                                    | <code>results['success'] is True; results['gauge']['gauge_id'] == "THAMES_TEDDINGTON" (+2 more)</code>            | Pass   | 0.003s |
| Handle non-existent gauge gracefully.                                     | <code>results['success'] is False; len(results['errors']) &gt; 0</code>                                           | Pass   | 0.001s |
| Analysis should include flood status.                                     | <code>'status' in results['flood_analysis']; results['flood_analysis']['status'] in ["NORMAL", "ALERT"]...</code> | Pass   | 0.003s |
| Test loading gauge metadata from JSON.                                    | <code>gauges is not None; len(gauges) == 4 (+2 more)</code>                                                       | Pass   | 0.001s |
| Test loading gauge time series.                                           | <code>readings is not None; "THAMES_TEDDINGTON" in readings (+1 more)</code>                                      | Pass   | 0.002s |
| Error message lists available ids                                         | <code>"GAUGE-test001" in str(exc_info.value)</code>                                                               | Pass   | 0.001s |
| Invalid gauge id raises                                                   | —                                                                                                                 | Pass   | 0.002s |

| Test Description                                                      | Acceptance Criteria                                                                                                             | Result | Time   |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Named file gauge ids                                                  | d["gauge_ids"] == ["GAUGE-test002"]                                                                                             | Pass   | 0.000s |
| Named file peaks length one                                           | all(len(r["peaks_m"]) == 1 for r in<br>d["sequences"])                                                                          | Pass   | 0.000s |
| Named file schema version                                             | d["schema_version"] ==<br>SCHEMA.VERSION.SPATIAL                                                                                | Pass   | 0.003s |
| Named file sequence count                                             | len(d["sequences"]) == 40                                                                                                       | Pass   | 0.000s |
| Named output file written                                             | (gauge_dir /<br>"sequence_gauge_GAUGE-test002.json").exists()                                                                   | Pass   | 0.000s |
| Summary num gauges is one                                             | single_run["num_gauges"] == 1                                                                                                   | Pass   | 0.000s |
| Single gauge classifier result dict                                   | isinstance(result, dict)                                                                                                        | Pass   | 6.842s |
| Single gauge + train_classifier=True<br>covers lines 353-361.         | result["num_gauges"] == 1;<br>stressm_dir.exists()                                                                              | Pass   | 6.710s |
| All mortgages fail returns empty list                                 | result["processing_stats"]["failed_mortgages"]<br>== 2; result["data"]["mortgages"] ==<br>[]                                    | Pass   | 0.001s |
| When _generate_single_mortgage<br>raises, the generator continues and | result["processing_stats"]["failed_mortgages"]<br>== 1;<br>result["processing_stats"]["successful_mortgages"]<br>== 2 (+1 more) | Pass   | 0.018s |
| Find nearest gauge                                                    | gauge_id == "THAMES.TEDDINGTON";<br>distance < 1.0                                                                              | Pass   | 0.000s |
| Full analysis high risk property                                      | results["success"] is True;<br>results["property"]["property_id"] ==<br>"THAMES_TEST_002" (+1 more)                             | Pass   | 0.004s |
| Full analysis low risk property                                       | results["success"] is True;<br>results["property"]["property_id"] ==<br>"THAMES_TEST_003" (+2 more)                             | Pass   | 0.003s |
| Load gauge data                                                       | gauge_meta is not None;<br>len(gauge_meta) == 3 (+2 more)                                                                       | Pass   | 0.002s |
| Load property data                                                    | properties is not None;<br>len(properties) == 3 (+3 more)                                                                       | Pass   | 0.002s |
| Clay medium high                                                      | "Medium" in r or "high" in r.lower()                                                                                            | Pass   | 0.000s |
| Peat high risk                                                        | "High" in r                                                                                                                     | Pass   | 0.000s |
| Rock keyword                                                          | "Low" in r                                                                                                                      | Pass   | 0.000s |
| Rock low risk                                                         | "Low" in r                                                                                                                      | Pass   | 0.000s |
| Sand medium risk                                                      | "Medium" in r                                                                                                                   | Pass   | 0.000s |
| Unknown soil                                                          | "Assessment" in r or "chalk" in r                                                                                               | Pass   | 0.000s |
| STORM-cccccccc (alert-only, no<br>warning/severe) must be last.       | sorted_storms[-1]["storm_id"] ==<br>"STORM-cccccccc"                                                                            | Pass   | 0.000s |
| STORM-bbbbbbbb (2 severe) must<br>rank before all others.             | sorted_storms[0]["storm_id"] ==<br>"STORM-bbbbbbbb"                                                                             | Pass   | 0.000s |

| Test Description                                                            | Acceptance Criteria                                       | Result | Time   |
|-----------------------------------------------------------------------------|-----------------------------------------------------------|--------|--------|
| gauges_severe must be non-increasing.                                       | severe_counts == sorted(severe_counts, reverse=True)      | Pass   | 0.000s |
| Defaults                                                                    | p.base_range_km == 40.0; p.nugget == 0.05 (+2 more)       | Pass   | 0.000s |
| From dict                                                                   | p.base_range_km == 50.0; p.nugget == 0.10 (+1 more)       | Pass   | 0.000s |
| Load from file                                                              | p.base_range_km == 40.0; p.nugget == 0.05                 | Pass   | 0.025s |
| At gauge location returns wse directly                                      | result == pytest.approx(3.0)                              | Pass   | 0.000s |
| Empty gauge data returns zero                                               | result == 0.0                                             | Pass   | 0.000s |
| Idw returns weighted value                                                  | 2.5 <= result <= 4.5                                      | Pass   | 0.000s |
| Single gauge                                                                | result > 0                                                | Pass   | 0.000s |
| Lines 119-120: distance ; 1m → direct return.                               | result == pytest.approx(7.0)                              | Pass   | 0.000s |
| Line 97-98: no gauge data → 0.0.                                            | spatial_interpolate_wse(51.5, -0.1, {}, {}) == 0.0        | Pass   | 0.000s |
| Lines 113-124: IDW with two equidistant gauges.                             | result == pytest.approx(2.0, rel=0.01)                    | Pass   | 0.000s |
| Detailed report uses more pages than basic, so should produce larger file.  | detailed.stat().st_size >= basic.stat().st_size           | Pass   | 0.033s |
| Generate analysis report custom filename                                    | path.name == "analysis.pdf"                               | Pass   | 0.004s |
| Generate analysis report custom filename                                    | path.name == "analysis.pdf"                               | Pass   | 0.010s |
| Generate analysis report exists                                             | gen.generate_analysis_report(gauge_data).exists()         | Pass   | 0.004s |
| Generate analysis report exists                                             | gen.generate_analysis_report(minimal_flood_data).exists() | Pass   | 0.012s |
| Generate analysis report with time-series                                   | path.exists()                                             | Pass   | 0.003s |
| Generate basic report custom filename                                       | path.name == "basic.pdf"                                  | Pass   | 0.017s |
| Generate basic report exists                                                | gen.generate_basic_report(gauge_data).exists()            | Pass   | 0.017s |
| Generate basic report exists                                                | gen.generate_basic_report(minimal_flood_data).exists()    | Pass   | 0.026s |
| Generate basic report non empty                                             | path.stat().st_size > 0                                   | Pass   | 0.017s |
| Generate basic report non empty                                             | path.stat().st_size > 0                                   | Pass   | 0.015s |
| Generate basic with timeseries                                              | path.exists()                                             | Pass   | 0.021s |
| Generate detailed report exists                                             | gen.generate_detailed_report(minimal_flood_data).exists() | Pass   | 0.015s |
| Detailed report includes mortgage_analysis — verify call reaches that page. | —                                                         | Pass   | 0.020s |

| Test Description                                                    | Acceptance Criteria                                                                                           | Result | Time   |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|--------|
| Generate monitoring report custom filename                          | <code>path.name == "mon.pdf"</code>                                                                           | Pass   | 0.004s |
| Generate monitoring report exists                                   | <code>gen.generate_monitoring_report(gauge_data, ts_data).exists()</code>                                     | Pass   | 0.003s |
| Generate monitoring report no timeseries                            | <code>path.exists()</code>                                                                                    | Pass   | 0.003s |
| Generate mortgage focused report custom filename                    | <code>path.name == "mort_focused.pdf"</code>                                                                  | Pass   | 0.019s |
| Generate mortgage focused report exists                             | <code>gen.generate_mortgage_focused_report(prop_data, mort_data...)</code>                                    | Pass   | 0.019s |
| Generate property only report custom filename                       | <code>path.name == "prop_only.pdf"</code>                                                                     | Pass   | 0.017s |
| Generate property only report exists                                | <code>gen.generate_property_only_report(prop_data).exists()</code>                                            | Pass   | 0.017s |
| Generate risk focused report custom filename                        | <code>path.name == "risk_only.pdf"</code>                                                                     | Pass   | 0.011s |
| Generate risk focused report with mortgage                          | <code>path.exists()</code>                                                                                    | Pass   | 0.013s |
| Generate risk focused report without mortgage                       | <code>path.exists()</code>                                                                                    | Pass   | 0.011s |
| Generate summary report custom filename                             | <code>path.name == "summary.pdf"</code>                                                                       | Pass   | 0.009s |
| Generate summary report exists                                      | <code>gen.generate_summary_report(minimal_flood_data).exists()</code>                                         | Pass   | 0.009s |
| Mortgage focused includes mortgage context pages                    | —                                                                                                             | Pass   | 0.019s |
| Risk focused with mortgage includes current status                  | —                                                                                                             | Pass   | 0.013s |
| Risk focused without mortgage excludes current status               | —                                                                                                             | Pass   | 0.015s |
| Only gauge_GAUGE-*_hd.json files are removed, not other files.      | <code>other.exists()</code>                                                                                   | Pass   | 0.224s |
| Multiple stale files are all removed.                               | <code>not s.exists()</code>                                                                                   | Pass   | 0.228s |
| Lines 37-39: old gauge_GAUGE-*_hd.json files are removed.           | <code>stale.exists(); not stale.exists()</code>                                                               | Pass   | 0.231s |
| Every dataset key must appear in _startupDetail — no silent blanks. | <code>f"key === '{k}'" in src</code>                                                                          | Pass   | 0.000s |
| Blotter stat uses trades length                                     | <code>"key === '_tdPreBlotter'" in src;</code><br><code>"data.trades.length + ' trades'" in src</code>        | Pass   | 0.000s |
| Eod history stat uses history length                                | <code>"key === '_tdPreEodHistory'" in src;</code><br><code>"data.history.length + ' snapshots'" in src</code> | Pass   | 0.000s |

| Test Description                                                | Acceptance Criteria                                                                                | Result | Time   |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------|--------|
| Gauge hist stat uses count                                      | "key === '_preGaugeHist'" in src;<br>'data.count' in src                                           | Pass   | 0.000s |
| Gauges stat uses gauges length                                  | "key === '_tdPreGauges'" in src;<br>"data.gauges.length + ' gauges'" in<br>src                     | Pass   | 0.000s |
| Gov audit stat uses events length                               | "key === '_preGovAudit'" in src;<br>"data.events.length + ' events'" in<br>src                     | Pass   | 0.000s |
| Gov bib stat uses references length                             | "key === '_preGovBib'" in src;<br>"data.references.length + ' refs'"<br>in src                     | Pass   | 0.000s |
| Gov docs stat uses documents<br>length                          | "key === '_tdPreGovDocs'" in src;<br>"data.documents.length + ' docs'" in<br>src                   | Pass   | 0.000s |
| Market stat uses object keys gauges                             | "key === '_tdPreMarket'" in src;<br>"Object.keys(data.gauges).length" in<br>src (+1 more)          | Pass   | 0.000s |
| Mortgages stat uses mortgages<br>length                         | "key === '_preMortgages'" in src;<br>"data.mortgages" in src                                       | Pass   | 0.000s |
| Port storms stat uses storms length                             | "key === '_tdPrePortStorms'" in src;<br>"data.storms.length + ' storms'" in<br>src                 | Pass   | 0.000s |
| Pre storms stat uses storms length                              | "key === '_preStorms'" in src;<br>"data.storms.length + ' storms'" in<br>src                       | Pass   | 0.000s |
| Property ts stat uses total proper-<br>ties                     | "key === '_prePropertyTS'" in src;<br>'data.data.summary.total.properties'<br>in src (+1 more)     | Pass   | 0.000s |
| Stress storms stat uses storms<br>length                        | "key === '_tdPreStressStorms'"<br>in src; "data.storms.length + '<br>scenarios'" in src            | Pass   | 0.000s |
| Yield curve stat uses object keys                               | "key === '_tdPreYieldCurve'" in src;<br>"Object.keys(data.yield_curve).length"<br>in src (+1 more) | Pass   | 0.000s |
| Add to map calls add child                                      | —                                                                                                  | Pass   | 0.001s |
| Add to map output contains script                               | '<script>' in rendered;<br>'window._tdPreloadDone' in rendered                                     | Pass   | 0.001s |
| Add to map method defined                                       | 'def add_to_map' in src                                                                            | Pass   | 0.000s |
| All fourteen endpoints present                                  | ep in js                                                                                           | Pass   | 0.000s |
| Must NOT defer to a button click<br>— runs on DOMContentLoaded. | 'DOMContentLoaded' in js                                                                           | Pass   | 0.000s |

| Test Description                                                     | Acceptance Criteria                                                                       | Result | Time   |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| 'Market state' was renamed to 'Hazard curves'.                       | 'Hazard curves' in js;<br>'"Market state'" not in js and '"Market state"' not in js       | Pass   | 0.000s |
| Sets window._tdPreloadDone so any IIFE can read it via global scope. | 'window._tdPreloadDone = true' in                                                         | Pass   | 0.000s |
| Shows progress popup                                                 | _src('src/visual/intera...<br>'startup-preloader-popup' in js;<br>'startup-pre-bar' in js | Pass   | 0.000s |
| Startup preloader class defined                                      | 'class StartupPreloader' in src                                                           | Pass   | 0.000s |
| Declares pre gauge hist                                              | 'window._preGaugeHist' in<br>_src('src/visual/interactivity/...                           | Pass   | 0.000s |
| Declares pre gov audit                                               | 'window._preGovAudit' in<br>_src('src/visual/interactivity/s...                           | Pass   | 0.000s |
| Declares pre gov bib                                                 | 'window._preGovBib' in<br>_src('src/visual/interactivity/sta...                           | Pass   | 0.000s |
| Declares pre property ts                                             | 'window._prePropertyTS' in<br>_src('src/visual/interactivity...                           | Pass   | 0.000s |
| Declares pre storms                                                  | 'window._preStorms' in<br>_src('src/visual/interactivity/sta...                           | Pass   | 0.000s |
| Declares td pre blotter                                              | 'window._tdPreBlotter' in<br>_src('src/visual/interactivity/...                           | Pass   | 0.000s |
| Declares td pre eod history                                          | 'window._tdPreEodHistory' in<br>_src('src/visual/interactivi...                           | Pass   | 0.000s |
| Declares td pre gauges                                               | 'window._tdPreGauges' in<br>_src('src/visual/interactivity/s...                           | Pass   | 0.000s |
| Declares td pre gov docs                                             | 'window._tdPreGovDocs' in<br>_src('src/visual/interactivity/...                           | Pass   | 0.000s |
| Declares td pre market                                               | 'window._tdPreMarket' in<br>_src('src/visual/interactivity/s...                           | Pass   | 0.000s |
| Declares td pre port storms                                          | 'window._tdPrePortStorms' in<br>_src('src/visual/interactivi...                           | Pass   | 0.000s |
| Declares td pre stress storms                                        | 'window._tdPreStressStorms' in<br>_src('src/visual/interacti...                           | Pass   | 0.000s |
| Declares td pre yield curve                                          | 'window._tdPreYieldCurve' in<br>_src('src/visual/interactivi...                           | Pass   | 0.000s |
| Line 46: GET /apple-touch-icon.png returns the PNG.                  | r.status_code == 200; "image/png" in<br>r.content_type                                    | Pass   | 0.002s |
| Line 51: GET /apple-touch-icon-precomposed.png returns the PNG.      | r.status_code == 200; "image/png" in<br>r.content_type                                    | Pass   | 0.002s |
| Line 41: GET /favicon.ico returns the icon file.                     | r.status_code == 200;<br>r.content_type in ("image/x-icon",<br>"image/vnd.microsoft.i...  | Pass   | 0.002s |

| Test Description                                                                                               | Acceptance Criteria                                                                                                                        | Result | Time   |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| <code>_navigateToStress</code> function must be defined.                                                       | ' <code>_navigateToStress</code> ' in <code>hist_js</code>                                                                                 | Pass   | 0.000s |
| Click handlers use <code>addEventListener</code> , not inline <code>onclick</code> .                           | ' <code>addEventListener</code> ' in <code>hist_js</code> ;<br>' <code>hist-storm-row</code> ' in <code>hist_js</code>                     | Pass   | 0.000s |
| Must call <code>switchTab(5)</code> , NOT <code>switchTab(4)</code> or any other index.                        | ' <code>switchTab(5)</code> ' in <code>hist_js</code> ;<br>' <code>switchTab(4)</code> ' not in<br><code>hist_js.replace( 'switc...</code> | Pass   | 0.000s |
| <code>_navigateToStress</code> must set <code>window._stressStormHint</code> before switching tab.             | ' <code>window._stressStormHint</code> ' in <code>hist_js</code>                                                                           | Pass   | 0.000s |
| Hint must be set BEFORE <code>switchTab</code> is called — otherwise Stress tab                                | <code>idx_hint &gt; 0</code> ; <code>idx_switch &gt; 0</code> (+1 more)                                                                    | Pass   | 0.000s |
| <code>data-storm-id</code> attribute used to pass <code>storm_id</code> to <code>_navigateToStress</code> .    | ' <code>data-storm-id</code> ' in <code>hist_js</code> ;<br>' <code>getAttribute</code> ' in <code>hist_js</code>                          | Pass   | 0.000s |
| Every gauge in <code>gaugehc.json</code> should have a trained classifier.                                     | not missing                                                                                                                                | Skip   | 0.000s |
| Every gauge referenced in <code>stress_storms</code> should have a trained classifier.                         | not missing                                                                                                                                | Skip   | 0.000s |
| Every gauge in <code>stress_storms</code> should have a hazard curve in <code>gaugehc.json</code> .            | not missing                                                                                                                                | Skip   | 0.000s |
| Storm IDs in <code>stress_storms</code> must overlap with storm IDs in <code>propertyts/</code> .              | <code>len(overlap) &gt; 0</code>                                                                                                           | Skip   | 0.000s |
| Every <code>storm_id</code> in <code>stress_storms.json</code> must exist in <code>gauges/GAUGE*.json</code> . | not missing                                                                                                                                | Skip   | 0.000s |
| Every storm dropdown must show the <code>storm_id</code> in the option text.                                   | not violations                                                                                                                             | Pass   | 0.001s |
| All dropdown option builders must use sentinel or pipe separators.                                             | not violations                                                                                                                             | Pass   | 0.001s |
| The storm display format is centralised in <code>config/format.py</code> .                                     | ' <code> </code> ' in <code>js</code> ; ' <code>storm_id</code> ' in <code>js</code> (+2 more)                                             | Pass   | 0.008s |
| 'precipitation' as a label suffix must not appear.                                                             | not violations                                                                                                                             | Pass   | 0.001s |
| 'severe levels' (old space-separated format) must not appear.                                                  | not violations                                                                                                                             | Pass   | 0.036s |
| <code>port_stress/setup.py</code> uses sentinel with <code>show_warning=True</code> .                          | ' <code>_storm_opt</code> ' in <code>src</code> ;<br>' <code>show_warning=True</code> ' in <code>src</code>                                | Pass   | 0.000s |



| Test Description                                                       | Acceptance Criteria                                                            | Result | Time   |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|--------|
| sp_table.py delegates storm label to config.format via sentinel.       | '_storm_opt' in src; '_STORM_OPT_' in src (+1 more)                            | Pass   | 0.000s |
| Custom variable name is used throughout — gsa_timeline uses 'r'.       | js.startswith('(r.name'); 's.storm_id' not in js                               | Pass   | 0.000s |
| show_peak=True adds peak water level after mm.                         | 'peak' in js_peak                                                              | Pass   | 0.000s |
| show_warning=True adds gauges_warning count between severe and mm.     | 'warn' in js_warn; 'warn' not in js_base                                       | Pass   | 0.000s |
| trading/stress/setup.py uses sentinel with show_peak=True.             | '_storm_opt' in src; 'show_peak=True' in src                                   | Pass   | 0.000s |
| Visual files must not contain hard-coded storm display format strings. | pat not in src                                                                 | Pass   | 0.001s |
| Unknown intensity profile falls to else: intensity = peak * t_frac.    | storm.storm_id.startswith("STORM-"); len(storm.track) == 5 (+1 more)           | Pass   | 0.000s |
| Both tabs use 'effective_precipitation_mm' field.                      | 'effective_precipitation_mm' in hist_js                                        | Pass   | 0.000s |
| Both tabs use 'gauges_severe' field.                                   | 'gauges_severe' in hist_js                                                     | Pass   | 0.000s |
| Both tabs use 'intensity_category' field.                              | 'intensity_category' in hist_js                                                | Pass   | 0.000s |
| Both tabs use 'peak_level_m' field.                                    | 'peak_level_m' in hist_js                                                      | Pass   | 0.000s |
| Both tabs use 'storm_id' (not 'id' or 'stormId').                      | 'storm_id' in hist_js                                                          | Pass   | 0.000s |
| Storm IDs in gaugets/ must come from storm_sequences.json.             | len(overlap) > 0                                                               | Pass   | 0.583s |
| Storm IDs in propertyts must come from storm_sequences.json.           | len(overlap) > 0                                                               | Pass   | 3.224s |
| storm_multi.make_storm_id() produces STORM-{8hex} IDs.                 | STORM_ID.RE.match(sid)                                                         | Pass   | 0.001s |
| storm generator must NOT use STORM-{i:03d} or STORM-{i:04d}.           | 'STORM-{i:' not in src; 'STORM-{i+' not in src                                 | Pass   | 0.000s |
| Single storm generator uses storm prefix                               | 'STORM-' in src or 'STORM_ID_PREFIX' in src                                    | Pass   | 0.052s |
| Single storm generator uses uuid                                       | 'uuid' in src                                                                  | Pass   | 0.000s |
| Uuid4 produces protocol matching ids                                   | GAUGE_ID.RE.match(gauge_id); PROP_ID.RE.match(prop_id) (+2 more)               | Pass   | 0.002s |
| Accepts valid                                                          | PROP_ID.RE.match('PROP-a3f7bc12'); PROP_ID.RE.match('PROP-00000000') (+6 more) | Pass   | 0.000s |

| Test Description                                                  | Acceptance Criteria                                                                       | Result | Time   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| Human-readable names like STORM-FLOOD are not protocol-compliant. | not STORM_ID_RE.match('STORM-FLOOD');<br>not STORM_ID_RE.match('STORM-SAFE')<br>(+1 more) | Pass   | 0.000s |
| STORM-001, STORM-0001 violate the agreed protocol.                | not GAUGE_ID_RE.match('GAUGE-001');<br>not GAUGE_ID_RE.match('GAUGE-002')<br>(+5 more)    | Pass   | 0.000s |
| Rejects uppercase                                                 | not PROP_ID_RE.match('PROP-A3F7BC12');<br>not STORM_ID_RE.match('STORM-C7F4B8E2')         | Pass   | 0.000s |
| STORM-PORT-001 uses a compound prefix — not protocol-compliant.   | not STORM_ID_RE.match('STORM-PORT-001');<br>not STORM_ID_RE.match('STORM-PORT-002')       | Pass   | 0.000s |
| Storm from list is runnable                                       | resp.status_code == 200;<br>len(data['storms']) > 0 (+2 more)                             | Pass   | 0.016s |
| Worst storm auto selection is runnable                            | run_resp.status_code == 200                                                               | Pass   | 0.010s |
| Storm list endpoint must be /api/v1/propertyts/storms.            | '/api/v1/propertyts/storms' in table.js                                                   | Pass   | 0.000s |
| Empty storm list must display a user-readable message.            | 'No flooding storms found' in table.js                                                    | Pass   | 0.000s |
| JS must check window._preStorms cache before fetching.            | '_preStorms' in table.js                                                                  | Pass   | 0.000s |
| When more than 30 storms exist, shows '+ N more' message.         | 'more storms' in hist.js or '+ ' in hist.js                                               | Pass   | 0.000s |
| Historical tab limits display to 30 storms.                       | '30' in hist.js                                                                           | Pass   | 0.000s |
| Comment or logic references peak_level sort for display.          | 'sorted by peak level' in hist.js or<br>{ 'pe...                                          | Pass   | 0.000s |
| Storm with empty name falls back to storm_id.                     | resp.status_code == 200; s['name'] == s['storm_id']                                       | Pass   | 0.031s |
| Apply storm list function exists                                  | '_applyStormList' in src                                                                  | Pass   | 0.000s |
| Checks pre storms cache                                           | 'window._preStorms' in src                                                                | Pass   | 0.000s |
| Cache var should be nulled after consuming to allow refresh.      | 'window._preStorms = null' in src;<br>'window.tdPreGovDocs = null' in src                 | Pass   | 0.000s |
| Storm display shows effective-precipitation.mm.                   | 'effective-precipitation.mm' in hist.js                                                   | Pass   | 0.000s |
| Storm display shows gauges_severe count.                          | 'gauges_severe' in hist.js                                                                | Pass   | 0.000s |
| Storm display shows intensity_category.                           | 'intensity_category' in hist.js                                                           | Pass   | 0.000s |
| Storm rows use max_trigger field for colour coding.               | 'max_trigger' in hist.js                                                                  | Pass   | 0.000s |

| Test Description                                                               | Acceptance Criteria                                                             | Result | Time   |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|--------|
| Storm display shows peak_level_m — the primary sort key.                       | 'peak_level_m' in hist_js                                                       | Pass   | 0.000s |
| Storm display uses storm_id.                                                   | 'storm_id' in hist_js                                                           | Pass   | 0.000s |
| Storm display uses 'name' field from the stress/storms response.               | 's.name' in hist_js or 'storm.name' in hist_js                                  | Pass   | 0.000s |
| Historical tab must NOT use /gauges/{id}/storms for the storm list.            | 'loadStormScenarios' in hist_js; fn_idx >= 0 (+1 more)                          | Pass   | 0.000s |
| _loadStormScenarios function must be defined.                                  | 'loadStormScenarios' in hist_js                                                 | Pass   | 0.000s |
| Storm list renders in hist-storms-list element.                                | 'hist-storms-list' in hist_js; 'hist-storms-panel' in hist_js                   | Pass   | 0.000s |
| Storm list must be loaded from the same endpoint as Stress tab.                | '/api/v1/trading/stress/storms' in hist_js                                      | Pass   | 0.000s |
| Sorts by peak level for auto-selection.                                        | 'peak_level_m' in stress_js                                                     | Pass   | 0.000s |
| Storm select element present.                                                  | 'stress-storm-select' in stress_js                                              | Pass   | 0.000s |
| Custom kernel                                                                  | d["decay_kernel"] == "exponential"                                              | Pass   | 0.000s |
| Decay kernel serialised as string                                              | isinstance(d["decay_kernel"], str); d["decay_kernel"] == "gaussian"             | Pass   | 0.000s |
| Has storm id                                                                   | storm.to_dict()["storm_id"] == "STORM-0001"                                     | Pass   | 0.000s |
| Has track                                                                      | "track" in d; len(d["track"]) == 2                                              | Pass   | 0.000s |
| Intensity profile serialised as string                                         | isinstance(d["intensity_profile"], str); d["intensity_profile"] == "triangular" | Pass   | 0.000s |
| Returns dict                                                                   | isinstance(d, dict); isinstance(d, dict)                                        | Pass   | 0.000s |
| Track has coords                                                               | "longitude" in point; "latitude" in point (+2 more)                             | Pass   | 0.000s |
| storms.py lines 91-92: corrupted gaugets file -i skipped in severe count loop. | r.status_code == 200                                                            | Pass   | 0.032s |
| storms.py line 80: gaugets dir doesn't exist -i skip severe counting.          | r.status_code == 200                                                            | Pass   | 0.030s |
| storms.py line 61: storm_sequences.json doesn't exist -i skip enrichment.      | r.status_code == 200; data["storm_responses"]["num_storms"] == 1                | Pass   | 0.030s |
| storms.py lines 75-76: corrupted storm_sequences.json -i silently skipped.     | r.status_code == 200                                                            | Pass   | 0.033s |

| Test Description                                                                                                                    | Acceptance Criteria                                                                  | Result | Time   |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|
| storms.py lines 113-115: exception in storm loading -i 500.                                                                         | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code> | Pass   | 0.031s |
| <code>_stream_proc</code> should call <code>proc.wait()</code> after exhausting <code>stdout</code> .                               | —                                                                                    | Pass   | 0.001s |
| <code>_stream_proc</code> should push each line from <code>proc.stdout</code> into <code>_output_lines</code> .                     | "line one" in lines; "line two" in lines (+1 more)                                   | Pass   | 0.001s |
| <code>_stream_proc</code> should strip trailing newlines before appending.                                                          | "stripped" in lines; "stripped\\n" not in lines                                      | Pass   | 0.000s |
| <code>_get_predictor</code> returns cached value without re-importing.                                                              | result is sentinel                                                                   | Pass   | 0.000s |
| <code>_get_predictor</code> returns <code>None</code> when both <code>stressm</code> and <code>legacy stress</code> dirs are empty. | pred is None                                                                         | Pass   | 0.001s |
| All gauges trained                                                                                                                  | <code>n.skipped == 0; n.trained == n.gauges</code>                                   | Pass   | 0.000s |
| Avg auc above threshold                                                                                                             | <code>auc &gt;= MIN_AUC</code>                                                       | Pass   | 0.000s |
| Classifier count                                                                                                                    | <code>len(classifier_files) &gt;= MIN_CLASSIFIERS</code>                             | Pass   | 0.000s |
| Classifier count matches summary                                                                                                    | <code>n_files &gt;= n.summary</code>                                                 | Pass   | 0.000s |
| Classifier filenames well formed                                                                                                    | not bad                                                                              | Pass   | 0.000s |
| Classifiers not empty                                                                                                               | not empty                                                                            | Pass   | 0.002s |
| Stress model dir exists                                                                                                             | <code>STRESS_MODEL_DIR.exists()</code>                                               | Pass   | 0.000s |
| Training summary exists                                                                                                             | <code>TRAINING_SUMMARY.exists()</code>                                               | Pass   | 0.000s |
| Training summary has required keys                                                                                                  | not missing                                                                          | Pass   | 0.000s |
| Hits /trading/stress/run.                                                                                                           | <code>'/trading/stress/run' in stress_js</code>                                      | Pass   | 0.000s |
| Hits /trading/stress/storms.                                                                                                        | <code>'/trading/stress/storms' in stress_js</code>                                   | Pass   | 0.000s |
| <code>stress/run</code> returns 500 on engine error.                                                                                | <code>resp.status_code == 500</code>                                                 | Pass   | 0.017s |
| <code>stress/gauges</code> returns 500 when gauge loader raises.                                                                    | <code>resp.status_code == 500</code>                                                 | Pass   | 0.006s |
| <code>stress/storms</code> returns 500 when storm loader raises.                                                                    | <code>resp.status_code == 500</code>                                                 | Pass   | 0.011s |
| <code>_tdPopulateGaugeDropdown</code> must survive f-string rendering.                                                              | <code>'_tdPopulateGaugeDropdown' in full_js</code>                                   | Pass   | 0.000s |
| <code>tdStressGaugeChanged</code> must survive f-string rendering in <code>tradingdesk.py</code> .                                  | <code>'tdStressGaugeChanged' in full_js</code>                                       | Pass   | 0.000s |
| Rendered JS must not have unescaped f-string brace artifacts.                                                                       | <code>len(full_js) &gt; 10000</code>                                                 | Pass   | 0.000s |
| <code>td-stress-gauge</code> element must be present for JS to find it.                                                             | <code>"getElementById('td-stress-gauge')"</code> in <code>stress_js</code>           | Pass   | 0.000s |

| Test Description                                                                                                   | Acceptance Criteria                                                                                                          | Result | Time   |
|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| <code>_tdPopulateGaugeDropdown</code> must exist — builds gauge selector.                                          | <code>'_tdPopulateGaugeDropdown'</code> in <code>stress_js</code>                                                            | Pass   | 0.000s |
| <code>_tdPopulateStormDropdown</code> must exist — builds storm list for selected gauge.                           | <code>'_tdPopulateStormDropdown'</code> in <code>stress_js</code>                                                            | Pass   | 0.000s |
| <code>td-stress-storm</code> element must be present for JS to find it.                                            | <code>"getElementById('td-stress-storm')"</code> in <code>stress_js</code>                                                   | Pass   | 0.000s |
| Stress functions must be INSIDE the IIFE, not after it.                                                            | <code>idx_gauge_changed &gt; 0;</code><br><code>idx_gauge_changed &lt; iife_close</code>                                     | Pass   | 0.000s |
| gauge onchange must call <code>tdStressGaugeChanged</code> .                                                       | <code>'tdStressGaugeChanged()'</code> in <code>stress_js</code>                                                              | Pass   | 0.000s |
| <code>tdStressGaugeChanged</code> must exist — called when gauge is selected.                                      | <code>'function tdStressGaugeChanged'</code> in <code>stress_js</code>                                                       | Pass   | 0.000s |
| storm onchange must call <code>tdStressStormChanged</code> .                                                       | <code>'tdStressStormChanged()'</code> in <code>stress_js</code>                                                              | Pass   | 0.000s |
| <code>tdStressStormChanged</code> must exist — called when storm is selected.                                      | <code>'function tdStressStormChanged'</code> in <code>stress_js</code>                                                       | Pass   | 0.000s |
| 'First with trades' fallback must only run when <code>sel.value</code> is still empty.                             | <code>"if (!sel.value &amp;&amp; tdStressGauges.length &gt; 0)"</code> in <code>js</code>                                    | Pass   | 0.000s |
| Hint selection must happen before the 'first with trades' fallback.                                                | <code>idx_hint &gt; 0; idx_fallback &gt; idx_hint</code>                                                                     | Pass   | 0.000s |
| <code>tdStressGaugeHint</code> must be set to null after use to prevent stale hint.                                | <code>'tdStressGaugeHint = null'</code> in <code>js</code>                                                                   | Pass   | 0.000s |
| When no market hint, <code>tdBlotterFilters.gauge_id</code> must be the fallback.                                  | <code>'tdBlotterFilters'</code> in <code>js;</code><br><code>'tdBlotterFilters.gauge_id'</code> in <code>js</code>           | Pass   | 0.000s |
| <code>loadStressData</code> must accept a <code>gaugeHint</code> parameter.                                        | <code>'function loadStressData(gaugeHint)'</code> in <code>js</code>                                                         | Pass   | 0.000s |
| <code>loadStressData</code> must store <code>gaugeHint</code> in <code>tdStressGaugeHint</code> .                  | <code>'if (gaugeHint) tdStressGaugeHint = gaugeHint'</code> in <code>js</code>                                               | Pass   | 0.000s |
| <code>_tdPopulateGaugeDropdown</code> must use <code>tdStressGaugeHint</code> for pre-selection.                   | <code>'tdStressGaugeHint'</code> in <code>js;</code><br><code>idx_hint.use &gt; idx_populate</code>                          | Pass   | 0.000s |
| <code>tdStressGaugeHint</code> state variable must exist in stress JS.                                             | <code>'tdStressGaugeHint'</code> in <code>js</code>                                                                          | Pass   | 0.000s |
| <code>switchTab('stress')</code> must use blotter filter as primary hint.                                          | <code>idx_blotter &gt; 0; idx_market &gt; 0</code> (+4 more)                                                                 | Pass   | 0.000s |
| <code>switchTab('stress')</code> in <code>tradingdesk</code> must pass gauge hint to <code>loadStressData</code> . | <code>"loadStressData(stressHint)"</code> in <code>js;</code><br><code>"tdSelectedGauge"</code> in <code>js</code> (+1 more) | Pass   | 0.000s |

| Test Description                                                                    | Acceptance Criteria                                                                | Result | Time   |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------|--------|
| switchTab('stress') must fall back to first blotter trade gauge when no filter set. | "tdBlotterData" in js;<br>"tdBlotterData[0].gauge_id" in js                        | Pass   | 0.000s |
| peak_position=0 uses decay-only path.                                               | len(levels) == 10; all(isinstance(v, float) for v in levels)                       | Pass   | 0.000s |
| remaining=0 uses frac=1.0 path.                                                     | len(levels) == 5                                                                   | Pass   | 0.000s |
| Both stress charts must show KO vertical annotation line.                           | 'koLine' in js; 'first_trigger_hour' in js                                         | Pass   | 0.000s |
| Stress table must show KO status per trade.                                         | 'triggered_hour' in js; 'KO H' in js                                               | Pass   | 0.000s |
| Stats bar must show knock-out count.                                                | 'num_triggered' in js; 'Knocked Out' in js                                         | Pass   | 0.000s |
| Storm id exact match required                                                       | resp.status_code == 200;<br>resp.status_code == 404                                | Pass   | 0.008s |
| Storm list has all run inputs                                                       | data['gauge_id'] == 'GAUGE-001';<br>'storm_id' in s                                | Pass   | 0.003s |
| Response includes trigger levels for threshold annotation on charts.                | 'alert_level' in data; 'warning_level' in data (+1 more)                           | Pass   | 0.179s |
| Response includes hydrograph_source string describing the data origin.              | 'hydrograph_source' in data                                                        | Pass   | 0.007s |
| Response includes storm_name and intensity_category for header display.             | data['storm_name'] == 'Severe Storm Alpha'; data['intensity_category'] == 'severe' | Pass   | 0.008s |
| Hourly array must be 168 elements (7-day storm window).                             | len(data['hourly']) == 168                                                         | Pass   | 0.009s |
| Surface table must cover the full 168h window regardless of knock-out.              | max(surface['hours']) >= 164                                                       | Pass   | 0.007s |
| If the index has not changed, the cached object is returned (no re-read).           | id(data1) == id(data2)                                                             | Pass   | 0.001s |
| Replacing stress_storms/_index.json on disk is picked up on next                    | len(data1['storms']) == 1;<br>len(data2['storms']) == 2                            | Pass   | 0.015s |
| Thames catchment has 52 gauges; all must be represented.                            | len(gauge_ids) == 52                                                               | Pass   | 0.014s |
| All gauge_id values in gauge_responses must match GAUGE-xxxx format.                | not bad                                                                            | Pass   | 0.031s |
| All storm_id values must use STORM- prefix.                                         | not bad                                                                            | Pass   | 0.002s |

| Test Description                                                          | Acceptance Criteria                                                                      | Result | Time   |
|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|--------|
| At least one storm exceeds alert                                          | <code>any_alert</code>                                                                   | Pass   | 0.000s |
| Each storm has gauge responses                                            | <code>not empty</code>                                                                   | Pass   | 0.001s |
| Each storm has required fields                                            | <code>not missing</code>                                                                 | Pass   | 0.004s |
| File exists                                                               | <code>STRESS_STORMS_INDEX.exists()</code> or<br><code>STRESS_STORMS_PATH.exists()</code> | Pass   | 0.000s |
| Gauge ids well formed                                                     | <code>not bad</code>                                                                     | Pass   | 0.020s |
| Gauge responses have required fields                                      | <code>not missing</code>                                                                 | Pass   | 0.098s |
| Has storms                                                                | <code>storms is not None and len(storms) &gt; 0</code>                                   | Pass   | 2.684s |
| Intensity categories present                                              | <code>cats</code>                                                                        | Pass   | 0.003s |
| Minimum storm count                                                       | <code>len(storms) &gt;= MIN_STORM_COUNT</code>                                           | Pass   | 0.000s |
| Peak level exceeds base                                                   | <code>not violations</code>                                                              | Pass   | 0.022s |
| Peak levels positive                                                      | <code>not violations</code>                                                              | Pass   | 0.017s |
| Storm ids unique                                                          | <code>not dupes</code>                                                                   | Pass   | 0.608s |
| Storm ids well formed                                                     | <code>not bad</code>                                                                     | Pass   | 0.002s |
| Storms cover multiple gauges                                              | <code>len(gauges) &gt;= 10</code>                                                        | Pass   | 0.018s |
| Storms must be sorted:<br>gauges_severe DESC,<br>gauges_warning DESC.     | <code>severe_counts == sorted(severe_counts, reverse=True)</code>                        | Pass   | 0.001s |
| Total storms key matches list length                                      | <code>data["total_storms"] == len(data["storms"])</code>                                 | Pass   | 0.040s |
| Trigger summary max trigger valid                                         | <code>not bad</code>                                                                     | Pass   | 0.002s |
| Every storm must have a fully-populated trigger_summary dict.             | <code>not missing</code>                                                                 | Pass   | 0.004s |
| No gauge id response has count                                            | <code>isinstance(data['count'], int); data['count'] &gt; 0</code>                        | Pass   | 0.002s |
| Startup preloader fetches without gauge_id — must not return 400.         | <code>r.status_code == 200</code>                                                        | Pass   | 0.002s |
| All storms in file returned — not filtered by gauge.                      | <code>'storms' in data; data['count'] == len(data['storms']) (+1 more)</code>            | Pass   | 0.002s |
| No gauge id returns success                                               | <code>data['status'] == 'success'</code>                                                 | Pass   | 0.002s |
| gauge_id param still filters to matching storms.                          | <code>data['status'] == 'success'; all(True for _ in data['storms'])</code>              | Pass   | 0.003s |
| When stress_storms.json is absent, returns 404.                           | <code>resp.status_code == 404; 'not found' in data.get('message', '').lower()</code>     | Pass   | 0.001s |
| ghc_stress.get_js() returns valid JS.                                     | <code>len(stress_js) &gt; 1000</code>                                                    | Pass   | 0.000s |
| Chart canvas element must use td-prefixed ID.                             | <code>'td-stress-chart-canvas' in js</code>                                              | Pass   | 0.000s |
| Chart sub-tabs must exist for Flood Probability, Stress P&L, and Surface. | <code>'td-stress-ctab-0' in js; 'td-stress-ctab-1' in js (+3 more)</code>                | Pass   | 0.000s |



| Test Description                                                   | Acceptance Criteria                                          | Result | Time   |
|--------------------------------------------------------------------|--------------------------------------------------------------|--------|--------|
| Stress tab must have both chart renderers.                         | '_tdRenderProbabilityChart' in js;                           | Pass   | 0.000s |
|                                                                    | '_tdRenderStressPnlChart' in js                              |        |        |
| Stress tab must have cleanup function.                             | 'tdCleanupStressCharts' in js                                | Pass   | 0.000s |
| Stress tab must have gauge selector dropdown.                      | 'td-stress-gauge' in js;                                     | Pass   | 0.000s |
|                                                                    | 'tdStressGaugeChanged' in js                                 |        |        |
| Stress tab must fetch from stress/-gauges endpoint.                | '/api/v1/trading/stress/gauges' in js                        | Pass   | 0.000s |
| Stress tab must have createStressView and loadStressData.          | 'createStressView' in js;                                    | Pass   | 0.000s |
|                                                                    | 'loadStressData' in js                                       |        |        |
| Charts must have KO annotation line.                               | 'koLine' in js; 'first_trigger_hour' in js                   | Pass   | 0.000s |
| Stress tab must have Property Damage link.                         | 'tdStressOpenDamage' in js; 'Property Damage' in js          | Pass   | 0.000s |
| Stress tab must have Storm Simulation link.                        | 'tdStressOpenGauge' in js; 'Storm Simulation' in js          | Pass   | 0.000s |
| Stress sub-module should render without errors.                    | len(js) > 0                                                  | Pass   | 0.000s |
| Stress tab must have stats bar with KO info.                       | '_tdRenderStressStats' in js;                                | Pass   | 0.000s |
|                                                                    | 'td-stress-stats-bar' in js (+1 more)                        |        |        |
| Stress tab must have storm selector dropdown.                      | 'td-stress-storm' in js;                                     | Pass   | 0.000s |
|                                                                    | 'tdStressStormChanged' in js                                 |        |        |
| Stress tab must render probability surface table.                  | 'probability_surface' in js;                                 | Pass   | 0.000s |
|                                                                    | 'td-stress-surface-wrap' in js (+2 more)                     |        |        |
| Stress tab must have trade table renderer.                         | '_tdRenderStressTable' in js;                                | Pass   | 0.000s |
|                                                                    | 'td-stress-table-wrap' in js                                 |        |        |
| Surface table must be horizontally scrollable for 168h of columns. | 'overflow:auto' in js; 'width:100%' not in snippet           | Pass   | 0.000s |
| Should fetch from /api/v1/trading/stress/storms.                   | '/api/v1/trading/stress/storms' in js                        | Pass   | 0.000s |
| Should use cash pricing formula.                                   | 'cash_price' in js or 'cashPrice' in js or 'portfolio.cas... | Pass   | 0.000s |
| Should have two chart sub-tabs: Flood Probability and Stress P&L.  | 'Flood Probability' in js; 'Stress P' in js (+2 more)        | Pass   | 0.000s |
| Should have storm selector dropdown.                               | 'stress-storm-select' in js;                                 | Pass   | 0.000s |
|                                                                    | 'populateStormSelect' in js                                  |        |        |
| Stress test JS should render without errors.                       | len(js) > 0                                                  | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                       | Result | Time   |
|------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Should POST to /api/v1/trading/stress/run.                       | '/api/v1/trading/stress/run' in js;<br>'_runStressScenario' in js         | Pass   | 0.000s |
| Should show P(flood) in chart.                                   | 'p_flood' in js                                                           | Pass   | 0.000s |
| Should pick up _stressStormHint from Historical tab.             | '_stressStormHint' in js;<br>'_stressStormHint = null' in js              | Pass   | 0.000s |
| Should render trade table at peak hour.                          | 'stress-table-wrap' in js;<br>'_renderStressTable' in js                  | Pass   | 0.000s |
| Chart tab clicks should use addEventListener (IIFE-safe).        | 'addEventListener' in js                                                  | Pass   | 0.000s |
| When stressm dir has .joblib files, predictor is loaded from it. | result is mock_predictor                                                  | Pass   | 0.001s |
| String with options picks from list                              | result in ["Brick", "Stone",<br>"Timber"]                                 | Pass   | 0.000s |
| String without options returns empty                             | result == ""                                                              | Pass   | 0.000s |
| No data returns 404                                              | r.status_code == 404;<br>r.get_json()["status"] == "error"                | Pass   | 0.000s |
| Options returns ok                                               | r.status_code == 200; r.status_code ==<br>200 (+2 more)                   | Pass   | 0.000s |
| With data returns success                                        | r.status_code == 200;<br>r.get_json()["status"] == "success"<br>(+3 more) | Pass   | 0.004s |
| Surface footer references KO trimming.                           | 'trimmed at KO' in stress_js                                              | Pass   | 0.000s |
| Surface table reads probability_surface from data.               | 'probability_surface' in stress_js                                        | Pass   | 0.000s |
| Renders an HTML table with monospace font.                       | 'border-collapse' in stress_js;<br>'font-family:monospace' in stress_js   | Pass   | 0.000s |
| Surface table applies trigger-band background colours.           | '#FFEBEE' in stress_js; '#FFF3E0' in<br>stress_js (+1 more)               | Pass   | 0.000s |
| Surface tab creates a wrapping div.                              | 'stress-surface-wrap' in stress_js                                        | Pass   | 0.000s |
| Lines 72-76: water_level != severe_level → no flooding.          | —                                                                         | Pass   | 0.000s |
| Line 96: flood_depths[i] = min(depth, 5.0).                      | np.all(depths <= 5.0)                                                     | Pass   | 0.000s |
| Lines 89-92: distance_factor reduces depth proportionally.       | depths[0] > 0; depths[1] > 0                                              | Pass   | 0.000s |
| Lines 72: len(gauge_data) == 0.                                  | —                                                                         | Pass   | 0.000s |
| Lines 93-94: distance != max_distance → depth = 0.0.             | depths[0] == 0.0                                                          | Pass   | 0.000s |

| Test Description                                                         | Acceptance Criteria                                                       | Result | Time   |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------|--------|--------|
| Lines 72-96: water_level is severe_level, close to Thames.               | np.any(depths > 0)                                                        | Pass   | 0.000s |
| Lines 72: gauge_data is None.                                            | —                                                                         | Pass   | 0.000s |
| loadPortfolioImpact() must be defined.                                   | 'loadPortfolioImpact' in table_js                                         | Pass   | 0.000s |
| Portfolio impact fetch must use the correct endpoint path.               | 'portfolio-impact' in table_js                                            | Pass   | 0.000s |
| Sort state variables must be initialised.                                | 'spSortCol' in table_js; 'spSortAsc' in table_js                          | Pass   | 0.000s |
| Summary card labels must be present.                                     | 'Properties Affected' in table_js; 'Total Damage' in table_js             | Pass   | 0.000s |
| any_flood has a lower threshold than severe - is higher or equal spread. | any_flood.5yr >= severe.5yr - 1e-6                                        | Pass   | 0.234s |
| PRS spread should be non-negative for all tenors.                        | all(s >= 0 for s in spreads)                                              | Pass   | 0.260s |
| Survival between zero and one                                            | 0.0 <= s <= 1.0                                                           | Pass   | 0.260s |
| Coverage booleans are bool                                               | isinstance(val, bool)                                                     | Pass   | 0.000s |
| Coverage has required keys                                               | key in tc                                                                 | Pass   | 0.000s |
| Coverage has file                                                        | "test_file" in tc; isinstance(tc["test_file"], str)                       | Pass   | 0.000s |
| GAUGE-001, GAUGE-002, GAUGE-9042bd95 are backward-compat exceptions.     | compat_ids == {'GAUGE-001', 'GAUGE-002', 'GAUGE-9042bd95'}                | Pass   | 0.000s |
| gauge.json and gaugehc.json gauge IDs must be identical sets.            | gauge.json_ids == gaugehc_ids                                             | Pass   | 0.000s |
| The new gauge entries in SAMPLE_GAUGEHC follow GAUGE-{8hex}.             | not violations                                                            | Pass   | 0.000s |
| New gauge IDs added in confest (GAUGE-{8hex}) follow the protocol.       | GAUGE_ID_RE.match(gid)                                                    | Pass   | 0.000s |
| Gauge IDs in SAMPLE_STRESS_STORMS gauge_responses are from _data.        | not violations                                                            | Pass   | 0.000s |
| _data.py STORM_PORT_* constants follow the protocol.                     | STORM_ID_RE.match(STORM_PORT_SEVERE); STORM_ID_RE.match(STORM_PORT_ALERT) | Pass   | 0.000s |
| All storm_id values in SAMPLE_PORT_STRESS_STORMS follow STORM-{8hex}.    | STORM_ID_RE.match(sid)                                                    | Pass   | 0.000s |

| Test Description                                                 | Acceptance Criteria                                                  | Result | Time   |
|------------------------------------------------------------------|----------------------------------------------------------------------|--------|--------|
| All storm_id values in SAMPLE_STRESS_STORMS follow STORM-{8hex}. | STORM_ID.RE.match(sid)                                               | Pass   | 0.000s |
| _data.py STORM_SEVERE and STORM_WARNING follow the protocol.     | STORM_ID.RE.match(STORM_SEVERE);<br>STORM_ID.RE.match(STORM_WARNING) | Pass   | 0.000s |
| Has lines field                                                  | "lines" in data;<br>isinstance(data["lines"], list)                  | Pass   | 0.001s |
| Has running field                                                | "running" in data;<br>isinstance(data["running"], bool)              | Pass   | 0.001s |
| Lines includes appended                                          | "test line" in data["lines"]                                         | Pass   | 0.001s |
| Not running initially                                            | data["running"] is False                                             | Pass   | 0.001s |
| Returns 200                                                      | resp.status_code == 200                                              | Pass   | 0.001s |
| Returns not found when no file                                   | data["status"] == "not_found"                                        | Pass   | 0.001s |
| Returns ok when file exists                                      | data["status"] == "ok"                                               | Pass   | 0.002s |
| Returns summary fields                                           | "summary" in data; "failures" in data<br>(+3 more)                   | Pass   | 0.002s |
| Running field is bool                                            | "running" in data;<br>isinstance(data["running"], bool)              | Pass   | 0.001s |
| Historical high exceeds severe by less than 1m                   | "Elevated risk" in str(elements)                                     | Pass   | 0.001s |
| Historical high exceeds severe by more than 1m                   | "High risk" in str(elements)                                         | Pass   | 0.001s |
| Historical high within severe                                    | "within severe warning levels" in<br>str(elements)                   | Pass   | 0.001s |
| Moderate alert range                                             | "Moderate alert range" in<br>str(elements)                           | Pass   | 0.001s |
| Narrow alert range                                               | "Narrow alert range" in<br>str(elements)                             | Pass   | 0.001s |
| Wide alert range                                                 | "Wide alert range" in str(elements)                                  | Pass   | 0.001s |
| Alert status                                                     | "ELEVATED RISK" in<br>str(self._elements("Alert"))                   | Pass   | 0.001s |
| Normal status                                                    | "NORMAL" in<br>str(self._elements("Normal"))                         | Pass   | 0.001s |
| Severe status                                                    | "CRITICAL RISK" in<br>str(self._elements("Severe"))                  | Pass   | 0.001s |
| Timeseries with no level                                         | isinstance(elements, list)                                           | Pass   | 0.000s |
| Timeseries with no readings                                      | isinstance(elements, list)                                           | Pass   | 0.000s |
| Warning status                                                   | "HIGH RISK" in<br>str(self._elements("Warning"))                     | Pass   | 0.001s |

| Test Description                                                    | Acceptance Criteria                                                        | Result | Time   |
|---------------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------|
| Test getting timeseries for a gauge from portfolio.                 | <code>len(active-gauges) &gt; 0; len(readings) &gt; 0</code>               | Pass   | 0.002s |
| Test that timeseries gauge IDs exist in portfolio.                  | <code>gauge is not None</code>                                             | Pass   | 0.003s |
| Count returns number of gauge files                                 | <code>TimeseriesLoader(tmp_path).count() == 2</code>                       | Pass   | 0.001s |
| Get available gauge ids                                             | <code>"GAUGE-001" in ids; "GAUGE-002" in ids</code>                        | Pass   | 0.001s |
| Get available gauge ids missing dir                                 | <code>TimeseriesLoader(tmp_path / "nonexistent").get_available...</code>   | Pass   | 0.000s |
| List all returns summaries                                          | <code>isinstance(TimeseriesLoader(tmp_path).list_summaries(), list)</code> | Pass   | 0.001s |
| Load all from directory                                             | <code>len(TimeseriesLoader(tmp_path).load_all()) &gt;= 1</code>            | Pass   | 0.001s |
| Missing dir returns empty                                           | <code>TimeseriesLoader(tmp_path / "nonexistent").load_all() == []</code>   | Pass   | 0.000s |
| Test that entity ID is the timestamp.                               | <code>entity_id is not None; "2025-01-01" in entity_id</code>              | Pass   | 0.004s |
| Test timeseries summary generation.                                 | <code>len(summaries) == 3; 'timestamp' in summary (+3 more)</code>         | Pass   | 0.003s |
| Test loading timeseries data.                                       | <code>len(records) == 3</code>                                             | Pass   | 0.002s |
| Clear cache                                                         | <code>not loader._cache_valid</code>                                       | Pass   | 0.001s |
| Force reload                                                        | <code>"GAUGE-002" in loader.get_available-gauge-ids()</code>               | Pass   | 0.001s |
| Invalidate cache                                                    | <code>not loader._cache_valid</code>                                       | Pass   | 0.000s |
| Test getting most recent reading.                                   | <code>latest is not None; latest['hour'] == 2 (+1 more)</code>             | Pass   | 0.001s |
| Test getting peak (highest) reading.                                | <code>peak is not None; peak['level'] == 4.8</code>                        | Pass   | 0.001s |
| Test getting readings for a specific gauge.                         | <code>len(readings) == 3; 'timestamp' in reading (+1 more)</code>          | Pass   | 0.001s |
| Test getting readings using gauge name.                             | <code>len(readings) == 3</code>                                            | Pass   | 0.001s |
| Test getting readings for non-existent gauge.                       | <code>readings == []</code>                                                | Pass   | 0.001s |
| Readings with unparseable timestamps are skipped (all-gauges path). | <code>len(result) == 1</code>                                              | Pass   | 0.001s |
| Readings with unparseable timestamps are skipped (gauge_id path).   | <code>len(result) == 1</code>                                              | Pass   | 0.001s |
| Get readings in range all gauges                                    | <code>isinstance(result, list)</code>                                      | Pass   | 0.001s |
| Get readings in range for gauge                                     | <code>isinstance(result, list)</code>                                      | Pass   | 0.001s |

| Test Description                                            | Acceptance Criteria                                                                   | Result | Time   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------|--------|--------|
| Get lareading                                               | TimeseriesLoader(tmp_path).get_latest_reading                                         | Pass   | 0.001s |
| Get lareading missing returns none                          | TimeseriesLoader(tmp_path).get_latest_reading                                         | Pass   | 0.000s |
| Get peak reading                                            | TimeseriesLoader(tmp_path).get_peak_reading                                           | Pass   | 0.001s |
| ...                                                         | ...                                                                                   |        |        |
| Get peak reading missing returns none                       | TimeseriesLoader(tmp_path).get_peak_reading                                           | Pass   | 0.000s |
| ...                                                         | ...                                                                                   |        |        |
| Get readings for gauge                                      | isinstance(result, list) and<br>len(result) >= 1                                      | Pass   | 0.001s |
| Gauge ID not found — fallback to gauge_name search.         | isinstance(result, list)                                                              | Pass   | 0.003s |
| Get readings missing gauge returns empty                    | TimeseriesLoader(tmp_path).get_readings_for_gauge("GAUGE-001")                        | Pass   | 0.000s |
| Get storm responses                                         | len(responses) == 1;<br>responses[0]["storm_id"] ==<br>"STORM-0001"                   | Pass   | 0.001s |
| Get storm responses missing gauge                           | TimeseriesLoader(tmp_path).get_storm_responses                                        | Pass   | 0.000s |
| Test getting list of gauge IDs in timeseries.               | len(gauge_ids) == 2; "GAUGE-001" in<br>gauge_ids (+1 more)                            | Pass   | 0.001s |
| Test calculating gauge statistics.                          | stats['gauge_id'] == "GAUGE-001";<br>stats['reading_count'] == 3 (+4 more)            | Pass   | 0.002s |
| Get gauge statistics missing gauge                          | stats["reading_count"] == 0; "error"<br>in stats                                      | Pass   | 0.000s |
| Readings with no numeric level field return error in stats. | "error" in stats                                                                      | Pass   | 0.001s |
| Test statistics for non-existent gauge.                     | stats['reading_count'] == 0; 'error'<br>in stats                                      | Pass   | 0.000s |
| Get gauge statistics with data                              | "reading_count" in stats;<br>stats["reading_count"] >= 1                              | Pass   | 0.002s |
| Exists by timestamp                                         | loader.exists(result[0]["timestamp"])<br>is True                                      | Pass   | 0.001s |
| Find by id not found returns none                           | TimeseriesLoader(tmp_path).find_by_id("1999-01-01T00:00:00")                          | Pass   | 0.001s |
| Get file status alias                                       | "entity_name" in<br>TimeseriesLoader(tmp_path).get_file_stat...                       | Pass   | 0.000s |
| Get status returns dict                                     | "entity_name" in status; "exists" in<br>status                                        | Pass   | 0.000s |
| Test getting readings in time range.                        | len(readings) == 2                                                                    | Skip   | 0.000s |
| Test getting readings in range for specific gauge.          | len(readings) >= 2;<br>reading.get('gaugeId') ==<br>"GAUGE-001" or 'level' in reading | Skip   | 0.000s |
| Bad gauge data is skipped                                   | len(result["readings"]) >= 1                                                          | Pass   | 0.000s |

| Test Description                                              | Acceptance Criteria                                                                       | Result | Time   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------|--------|
| ConstructionYear triggers building age row in summary table.  | <code>isinstance(result, list); len(result) &gt; 0</code>                                 | Pass   | 0.001s |
| Empty property does not crash                                 | <code>isinstance(result, list)</code>                                                     | Pass   | 0.000s |
| Missing nested key falls back to 'Unknown Property'.          | <code>result == "Unknown Property"</code>                                                 | Pass   | 0.000s |
| Get property id present                                       | <code>result == "PROP-001"</code>                                                         | Pass   | 0.000s |
| Has tables                                                    | <code>any(isinstance(e, Table) for e in result)</code>                                    | Pass   | 0.000s |
| Has title paragraph                                           | <code>any("Comprehensive Property Analysis Report" in t for t i...</code>                 | Pass   | 0.000s |
| Property id in output                                         | <code>any("PROP-001" in t for t in texts)</code>                                          | Pass   | 0.000s |
| Property overview section header                              | <code>any("Property Overview" in t for t in texts)</code>                                 | Pass   | 0.000s |
| Property summary section header                               | <code>any("Property Summary" in t for t in texts)</code>                                  | Pass   | 0.000s |
| Returns list                                                  | <code>isinstance(result, list); len(result) &gt; 0</code>                                 | Pass   | 0.001s |
| Unknown property id fallback                                  | <code>any("Unknown Property" in t for t in texts)</code>                                  | Pass   | 0.000s |
| With mortgage data does not crash                             | <code>isinstance(result, list)</code>                                                     | Pass   | 0.001s |
| Posts to /api/v1/prs/commit.                                  | <code>'/api/v1/prs/commit' in prs_js</code>                                               | Pass   | 0.000s |
| Commit payload includes payer flag.                           | <code>'isPayer' in prs_js</code>                                                          | Pass   | 0.000s |
| Panel should add to a Folium map without error.               | <code>len(html._children) &gt; 0</code>                                                   | Pass   | 0.022s |
| Blotter sub-module should render with FS01 column.            | <code>'createBlotterView' in js;</code><br><code>'loadBlotterData' in js (+2 more)</code> | Pass   | 0.000s |
| EOD sub-module should render.                                 | <code>'createEodView' in js;</code> <code>'loadEodData' in js</code>                      | Pass   | 0.000s |
| Aggregate View sub-module should render.                      | <code>'createMapView' in js;</code> <code>'loadMapData' in js (+2 more)</code>            | Pass   | 0.000s |
| Market sub-module should render.                              | <code>'createMarketView' in js;</code><br><code>'loadMarketData' in js</code>             | Pass   | 0.000s |
| Ensure no unescaped single braces that could cause NameError. | <code>len(js) &gt; 1000</code>                                                            | Pass   | 0.000s |
| Panel should contain all 7 tab IDs and labels.                | <code>'blotter' in js;</code> <code>'market' in js (+7 more)</code>                       | Pass   | 0.000s |
| Panel should export TradingDesk to window.                    | <code>'window.TradingDesk' in js;</code><br><code>'window.showTradingDesk' in js</code>   | Pass   | 0.000s |
| Panel should contain the Pi button.                           | <code>'&amp;Pi;' in js</code>                                                             | Pass   | 0.000s |
| TradingDeskPanel JS should render without errors.             | <code>len(js) &gt; 0;</code> <code>'&lt;script&gt;' in js (+1 more)</code>                | Pass   | 0.000s |



| Test Description                                                     | Acceptance Criteria                                                                                    | Result | Time   |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------|--------|
| FS01 Matrix sub-module should render.                                | 'createRiskView' in js;                                                                                | Pass   | 0.000s |
| get_statistics() returns panel_width and panel_height.               | 'loadRiskData' in js (+1 more)<br>stats["panel_width"] == "420px";<br>stats["panel_height"] == "600px" | Pass   | 0.000s |
| Td preloader does not redeclare td pre blotter                       | 'window._tdPreBlotter = null' not in src                                                               | Pass   | 0.000s |
| Td preloader does not redeclare td pre market                        | 'window._tdPreMarket = null' not in src                                                                | Pass   | 0.000s |
| Td preloader does not redeclare td preload done                      | 'var _tdPreloadDone = false' not in src                                                                | Pass   | 0.000s |
| severe_warning=100m ensures no events breach it → fallback to alert. | result["label_threshold"] == "flood_alert"                                                             | Pass   | 6.612s |
| Model file written                                                   | stressm_dir.exists()                                                                                   | Pass   | 7.464s |
| Result has gauge id                                                  | result.get("gauge_id") == "GAUGE-clf001"                                                               | Pass   | 7.211s |
| Result has label threshold                                           | "label_threshold" in result;<br>result["label_threshold"] in ("severe_warning", "flood_al...           | Pass   | 7.108s |
| Returns result dict                                                  | isinstance(result, dict)                                                                               | Pass   | 6.194s |
| severe_warning=0 exercises the else branch in log_hs computation.    | isinstance(result, dict)                                                                               | Pass   | 6.945s |
| Monthly rent and annual rent                                         | isinstance(result, list)                                                                               | Pass   | 0.000s |
| No transaction data                                                  | isinstance(self._page().generate_elements(prop, list)                                                  | Pass   | 0.000s |
| Rental yield computed                                                | isinstance(result, list)                                                                               | Pass   | 0.000s |
| Sale date shown                                                      | isinstance(result, list)                                                                               | Pass   | 0.000s |
| Sale price and appreciation                                          | any(isinstance(e, Table) for e in result)                                                              | Pass   | 0.000s |
| Sales section rendered                                               | any("Transaction" in t for t in texts)                                                                 | Pass   | 0.000s |
| Assess yield average                                                 | page._assess_yield(5.0) == "Average yield"                                                             | Pass   | 0.000s |
| Assess yield below average                                           | page._assess_yield(3.0) == "Below average yield"                                                       | Pass   | 0.000s |
| Assess yield excellent                                               | page._assess_yield(9.0) == "Excellent yield"                                                           | Pass   | 0.000s |
| Assess yield good                                                    | page._assess_yield(7.0) == "Good yield"                                                                | Pass   | 0.000s |
| Assess yield poor                                                    | page._assess_yield(1.0) == "Poor yield"                                                                | Pass   | 0.000s |
| Empty property does not crash                                        | isinstance(result, list)                                                                               | Pass   | 0.000s |

| Test Description                   | Acceptance Criteria                                                                                      | Result | Time   |
|------------------------------------|----------------------------------------------------------------------------------------------------------|--------|--------|
| Empty transactions                 | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Has transactions header            | <code>any("Transaction" in t or "History" in t for t in texts)</code>                                    | Pass   | 0.000s |
| Market analysis appreciation       | <code>"Capital Appreciation" in result</code>                                                            | Pass   | 0.000s |
| Market analysis few vacancies      | <code>"Good" in result["Rental Stability"]</code>                                                        | Pass   | 0.000s |
| Market analysis many vacancies     | <code>"Concerning" in result["Rental Stability"]</code>                                                  | Pass   | 0.000s |
| Market analysis marketing above 30 | <code>"Marketing Efficiency" in result;</code><br><code>"Good" in result["Marketing Efficiency"]</code>  | Pass   | 0.000s |
| Market analysis marketing above 90 | <code>"Below average" in result["Marketing Efficiency"]</code>                                           | Pass   | 0.000s |
| Market analysis never rented       | <code>"Rental History" in result;</code> "No<br><code>rental history" in result["Rental History"]</code> | Pass   | 0.000s |
| Market analysis no vacancy         | <code>"Rental Stability" in result;</code><br><code>"Excellent" in result["Rental Stability"]</code>     | Pass   | 0.000s |
| Market analysis yield              | <code>"Gross Rental Yield" in result</code>                                                              | Pass   | 0.000s |
| No transaction history             | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Returns list                       | <code>isinstance(result, list); len(result) &gt; 0</code>                                                | Pass   | 0.000s |
| Sales extended marketing           | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Sales quick sale                   | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| Sales very long marketing          | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| With mortgage data                 | <code>isinstance(result, list)</code>                                                                    | Pass   | 0.000s |
| With rental data                   | <code>isinstance(result, list);</code><br><code>any(isinstance(e, Table) for e in result)</code>         | Pass   | 0.000s |
| With sales data                    | <code>isinstance(result, list);</code><br><code>any(isinstance(e, Table) for e in result)</code>         | Pass   | 0.000s |
| Gauges alert count                 | <code>single_storm_data["trigger_summary"]["gauges_alert"] == 2</code>                                   | Pass   | 0.000s |
| Gauges impacted matches alert      | <code>ts["gauges_impacted"] ==</code><br><code>ts["gauges_alert"]</code>                                 | Pass   | 0.000s |
| Gauges severe count                | <code>single_storm_data["trigger_summary"]["gauges_severe"] == 1</code>                                  | Pass   | 0.000s |
| Gauges warning count               | <code>single_storm_data["trigger_summary"]["gauges_warning"] == 1</code>                                 | Pass   | 0.000s |

| Test Description                                        | Acceptance Criteria                                                          | Result | Time   |
|---------------------------------------------------------|------------------------------------------------------------------------------|--------|--------|
| Max trigger is alert when alert only                    | data["storms"][0]["trigger_summary"]["max_trigger"] == "a..."                | Pass   | 0.002s |
| Max trigger is severe                                   | single_storm_data["trigger_summary"]["max_trigger"] == "s..."                | Pass   | 0.000s |
| Max trigger is warning when no severe                   | data["storms"][0]["trigger_summary"]["max_trigger"] == "w..."                | Pass   | 0.003s |
| Trigger summary present                                 | "trigger_summary" in single_storm_data                                       | Pass   | 0.000s |
| Assessment date updated                                 | r.get_json()["assessment"]["assessment_date"] is not None                    | Pass   | 0.002s |
| No data file returns 404                                | r.status_code == 404                                                         | Pass   | 0.001s |
| Principle not found returns 404                         | r.status_code == 404                                                         | Pass   | 0.002s |
| Score 1 derives non compliant status                    | r.status_code == 200; p["status"] == "Non-compliant"                         | Pass   | 0.005s |
| Score 4 derives fully compliant status                  | p["status"] == "Fully Compliant"                                             | Pass   | 0.002s |
| Update evidence field                                   | r.status_code == 200; p["evidence"] == "New evidence gathered"               | Pass   | 0.002s |
| Update gaps and remediation                             | p["gaps"] == "Minor gap";<br>p["remediation"] == "Action plan created"       | Pass   | 0.002s |
| Valid update returns 200                                | r.status_code == 200;<br>r.get_json()["status"] == "success"                 | Pass   | 0.002s |
| Not found returns 404                                   | r.status_code == 404                                                         | Pass   | 0.002s |
| Update changes field                                    | ref["author"] == "New Author"                                                | Pass   | 0.004s |
| Update returns success                                  | r.status_code == 200;<br>r.get_json()["status"] == "success"                 | Pass   | 0.003s |
| Update with new category adds to list                   | "NewCat" in r.get_json()["categories"]                                       | Pass   | 0.003s |
| Update sets updated at                                  | r.get_json()["meeting"]["updated_at"] is not None                            | Pass   | 0.003s |
| Update title                                            | r.status_code == 200;<br>r.get_json()["meeting"]["title"] == "Updated Title" | Pass   | 0.002s |
| Update unknown id returns 404                           | r.status_code == 404                                                         | Pass   | 0.001s |
| Line 211: update_action with non-existent meeting.      | r.status_code == 404                                                         | Pass   | 0.002s |
| Line 147: update_decision with non-existent meeting.    | r.status_code == 404                                                         | Pass   | 0.002s |
| Line 333: update_minute_item with non-existent meeting. | r.status_code == 404                                                         | Pass   | 0.002s |

| Test Description                                        | Acceptance Criteria                                                                                     | Result | Time   |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------|--------|
| Line 275: update_participant with non-existent meeting. | <code>r.status_code == 404</code>                                                                       | Pass   | 0.002s |
| Empty value returns 400                                 | <code>r.status_code == 400</code>                                                                       | Pass   | 0.002s |
| Invalid choice value returns 400                        | <code>r.status_code == 400</code>                                                                       | Pass   | 0.002s |
| Invalid field returns 400                               | <code>r.status_code == 400;</code><br><code>r.get_json()["status"] == "error"</code>                    | Pass   | 0.002s |
| Model not found returns 404                             | <code>r.status_code == 404; r.status_code == 404 (+1 more)</code>                                       | Pass   | 0.002s |
| No reason returns 400                                   | <code>r.status_code == 400</code>                                                                       | Pass   | 0.002s |
| Response has audit entry                                | <code>"audit_entry" in data;</code><br><code>data["audit_entry"]["event_type"] == "field.update"</code> | Pass   | 0.003s |
| Update lifecycle stage                                  | <code>r.status_code == 200</code>                                                                       | Pass   | 0.004s |
| Update validation status                                | <code>r.status_code == 200</code>                                                                       | Pass   | 0.003s |
| Valid choice field update returns success               | <code>r.status_code == 200;</code><br><code>r.get_json()["new_value"] == "Amber"</code>                 | Pass   | 0.005s |
| Valid text field update returns success                 | <code>r.status_code == 200; data["status"] == "success" (+2 more)</code>                                | Pass   | 0.003s |
| No inventory returns 404                                | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error" (+3 more)</code>          | Pass   | 0.000s |
| Save failure returns 500                                | <code>r.status_code == 500;</code><br><code>r.get_json()["status"] == "error"</code>                    | Pass   | 0.002s |
| Activity not found returns 404                          | <code>r.status_code == 404</code>                                                                       | Pass   | 0.002s |
| Invalid role id returns 400                             | <code>r.status_code == 400</code>                                                                       | Pass   | 0.002s |
| Notes updated                                           | <code>activities[0]["notes"] == "New note"</code>                                                       | Pass   | 0.003s |
| Setting C or I to None (null) is valid.                 | <code>r.status_code == 200</code>                                                                       | Pass   | 0.003s |
| Valid activity update returns success                   | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>                  | Pass   | 0.003s |
| Escalation tier threshold updated                       | <code>triggers[0]["tier_threshold"]["1"] == "Immediate"</code>                                          | Pass   | 0.002s |
| Response has audit entry                                | <code>"audit_entry" in r.get_json();</code><br><code>"audit_entry" in r.get_json()</code>               | Pass   | 0.004s |
| Trigger not found returns 404                           | <code>r.status_code == 404</code>                                                                       | Pass   | 0.002s |
| Valid escalation update returns success                 | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>                  | Pass   | 0.003s |
| Backup field updated                                    | <code>owner["backup"] == "Owner B"</code>                                                               | Pass   | 0.002s |
| No raci file returns 404                                | <code>r.status_code == 404</code>                                                                       | Pass   | 0.001s |
| Response has audit entry                                | <code>"audit_entry" in r.get_json();</code><br><code>"audit_entry" in r.get_json()</code>               | Pass   | 0.002s |
| Role not found returns 404                              | <code>r.status_code == 404</code>                                                                       | Pass   | 0.002s |

| Test Description                                                           | Acceptance Criteria                                                                               | Result | Time   |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------|--------|
| Valid role update returns success                                          | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>            | Pass   | 0.003s |
| All valid statuses                                                         | <code>r.status_code == 200</code>                                                                 | Pass   | 0.009s |
| Invalid status returns 400                                                 | <code>r.status_code == 400;</code><br><code>r.get_json()["status"] == "error"</code>              | Pass   | 0.002s |
| Model not found returns 404                                                | <code>r.status_code == 404;</code><br><code>r.get_json()["status"] == "error"</code><br>(+3 more) | Pass   | 0.002s |
| Question not found returns 404                                             | <code>r.status_code == 404</code>                                                                 | Pass   | 0.002s |
| Risk rating recalculated                                                   | <code>rr["last_calculated"] is not None</code>                                                    | Pass   | 0.003s |
| Valid update has model and audit entry                                     | <code>"model" in data; "audit_entry" in data</code>                                               | Pass   | 0.003s |
| Valid update returns success                                               | <code>r.status_code == 200;</code><br><code>r.get_json()["status"] == "success"</code>            | Pass   | 0.003s |
| update_yield_curve creates yield_curve from default when missing.          | <code>'1' in updated['yield_curve'];</code><br><code>updated['yield_curve']['1'] == 0.055</code>  | Pass   | 0.002s |
| Empty filename returns 400                                                 | <code>r.status_code == 400</code>                                                                 | Pass   | 0.001s |
| No file returns 400                                                        | <code>r.status_code == 400; data["status"] == "error"</code>                                      | Pass   | 0.000s |
| Upload appears in list                                                     | <code>len(data["documents"]) == 1</code>                                                          | Pass   | 0.003s |
| Upload response has id                                                     | <code>"id" in data["document"]</code>                                                             | Pass   | 0.002s |
| Upload stores description                                                  | <code>data["document"]["description"] == "My description"</code>                                  | Pass   | 0.002s |
| Valid upload returns 200                                                   | <code>r.status_code == 200; data["status"] == "success" (+1 more)</code>                          | Pass   | 0.002s |
| Upload save failure returns 500                                            | <code>r.status_code == 500; data["status"] == "error" (+1 more)</code>                            | Pass   | 0.031s |
| Get page categories contains expected keys                                 | <code>key in cats</code>                                                                          | Pass   | 0.001s |
| Get page categories contains expected keys                                 | <code>key in cats</code>                                                                          | Pass   | 0.001s |
| Get page categories returns copy                                           | <code>"_new_key_" not in gen.categories</code>                                                    | Pass   | 0.001s |
| Get page categories returns copy                                           | <code>"_new_key_" not in gen.categories</code>                                                    | Pass   | 0.002s |
| Mutating the returned dict must not affect the generator's internal state. | <code>"_new_key_" not in gen.categories</code>                                                    | Pass   | 0.001s |
| Get page categories returns dict                                           | <code>isinstance(cats, dict)</code>                                                               | Pass   | 0.001s |
| Get page categories returns dict                                           | <code>isinstance(cats, dict)</code>                                                               | Pass   | 0.002s |
| Get page categories returns dict                                           | <code>isinstance(cats, dict)</code>                                                               | Pass   | 0.001s |
| List available pages contains all expected                                 | <code>expected &lt;= set(pages)</code>                                                            | Pass   | 0.002s |

| Test Description                                                      | Acceptance Criteria                                                                                                               | Result | Time   |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| List available pages contains all expected                            | <code>expected &lt;= set(pages)</code>                                                                                            | Pass   | 0.001s |
| List available pages contains expected                                | <code>expected &lt;= set(pages)</code>                                                                                            | Pass   | 0.001s |
| List available pages non empty                                        | <code>len(gen.list_available_pages()) &gt; 0</code>                                                                               | Pass   | 0.001s |
| List available pages non empty                                        | <code>len(gen.list_available_pages()) &gt; 0</code>                                                                               | Pass   | 0.001s |
| List available pages returns list                                     | <code>isinstance(pages, list)</code>                                                                                              | Pass   | 0.001s |
| List available pages returns list                                     | <code>isinstance(pages, list)</code>                                                                                              | Pass   | 0.002s |
| List available pages returns list                                     | <code>isinstance(pages, list)</code>                                                                                              | Pass   | 0.001s |
| Validate pages all invalid                                            | <code>valid == []; "bogus_one" in invalid</code><br>(+1 more)                                                                     | Pass   | 0.001s |
| Validate pages all invalid                                            | <code>valid == []; "no_such_page" in invalid</code><br>(+1 more)                                                                  | Pass   | 0.002s |
| Validate pages all invalid                                            | <code>valid == []; "bogus_one" in invalid</code><br>(+1 more)                                                                     | Pass   | 0.001s |
| Validate pages all valid                                              | <code>"title_overview" in valid;</code><br><code>"sensor_details" in valid</code> (+1 more)                                       | Pass   | 0.001s |
| Validate pages all valid                                              | <code>"title_overview" in valid;</code><br><code>"data_summary" in valid</code> (+1 more)                                         | Pass   | 0.002s |
| Validate pages all valid                                              | <code>"title" in valid; "appendix" in valid</code><br>(+1 more)                                                                   | Pass   | 0.001s |
| Validate pages empty list                                             | <code>valid == []; invalid == []</code>                                                                                           | Pass   | 0.001s |
| Validate pages empty list                                             | <code>valid == []; invalid == []</code>                                                                                           | Pass   | 0.002s |
| Validate pages empty list                                             | <code>valid == []; invalid == []</code>                                                                                           | Pass   | 0.001s |
| Validate pages mixed                                                  | <code>"title_overview" in valid;</code><br><code>"nonexistent_page" in invalid</code>                                             | Pass   | 0.001s |
| Validate pages mixed                                                  | <code>"title_overview" in valid;</code><br><code>"nonexistent_page" in invalid</code>                                             | Pass   | 0.002s |
| Validate pages mixed                                                  | <code>"title" in valid; "nonexistent_page"</code><br><code>in invalid</code>                                                      | Pass   | 0.001s |
| The error message should be a string representation of the exception. | <code>"validation_error" in result;</code><br><code>isinstance(result["validation_error"][0],</code><br><code>str)</code>         | Pass   | 0.000s |
| Another path to trigger the except branch.                            | <code>"validation_error" in result</code>                                                                                         | Pass   | 0.000s |
| Lines 251-252: passing data that causes an exception in the try       | <code>"validation_error" in result;</code><br><code>isinstance(result["validation_error"],</code><br><code>list)</code> (+1 more) | Pass   | 0.000s |
| Alt key structure                                                     | <code>validate_gauge_exists("GAUGE-002",</code><br><code>gauge_file) is True</code>                                               | Pass   | 0.001s |

| Test Description                                                                          | Acceptance Criteria                                                                                                     | Result | Time   |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Existing gauge returns true                                                               | <code>validate_gauge_exists("GAUGE-001", gauge_file) is True</code>                                                     | Pass   | 0.001s |
| Missing file returns false                                                                | <code>validate_gauge_exists("GAUGE-001", tmp_path / "missing.js...)</code>                                              | Pass   | 0.000s |
| Nonexistent gauge returns false                                                           | <code>validate_gauge_exists("GAUGE-XXX", gauge_file) is False</code>                                                    | Pass   | 0.001s |
| Binary file returns false                                                                 | <code>result is False</code>                                                                                            | Pass   | 0.001s |
| Corrupt json returns false                                                                | <code>result is False</code>                                                                                            | Pass   | 0.001s |
| Line 196: audit log overflow during VQ update.                                            | <code>r.status_code == 200</code>                                                                                       | Pass   | 0.036s |
| Line 130: inventory not found → 404.                                                      | <code>r.status_code == 404</code>                                                                                       | Pass   | 0.001s |
| Line 176: save failure → 500.                                                             | <code>r.status_code == 500</code>                                                                                       | Pass   | 0.002s |
| Update triggers risk recalculation                                                        | <code>rr["calculated_score"] is not None; rr["calculated_rating"] in ["Acceptable", "Conditional", ... (+1 more)</code> | Pass   | 0.005s |
| Update validation question                                                                | <code>response.status_code == 200; data["status"] == "success" (+5 more)</code>                                         | Pass   | 0.003s |
| Update validation question invalid status                                                 | <code>response.status_code == 400</code>                                                                                | Pass   | 0.003s |
| Update validation question not found                                                      | <code>response.status_code == 404</code>                                                                                | Pass   | 0.003s |
| When verbose=False, the per-counterparty log lines are absent.                            | <code>"[1/2]" not in caplog.text; "Wrote 2 counterparties" not in caplog.text</code>                                    | Pass   | 0.000s |
| Lines 169-170: when verbose=True, each counterparty is logged.                            | <code>len(result["data"]) == 3; "[1/3]" in caplog.text (+2 more)</code>                                                 | Pass   | 0.001s |
| Line 187: verbose=True logs final summary with count and path.                            | <code>"Wrote 2 counterparties" in caplog.text</code>                                                                    | Pass   | 0.000s |
| With count i 2000 and verbose=False, no mid-loop progress line.                           | <code>"ETA" not in out or "[" not in out</code>                                                                         | Pass   | 0.078s |
| count=501 with verbose=True triggers the progress line at i=500.                          | <code>result["num_sequences"] == 501; "[500" in out or "500/" in out</code>                                             | Pass   | 0.665s |
| <code>create_visualization()</code> returns a <code>TCEventVisualization</code> instance. | <code>isinstance(viz, TCEventVisualization)</code>                                                                      | Pass   | 0.000s |
| All addressed                                                                             | <code>_vq_summary(model) == "3/3"</code>                                                                                | Pass   | 0.000s |
| Empty questions                                                                           | <code>_vq_summary(model) == "0/0"</code>                                                                                | Pass   | 0.000s |
| None addressed                                                                            | <code>_vq_summary(model) == "0/2"</code>                                                                                | Pass   | 0.000s |



Some not applicable

`_vq_summary(model) == "1/2"`

Pass

0.000s

## Failed Test Details

**test\_eod\_snapshots\_exist** File: tests/data/test\_storm\_data.py

```
1 tests/data/test_storm_data.py:637: in test_eod_snapshots_exist
2     assert len(eod_files) >= 50, (
3 E   AssertionError: Only 0 EOD snapshots (expected ~63). Run 'python app.py port --blotter' to
      regenerate.
4 E   assert 0 >= 50
5 E   +   where 0 = len([])
```

**test\_returns\_script\_tag** File: tests/visual/interactivity/test\_backend\_handler.py

```
1 tests/visual/interactivity/test_backend_handler.py:56: in test_returns_script_tag
2     assert 'http://localhost:5013' in js
3 E   assert 'http://localhost:5013' in '<script>window.__BACKEND_CONFIG = {"url": "", "
      endpoints": {"health": "/api/v1/health"}, "timeout": 30000};\n// Copyr...n    }\n\n})(
      typeof window !== \'undefined\' ? window : (typeof global !== \'undefined\' ? global :
      this));\n</script>'
```